Dell PowerFlex Manager 5.1.x

User Guide

August 2026
Rev. 1.1

DELL Technologies

Notes, cautions, and warnings

NOTE: A NOTE indicates important information that helps you make better use of your product.

CAUTION: A CAUTION indicates either potential damage to hardware or loss of data and tells you how to avoid the problem.

WARNING: A WARNING indicates a potential for property damage, personal injury, or death.

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1

Introduction

This document provides procedures on how to manage PowerFlex rack, PowerFlex appliance, or Powerflex software through PowerFlex Manager.

The target audience for this document includes Dell Technologies customers and services.

This document provides information for PowerFlex nodes that are configured with erasure coding, mirroring, or both protection schemes. When content is specific to a single protection scheme, the applicable method (erasure coding or mirroring) is explicitly identified in the section title. If a section title does not specify a protection method, the content applies to both erasure-coded and mirrored configurations.

Erasure coding is supported with PowerFlex core software 5.0.x and above. Mirroring is supported with PowerFlex core software 4.5.x.

SupportAssist is now known as Dell Technologies connectivity services. References in the documentation will be updated over time.

For additional PowerFlex software documentation, go to Dell PowerFlex Software: Technical Documentation.

Common PowerFlex terms

The following table identifies common PowerFlex terms and associated acronyms.

Table 1. Common PowerFlex terms

Term	Acronym
device gateway target	DGWT
management virtual machine	MVM
metadata manager	MDM
PowerFlex data server	PDS
PowerFlex management cluster	PFMC
PowerFlex management platform	PFMP
storage data client	SDC
storage data target	SDT

See the PowerFlex glossary for more information about PowerFlex terms and acronyms.

2

Revision history

Table 2. Revisions

Date	Document revision	Description of changes
August 2026	1.1	Updates to the security content in the **Configuring system settings** section for PowerFlex nodes using erasure coding or mirroring.
June 2026	1.0	Initial release

3

Getting started with a PowerFlex deployment (mirroring)

This section provides instructions on how to get started with your PowerFlex system deployment through the initial Welcome wizard.

The initial Welcome wizard is a step-by-step wizard that guides you through the process of deploying PowerFlex nodes. Once you complete the initial setup script, the PowerFlex Manager user interface is configured and you can access the initial Welcome wizard through logging into the PowerFlex Manager user interface.

You must complete the initial setup script to access the initial Welcome wizard to start deploying PowerFlex nodes.

The purpose of the initial setup script is to gather minimal information to configure basic networking. The initial setup script is completed through the CLI when the first node boots with the pre-loaded operating system. The initial setup script configures the installer node, VM, and PowerFlex Manager ingress IP address with the appropriate networking. You can then transition to the web browser, log in to the PowerFlex Manager user interface, and access the initial Welcome wizard to configure your system.

Before you begin the deployment, review the prerequisite checklist to find out what information is required.

Plan and prepare

Before you start the deployment, you must:

- Review the prerequisite checklist to ensure that you have the information to configure your PowerFlex system.
- Configure the initial setup script to set up the basic networking to enable you to log in to the initial Welcome wizard to configure your PowerFlex system.
- If you plan to use an input properties file to automatically populate specific properties in the initial setup script, review the *Input properties file* topic before the *Configuring the initial setup* topic.
- If you plan to download a sample configuration file to automatically populate fields in the initial Welcome wizard, review the *Sample configuration file entries* under *Choose deployment type*.

Related information

Prerequisite checklist
Input properties file
Sample configuration file entries
Terminology (mirroring)

Prerequisite checklist

To ensure that the PowerFlex Manager deployment is complete, efficient, and free from unexpected issues, review the following checklist before you begin the deployment process:

Table 3. Prerequisite checklist

Item	Check
To log in to PowerFlex Manager, you need the following information:	
PowerFlex Manager username	
PowerFlex Manager password	
To upload a completed configuration JSON file, you must know the location of the configuration JSON file	

Table 3. Prerequisite checklist (continued)

Item	Check
To connect to Dell Technologies connectivity services by the gateway server, you need the following information:	
Primary gateway IP address	
Gateway port	
Secondary gateway IP address	
To authenticate to Dell Technologies connectivity services, you need the following information:	
Access key	
Pin	
ELMS eSWUID (unique software license ID for PowerFlex system)	
Solution serial number	
Site ID	
To enable support cases to be created, you need the following contact information:	
First name	
Last name	
Email address	
Phone number	
To upload a compliance file and or a compatibility file from a local network path, you need the following information:	
Release Certificate Matrix (RCM) or Intelligent Catalog (IC) version	
SMB share or HTTPS share	
File path	
Username	
Password	
To define your network requirements, you need to know the following information:	
Network architecture (either LACP or trunk)	
PowerFlex node types	
Number and types of networks	
Network automation type	
To configure the environment, you need the following information:	
Time zone	
IP address of the Network time protocol (NTP)	
PowerFlex Manager ingress hostname	
To configure a network, you need the following information for each network that you add:	
Name of the network	
Network type	
VLAN ID	
Subnet mask	
Gateway	
Four internal IP addresses for services within PowerFlex Manager	

Table 3. Prerequisite checklist (continued)

Item	Check
FQDN for the external ingress IP address	
To connect the installer node, you need the following information:	
IDRAC IP address configured on the installer node	
iDRAC credential username configured on the installer node	
iDRAC credential password configured on the installer node	
Know which version (V2 or V3) of SNMP that you will use	
SNMP community string for V2 or SNMP username and security level for V3	
Whether you want PowerFlex nodes to be automatically configured to send alerts to PowerFlex Manager:	
Operating system IP address configured on the installer node	
Operating system username configured on the installer node	
Operating system password configured on the installer node	
To connect the installer node to each network that you defined, you need the following information:	
IP address to be assigned to the installer node for each network	
IP address to be assigned to the management VM for each network	
To define the ports (teaming protocol LACP), you must know the two ports that you will use for the second interface pair	
To map the ports (teaming protocol LACP), you must know which networks should be hosted on which interface pair	
To map the ports (teaming protocol TRUNK), you must know which networks should be hosted on which interface	
To build the deployment, you need the following information:	
Hostname to assign to the installer node	
Hostname to assign to the management VM	
To configure the two PowerFlex nodes, in addition to the installer node, you need the following information for each PowerFlex node:	
IDRAC hostname or IP address	
IDRAC credential	
PowerFlex node hostname	
PowerFlex node credential	
Management VM hostname	
Management VM credential	
If you are using full network automation, to add the switches that are connected to the PowerFlex nodes, you need the following information:	
Hostname or IP address of the switch	
Switch credential	
To add PowerFlex node IP addresses for each network, you need the following information for each PowerFlex node:	
IP address to be assigned to the PowerFlex node for each network	
IP address to be assigned to the management VM for each network	

Related information

Prerequisite checklist
Input properties file
Sample configuration file entries
Terminology (mirroring)
Plan and prepare

Input properties file

The input properties file defines values for specific properties. You can use the input properties file to automatically populate specific properties in the initial setup script to avoid having to manually input the data.

Table 4. Properties and values

Property	Description	Example value
Switch port teaming protocol	Enter 1 for LACP and 2 for Trunk port.	`1`
Management network VLAN ID	Enter an integer 1–4094, which is assigned to the management network.	`546`
Management network IPv4 for host	Enter the wanted IP address for the KVM host in the management network.	`100.22.108.111`
Management network IPv4 for the PowerFlex Manager MVM	Enter the wanted IP address for the KVM management VM domain in the management network.	`100.22.108.112`
PowerFlex Manager UI (ingress) ingress IPv4 address	Enter the wanted IP address for the ingress IP address of the PowerFlex Manager services in the management network.	`100.22.108.113`
Management network subnet mask	Enter the CIDR prefix length or the netmask value for the management network, which is used for routing. For example, 24 (prefix length) or 255.255.255.0 (netmask).	`255.255.255.192`
Management network default IPv4 gateway	Enter the IP address of the gateway in the management network.	`100.22.108.124`
Management network main DNS server IPv4 address	Enter one or more comma-separated value (s) of the DNS name server IP address for the management network.	`100.68.68.1,100.68.68.2`
Management network DNS suffix	Enter the DNS domain names used for the hostname lookup for the management network.	`dell.com`
Nic devices	Enter one or more comma-separated values of the target NIC devices to use (if they are different than the default ones).	`p2p1, p3p1`

Related information

Prerequisite checklist
Input properties file
Sample configuration file entries
Terminology (mirroring)

Sample configuration file entries

You can choose to download an empty configuration json file, enter the values for your configuration, and then upload the file through the initial Welcome wizard. This saves you time entering the data manually in the initial Welcome wizard. You must verify the data and save the data on each screen in the initial Welcome wizard.

Table 5. Sample configuration JSON file entries

Step	Sample entry

Download compliance file from Dell Technologies connectivity services	Use preloaded Intelligent Catalog:
	`"compliance": {`
	`"compliance": {`
	`"type": "preloaded",`
	`"path": "",`
	`"username": "",`
	`"password": ""`
	`},`
	`"compatibility": {`
	`"type": "preloaded",`
	`"data": "",`
	`"compatibilityFilename": ""`
	`}`
	`},`

	Upload compliance file from the local network path:
	`"compliance": {`
	`"type": "network",`
	`"path":`
	`"https://100.xx.xx.xx/folder//RCM//RCMs/Flex_RCM_4_8_x_x_r1.zip",`
	`"username": "",`
	`"password": ""`
	`}`

Define network requirements	Configure a PowerFlex Management network:
	`networkConfiguration": {`
	`"networkMode": "FULL",`
	`"networks": [`
	`{`
	`"id": "management",`
	`"name": "PowerFlex Management",`
	`"description": "",`
	`"type": "SCALEIO_MANAGEMENT",`
	`"vlanId": 547,`
	`"subnetMask": "xxx.xxx.xxx.xxx",`
	`"gatewayIp": "xxx.xx.xxx.xxx",`
	`"internalIps": [`
	`"xxx.xx.xxx.xxx",`
	`"xxx.xx.xxx.xxx",`
	`"xxx.xxx.xxx.xxx",`
	`"xxx.xx.xxx.xxx"`
	`],`
	`"externalIngressIp":`
	`"100.68.126.250"`
	`},`

Related information

Configure PowerFlex nodes through the initial Welcome wizard
Plan and prepare
Prerequisite checklist
Input properties file

Retrieve the configuration file

You must have the configuration file in order to restore PowerFlex Manager from a backup. You can generate the configuration file through an API call.

1. Connect to the management virtual machine using the **delladmin** credentials.
2. To generate the configuration file, run:

```
svc_ip=$(kubectl get svc -n powerflex thin-deployer | grep -v 'CLUSTER' | awk -F ' '
'{print $3}')
# Collect the file in JSON File
curl -X GET https://${svc_ip}:9433/asm/pfmp_config -k -o PFMP_CONFIG.json
```

Here is an example of the configuration file output:

```
 "Nodes": [
    {
      "hostname": "abcde-1",
      "ipaddress": "100.xx.xxx.xxx"
    },
    {
      "hostname": "abcde-2",
      "ipaddress": "100.xx.xxx.xxx"
    },
    {
      "hostname": "abcde-3",
      "ipaddress": "100.xx.xxx.xxx"
    }
  ],
  "ClusterReservedIPPoolCIDR": "10.xx.0.0/xx",
  "ServiceReservedIPPoolCIDR": "10.xx.0.0/xx",
  "RoutableIPPoolCIDR": [
    {
      "flex-data1": "xxx.xxx.xxx.1-xxx.xxx.xxx.4"
    },
    {
      "flex-data2": "xxx.xxx.xxx.1-xxx.xxx.xxx.4"
    },
    {
      "flex-stor-mgmt": "xxx.xx.xxx.xxx-xxx.xx.xxx.xxx"
    }
  ],
  "PFMPHostIP": "100.xx.101.xxx"
```

NOTE: The network name is what was defined.

3. Run the restore script that is included with the installer bundle:

```
./restore_backup.sh
```

4. Provide details to complete the execution of the script.

   You are prompted to provide the SSH username and password. In addition, you are asked to specify whether the passwords are the same for all nodes, and also provide the location of the backup file and the encrypted password for the file. You may also be asked to provide the CIFS username and password. The CIFS credentials may not be required for a CIFS share that allows for anonymous access.

   NOTE: The passwords must be in base64 encoded format.

   To complete the execution of the restore script, you must specify whether the restore operation will be performed on an existing cluster or a new cluster.

   Here is a snippet that shows a sample run of the restore script:

```
[root@sheetal-installer scripts]# ./restore_backup.sh

/usr/local/lib/python3.8/site-packages/paramiko/transport.py:236:
CryptographyDeprecationWarning: Blowfish has been deprecated "class":
```

```
algorithms.Blowfish, Installation logs are available at <Bundle root>/PFMP_Installer/
logs/ directory. More detailed logs are available at <Bundle root>/atlantic/logs/
directory. PFMP Installer is about to reset a PFMP cluster based on the configuration
specified in the PFMP_Config.json.

Please enter the ssh username for the nodes specified in the
PFMP_Config.json[root]:root

Are passwords the same for all the cluster nodes[Y]?:Y

Please enter the ssh password for the nodes specified in the PFMP_Config.json.
Password:

Please enter backup file location - Example CIFS: \\1.2.3.4\Lab\backup\file.tgz

Please enter CIFS username. Press enter to skip if username is not
required:administrator

Please enter CIFS password(base64 encoded). Press enter to skip if username is not
required: UmFpZDR1cyE=

Please enter encryption password for backup file (base64 encoded): UmFpZDR1cyE=

Perform restore on existing cluster? Please enter yes/y or no/n :
```

The restore process prints status information until the restore is complete.

Related information

Restoring PowerFlex Manager

Configure the initial setup

The initial setup script collects basic networking information to set up the management interface. Then, the node, VM, and PowerFlex Manager are configured with the appropriate networking. You can log in to the initial Welcome wizard to complete the configuration of your system.

Review the *Input properties file* topic to ensure that you have the information to complete the initial setup script.

Ensure that the switches are correctly configured.

The first step in the initial setup process is to configure the management network with a static IP address. The management network is configured with a bond through a preconfigured LACP or Trunk port in the switches.

To configure the management network, you can use an input properties file or you can enter the data manually. This procedure provides steps on how to manually add data to the initial setup script.

1. Log in to IDRAC with the default username and password. The default username is: **root**.
2. Open the virtual console and log in to the operating system using the default username and password.
3. You can choose to a predefined input properties file or manually enter the data.
   a. To use a predefined input properties file, type **y** and press **Enter**.
   b. If you do not have an input properties file created, when prompted to enter the value to use for the full path of the input properties file, stop the script.

   Otherwise, type **n** and press **Enter** to continue to the manual data entering process.
4. Enter a value for the `teaming protocol` that is preconfigured on the switches, choose one of the following:
   - Select option **1** for LACP (mode 802.3ad) and press **Enter**.
   - Select option **2** for Trunk (mode 1) and press **Enter**.

   NOTE: If the default value is incorrect, type **n** and press **Enter**. You can reenter the correct value for any of the properties.
5. To confirm your choice, (**y/n**), type **Y** and press **Enter**.
6. Enter the value to use for the `management network VLAN ID` and press **Enter**.
7. To confirm your choice, (**y/n**), type **Y** and press **Enter**.
8. Enter the value for the `management network IPv4 for host` and press **Enter**.

9. Confirm that the default value is correct, type **y** and press **Enter**.
10. Enter the value for the `management network IPv4 for the PowerFlex Manager MVM` and press **Enter**.
11. Confirm that the value is correct, type **y** and press **Enter**.
12. Enter a value for the `Powerflex Manager UI (ingress) ingress IPv4 address` and press **Enter**.
13. Confirm that the value is correct, type **y** and press **Enter**.
14. Enter a value for the `management network subnet mask` and press **Enter**.

    NOTE: If you enter an invalid notation, you have three attempts in total to correct it.

15. Confirm that the value is correct, type **y** and press **Enter**.
16. Enter a value for the `management network default IPv4 gateway`, and press **Enter**.
17. Confirm that the value is correct, type **y** and press **Enter**.
18. Enter a value for the `management network main DNS server IPv4 address` and press **Enter**.
19. Confirm that the value is correct, type **y** and press **Enter**.
20. Enter a value for the `secondary DNS server` and press **Enter**.
21. Confirm that the value is correct, type **y** and press **Enter**.
22. Enter a value for the `management network DNS suffix` and press **Enter**.
23. Confirm that the default value is correct, type **y** and press **Enter**.
24. A list of available NICs is displayed along with the default NICs. Confirm the usage of the default NICs, type **y** and press **Enter**.
25. To validate the management network information, confirm that the values are correct, type **y** and press **Enter**.

    NOTE: If the default values are incorrect, type **n** - repeat the configuration process.

26. The script runs and assigns the IP address to the management MVM and allows the MVM to start running.

    NOTE: The script takes ten to thirty minutes to complete.

    Once the script is complete, an IP address is allocated to the management MVM and the MVM starts to run. The components for PowerFlex Manager are on the MVM.
27. Log in to the PowerFlex Manager user interface with the IP address provided.
    The initial Welcome wizard opens to enable you to complete the configuration.

Related information

Input properties file

Configure PowerFlex nodes through the initial Welcome wizard

Once the initial setup script is completed, you can log in to PowerFlex Manager and access the initial Welcome wizard to configure the installer node and any additional nodes that you want to add to your PowerFlex system.

Related information

Sample configuration file entries
Terminology (mirroring)

Choose deployment type

You can choose to deploy PowerFlex nodes with an integrated management cluster, or as a PowerFlex management cluster, through the initial Welcome wizard.

Ensure you review the prerequisite checklist to find out what information is required before you start the deployment.

The integrated management cluster deployment option deploys the PowerFlex management platform onto three PowerFlex storage nodes where PowerFlex Manager and PowerFlex storage co-reside on the same node on your system.

The PowerFlex management cluster deployment option deploys PowerFlex nodes as a dedicated PowerFlex management cluster. The PowerFlex management cluster hosts the management VMs and applications that manage your PowerFlex rack or PowerFlex appliance nodes which in turn host your production storage.

1. Log in to PowerFlex Manager using your default credentials. For more information, see the *Dell PowerFlex Family 4.x Security Configuration Guide*.
   The initial Welcome wizard opens.
2. You must change your password to activate your account.
3. On the **Welcome** screen, click the tile of your chosen deployment option.
   A configuration window displays.
4. You can choose how you want to start configuring your system. Select one of the following:
   - **Upload a configuration**: With this option, you can upload a completed configuration `json` file that is saved. The information is autofilled for each step of the initial welcome wizard. The information is validated at each step, and the system is deployed.
   - **Download an example configuration json file**: With this option, you can download an example configuration `json` file and edit it for your own system. You can then upload the completed configuration. The information is autofilled for each step of the initial welcome wizard. The information is validated at each step, and the system is deployed. This method of configuration saves time, and you can reuse the configuration for future deployments.
   - **New configuration**: With this option you enter your configuration details step by step. The information is validated at each step, and the system is deployed.

   NOTE: Once you click **Next**, you cannot change the configuration type.
5. Click **Next**.

Enable Dell Technologies connectivity services

Dell Technologies connectivity services uses the secure connect gateway to automate issue detection and support case creation, resolve issues remotely, and provide critical information to Dell remote support to resolve complex issues.

If you do not want to enable Dell Technologies connectivity services, you can skip this step and enable Dell Technologies connectivity services later through adding it as a destination to a notification policy in Events and Alerts.

The secure connect gateway enables the following:

- Dell Technologies connectivity services to open service requests automatically because alerts are passed from PowerFlex Manager to Dell Technologies connectivity services through the secure connect gateway.
- Dell Technologies Support to leverage a capability using Dell Technologies connectivity services backend to connect through the secure connect gateway to the system and use Remote Connect for troubleshooting issues remotely.
- Configuration data and metrics to be sent to Dell AIOps (formerly known as CloudIQ) to enhance monitoring (if Dell AIOps is enabled).

1. On the **Overview** screen, click **Next**.
2. On the **Telemetry Agreement** screen, select the checkbox to accept the agreement and click **Next**.
3. On the **Connection Type** screen, you can choose either of the following options:
   a. To connect directly to Dell Technologies, click **Connect directly**.
   b. To connect to a Dell Technologies connectivity services gateway server by remote access, click **Connect via gateway server** and provide the primary gateway IP address or hostname.

      The port number is auto populated to `9443`.

   c. Optionally, to include an additional gateway server, click **Add Gateway**. The maximum number of gateways is eight.
   d. Provide the secondary gateway IP address or hostname.
   e. To enable PowerFlex Manager to send telemetry data, alerts, and analytics through Dell Technologies connectivity services, turn **Connect to Dell AIOps** to on.
   f. Click **Mask IP Addresses** to replace all your IP addresses with xxx.xxx.xxx.xxx before any information is sent about your environment to Dell Technologies Support.
   a. To enable Dell Technologies support engineers to troubleshoot your system remotely, turn **Enable remote support** to on.

NOTE: If you choose to enable this feature now, you can disable it in the future.

b. Click **Next**.

4. On the **Authentication** screen, details:
   a. Under **Authentication Details**, provide the following detail to verify a secure connection:
      - Access key
      - Pin
   b. Under **License Type**, choose either of the following options:
      - If you are using dynamic licensing, select **I have a license authorization code (LAC)**.
      - If you are using a key-based license, select **I have a license file in xml format**.
   c. Under **Device Registration**, provide the details of your device to register with Dell Technologies connectivity services.

Table 6. Device registration details

Device type	Information required
PowerFlex rack	• Type • Solution serial number • Castle ID (Site ID) • ELMS eSWUID (required only for key-based licensing)
PowerFlex appliance	• Type • Solution serial number • Castle ID (Site ID) • ELMS eSWUID (required only for key-based licensing
PowerFlex software	• Type • Deployment environment - one of the following: ○ Amazon Web Services ○ Microsoft Azure ○ On-premises PowerFlex custom node ○ On-premises other node type

   d. Click **Next**.
5. On the **Support Contacts** screen, identify the primary support contact by providing their first name, last name, email, and phone details. Optionally, you can add up to two additional support contacts.
6. Click **Next**.

Ensure compliance

To ensure that your system is compliant, you must first download the compliance file and then upload the compliance and compatibility files.

Ensure that you select the correct option for the compliance and compatibility file because you cannot change the selection after you click **Next**.

If Dell Technologies connectivity services is not configured, the compliance and compatibility files must be uploaded from the local network path or a preloaded intelligent catalog must be used.

1. On the **Ensure Compliance** screen, to monitor resource firmware and hardware compliance, select one of the following options:
   - Download compliance file from Dell Technologies connectivity services
   - Use pre-loaded Intelligent Catalog
   - Upload compliance file from the local network path. You must enter a file path, username, and password.

   You can test the status of the connection by clicking **Test connection**.
2. To facilitate upgrades, select one of the following options:
   - Download compatibility file from Dell Technologies connectivity services
   - Use pre-loaded compatibility file version

- Upload compatibility file from local. You can browse to the file location.

3. Click **Next**.

Define network requirements

The minimum network requirements for deployment are a management network and two data networks. The mandatory networks are already added but must be configured.

There are six network types:

- PowerFlex management
- Hypervisor (operating system) management
- General purpose LAN
- PowerFlex data
- PowerFlex data (client traffic only)
- PowerFlex data (server traffic only)

In some environments, the minimum network requirements of a management network and two data networks are sufficient. But in other cases, adding additional networks may be required. Your networking setup determines the requirements.

If additional networks are required but not configured, it is assumed that the operating system management IP address and the hardware management IP addresses (iDRACs) are accessible from the PowerFlex management network.

If the operating system management IP address and the hardware management IP addresses (iDRACs) are not accessible from the PowerFlex management network, ensure that you define additional networks for operating system and hardware management. To determine which networks to add, you must consider you needs and your physical network configuration.

1. On the **Define Network Requirements** screen, select the checkbox beside one of the following network automation types:

   Full network automation: If you have supported switches, select this option. PowerFlex Manager works with fully integrated supported and managed switches requiring less manual configuration. Full network automation also provides better error handling since PowerFlex Manager communicates with the switches and identifies any problems that may exist with the switch configurations.

   Partial network automation: If you have unsupported switches or your network management does not allow for automation of switch configuration, select this option. The unsupported switches are not managed. You must ensure that the switches are configured correctly, since PowerFlex Manager does not configure the switches for you. If your switch is not configured correctly, the deployment may fail. PowerFlex Manager cannot provide information about why the deployment failed and/or switch misconfigurations.

2. To configure the environment, click **Environment**. A **Configure Environment** screen opens.
   a. Select a time zone.
   b. Enter the IP address or fully qualified domain name (FQDN) of one or more NTP servers. Use commas to separate multiple entries.
   c. Enter the PowerFlex Manager ingress hostname. For example, **`dellpowerflex.com`**.

   NOTE: The primary DNS, secondary DNS, and DNS suffix are read-only because they contain the values you entered in the initial setup script.

3. Click **Save**. To enable PowerFlex Manager to automatically configure nodes, click **+ Add network**.
4. To configure a PowerFlex management network, from the list of networks, click **PowerFlex Management**, and enter the following network information:
   a. Specify four internal IP addresses for PowerFlex Manager.
   b. Specify the external ingress IP address for PowerFlex Manager.
   c. Click **Save**.

   NOTE: The fields that are pre-populated for the PowerFlex management network can be edited. The PowerFlex management network cannot be deleted from the list of networks.

5. To configure a data 1 and data 2 network, enter the following information for each of the data networks:
   a. Enter a VLAN ID which can be any number between 1 and 4094.
   b. Enter a subnet mask.
   c. (Optional) Enter a gateway.
   d. Specify four internal IP addresses for PowerFlex Manager.

e. Specify the external ingress IP address for PowerFlex Manager.
f. Click **Save**.

NOTE: The fields that are pre-populated for the data networks can be edited. The data networks can be deleted from the list of networks.

6. If you are deploying a PowerFlex management cluster (KVM), you must add an out-of-band-management network, enter the following network information:
   a. Enter a name. For example, **`Out of Band`**.
   b. (Optional) Enter a description. For example, **`Used for accessing PowerFlex node IDRAC`**.
   c. Select **General Purpose LAN** from the dropdown.
   d. Enter a VLAN ID which can be any number between 1 and 4094.
   e. Enter a subnet mask.
   f. (Optional) Enter a gateway.
   g. Specify four internal IP addresses for PowerFlex Manager.
   h. Specify the external ingress IP address for PowerFlex Manager.
   i. Click **Save**.

   NOTE: The out-of-band management network can be deleted from the list of networks.

7. Ensure that the networking information is correct because once you click **Next**, you cannot change the networking details.
8. A message is displayed verifying that you want to continue to the installer node step. Click **Continue**.

Connect installer node

The information that you provide for the installer node is configured and validated in each step of the initial Welcome wizard. The installer node is then deployed on the Deploy step and cannot be modified thereafter.

The connect the installer node consists of five sub-steps:

1. Connect node
2. Connect node to the network
3. Define ports
4. Map ports
5. Deploy

Connect node

To connect the installer node, which is the current node that is running, PowerFlex Manager requires the iDRAC and operating system information that you assigned to the installer node when you powered on the node.

1. On the **Connect Installer Node** screen, from the **Operating system** section, enter the following information that you assigned to the installer node when you powered on the node:
   a. Operating system IP address
   b. Node hostname
   c. Management VM hostname
2. From the **Operating system credentials** section, enter the information that you assigned to the installer node when you powered on the node:
   a. Credential name
   b. Username
   c. Password
   d. Confirm password
3. Click **Next**.

Connect nodes to networks

You must enter the operating system IP address of both the node and the management VM that you want assigned to the networks. The IP addresses are pre-populated and assigned to the PowerFlex Management network through completing the initial setup script.

Once validation is successful, you cannot change the values that you entered.

1. On the **Connect Node to Networks** screen, for the **Data 1** network, enter the node IP address and the management VM IP address of the installer node.
2. For the **Data 2** network, enter the node IP address and the management VM IP address of the installer node.
3. Wait while the validation completes. Validation ensures that the installer node can be connected to each network.
4. Once validation is successful, you will see a message `The network is successfully connected`.
5. Click **Next** to define the ports.

Define ports (LACP)

You must create and configure two interface pairs for the teaming protocol (LACP). You already created port pair one in the initial setup script. You must now create port pair two by selecting two interfaces for pair two.

The port pair one that is already configured in the initial setup script is displayed in the diagram as pre-populated and disabled. Irrespective of how many ports you have, only four are used. When selecting the two ports for port pair two, keep in mind that the same port cannot be used more than once.

You can choose to select a separate physical interface to access the hardware (out-of-band) management network.

1. On the **Define Ports** screen, from the first **OS interface** drop-down, select an interface for port pair two.
2. From the second **OS interface** drop-down, select a second interface for port pair two.
3. Wait while the validation completes. Validation is only on the values that you entered.
4. Click **Next** to map the ports.

Define ports (Trunk)

The PowerFlex management network is already configured to run on a group of interfaces. The data networks are configured on standalone interfaces.

When you completed the initial setup script, you selected four ports that will be used on this node. These ports are already configured and display in the diagram as pre-populated and disabled.

You can choose to select a separate physical interface to access the hardware (out-of-band) management network.

1. On the **Define Ports** screen, from the first **OS interface** drop-down, select an interface for port pair two.
2. From the second **OS interface** drop-down, select a second interface for port pair two.
3. Wait while the validation completes. Validation is only on the values that you entered.
4. Click **Next** to map the ports.

Map ports (LACP)

You must specify the bonded port pairs for an LACP configuration. Only two port pairs are supported. The two ports you selected in the initial setup script become one pair. The remaining two ports become the second pair.

PowerFlex management is already configured to run on port pair one and is read-only.

1. On the **Map Ports** screen, from the dropdown select a port pair for the data 1 network.
2. From the dropdown, select a port pair for the data 2 network.
3. Wait while the validation completes. Validation is only on the values that you entered.
4. Click **Next** to deploy.

Map ports (Trunk)

The PowerFlex management network is already configured to run on the ports. You must identify the ports for the remaining networks.

PowerFlex management is already configured to run on port pair one and is read-only.

1. On the **Map Ports** screen, from the dropdown select a port pair for the **Data 1** network.
2. From the dropdown, select a port pair for the **Data 2** network.
3. From the dropdown, select a port pair for the **Data 3** network.
4. From the dropdown, select a port pair for the **Data 4** network.
5. Wait while the validation completes. Validation is only on the values that you entered.
6. Click **Next** to deploy.

View installer node deployment status

You can view the status of the installer node deployment step-by-step.

During the deployment, there will be a temporary loss of access to the user interface. After the deployment, your browser redirects to a new page. You are required to access the new certificate and log into PowerFlex Manager user interface again.

1. On the **Deploy** screen, view the deployment status at each step.
2. Click **Next**.
3. A message is displayed to let you know the node is restarting. After the node restarts, log in to PowerFlex Manager to view the deployment status.

You may have to wait while the node is being recovered.

Build deployment

You must add a minimum of three nodes to the deployment. The installer node is already added. Therefore, you must add two additional nodes. After a node is added, it becomes a management node. If you are using full network automation, you can also add additional switches.

It may take several minutes for a node to be added because of additional validation.

1. On the **Build Deployment** screen, select the installer node. The **Installer Node** screen opens.
   a. Enter the hostname or IP address that you assigned to the iDRAC when completing the initial setup script.
   b. To assign an iDRAC credential, select an existing credential from the dropdown list or create a credential by clicking **+**.
   c. If you want to auto-configure the node to send alerts to PowerFlex Manager, turn the toggle to on.
   d. Under the **Node Operating System** section, enter the node hostname that will be assigned to the operating system,
   e. To assign a credential to the operating system of the node, either select an existing credential from the dropdown or create a credential by clicking **+**.
   f. Under the **Management VM Operating System** section, enter a management VM hostname that will be assigned to the operating system.
   g. Click **Save**.
2. On the **Build Deployment** screen, select a second node by clicking **Configure node**. A **Configure Node** screen opens.
   a. Enter the hostname or IP address that you assigned to the iDRAC when completing the initial setup script.
   b. To assign an iDRAC credential, select an existing credential from the dropdown list or create a credential by clicking **+**.
   c. If you want to auto-configure the node to send alerts to PowerFlex Manager, turn the toggle to on.
   d. Under the **Node Operating System** section, enter the node hostname that will be assigned to the operating system,
   e. To assign a credential to the operating system of the node, either select an existing credential from the dropdown or create a credential by clicking **+**.
   f. Under the **Management VM Operating System** section, enter a management VM hostname that will be assigned to the operating system.
   g. To add a management VM credential that will be assigned to the operating system, select an existing credential from the dropdown or create a credential by clicking +.
   h. Click **Save**.
3. If you selected full network automation when you defined your networks, you must add one or more switches by completing the following steps:

a. From the **Switches** section, click **Configure switch**.
   A **Configure Switch** opens.
b. To add an IP address for the switch, enter an IP address in the field.
c. To add a credential to the switch, select a credential from the dropdown or create a credential by clicking **+**.
d. If you want to configure your Dell operating system 10 switch to send alerts to PowerFlex Manager, turn the toggle to on.
e. Click **Save**.
f. To add additional switches, click **+ Add switch**.

NOTE: You are required to configure the first switch for a deployment with full network automation. The first switch cannot be deleted.

Add IP addresses

You can enter the node IP address and the management VM IP address for each node that you added on each specific network. The IP addresses are pre-populated and already assigned to the PowerFlex Management network through completing the initial setup script.

The installer node is ready only because the values were already added in the connect installer node step.

1. From the **Add IP Addresses** screen, select the node that you want to add both the node and management VM operating system IP address to.
2. To add the IP addresses to the **Data 1** network, enter the operating system IP address in the **Node IP address** and the **Management VM IP address** fields.
3. To add the IP addresses to the **Data 2** network, enter the operating system IP address in the **Node IP address** and the **Management VM IP address** fields.
4. Wait while validation completes and click **Next**.

Deploy system

You can review the deployment and export your configuration in the deploy system step.

The compliance file and compatibility files that are downloaded in the **Ensure Compliance** step must be fully downloaded before you can deploy. These files are large and can take time to download.

If the files are still downloading when you reach the **Deploy system** step, you will see a loading icon on the **Ensure Compliance** step and the **Deploy** button is disabled.

1. From the **Deploy System** screen, you can review the deployment summary of your chosen deployment type.
2. To view more details of each step, click **>** to expand the section and view more details.
3. Click **Deploy**.

   NOTE: During the deployment, there is a temporary loss of access to the user interface.

4. To export the configuration, click **Export**.
5. To view the latest 200 lines of the log file, click **Log Messages**.

The **Deployment Status** screen displays the status of each step of the deployment along with a start and elapsed time. The progress bar displays the percentage completeness.

Next steps

Once the discovery is complete, you should see the PowerFlex system resource that you deployed.

You must then add PowerFlex settings to your deployment.

NOTE: If you delete the PowerFlex system resource group before you apply PowerFlex settings, you must start the PowerFlex deployment process again.

To see the PowerFlex system, go to **Lifecycle** > **Resources** > **All Resources** > **View Details** page.

The **Details** page displays detailed information about the PowerFlex system and the associated components.

Related information

Viewing resource details
Add PowerFlex settings to a PowerFlex management cluster (KVM)
Add PowerFlex settings to an integrated PowerFlex management (KVM) environment

4

Welcome to PowerFlex Manager

This section introduces the main features of PowerFlex Manager.

PowerFlex Manager provides a level of IT operations management based on your chosen consumption model (Dell PowerFlex rack, Dell PowerFlex appliance, PowerFlex custom node, or just PowerFlex software). It increases efficiency by reducing time-consuming manual tasks that are required to manage system operations. Use PowerFlex Manager to deploy and manage new and existing environments.

PowerFlex Manager includes the following features:

- A dashboard that provides system configuration details and communicates health status for infrastructure elements and resource groups
- Firmware and software compliance monitoring and reporting
- Compliance remediation for nodes, switches, VMware ESXi, PowerFlex, and CloudLink
- Hardware monitoring and alerting through either Dell Technologies connectivity services, email, or SNMP and syslog to Dell Technologies Support
- Aggregated logging for troubleshooting with the ability to send logs through Dell Technologies connectivity services to Dell Technologies Support

Use PowerFlex Manager to define and capture infrastructure requirements to support a specific business outcome. Once defined, the process can be repeated through automation.

New features

This PowerFlex release introduces the following new features:

Table 7. New features

Feature	Details
Allocation granularity of 1 GB	The minimum size of a volume is 1 GB. In this release, you can increase the volume size in increments of 1 GB. For example, if you have a volume that is 10 GB, you can increase it to 11 GB.
CloudLink	Added support for CloudLink 8.3. CloudLink 8.3 and later versions are supported only on PowerFlex platforms. **NOTE:** CloudLink 8.2 The Key Management Interoperability Protocol (KMIP) server and machine encryption functionalities are available. CloudLink 8.3 KMIP server and machine encryption functionalities remain operational but are deprecated. They are no longer covered under standard support and are planned to be disabled in a future release. Dell Technologies strongly recommends planning a transition to supported alternative solutions to ensure continued functionality and support. For guidance on available options or migration planning, contact your Dell Technologies account representative.
Dynamic licensing	In this release, PowerFlex supports dynamic licensing. Dynamic Licensing offers an enhanced licensing experience by automating license management activities, such as registrations, capacity expansions, and subscription renewals. You can register

Table 7. New features (continued)

Feature	Details
	software instances and share licensing information between the product instance and Dell. Dynamic licensing introduces the ability to manage license allocations more easily across product registrations, enabling a flexible and keyless experience. Through PowerFlex Manager, you can register your device and migrate from key-based XML format licensing to dynamic licensing. See the *Dell PowerFlex Family Licensing Guide* for more information.
Modify the properties of a storage node (for erasure coding only)	In this release of PowerFlex Manager, you can modify the properties of a storage node. You can complete the following: ● Rename a storage node ● Add or remove an IP address from a storage node ● Modify the role associated with an IP address of a storage node
Pause inventory jobs	In this release, PowerFlex Manager introduces the option to pause scheduled inventory jobs for up to 24 hours, ensuring upgrades can complete without interruption.
Protection schemes	Added the 4+2 protection scheme.
Storage data clients on VMware Cloud Foundation platform	You can install and upgrade storage data clients on a VMware Cloud Foundation (VCF)-based node or a VMware ESXi 9.x-based node.
User experience enhancements	In this release, PowerFlex Manager delivers a simplified user experience that is related to storage pools, volumes, and snapshots and enhanced capacity metrics reporting. ● You can view the security and expiration that is associated with a volume snapshot. ● The following capacity metrics are now available through the PowerFlex Manager dashboard: ○ Reducible DRR: The data reduction ratio for the reducible data within the system or protection domain. This metric ignores any unreducible data in the calculation. ○ Unreducible data: The amount of data that is written within the system or protection domain which cannot be reduced by compression or the pattern detection logic PowerFlex uses.

Terminology (erasure coding)

The following table describes important PowerFlex Manager terms and concepts.

Table 8. Terminology

Term	How used in PowerFlex Manager
Compliance	Compliance is the comparison between current firmware and software levels and target compliance versions, which contain the baseline firmware and software versions. PowerFlex Manager shows any deviation from the baseline in the compliance status of the resources. You can use PowerFlex Manager to initiate updates to bring the resources to a compliant state. To facilitate upgrades of the management software, and compliance upgrades. PowerFlex Manager steers you toward valid upgrade paths. If the current version of the software is incompatible with the target version, when you attempt an upgrade, PowerFlex Manager displays a warning. PowerFlex Manager also warns you if any of the RCM or IC versions that are loaded are incompatible with the target compliance versions.
Destination	A destination defines the external location where PowerFlex Manager sends events and alerts to, for example, SNMP.

Table 8. Terminology (continued)

Term	How used in PowerFlex Manager
Device group	A device group consists of devices that come from five or more nodes that are contained in a single protection domain. The nodes that belong to a device group have the same media type.
Device	A storage device or persistent memory device that is added to a storage system.
Device Gateway Target	A Device Gateway Target (DGWT) is a distributed capacity service running on the storage node. The DGWT service processes and saves the data to the devices. The component is accountable for placing data on the storage drives using erasure coding protection schemes.
Host	A host is an entity that consumes PowerFlex storage for application usage. There are two methods of consuming PowerFlex block storage: using the SDC kernel driver, or using NVMe over TCP connectivity. Therefore, a host is either an SDC or an NVMe host.
NVMe target	A frontend component that translates NVMe over Fabric protocol into internal PowerFlex protocols. The NVMe target provides I/O and discovery services to NVMe hosts configured on the PowerFlex system. A minimum of two NVMe targets must be assigned to a protection domain before it can serve NVMe hosts, to provide minimal path resiliency to hosts.
PowerFlex data server	A PowerFlex data server(PDS) is a storage service that is responsible for processing host I/Os. The PDS service organizes the data within the storage node and divides it into smaller chunks of data of equal size.
Protection domain	A protection domain is a logical entity that consists of and manages a group of storage nodes each of which contains devices. A storage node contains both the PowerFlex data server (PDS) and a Device Gateway Target (DGWT). The PDS and DGWT services enable the storage and processing of data to be separated. Data is logically isolated into specific datasets and performance capabilities within a specified protection domain. The effect of any specific device failure is limited.
Repository	The location of firmware and software packages from which updates can be performed. You can create more than one repository, selecting one as the default. This default repository is used to calculate compliance for all undeployed resources. PowerFlex Manager lets you specify other repositories for deployed resources. The operating system is included with the compliance version, and you can select the operating system that you want to use from the compliance version.
Resource	The PowerFlex nodes, network switches, and hypervisors in the system. When PowerFlex Manager discovers resources, it identifies each component and how to talk to its management interface.
Resource group	A group of resources that are managed as a cluster. It is the state of your deployed configuration. PowerFlex Manager performs health monitoring, compliance monitoring and remediation, and the ability to add or remove resources. A resource group is the finished outcome.
Snapshot	Snapshots are read-only copies of volumes that are used for local data protection. They provide a method of recovery for data that has been corrupted or accidentally deleted.
Software management	Software management is the term that is used to deploy and configure PowerFlex software on a set of nodes on an existing operating system. The term is also referred to as software-only or just PowerFlex software.
Source	A source defines what PowerFlex Manager should listen for in order to forward events and alerts to a destination, for example, PowerFlex alerts or SNMP forwarding. In PowerFlex Manager, PowerFlex internal sources are treated as sources within the context of notification policy rules.
Storage pool	A logical structure where volumes are created.

Table 8. Terminology (continued)

Term	How used in PowerFlex Manager
Template	A template contains the configuration requirements that must be applied to a resource or group of resources during deployment. These requirements include firmware and software, operating system and or hypervisor, and PowerFlex. A template represents the wanted state of your deployed configuration. PowerFlex Manager provides sample templates that represent the standard configuration options. You can clone a sample and use it for your implementation. You can also define a custom template if necessary.
Thin clone	A thin clone is a read/write copy of a volume or a snapshot that allows changes to content using writes. Thin clones are thin copies of the object from which they were created.
Volume	A unit of storage that can be identified by a unique ID or name. PowerFlex objects are assigned a unique ID that can be used to identify the object. The default name for each volume object is its ID. Dell highly recommends giving each volume a meaningful name that is associated with its operational role.

Terminology (mirroring)

The following table describes important PowerFlex Manager terms and concepts.

Table 9. Terminology

Term	How it is used in PowerFlex Manager
Acceleration pool	An acceleration pool is a group of acceleration devices within a protection domain. PowerFlex only supports acceleration of fine granularity storage pools.
Compliance	Compliance is the comparison between current firmware and software levels and target compliance versions, which contain the baseline firmware and software versions. PowerFlex Manager shows any deviation from the baseline in the compliance status of the resources. You can use PowerFlex Manager to initiate updates to bring the resources to a compliant state. To facilitate upgrades of the management software, and compliance upgrades. PowerFlex Manager steers you toward valid upgrade paths. If the current version of the software is incompatible with the target version, when you attempt an upgrade, PowerFlex Manager displays a warning. PowerFlex Manager also warns you if any of the RCM or IC versions that are loaded are incompatible with the target compliance versions.
Destination	A destination defines the external location where PowerFlex Manager sends events and alerts to, for example, Dell OpenManage Enterprise (OME).
Device	A storage device or acceleration device that is added to an SDS in the system.
Host	A host is an entity that consumes PowerFlex storage for application usage. There are two methods of consuming PowerFlex block storage: using the SDC kernel driver, or using NVMe over TCP connectivity. Therefore, a host is either an SDC or an NVMe host.
Installer node	The installer node is the initial node that is powered on to setup and initialize the deployment. The installer node deploys PowerFlex nodes and sets up the management infrastructure to enable you to manage the PowerFlex system.
Initial setup script	The purpose of the initial setup script is to gather minimal information to configure basic networking. The initial setup script is completed through the CLI when the first node boots with the pre-loaded operating system. The initial setup script configures the installer node, VM, and PowerFlex Manager ingress IP address with the appropriate networking. You can then transition to the web browser, log in to

Table 9. Terminology (continued)

Term	How it is used in PowerFlex Manager
	the PowerFlex Manager user interface, and access the initial Welcome wizard to configure your system.
initial Welcome wizard	The initial Welcome wizard is a step-by-step wizard that guides you through the process of deploying PowerFlex nodes. Once you complete the initial setup script, the PowerFlex Manager user interface is configured and you can access the initial Welcome wizard through logging into the PowerFlex Manager user interface.
NVMe target	A frontend component that translates NVMe over Fabric protocol into internal PowerFlex protocols. The NVMe target provides I/O and discovery services to NVMe hosts configured on the PowerFlex system. A minimum of two NVMe targets must be assigned to a protection domain before it can serve NVMe hosts, to provide minimal path resiliency to hosts.
Protection domain	A set of SDSs, with (optionally) SDTs and SDRs, configured in separate logical groups. These logical groups allow you to physically and/or logically isolate specific datasets and performance capabilities within specified protection domains and limit the effect of any specific device failure.
PowerFlex management cluster (KVM)	The management cluster runs a dedicated instance of PowerFlex Manager to manage it. This cluster runs on KVM with a minimum of three nodes.
Repository	The location of firmware and software packages from which updates can be performed. You can create more than one repository, selecting one as the default. This default repository is used to calculate compliance for all undeployed resources. PowerFlex Manager lets you specify other repositories for deployed resources. The operating system is included with the compliance version, and you can select the operating system that you want to use from the compliance version.
Resource	The PowerFlex nodes, network switches, and hypervisors in the system. When PowerFlex Manager discovers resources, it identifies each component and how to talk to its management interface.
Resource group	A group of resources that are managed as a cluster. It is the state of your deployed configuration. PowerFlex Manager performs health monitoring, compliance monitoring and remediation, and the ability to add or remove resources. A resource group is the finished outcome. Resource groups were referred to as services in previous releases of PowerFlex Manager.
Software management	Software management is the term that is used to deploy and configure PowerFlex software on a set of nodes on an existing operating system. The term is also referred to as software-only or just PowerFlex software.
Source	A source defines what PowerFlex Manager should listen for in order to forward events and alerts to a destination, for example, PowerFlex alerts or SNMP forwarding. In PowerFlex Manager, PowerFlex internal sources are treated as sources within the context of notification policy rules.
Storage Data Server (SDS)	An object that manages the capacity of a single server and acts as a backend for data access. The SDS is installed on all servers contributing storage devices to PowerFlex. These devices are accessed through the SDS.
Storage pool	A set of physical storage devices in a protection domain. A volume is distributed over all devices residing in the same storage pool. Add, modify, or remove a storage pool in the PowerFlex system.
Template	A template contains the configuration requirements that must be applied to a resource or group of resources during deployment. These requirements include firmware and software, operating system and or hypervisor, and PowerFlex. A template represents the wanted state of your deployed configuration.

Table 9. Terminology (continued)

Term	How it is used in PowerFlex Manager
	PowerFlex Manager provides sample templates that represent the standard configuration options. You can clone a sample and use it for your implementation. You can also define a custom template if necessary.
Volume	A unit of storage that can be identified by a unique ID or name. PowerFlex objects are assigned a unique ID that can be used to identify the object. The default name for each volume object is its ID. Dell highly recommends giving each volume a meaningful name that is associated with its operational role.

Related information

Configure PowerFlex nodes through the initial Welcome wizard

User roles (erasure coding)

User roles control the activities that different types of users can perform when using PowerFlex Manager.

Ensure that you configure the active directory before assigning roles. The roles that can be assigned to local users and LDAP users are identical. Each user can only be assigned one role. If an LDAP user is assigned directly to a user role and also to a group role, the LDAP user is provided with permissions of both roles.

The following table summarizes the activities that can be performed for each user role:

Table 10. User roles and activities

Role	Description	Activities
SuperUser	A SuperUser can perform all system operations.	• Manage storage resources • Manage life cycle operations, resource groups, templates, deployment, backend operations • Manage snapshots, snapshot policies • Manage users, certificates • Replace drives • Hardware operations • View storage configurations, resource details • View platform configuration, resource details • System monitoring (events, alerts) • Perform serviceability operations • Update system settings
SystemAdmin	A SystemAdmin can perform all operations, except for user management and security ones.	• Manage storage resources • Manage life cycle operations, resource groups, templates, deployment, backend operations • Manage snapshots, snapshot policies • Replace drives • Hardware operations • View storage configurations, resource details • View platform configuration, resource details • System monitoring (events, alerts) • Perform serviceability operations • Update system settings
StorageAdmin	A StorageAdmin can perform all storage-related front-end operations including block system operations, such as create a volume. **NOTE:** Operations such as create storage pool cannot be performed by the StorageAdmin	• Manage storage resources • Replace drives • Hardware operations • View storage configurations, resource details • View platform configuration, resource details • System monitoring (events, alerts)

Table 10. User roles and activities (continued)

Role	Description	Activities
	role, but can be performed by the LifecycleAdmin role.	
LifecycleAdmin	A LifecycleAdmin can manage the life cycle of hardware and PowerFlex systems.	• Manage life cycle operations, resource groups, templates, deployment, backend operations • Replace drives • Hardware operations • View resource groups and templates • System monitoring (events, alerts)
SnapshotManager	SnapshotManager is a subset of StorageAdmin, working only on existing systems. This role includes all operations that are required to set up and manage snapshots.	• View storage configurations, resource details • System monitoring (events, alerts)
SecurityAdmin	The SecurityAdmin manages PowerFlex role-based access control (RBAC), and LDAP user federation. It includes all security aspects of the system.	• Manage users, certificates • System monitoring (events, alerts)
Technician	The Technician can perform all hardware FRU operations on the system, including entering a node into maintenance mode.	• Replace drives • Hardware operations • System monitoring (events, alerts) • Perform serviceability operations
DriveReplacer	This is a subset of the technician role. The DriveReplacer can perform only the operations that are required for a drive replacement.	• Replace drives • System monitoring (events, alerts)
Monitor	The monitor role has read-only access to the system, including topology, alerts, events, and metrics.	• View storage configurations, resource details • View platform configuration, resource details • System monitoring (events, alerts)
Support	The support role is a special kind of SystemAdmin (all activities except for user/security management operations) to be used only by Dell Technologies support staff and developers. This user role has access to undocumented, special operations and options for common operations, required only for support purposes. **NOTE:** This special role should be used only by Dell Technologies support. It opens special, often dangerous, commands for advanced troubleshooting.	• Manage storage resources • Manage life cycle operations, resource groups, templates, deployment, backend operations • Replace drives • Hardware operations • View storage configurations, resource details • View platform configuration, resource details • System monitoring (events, alerts) • Perform serviceability operations • Special Dell Technologies Support operations

LDAP users and security permissions (erasure coding)

LDAP security permissions are managed through group memberships, allowing permissions to be assigned to groups instead of individual users. This streamlines administration and improves security by ensuring only authorized users can access specific resources.

You cannot assign multiple roles to a group. All roles in a group have a proximate hierarchy.

Table 11. LDAP hierarchy

Ranking	PowerFlex group role without security officer permissions	PowerFlex group role with security officer permissions
1	N/A (local user)	N/A (local user)
2	Admin	SuperUser

Table 11. LDAP hierarchy (continued)

Ranking	PowerFlex group role without security officer permissions	PowerFlex group role with security officer permissions
3	Admin	SuperUser
4	Admin	SuperUser
5	StorageAdmin or LifecycleAdmin	SuperUser
6	Monitor	SecurityAdmin
Not ranked	SecurityAdmin	SecurityAdmin

User roles (mirroring)

User roles control the activities that different types of users can perform when using PowerFlex Manager.

Ensure that you configure the active directory before assigning roles. The roles that can be assigned to local users and LDAP users are identical. Each user can only be assigned one role. If an LDAP user is assigned directly to a user role and also to a group role, the LDAP user is provided with permissions of both roles.

NOTE: User definitions are not imported from earlier versions of PowerFlex and must be configured again.

The following table summarizes the activities that can be performed for each user role:

Table 12. User roles and activities

Role	Description	Activities
SuperUser	A SuperUser can perform all system operations.	• Manage storage resources • Manage life cycle operations, resource groups, templates, deployment, backend operations • Manage replication operations, peer systems, replication consistent groups (RCGs) • Manage snapshots, snapshot policies • Manage users, certificates • Replace drives • Hardware operations • View storage configurations, resource details • View platform configuration, resource details • System monitoring (events, alerts) • Perform serviceability operations • Update system settings
SystemAdmin	A SystemAdmin can perform all operations, except for user management and security ones.	• Manage storage resources • Manage life cycle operations, resource groups, templates, deployment, backend operations • Manage replication operations, peer systems, replication consistent groups (RCGs) • Manage snapshots, snapshot policies • Replace drives • Hardware operations • View storage configurations, resource details • View platform configuration, resource details • System monitoring (events, alerts) • Perform serviceability operations • Update system settings
StorageAdmin	A StorageAdmin can perform all storage-related front-end operations including block systems, such as create volume.	• Manage storage resources • Manage replication operations, peer systems, replication consistent groups (RCGs)

Table 12. User roles and activities (continued)

Role	Description	Activities
	NOTE: Operations such as create storage pool cannot be performed by StorageAdmin, but can be performed by the LifecycleAdmin role.	• Manage snapshots, snapshot policies • Replace drives • Hardware operations • View storage configurations, resource details • View platform configuration, resource details • System monitoring (events, alerts)
LifecycleAdmin	A LifecycleAdmin can manage the life cycle of hardware and PowerFlex systems.	• Manage life cycle operations, resource groups, templates, deployment, backend operations • Replace drives • Hardware operations • View resource groups and templates • System monitoring (events, alerts)
ReplicationManager	The ReplicationManager is a subset of the StorageAdmin role, for work on existing systems for setup and management of replication and snapshots.	• Manage replication operations, peer systems, replication consistent groups (RCGs) • Manage snapshots, snapshot policies • View storage configurations, resource details (volume, snapshot, replication views) • System monitoring (events, alerts)
SnapshotManager	The SnapshotManager is a subset of StorageAdmin, working only on existing systems. This role includes all operations that are required to set up and manage snapshots.	• Manage snapshots, snapshot policies • View storage configurations, resource details • System monitoring (events, alerts)
SecurityAdmin	The SecurityAdmin manages PowerFlex role-based access control (RBAC), and LDAP user federation. It includes all security aspects of the system.	• Manage users, certificates • System monitoring (events, alerts)
DriveReplacer	This is a subset of the Technician role. The DriveReplacer can perform only the operations that are required for a drive replacement.	• Replace drives • System monitoring (events, alerts)
Technician	The Technician can perform all hardware FRU operations on the system, including entering a node into maintenance mode.	• Replace drives • Hardware operations • System monitoring (events, alerts) • Perform serviceability operations
Monitor	The Monitor role has read-only access to the system, including topology, alerts, events, and metrics.	• View storage configurations, resource details • View platform configuration, resource details • System monitoring (events, alerts)
Support	The Support role is a special type of SystemAdmin (all activities except for user/security management operations) to be used only by Dell Technologies support staff and developers. This user role has access to undocumented, special operations and options for common operations, required only for support purposes. **NOTE:** This special role should be used only by Dell Technologies support. It opens special, often dangerous, commands for advanced troubleshooting.	• Manage storage resources • Manage life cycle operations, resource groups, templates, deployment, backend operations • Manage replication operations, peer systems, replication consistent groups (RCGs) • Manage snapshots, snapshot policies • Replace drives • Hardware operations • View storage configurations, resource details • View platform configuration, resource details • System monitoring (events, alerts) • Perform serviceability operations • Special Dell Technologies Support operations

Mapping PowerFlex 4.0.x or later roles to legacy roles (mirroring)

For reference, the tables below map user and group roles from earlier releases to the roles used in the current release.

Role mapping per legacy user interface

Table 13. Role mapping

Legacy role before version PowerFlex 4.0	New role after PowerFlex 4.0.x and later
PowerFlex Monitor	Monitor
PowerFlex Back-end configurator	LifecycleAdmin
PowerFlex frontend configurator	StorageAdmin
PowerFlex Configurator	SystemAdmin
PowerFlex Security	SecurityAdmin
PowerFlex Administrator	StorageAdmin
PowerFlex local Super User	SuperUser
PowerFlex technician commands	Support
PowerFlex Manager Administrator	SuperUser
PowerFlex Manager read-only	Monitor
PowerFlex Manager Standard owner	LifecycleAdmin
PowerFlex Manager Standard member	LifecycleAdmin
PowerFlex Manager Operator	DriveReplacer

Legacy LDAP users and groups

LDAP user role permissions were modified in PowerFlex 4.0. For reference purposes, the following table shows the mapping of legacy users and groups to the roles now supported by PowerFlex.

Before PowerFlex 4.0, PowerFlex mapped LDAP users and groups to roles. This mapping allowed assignment of multiple roles to a single group. Assignment of multiple roles to a group is no longer supported.

All legacy roles have a proximate hierarchy. The group's single role is the highest ranking role.

Table 14. LDAP hierarchy

Ranking	Legacy PowerFlex LDAP role earlier than PowerFlex 4.0	PowerFlex 4.0 or later group role without security officer permissions	PowerFlex 4.0 or later group role with security officer permissions
1	SuperUser	N/A (local user)	N/A (local user)
2	Administrator	Admin	SuperUser
3	Configurator	Admin	SuperUser
4	Frontend configuration and backend configuration	Admin	SuperUser
5	Frontend configuration or backend configuration	StorageAdmin or LifecycleAdmin	SuperUser
6	Monitor	Monitor	SecurityAdmin
Not ranked	Security Officer (SE)	SecurityAdmin	SecurityAdmin

5

Initial configuration and setup (erasure coding)

This section includes tasks you need to perform when you first begin using PowerFlex Manager.

Initial configuration wizard

The first time you log in to PowerFlex Manager, you are prompted with an **Initial Configuration Wizard**, which prompts you to configure the basic settings that are required to start using PowerFlex Manager.

Before you begin, have the following information available:

- Dell Technologies connectivity services configuration details. Dell Technologies connectivity services uses secure connect gateway technology, which enables call home functionality and remote connectivity.
- Information about whether you intend to use a Release Certification Matrix (RCM) or Intelligent Catalog (IC).
- Information about the type of installation you want to perform, including details about your existing PowerFlex instance when you intend to import from another PowerFlex instance.

To configure the basic settings:

1. On the **Welcome** page, read the instructions and click **Next**.
2. On the **Dell Technology connectivity services (Optional)** page, optionally enable Dell Technologies connectivity services . If you enable Dell Technologies connectivity services, you must accept the telemetry agreement, and click **Next**.

   NOTE: If you are enabling Dell Technologies connectivity services, the secure connect gateway and policy manager must be deployed and configured before you can enable Dell Technologies connectivity services.

3. On the **Connect to Dell Technologies connectivity services** page, enter the required information and click **Next**.
4. On the **Installation Type** page, specify whether you want to deploy a new instance of PowerFlex or import an existing instance, and click **Next**.
5. On the **Summary** page, verify all settings for Dell Technologies connectivity services and installation type. Click **Finish** to complete the initial configuration.

After completing the **Initial Configuration Wizard**, you can get started using PowerFlex Manager from the **Getting Started** page.

Dell Technologies connectivity services

Dell Technologies connectivity services is a secure support technology for the data center. Dell Technologies connectivity services uses secure connect gateway, which enables call home functionality and remote connectivity.

PowerFlex Manager provides support through integration with the secure connect gateway, ensuring better alignment with Dell Technologies services initiatives to enhance the user experience. Dell Technologies connectivity services is the functionality that enables this connection.

When you enable Dell Technologies connectivity services, you can take advantage of the following benefits, depending on the service agreement on your device:

- Automated issue detection - Dell Technologies connectivity services monitors your Dell Technologies devices and automatically detects hardware issues, both proactively and predictively.
- Automated case creation - When an issue is detected, Dell Technologies connectivity services automatically opens a support case with Dell Technologies Support.
- Automated diagnostic collection - Dell Technologies connectivity services automatically collects system state information from your devices and uploads it securely to Dell Technologies. Dell Technologies Support uses this information to troubleshoot the issue.
- Proactive contact - A Dell Technologies Support agent contacts you about the support case and helps you resolve the issue.

Enabling Dell Technologies connectivity services

You can enable Dell Technologies connectivity services as part of the initial configuration wizard. Alternatively, you can enable it later by adding it as a destination to a notification policy in Events and Alerts.

- Ensure secure connect gateway, and policy manager are deployed and configured.
- Ensure that you have the information you require to configure Dell Technologies connectivity services.

If you do not enable Dell Technologies connectivity services in the initial setup, you can add it later as a destination when adding a notification policy through **Settings** > **Events and Alerts** > **Notification Policies**.

The first time that you access the initial configuration wizard, the connection status displays as not configured.

1. On the **Overview** screen, click **Next**.
2. On the **Dell Technologies connectivity services (Optional)** screen, select the checkbox to accept the telemetry agreement, and click **Next**.
3. On the **Connect to Dell Technologies connectivity services** screen, compete the following steps:
   a. Under **Connection Type**, choose either of the following options:
      - To connect directly to Dell Technologies, click **Connect directly**.
      - To connect to a Dell Technologies connectivity services gateway server by remote access, click **Connect via gateway server** and provide the following information:
        - Primary gateway IP address or and hostname
        - Secondary gateway IP address or and hostname
        - The port number is auto populated to **9443**. Optionally, to include an additional gateway server, click **Add Gateway**. The maximum number of gateways is eight.

   b. Under **License Type**, choose either of the following options:
      - If you are using dynamic licensing, select **I have a license authorization code (LAC)**.
      - If you are using a key-based license, select **I have a license file in xml format**.

   c. Under **Device Registration**, provide the details of your device to register with Dell Technologies connectivity services.

Table 15. Device registration details

Device type	Information required
PowerFlex rack	• Type • Solution serial number • Castle ID (Site ID) • ELMS eSWUID (required only for key-based licensing)
PowerFlex appliance	• Type • Solution serial number • Castle ID (Site ID) • ELMS eSWUID (required only for key-based licensing
PowerFlex software	• Type • Deployment environment - one of the following: ○ Amazon Web Services ○ Microsoft Azure ○ On-premises PowerFlex custom node ○ On-premises other node type

   d. Under **Customize Connection to Dell Technologies connectivity services**, you can choose the following:
      - To enable PowerFlex Manager to send telemetry data, alerts, and analytics through Dell Technologies connectivity services, turn **Connect to Dell AIOps** to on.
      - To replace the IP addresses with xxx.xxx.xxx.xxx before any information is sent about your environment to Dell Technologies Support, click **Mask IP Addresses**.
      - To enable Dell Technologies support engineers to troubleshoot your system remotely, turn **Enable remote support** to on.

        NOTE: If you choose to enable this feature now, you can disable it in the future.

e. Under **Authenicate to Dell Technologies connectivity services**, provide the access key and pin to verify a secure connection. You can generate an access key and self select a pin from Support for PowerFlex Software .
f. Click **Next**.

4. On the **Support Contacts** screen, identify the primary support contact by providing their first name, last name, email, and phone details. Optionally, you can add up to two additional support contacts.
5. Click **Finish**.
6. Select the Dell Technologies connectivity services that you created.
   a. In **Authentication Details** optionally, click **Test Connection**.

      A notification message displays the Dell Technologies connectivity services connection status. If the connection fails, verify your authentication details and test the connection again.
   b. Optionally, click **Send Test Alert** to send a test alert.

After you complete the initial configuration wizard, go to the **Settings** page and verify that Dell Technologies connectivity services has been turned on. You may need to configure the Dell Technologies connectivity services policy explicitly on the **Settings** > **Events and Alerts** > **Notification Policies** page.

Related information

Events
Viewing an event
Alerts
Add a notification policy
Monitoring events and alerts
Events and alerts
License management
Add a notification policy
Licensing PowerFlex for dynamic licensing (connected method)
Migrating from key-based licensing to dynamic licensing (connected method)

Specifying the installation type

If you have an existing PowerFlex deployment that you would like to import, you can specify details about this deployment as part of the initial configuration. Alternatively, if you would like to deploy a new instance, you can bypass the import step as part of the initial configuration.

The initial configuration wizard supports two different installation workflows; a new instance of PowerFlex or the import of a PowerFlex instance.

1. Specify the type of installation that you want to perform:
   - If you do not have an existing PowerFlex deployment and would like to bypass the import step, click **I want to deploy a new instance of PowerFlex** > **NextFinish**.
   - If you would like to import an existing PowerFlex directly from an existing PowerFlex Manager virtual appliance, click **I have a PowerFlex instance to import**
2. If you are selected to import a PowerFlex instance, complete the following steps:
   a. Select whether the PowerFlex instance is to be used for the production storage or the management cluster.
   b. Enter the IP addresses for the MDM.
   c. Enter the system ID for the cluster.
   d. Create the credential name and LIA password for the PowerFlex Management system.
   e. Confirm the LIA password.
   f. Click **Save**.
3. Click **Next** to proceed.

Deploy a PowerFlex cluster using a CSV topology file

Use the following procedure to deploy a PowerFlex cluster using a CSV topology file.

1. Log in to PowerFlex Manager.
2. On the **Getting Started** page, click **Deploy With Installation File** to provide an installation topology file for deployment of nodes.

3. In the **PowerFlex Installer** window, click **Browse** to upload the appropriate artifacts (RPMs) according to the operating system of your PowerFlex operating system machines.
4. Click **Upload Files** and click **Next** when the RPMs have finished uploading.
5. Click **Browse** to upload an installation CSV topology file that describes the desired topology of PowerFlex.

   To view sample files, click the links to the following two example files: **PowerFlex_config_example_all fields.csv (all fields)** and **PowerFlex_basic_example_minimum fields.csv (minimum fields).**
6. Click **Upload Installation CSV**.
7. Select the **I accept the terms of the End User License Agreement (mandatory)** check box.
8. Click **Next**.
9. In the **Advanced Settings** tab, you can select how you would like to install PowerFlex.
   - You can skip options to upload, install, configure, and / or validate a device.
   - To overwrite data, select **Force overwrite of existing data on devices used for PowerFlex storage**.

     PowerFlex can be installed on new or clean disks (storage node devices). If the installation is done on a previously deployed PowerFlex cluster, or used disks, you must tell the installer to overwrite the older data.

     NOTE: If you select this option, the data is erased.
   - The device awareness check is enabled by default. If you disable this setting, it may cause potential device failures not to be detected early enough. To disable the diagnostic, turn **Enable Diagnostics Check** to off.
10. Click **Next**.
11. In the **Summary** tab, verify all the configurations and click **Finish**.
12. Click the **Go to Events Page** button to track the installation progress.

Verifying the initial configuration

Before you complete the initial configuration, you can review the settings that you provided for Dell Technologies connectivity services, compliance, and installation type.

1. On the **Summary** page, verify the settings that you configured on the previous pages.
2. To edit any information, click **Back** or click the corresponding page name in the left pane.
3. If you are importing an existing PowerFlex instance from PowerFlex Manager, type `IMPORT POWERFLEX MANAGER`.
4. If the information is correct, click **Finish** to complete the initial configuration.

   If you are importing an existing environment, PowerFlex Manager displays a message indicating that the import operation is in progress. When the import operation is complete, PowerFlex Manager displays the **Getting Started** page.

After completing the **Initial Configuration Wizard**, you must perform some additional steps. The steps vary depending on which installation type you selected:

- If you clicked **I want to deploy a new instance of PowerFlex** on the **Installation Type** page, you now must deploy a new instance of PowerFlex. The **Getting Started** page provides more information about what to do next.
- If you clicked **I have a PowerFlex instance managed by PowerFlex Manager to import** on the **Installation Type** page, you must perform these steps:
  1. On the **Settings** page, upload the compatibility matrix file and upload the latest repository catalog (RCM or IC).
  2. On the **Resources** page, select the PowerFlex entry, and perform a nondisruptive update.
  3. On the **Resource Groups** page, perform an RCM/IC upgrade on any migrated service that must be upgraded.

     CAUTION: Check the Alerts page before performing the upgrade. Look for major and critical alerts that are related to PowerFlex Block to be sure that the MDM cluster is healthy before proceeding.

Getting started

The **Getting Started** page guides you through the common configurations that are required to prepare a new PowerFlex Manager environment. A green check mark on a step indicates that you have completed the step. Only super users have access to the **Getting Started** page.

The following table describes each step:

Table 16. Getting Started steps

Step	Description
Upload Compliance File	Provide compliance file location and authentication information for use within PowerFlex Manager. The compliance file defines the specific hardware components and software version combinations that are tested and certified by Dell for Dell products. This step enables you to choose a default compliance version for compliance or add new compliance versions. You can also click **Settings** > **Repositories** > **Compliance Versions**. **NOTE:** Before you make an RCM or IC the default compliance version, you must first upload a suitable compatibility management file under **Settings** > **Repositories** > **Compatibility Management**.
Define Networks	If you plan to perform an advanced CSV-based deployment, this step can be skipped. Enter detailed information about the available networks in the environment. This information is used later during deployments that are based on templates and resource groups. These deployments use the network information to configure nodes and switches to have the right network connectivity. PowerFlex Manager uses the defined networks in templates to specify the networks or VLANs that are configured on nodes and switches for your resource groups. This step is enabled immediately after you perform an initial configuration for PowerFlex Manager. You can also click **Settings** > **Networking** > **Networks**.
Discover Resources	If you plan to perform an advanced CSV-based deployment, this step can be skipped. Grant PowerFlex Manager access to resources (nodes, switches, virtual machine managers) in the environment by providing the management IP and credential for the resources to be discovered. This step is not enabled until you define your networks. You can also click **Resources** > **Discover Resources**.
Manage Deployed Resources (Optional)	If you plan to perform an advanced CSV-based deployment, this step can be skipped. Add an existing resource group for a cluster that is already deployed and manage the resources within PowerFlex Manager. This step is not enabled until you define your networks. You can also click **Lifecycle** > **Resource Groups** > **Add Existing Resource Group**.
Deploy Resources	If you plan to perform an advanced CSV-based deployment, this step can be skipped. Create a template with requirements that must be followed during a deployment. Templates enable you to automate the process of configuring and deploying infrastructure and workloads. For most environments, you can clone one of the sample templates that are provided with PowerFlex Manager and make modifications as needed. Choose the sample template that is most appropriate for your environment. For a two-layer deployment, clone the compute-only templates. Then clone one of the storage templates. This step is not enabled until you define your networks.

Table 16. Getting Started steps (continued)

Step	Description
	You can also click **Lifecycle** > **Templates**.
Manage PowerFlex License	Configure licensing for PowerFlex. You can also click **Settings** > **License Management**.

To revisit the **Getting Started** page, click **Getting Started** on the help menu.

Related information

Networking
Discover a resource
Adding an existing resource group
Templates
Add a storage node

Changing your password

When you first log in to PowerFlex Manager, you must set your password. You can also change your password at any time after the first login.

1. Click the user icon in the upper right corner of PowerFlex Manager.
2. Click **Change password**.
3. Type the password in the **New Password** field.

   Password rules are:

   - Includes a minimum of eight and a maximum of 32 characters
   - Includes only alphanumeric and punctuation characters
4. Type the password again in the **Verify Password** field.
5. Click **Apply**.

6

Initial configuration and setup (mirroring)

This section includes tasks you need to perform when you first begin using PowerFlex Manager.

Initial configuration wizard

The first time you log in to PowerFlex Manager, you are prompted with an **Initial Configuration Wizard**, which prompts you to configure the basic settings that are required to start using PowerFlex Manager.

Before you begin, have the following information available:

- Dell Technologies connectivity services configuration details. Dell Technologies connectivity services uses secure connect gateway technology, which enables call home functionality and remote connectivity.
- Information about whether you intend to use a Release Certification Matrix (RCM) or Intelligent Catalog (IC).
- Information about the type of installation you want to perform, including details about your existing PowerFlex instance when you intend to import from another PowerFlex instance.

To configure the basic settings:

1. On the **Welcome** page, read the instructions and click **Next**.
2. On the **Dell Technology connectivity services (Optional)** page, optionally enable Dell Technologies connectivity services . If you enable Dell Technologies connectivity services, you must accept the telemetry agreement, and click **Next**.

   NOTE: If you are enabling Dell Technologies connectivity services, the secure connect gateway and policy manager must be deployed and configured before you can enable Dell Technologies connectivity services.

3. On the **Connect to Dell Technologies connectivity services** page, enter the required information and click **Next**.
4. On the **Installation Type** page, specify whether you want to deploy a new instance of PowerFlex or import an existing instance, and click **Next**.
5. On the **Summary** page, verify all settings for Dell Technologies connectivity services and installation type. Click **Finish** to complete the initial configuration.

After completing the **Initial Configuration Wizard**, you can get started using PowerFlex Manager from the **Getting Started** page.

Dell Technologies connectivity services

Dell Technologies connectivity services is a secure support technology for the data center. Dell Technologies connectivity services uses secure connect gateway, which enables call home functionality and remote connectivity.

PowerFlex Manager provides support through integration with the secure connect gateway, ensuring better alignment with Dell Technologies services initiatives to enhance the user experience. Dell Technologies connectivity services is the functionality that enables this connection.

When you enable Dell Technologies connectivity services, you can take advantage of the following benefits, depending on the service agreement on your device:

- Automated issue detection - Dell Technologies connectivity services monitors your Dell Technologies devices and automatically detects hardware issues, both proactively and predictively.
- Automated case creation - When an issue is detected, Dell Technologies connectivity services automatically opens a support case with Dell Technologies Support.
- Automated diagnostic collection - Dell Technologies connectivity services automatically collects system state information from your devices and uploads it securely to Dell Technologies. Dell Technologies Support uses this information to troubleshoot the issue.
- Proactive contact - A Dell Technologies Support agent contacts you about the support case and helps you resolve the issue.

Enabling Dell Technologies connectivity services

You can enable Dell Technologies connectivity services as part of the initial configuration wizard. Alternatively, you can enable it later by adding it as a destination to a notification policy in Events and Alerts.

- Ensure secure connect gateway, and policy manager are deployed and configured.
- Ensure that you have the information you require to configure Dell Technologies connectivity services.

If you do not enable Dell Technologies connectivity services in the initial setup, you can add it later as a destination when adding a notification policy through **Settings** > **Events and Alerts** > **Notification Policies**.

The first time that you access the initial configuration wizard, the connection status displays as not configured.

1. On the **Overview** screen, click **Next**.
2. On the **Dell Technologies connectivity services (Optional)** screen, select the checkbox to accept the telemetry agreement, and click **Next**.
3. On the **Connect to Dell Technologies connectivity services** screen, compete the following steps:
   a. Under **Connection Type**, choose either of the following options:
      - To connect directly to Dell Technologies, click **Connect directly**.
      - To connect to a Dell Technologies connectivity services gateway server by remote access, click **Connect via gateway server** and provide the following information:
        - Primary gateway IP address or and hostname
        - Secondary gateway IP address or and hostname
        - The port number is auto populated to **9443**. Optionally, to include an additional gateway server, click **Add Gateway**. The maximum number of gateways is eight.

   b. Under **License Type**, choose either of the following options:
      - If you are using dynamic licensing, select **I have a license authorization code (LAC)**.
      - If you are using a key-based license, select **I have a license file in xml format**.

   c. Under **Device Registration**, provide the details of your device to register with Dell Technologies connectivity services.

Table 17. Device registration details

Device type	Information required
PowerFlex rack	• Type • Solution serial number • Castle ID (Site ID) • ELMS eSWUID (required only for key-based licensing)
PowerFlex appliance	• Type • Solution serial number • Castle ID (Site ID) • ELMS eSWUID (required only for key-based licensing
PowerFlex software	• Type • Deployment environment - one of the following: ○ Amazon Web Services ○ Microsoft Azure ○ On-premises PowerFlex custom node ○ On-premises other node type

   d. Under **Customize Connection to Dell Technologies connectivity services**, you can choose the following:
      - To enable PowerFlex Manager to send telemetry data, alerts, and analytics through Dell Technologies connectivity services, turn **Connect to Dell AIOps** to on.
      - To replace the IP addresses with xxx.xxx.xxx.xxx before any information is sent about your environment to Dell Technologies Support, click **Mask IP Addresses**.
      - To enable Dell Technologies support engineers to troubleshoot your system remotely, turn **Enable remote support** to on.

        NOTE: If you choose to enable this feature now, you can disable it in the future.

e. Under **Authenicate to Dell Technologies connectivity services**, provide the access key and pin to verify a secure connection. You can generate an access key and self select a pin from Support for PowerFlex Software .

f. Click **Next**.

4. On the **Support Contacts** screen, identify the primary support contact by providing their first name, last name, email, and phone details. Optionally, you can add up to two additional support contacts.
5. Click **Finish**.
6. Select the Dell Technologies connectivity services that you created.
   a. In **Authentication Details** optionally, click **Test Connection**.

      A notification message displays the Dell Technologies connectivity services connection status. If the connection fails, verify your authentication details and test the connection again.

   b. Optionally, click **Send Test Alert** to send a test alert.

After you complete the initial configuration wizard, go to the **Settings** page and verify that Dell Technologies connectivity services has been turned on. You may need to configure the Dell Technologies connectivity services policy explicitly on the **Settings** > **Events and Alerts** > **Notification Policies** page.

Related information

Events
Viewing an event
Alerts
Add a notification policy
Monitoring events and alerts
Events and alerts
License management
Add a notification policy
Licensing PowerFlex for dynamic licensing (connected method)
Migrating from key-based licensing to dynamic licensing (connected method)

Specifying the installation type (mirroring)

If you have an existing PowerFlex deployment that you would like to import, you can specify details about this deployment as part of the initial configuration. Alternatively, if you would like to deploy a new instance, you can bypass the import step as part of the initial configuration.

If you are importing an existing PowerFlex deployment that was not managed by PowerFlex Manager, make sure you have the IP address, username, and password for the primary and secondary MDMs. If you are importing an existing PowerFlex deployment that was managed by PowerFlex Manager, make sure you have the IP address, username, and password for the PowerFlex Manager virtual appliance.

The initial configuration wizard supports three different installation workflows:

- No migration required

  You would like to deploy a new instance of PowerFlex.

  If you would like to use the legacy PowerFlex Installer to provide an installation topology file, click **Deploy With Installation File** on the **Getting Started** page after you complete the initial configuration. See the topic on deploying a PowerFlex cluster using a CSV topology file in the *PowerFlex and PowerFlex Manager Software Install and Upgrade Guide* for more information.

- Migration from a core PowerFlex instance that was not managed with PowerFlex Manager

  In this workflow, you need to provide the IP and credentials for the PowerFlex MDM cluster. This workflow allows you to import a PowerFlex 3.6.x instance into PowerFlex Manager.

- Migration from a full PowerFlex Manager environment that had previously been used to manage a PowerFlex instance

  In this workflow, you need to provide the IP and credentials for the PowerFlex Manager virtual appliance instance

1. Specify the type of installation that you want to perform:
   - Click **I want to deploy a new instance of PowerFlex** if you do not have an existing PowerFlex deployment and would like to bypass the import step.

- Click **I have a PowerFlex instance to import** if you would like to import an existing PowerFlex instance that was not managed by PowerFlex Manager.

  Specify whether you are running **PowerFlex 3.x** or **PowerFlex 4.x**.

  For a PowerFlex 3.x system, provide the following details about the existing PowerFlex instance:

  - IP addresses for the primary and secondary MDMs (separated by a comma with no spaces)
  - Admin username and password for the primary MDM
  - Operating system username and password for the primary MDM
  - LIA password

  For a PowerFlex 4.x system, indicate whether the PowerFlex instance is used for **Production Storage** or a **Management Cluster**. The management cluster use case is applicable for PowerFlex appliance and PowerFlex rack install types that have a dedicated management cluster with PowerFlex as a shared storage for the datastore hosting the management VMs.

  Then, provide the following details about the existing PowerFlex instance:

  - IP addresses for all nodes with a primary and secondary MDM
  - System ID for the cluster
  - LIA password
- Click **I have a PowerFlex instance managed by PowerFlex Manager to import** if you would like to import an existing PowerFlex directly from an existing PowerFlex Manager virtual appliance.

  Provide the following details about the existing PowerFlex Manager virtual appliance:

  - IP address or DNS name for the virtual appliance
  - Username and password for the virtual appliance

  NOTE: The Block tab in the PowerFlex Manager user interface does not display information about the PowerFlex instance until the upgrade to 4.x is complete.

2. Click **Next** to proceed.

For a full PowerFlex Manager migration, the import process backs up and restores information from the old PowerFlex Manager virtual appliance. The migration process for the full PowerFlex Manager workflow imports all resources, templates, and services from a 3.8 instance of PowerFlex Manager. The migration also connects the legacy PowerFlex gateway to the MDM cluster, which enables the **Block** tab in the user interface to function.

The migrated environment includes the PowerFlex gateway resource. The operating system hostname and asset or service tag are set to **`powerflex`**.

For a system that has just the Dell PowerFlex software, there is no PowerFlex Manager information available after the migration completes. The migrated environment does not include resources, templates, and services.

Deploy a PowerFlex cluster using a CSV topology file

Use the following procedure to deploy a PowerFlex cluster using a CSV topology file.

1. Log in to PowerFlex Manager.
2. On the **Getting Started** page, click **Deploy With Installation File** to provide an installation topology file for deployment of nodes.
3. In the **PowerFlex Installer** window, click **Browse** to upload the appropriate artifacts (RPMs) according to the operating system of your PowerFlex operating system machines.
4. Click **Upload Files** and click **Next** when the RPMs have finished uploading.
5. Click **Browse** to upload an installation CSV topology file that describes the desired topology of PowerFlex.

   To view sample files, click the links to the following two example files: **PowerFlex_config_example_all fields.csv (all fields)** and **PowerFlex_basic_example_minimum fields.csv (minimum fields).**
6. Click **Upload Installation CSV**.
7. Select the **I accept the terms of the End User License Agreement (mandatory)** check box.
8. Click **Next**.
9. In the **Advanced Settings** tab, you can select how you would like to install PowerFlex.
   - You can skip options to upload, install, configure, and / or validate a device.
   - To overwrite data, select **Force overwrite of existing data on devices used for PowerFlex storage**.

PowerFlex can be installed on new or clean disks (storage node devices). If the installation is done on a previously deployed PowerFlex cluster, or used disks, you must tell the installer to overwrite the older data.

NOTE: If you select this option, the data is erased.

- The device awareness check is enabled by default. If you disable this setting, it may cause potential device failures not to be detected early enough. To disable the diagnostic, turn **Enable Diagnostics Check** to off.

10. Click **Next**.
11. In the **Summary** tab, verify all the configurations and click **Finish**.
12. Click the **Go to Events Page** button to track the installation progress.

Verifying the initial configuration (mirroring)

Before you complete the initial configuration, you can review all the settings that you provided for Dell Technologies connectivity services, compliance, and installation type.

1. On the **Summary** page, verify the settings that you configured on the previous pages.
2. To edit any information, click **Back** or click the corresponding page name in the left pane.
3. If you are importing an existing PowerFlex instance from PowerFlex Manager, type `IMPORT POWERFLEX MANAGER`.
4. If the information is correct, click **Finish** to complete the initial configuration.

   If you are importing an existing environment, PowerFlex Manager displays a message indicating that the import operation is in progress. When the import operation is complete, PowerFlex Manager displays the **Getting Started** page.

After completing the **Initial Configuration Wizard**, you may need to perform some additional steps. The steps vary depending on which installation type you selected:

- If you clicked **I want to deploy a new instance of PowerFlex** on the **Installation Type** page, you now must deploy a new instance of PowerFlex. The **Getting Started** page provides more information about what to do next.
- If you clicked **I have a PowerFlex instance to import** on the **Installation Type** page, you must perform these steps:
  1. On the **Settings** page, upload the compatibility matrix file and upload the latest PowerFlex software catalog.

     This catalog only includes the components that are required for an upgrade of PowerFlex.
  2. On the **Resources** page, select the PowerFlex entry, and perform a nondisruptive update.

     You do not need a resource group (service) to perform an upgrade of the PowerFlex environment. PowerFlex Manager does not support **Add Existing Resource Group** operations for a PowerFlex software migration. If you want to be able to perform any deployments, you need a new resource group. Therefore, you must create a new template (or clone a sample template), and deploy a new resource group from the template.
- If you clicked **I have a PowerFlex instance managed by PowerFlex Manager to import** on the **Installation Type** page, you must perform these steps:
  1. On the **Settings** page, upload the compatibility matrix file, resynchronize the existing RCM, download the latest repository catalog (RCM or IC), and make it the default catalog.
  2. On the **Resources** page, select the PowerFlex entry, and perform a nondisruptive update.
  3. On the **Resource Groups** page, perform an RCM/IC upgrade on any migrated service that must be upgraded.

     The migrated resource groups are initially non-compliant, because PowerFlex Manager is running a later RCM that includes PowerFlex 4.x. These resource groups must be upgraded to the latest RCM before they can be expanded or managed with automation operations.

     CAUTION: Check the Alerts page before performing the upgrade. Look for major and critical alerts that are related to PowerFlex Block to be sure that the MDM cluster is healthy before proceeding.
  4. Power off the old PowerFlex Manager VM, the old PowerFlex gateway VM, and the presentation server VM.

     The upgrade of the cluster causes the old PowerFlex Manager virtual appliances to stop working.
  5. After validating the upgrade, decommission the old instances of PowerFlex Manager, the PowerFlex gateway, and the presentation server.

     Do not delete the old instances until you have had a chance to review the initial configuration and confirm that the old environment was migrated successfully.

Getting started (mirroring)

The **Getting Started** page guides you through the common configurations that are required to prepare a new PowerFlex Manager environment. A green checkmark on a step indicates that you have completed the step. Only super users have access to the **Getting Started** page.

The following table describes each step:

Table 18. Getting Started steps

Step	Description
Upload Compliance File	Provide compliance file location and authentication information for use within PowerFlex Manager. The compliance file defines the specific hardware components and software version combinations that are tested and certified by Dell for hyperconverged infrastructure and other Dell products. This step enables you to choose a default compliance version for compliance or add new compliance versions. You can also click **Settings** > **Repositories** > **Compliance Versions**. **NOTE:** Before you make an RCM or IC the default compliance version, you must first upload a suitable compatibility management file under **Settings** > **Repositories** > **Compatibility Management**.
Define Networks	If you plan to perform an advanced CSV-based deployment, this step can be skipped. Enter detailed information about the available networks in the environment. This information is used later during deployments that are based on templates and resource groups. These deployments use the network information to configure nodes and switches to have the right network connectivity. PowerFlex Manager uses the defined networks in templates to specify the networks or VLANs that are configured on nodes and switches for your resource groups. This step is enabled immediately after you perform an initial configuration for PowerFlex Manager. You can also click **Settings** > **Networking** > **Networks**.
Discover Resources	If you plan to perform an advanced CSV-based deployment, this step can be skipped. Grant PowerFlex Manager access to resources (nodes, switches, virtual machine managers) in the environment by providing the management IP and credential for the resources to be discovered. This step is not enabled until you define your networks. You can also click **Resources** > **Discover Resources**.
Manage Deployed Resources (Optional)	If you plan to perform an advanced CSV-based deployment, this step can be skipped. Add an existing resource group for a cluster that is already deployed and manage the resources within PowerFlex Manager. This step is not enabled until you define your networks. You can also click **Lifecycle** > **Resource Groups** > **Add Existing Resource Group**.
Deploy Resources	If you plan to perform an advanced CSV-based deployment, this step can be skipped. Create a template with requirements that must be followed during a deployment. Templates enable you to automate the process of configuring and deploying infrastructure and workloads. For most environments, you can clone

Table 18. Getting Started steps (continued)

Step	Description
	one of the sample templates that are provided with PowerFlex Manager and modify as needed. Choose the sample template that is most appropriate for your environment. For example, for a hyperconverged deployment, clone one of the hyperconverged templates. For a two-layer deployment, clone the compute-only templates. Then clone one of the storage templates. This step is not enabled until you define your networks. You can also click **Lifecycle** > **Templates**.
Manage PowerFlex License	Configure licensing for PowerFlex. You can also click **Settings** > **License Management**.

If you would like to deploy PowerFlex with a CSV file, click **Deploy With Installation File**. See the topic on deploying a PowerFlex cluster using a CSV topology file in the *PowerFlex and PowerFlex Manager Software Install Guide* for more information.

Related information

Networking
Discover a resource
Adding an existing resource group (mirroring)
Templates
License management
Adding an existing resource group

Changing your password (mirroring)

When you first log in to PowerFlex Manager, you must set your password. You can also change your password at any time after the first login.

1. Click the user icon in the upper right corner of PowerFlex Manager.
2. Click **Change password**.
3. Enter the **Current Password**.
   If you are changing a password for another user entering the current password is not required.
4. Type the password in the **New Password** field.
   Password rules are:
   - Includes fewer than 32 characters
   - Includes only alphanumeric and punctuation characters
5. Type the password again in the **Verify Password** field.
6. Click **Apply**.

7

Performing common tasks (erasure coding)

This section includes tasks that you perform regularly when using PowerFlex Manager.

Deploying and provisioning

The following table describes common tasks for deploying and provisioning the system and what steps to take in PowerFlex Manager to initiate each task:

Table 19. Common tasks

If you want to...	Do this in PowerFlex Manager...
Define a resource group and infrastructure requirements	1. If you are managing and updating components other than PowerFlex, upload the compliance file on the **Settings** > **Repositories** > **Compliance Versions** page. 2. Define the networks that you need on the **Settings** > **Networking** > **Networks** page. 3. Discover the resources required for your deployment on the **Resources** > **Discover Resources** page. 4. Clone a sample template and make changes as needed. Click **Lifecycle** > **Templates** and click **Sample templates**. Open a sample template and click **More Actions** > **Clone** on the right side of the screen. For example, for a two-layer deployment, clone one of the PowerFlex storage-only templates. Then clone one of the PowerFlex compute-only templates. You can also create a template. However, for most environments, you can clone one of the sample templates that are provided with PowerFlex Manager.
Deploy a new resource group	Click **Lifecycle** > **Resource Groups**. On the **Resource Groups** page, click **Deploy New Resource Group**. You can only deploy a resource group using a published template.
Perform an advanced PowerFlex CSV deployment	If you would like to use the legacy PowerFlex installer to provide an installation topology file, click **Deploy With Installation File** on the **Getting Started** page. See *Deploy a PowerFlex cluster using a CSV topology file*.

Related information

Repositories
Networking
Resources
Preparing a template for a resource group deployment
Deploying a resource configuration in a resource group

Configuring block storage

The following table describes common tasks for deploying and managing block storage and what steps to take in PowerFlex Manager to initiate each task:

Table 20. Common tasks

If you want to...	Do this in PowerFlex Manager...
Deploy a new block storage configuration	1. If you are managing and updating components other than PowerFlex, upload the compliance file on the **Settings** > **Repositories** > **Compliance Versions** page. 2. Define the networks that you need on the **Settings** > **Networking** > **Networks** page. 3. Discover the resources required for your deployment on the **Resources** > **Discover Resources** page. 4. Clone a sample template and make changes as needed. Click **Lifecycle** > **Templates** and click **Sample templates**. Open a sample template and click **More Actions** > **Clone** on the right side of the screen. 5. Click **Lifecycle** > **Resource Groups**. On the **Resource Groups** page, click **Deploy New Resource Group**. For a two-layer deployment, clone one of the PowerFlex storage-only templates. Then, clone one of the PowerFlex compute-only templates. You can also create a template. However, for most environments, you can clone one of the sample templates that are provided with PowerFlex Manager.
Import an existing block storage configuration	Click **Lifecycle** > **Resource Groups** and click **Add Existing Resource Group**. Be sure to upload a compliance file (if necessary), define the networks, and discover resources before adding the existing resource group.
Manage block storage	On the menu bar, click **Block**, and choose the type of block storage components you want to manage: • **Storage System** • **SDSs** • **Volumes** • **Hosts** If you create objects on the **Block** tab, you must update the inventory on the **Resources** page, and then click **Update Resource Group Details** on the **Lifecycle** > **Resource Groups** page for any resource group that requires the updates.

Related information

Repositories
Networking
Discover a resource
Clone a template
Deploying a resource configuration in a resource group
Discover a resource

Managing components

Once PowerFlex Manager is configured, you can use it to manage the system.

The following table describes common tasks for managing system components and what steps to take in PowerFlex Manager to initiate each task:

Table 21. Common tasks

If you want to...	Do this in PowerFlex Manager...
Add an existing resource group	Click **Lifecycle** > **Resource Groups** and click **Add Existing Resource Group**.

Table 21. Common tasks (continued)

If you want to...	Do this in PowerFlex Manager...
	Be sure to upload a compliance file (if necessary), define the networks, and discover resources before adding the existing resource group.
Enter maintenance mode (erasure coding)	1. Click **Lifecycle** > **Storage Nodes**. 2. From the list of storage nodes, select the checkbox beside the storage node that you want to enter into maintenance mode and click **More Actions** > **Enter Maintenance Mode**. 3. In the **Enter Storage Node into Maintenance Mode** dialog box, verify that you are selecting the wanted storage node, and click **Enter Maintenance Mode**.
Exit maintenance mode (erasure coding)	1. Click **Lifecycle** > **Storage Nodes**. 2. From the list of storage nodes, select the checkbox beside the storage node that you want to exit from maintenance mode and click **More Actions** > **Exit Maintenance Mode**. 3. In the **Exit Maintenance Mode on Storage Node** dialog box, verify that you are selecting the wanted storage node, and click **Exit Maintenance Mode**.
Enter service mode (applicable for PowerFlex appliance and PowerFlex rack only)	1. Click **Lifecycle** > **Resource Groups**. 2. On the **Resource Groups** page, select a resource group. 3. On the **Resource Group Details** tab, under **More Actions**, click **Enter Service Mode**.
Exit service mode (applicable for PowerFlex appliance and PowerFlex rack only)	1. Click **Lifecycle** > **Resource Groups**. 2. On the **Resource Groups** page, select a resource group. 3. On the **Resource Group Details** tab, under **More Actions**, click **Exit Service Mode**.
Perform node expansion	1. Click **Lifecycle** > **Resource Groups**. On the **Resource Groups** page, select a resource group. 2. On the **Resource Group Details** tab, under **Add Resources**, click **Add Nodes**. The procedure is the same for new resource groups and existing resource groups.
Remove a node	1. Click **Lifecycle** > **Resource Groups**. 2. On the **Resource Groups** page, select a resource group. 3. On the **Resource Group Details** tab, under **More Actions**, click **Remove Resource**. 4. Select **Delete Resource** for the **Resource removal type**.
Replace a drive	1. Click **Lifecycle** > **Resource Groups**. 2. On the **Resource Groups** page, select a resource group. 3. Under **Physical Nodes**, click **Drive Replacement**.
Reconfigure MDM roles	1. Click **Lifecycle** > **Resource Groups**. 2. On the **Resource Groups** page, select a resource group. 3. On the **Resource Group Details** tab, click **Reconfigure MDM Roles** under **More Actions**. You can also reconfigure MDM roles from the **Resources** page. Select a PowerFlex gateway and click **View Details**. Then, click **Reconfigure MDM Roles**.
Run inventory (nodes, switches, PowerFlex gateway, and VMware vCenter cluster)	1. Click **Resources** and click the **All Resources** tab. 2. Click the checkbox for the resource you want to update and then click **Run Inventory**. 3. After running the inventory, click **Update Resource Group Details** under **More Actions** on the **Resource Groups** page for any resource group that requires the updated resource data.
View network topology	1. Click **Lifecycle** > **Resource Groups**. 2. On the **Resource Groups** page, select a resource group. 3. On the **Resource Group Details** tab, click the **Port View** tab.

Related information

Life cycle
Resource groups
Templates
Resources

Monitoring system health

Once PowerFlex Manager is configured, you can monitor system health.

The following table describes common tasks for monitoring system health and managing software and firmware compliance and what steps to take in PowerFlex Manager to initiate each task:

Table 22. Common tasks

If you want to...	Do this in PowerFlex Manager...
Download a report that lists compliance details for all resources	1. Click **Resources**. 2. Click **Export Report** and download a compliance report (PDF or CSV) or a configuration report (PDF).
Generate a troubleshooting bundle	1. Click **Settings** and click **Serviceability**. 2. Click **Generate Troubleshooting Bundle**. You can also generate a troubleshooting bundle from the **Resource Groups** page: 1. Click **Lifecycle** > **Resource Groups**. 2. On the **Resource Groups** page, select a resource group. 3. On the **Resource Group Details** page, click **Generate Troubleshooting Bundle**.
Monitor software and firmware compliance	1. Click **Lifecycle** > **Resource Groups**. 2. On the **Resource Groups** page, select a resource group. 3. On the **Resource Group Details** page, click **View Compliance Report**.
Monitor system resources and health	On the **Dashboard**, look at the **Overall Performance**, **Usable Capacity**, **Raw Capacity**, **Data Savings**, and **Resources / Inventory** sections. For information about which resource groups are healthy and in compliance and which are not, look at the **Resource Groups** section.
Monitor system resources and health	On the **Dashboard**, look at the **Overall Performance**, **Usable Capacity**, ,**Data Savings** sections. For information about which resource groups are healthy and in compliance, and which are not, look at the **Resource Groups** section.
Perform software and firmware remediation	From the compliance report, view the firmware or software components. Click **Update Resources** to update non-compliant resources.
View alerts	On the menu bar, click **Monitoring** > **Alerts**.

Related information

Life cycle
Resources

Managing external changes

If you make manual changes outside of PowerFlex Manager, you might need to perform some steps within PowerFlex Manager to ensure that the external changes are reflected within the user interface and the environment is kept in a healthy state.

The following table describes common tasks for managing external changes and what steps to take in PowerFlex Manager for each task:

Table 23. Common tasks

If you have...	Do this in PowerFlex Manager...
Created objects on the **Block** tab	Click **Run Inventory** on the **Resources** page to update the inventory. Then, click **Update Resource Group Details** on the **Lifecycle** > **Resource Groups** page for any resource group that requires the updates.
Manually added nodes outside of PowerFlex Manager	Click **Run Inventory** on the **Resources** page to update the inventory. Then, click **Update Resource Group Details** on the **Lifecycle** > **Resource Groups** page for any resource group that requires the updated nodes.
Manually added VMware NSX to a cluster outside of PowerFlex Manager	Click **Update Resource Group Details** on the resource group, so it will correctly reflect the environment with VMware NSX, which will be in life cycle mode.
Manually changed the IP address for a switch	1. Remove the switch. On the **Resources** page, click the **All Resources** tab. From the list of resources, select the switch, and click **Remove**. 2. Rediscover the switch. On the **Resources** page, click the **All Resources** tab and click **Discover**. 3. If the IP address is in use, remove the resource group and add it again. On the **Lifecycle** > **Resource Groups** page, click **+Add Existing Resource Group**.
Manually created additional VLANs on the VDS in vCenter outside of PowerFlex Manager	1. Remove the resource group. On the **Resource Group Details** page, click **Remove Resource Group** under **More Actions**. When you click **Remove Resource Group**, do not select **Delete Resource Group** as the **Resource Group Removal Type**. This action completely tears down the resource group and makes configuration changes to the nodes, switch ports, virtual machine managers, and PowerFlex. Instead, select **Remove Resource Group** as the **Resource Group Removal Type**. Then, to keep the nodes in the inventory, select **Leave nodes in PowerFlex Manager inventory and set state to** and choose **Managed**. 2. Rediscover the resource group. On the **Lifecycle** > **Resource Groups** page, click **+Add Existing Resource Group**. If you do not choose the correct options when you remove the resource group, you could tear down the resource group and destroy it, or leave the servers in an unmanaged state, and not be able to add the existing resource group.
Manually created volumes outside of PowerFlex Manager	Click **Run Inventory** on the **Resources** page to update the inventory. Then, click **Update Resource Group Details** on the **Lifecycle** > **Resource Groups** page for any resource group that requires the updated volumes.
Manually deleted volumes outside of PowerFlex Manager	Click **Run Inventory** on the **Resources** page to update the inventory. Then, click **Update Resource Group Details** on the **Lifecycle** > **Resource Groups** page for any resource group that requires updated information about volumes deleted. PowerFlex Manager displays a message indicating that the volumes have been removed from the resource group.
Manually removed nodes outside of PowerFlex Manager	Click **Run Inventory** on the **Resources** page to update the inventory. Then, click **Update Resource Group Details** on the **Resource Groups** page for any resource group that requires updated information about nodes deleted. Then, manually remove the resource on the **Resources** page by clicking the **All Resources** tab, selecting the resource, and clicking **Remove**.
Manually replaced a node outside of PowerFlex Manager for a resource group in PowerFlex Manager	1. Remove the node from the resource group.

Table 23. Common tasks (continued)

If you have...	Do this in PowerFlex Manager...
	On the **Lifecycle** > **Resource Groups** page, select a resource group. On the **Resource Group Details** page, under **More Actions**, click **Remove Resource**. 2. Remove the node from the list of resources. On the **Resources** page, click the **All Resources** tab. From the list of resources, select the resource, and click **Remove**. 3. Discover the new node. On the **Resources** page, click **Discover** on the **All Resources** tab. 4. Update the resource group details. On the **Lifecycle** > **Resource Groups** page, click **Update Resource Group Details**.
Renamed objects such as VMware ESXi host, volume, datastore, VDS, port group, data center, cluster, and so forth	Click **Run Inventory** on the **Resources** page to update the inventory. Then, click **Update Resource Group Details** on the **Lifecycle** > **Resource Groups** page for any resource group that requires the updates.

Related information

Life cycle
Resource groups

8

Performing common tasks (mirroring)

This section includes tasks that you perform regularly when using PowerFlex Manager.

Deploying and provisioning

The following table describes common tasks for deploying and provisioning the system and what steps to take in PowerFlex Manager to initiate each task:

Table 24. Common tasks

If you want to...	Do this in PowerFlex Manager...
Perform an advanced PowerFlex CSV deployment	If you would like to use the legacy PowerFlex Installer to provide an installation topology file, click **Deploy With Installation File** on the **Getting Started** page. See the topic on deploying a PowerFlex cluster using a CSV topology file in the *Dell PowerFlex and PowerFlex Manager Install and Upgrade Guide* for more information.
Define a resource group and infrastructure requirements	1. If you will be managing and updating components other than PowerFlex, upload the compliance file on the **Settings** > **Repositories** > **Compliance Versions** page. 2. Define the networks you need on the **Settings** > **Networking** > **Networks** page. 3. Discover the resources required for your deployment on the **Resources** > **Discover Resources** page. 4. Clone a sample template and make changes as needed. Click **Lifecycle** > **Templates** and click **Sample templates**. Open a sample template and click **More Actions** > **Clone** on the right side of the screen. For example, for a PowerFlex hyperconverged deployment, clone one of the PowerFlex hyperconverged templates. For a two-layer deployment, clone one of the PowerFlex storage-only templates. Then clone one of the PowerFlex compute-only templates. You can also create a new template. However, for most environments, you can simply clone one of the sample templates that are provided with PowerFlex Manager.
Deploy a new resource group	Click **Lifecycle** > **Resource Groups**. On the **Resource Groups** page, click **Deploy New Resource Group**. You can only deploy a resource group using a published template.

Configuring block storage

The following table describes common tasks for deploying and managing block storage and what steps to take in PowerFlex Manager to initiate each task:

Table 25. Common tasks

If you want to...	Do this in PowerFlex Manager...
Deploy a new block storage configuration	1. If you will be managing and updating components other than PowerFlex, upload the compliance file on the **Settings** > **Repositories** > **Compliance Versions** page. 2. Define the networks you need on the **Settings** > **Networking** > **Networks** page. 3. Discover the resources required for your deployment on the **Resources** > **Discover Resources** page. 4. Clone a sample template and make changes as needed. Click **Lifecycle** > **Templates** and click **Sample templates**. Open a sample template and click **More Actions** > **Clone** on the right side of the screen. 5. Click **Lifecycle** > **Resource Groups**. On the **Resource Groups** page, click **Deploy New Resource Group**. For example, for a PowerFlex hyperconverged deployment, clone one of the PowerFlex hyperconverged templates. For a two-layer deployment, clone one of the PowerFlex storage-only templates. Then, clone one of the PowerFlex compute-only templates. You can also create a new template. However, for most environments, you can simply clone one of the sample templates that are provided with PowerFlex Manager.
Import an existing block storage configuration	Click **Lifecycle** > **Resource Groups** and click **+Add Existing Resource Group**. Be sure to upload a compliance file (if necessary), define the networks, and discover resources before adding the existing resource group.
Manage block storage	On the menu bar, click **Block**, and choose the type of block storage components you want to manage: • **Protection Domains** • **Fault Sets** • **SDSs** • **Storage Pools** • **Acceleration Pools** • **Devices** • **Volumes** • **NVMe Targets** • **Hosts** If you create new objects on the **Block** tab, you need to update the inventory on the **Resources** page, and then click **Update Resource Group Details** on the **Lifecycle** > **Resource Groups** page for any resource group that requires the updates.

Managing components

Once PowerFlex Manager is configured, you can use it to manage the system.

The following table describes common tasks for managing system components and what steps to take in PowerFlex Manager to initiate each task:

Table 26. Common tasks

If you want to...	Do this in PowerFlex Manager...
View network topology	1. Click **Lifecycle** > **Resource Groups**. 2. On the **Resource Groups** page, select a resource group.

Table 26. Common tasks (continued)

If you want to...	Do this in PowerFlex Manager...
	3. On the **Resource Group Details** tab, click the **Port View** tab.
Run inventory (nodes, switches, PowerFlex gateway, PowerFlex management cluster (KVM), and VMware vCenter cluster)	1. Click **Resources** and click the **All Resources** tab. 2. Click the checkbox for the resource you want to update and then click **Run Inventory**. 3. After running the inventory, click **Update Resource Group Details** under **More Actions** on the **Resource Groups** page for any resource group that requires the updated resource data.
Add an existing resource group	Click **Lifecycle** > **Resource Groups** and click **+Add Existing Resource Group**. Be sure to upload a compliance file (if necessary), define the networks, and discover resources before adding the existing resource group.
Perform node expansion	1. Click **Lifecycle** > **Resource Groups**. On the **Resource Groups** page, select a resource group. 2. On the **Resource Group Details** tab, under **Add Resources**, click **Add Nodes**. The procedure is the same for new resource groups and existing resource groups.
Remove a node	1. Click **Lifecycle** > **Resource Groups**. 2. On the **Resource Groups** page, select a resource group. 3. On the **Resource Group Details** tab, under **More Actions**, click **Remove Resource**. 4. Select **Delete Resource** for the **Resource removal type**.
Enter service mode (applicable for PowerFlex appliance and PowerFlex rack only)	1. Click **Lifecycle** > **Resource Groups**. 2. On the **Resource Groups** page, select a resource group. 3. On the **Resource Group Details** tab, under **More Actions**, click **Enter Service Mode**.
Exit service mode (applicable for PowerFlex appliance and PowerFlex rack only)	1. Click **Lifecycle** > **Resource Groups**. 2. On the **Resource Groups** page, select a resource group. 3. On the **Resource Group Details** tab, under **More Actions**, click **Exit Service Mode**.
Replace a drive	1. Click **Lifecycle** > **Resource Groups**. 2. On the **Resource Groups** page, select a resource group. 3. Under **Physical Nodes**, click **Drive Replacement**.
Reconfigure MDM roles	1. Click **Lifecycle** > **Resource Groups**. 2. On the **Resource Groups** page, select a resource group. 3. On the **Resource Group Details** tab, click **Reconfigure MDM Roles** under **More Actions**. You can also reconfigure MDM roles from the **Resources** page. Select a PowerFlex gateway and click **View Details**. Then, click **Reconfigure MDM Roles**.

Related information

Life cycle
Resource groups
Templates
Resources

Monitoring system health

Once PowerFlex Manager is configured, you can monitor system health.

The following table describes common tasks for monitoring system health and managing software and firmware compliance and what steps to take in PowerFlex Manager to initiate each task:

Table 27. Common tasks

If you want to...	Do this in PowerFlex Manager...
Download a report that lists compliance details for all resources	1. Click **Resources**. 2. Click **Export Report** and download a compliance report (PDF or CSV) or a configuration report (PDF).
Generate a troubleshooting bundle	1. Click **Settings** and click **Serviceability**. 2. Click **Generate Troubleshooting Bundle**. You can also generate a troubleshooting bundle from the **Resource Groups** page: 1. Click **Lifecycle** > **Resource Groups**. 2. On the **Resource Groups** page, select a resource group. 3. On the **Resource Group Details** page, click **Generate Troubleshooting Bundle**.
Monitor software and firmware compliance	1. Click **Lifecycle** > **Resource Groups**. 2. On the **Resource Groups** page, select a resource group. 3. On the **Resource Group Details** page, click **View Compliance Report**.
Monitor system resources and health	On the **Dashboard**, look at the **Overall Performance**, **Usable Capacity**, **Raw Capacity**, **Data Savings**, and **Resources / Inventory** sections. For information about which resource groups are healthy and in compliance, and which are not, look at the **Resource Groups** section.
Monitor system resources and health	On the **Dashboard**, look at the **Overall Performance**, **Usable Capacity**, **Data Savings** sections. For information about which resource groups are healthy and in compliance, and which are not, look at the **Resource Groups** section.
Perform software and firmware remediation	From the compliance report, view the firmware or software components. Click **Update Resources** to update non-compliant resources.
View alerts	On the menu bar, click **Monitoring** > **Alerts**.

Related information

Life cycle
Resources

Managing external changes

If you make manual changes outside of PowerFlex Manager, you might need to perform some steps within PowerFlex Manager to ensure that the external changes are reflected within the user interface and the environment is kept in a healthy state.

The following table describes common tasks for managing external changes and what steps to take in PowerFlex Manager for each task:

Table 28. Common tasks

If you have...	Do this in PowerFlex Manager...
Created objects on the **Block** tab	Click **Run Inventory** on the **Resources** page to update the inventory. Then, click **Update Resource Group Details** on the **Lifecycle** > **Resource Groups** page for any resource group that requires the updates.
Manually added nodes outside of PowerFlex Manager	Click **Run Inventory** on the **Resources** page to update the inventory. Then, click **Update Resource Group Details** on the **Lifecycle** > **Resource Groups** page for any resource group that requires the updated nodes.
Manually added VMware NSX to a cluster outside of PowerFlex Manager	Click **Update Resource Group Details** on the resource group, so it will correctly reflect the environment with VMware NSX, which will be in life cycle mode.

Table 28. Common tasks (continued)

If you have...	Do this in PowerFlex Manager...
Manually changed the IP address for a switch	1. Remove the switch. On the **Resources** page, click the **All Resources** tab. From the list of resources, select the switch, and click **Remove**. 2. Rediscover the switch. On the **Resources** page, click the **All Resources** tab and click **Discover**. 3. If the IP address is in use, remove the resource group and add it again. On the **Lifecycle** > **Resource Groups** page, click **+Add Existing Resource Group**.
Manually created additional VLANs on the VDS in vCenter outside of PowerFlex Manager	1. Remove the resource group. On the **Resource Group Details** page, click **Remove Resource Group** under **More Actions**. When you click **Remove Resource Group**, do not select **Delete Resource Group** as the **Resource Group Removal Type**. This action completely tears down the resource group and makes configuration changes to the nodes, switch ports, virtual machine managers, and PowerFlex. Instead, select **Remove Resource Group** as the **Resource Group Removal Type**. Then, to keep the nodes in the inventory, select **Leave nodes in PowerFlex Manager inventory and set state to** and choose **Managed**. 2. Rediscover the resource group. On the **Lifecycle** > **Resource Groups** page, click **+Add Existing Resource Group**. If you do not choose the correct options when you remove the resource group, you could tear down the resource group and destroy it, or leave the servers in an unmanaged state, and not be able to add the existing resource group.
Manually created volumes outside of PowerFlex Manager	Click **Run Inventory** on the **Resources** page to update the inventory. Then, click **Update Resource Group Details** on the **Lifecycle** > **Resource Groups** page for any resource group that requires the updated volumes.
Manually deleted volumes outside of PowerFlex Manager	Click **Run Inventory** on the **Resources** page to update the inventory. Then, click **Update Resource Group Details** on the **Lifecycle** > **Resource Groups** page for any resource group that requires updated information about volumes deleted. PowerFlex Manager displays a message indicating that the volumes have been removed from the resource group.
Manually removed nodes outside of PowerFlex Manager	Click **Run Inventory** on the **Resources** page to update the inventory. Then, click **Update Resource Group Details** on the **Resource Groups** page for any resource group that requires updated information about nodes deleted. Then, manually remove the resource on the **Resources** page by clicking the **All Resources** tab, selecting the resource, and clicking **Remove**.
Manually replaced a node outside of PowerFlex Manager for a resource group in PowerFlex Manager	1. Remove the node from the resource group. On the **Lifecycle** > **Resource Groups** page, select a resource group. On the **Resource Group Details** page, under **More Actions**, click **Remove Resource**. 2. Remove the node from the list of resources. On the **Resources** page, click the **All Resources** tab. From the list of resources, select the resource, and click **Remove**. 3. Discover the new node. On the **Resources** page, click **Discover** on the **All Resources** tab. 4. Update the resource group details.

Table 28. Common tasks (continued)

If you have...	Do this in PowerFlex Manager...
	On the **Lifecycle** > **Resource Groups** page, click **Update Resource Group Details**.
Renamed objects such as VMware ESXi host, volume, datastore, VDS, port group, data center, cluster, and so forth	Click **Run Inventory** on the **Resources** page to update the inventory. Then, click **Update Resource Group Details** on the **Lifecycle** > **Resource Groups** page for any resource group that requires the updates.

9

Displaying system information at a glance (erasure coding)

This section introduces the dashboard, which displays a graphical view of the overall health of the system.

The dashboard displays at-a-glance information about overall performance, usable and raw capacity, data savings, resources and inventory, resource groups, and alerts.

Interpreting overall performance and bandwidth

This section displays a graph that shows the overall performance (Input/Output Operations Per Second (IOPS) and bandwidth) for a block configuration within the system.

IOPS measures how many read and write operations the system can handle per second. Bandwidth measures the total amount of data that is transferred per second in GB/s.

The relationship between IOPS and bandwidth can be summarized as follows:

- High IOPS, low bandwidth: This scenario (which is shown in Figure 1: Overall performance) occurs when there are many small I/O operations. Each operation transfers a small amount of data, resulting in high IOPS (7265 K) but relatively low bandwidth (4.84 GB/s).
- Low IOPS, Hhgh bandwidth: This scenario occurs when there are fewer, larger I/O operations. Each operation transfers a large amount of data, leading to lower IOPS but higher bandwidth.

You can hover over the graph at any time to get the exact metrics for that point in time.

The following graph is an example of how overall performance (IOPS and bandwidth) for a block configuration is represented in the dashboard:

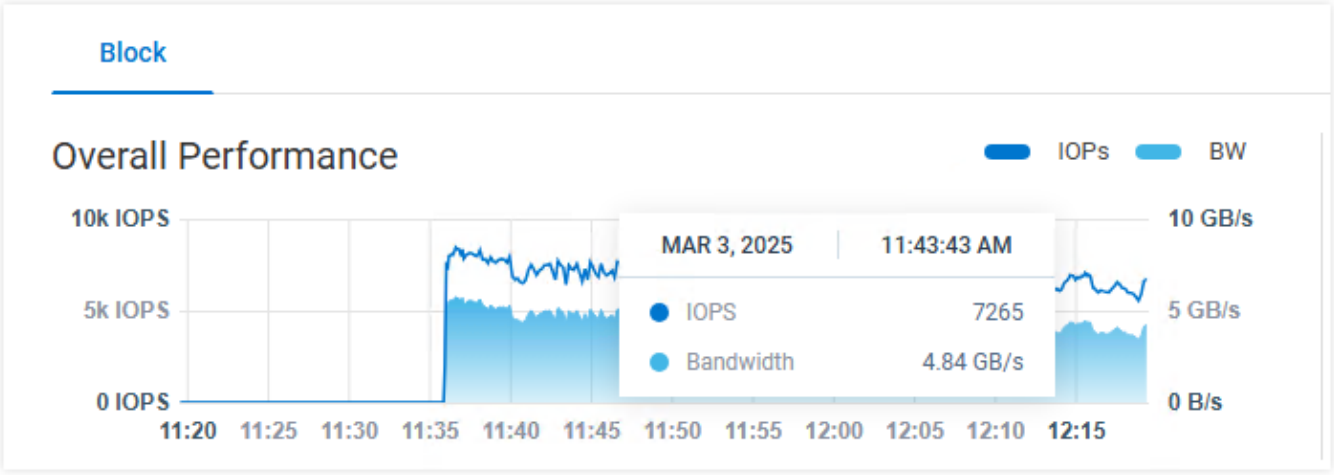


Figure 1. Overall performance

Interpreting latency

This section displays a graph that shows latency levels for a block configuration within the system.

Latency is the time that it takes for a single I/O operation to complete typically measured in milliseconds (ms). It represents the delay between when a request (reading or writing) for data is made and the data is delivered.

The relationship between latency, IOPS, and bandwidth can be summarized as follows:

- IOPS and latency: Higher IOPS means lower latency (faster response times) but only up to the limits of the system.
- Bandwidth and latency: High bandwidth does not necessarily mean low latency. A system with high bandwidth but high latency may still feel slow for small, frequent operations.

You can hover over the graph at any time to get the exact metrics for that point in time.

The following graph is an example of how latency for a block configuration is represented in the dashboard:

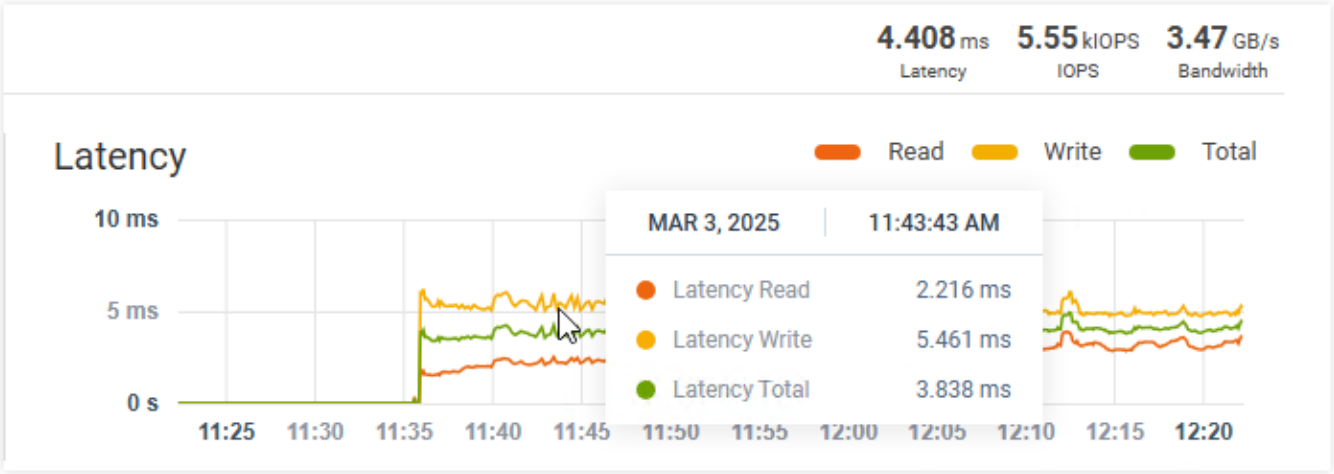


Figure 2. Latency

Interpreting usable capacity

This section displays a graph that shows the usable capacity within the system. Usable capacity is the physical capacity available for user data after applying a protection scheme and accounting for spare capacity.

For example, if you had 11 PowerFlex nodes with each PowerFlex node having 10 TB of capacity. Then, the raw capacity is 110 TB (11 x 10 TB = 110 TB). If you then allocate one PowerFlex node as a spare, the raw capacity reduces to 100 TB. If you apply the protection scheme (8 + 2), the total raw capacity reduces to 80 TB (80% of 100).

Used capacity is calculated as the amount of system capacity that is used as a percentage of the total system capacity.

The following diagram is an example of how usable capacity is represented in the dashboard:

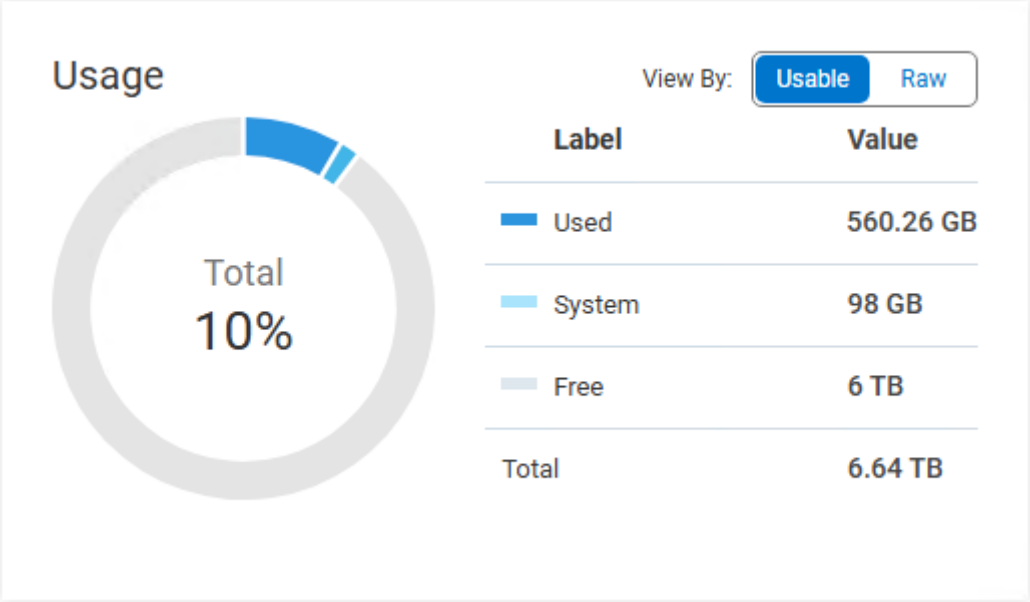


Figure 3. Usable capacity

Table 29. Usable capacity

Capacity	Description
Total	The total protected capacity that is available for user data and system (internal) usage.
Used	The physical capacity used by user data in the system.
System	The physical capacity in use by the system for internal use.
Free	The free physical capacity in the system.

Table 30. Raw capacity

Capacity	Description
Total raw	The total sum of devices in the system.
Raw used	The allocated raw capacity that is used by user data.
Raw system	The raw capacity that is in use by the system for internal use by the system.
Raw free	The free raw capacity that is available to allocate in the system.

Interpreting data savings

This section displays a graph that shows the data savings within the system. It provides details about the overall savings, thin and snapshot savings, and the data reduction ratio.

The following terms around data savings are used in the dashboard:

- Provisioned: The total size of volumes and snapshots in the system.
- Logical: The logical capacity in use by user data and snapshots in the system.
- Physical: The amount of storage that the user data and snapshots take after data reduction in the system. The physical resides on the storage pools.

The following graph is an example of how data savings are represented in the dashboard:

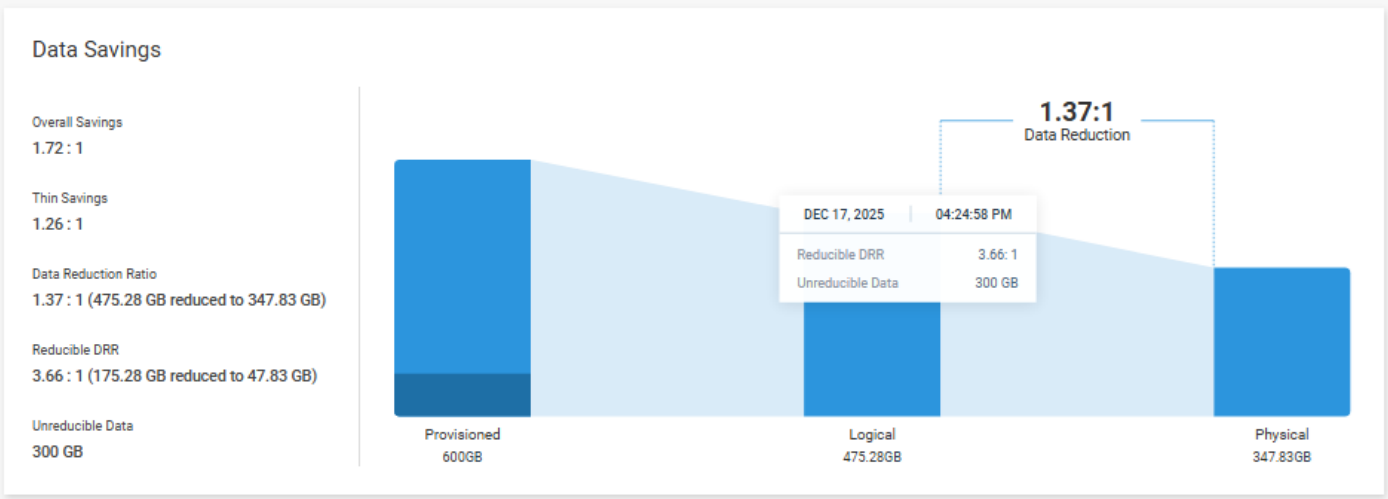


Figure 4. Data savings

Table 31. Data saving type

Savings	Description	Example
Overall savings	The ratio between provisioned and physical storage.	600 GB / 347.83 GB = a ratio of 1.72:1.
Thin and snapshot savings	The ratio between provisioned and logical used capacity.	600 GB / 475.28 GB = a ratio of 1.26:1.

Table 31. Data saving type (continued)

Savings	Description	Example
Data reduction ratio	The ratio of the logical used space to the physical used space after data reduction, for example, compression.	If the original data size is 475.28 GB and the reduced data size after compression is 347.83 GB, the data reduction ratio is: 1.37:1.
Unreducible data	The amount of data that is written within the system or protection domain that could not be reduced by PowerFlex's compression or pattern detection logic.	In this example, the unreducible data is 300 GB.
Reducible data reduction ratio (DRR)	The data reduction ratio for the reducible data within the system or protection domain. This metric ignores any unreducible data in the calculation.	The logical reducible data in the system is 175.28 GB, and the physical reducible data is 47.83 GB. Then, the reducible DRR is the ratio between the two: 3.66:1.

Resources/Inventory

This section displays the number of resources that are in the current inventory.

The list of resources include the following:

- VM managers
- Nodes
- Switches
- Protection domains
- Device groups
- Devices
- Storage pools
- Volumes
- Hosts

Resource groups

This section displays a graphical representation of the resource groups deployed based on status (for PowerFlex rack and PowerFlex appliance users only). The number next to each icon indicates the number of resource groups in a particular state.

The following diagram is an example of how the health and compliance of the resource groups are represented in the dashboard:

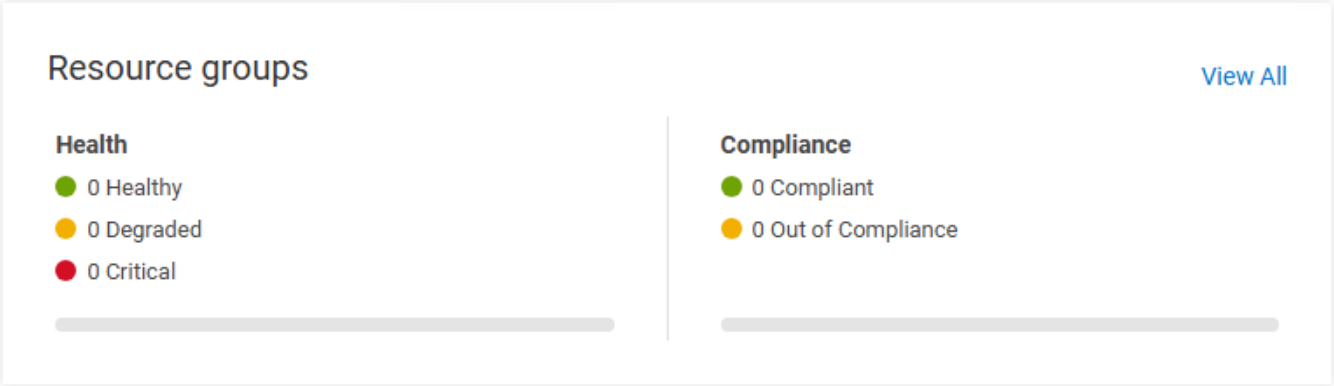


Figure 5. Resource group states

The resource groups are categorized into the following states:

Table 32. Resource group status

Tab	Status	Description
Health	**Healthy**	The green band on the graphic indicates that the resource group is successfully deployed.
	Degraded	The yellow band indicates that resources in a resource group require corrective action, but does not affect overall system health. For example, the firmware version that is installed on a resource in the resource group is not compliant.
	Critical	The red band indicates resource groups that have critical health problems.
	Service Mode	The yellow band indicates that the resource group has been placed in service mode.
Compliance	**Compliant**	The green band indicates that the resource group is compliant with the target compliance version.
	Out of compliance	The yellow band indicates that the resource group is non-compliant with the target compliance version.

You can monitor node health by viewing the status of the resource group on the **Resource Groups** page.

If a resource group is in a yellow (or warning) state, it means that one or more nodes is in a warning or failed state. If a resource group is in a red (or error) state, it indicates that the resource group has fewer than two nodes that are not in a failed state.

To view the status of a failed node component, hover the cursor on the image of the failed component in the resource group.

Alerts

This section lists the current alerts within the system that are categorized by the severity level.

The severity levels are:

- **Critical**: A critical level informs you that the system requires immediate attention. For example, the MDM certificate has expired.
- **Major**: A major level informs you that action is required soon. For example, the system is unable to receive MDM events.
- **Minor**: A minor level informs you that action is required but is not urgent. For example, PowerFlex is using a trial license.

The following diagram is an example of how the system alerts are represented in the dashboard:

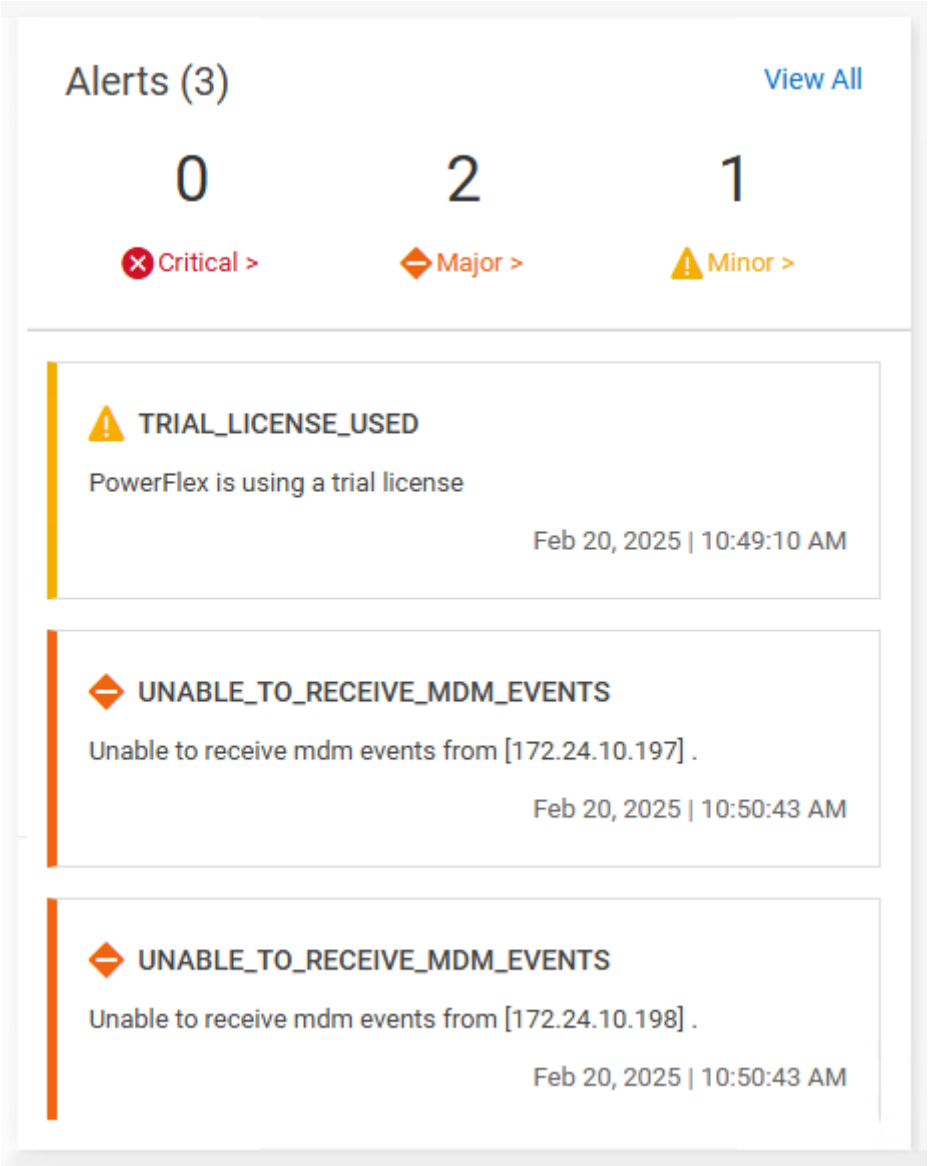


Figure 6. Alerts

For more detailed information about alerts, click **View All** to be redirected to the **Alerts** section under the **Monitoring** tab.

10

Displaying system information at a glance (mirroring)

The following section introduces the dashboard, which displays a graphical view of the overall health of the system.

The dashboard displays at-a-glance information about overall performance, usable and raw capacity, data savings, resources and inventory, resource groups, and alerts.

Overall performance and latency

This section of the dashboard displays graphs that show the overall performance (IOPS and bandwidth) and latency for the block configuration within the system.

Interpreting usable capacity

This section of the dashboard displays the usable capacity within the system. Used capacity is calculated as the amount of system capacity that is used as a percentage of the total system capacity. The following diagram is an example of how usable capacity is represented in the dashboard:

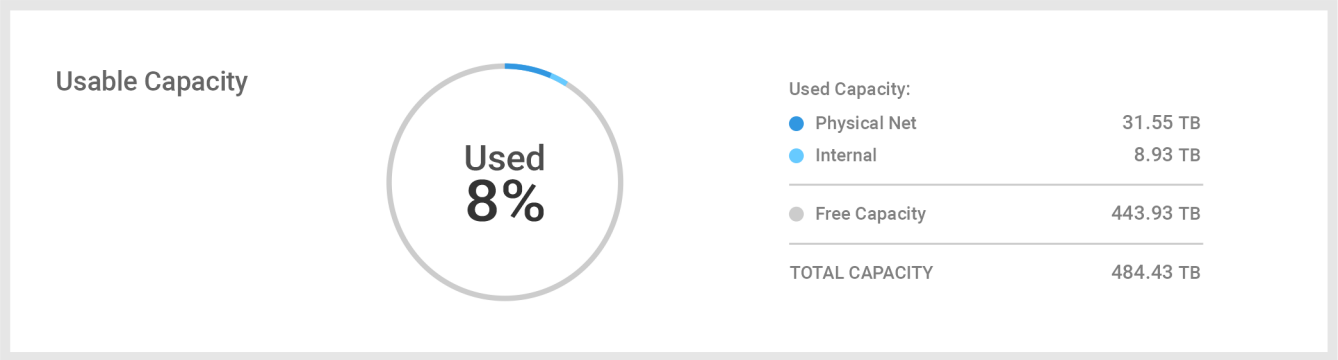


Figure 7. Usable capacity

The following table describes the terminology around capacity that is used in the dashboard:

Table 33. Usable capacity

Capacity	Description	Example (TB)
Total capacity	The total protected (mirrored) space that is available for user data and internal usage.	484.43
Physical net	The physical space on the storage pool after data reduction (if applicable). This includes the snapshot space. The physical net is an indication of the quantity of data that is stored in the system on the storage pool.	31.55
Internal	The physical space in use by the system for internal use.	8.93
Free capacity	The free physical space in the system.	443.93

Interpreting data savings

This section displays the data savings within the system.

The following terms around data savings are used in the dashboard:

- Provisioned: The total size of volumes and snapshots including over-provisioning (that is the total logical address range) in the system.
- Allocated: The logical capacity in use by user data and snapshots in the system.
- Physical net: The amount of storage that the user data and snapshots take after data reduction in the system. The physical net resides on the storage pools.

The following diagram is an example of how data savings are represented in the dashboard for medium granularity:

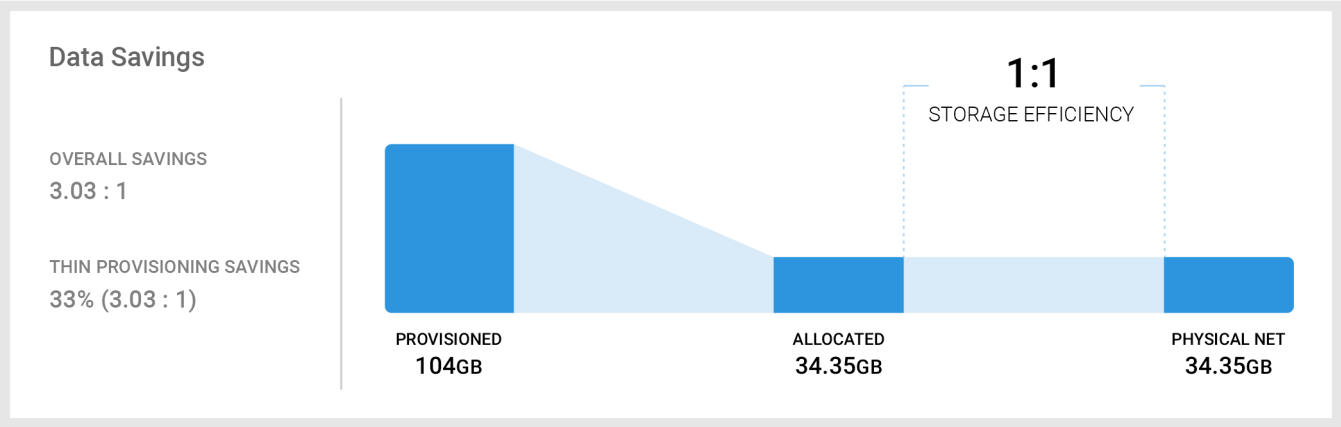
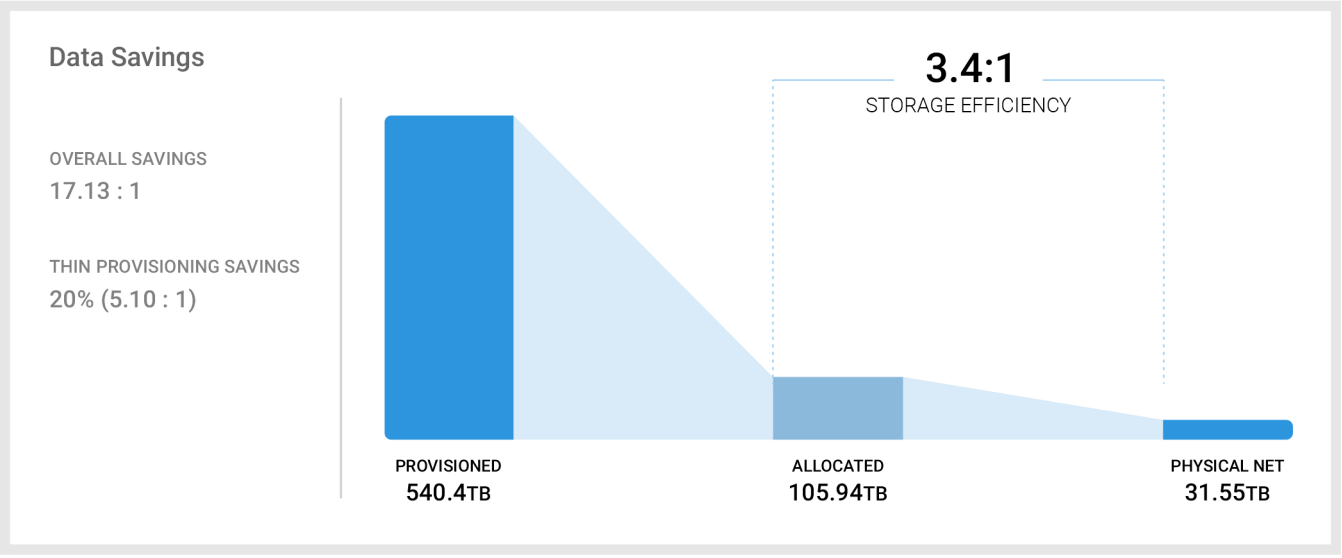


Figure 8. Data savings medium granularity

The following diagram is an example of how data savings are represented in the dashboard for fine granularity:

Figure 9. Data savings fine granularity



The following table describes the types of savings with examples:

Table 34. Data saving type

Savings type	Description	Example
Overall savings	The ratio between provisioned and physical net storage.	3.03% or 33%

Table 34. Data saving type (continued)

Savings type	Description	Example
		In medium granularity, this is 104 GB / 34.35 GB = 3.03. In fine granularity, this is 540.4 TB / 31.55 TB = 17.13.
Thin provisioning savings	The ratio between provisioned and allocated storage.	If 104 GB of storage is provisioned to VMs but only 34.35 GB is used on the volumes, the thin provisioning ratio is 3.03:1 (104 GB/34.35 GB) or 33 % in medium granularity. If 540.4 TB of storage is provisioned to VMs but only 105.94 TB is used on the volumes, the thin provisioning ratio is 5.10:1 (540 TB / 105.94 TB) or 20 % in fine granularity.
Storage efficiency	The ratio between the logical space needed (written by the operating system) compared with the actual physical space that is written to the storage after data reduction (if applicable).	The logical space that is written is 105.94 TB. After data reduction, it actually writes only 31.55 TB to the storage pool which is a 3.4 reduction ratio (105.94 TB / 31.55 TB).

Resources/Inventory

This section shows the number of resources in the current inventory:

- VM managers
- Nodes
- Switches
- Protection domains
- Storage pools
- Volumes
- Hosts

Resource groups

This section displays a graphical representation of the resource groups deployed based on status. The number next to each icon indicates the number of resource groups in a particular state. The resource groups are categorized into the following states:

Table 35. Resource group status

Tab	Status	Description
Health	**Healthy**	The green band on the graphic indicates that the resource group is successfully deployed.
	Degraded	The yellow band indicates that resources in a resource group require corrective action, but does not affect overall system health. For example, the firmware version

Table 35. Resource group status (continued)

Tab	Status	Description
		that is installed on a resource in the resource group is not compliant.
	Critical	The red band indicates resource groups that have critical health problems.
	Service Mode	The yellow band indicates that the resource group has been placed in service mode.
Compliance	**Compliant**	The green band indicates that the resource group is compliant with the target compliance version.
	Out of compliance	The yellow band indicates that the resource group is non-compliant with the target compliance version.

You can monitor node health by viewing the status of the resource group on the **Resource Groups** page.

If a resource group is in a yellow (or warning) state, it means that one or more nodes is in a warning or failed state. If a resource group is in a red (or error) state, it indicates that the resource group has fewer than two nodes that are not in a failed state.

To view the status of a failed node component, hover the cursor on the image of the failed component in the resource group.

Alerts

This section lists the current alerts within the system that is categorized by the severity level:

- Critical
- Major
- Minor

11

Managing block storage (erasure coding)

Block storage management involves the configuration of the storage system components. The components includes the protection domain, device groups, devices, storage pools and NVMe targets. In addition, block storage management involves configuring volumes and hosts.

Storage system

A storage system consists of logical entities, such as devices and device groups, storage pools, and NVMe targets, which form the protection domain.

The devices are organized into a device group. A device group consists of devices that have the same drive type, either a solid-state drive (SSD) or persistent memory (PMEM), for example software-defined persistent memory (SDPM). It is from within the device group that storage pools are allocated.

A storage pool contains volumes and snapshots that hold user data.

Protection domains

A protection domain is a logical entity that contains a group of storage nodes and the devices in the storage nodes. The protection domain manages the storage of those storage nodes. It may also contain storage data targets (SDTs) and storage data clients (SDCs). These logical groups allow you to physically or logically isolate specific datasets and performance capabilities within specified protection domains and limit the effect of any specific device failure. You can add, modify, activate, inactivate, or remove a protection domain in the PowerFlex system.

Add a protection domain

A protection domain consists of storage nodes, with (optionally) SDTs and SDCs, configured in separate logical groups. You can add protection domains to a PowerFlex system.

1. Go to **Block** > **Storage System** > **Protection Domain**.
2. Click **Add Protection Domain**.
3. In the **Create Protection Domain** dialog box, enter the name of the protection domain and click **Create**.
4. Verify that the operation is finished and is successful, and click **Dismiss**.

You can now add storage nodes, device groups, devices, and storage pools to the protection domain.

Activate a protection domain

You must activate the protection domain to enable access to the data.

Before activating the protection domain, ensure that the storage nodes and the storage data targets (SDTs) in the protection domain are connected and have no alerts. Otherwise, an unnecessary rebuild may occur. If necessary, contact Dell Technologies Support for guidance.

1. On the menu bar, click **Block** > **Storage System** > **Protection Domains**.
2. In the list of protection domains, select the relevant protection domain, and click **More Actions** > **Activate**.
3. In the **Activate Protection Domain** dialog box, click **Yes** for **Force activate** and then click **Activate** to enable access to the data on the protection domain.
4. Verify that the operation is finished and is successful, and click **Dismiss**.
5. Repeat as necessary for each additional protection domain that you want to activate.

Inactivate a protection domain

Use this procedure for a graceful system shutdown.

NOTE: When you inactivate a protection domain, the data remains on the storage node. If you no longer need it, it is preferable to remove a protection domain.

While a protection domain is inactivated, the following activities can take place behind the scenes:

- Determine if there are any current rebuild/rebalance activities taking place. If so, the shutdown will be delayed (unless it is forced) until they are finished.
- Block future rebuild/rebalance activities.
- Temporarily disable application I/O and disable access to volumes.
- Move the DRL mode of the storage nodes to harden, in preparation for restarting the server.
- Reload of the storage nodes before re-enabling data access.

1. On the menu bar, click **Block** > **Storage System** > **Protection Domains**.
2. In the list of protection domains, select the relevant protection domain, and click **More Actions** > **Inactivate**.
3. In the **Inactivate Protection Domain**, enter the user password, and click **Inactivate**.
4. Verify that the operation is finished and is successful, and click **Dismiss**.
5. Repeat as necessary for each additional protection domain that you want to inactivate.

Configure I/O priorities and bandwidth use

PowerFlex includes advanced settings that control I/O priorities and bandwidth use and can be used to fine-tune system performance. The fine-tuning includes the ability to control secondary I/Os which affects the garbage collector, rebuild, and rebalance.

NOTE: These features affect system performance, and should only be configured by an advanced user. It is recommended to retain the default settings, unless you are an advanced user.

1. On the menu bar, click **Block** > **Storage System** > **Protection Domains**.
2. From the list of protection domains, select the relevant checkbox, and click **More Actions** > **Set I/O priority**.
3. In the **Set IO Priority for Protection Domain** dialog box, select one of the following secondary I/O priority policies that are described in the table below:

Table 36. Policies for the secondary I/O priority

Policy	Description
Unlimited	Secondary I/Os are not limited.
Favor Application	Limits the number of allowed concurrent I/Os to the values entered in the **Concurrent I/O limit** and **Bandwidth I/O limit** fields, regardless of user I/O. If you select the Favour Application IO policy, you must set the overall limit and the advanced bandwidth limits for each protection domain on each device. The value of the limit can range from **1** to **10**. The value of the advanced bandwidth can range from **1** to **1024**. The default value is **10**.

4. Click **Apply** and click **Dismiss**.

Remove a protection domain

Remove a protection domain from the PowerFlex system.

Ensure that the storage nodes and device group are removed from the protection domain before removing it from the system.

1. On the menu bar, click **Block** > **Storage System** > **Protection Domains**.
2. From the list of protection domains, select the relevant protection domain and click **More Actions** > **Delete**.
3. In the **Delete Protection Domain**, click **Delete**.

4. Verify that the operation is finished and is successful, and click **Dismiss**.

Device group

A device group consists of devices that come from five or more nodes that are contained in a single protection domain. The nodes that belong to a device group have the same media type.

The capacity of a device group is derived from the capacity of the healthy storage devices that it contains. Storage pools are allocated through the device group and consume the capacity of the device group.

Add a device group

You should add one device group of type SSD per protection domain. There is an option to add additional persistent memory type device groups to a protection domain. The persistent memory device group is used for write cache.

NOTE: This feature is intended for advanced users only.

1. On the menu bar, click **Block** > **Storage System** > **Device Groups**.
2. Click **Add Device Group**.
3. In the **Create Device Group** dialog box, define the following settings:
   a. **Name**: Enter a name for the device group.
   b. **Protection Domain**: Select the protection domain associated with the device group.
   c. **Type**: Select the media type of the devices in the group: **SSD** or **PMEM**.
4. Click **Create**.
5. Verify that the operation is finished and is successful, and click **Dismiss**.

Rename a device group

You can change the name of a device group. You can only rename one device group at a time.

1. On the menu bar, click **Block** > **Storage System** > **Device Groups**.
2. From the list of device groups, select the check box of the device group that you want to rename, and click **More Actions** > **Rename**.
3. In the **Rename Device Group** dialog box, enter the new name, and click **Apply**.
4. Verify that the operation is finished and is successful, and click **Dismiss**.

Set spare capacity of a device group

You must set the spare capacity of a device group in terms of nodes for this release. The default spare capacity is one node, and the maximum is two nodes.

Utilizing spare devices is a feature that is recommended only for advanced configurations.

1. On the menu bar, click **Block** > **Storage System** > **Device Groups**.
2. From the list of device groups, select the check box of the device group that you want to set the capacity for, and click **More Actions** > **Spare Capacity**.
3. In the **Set Spare Capacity for Device Group** dialog box, select the spare nodes for the device group.
4. Click **Apply**.
5. Verify that the operation is finished and is successful, and click **Dismiss**.

Enable the device diagnostic check

You can enable the device awareness check to detect an issue with a device in advance. If there is an issue with a device, an alert is sent to the dashboard.

The device awareness check is enabled by default. If you disable this setting, it may cause potential device failures not to be detected early enough. To disable the diagnostic, turn **Enable Diagnostics Check** to off.

Device diagnostics is an advanced column and is not set to display by default in the PowerFlex Manager dashboard. You can add the device diagnostics setting to display through the table column arrangement icon at the upper right of the table.

1. On the menu bar, click **Block** > **Storage System**.
2. Click **Device Groups**.
3. Select the checkbox beside the device group to enable the device diagnostic check.
4. Click **More Actions** > **Advanced Settings**.
   The **Advanced Settings for Device Group** dialog box displays.
5. To enable the diagnostic, turn **Enable Diagnostics Check** to on and click **Apply**.
6. Verify that the operation is finished and is successful, and click **Dismiss**.

Remove a device group

You can remove a device group from a protection domain.

1. On the menu bar, click **Block** > **Storage System** > **Device Groups**.
2. From the list of device groups, select the checkbox of the device group that you want to remove, and select **More Actions** > **Delete**.
3. In the **Delete Device Group** dialog box, verify that you selected the wanted device group, and click **Remove**.
4. Verify that the operation is finished and is successful, and click **Dismiss**.

Devices

A device is a single storage medium, for example, a SSD, which is added to a device group in the storage system.

Activate a device

You must activate a device after it is added to the device group in order to start using it.

1. Go to **Block** > **Storage System** > **Devices**.
2. From the list of devices, select the checkboxes of the required devices, and click **More Actions** > **Activate**.
3. In the **Activate Device** dialog box, click **Activate**.
4. Verify that the operation is finished and is successful, and click **Dismiss**.

Update the device path

You can manually update the device path.

1. Go to **Block** > **Storage System** > **Devices**.
2. From the list of devices, select the check boxes of the required devices, and click **More Actions** > **Update Device Path**.
3. In the **Update Device Path** dialog box, enter the new path and click **Update Path**.
4. Verify that the operation is finished and is successful, and click **Dismiss**.

Clear a device error

If the system does not automatically clear the system, you can rectify the error by clearing it.

1. On the menu bar, click **Block** > **Storage System** > **Devices**.
2. From the list of devices, select the check boxes of the relevant devices, and click **More Actions** > **Clear Errors**.
3. In the **Clear device errors** dialog box, click **Clear Errors**.
4. Verify that the operation is finished and is successful, and click **Dismiss**.

Rename a device

You can change the name of a device.

You can view the current device name by displaying the name column in the device list. The name column is hidden, by default. When no device name has been defined, the name is set by default to the path name of the device.

The device name must adhere to the following rules:

- Contains less than 32 characters
- Contains only alphanumeric and punctuation characters
- Is unique within the object type

1. On the menu bar, click **Block** > **Storage System** > **Devices**.
2. From the list of devices, select the required device, and click **Modify** > **Rename**.
3. In the **Rename Device** dialog box, enter the new name, and click **Apply**.
4. Verify that the operation is finished and is successful, and click **Dismiss**.

Set device capacity limits

Prior to replacing a device with a device of a smaller capacity, set the capacity limit of the device being removed to the capacity of the new device. Capacity will be decreased, but the size of the disk remains unchanged.

NOTE: The capacity assigned to the storage device must be smaller than its actual physical size.

1. On the menu bar, click **Block** > **Storage System** > **Devices**.
2. From the list of devices, select the relevant device, and click **Modify** > **Set Capacity Limit**.
3. In the **Set Capacity Limit** dialog box, enter the capacity and select a unit type (TB).
   This dialog box displays the maximum capacity available for the device.
4. Click **Apply**.
5. Verify that the operation is finished and is successful, and click **Dismiss**.

Modify device LED settings

You can set a device's LED to blink or turn it off to help you physically identify a device in the system chassis.

1. On the menu bar, click **Block** > **Storage System** > **Devices**.
2. From the list of devices, select the relevant device, and click **Modify** > **LED Settings**.
3. In the **Set device LED settings** dialog box, do one of the following:
   - To turn on the device's LED, select the **On** check box.
   - To turn off the device's LED, clear the **On** check box.
4. Click **Apply**.
5. Verify that the operation is finished and is successful, and click **Dismiss**.

Storage pools

A storage pool is a container of volumes that uses one device group and, if available, one PMEM device group. A storage pool consumes the capacity from the device group. A storage pool is a container of volumes that belongs to a single device group and consumes the capacity from the device group.

Configure storage pool general settings

You can set the capacity alert thresholds that specify the amount of capacity which must be reserved as free space. The thresholds apply to both the capacity and the over-provisioning limits selected.

1. On the menu bar, click **Block** > **Storage System** > **Storage Pools**.
2. Select the storage pool, and click **Modify** > **General Settings**.

3. In the **Storage Pool Settings** dialog box, configure the following settings for the storage pool:

Option	Description
Enable compression	For storage pools, inline compression allows you to gain more effective capacity.
Enable Over-Provisioning Control	For storage pools, enabling over-provisioning enables you to limit the size of the thin clones that are created by percentages. For example, if the storage pool is 10 TB and the over-provisioning factor is 200%, you can create thin volumes up to 20 TB. The default is unlimited.

4. Click **Apply**.
5. Verify that the operation is finished and is successful, and click **Dismiss**.

Configure storage pool capacity

You can set a storage pool capacity threshold which ensures that you are notified when consumption exceeds the threshold. If over-provisioning is enabled, the same threshold applies.

1. On the menu bar, click **Block** > **Storage System** > **Storage Pools**.
2. Select the storage pool, and click **Modify** > **Capacity**.
3. Set the capacity alert thresholds.
   - The default high alert threshold is 80%. If the used capacity exceeds 80% or if the over-provisioning limit exceeds 80%, a high alert is raised.
   - The default critical alert threshold is 90%. If the used capacity exceeds 90% or if the over-provisioning limit exceeds 90%, a critical alert is raised.
4. Click **Apply**.
5. Verify that the operation is finished and is successful, and click **Dismiss**.

Expand storage pool capacity

You can choose to use all the available capacity, or you can choose to customize the capacity size of the storage pool. If you choose to use all the capacity, the system calculates the new size of the capacity.

1. On the menu bar, click **Block** > **Storage System** > **Storage Pools**.
2. Select the storage pool, and click **Modify** > **Expand**.
3. You can choose either to use the available capacity or customize the capacity.
4. Click **Apply**.
5. Verify that the operation is finished and is successful, and click **Dismiss**.

NVMe targets

NVMe targets (or SDT components) must be configured on the PowerFlex system side, in order to use NVMe over TCP technology.

The NVMe target (or SDT component) is a frontend component that translates NVMe over TCP protocol into internal PowerFlex protocols. The NVMe target provides I/O and discovery services to NVMe hosts configured on the PowerFlex system. A minimum of two NVMe targets must be assigned to a protection domain before it can serve NVMe hosts, to provide minimal path resiliency to hosts.

TCP ports, IP addresses, and IP address roles must be configured for each NVMe target (or SDT component). You can assign both storage and host roles to the same target IP addresses. Alternatively, assign the storage role to one target IP address, and add another target IP address for the host role. Both roles must be configured on each NVMe target.

- The host port listens for incoming connections from hosts, over the NVMe protocol.
- The storage port listens for connections from the MDM.

Once the NVMe targets have been configured, add hosts to PowerFlex, and then map volumes to the hosts. Connect hosts to NVMe targets, preferably using the discovery feature.

On the operating system of the compute nodes, NVMe initiators must be configured. Network connectivity is required between the NVMe targets and the NVMe initiators, and between NVMe targets (or SDT components) and storage nodes.

Add an NVMe target

Add an NVMe target (or SDT component).

1. On the menu bar, click **Block** > **Storage System** > **NVMe Targets**.
2. Click **Add NVMe Target**.
3. From the **Add NVMe Target** dialog box, configure the following settings:
   a. For **Target Name**, enter a name for the NVMe target (or SDT component).
   b. For **Protection Domain**, select a protection domain.
   c. Accept the default TCP ports, or modify them if necessary.
      - **Storage Port** listens for incoming connections from the MDM, and is only open on an IP address configured with the storage role. Default=12200
      - **I/O Port** listens for incoming connections for I/O, and is only open on an IP address configured with the host role. Default=4420 (which conforms to the NVMe over TCP standard)
      - **Discovery Port** listens for incoming discovery connections, and is only open on an IP address configured with the host role. Default=8009 (which conforms to the NVMe over TCP standard. Use zero to disable the discovery service.)
   d. For **Target IPs**, enter an IP address, and then select a role for the IP address from the **Target IPs Roles** menu. Click **Add Target IP**, and repeat this step until all IP addresses are configured.
4. Click **Add**.
5. Verify that the operation is finished and is successful, and click **Dismiss**.

Modify an NVMe target

Modify the configuration of an NVMe target (or SDT component).

1. On the menu bar, click **Block** > **Storage System** > **NVMe Targets**.
2. From the list of NVMe targets, select the required NVMe target, and click **Modify**.
3. In the **Modify NVMe Target** dialog box, modify the desired fields.
4. Optionally, add more target IP addresses, by clicking **Add Target IP**.
5. Click **Modify NVMe Target**.
6. Verify that the operation is finished and is successful, and click **Dismiss**.

Remove an NVMe target

Remove an NVMe target (or SDT component).

1. On the menu bar, click **Block** > **Storage System** > **NVMe Targets**.
2. From the list of NVMe targets, select the required NVMe target, and click **Remove**.
3. In the **Remove NVMe Target** dialog box, verify that you are removing the wanted NVMe target, and click **Remove**.
4. Verify that the operation is finished and is successful, and click **Dismiss**.

Enter an NVMe target into maintenance mode

You can enter multiple NVMe targets to maintenance mode at one time. You cannot enter an NVMe target that is already in maintenance mode. You must enter an NVMe target into maintenance mode before a storage node and exit the NVME target after a storage node.

1. On the menu bar, click **Block** > **Storage System** > **NVMe Targets**.
2. From the list of NVMe targets, select the required NVMe target, and click **More Actions**.
3. In the **Enter NVMe Target to Maintenance Mode** dialog box, verify that you are selecting the wanted NVMe target, and click **Enter Maintenance Mode**.
4. Verify that the operation is finished and is successful, and click **Dismiss**.

Exit an NVMe target from maintenance mode

You can exit an NVMe target from maintenance mode

1. On the menu bar, click **Block** > **Storage System** > **NVMe Targets**.
2. From the list of NVMe targets that are in maintenance mode, select the required NVMe target, and click **More Actions**.
3. In the **Exit NVMe Target from Maintenance Mode** dialog box, verify that you are selecting the wanted NVMe target, and click **Exit Maintenance Mode**.
4. Verify that the operation is finished and is successful, and click **Dismiss**.

Volumes

Define, configure and manage volumes in the PowerFlex system.

Create a volume

A volume is distributed over the devices residing in the same storage pool. You can add one or multiple volumes at one time.

The system must have the following:

- At least five storage nodes
- Sufficient initial capacity is available to create thin volumes.

Dell Technologies highly recommends giving each volume that a meaningful name associated with its operational role.

1. On the menu bar, click **Block** > **Volumes**.
2. Click **Create Volume**.
3. In the **Create Volume** dialog box, configure the following items:
   a. Select a protection domain. You can search for the protection domain by name. If there is only one protection domain, it is pre-populated.
   b. Enter the number of volumes.
   c. Enter a name for the volume.
   d. Enter the volume size in TB or GB (basic allocation granularity is 1 GB).
4. Click **Create**.
5. Verify that the operation is finished and is successful, and click **Dismiss**.

To use the created volume, you must map it to at least one host. If the restricted SDC mode is enabled for the system, you must approve SDCs prior to mapping volumes to them.

Map a volume to a host

Mapping exposes the volume to the specified host, creating a block device on the host. You can map a volume to one or more hosts.

Volumes can only be mapped to one type of host either an SDC or an NVMe. Ensure that you know which type of hosts are being used for each volume, to avoid mixing host types.

1. On the menu bar, click **Block** > **Volumes**.
2. From the list of volumes, select one or more volumes, and click **Provision** > **Map**.
3. In the **Map Volume to Hosts** dialog box, select one or more hosts to which you want to map to the volumes.
   - If a volume is already mapped to a host, only hosts of the same type, SDC or NVMe, are listed.
   - If the volume is not mapped to a host, click **SDC** or **NVMe** to set the type of hosts to be listed.
4. Click **Map**.
5. Verify that the operation is finished and is successful, and click **Dismiss**.

Unmap a volume

You can unmap one or more volumes from a host.

1. On the menu bar, click **Block** > **Volumes**.
2. From the list of volumes, select the relevant volumes, and then click **Provision** > **Unmap**.
3. Select the host from which to remove mapping to the volumes.
4. Click **Unmap**.
5. Verify that the operation is finished and is successful, and click **Dismiss**.

Create a volume snapshot

You can create instantaneous snapshots of one or more volumes.

1. On the menu bar, click **Block** > **Volumes**.
2. From the list of volumes, select the relevant volumes, and then click **Protect** > **Create Snapshot**.
3. In the **Create snapshot of volume** dialog box, enter the following details:
   a. The name of the snapshot. You can accept the default name, or give the snapshot a name according to the following rules:
      - Contains less than 32 characters
      - Contains only alphanumeric and punctuation characters
      - Is unique within the object type
   b. Enter an index value.
4. (Optional) If you want to prohibit the deletion of snapshots until a specific time, select **Use secure snapshots** option. The use secure snapshots option prohibits deletion of the snapshots until the defined expiration period has elapsed.
   a. To define the expiration period, enter the number in **Expiration Time** and select the units from the dropdown menu.
5. Click **Create**.
6. Verify that the operation is finished and is successful, and click **Dismiss**.

Restore volume data

The volume on the host must be taken offline to prevent data integrity issues.

1. On the menu bar, click **Block** > **Volumes**.
2. From the list of volumes, select the relevant volume, and then click **Protect** > **Restore Data**.
3. In the **Restore Data of Volume** window, select the snapshot to restore from.
4. Click **Restore Volume**.
5. Verify that the operation is finished and is successful, and click **Dismiss**.

Assign a snapshot policy

You can assign a snapshot policy which is a set of rules that controls the creation, retention, and expiration of snapshots.

1. On the menu bar, click **Block** > **Volumes**.
2. Select the volumes or snapshots that you want to assign to the policy, and click **Protect** > **Assign Snapshot Policy**.
3. In the **Assign Snapshot Policy** dialog box, from the list of snapshot policies, select the relevant policy to assign and click **Assign Policy**.
4. Verify that the operation is finished and is successful, and click **Dismiss**.

Unassign a volume from a snapshot policy

You can unassign a volume from a snapshot policy to detach the source volume from the policy and delete associated auto snapshots. This process helps manage snapshots and ensures that only relevant data is retained.

If the auto snapshots are locked, unassigning them from a policy only detaches them. The auto snapshots must be manually deleted.

1. On the menu bar, click **Block** > **Volumes**.
2. Select the volume that you want to unassign from the policy, and click **Protect** > **Unassign Snapshot Policy**.
3. (Optional) If you want the snapshot to be removed, in the **Unassign Volumes from Snapshot Policy** dialog box, select the **Delete related snapshots** checkbox and click **Unassign Policy**.
4. In the **Unassign Volumes From all Snapshot Policies** dialog box, click **Yes**.
5. Verify that the operation is finished and is successful, and click **Dismiss**.

Create a thin clone of a volume

Thin clones are writable copies of a volume or a snapshot.

A thin clone is a space efficient copy that shares data blocks with its parent object and not a full backup of the original resource.

Thin clones retain full read access to the original resource. You can modify the data within the thin clone while preserving the original data on the parent resource.

Using thin clones, you can establish hierarchical points in time to preserve data over different stages of data modifications. If the parent resource (volume or snapshot) is deleted, the thin clone is unaffected.

1. On the menu bar, click **Block** > **Volumes**.
2. Select the volume that you want to create a thin clone from, and click **Repurpose** > **Create Thin Clone**.
3. In the **Create thin clone of volume** dialog box, enter a thin clone name, and click **Create**.
4. Verify that the operation is finished and is successful, and click **Dismiss**.

Refresh volume data

Ensure that you have the most up-to-date data by refreshing.

1. On the menu bar, click **Block** > **Volumes**.
2. From the list of volumes, select the relevant volume, and then click **Repurpose** > **Refresh Data**.
3. In the **Refresh Data of Volume** window, select the volume, thin clone, or snapshot to refresh from.
4. Click **Refresh Volume**.
5. Verify that the operation is finished and is successful, and click **Dismiss**.

Increase a volume size

You can increase, but not decrease, the logical size of one volume at any time, as long as there is enough capacity for the volume size to grow.

1. On the menu bar, click **Block** > **Volumes**.
2. From the list of volumes, select the volume, and click **Modify** > **Resize**.
3. In the **Resize Volume** dialog box, enter the new volume size, and select a unit type. You can set the volume to any size (1 GB granularity).
4. Click **Apply**.
5. Verify that the operation is finished and is successful, and click **Dismiss**.

Rename a volume

You can use the default volume name or create a name.

1. On the menu bar, click **Block** > **Volumes**.
2. From the list of volumes, select the relevant volume, and click **More Actions** > **Rename**.
3. In the **Rename Volume** dialog box, enter a new name and click **Apply**.
4. Verify that the operation is finished and is successful, and click **Dismiss**.

Delete a volume

You can remove a volume from PowerFlex provided the volume is not mapped to a host and it is not assigned to a snapshot policy.

Before deleting a volume, ensure the following:

- If the volume is mapped to a host, unmap it before deleting the volume.
- If the volume is assigned to a snapshot policy, unassign it from the snapshot policy before deleting the volume.

To prevent causing a data unavailability scenario, avoid deleting volumes or snapshots while the MDM cluster is being upgraded.

CAUTION: All data is erased from a deleted volume. All direct decedent snapshots of the volume are deleted with it. Thin clones and snapshots of thin clones are unaffected.

1. On the menu bar, click **Block** > **Volumes**.
2. From the list of volumes, select the required volume, and click **More Actions** > **Delete**.
3. In the **Delete Volume** dialog box, verify the volumes to be removed, and click **Delete**.
4. Verify that the operation is finished and is successful, and click **Dismiss**.

Set a volume bandwidth and IOPS limits

Setting bandwidth and IOPS limits for volumes lets you control the quality of service (QoS). Bandwidth and IOPS limits are set on a per-host basis.

Before you set these limits, ensure that the volumes are mapped .

1. On the menu bar, click **Block** > **Volumes**.
2. From the list of volumes, select the relevant volume, and then click **More Actions** > **Set Limits**.
3. In the **Set IO limits for volume** dialog box, enter the required values for **Bandwidth Limits** and **IOPS Limits**, or select the corresponding **Unlimited** check box.
   - The number of IOPS must be larger than 10.
   - The volume network bandwidth is in MB/s.
   - The I/O limits are applied to every mapped SDC.
4. Click **Apply**.
5. Verify that the operation is finished and is successful, and click **Dismiss**.

Rename a snapshot

You can rename a snapshot for clarity, version control, organization and categorization, and for audit purposes.

1. On the menu bar, click **Block** > **Volumes**.
2. From the list of volumes, select the relevant volume, and then click **View Details**.
3. Click **Snapshots**.
4. Select the checkbox beside the snapshot that you want to rename.
5. Go to **Modify** > **Rename**.
6. In the **Rename Snapshot** dialog box, enter a new name and click **Apply**.
7. Verify that the operation is finished and is successful, and click **Dismiss**.

Delete a snapshot

You can delete a snapshot if it is no longer relevant, or if you want to free up storage space.

1. On the menu bar, click **Block** > **Volumes**.
2. From the list of volumes, select the relevant volume, and then click **View Details**.
3. Click **Snapshots**.
4. Select the checkbox beside the snapshot that you want to delete.
5. Go to **Modify** > **Delete**.
6. Verify that the operation is finished and is successful, and click **Dismiss**.

View volume details

You can view details about volumes including snapshots, host , and vTree details.

1. On the menu bar, click **Block** > **Volumes**.
2. Select the volume from the list that you want to view details about and click **View Details** in the lower right pane.

   The following details display:

   - Objects related to the volume that you are viewing.
   - General information:
     - Type: Primary or thin clone
     - Size
     - Mapped: Whether the volume is mapped or not.
     - Created time: The time the volume was created.
     - Read-only: The volume is access mode.
     - VTree logical written: The total sum of logical capacity used by the volumes, snapshots, and thin-clones that belong to the same family.
   - Performance:
     - Latency: The time that it takes for a single I/O operation to complete and is typically measured in milliseconds (ms).
     - IOPs limit: The maximum number of read and write operations that the system can handle per second
     - Bandwidth: The maximum of data that is transferred per second in GB/s.

   The following specific details are available for a host:

   - Name: The name assigned during configuration.
   - IP address: The IP address assigned during configuration.
   - Storage access protocol: Either SDC or NVMe/TCP.
   - Host access: The default access mode for newly mapped volumes.
   - Version: The SDC software version, applicable for SCD only.
   - Operating system(OS): The operating system of the host.
   - BW limit: The maximum of data that is transferred per second in GB/s.
   - IOPs limit: The maximum number of read and write operations that the system can handle per second.
   - Latency: The time that it takes for a single I/O operation to complete and is typically measured in milliseconds (ms).

   The following specific details are available for a snapshot:

   - Name: The name assigned during configuration.
   - Size: The size of the object is measured in GB.
   - Creation time: The time the object was created,
   - Parent ID: A unique identifier of the parent snapshot that is used to link snapshots together.
   - Secured: The snapshot is lprotected against deletion or modification for a defined retention period.
   - Expiration time: The point in time when the snapshot security expires.
   - VTree ID: The ID of the vTree the object belongs to.

   The following specific details are available for a vTree:

   - Name: The name assigned during configuration.
   - Storage Pool: The storage pool to which the object belongs.
   - Provisioned: The total provisioned size of each object.
   - Logical: The logical capacity in use by each object.
   - Type: The type of object.

Hosts

Hosts are entities that consume PowerFlex storage for application usage. There are two methods of consuming PowerFlex block storage: using the SDC kernel driver, or using NVMe over TCP connectivity. Therefore, a host is either an SDC or an NVMe host.

Once a host is configured, volumes may be mapped to it. In each case, hosts must be mapped to volumes.

Add an NVMe host

Add an NVMe host to PowerFlex.

Ensure the following:

- You know the NVMe Qualified Name (NQN) of the host. If you do not know the NQN, see the documentation for the host operating system.
- The host is connected to the Ethernet switch.
- The host is configured with the correct VLAN ID and routing rules.

1. On the menu bar, click **Block** > **Hosts**.
2. Click **+ Add Host**.
3. In the **Add NVMe Host** dialog box, enter a hostname in the **Host Name** field.
4. In the **Host NQN** field, enter the NQN string for the host.
5. In the **Number of Paths Per Volume** field, enter the maximum number of paths to be provided between the host and each volume (default is 4).
6. In the **Number of System Ports per Protection Domain** field, enter the maximum number of ports to be provided between the host and each protection domain (default is 10).
7. Click **Add**.
8. Verify that the operation is finished and is successful, and click **Dismiss**.
9. Configure the host operating system to connect to the cluster's NVMe discovery service on one or more of the NVMe targets.

Map volumes to hosts

Map hosts to volumes to ensure that the host can access the volumes.

Volumes can only be mapped to one type of host: either SDC or NVMe. Ensure that you know which type of hosts are being used for each volume, to avoid mixing host types.

1. On the menu bar, click **Block** > **Hosts**.
2. From the list of hosts, select the relevant host, and click **Provision** > **Map**.
3. In the **Map Hosts to Volumes** dialog box, select the volumes to be mapped to the selected hosts, and click **Map**.
4. Verify that the operation is finished and is successful, and click **Dismiss**.

Unmap volumes from hosts

Remove mapping between volumes and hosts.

1. On the menu bar, click **Block** > **Hosts**.
2. From the list of hosts, select the relevant host, and click **Provision** > **Unmap**.
3. In the **Unmap** dialog box, ensure that the correct host is selected.
4. Click **Unmap**.
5. Verify that the operation is finished and is successful, and click **Dismiss**.

Remove hosts

You can only remove hosts with unmapped volumes. For SDCs, the host must be disconnected from the PowerFlex cluster before it can be removed.

1. On the menu bar, click **Block** > **Hosts**.
2. From the list of hosts, select the relevant host, and click **Remove**.
3. In the **Remove Host** dialog box, ensure that you have selected the correct host for removal.
4. Click **Remove**.
5. Verify that the operation is finished and is successful, and click **Dismiss**.

Configure or modify approved host IP addresses

When the restricted host (SDC) mode is set to `approved IP restriction`, configure the host IP addresses before mapping volumes to the hosts.

Ensure that the hosts are approved by GUID.

You can add up to four IP addresses.

1. On the menu bar, click **Block** > **Hosts**.
2. From the list of hosts, select the relevant host, and click **Modify** > **Modify Approved IP Addresses**.
3. In the **Modify Approved IP addresses** dialog box, enter the IP address of the hosts, and click **Add IP Address**. Repeat this step for additional addresses.
4. Click **Apply**.
5. Verify that the operation is finished and is successful, and click **Dismiss**.

Approve SDCs

When the restricted host (SDC) mode is set to GUID restriction, approve the SDCs before mapping the SDCs to volumes.

For more information about the restricted host feature, see the *Dell PowerFlex Technical Overview* or *Dell PowerFlex Family 5.x Security Configuration Guide*.

1. On the menu bar, click **Block** > **Hosts**.
2. From the list of hosts, select one or more hosts and click **Modify** > **Approve**.
3. In the **Approve host** dialog box, verify that the hosts listed are the ones that you want to approve, and click **Approve**.
4. Verify that the operation is finished and is successful, and click **Dismiss**.

Rename hosts

You can rename a host following the specific naming rules.

The hostname must adhere to the following rules:

- Contains less than 32 characters
- Contains only alphanumeric and punctuation characters
- Is unique within the object type

1. On the menu bar, click **Block** > **Hosts**.
2. From the list of hosts, select the relevant host, and click **Modify** > **Rename**.
3. In the **Rename Host** dialog box, enter the new name, and click **Apply**.
4. Verify that the operation is finished and is successful, and click **Dismiss**.

Modify an SDC performance profile

SDC performance profiles are set by default to high and can be changed to compact. The compact setting may impact the system performance.

The default setting configures a predefined set of parameters for high-performance use cases.

NOTE: Performance tuning is case-specific. To prevent undesirable effects, Dell Technologies highly recommends that you thoroughly test all changes. For further assistance, contact Dell Technologies Support.

1. On the menu bar, click **Block** > **Hosts**.
2. From the list of hosts, select the relevant host, and click **Modify** > **Modify Performance Profile**.
3. In the **Modify Performance Profile Host** dialog box, select the wanted performance profile: **High** or **Compact**.
4. Click **Apply**.
5. Verify that the operation is finished and is successful, and click **Dismiss**.

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Managing block storage (mirroring)

Block storage management involves the configuration of protection domains, fault sets, storage data servers, and storage pools. In addition, block storage management involves configuring acceleration pools, devices, and volumes, as well as NVMe targets and hosts.

Protection domains

A protection domain is a set of storage data servers (SDSs) configured in separate logical groups. It may also contain NVMe target components known as storage data targets (SDTs) and replication components known as storage data replicators (SDRs). These logical groups allow you to physically and/or logically isolate specific data sets and performance capabilities within specified protection domains and limit the effect of any specific device failure. You can add, modify, activate, inactivate, or remove a protection domain in the PowerFlex system.

Add a protection domain (mirroring)

A protection domain is a set of SDSs, with (optionally) SDTs and SDRs, configured in separate logical groups. You can add protection domains to a PowerFlex system.

1. On the menu bar, click **Block** > **Storage System** > **Protection Domains**.
2. Click **Add Protection Domain**.
3. In the **Create Protection Domain** dialog box, enter the name of the protection domain and click **Create**.
4. Verify that the operation has finished successfully, and click **Dismiss**.

You can now add SDSs, fault sets, storage pools, and acceleration pools to the protection domain. Replication can also be set up to ensure the data is protected and saved to a remote cluster.

Configure network throttling

Configure network throttling to control the flow of traffic over the network.

Network throttling is configured separately for each protection domain. The SDS nodes transfer data between themselves. This data consists of user data being replicated as part of the RAID protection, and data copied for internal rebalancing and recovery from failures. You can modify the balance between these types of data loads by limiting the data copy bandwidth. This change affects all SDSs in the specified protection domain.

NOTE: These features affect system performance, and should only be configured by an advanced user. Contact Dell Technologies Support before you change this configuration.

1. On the menu bar, click **Block** > **Storage System** > **Protection Domains**.
2. In the list of protection domains, select the relevant protection domain checkbox, and click **Modify** > **Network Throttling**.
3. In the **Set Network Throttling for PD** dialog box, enter the bandwidth for the following settings, or select **Unlimited** to allow for unlimited throughput for that setting:
   - **Rebalance throughput limit per SDS**
   - **Rebuild throughput limit per SDS**
   - **vTree migration throughput limit per SDS**
   - **Overall throughput limit per SDS**
4. Click **Apply**.
5. Verify that the operation has finished successfully, and click **Dismiss**.

Inactivate a protection domain (mirroring)

Use this procedure for a graceful system shutdown.

NOTE: When you inactivate a protection domain, the data remains on the storage data servers (SDS) SDSs. It is therefore preferable to remove a protection domain if you no longer need it.

While a protection domain is inactivated, the following activities can take place behind the scenes:

- Determine if there are any current rebuild/rebalance activities taking place. If so, the shutdown will be delayed (unless it is forced) until they are finished.
- Block future rebuild/rebalance activities.
- Temporarily disable application I/O and disable access to volumes.
- Move the DRL mode of all SDSs to harden, in preparation for restarting the server.
- Reload of all SDSs before re-enabling data access.

1. On the menu bar, click **Block** > **Storage System** > **Protection Domains**.
2. In the list of protection domains, select the relevant protection domain, and click **More Actions** > **Inactivate**.
3. In the **Inactivate Protection Domain**, enter the user password, and click **Inactivate**.
4. Verify that the operation has finished successfully, and click **Dismiss**.
5. Repeat as necessary for each additional protection domain that you want to inactivate.

Activate a protection domain (mirroring)

You must activate the protection domain to enable access to the data.

Ensure that all storage data servers (SDS), storage data targets (SDTs), and storage data replicators (SDRs) in the protection domain are connected and have no alerts before activating the protection domain. Otherwise, an unnecessary rebuild may occur. If necessary, contact Dell Technologies Support for guidance.

1. On the menu bar, click **Block** > **Storage System** > **Protection Domains**.
2. In the list of protection domains, select the relevant protection domain, and click **More Actions** > **Activate**.
3. In the **Activate Protection Domain** dialog box, click **Yes** for **Force activate** and then click **Activate** to enable access to the data on the protection domain.
4. Verify that the operation is finished and is successful, and click **Dismiss**.
5. Repeat as necessary for each additional protection domain that you want to activate.

Remove a protection domain (mirroring)

Remove a protection domain from the PowerFlex system.

Ensure that all SDSs, storage pools, acceleration pools, and fault sets have been removed from the protection domain before removing it from the system.

1. On the menu bar, click **Block** > **Storage System** > **Protection Domains**.
2. From the list of protection domains, select the relevant protection domain and click **More Actions** > **Delete**.
3. In the **Delete Protection Domain**, click **Delete**.
4. Verify that the operation is finished and is successful, and click **Dismiss**.

Fault sets

Fault sets are logical entities that contain a group of SDSs within a protection domain. A fault set can be defined for a set of servers that are likely to fail together, for example, an entire rack full of servers. PowerFlex maintains mirrors of all chunks within a fault set on SDSs that are outside of this fault set so that data availability is assured even if all the servers within one fault set fail simultaneously.

NOTE: Fault sets are not supported on the PowerFlex management cluster (KVM).

Add fault sets

Add a fault set to a protection domain to prevent data loss in case of a single failure.

You must adhere to the following process for creating fault sets:

1. Ensure that a protection domain exists, or add a new one.
2. Ensure that a storage pool and fault sets (with a minimum of three fault units) exist, or add new ones.
3. Add the SDS and designate the protection domain and fault set. At the same time, add the SDS devices into a storage pool.

   If you use the automated deployment and installation tools, they follow this order automatically.

   You can only create and configure fault sets before adding SDSs to the system. Configuring fault sets incorrectly may prevent the creation of volumes. An SDS can only be added to a fault set during the creation of the SDS.

1. On the menu bar, click **Block** > **Storage System** > **Fault Sets**.
2. Click **Add Fault Set**.
3. In the **Create Fault Set** dialog box, enter a name and select the protection domain, and click **Create**.
4. Verify that the operation has finished and was successful, and then click **Dismiss**.
   The new fault set is a part of the protection domain.

Place a fault set in maintenance mode

Place a fault set into maintenance mode in order to perform non-disruptive maintenance on the group of storage data servers (SDSs).

1. On the menu bar, click **Block** > **Storage System** > **Fault Sets**.
2. In the list of fault sets, select the relevant fault set checkbox and click **More Actions** > **Enter Maintenance Mode**.
3. In the **Enter Fault Set to Maintenance Mode** dialog box, select one of the following maintenance mode options:
   - **Instant**—a node is temporarily removed without building a new copy of the data. During maintenance, the system only mirrors new writes. After maintenance is complete, the system applies the new writes to the node that was under maintenance.
   - **Protected**—a third copy is created before entering maintenance mode. Ensuring that if there is a node failure where the second copy of SDS is required, there is still a full backup of the SDS. Leaving no room for discrepancy between the copies. More storage capacity from the node is required.
4. Click **Enter Maintenance Mode**.
5. Verify that the operation has finished and was successful, and click **Dismiss**.
6. Optionally, at the upper right side of the toolbar, click the **Running Storage Jobs** icon to check maintenance mode status.

Exit fault set from maintenance mode

Once you have finished performing maintenance tasks on the fault set, you can manually exit maintenance mode and return the fault set to normal mode.

1. On the menu bar, click **Block** > **Storage System** > **Fault Sets**.
2. In the list of fault sets, select the relevant fault set checkbox and click **More Actions** > **Exit Maintenance Mode**.
3. In the **Exit Fault Set from Maintenance Mode** dialog box, click **Exit Maintenance Mode**.
4. Verify that the operation has finished and was successful, and then click **Dismiss**.

Cancel protected maintenance mode for fault set

A third copy of the SDS is created when entering protected maintenance mode, which can take a long time. You can back out of this process before entering protected maintenance mode is complete, such as when there is insufficient capacity to create the third copy.

1. On the menu bar, click **Block** > **Storage System** > **Fault Sets**.
2. In the list of fault sets, select the relevant fault set checkbox and click **More Actions** > **Abort Enter Protected Maintenance Mode**.
3. In the **Abort Enter Maintenance Mode** dialog box, click **Abort**.
4. Verify that the operation has finished and was successful, and then click **Dismiss**.

Delete fault sets

Use the following procedure to delete and verify that the desired fault set is deleted.

Ensure that any configured SDSs have been removed from the fault set that is to be deleted.

1. On the menu bar, click **Block** > **Storage System** > **Fault Sets**.
2. From the list of fault sets, select the fault set that you want to delete.
3. Select **More Actions** > **Delete**.
4. In the **Delete Fault Set** dialog box, verify that the desired fault set will be deleted, and click **Delete** > **Dismiss**.

Storage data servers

The storage data server (SDS) manages the capacity of a single server and acts as a back-end for data access. The SDS is installed on all servers contributing storage devices to PowerFlex. These devices are accessed through the SDS.

SDSs and their devices can be added to a system one by one or in bulk operations. Up to eight IP addresses can be associated with each SDS. You can associate different types of cache with SDSs. SDSs can be entered into maintenance mode to perform maintenance operations on the PowerFlex system.

Add storage data servers

Add storage data servers to the PowerFlex system.

- Ensure that at least one suitable storage pool is defined in the required protection domain.
- All devices in a storage pool must be the same media type. Ensure that you know the type of devices you are adding to the system.
- Ensure that the storage pool to which you are adding devices is configured to receive that media type.
- If you want to add acceleration devices now, ensure that at least one acceleration pool is defined.

Device data is erased when devices are added to an SDS. When adding a device to an SDS, PowerFlex checks that the device is clear before adding it. If the device is not clear, an error is returned. A device that has been used in the past can be added to the SDS by using the `Force Device Takeover` option. When this option is used, any data that was previously saved on the device is erased.

You can assign a name to the SDS, as well as to the devices. This name can assist in future object identification. This can be particularly helpful for SDS devices, because the names remain constant, even if the path changes. SDS and device names must meet the following requirements:

- Contains less than 32 characters
- Contains only alphanumeric and punctuation characters
- Is unique within the object type

NOTE: Devices can be tested before going online. Various testing options are available the **Advanced** area of the window (default: **Test and Activate**).

NOTE: Acceleration settings can be configured later from the **Block** > **Storage System** > **Acceleration Pools** page.

NOTE: You cannot enable zero padding after adding the devices.

1. On the menu bar, click **Block** > **SDSs**.
2. Click **Add SDS**.
3. Configure the following settings:
   - Enter SDS name
   - Select a protection domain
   - Select a fault set, if applicable
   - Enter SDS port used for communication
   - Enter the IP address for SDC, SDS or both and click **Add IP**.
4. For additional IP addresses, enter the IP address, select the communication role and click **Add IP**.
5. Expand **Advanced** for more options. Configure the following options (for advanced users):

- To enable RMcache, select **Use Read RAM Cache** and enter the size in MB.
- If PowerFlex is running on older systems, click one of the options for **Performance Profile**: Compact or High. Otherwise, keep the default as High.
- To force clean a node, select **Force Clean SDS**.

6. Click **Add**.

An SDS is added to the system.

Add storage devices

Add storage devices to the PowerFlex system one by one, or in bulk operations. By default, performance tests are performed on the added devices and the results are saved in the system.

Ensure that at least one suitable storage pool is defined in the required protection domain.

All devices in a storage pool must be the same media type. Ensure that you know the type of devices that you are adding to the system, and that the storage pool to which you are adding devices is configured to receive that media type.

1. On the menu bar, click **Block** > **SDSs**.
2. Select the relevant SDS from the SDS list.
3. Click **Add Device** > **Storage Device**.
4. In the **Add Storage Device to SDS** dialog box, enter the following required parameters for the storage device:
   a. Device path

      The length of the device path must not exceed 63 characters.

      For example, this path for an NVMe drive is not supported because it is too long:

      `/dev/disk/by-id/Dell_Express_Flash_NVMe_PM1725_1.6TB_SFF_______S2JPNA0J500141`

      Alternatively, this equivalent name is supported: `/dev/disk/by-id/nvme-eui.002538957100082e`

   b. Name of the device

      Assigning a name to a storage device can be particularly helpful for identifying devices in the future because the name remains constant, even if the path changes. Device names must meet the following requirements:
      - Contains less than 32 characters
      - Contains only alphanumeric and punctuation characters
      - Is unique within the object type

   c. Select the storage pool.
   d. Select the media type of the device: `HDD` or `SSD`.
   e. Click **Add Device**.
      The device is added to the Devices list.
5. Repeat the process for each additional storage device you wish to add.
6. Expand **Advanced** for more options; recommended for advanced users only.
7. Under **Device tests**, select the test option:
   - `Test and activate device` — Read and write test will be run on the device before it is capacity is used.
   - `Test only` — Devices will be tested, but not used.
   - `Activate without test` — The device capacity will be used without any device testing.

   By default, PowerFlex tests the performance of the device being added before its capacity can be used, and saves the results. Two tests are performed: random writes and random reads. When the tests are complete, the device capacity is added automatically to the storage pool used by the MDM. To modify this behavior, specify one of the test options.
8. Define the device test timeout.

   This value is the maximum test run time in seconds. The test stops when it reaches either this limit, or the time it takes to complete 128 MB of data read/write, whichever is first. When `Activate without test` is selected, this timeout is ignored.
9. Select whether to force device takeover.

   When devices are added to an SDS, PowerFlex checks that the device is clear before adding it. If the device is not clear, an error message is returned, and the command fails for that device. If you would like to overwrite existing data on the device by forcing the command, set **Force device takeover** to `YES`.

   CAUTION: Select `YES` with caution, because all data on the device will be destroyed.

10. Click **Add Devices**.
11. Verify that the operation has finished and was successful, and click **Dismiss**.

Add acceleration devices

Add acceleration devices to the PowerFlex system one by one or in bulk operations. By default, performance tests are performed on the added devices and the results are saved in the system.

Ensure that at least one suitable acceleration pool is defined.

All devices in a storage pool must be the same media type. Ensure that you know the type of devices that you are adding to the system and that the storage pool to which you are adding devices is configured to receive that media type.

1. On the menu bar, click **Block** > **SDSs**.
2. Select the relevant SDS from the SDS list.
3. Click **Add Device** > **Acceleration Device**.
4. In the **Add Acceleration Device to SDS** dialog box, enter the following required parameters for the acceleration device:
   a. Device path

      The length of the device path name must not exceed 63 characters.
   b. Name of the device

      Assigning a name to a storage device can be particularly helpful for identifying devices in the future, because the name remains constant, even if the path changes. Device names must meet the following requirements:
      - Contains less than 32 characters
      - Contains only alphanumeric and punctuation characters
      - Is unique within the object type
   c. Select the acceleration pool.
   d. Click **Add Device**.
      The device is added to the Devices list.
5. Repeat the process for each additional acceleration device you wish to add.
6. Expand **Advanced** for more options; recommended for advanced users only.
7. Under **Device tests**, select the test option:
   - `Test and activate device` — Read and write test will be run on the device before it is capacity is used.
   - `Activate without test` — The device capacity will be used without any device testing.

   By default, PowerFlex tests the performance of the device being added before its capacity can be used, and saves the results. Two tests are performed: random writes and random reads. When the tests are complete, the device capacity is added automatically to the storage pool used by the MDM. To modify this behavior, specify one of the test options.
8. Define the device test timeout.

   This value is the maximum test run time in seconds. The test stops when it reaches either this limit, or the time it takes to complete 128 MB of data read/write, whichever is first. When `Activate without test` is selected, this timeout is ignored.
9. Select whether to force device takeover.

   When devices are added to an SDS, PowerFlex checks that the device is clear before adding it. If the device is not clear, an error message is returned, and the command fails for that device. If you would like to overwrite existing data on the device by forcing the command, set **Force device takeover** to `YES`.

   CAUTION: Select `YES` with caution, because all data on the device will be destroyed.
10. Click **Add Devices**.
11. Verify that the operation has finished and was successful, and click **Dismiss**.

Rename SDS (mirroring)

You can rename a SDS following the specific naming rules.

The SDS name must adhere to the following rules:

- Contains less than 32 characters
- Contains only alphanumeric and punctuation characters

- Is unique within the object type

1. On the menu bar, click **Block** > **SDS**.
2. From the list of SDSs, select the relevant SDS, and click **Modify** > **Rename**.
3. In the **Rename SDS** dialog box, enter the new name, and click **Apply**.
4. Verify that the operation is finished and is successful.

Configure RMcache

RMcache uses RAM that is allocated for caching. Its size is limited to the amount of allocated RAM. By default, RMcache caching is disabled.

- Enable RMcache at the storage pool level for all of the SDSs in the storage pool.
- RMcache must also be enabled at the SDS level.

For a read to be stored in the RAM of a specific SDS, the RMcache feature on that SDS must be enabled, and the relevant storage pool and the relevant volume must both be configured to use RMcache. Caching only begins after one or more devices are added to the SDS. The amount of RAM that you may allocate for RMcache is limited and can never be the maximum available RAM.

Enabling RMcache at the storage pool level allows you to control the cache settings for all SDSs in the storage pool. You can enable RAM caching for a storage pool and then disable caching on one or more SDSs individually.

NOTE: Only I/Os that are multiples of 4k bytes can be cached.

1. On the menu bar, click **Block** > **SDSs**.
2. In the list of SDSs, select the relevant SDS, and click **Modify** > **Cache Settings**.
3. In the **SDS Cache Settings** dialog box, select **Enable Read RMcache**. Enter the RMcache cache size.
   The minimum RMcache cache size is 128 MB.
4. Click **Apply**.

Modify an SDS performance profile

SDS performance profiles are set by default to high and can be changed to compact. The compact setting may impact the system performance.

The default setting configures a predefined set of parameters for high-performance use cases.

NOTE: Performance tuning is case-specific. To prevent undesirable effects, Dell Technologies highly recommends that you thoroughly test all changes. For further assistance, contact Dell Technologies Support.

1. On the menu bar, click **Block** > **SDS**.
2. From the list of SDSs, select the relevant SDS, and click **Modify** > **Modify Performance Profile**.
3. In the **Modify Performance Profile Host** dialog box, select the wanted performance profile: **High** or **Compact**.
4. Click **Apply**.
5. Verify that the operation is finished and is successful.

Add an IP address

Add an additional IP address to a storage data server.

1. On the menu bar, click **Block** > **SDSs**.
2. In the list of storage data servers, select the relevant SDS checkbox and click **More Actions** > **Add IP Address**.
3. In the **Add IP** dialog box, enter the IP address.
4. Select the communication role.
5. Click **Add IP**.
6. Verify that the operation has finished and was successful.

Remove an IP address

Remove an IP address from a storage data server.

1. On the menu bar, click **Block** > **SDSs**.
2. In the list of storage data servers, select the relevant SDS checkbox and click **More Actions** > **Remove IP Address**.
3. In the **Remove IP Address** dialog box, select the IP address to remove from the cluster.
4. Click **Remove**.
5. Click **Remove selected IPs**.
6. Verify that the operation has finished and was successful.

Remove SDSs

Remove SDSs and devices gracefully from a system. The removal of some objects in the system can take a long time, because removal may require data to be moved to other storage devices in the system.

If you plan to replace a device with a device containing less storage capacity, you can configure the device to a smaller capacity than its actual capacity, in preparation for replacement. This reduces rebuild and rebalance operations in the system later on.

The system has job queues for operations that take a long time to run. You can view jobs by clicking the **Running Storage Jobs** icon on the right side of the toolbar. Operations that are waiting in the job queue are shown as Pending. If a job in the queue will take a long time, and you do not want to wait, you can cancel the operation using the **Abort** button in the **Remove** command window (if you left it open), or using the `Abort entering Protected Maintenance Mode` command from the **More Actions** menu.

CAUTION: The `Remove` command deletes the specified objects from the system. Use the `Remove` command with caution.

1. On the menu bar, click **Block** > **SDSs**.
2. In the list of storage data servers, select the relevant SDS checkbox and click **More Actions** > **Remove**.
3. In the **Remove SDS** dialog box, click **Remove**.
4. Verify that the operation has finished and was successful, and click **Dismiss**.

Place storage data server in maintenance mode

Place an storage data server in maintenance mode to perform non-disruptive maintenance on the storage data server.

1. On the menu bar, click **Block** > **SDSs**.
2. In the list of storage data servers, select the relevant storage data server and click **More Actions** > **Enter Maintenance Mode**.
3. In the **Enter SDS into Maintenance Mode** dialog box, select one of the following options:
   - **Instant** — a node is temporarily removed without building a new copy of the data. During maintenance, the system only mirrors new writes. After maintenance is complete, the system applies the new writes to the node that was under maintenance.
   - **Protected** — a third copy is created before entering maintenance mode. This ensures that if there is a node failure where the second copy of storage data server is required, there is still a full backup of the storage data server. This leaves no room for discrepancy between the copies. More storage capacity from the node is required.
4. Click **Enter Maintenance Mode**.
5. Verify that the operation has finished and was successful, and click **Dismiss**.
6. Optionally, at the top right side of the toolbar, click the **Running Storage Jobs** icon to check maintenance mode status.

Exit storage data server from maintenance mode

After completing maintenance tasks, remove the storage data server from maintenance mode.

Once you have finished performing maintenance tasks on the storage data server, you can manually exit maintenance mode and return the storage data server to normal mode.

1. On the menu bar, click **Block** > **SDSs**.

2. In the list of storage data servers, select the relevant storage data server and click **More Actions** > **Exit Maintenance Mode**.
3. In the **Exit SDS from Maintenance Mode** dialog box, click **Exit Maintenance Mode**.
4. Verify that the operation has finished successfully, and click **Dismiss**.

After the operation has been completed successfully, the storage data server returns to normal operation, and data deltas collected on other storage data servers during the maintenance period are copied back to the storage data server.

Cancel entering protected maintenance mode for SDS

A third copy of the SDS is created when entering protected maintenance mode (PMM), which can take a long time. You can back out of this process before entering PMM is complete, such as when there is insufficient capacity to create the third copy.

1. On the menu bar, click **Block** > **SDSs**.
2. In the list of SDSs, select the relevant fault set checkbox and click **More Actions** > **Abort entering Protected Maintenance Mode**.
3. In the **Abort entering Maintenance Mode** dialog box, click **Abort**.
4. Verify that the operation has finished successfully, and then click **Dismiss**.

Storage pools (mirroring)

A storage pool is a container of volumes that uses one device group and, if available, one PMEM device group. A storage pool consumes the capacity from the device group.

A storage pool is a container of volumes that belongs to a single device group and consumes the capacity from the device group.

The capacity of a storage pool is based on the capacity of the device group. You can create one storage pool per device group and allocate the full capacity of the device group to that storage pool.

Add, modify, or remove a storage pool from the storage system. You can connect a storage pool to a persistent memory device group to enhance write performance by using the persistent memory as a write cache.

Add storage pools

A storage pool is a group of storage devices within a protection domain. Each time that you add devices to the system, you must map them to either storage pools or to acceleration pools. Create storage pools before you start adding SDSs and storage devices to the system.

- Familiarize yourself with the types of storage pools that are available, and ensure that you know the media type of the devices that will be used in the storage pool. Each storage pool must contain devices of only one media type.
- A storage pools with fine granularity data layout, requires an acceleration pool which contains at least one NVDIMM configured as a DAX device is required. Ensure that you have configured an NVDIMM acceleration pool prior to creating a fine granularity storage pool.

1. On the menu bar, click **Block** > **Storage System** > **Storage Pools**.
2. Click **Add Storage Pool**.
3. In the **Create Storage Pool** dialog box, define the following settings:
   a. Define the storage pool name according to the following rules:
      - Contains less than 32 characters
      - Contains only alphanumeric and punctuation characters
      - Is unique within the object type
   b. Select the relevant protection domain.
   c. Select the media type of the devices in the storage pool: `HDD` or `SSD`.

      All devices for this storage pool must be of the same media type.
   d. If you select SSD, select the **Data Layout - Granularity** type: **Medium** or **Fine**.
   e. For a fine granularity storage pool, select the relevant **Acceleration Pool**.
4. To use RMcache for caching, select **Use Read RAM Cache**.
   a. If you are using RMcache, select the **Write Handling Mode**: `Cached` or `PassThrough`.

This defines whether the system stores the data of this storage pool's writes in the SDS RMcache, or not. The default is to store the write data in cache (cached).

NOTE: The RMcache features are advanced features, and it is usually recommended to accept the default values. You can configure these features later, if necessary, by clicking **Modify** > **Cache**.

5. To enable validation of the checksum value of in-flight data reads and writes, select **Use Inflight Checksum**.
6. By default, the **Use Persistent Checksum** is selected to ensure persistent checksum data validation.

   NOTE: This option is enabled only when HDD or SSD with medium granularity is selected.

7. Click **Create**.
8. Verify that the operation has finished successfully, and click **Dismiss**.

Configure storage pool general settings (mirroring)

Configure storage pool settings, including checksum, zero padding, and compression.

1. On the menu bar, click **Block** > **Storage System** > **Storage Pools**.
2. In the list of storage pools, select the relevant storage pool, and click **Modify** > **General Settings**.
3. In the **Storage Pool Settings** dialog box, configure the following settings for the storage pool:

Option	Description
Enable Rebuild/ Rebalance	By default, the rebuild/rebalance features are enabled in the system because they are essential for system health, optimal performance, and data protection. **CAUTION: Rebuilding and rebalancing are essential parts of PowerFlex and should only be disabled temporarily, in special circumstances. If rebuilds are disabled, redundancy will not be restored after failures. Disabling rebalance may cause the system to become unbalanced even if no capacity is added or removed.**
Enable Inflight Checksum	Inflight checksum protection mode can be used to validate data reads and writes in storage pools, in order to protect data from data corruption.
Enable Persistent Checksum	Persistent checksum can be used to support the medium granularity layout in protecting the storage device from data corruption. Select **validate on read** to validate data reads in the storage pool. **NOTE:** If you want to enable or disable persistent checksum, you must first disable the background device scanner from the storage pool.
Enable Zero Padding Policy	When zero padding policy is enabled, it ensures that every read from an area previously not written to returns zeros. Zero padding policy is enabled by default on fine granularity-based storage pools and cannot be changed. For medium granularity storage pools, the zero padding policy can be modified only if the storage pool is empty and has no SDS devices added to it.
Enable Compression	For fine granularity storage pools, inline compression allows you to gain more effective capacity.

4. Click **Apply**.

Configure RMcache for the storage pool

RMcache uses RAM that is allocated for caching. The size is limited to the amount of allocated RAM. By default, RMcache caching is disabled.

RMcache must also be enabled for each SDS in the storage pool.

RMcache caching only begins once storage devices have been added to the SDSs. It is possible to enable RMcache for a storage pool and then disable caching on one or more SDSs individually.

NOTE: Only I/Os that are multiples of 4K bytes can be cached.

1. On the menu bar, click **Block** > **Storage System** > **Storage Pools**.
2. In the list of storage pools, select the checkbox of the relevant storage pool, and click **Modify** > **Cache**.

3. In the **Storage Pool Cache Settings** dialog box, select **Use Read RAM Cache** .
4. Click **Apply**.

Using the background device scanner

The background device scanner scans devices in the system to check for errors.

You can enable and disable the background device scanner, as well as reset the background device scanner counters. Information about errors is provided in event reports.

Reset error counters (mirroring)

Reset the background device scanner error counters for the specified storage pools. There are counters for data comparison errors and fixed read errors.

1. On the menu bar, click **Block** > **Storage System** > **Storage Pools**.
2. In the list of storage pools, select the relevant storage pool checkbox, and click **More Actions** > **Reset Error Counters**.
3. In the **Reset Error Counters** dialog box, select the relevant option:
   - **Reset Fixed Read Error Counters**—This counter tracks errors that are automatically fixed.
   - **Reset Compare Error Counters**—This counter tracks read errors.
4. Click **Apply**.
5. Verify that the operation completed successfully and click **Dismiss**.

Enable the background device scanner

Enable the background device scanner to check for errors on the devices in the specified storage pool.

1. On the menu bar, click **Block** > **Storage System** > **Storage Pools**.
2. In the list of storage pools, select the relevant storage pool check box, and click **More Actions** > **Background Device Scanner**.
3. In the **Set Background Device Scanner for Storage Pool** dialog box, select the relevant option for the following settings. By default, all options are selected.
   - **Enable Background Device Scanner**
   - **Fix Local Device Errors**— automatically fixes device errors.
   - **Compare Data**— compare between primary and secondary copies of data.

     NOTE: Zero padding must be enabled in order to set the background device scanner to data compare mode.

   - If **Compare Data** is selected, select or clear **Fix Local Device Errors**.
   - **Bandwith Limit** in KB/S. Default=3072 KB/S

     NOTE: High bandwidth may create negative impact on system performance and should be used carefully and in extreme cases only—for example, when there is an urgent need to check certain devices. When setting the background device scanner bandwidth, you should consider the maximum bandwidth of the devices.

4. Click **Apply**.
5. Verify that the operation has finished and was successful, and click **Dismiss**.

Set media type for storage pool

Update the media type of the storage pool. All devices in the storage pool will be this media type.

1. On the menu bar, click **Block** > **Storage System** > **Storage Pools**.
2. In the list of storage pools, select the relevant storage pool checkbox, and click **Modify** > **Media Type**.
3. In the **Set Media Type for Storage Pool** dialog box, from the **Media Type** list, select the media type for the storage pool.
   - **HDD**
   - **SSD**

- **Transitional**—The media type is defined per device rather than at the storage pool level, to allow for migration of devices from one storage pool to another.

4. Select the **Overwrite SDS device media type** checkbox to overwrite the current device media type set for the SDS.
5. Click **Apply**.
6. Verify that the operation has finished and was successful, and click **Dismiss**.

Configure I/O priorities and bandwidth use (mirroring)

PowerFlex includes advanced settings that control I/O priorities and bandwidth use and can be used to fine-tune system performance. The fine-tuning includes the ability to control secondary I/Os which affects the garbage collector, rebuild, and rebalance.

NOTE: These features affect system performance, and should only be configured by an advanced user. It is recommended to retain the default settings, unless you are an advanced user.

1. On the menu bar, click **Block** > **Storage System** > **Storage Pools**.
2. In the list of storage pools, select the relevant storage pool checkbox, and click **Modify** > **IO Priority**.
3. In the **Set IO Priority for Storage Pool** dialog box, for each of the tabs—`Rebuild`, `Rebalance`, `Migration`, and `Maintenance Mode`—select one of the following IO Priority options:
   - **Unlimited**—I/Os are not limited
   - **Limit concurrent IO**—limit the number of allowed concurrent I/Os to the value entered in the **Concurrent I/O limit** field
   - **Favor Application IO**—limit the number of allowed concurrent I/Os to the values entered in the **Concurrent I/O limit** and **Bandwidth I/O limit** fields, regardless of user I/O
4. Click **Apply** and click **Dismiss**.

Devices (mirroring)

Storage devices or acceleration devices are added to an SDS or to all SDSs in the system.

There are two types of devices: storage devices and acceleration devices.

Activate a device (mirroring)

Activate one or more devices that were added to a system using the `Test only` option for device tests.

1. On the menu bar, click **Block** > **Storage System** > **Devices**.
2. In the list of devices, select the checkboxes of the required devices, and click **More Actions** > **Activate**.
3. In the **Activate Device** dialog box, click **Activate**.
4. Verify that the operation has finished and was successful and click **Dismiss**.

Clear a device error (mirroring)

Perform the following procedure when device errors have been rectified, but the errors have not been cleared automatically by the system.

1. On the menu bar, click **Block** > **Storage System** > **Devices**.
2. In the list of devices, select the checkboxes of the relevant devices, and click **More Actions** > **Clear Errors**.
3. In the **Clear device errors** dialog box, click **Clear Errors**.
4. Verify that the operation has finished and was successful and click **Dismiss**.

Remove a device (mirroring)

Use the following procedure to remove a storage or acceleration device.

Before removing an NVDIMM or SDPM acceleration device, remove all storage devices that are being accelerated by the persistent memory. Then, remove the NVDIMM or SDPM from its acceleration pool.

1. On the menu bar, click **Block** > **Storage System** > **Devices**.
2. In the list of devices, find the device that you want to remove, make a note of the SDS in which it is installed, and the storage pool or acceleration pool to which it belongs.

   This information will be useful for adding the device back to the system later.
3. Select the required device, and click **More Actions** > **Remove**.
4. In the **Remove Device** dialog box, verify that you have selected the desired device, and click **Remove**.
5. Verify that the operation has finished and was successful, and click **Dismiss**.
   A rebuild/rebalance occurs. For each device that was removed from the SDS, the corresponding cell in the **Used Size** column counts down to zero, and then the device disappears from the **Devices** list.

Rename a device (mirroring)

Use this procedure to change the name of a device.

You can view the current device name by displaying the **Name** column in the device list. The **Name** column is hidden, by default. When no device name has been defined, the name is set by default to the device's path name.

The device name must adhere to the following rules:

- Contains less than 32 characters
- Contains only alphanumeric and punctuation characters
- Is unique within the object type

1. On the menu bar, click **Block** > **Storage System** > **Devices**.
2. In the list of devices, select the required device, and click **Modify** > **Rename**.
3. In the **Rename Device** dialog box, enter the new name, and click **Apply**.
4. Verify that the operation has finished and was successful, and click **Dismiss**.

Set media type (mirroring)

You can set a device media type as SSD or HDD.

Ensure that you are adding the correct device to the system.

All devices in a storage pool must be the same media type. Set the media type for a device before adding it to a storage pool.

1. On the menu bar, click **Block** > **Storage System** > **Devices**.
2. In the list of devices, select the relevant device, and click **Modify** > **Set Media Type**.
3. Select the media type that corresponds to the device.
4. Click **Apply**.
5. Verify that the operation has finished and was successful, and click **Dismiss**.

Set device capacity limits (mirroring)

Prior to replacing a device with a device of a smaller capacity, set the capacity limit of the device being removed to the capacity of the new device. Capacity will be decreased, but the size of the disk remains unchanged.

NOTE: The capacity assigned to the storage device must be smaller than its actual physical size.

1. On the menu bar, click **Block** > **Storage System** > **Devices**.
2. In the list of devices, select the relevant device, and click **Modify** > **Set Capacity Limit**.
3. In the **Set Capacity Limit** dialog box, enter the capacity and select a unit type (MB or GB).

   This dialog box displays the maximum capacity available for the device.

4. Click **Apply**.
5. Verify that the operation has finished and was successful, and click **Dismiss**.

Modify device LED settings (mirroring)

Set a device's LED to blink or turn it off. This feature can help you physically identify a device in the system chassis.

1. On the menu bar, click **Block** > **Storage System** > **Devices**.
2. In the list of devices, select the relevant device, and click **Modify** > **LED Settings**.
3. In the **Set device LED settings** dialog box, do one of the following:
   - To turn on the device's LED, select the **On** checkbox.
   - To turn off the device's LED, clear the **On** checkbox.
4. Click **Apply**.
5. Verify that the operation has finished and was successful, and click **Dismiss**.

Acceleration pools

An acceleration pool is a group of acceleration devices within a protection domain. PowerFlex only supports acceleration of fine granularity storage pools.

Fine granularity acceleration uses NVDIMMs devices configured to fine granularity storage pools. Configure NVDIMM acceleration pools for fine granularity acceleration.

Add an acceleration pool

Add an acceleration pool to a protection domain to accelerate fine granularity storage performance.

Ensure that PowerFlex and acceleration devices are prepared before adding acceleration pools. SDPM or NVDIMM must be configured as DAX (acceleration) devices.

1. On the menu bar, click **Block** > **Storage System** > **Acceleration Pools**.
2. Click **Add Acceleration Pool**.
3. In the **Create Acceleration Pool** dialog box, specify a name for the acceleration pool.
4. Select the following information from the relevant menus:
   - **Pool type**: Select **NVDIMM** or **SSD** for the acceleration pool type.

     NOTE: Dell PowerEdge R660 and R760 servers uses software-defined persistent memory (SDPM). If you are using SDPM, select NVDIMM for the pool type.

   - **Protection Domain**: Select the protection domain to be accelerated.
5. In the **Add Devices** area, select the devices that will be used for acceleration:

Option	Description
Add acceleration devices from all SDSs that contribute to the relevant acceleration pool.	Optionally, select the **Add Devices To All SDSs** checkbox to add acceleration devices from all SDSs in the protection domain. Otherwise, select acceleration devices one by one.
Add devices one by one.	Enter the path and name of each acceleration device, select the SDS on which the device is installed, and then click **Add Devices**. Repeat for all desired acceleration devices in the acceleration pool.

6. Click **Create**.
7. Verify that the operation has finished and was successful, and click **Dismiss**.

Rename an acceleration pool

Rename an acceleration pool.

1. On the menu bar, click **Block** > **Storage System** > **Acceleration Pools**.
2. In the acceleration pools list, select the desired acceleration pool.
3. Click **Rename**.
4. In the **Rename Acceleration Pool** dialog box, enter the new name for the acceleration pool, and click **Apply**.
5. Verify that the operation has finished and was successful, and click **Dismiss**.

Remove an acceleration pool

Delete an acceleration pool from PowerFlex.

Remove all acceleration devices from the acceleration pool before deleting the acceleration pool.

1. On the menu bar, click **Block** > **Storage System** > **Acceleration Pools**.
2. In the list of acceleration pools, select the desired acceleration pool.
3. Click **Delete**.
4. In the **Delete Acceleration Pool** dialog box, click **Delete**.
5. Verify that the operation has finished and was successful, and click **Dismiss**.

Volumes

Define, configure and manage volumes in the PowerFlex system.

Add volumes (mirroring)

Use the following procedure to add volumes. Dell Technologies highly recommends giving each volume a meaningful name that is associated with its operational role.

There must be at least three SDS nodes in the system and there must be sufficient capacity available for the volumes.

PowerFlex objects are assigned a unique ID that can be used to identify the object in CLI commands. The default name for each volume object is its ID. The ID is displayed in the volumes list or can be obtained using a CLI query. Define each volume name according to the following rules:

- Contains less than 32 characters
- Contains only alphanumeric and punctuation characters
- Is unique within the object type

Dell Technologies highly recommends giving each volume that a meaningful name associated with its operational role.

1. On the menu bar, click **Block** > **Volumes**.
2. Click **Create Volume**.
3. In the **Create Volume** dialog box, configure the following items:
   a. Select the storage pool name and protection domain where the volume will reside.
   b. Enter the number of volumes.
   c. Enter a name for the volume.

      If the number of volumes is greater than one, then enter a **Volume Prefix** and a **Starting Number** of the volume. For example, if the volume prefix is `Vol%i%` and the starting number value is `100`, the name of the volume created will be Vol100, the second volume will be Vol101 and so on.
   d. Enter the volume size in GB (basic allocation granularity is 8 GB).
   e. Select either **Thin** (default) or **Thick** provisioning options.
4. Click **Create**.
5. Verify that the operation is finished and is successful, and click **Dismiss**.

To use the created volume, you must map it to at least one host. If the restricted SDC mode is enabled for the system, you must approve SDCs before mapping volumes to them.

Map a volume to a host (mirroring)

Mapping exposes the volume to the specified host, creating a block device on the host. You can map a volume to one or more hosts.

Volumes can only be mapped to one type of host either an SDC or an NVMe. Ensure that you know which type of hosts are being used for each volume, to avoid mixing host types.

For Linux-based devices, the `scini` device name may change on reboot. Dell Technologies recommends that you mount a mapped volume to the PowerFlex unique ID, which is a persistent device name, rather than to the `scini` device name.

To identify the unique ID, run the command `ls -l /dev/disk/by-id/`.

You can also identify the unique ID using VMware. In the VMware management interface, the device is called **EMC Fibre Channel Disk**, followed by an ID number starting with the prefix **eui**.

NOTE: You cannot map a volume if the volume is an auto snapshot that is not locked.

NOTE: You cannot map the volume on the target of a peer system if it is connected to a replication consistency group.

1. On the menu bar, click **Block** > **Volumes**.
2. In the list of volumes, select one or more volumes, and click **Provision** > **Map**.
3. A list of the hosts that can be mapped to the selected volumes is displayed. If a volume is already mapped to a host, only hosts of the same type, NVMe or SDC, are listed. If the volume is not mapped to a host, click **NVMe** or **SDC** to set the type of hosts to be listed.
4. In the **Map Volume** dialog box, select one or more hosts to which you want to map to the volumes.
5. Click **Map**.
6. Verify that the operation has finished and was successful, and click **Dismiss**.

Unmap a volume (mirroring)

You can unmap one or more volumes from a host.

1. On the menu bar, click **Block** > **Volumes**.
2. In the list of volumes, select the relevant volumes, and then click **Provision** > **Unmap**.
3. Select the host from which to remove mapping to the volumes.
4. Click **Unmap**.
5. Verify that the operation has finished and was successful, and click **Dismiss**.

Snapshots

A snapshot is a copy of a volume at a specific point in time. You can create, restore, create thin clones, and delete snapshots.

Starting in PowerFlex Manager 5.1.x, the identification of volumes and snapshots have changed. Prior to PowerFlex Manager 5.1.x, only volumes were displayed by default in **Block** > **Volumes**, while all the snapshots (read-only or read/write) were listed in **Data Protection** > **Snapshots**. Beginning in PowerFlex Manager 5.1.x, both volumes and read/write snapshots are displayed in **Block** > **Volumes**, and read-only snapshots are listed in the **Snapshots** tab in the **View Details** screen of each individual volume. In addition, in PowerFlex Manager 5.1.x, snapshots that qualify for migration are treated as volumes. For example, when a snapshot becomes eligible for migration it moves into the volumes screen, where it appears as a standard volume and can be selected in migration workflows.

Create snapshots

PowerFlex lets you to create instantaneous snapshots of one or more volumes.

The **Use secure snapshots** option prohibits deletion of the snapshots until the defined expiration period has elapsed.

When you create a snapshot of more than one volume, PowerFlex generates a consistency group by default. The snapshots under the consistency group are taken simultaneously for all listed volumes, thereby ensuring their consistency. You can view the consistency group by clicking **View Details** in the right pane and then clicking the **Snapshots Consistency Group** tab in the left pane.

NOTE: The consistency group is for convenience purposes only. No protection measures are in place to preserve the consistency group. You can delete members from the group.

1. On the menu bar, click **Block** > **Volumes**.
2. In the list, select the relevant volumes, and click **Protect** > **Create Snapshot**.
3. In the **Create snapshot of volume** dialog box, enter the following details:
   a. The name of the snapshot. You can accept the default name, or give the snapshot a name according to the following rules:
      - Contains less than 32 characters
      - Contains only alphanumeric and punctuation characters
      - Is unique within the object type
   b. Enter an index value.
4. (Optional) If you want to prohibit the deletion of snapshots until a specific time, select **Use secure snapshots** option. The use secure snapshots option prohibits deletion of the snapshots until the defined expiration period has elapsed.
   a. To define the expiration period, enter the number in **Expiration Time** and select the units from the dropdown menu.
5. Click **Create**.
6. Verify that the operation has finished and was successful, and click **Dismiss**.

Restore a snapshot

Perform the following procedure to restore a snapshot to a previous point in time when the snapshot was taken.

Before proceeding with the restore operation, applications using the volume should be shut down and the volume must be offline on the hosts, to prevent data integrity issues.

1. On the menu bar, click **Block** > **Volumes**.
2. From the list of volumes, select the relevant volume, and then click **View Details**.
3. Click **Snapshots**.
4. Select the checkbox beside the snapshot that you want to restore.
5. Click **Restore**.
6. In the **Restore Using Snapshot** dialog box, verify the details. Click **Restore**.
7. Verify that the operation is finished and is successful, and click **Dismiss**.

Create a thin clone of a snapshot

Thin clones are writable copies of a volume or a snapshot.

A thin clone is a space efficient copy that shares data blocks with its parent object and not a full backup of the original resource.

Thin clones retain full read access to the original resource. You can modify the data within the thin clone while preserving the original data on the parent resource.

Using thin clones, you can establish hierarchical points in time to preserve data over different stages of data modifications. If the parent resource (volume or snapshot) is deleted, the thin clone is unaffected.

1. On the menu bar, click **Block** > **Volumes**.
2. From the list of volumes, select the relevant volume, and then click **View Details**.
3. Click **Snapshots**.
4. Select the checkbox beside the snapshot that you want to clone.
5. Click **Clone**.
6. In the **Create thin clone** dialog box, enter a new name and click **Create**.
7. Verify that the operation is finished and is successful, and click **Dismiss**.

Rename a snapshot

You can rename a snapshot for clarity, version control, organization and categorization, and for audit purposes.

1. On the menu bar, click **Block** > **Volumes**.
2. From the list of volumes, select the relevant volume, and then click **View Details**.

3. Click **Snapshots**.
4. Select the checkbox beside the snapshot that you want to rename.
5. Go to **Modify** > **Rename**.
6. In the **Rename Snapshot** dialog box, enter a new name and click **Apply**.
7. Verify that the operation is finished and is successful, and click **Dismiss**.

Delete snapshots

Remove snapshots of volumes from PowerFlex.

Ensure that the snapshot that you are removing is not mapped to any hosts. If the snapshot is mapped, unmap it before removing it. In addition, ensure that the snapshot is not the source volume of any snapshot policy. You must remove the volume from the snapshot policy before you can remove the snapshot.

To prevent causing a data unavailability scenario, avoid deleting volumes or snapshots while the MDM cluster is being upgraded.

CAUTION: Removing a snapshot erases all the data in the corresponding snapshot.

1. On the menu bar, click **Block** > **Volumes**.
2. In the list of volumes, select the volume that contains the snapshot.
3. In the right pane, click **View Details**.
4. Go to the **Snapshots** tab.
5. Select the checkbox beside the snapshot you want to delete.
6. Go to **Modify** > **Delete**.
7. In the **Delete Volume** dialog box, select an option:
   - Delete volume
   - Delete volume with all its snapshots
8. Click **Delete**.
9. In the next dialog box confirming the snapshots to remove, verify that you are deleting the correct snapshots, and click **Delete**.
10. Verify that the operation has finished and was successful, and click **Dismiss**.

Restore volume data

The volume on the host must be taken offline to prevent data integrity issues.

1. On the menu bar, click **Block** > **Volumes**.
2. From the list of volumes, select the relevant volume, and then click **Protect** > **Restore Data**.
3. In the **Restore Data of Volume** window, select the snapshot to restore from.
4. Click **Restore Volume**.
5. Verify that the operation is finished and is successful, and click **Dismiss**.

Assign a snapshot policy

You can assign a snapshot policy which is a set of rules that controls the creation, retention, and expiration of snapshots.

1. On the menu bar, click **Block** > **Volumes**.
2. Select the volumes or snapshots that you want to assign to the policy, and click **Protect** > **Assign Snapshot Policy**.
3. In the **Assign Snapshot Policy** dialog box, from the list of snapshot policies, select the relevant policy to assign and click **Assign Policy**.
4. Verify that the operation is finished and is successful, and click **Dismiss**.

Unassign a volume from a snapshot policy

You can unassign a volume from a snapshot policy to detach the source volume from the policy and delete associated auto snapshots. This process helps manage snapshots and ensures that only relevant data is retained.

If the auto snapshots are locked, unassigning them from a policy only detaches them. The auto snapshots must be manually deleted.

1. On the menu bar, click **Block** > **Volumes**.
2. Select the volume that you want to unassign from the policy, and click **Protect** > **Unassign Snapshot Policy**.
3. (Optional) If you want the snapshot to be removed, in the **Unassign Volumes from Snapshot Policy** dialog box, select the **Delete related snapshots** checkbox and click **Unassign Policy**.
4. In the **Unassign Volumes From all Snapshot Policies** dialog box, click **Yes**.
5. Verify that the operation is finished and is successful, and click **Dismiss**.

Create a thin clone of a volume

Thin clones are writable copies of a volume or a snapshot.

A thin clone is a space efficient copy that shares data blocks with its parent object and not a full backup of the original resource.

Thin clones retain full read access to the original resource. You can modify the data within the thin clone while preserving the original data on the parent resource.

Using thin clones, you can establish hierarchical points in time to preserve data over different stages of data modifications. If the parent resource (volume or snapshot) is deleted, the thin clone is unaffected.

1. On the menu bar, click **Block** > **Volumes**.
2. Select the volume that you want to create a thin clone from, and click **Repurpose** > **Create Thin Clone**.
3. In the **Create thin clone of volume** dialog box, enter a thin clone name, and click **Create**.
4. Verify that the operation is finished and is successful, and click **Dismiss**.

Refresh volume data

Ensure that you have the most up-to-date data by refreshing.

1. On the menu bar, click **Block** > **Volumes**.
2. From the list of volumes, select the relevant volume, and then click **Repurpose** > **Refresh Data**.
3. In the **Refresh Data of Volume** window, select the volume, thin clone, or snapshot to refresh from.
4. Click **Refresh Volume**.
5. Verify that the operation is finished and is successful, and click **Dismiss**.

Increase a volume size (mirroring)

You can increase, but not decrease, the logical size of one volume at any time, as long as there is enough capacity for the volume size to grow.

1. On the menu bar, click **Block** > **Volumes**.
2. In the list of volumes, select the volume, and click **More Actions** > **Resize**.
3. In the **Resize Volume** dialog box, enter the new volume size, and select a unit type. (The basic allocation granularity is 8 GB.)
4. Click **Apply**.
5. Verify that the operation has finished and was successful, and click **Dismiss**.

Rename a volume

You can use the default volume name or create a name.

1. On the menu bar, click **Block** > **Volumes**.

2. From the list of volumes, select the relevant volume, and click **More Actions** > **Rename**.
3. In the **Rename Volume** dialog box, enter a new name and click **Apply**.
4. Verify that the operation is finished and is successful, and click **Dismiss**.

Delete a volume (mirroring)

You can remove a volume from PowerFlex provided the volume is not mapped to a host and it is not assigned to a snapshot policy.

Ensure that the volume that you are deleting is not mapped to any hosts. If it is, unmap it before deleting it. In addition, ensure that the volume is not the source volume of any snapshot policy. You must remove the volume from the snapshot policy before you can remove the volume.

To prevent causing a data unavailability scenario, avoid deleting volumes or snapshots while the MDM cluster is being upgraded.

CAUTION: All data is erased from a deleted volume.

1. On the menu bar, click **Block** > **Volumes**.
2. In the list of volumes, select the required volume, and click **More Actions** > **Delete**.
3. In the **Delete Volume** dialog box, verify the volumes to be removed, and click **Delete**.
4. Verify that the operation has finished and was successful.

Set a volume bandwidth and IOPS limits (mirroring)

Setting bandwidth and IOPS limits for volumes lets you control the quality of service (QoS). Bandwidth and IOPS limits are set on every host.

Ensure that the volumes are mapped before you set these limits.

1. On the menu bar, click **Block** > **Volumes**.
2. In the list of volumes, select the relevant volume, and then click **More Actions** > **Set Limits**.
3. In the **Set IO limits for volume** dialog box, enter the required values for **Bandwidth Limits** and **IOPS Limits**, or select the corresponding **Unlimited** checkbox.
   - The number of IOPS must be larger than 10.
   - The volume network bandwidth is in MB/s.
   - The I/O limits are applied to every mapped SDC.
4. Click **Apply**.
5. Verify that the operation has finished successfully and click **Dismiss**.

Remove a snapshot consistency group

Remove a consistency group with all its snapshots.

NOTE: You cannot remove a consistency group that contains auto snapshots.

1. On the menu bar, click **Block** > **Volumes**.
2. In the list of volumes, select the relevant volume, and then in the right pane, click **View Details**.
3. Click the **Snapshots Consistency Group** tab.
4. Select the required consistency group.
5. Click **Remove SCG**.
6. In the **Remove SCG** dialog box, click **Remove SCG**.
7. Verify that the operation has finished and was successful, and click **Dismiss**.

Volume tree vTree migration

Migration of a volume tree (vTree) allows you to move a vTree to a different storage pool.

Migration of a vTree frees up capacity in the source storage pool. For example, you can migrate a vTree from an HDD-based storage pool to an SSD-based storage pool, or to a storage pool with different attributes such as `thin` or `thick`.

There are several possible reasons for migrating a vTree to a different storage pool:

- To move the volumes to a different storage pool type
- To move to a different storage pool or protection domain due to multitenancy
- To decrease the capacity of a system by moving out of a specific storage pool
- To change from a thin-provisioned volume to a thick-provisioned volume, or the reverse
- To move the volumes from a medium granularity storage pool to a fine granularity storage pool
- To clear a protection domain for maintenance purposes, and then return the volumes to it

During vTree migration, you can run other tasks such as creating snapshots, deleting snapshots, and entering maintenance mode.

NOTE: You cannot create snapshots when migrating a vTree from a medium granularity storage pool to a fine granularity storage pool.

When a user requests a vTree migration, the MDM begins the process by estimating whether the destination storage pool has enough capacity for a successful migration. The MDM bases the estimation on its information about the current capacity of the vTree. If there is insufficient capacity at the destination based on that estimate, migration does not start. (An advanced option allows you to force the migration even if there is insufficient capacity at the destination, with the intention to increase the capacity as required during the migration.) The MDM does not reserve the estimated capacity at the destination (since the capacity of the source volume can grow during migration and the reserved capacity does not guarantee success). The MDM does not retain source capacity once it has been migrated, but releases it immediately.

Use the following table to understand which vTree migrations are possible, and under what specific conditions:

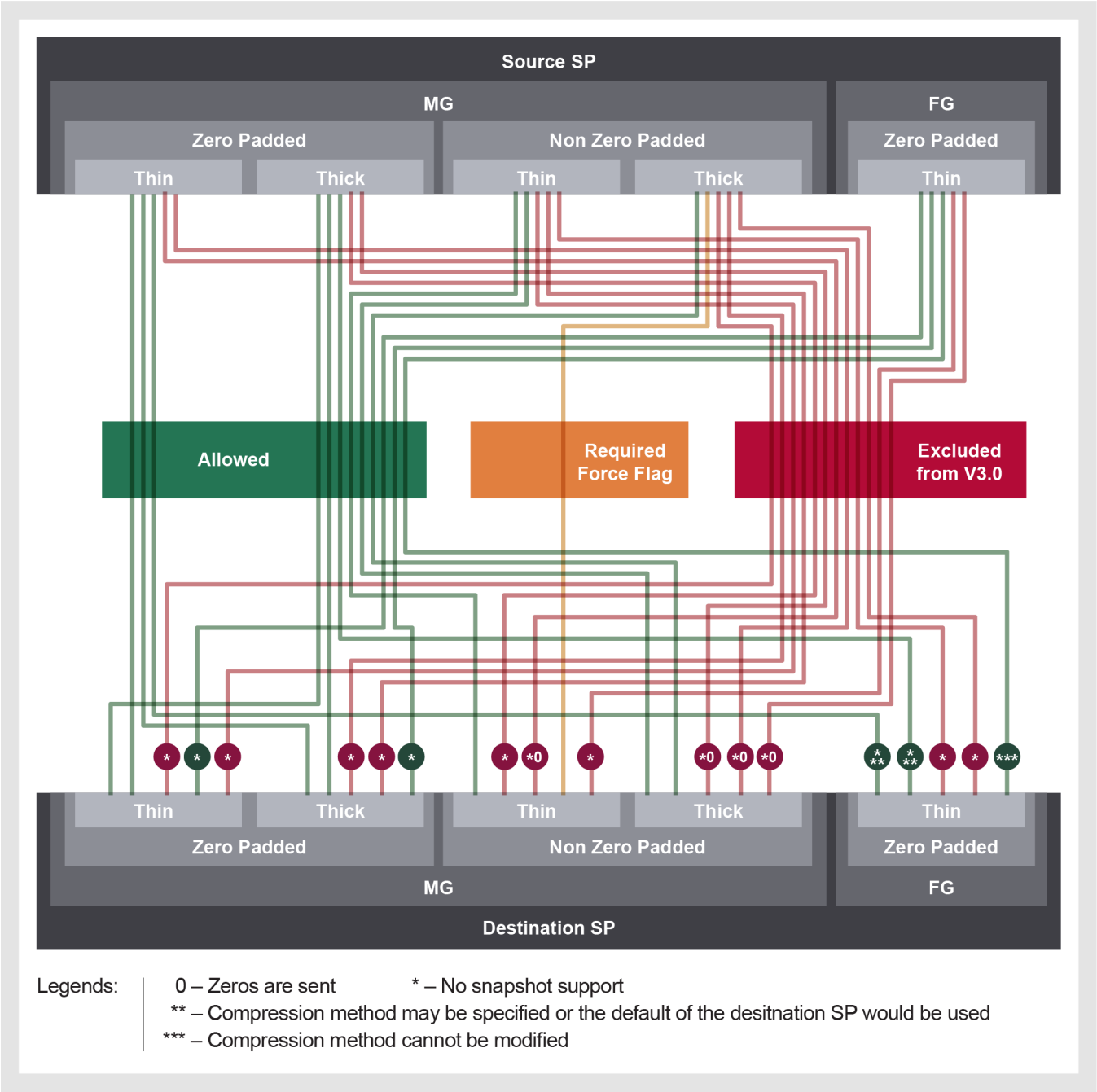


Figure 10. vTree Migrations

vTree migration can take a long time, depending on the size of the vTree and the system workload. During migration, the vTree is fully available for user I/O. vTree migration is done volume block by volume block. When a single block has completed its migration, the capacity of the block at the source becomes available, and it becomes active in the destination storage pool. During migration, the vTree has some of its blocks active in the source storage pool, and the remaining blocks active in the destination storage pool.

NOTE: You can speed up the migration by adjusting the volume migration I/O priority (QoS). The default favors applications with one concurrent I/O and 10 MB/sec per device. Increasing the 10 MB/sec setting increases the migration speed in most cases. The maximum value that can be reached is 25 MB/sec. The faster the migration, the higher the impact might be on applications. To avoid significant impact, the value of concurrent I/O operations per second should not be increased.

When migrating from a medium granularity storage pool to a fine granularity storage pool, volumes must be zero padded.

You can pause a vTree migration at any time, in the following ways:

- Gracefully—to allow all data blocks currently being migrated to finish before pausing.
- Forcefully—to stop the migration of all blocks currently in progress.

Once paused, you can choose to either resume the vTree migration, or to roll back the migration and have all volume blocks returned to the original storage pool.

vTree migration might also be paused internally by the system. System pauses happen when a rebuild operation begins at either the source or destination storage pool. If the migration is paused due to a rebuild operation, it remains paused until the rebuild ends. If the system encounters a communication error that prevents the migration from proceeding, it pauses the migration and periodically tries to resume it. After a configurable number of attempts to resume the migration, the migration remains paused, and no additional retries will be attempted. You can manually resume migrations that were internally paused by the system.

Concurrent vTree migrations are allowed in the system. These migrations are prioritized by the order in which they were invoked, or by manually assigning the migration to the head or the tail of the migration queue. You can update the priority of a migration while it is being run. The system strives to adhere to the priority set by the user, but it is possible that volume blocks belonging to migrations lower in priority are run before ones that are higher in priority. This can happen when a storage pool that is involved in migrating a higher priority block is busy with other incoming migrations, and the storage pools involved in lower priority migrations are available to run the migration.

Migrate a volume tree vTree

Migrate a volume and all of its snapshots to a different storage pool. Volumes undergoing migration remain available for I/O.

NOTE: vTree migration is a long process and can take days or weeks, depending on the size of the vTree.

The following limitations apply:

- Migration between storage pools with different data layouts is only allowed if there is a single volume in the vTree.
- vTrees containing a manually created snapshot cannot be migrated.
- You cannot migrate a volume that is a source volume of a snapshot policy between storage pools with different data layouts.
- Volumes involved in replication cannot be migrated.

1. On the menu bar, click **Block** > **Volumes**.
2. In the list of volumes, select the volume that you want to migrate.
3. In the right pane, click **View Details**.
4. Click the **VTree** tab.
5. Select **Migrate vTree**.
6. In the **Migrate vTree** dialog box, in the **TARGET** area, select the destination storage pool.
   Ensure that you select a storage pool with enough capacity for the vTree size.
7. Optionally, expand **Advanced** to select one or several of the following advanced options:

Option	Description
Add migration to the head of the migration queue	Give this vTree migration the highest priority in the migration priority queue.
Ignore destination capacity	Allow the migration to start regardless of whether there is enough capacity at the destination, or not.
Enable compression	Compression is done by applying a compression-algorithm to the data.
Convert vTree from...	Convert a thin-provisioned vTree to thick-provisioned, or vice-versa, at the destination, depending on the provisioning of the source volume. **NOTE:** SDCs with a version earlier than PowerFlex 3.0 do not fully support converting a thick-provisioned vTree to a thin-provisioned vTree during migration; after migration, the vTree will be thin-provisioned, but the SDC will not be able to trim it. These volumes can be trimmed by unmapping and then remapping them, or by restarting the SDC. The SDC version will not affect capacity allocation, and a vTree converted from thick to thin provisioning will be reduced in size accordingly in the system.
Save current vTree provisioning state during migration	The provisioning state is returned to its original state before the migration took place.

8. Click **Migrate vTree**.

   The vTree migration is initiated. The vTree appears in both the source and the destination storage pools.
9. At the top right of the window, click the **Running Storage Jobs** icon and check the progress of the migration of the vTree.
10. Verify that the operation has finished and was successful, and click **Dismiss**.

Pause vTree migration

You can pause a vTree migration at any time.

The following methods can be used to pause vTree migration:

- Gracefully—allows all data blocks currently being migrated to finish migration before pausing.
- Forcefully—stops the migration of all blocks currently in progress.

CAUTION: The forceful method carries a potential risk of data loss.

1. On the menu bar, click **Block** > **Volumes**.
2. In the list of volumes, select the volume for which you want to pause migration.
3. In the right pane, click **View Details**.
4. Click the **VTree** tab.
5. Click **Migration** and select **Pause migration**.
6. In the **Pause VTree Migration** dialog box, select the required option:
   - Gracefully
   - Forcefully
7. Click **Pause Migration**.

   If you selected to pause the migration gracefully, migration status is displayed. Once paused, you can choose to roll back the vTree migration, or resume the migration using the **VTree** menu.
8. Verify that the operation has finished and was successful, and click **Dismiss**.

Resume a vTree migration

You can resume a vTree migration that was paused at any time.

1. On the menu bar, click **Block** > **Volumes**.
2. In the list of volumes, select the volume where migration was paused.
3. In the right pane, click **View Details**.
4. Click the **vTree** tab.
5. Click **Migration** and select **Resume migration**.
6. In the **Resume VTree Migration** dialog box, click **Resume Migration**.
7. Verify that the operation has finished and was successful, and click **Dismiss**.

Roll back vTree migration

When a vTree migration is paused, you can roll back the migration so that the volume and all of its snapshots are returned to the source storage pool.

1. On the menu bar, click **Block** > **Volumes**.
2. In the list of volumes, select the volume for which you want to roll back migration.
3. In the right pane, click **View Details**.
4. Click the **VTree** tab.
5. Click **Migration** and select **Roll back Migration**.
6. In the **Migrate vTree** dialog box, verify the source and target for the rollback, and click **Roll back Migration**.
7. Verify that the operation has finished and was successful, and click **Dismiss**.

The migration resumes in the reverse direction. Any data already migrated to the destination storage pool is now migrating back to the source storage pool.

Set vTree migration priority

Specify whether a vTree migration will be at the beginning or at the end of the migration queue.

NOTE: This feature is available only when there is more than one vTree migration currently in the queue.

1. On the menu bar, click **Block** > **Volumes**.
2. In the list of volumes, select the relevant volume.
3. In the right pane, click **View Details**.
4. Click the **VTree** tab.
5. Click **Migration**, select **Set Priority**.
6. In the **Set VTree Migration Policy** dialog box, select whether to move the current vTree migration to the head or to the tail of the migration queue, and click **Set Priority**.
7. Verify that the operation has finished and was successful, and click **Dismiss**.

Remove a vTree

You can remove a vTree from PowerFlex, as long as it is unmapped.

Ensure that the vTree is unmapped.

1. On the menu bar, click **Block** > **Volumes**.
2. In the list of volumes, select the volume that you want to remove.
3. In the right pane, click **View Details**.
4. Click the **VTree** tab.
5. Click the **Migration**, select **Remove**.
6. In the **Remove VTree** dialog box, click **Remove VTree**.
7. Verify that the operation has finished and was successful, and click **Dismiss**.

NVMe targets (mirroring)

NVMe targets (or SDT components) must be configured on the PowerFlex system side, in order to use NVMe over TCP technology.

The NVMe target (or SDT component) is a frontend component that translates NVMe over TCP protocol into internal PowerFlex protocols. The NVMe target provides I/O and discovery services to NVMe hosts configured on the PowerFlex system. A minimum of two NVMe targets must be assigned to a protection domain before it can serve NVMe hosts, to provide minimal path resiliency to hosts.

TCP ports, IP addresses, and IP address roles must be configured for each NVMe target (or SDT component). You can assign both storage and host roles to the same target IP addresses. Alternatively, assign the storage role to one target IP address, and add another target IP address for the host role. Both roles must be configured on each NVMe target.

- The host port listens for incoming connections from hosts, over the NVMe protocol.
- The storage port listens for connections from the MDM.

Once the NVMe targets have been configured, add hosts to PowerFlex, and then map volumes to the hosts. Connect hosts to NVMe targets, preferably using the discovery feature.

On the operating system of the compute nodes, NVMe initiators must be configured. Network connectivity is required between the NVMe targets and the NVMe initiators, and between NVMe targets (or SDT components) and SDSs.

Add an NVMe target (mirroring)

Add an NVMe target (or SDT component) to PowerFlex.

1. On the menu bar, click **Block** > **Storage System** > **NVMe Targets**.
2. Click **Add NVMe Target**.
3. In the **Add NVMe Target** dialog box, configure the following settings:
   a. For **Target Name**, enter a name for the NVMe target (or SDT component).

b. For **Protection Domain**, select a protection domain.
c. Accept the default TCP ports, or modify them if necessary.
   - **Storage Port** listens for incoming connections from the MDM, and is only open on an IP address configured with the storage role. Default=12200
   - **I/O Port** listens for incoming connections for I/O, and is only open on an IP address configured with the host role. Default=4420 (which conforms to the NVMe over TCP standard)
   - **Discovery Port** listens for incoming discovery connections, and is only open on an IP address configured with the host role. Default=8009 (which conforms to the NVMe over TCP standard. Use zero to disable the discovery service.)
d. For **Target IPs**, enter an IP address, and then select a role for the IP address from the **Target IPs Roles** menu. Click **Add Target IP**, and repeat this step until all IP addresses are configured.

4. Click **Add**.
5. Verify that the operation has finished and was successful, and click **Dismiss**.

Modify an NVMe target (mirroring)

Modify the configuration of an NVMe target (or SDT component).

1. On the menu bar, click **Block** > **Storage System** > **NVMe Targets**.
2. In the list of NVMe targets, select the required NVMe target, and click **Modify**.
3. In the **Modify NVMe Target** dialog box, modify the desired fields.
4. Optionally, add more target IP addresses, by clicking **Add Target IP**.
5. Click **Modify NVMe Target**.
6. Verify that the operation has finished and was successful, and click **Dismiss**.

Remove an NVMe target (mirroring)

Remove an NVMe target (or SDT component) from PowerFlex.

1. On the menu bar, click **Block** > **Storage System** > **NVMe Targets**.
2. In the list of NVMe targets, select the required NVMe target, and click **Remove**.
3. In the **Remove NVMe Target** dialog box, verify that you are removing the desired NVMe target, and click **Remove**.
4. Verify that the operation has finished and was successful, and click **Dismiss**.

Hosts

Hosts are entities that consume PowerFlex storage for application usage. There are two methods of consuming PowerFlex block storage: using the SDC kernel driver, or using NVMe over TCP connectivity. Therefore, a host is either an SDC or an NVMe host.

Once a host is configured, volumes may be mapped to it. In each case, hosts must be mapped to volumes.

Add an NVMe host (mirroring)

Add an NVMe host to PowerFlex.

Ensure the following:

- You know the NVMe Qualified Name (NQN) of the host. If you do not know the NQN, see the documentation for the host operating system.
- The host is connected to the Ethernet switch.
- The host is configured with the correct VLAN ID and routing rules.

1. On the menu bar, click **Block** > **Hosts**.
2. Click **Add Host**.
3. In the **Add NVMe Host** dialog box, enter a hostname in the **Host Name** field.
4. In the **Host NQN** field, enter the NQN string for the host.

5. In the **Number of Paths Per Volume** field, enter the maximum number of paths to be provided between the host and each volume (default is 4).
6. In the **Number of System Ports per Protection Domain** field, enter the maximum number of ports to be provided between the host and each protection domain (default is 10).
7. Click **Add**.
8. Verify that the operation has finished and was successful, and click **Dismiss**.
9. Configure the host operating system to connect to the cluster's NVMe discovery service on one or more of the NVMe targets.

The CLI for PowerFlex also provides options for adding a host group:

```
sio@localhost [/home/sio] $ scli --add_nvme_host --help

Usage: scli --add_nvme_host --nvme_host_nqn <NQN> [--host_name <NAME>] [--
max_number_paths <VALUE>] [--max_number_sys_ports <VALUE>] [--host_os_full_type <OS>] [--
host_group_name <ID> | --host_group_id <NAME>] | --force
Description: Add a NVMe Host to the system
Parameters:
    --nvme_host_nqn <NQN>               NVMe Host NQN
    --host_name <NAME>                  Host name
    --max_number_paths <VALUE>          Maximal number of paths allowed per mapped volume
                                        Valid range: 2-8. Default: 4
    --max_number_sys_ports <VALUE>      Maximal number of System Ports allowed per
Protection Domain
                                        Valid range: 2-128. Default: 10
    --host_os_full_type <OS>            NVMe Host OS type
                                        Options: generic (default) or powerflex
    --host_group_id <ID>                Host Group ID
    --host_group_name <NAME>            Host Group name
    --force                             Force NQN, ignore all NQN formating rules.
```

Map volumes to hosts (mirroring)

Volumes can only be mapped to one type of host: either SDC or NVMe. Ensure that you know which type of hosts are being used for each volume, to avoid mixing host types.

1. On the menu bar, click **Block** > **Hosts**.
2. In the list of hosts, select the relevant host, and click **Mapping** > **Map**.
3. In the **Map Hosts to Volumes** dialog box, select the volumes to be mapped to the selected hosts, and click **Map**.
4. Verify that the operation has finished and was successful, and click **Dismiss**.

Unmap volumes from hosts (mirroring)

Remove mapping between volumes and hosts.

1. On the menu bar, click **Block** > **Hosts**.
2. In the list of hosts, select the relevant host, and click **Mapping** > **Unmap**.
3. In the **Unmap** dialog box, ensure that the correct host is selected.
4. Click **Unmap**.
5. Verify that the operation has finished and was successful, and click **Dismiss**.

Remove hosts (mirroring)

Remove hosts from PowerFlex.

1. On the menu bar, click **Block** > **Hosts**.
2. In the list of hosts, select the relevant host, and click **Remove**.
3. In the **Remove Host** dialog box, ensure that you have selected the correct host for removal.

NOTE: You can only remove hosts with unmapped volumes. For SDCs, the host must be disconnected from the PowerFlex cluster before it can be removed.

4. Click **Remove**.
5. Verify that the operation has finished and was successful, and click **Dismiss**.

Configure or modify approved host IP addresses (mirroring)

When the restricted host (SDC) mode is set to `approved IP restriction`, configure the host IP addresses before mapping volumes to the hosts.

Ensure that the hosts are approved by GUID.

1. On the menu bar, click **Block** > **Hosts**.
2. In the list of hosts, select the relevant host, and click **Modify** > **Modify Approved IP Addresses**.
3. In the **Modify Approved IP addresses** dialog box, enter the IP address of the hosts, and click **Add IP Address**. Repeat this step for additional addresses.

   You can add up to a total of four IP addresses.
4. Click **Apply**.
5. Verify that the operation has finished and was successful, and click **Dismiss**.

Approve SDCs (mirroring)

When the restricted host (SDC) mode is set to GUID restriction, approve the SDCs before mapping the SDCs to volumes.

For more information about the restricted host feature, see the *PowerFlex and PowerFlex Manager Technical Overview* or *Dell PowerFlex Family 5.x Security Configuration Guide*.

1. On the menu bar, click **Block** > **Hosts**.
2. In the list of hosts, select one or more hosts and click **Modify** > **Approve**.
3. In the **Approve host** dialog box, verify that the hosts listed are the ones that you want to approve, and click **Approve**.
4. Verify that the operation has finished and was successful, and click **Dismiss**.

Rename hosts (mirroring)

You can rename a host following the specific naming rules.

The hostname must adhere to the following rules:

- Contains less than 32 characters
- Contains only alphanumeric and punctuation characters
- Is unique within the object type

1. On the menu bar, click **Block** > **Hosts**.
2. From the list of hosts, select the relevant host, and click **Modify** > **Rename**.
3. In the **Rename Host** dialog box, enter the new name, and click **Apply**.
4. Verify that the operation is finished and is successful, and click **Dismiss**.

Modify an SDC performance profile

SDC performance profiles are set by default to high and can be changed to compact. The compact setting may impact the system performance.

The default setting configures a predefined set of parameters for high-performance use cases.

NOTE: Performance tuning is case-specific. To prevent undesirable effects, Dell Technologies highly recommends that you thoroughly test all changes. For further assistance, contact Dell Technologies Support.

1. On the menu bar, click **Block** > **Hosts**.
2. From the list of hosts, select the relevant host, and click **Modify** > **Modify Performance Profile**.

3. In the **Modify Performance Profile Host** dialog box, select the wanted performance profile: **High** or **Compact**.
4. Click **Apply**.
5. Verify that the operation is finished and is successful, and click **Dismiss**.

13

Protecting a block storage environment (erasure coding)

This section explains how to use local features to protect your block storage.

Snapshot policies

In addition to being created manually, snapshots can be created automatically by the system according to a defined interval. Within PowerFlex, snapshot policies are used to achieve automatic data protection on storage resources (volumes and clones).

A snapshots policy is a group of user-defined rules that are used to establish local data protection on an assigned storage resource. In PowerFlex, you can assign a snapshot policy to a storage resource which defines the level of protection. Protection policies are also created at the cluster level, and not per individual protection domain. This means that any storage resource on any protection domain in the system can leverage a snapshot policy once it is created.

You can define a rule. For example:

Rule: Take snapshots every 60 minutes.

Retention Rules:

- 24 snapshots
- Seven snapshots
- Four snapshots

After defining the rules, select the storage resource to assign to the snapshot policy. You can assign multiple source volumes to a snapshot policy, but only a single policy per storage resource is allowed. Only one volume per vTree may be assigned to a policy (any policy).

Create a snapshot policy

Create snapshot policies with a maximum of six retention rules that each specify a retention period and number of snapshots to retain, and then associate storage resources to the policy.

1. On the menu bar, click **Protection** > **Snapshot Policies**.
2. Click **Create Snapshot Policy**.
3. In the **Create Snapshot Policy** dialog box, assign the policy a name and the time interval between snapshots.
4. Create retention rules:
   a. In the **Retention** field, add the number of snapshots to retain.
   A retention period is created for the specified number of snapshots times the snapshot interval.
   b. Optionally, to create an additional retention rule, click **Add Retention**, and repeat the previous step. Each rule is based on the specified number of snapshots times the previous retention period.

   Up to six retention rules can be created.
5. Optionally, select the **Secured** checkbox to prevent the snapshots that are created by the policy from being modified or deleted before the end of the expiration date.
6. Click **Create** or **Create and Activate** to add the snapshot policy to PowerFlex.

   If created, but not activated, the snapshot policy shall be paused.
7. In the **Snapshot Policy was successfully added** dialog box, click **Assign volume to snapshot policy**.
8. In the **Select volumes you wish to assign to the policy** window, select one or more volumes, and click **Assign Policy**.
9. Verify that the operation has finished successfully, and click **Dismiss**.

Remove snapshot policy

Remove a snapshot policy to stop PowerFlex from automatically creating snapshots for the specified volumes. You cannot remove snapshot policies that are secured.

1. On the menu bar, click **Protection** > **Snapshot Policies**.
2. In the list of snapshot policies, select the relevant policy.
3. Ensure that all storage resources are unassigned from the selected snapshot policy, and click **More Actions** > **Delete**.
4. In the **Remove Snapshot Policy** dialog box, click **Remove**.
5. Verify that the operation has finished successfully, and click **Dismiss**.

Modify snapshot policy

PowerFlex allows you to modify the parameters of snapshot policies, when required.

1. On the menu bar, click **Protection** > **Snapshot Policies**.
2. In the list of snapshot policies, select the relevant policy, and then click **Modify** > **Modify Policy**.
3. In the **Modify Snapshot Policy** dialog box, change the desired parameters:
   a. In the **Retention** field, set the number of snapshots to retain.
   A retention period is created for the specified number of snapshots times the snapshot interval.
   b. Optionally, to create an additional retention rule, click **Add Retention** and repeat the previous step. Each rule is based on the specified number of snapshots times the previous retention period.
   Up to six retention rules can be created.
   c. Optionally, select the **Secured** checkbox to prevent the snapshots that are created by the policy from being modified or deleted before the end of the expiration date.
4. Click **Modify**.
5. Verify that the operation has finished successfully, and click **Dismiss**.

Rename snapshot policy

1. On the menu bar, click **Protection** > **Snapshot Policies**.
2. In the list of snapshot policies, select the relevant policy, and click **Modify** > **Rename**.
3. In the **Rename Snapshot Policy** dialog box, enter a new name and click **Apply**.
4. Verify that the operation has finished successfully, and click **Dismiss**.

Activate snapshot policy

1. On the menu bar, click **Protection** > **Snapshot Policies**.
2. In the list of snapshot policies, select the relevant policy, and click **Protect** > **Activate**.
3. In the **Activate Snapshot Policy** dialog box, click **Activate**.
4. Verify in the **State** column that the snapshot policy is active, and click **Dismiss**.

Pause snapshot policy

1. On the menu bar, click **Protection** > **Snapshot Policies**.
2. In the list of snapshot policies, select the relevant policy, and click **Protect** > **Pause**.
3. In the **Pause Snapshot Policy** dialog box, click **Pause**.
4. Verify in the **State** column that the snapshot policy is in paused state, and click **Dismiss**.

Assign volumes to a snapshot policy

1. On the menu bar, click **Protection** > **Snapshot Policies**.
2. In the list of snapshot policies, select the relevant policy, and click **Protect** > **Assign Volumes**.
3. In the **Assign Volumes to Snapshot Policy** dialog box, select the volumes that you want to assign to the policy, and click **Assign Policy**.
4. Verify that the operation has finished successfully, and click **Dismiss**.

Unassign a volume from a snapshot policy

When you unassign volumes from a snapshot policy, the storage resource is detached from the snapshot policy.

1. On the menu bar, click **Protection** > **Snapshot Policies**.
2. In the list of snapshot policies, select the relevant policy, and click **Protect** > **Unassign Volumes**.
3. In the **Unassign Volumes from Snapshot Policy** dialog box:
   a. To unassign the snapshot policy from a volume, select the checkbox of the relevant volume.
   b. Optionally, to delete the relevant snapshots of the selected volume, select the **Delete related snapshots** checkbox.
4. Click **Unassign Policy**.
5. Verify that the operation finished successfully, and click **Dismiss**.

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Protecting a block storage environment (mirroring)

This section explains how to use local features to protect your block storage.

Snapshot policies (mirroring)

Snapshot policies enable you to define policies for the number of snapshots that PowerFlex takes of one or more defined volumes at a given time.

Snapshots are taken according to the defined rules. You can define the time interval between two rounds of snapshots, and the number of snapshots to retain, in a multi-level structure. For example, take snapshots every x minutes/hours/days/weeks. You can define a maximum of six levels, with the first level having the most frequent snapshots.

For example:

Rule: Take snapshots every 60 minutes

Retention Levels:

- 24 snapshots
- 7 snapshots
- 4 snapshots

After defining the parameters, select the source volume to add to the snapshot policy. You can add multiple source volumes to a snapshot policy, but only a single policy per source volume is allowed. Only one volume per vTree may be used as a source volume of a policy (any policy).

When you remove the source volume from the policy, you must choose how to handle snapshots. Snapshots created by the policy are referred to as auto snapshots. This is indicated if a snapshot policy is displayed for the snapshot.

- If the source volume has auto snapshots, you cannot unassign the source volume from the snapshot policy. You can remove auto snapshots from **Snapshots**.
- If the source volume has auto snapshots but none of them are locked, you are prompted to confirm that you would like to delete all auto snapshots. If any of the auto snapshots are locked, the locked auto snapshots are just detached from the snapshot policy, but not deleted. You can manually delete auto snapshots. It doesn't matter if the auto snapshot is locked or not, you can still delete.

Create a snapshot policy

Create snapshot policies with a maximum of six retention rules that each specify a retention period and number of snapshots to retain, and then associate storage resources to the policy.

1. On the menu bar, click **Protection** > **Snapshot Policies**.
2. Click **Create Snapshot Policy**.
3. In the **Create Snapshot Policy** dialog box, assign the policy a name and the time interval between snapshots.
4. Create retention rules:
   a. In the **Retention** field, add the number of snapshots to retain.
   A retention period is created for the specified number of snapshots times the snapshot interval.
   b. Optionally, to create an additional retention rule, click **Add Retention**, and repeat the previous step. Each rule is based on the specified number of snapshots times the previous retention period.

   Up to six retention rules can be created.
5. Optionally, select the **Secured** checkbox to prevent the snapshots that are created by the policy from being modified or deleted before the end of the expiration date.
6. Click **Create** or **Create and Activate** to add the snapshot policy to PowerFlex.

   If created, but not activated, the snapshot policy shall be paused.

7. In the **Snapshot Policy was successfully added** dialog box, click **Assign volume to snapshot policy**.
8. In the **Select volumes you wish to assign to the policy** window, select one or more volumes, and click **Assign Policy**.
9. Verify that the operation has finished successfully, and click **Dismiss**.

Remove snapshot policy

Remove a snapshot policy to stop PowerFlex from automatically creating snapshots for the specified volumes. You cannot remove snapshot policies that are secured.

1. On the menu bar, click **Protection** > **Snapshot Policies**.
2. In the list of snapshot policies, select the relevant policy.
3. Ensure that all storage resources are unassigned from the selected snapshot policy, and click **More Actions** > **Delete**.
4. In the **Remove Snapshot Policy** dialog box, click **Remove**.
5. Verify that the operation has finished successfully, and click **Dismiss**.

Modify snapshot policy

PowerFlex allows you to modify the parameters of snapshot policies, when required.

1. On the menu bar, click **Protection** > **Snapshot Policies**.
2. In the list of snapshot policies, select the relevant policy, and then click **Modify** > **Modify Policy**.
3. In the **Modify Snapshot Policy** dialog box, change the desired parameters:
   a. In the **Retention** field, set the number of snapshots to retain.
   A retention period is created for the specified number of snapshots times the snapshot interval.
   b. Optionally, to create an additional retention rule, click **Add Retention** and repeat the previous step. Each rule is based on the specified number of snapshots times the previous retention period.

      Up to six retention rules can be created.
   c. Optionally, select the **Secured** checkbox to prevent the snapshots that are created by the policy from being modified or deleted before the end of the expiration date.
4. Click **Modify**.
5. Verify that the operation has finished successfully, and click **Dismiss**.

Rename snapshot policy

1. On the menu bar, click **Protection** > **Snapshot Policies**.
2. In the list of snapshot policies, select the relevant policy, and click **Modify** > **Rename**.
3. In the **Rename Snapshot Policy** dialog box, enter a new name and click **Apply**.
4. Verify that the operation has finished successfully, and click **Dismiss**.

Secure snapshot policy

Perform the following task to secure snapshots in snapshot policies.

1. On the menu bar, click **Protection** > **Snapshot Policies**.
2. In the list of snapshot policies, select the relevant policy, and click **Modify** > **Secure Policy**.
3. In the **Modify snapshot access** dialog box, optionally click the checkbox **Apply to existing snapshots (including locked snapshots)** to secure all existing snapshots.
4. Click **Approve**.
5. Securing snapshots is irreversible. Click **Approve** to confirm that you want to secure snapshots for the selected snapshot policy.
6. Verify that the operation has finished successfully, and click **Dismiss**.

Activate snapshot policy

1. On the menu bar, click **Protection** > **Snapshot Policies**.
2. In the list of snapshot policies, select the relevant policy, and click **Protect** > **Activate**.
3. In the **Activate Snapshot Policy** dialog box, click **Activate**.
4. Verify in the **State** column that the snapshot policy is active, and click **Dismiss**.

Pause snapshot policy

1. On the menu bar, click **Protection** > **Snapshot Policies**.
2. In the list of snapshot policies, select the relevant policy, and click **Protect** > **Pause**.
3. In the **Pause Snapshot Policy** dialog box, click **Pause**.
4. Verify in the **State** column that the snapshot policy is in paused state, and click **Dismiss**.

Assign volumes to a snapshot policy

1. On the menu bar, click **Protection** > **Snapshot Policies**.
2. In the list of snapshot policies, select the relevant policy, and click **Protect** > **Assign Volumes**.
3. In the **Assign Volumes to Snapshot Policy** dialog box, select the volumes that you want to assign to the policy, and click **Assign Policy**.
4. Verify that the operation has finished successfully, and click **Dismiss**.

Unassign a volume from a snapshot policy

When you unassign volumes from a snapshot policy, the storage resource is detached from the snapshot policy.

1. On the menu bar, click **Protection** > **Snapshot Policies**.
2. In the list of snapshot policies, select the relevant policy, and click **Protect** > **Unassign Volumes**.
3. In the **Unassign Volumes from Snapshot Policy** dialog box:
   a. To unassign the snapshot policy from a volume, select the checkbox of the relevant volume.
   b. Optionally, to delete the relevant snapshots of the selected volume, select the **Delete related snapshots** checkbox.
4. Click **Unassign Policy**.
5. Verify that the operation finished successfully, and click **Dismiss**.

Remote protection

PowerFlex enables native asynchronous replication for PowerFlex storage-only and hyperconverged configurations. Remote protection is an optional feature which ensures data protection of the PowerFlex environment. It creates a remote copy of one volume from one cluster to another.

Replication is used to quickly and easily recover from a physical or logical disaster, to migrate data, to test data at a remote site, or to offload backup. The PowerFlex implementation is designed to allow a sub-minute RPO reducing the data-loss to minimal if there is disaster recovery. As with all other PowerFlex data services, replication is elastic, can scale online by adding or removing nodes, is flexible, and easy to manage. It enables instant test and failover operations.

When replicating between systems it is recommended to have the same storage pool type, disk capacity and type, and performance capabilities between the local and remote sites. While replication between medium granularity (MG) to fine granularity (FG) storage pools is fully supported, account for the performance characteristics of each storage pool. PowerFlex does not support replication of volumes mapped to NVMe hosts.

The following workflow summarizes setting up remote replication. See specific procedures for detailed steps.

1. Install the remote (target) PowerFlex system.
2. Extract a certificate on each PowerFlex system and copy it to its peer system.

3. Configure journal capacity. Journal capacity is a percentage of the total storage capacity. There are several factors that should be considered when defining it:
   - Journal capacity must be allocated to be able to add SDRs to the system.
   - Before configuring journal capacity, ensure that there is enough space in the storage pool.
   - The journal capacity depends on the change rate and Recovery Point Objective (RPO). Data writes are accumulated in the journal until half the RPO time has been reached, to ensure that data is not lost and a consistent copy is maintained between the volumes.
   - When the total storage capacity in the system increases, a small percentage is needed for the journal capacity.
   - As application workload increases, more journal capacity must be added, accordingly.
4. Ensure that SDRs are installed and added to the source and peer systems.
   - The SDR component is typically installed during installation of PowerFlex. It manages replication activity within the system, and between the source and target. Dell Technologies recommends that you install at least three SDRs per system, for failure or backup purposes. Within the system, the SDR channels the I/O writes from the host to the source journal. Simultaneously, it also sends the data to the SDS to be logged. As data accumulates in the source journal, the system decides when to close the journal (this is usually at half the RPO time) in order to be ready to transfer the data to the target journal. At the target site, the SDR is also responsible for applying the data from the target journal to the target volume.
5. Define peer (target) systems from the source side.
6. Create and activate replication consistency groups (RCGs).
   - An RCG is a set of volumes which must maintain a consistent copy on the remote (target) site where write order is maintained between volumes. This pair shares the same attributes.
   - The volumes to be replicated must be the same size on source and target systems. If the network is up, the systems should be connected.

Extract and upload certificates

On both source and target replication systems, extract the root certificate and copy it to the peer system. This procedure is required to allow communication and data transfer between the systems.

Ensure that you have admin user credentials for command line access to both PowerFlex systems.

1. In the command line, log in to the source system:

   ```
   scli --login --username <M&O UI user> --password <M&O UI password> --
   management_system_ip <M&O UI IP>
   ```

2. Extract the root certificate:

   ```
   scli --extract_root_ca --certificate_file <FILE_PATH>
   ```

   where *<FILE_PATH>* is the location where the certificate will be saved, and the file name. For example: `/opt/source_sys.crt`
3. Copy the certificate file to the target system.
4. On the target system, perform steps 1 and 2.
5. Copy the target system's certificate file to the source system.
6. On the source system, add the certificate for the target system:

   ```
   scli --add_trusted_ca --certificate_file <PATH_TO_LOCAL_COPY_OF_TARGET_CERT>
   ```

   where *<PATH_TO_LOCAL_COPY_OF_TARGET_CERT>* is the copy of the target system's certificate that you copied to the source system.
7. On the target system, add the source system's certificate:

   ```
   scli --add_trusted_ca --certificate_file <PATH_TO_LOCAL_COPY_OF_SOURCE_CERT>
   ```

   where *<PATH_TO_LOCAL_COPY_OF_SOURCE_CERT>* is the copy of the source system's certificate that you copied to the target system.

Journal capacity

You should consider several factors when allocating journal capacity.

Journal capacity is defined as a percentage of the total storage capacity in the storage pool and must equal at least 28 GB per SDR. In general, journal capacity should be at least 5% of replicated usable capacity in the protection domain, including volumes used as source and targets. It is important to assign enough storage capacity for the replication journal.

The amount of capacity needed for the journal is based on the following factors:

- Minimal requirements—28 GB multiplied by the number of SDR sessions. The number of SDR sessions is equal to the number of SDRs plus one. The extra SDR session is to ensure that a new session can be allocated for an SDR during a system upgrade.
- The capacity needed to sustain an outage—application WAN bandwidth multiplied by the planned WAN outage. In general, journal capacity in the protection domain should be at least 5% of the application pool. If the application has a heavy I/O load, larger capacity should be used. Similarly, if a long outage is expected, a larger capacity should be allocated. If there are replicated volumes in more than one storage pool in the protection domain, this calculation should be repeated for each storage pool, and the allocated journal capacity in the protection domain must at least equal the sum of the size per application pool.

Use the following steps to calculate exactly how much journal capacity to allocate:

1. Select the storage pools from which to allocate the journal capacity. The journal is shared between all of the replicated RCGs in the protection domain. Journal capacity should be allocated from storage pools as fast as (or faster than) the storage pool of the fastest replicated application in the protection domain. It should use the same drive technology and about the same drive count and distribution in nodes.
2. Consider the minimal requirements needed (28 GB multiplied by the number of SDR sessions) and the capacity needed to sustain an outage. Journal capacity will be at least the maximum of these two factors.
3. Take into account the expected outage time. The minimal outage allowance is one hour, but at least three hours are recommended.
4. Calculate the journal capacity needed per application: maximal application throughput x maximum outage interval.
5. Since journal capacity is defined as a percentage of storage pool capacity, calculate the percentage of capacity based on the previously calculated needs.

For example:

- An application generates 1 GB/s of writes.
- The maximal supported outage is 3 hours (3 hours x 3600 seconds = 10800 seconds).
- The journal capacity needed for this application is 1 GB/s x 10800 s = ~10.547 TB.
- Since the journal capacity is expressed as a percentage of the storage pool capacity, divide the 10.547 TB by the size of the storage pool, which is 200 TB: 100 x 10.547 TB/200 TB = 5.27%. Round this up to 6%.
- Repeat this for each application being replicated.

When a protection domain has several storage pools and several replicated applications, the journal capacity should be calculated as in the example above, and the capacity can be divided among all the storage pools (provided they are fast enough). For higher availability, the journal capacity should be allocated from multiple storage pools.

NOTE: When storage pool capacity is critical, capacity cannot be allocated for new volumes or for expanding existing volumes. This behavior must be taken into account when planning the capacity available for journal usage. The volume usage must leave enough capacity available in the storage pool to allow provisioning of journal volumes. The plan should account for the storage pool staying below critical capacity even when the journal capacity is almost fully utilized.

It is important to note that since journal capacity is defined as a percentage of the total storage capacity in the storage pool, increasing the total storage capacity by adding devices will increase the journal capacity. Similarly, if you decrease the total storage capacity by removing devices from the storage pool, the journal capacity will automatically decrease.

Create journal capacity

Set the percentage of storage pool capacity that is allocated for journal capacity.

1. On the menu bar, click **Protection** > **Journal Capacity**.
2. Click **Create Journal Capacity**.
3. In the **Create Journal Capacity** dialog box, search for, and select the required storage pool.
4. Enter a percentage for the journal capacity.
5. Click **Create**.

6. Verify that the operation has finished and was successful, and click **Dismiss**.

Modify journal capacity

Reset the journal capacity according to the updated storage device size.

1. On the menu bar, click **Protection** > **Journal Capacity**.
2. In the list of storage pools, select the required storage pool and click **Modify**.
3. In the **Storage Pool** dialog box, enter the updated percentage for journal capacity.
4. Click **Modify** to update the journal capacity.
5. Verify that the operation has finished and was successful, and click **Dismiss**.

Delete journal capacity

Delete journal capacity from a storage pool.

1. On the menu bar, click **Protection** > **Journal Capacity**.
2. Select the required storage pool from the list, and click **Delete**.
3. In the **Delete Journal Capacity** dialog box, click **Delete**.
4. Verify that the operation has finished and was successful, and click **Dismiss**.

Add a peer system

On the source system, add the connection information of the target system (remote site) to enable replication between the peer systems.

Ensure the following are complete.

- PowerFlex storage nodes or PowerFlex hyperconverged nodes are deployed with a replication resource group at each site
- Certificates are generated and copied to each participating system

1. Log in to PowerFlex Manager.
2. Select **Protection** > **Peer Systems**.
3. Click **+ Add Peer System**.
4. Enter the following:
   - Peer system name
   - ID
   - PowerFlex management IP addresses (primary MDM, secondary MDM, and standby MDM - if in use)
5. Click **Add IP for each additional replication IP in the target Replication Group** and click **Add**.
6. Repeat the steps for the other peer system.

   NOTE: The connection state is **Not Authorized** until the peer system is also configured.

   After a few moments, the target system displays the state as **Connected**.

Restart the replication cluster

This task describes how to restart the nodes in the replication cluster.

Inactivate the protection domain first on the source and then on the target system.

In case of 2-way replication, inactivate the protection domains in any order. Ignore error messages that appear in the active protection domain. To avoid lengthy synchronization times or running out of journal capacity, it is advisable to activate the replication cluster quickly.

1. Shut down all nodes within the replication cluster.
2. Power on all nodes within the replication cluster.
3. Activate the protection domain first on the target and then on the source system.
4. Validate that the RCG in the replication cluster returns to a working state.

SDRs

Storage data replicators (SDRs) are responsible for processing all I/Os of replication volumes.

All application I/Os of replicated volumes are processed by the source SDRs. At the source, application I/Os are sent by the SDC to the SDR. The I/Os are sent to the target SDRs and stored in their journals. The target SDRs' journals apply the I/Os to the target volumes. A minimum of two SDRs are deployed at both the source and target systems to maintain high availability. If one SDR fails, the MDM directs the SDC to send the I/Os to an available SDR.

Add SDR

Add an additional SDR to an existing PowerFlex system, or add back a new SDR if a previously existing SDR was removed. A minimum of two SDRs are required on each replication system. Each SDR must be configured with one or more IP addresses and roles.

The SDR communicates with several components, including: SDC (application), SDS (storage) and remote SDR (external). When an IP address is added to an SDR, the role or roles of the IP address must be defined. The IP address role determines the component with which that IP address communicates. For example, the application role means that the associated IP address is used for SDR-SDC communication. By default, all the roles are selected for an IP address.

SDR components must be deployed as resources before you can add them using this procedure.

1. On the menu bar, click **Protection** > **SDRs**.
2. Click **Add SDR**.
3. In the **Add SDR** dialog box, enter the connection information of the SDR:
   a. Enter the SDR name.
   b. If necessary, modify the SDR port number.
   c. Select the relevant protection domain.
   d. Enter the IP address of the SDR.
   e. Select one or more roles, for example, default: all roles are selected.
   f. If the SDR has more than one IP address, click **Add IP** to add more IP addresses and their roles.
   g. Click **Add SDR** to initiate a connection with the peer system.
4. Verify that the operation has finished and was successful, and click **Dismiss**.

Modify IP address role

Update the role of an IP address configured on the SDR.

The SDR communicates with several components, including: SDC (application), SDS (storage) and remote SDR (external). When an IP address is added to an SDR, the role or roles of the IP address must be defined. The IP address role determines the component with which that IP address communicates. For example, the application role means that the associated IP address is used for SDR-SDC communication. The default setting is all roles.

1. On the menu bar, click **Protection** > **SDRs**.
2. In the list of SDRs, select the relevant SDR, and click **Modify** > **Modify IP Role**
3. In the **Modify IPs Role** dialog box, select or clear the desired roles for the IP address.
4. Click **Apply**.
5. Verify that the operation has finished and was successful, and click **Dismiss**.

Modify an SDR performance profile

The SDR performance profile is set to high by default, but can be modified. Ensure that your system's architecture is equipped to handle enhanced system performance of SDR activities.

1. On the menu bar, click **Protection** > **SDRs**.
2. In the list of SDRs, select the relevant SDR, and click **Modify** > **Modify Performance Profile**.
3. In the **Modify Performance Profile SDR** dialog box, click either **High** or **Compact**.
4. Click **Apply**.
5. Verify that the operation has finished and was successful, and click **Dismiss**.

Enter SDR into maintenance mode

Place the SDR in maintenance mode prior to performing maintenance on the SDR.

There must be at least two SDRs installed on each system.

When maintenance mode is activated, the MDM changes the mapping and distributes the I/Os to the remaining SDRs.

1. On the menu bar, click **Protection** > **SDRs**.
2. In the list of SDRs, select the desired SDR, and click **More Actions** > **Enter Maintenance Mode**.
3. In the **Enter SDR to Maintenance Mode** dialog box, click **Enter Maintenance Mode**.
4. Verify that the operation has finished and was successful, and click **Dismiss**.

Exit SDR from maintenance mode

Remove the SDR from maintenance mode. This returns the SDR to regular operation.

When maintenance mode is deactivated, the MDM changes the mapping and resumes distribution of the I/Os to the SDR.

1. On the menu bar, click **Protection** > **SDRs**.
2. In the list of SDRs, select the relevant SDR checkbox, and click **More Actions** > **Exit Maintenance Mode**.
3. In the **Exit SDR from Maintenance Mode** dialog box, click **Exit Maintenance Mode**.
4. Verify that the operation has finished and was successful, and click **Dismiss**.

Replication consistency group

Replication consistency group (RCG) is an entity that includes a set of consistent volume pairs. The volume on the source from a single protection domain is replicated to a remote volume from a single protection domain on the target. This creates a consistent pair of volumes.

When replication is first activated for an RCG, the target volumes need to be synchronized with the source volumes. For each volume pair, the entire contents of each source volume are copied to the corresponding target volume. When there is more than one volume pair in the RCG, the order in which the volumes are synchronized is determined by the order in which the volume pairs were created. The initial synchronization occurs while all applications are running and performing I/O. Any writes to an area of the volume that has already been synchronized will be sent to the journal. Writes to an area of the volume that has not already been synchronized will be ignored, as the updated content will be copied over eventually as part of the synchronization.

The initial synchronization can also take place while the system is offline, however the application I/O must first be paused.

You can add and manage RCGs on both the source and target systems.

Replication direction and mapping

Use this table as a reference for replication direction and default access to volumes according to a subsequent RCG operation and action of the PowerFlex system.

The replication involves two peers, system A and system B. When the replication is set up, system A is set as source and system B is set as target. The following replication direction refers to the initial direction A->B and to changes to that direction.

Table 37. Replication direction and mapping

Subsequent RCG operations	Possible actions	Replication direction/access	Access to volumes
Normal	Switchover/test failover/ failover Remove	A to B	Access to the volumes is allowed only through the source (system A)
After failover	Reverse/restore Remove	N/A - data is not replicated	By default access to the volume is allowed through the original target (system B).

Table 37. Replication direction and mapping (continued)

Subsequent RCG operations	Possible actions	Replication direction/access	Access to volumes
			It is possible to enable access through the original source (system A).
After failover + reverse **NOTE:** Switchover and test failover are only possible after the peers are synchronized.	Switchover/test failover/ failover Remove	B to A	Access to the volumes is allowed only through the original target (system B)
After failover + restore **NOTE:** Switchover and test failover are only possible after the peers are synchronized.	Switchover/test failover/ failover Remove	A to B	Access to the volumes is allowed only through the source (system A)
After switchover	Switchover/test failover stop/failover Remove	B to A	Access to the volumes is allowed only through the original target (system B)
After test failover	Switchover/test failover/ failover Remove	A to B	Access to the volumes is allowed through both systems (system A and system B)

Create a replication consistency group

Create a replication consistency group (RCG) and add it to the system. Replication occurs between volumes, and RCGs maintain consistency between volume pairs in an RCG.

- Volumes in an RCG pair must be exactly the same size.
- Protection domains must be configured on both source and target systems.

The recovery point objective (RPO) configured in RCGs defines the maximum amount of time during which data is lost. Dell Technologies recommends setting a low RPO to ensure that minimal data is lost during a possible compromise of data transfer from source to target. If for example, if one minute is set as the RPO, you will not lose more than 30 seconds of data. There is no replication unless the source volume is consistent with the data from the target volume.

1. On the menu bar, click **Protection** > **RCGs**.
2. Click **Add RCG**.
3. In the **Add RCG** wizard, enter the information for the RCG:
   a. On the **Properties** page:
      - Enter the **RCG Name**.
      - Enter the number of **RPO** (recovery point objective) seconds or minutes. This is the amount of time of data loss is tolerated if replication between the systems is compromised.

        NOTE: It is recommended to enter the minimum amount of time the feature allows, which is 15 seconds.

      - Select the **Source Protection Domain**.
      - Select the **Target System**.
      - Select the **Target Protection Domain**.
      - Click **Next** .
   b. On the **Provisioning Type** page:
      - If you have not defined volumes to receive replicated data in the target system, select **Auto Provisioning**.
      - If you have already defined volumes to receive replicated data in the target system, select **Manual Provisioning**.

c. For auto provisioning only, on the **Replication Pairs** page, perform the following:
   - Click the desired volume in the **Source** column.
   - Select a type: **Thick** or **Thin**.
   - Select a storage pool from the target system.
   - Click **Add Pair**. The volume pair is added.
   - Click **Next**.
   - On the **Map Volumes** page, select a volume on the target side.
   - Select a host to which to map the volume.
   - Click **Next**, and go to step 3(e).

d. For manual provisioning only, on the **Replication Pairs** page, perform the following:
   - Select the desired volume in the **Source** column, and then in the **Target** column, select the target volume (only volumes of the same size are displayed).
   - Click **Add Pair**. The volume pair is added.
   - Click **Next**.

e. On the **Summary** page, ensure that the correct source and volume pair are defined, and then click **Create and Activate** or **Create**.

   The **Create and Activate** option creates the RCG and starts replication. The **Create** option creates the RCG but does not activate it. You can activate it later from **Protection** > **RCGs** > **More Actions** > **Activate**.

4. Optionally, at the top right of the window, click the **Running Jobs** icon, and check the progress of the initial copy state.
5. Verify that the operation has finished and was successful, and click **Dismiss**.

Map RCG target volumes to hosts

Designate which hosts can access the RCG from the target volumes.

This option is only enabled from the target RCG.

1. On the menu bar, click **Protection** > **RCGs**.
2. In the **Map RCG Target Volume** dialog box, click the relevant RCG checkbox, and click **Mapping** > **Map**.
3. Select one or more volumes on the left and click **>>** to add them to the selection list on the right.

   Optionally, filter the list of the volumes by entering a search string in the **Filter** box.
4. Click **Next**.
5. Select the hosts that you want to map to the selected volumes.

   Optionally, filter the list of the hosts by entering a search string in the **Filter** box.
6. Click **Next**.
7. Assign the **Access Mode** to the hosts that will be mapped to the target volumes.
   a. To assign the Read-Only mode to a set of mappings, select the mappings and click **Read Access** at the top of the page.
   b. To assign the No Access mode to a set of mappings, select the mappings and click **No Access** at the top of the page.

   NOTE: **Read Access** mode applies to all platforms, except Windows clusters, which require the **No Access** mode.
8. Click **Finish**.

Unmap a host from RCG target volumes

Unmap a host from RCG target volumes.

This option is only enabled from the target RCG.

1. On the menu bar, click **Protection** > **RCGs**.
2. Click the relevant RCG checkbox, and click **Mapping** > **Unmap**.
3. In the **Unmap RCG** dialog box, select the relevant host checkboxes to unmap them from the RCG volumes, and click **Unmap**.

Set the target to inconsistent mode

Set the target to inconsistent mode to pause the apply process from target journal to target volume until the source journal has completed sending data to the target journal. If there is no consistent image on the target journal, the system does not apply it.

NOTE: Dell Technologies recommends taking a snapshot of the target before setting the target to inconsistent mode. The snapshot can later be used for recovery purposes, using a consistent snapshot.

1. On the menu bar, click **Protection** > **RCGs**.
2. Click the relevant RCG checkbox, and click **Modify** > **Set Target to Inconsistent Mode**.
3. In the **Set Target to Inconsistent Mode RCG** dialog box, click **Apply**.
4. Verify that the operation has finished and was successful, and click **Dismiss**.

Set the target to consistent mode

If the target is set to inconsistent mode, set it back to consistent mode. As data is transferred from source to target, the SDR verifies that the data in the journal is consistent with the data from the source. The SDR then sends an apply command to the journal to prompt the SDR to send data to the volume.

1. On the menu bar, click **Protection** > **RCGs**.
2. Click the relevant RCG checkbox, and click **Modify** > **Set Target to Consistent Mode**.
3. In the **Set Target to Consistent Mode RCG** dialog box, click **Apply**.
4. Verify that the operation has finished and was successful, and click **Dismiss**.

Modify the recovery point objective

Update the recovery point objective time as required.

The recovery point objective defines the maximum amount of time during which data is lost. Dell Technologies recommends setting a low recovery point objective to ensure that minimal data is lost during a possible compromise of data transfer from source to target. If for example, you set one minute as the recovery point objective, you will not lose more than 30 seconds of data.

1. On the menu bar, click **Protection** > **RCGs**.
2. Click the relevant replication consistency group checkbox, and click **Modify** > **Modify RPO**.
3. In the **Modify RPO for RCG** dialog box, modify the RPO time, and click **Apply**.
4. Verify that the operation has finished and was successful, and click **Dismiss**.

Add a replication pair to an replication consistency group

Add a replication pair to an existing replication consistency group.

1. On the menu bar, click **Protection** > **RCGs**.
2. Click the relevant replication consistency group checkbox, and then click **Modify** > **Add Pair**.
3. In the **Add Pairs** wizard, perform the following:
   a. On the **Provisioning Type** page:
      - If you have not defined volumes to receive replicated data in the target system, select **Auto Provisioning**.
      - If you have already defined volumes to receive replicated data in the target system, select **Manual Provisioning**.
   b. For auto provisioning only, on the **Replication Pairs** page, perform the following:
      - Click the desired volume in the **Source** column.
      - Select a type: **Thick** or **Thin**.
      - Select a storage pool from the target system.
      - Click **Add Pair**. The volume pair is added.
      - Click **Next**.
      - On the **Map Volumes** page, select a volume on the target side.
      - Select a host to which to map the volume.
      - Click **Next**, and go to step 3d.
   c. For manual provisioning only, on the **Replication Pairs** page, perform the following:

- Select the desired volume in the **Source** column, and then in the **Target** column, select the target volume (only volumes of the same size are displayed).
- Click **Add Pair**. The volume pair is added.
- Click **Next**.

d. On the **Summary** page, ensure that the correct source and volume pair are defined, and click **Add Pairs**.

4. Optionally, at the upper right of the window, click the **Running Jobs** icon, and check the progress of the initial copy state.
5. Verify that the operation has finished and was successful, and then click **Dismiss**.

Pause replication for an RCG

Pause replication for a replication consistency group (RCG). The pause command stops the transfer of data from the source to the target.

1. On the menu bar, click **Protection**.
2. Click the relevant RCG checkbox, and click **More Actions** > **Pause**.
3. In the **Pause RCG** dialog box, click one of the following options:
   - **Stop Data Transfer**—this option saves all the data in the source journal volume until there is no longer any available capacity
   - **Track Changes**—this option enables manual slim mode where only metadata in the source journal volumes is saved
4. Click **Pause**.
5. Verify that the operation has finished and was successful, and click **Dismiss**.

Resume replication for an RCG

Resume replication for a replication consistency group (RCG). The resume command resumes the transfer of data from the source to the target.

1. On the menu bar, click **Protection**.
2. Click the relevant RCG checkbox, and click **More Actions** > **Resume**.
3. In the **Resume RCG** dialog box, click **Resume**.
4. Verify that the operation has finished and was successful, and click **Dismiss**.

Create a snapshot of an RCG volume

Create a local snapshot of a volume that is in a replication consistency group (RCG). The latest image of the volume is used for the snapshot. When creating a snapshot, the RCG enters freeze mode.

1. On the menu bar, click **Protection** > **RCGs**.
2. In the right pane, select the relevant RCG checkbox, and click **More Actions** > **Create Snapshots**.
3. In the **Create Snapshots RCG** dialog box, click **Create Snapshots**.
4. Verify that the operation has finished and was successful, and click **Dismiss**.

Add an RCG snapshot policy

Create snapshot policies with a maximum of six retention rules that each specify a retention period and number of snapshots to retain, and then add source volumes from the replication consistency group (RCG) to the policy.

1. On the menu bar, click **Protection** > **RCGs**.
2. From the RCGs list, select the desired RCG, and click **More Actions** > **Add Snapshot Policy**.
3. In the **Create Snapshot Policy** dialog box, assign the policy a name and the time interval between snapshots.
4. Create retention rules:

   a. In the **Retention Rules** area, add the number of snapshots to retain.
   A retention period is created for the specified number of snapshots times the snapshot interval.

   b. Optionally, to create an additional retention rule, click **Add Retention** and repeat the previous step. Each rule is based on the specified number of snapshots times the previous retention period.

   Up to six retention rules can be created.

5. Optionally, clear the **Read Only** checkbox to cancel read-only access. This checkbox is selected by default.
6. Optionally, select the **Secured** checkbox to prevent the snapshots created by the policy from being modified or deleted before the end of the expiration date.
7. Click **Create** or **Create and Activate** to add the snapshot policy to PowerFlex.
8. In the **Snapshot Policy was successfully added** dialog box, click **Assign volume to snapshot policy**.
9. In the **Select volumes you wish to assign to the policy** window, select the desired volumes, and then click **Assign Policy**.
10. Verify that the operation has finished successfully, and click **Dismiss**.

Perform a failover

If the system is not healthy, you can fail over the source role to the target system. When the source is compromised, data from the host stops sending I/Os to the source volume, replication is then stopped and the target system takes on the source role. The host on the target starts sending I/Os to the volume. The target takes on the role of source, and the source takes on the role of target.

Before performing a failover, stop the application and unmount the file systems at the source (if the source is available). Target volumes can only be mapped after performing a failover.

There are two options when choosing to fail over an RCG:

- Switchover—This option is a complete synchronization and failover between the source and the target. Application I/Os are stopped at the source, and then source and target volumes are synchronized. Access mode is changed of the target volumes to the target host, roles are switched, and finally, the access mode of the new source volumes is changed to read/write.
- Latest PIT—The system prevents any writes to the source volumes.

1. On the menu bar, click **Protection** > **RCGs**.
2. Click the relevant RCG checkbox, and then click **More Actions** > **Failover**.
3. In the **Failover RCG** dialog box, select one of the following options: **Switchover (Sync & Failover)** or **Latest PIT: (date & time)**.
4. Click **Apply Failover**.
5. In the **RCG Sync & Failover** dialog box, click **Proceed**.
6. Verify that the operation has finished and was successful, and click **Dismiss**.
7. From the upper right, click the **Running Jobs** icon and check the progress of the failover.

Reverse replication

When the replication consistency group is in failover or switchover mode, you can reverse or restore the replication. Reversing replication changes the direction, so that the original target becomes the source. All data at the original source is overwritten by the data at the target side. This option may be selected from either source or target systems.

This option is available when the replication consistency group is in failover mode, or when the target system is not available. It is recommended to take a snapshot of the original source before reversing the replication for backup purposes.

1. On the menu bar, click **Protection** > **RCGs**.
2. Click the relevant RCG checkbox, and click **More Actions** > **Reverse**.
3. In the **Reverse Replication RCG** dialog box, click **Apply**.
4. Verify that the operation has finished and was successful, and click **Dismiss**.

Restore replication

When the replication consistency group is in failover or switchover mode, you can reverse or restore the replication. Restoring replication maintains the replication direction from the original source and overwrites all data at the target side. This option may be selected from either source or target systems.

This option is available when a replication consistency group is in failover mode, or when the target system is not available. Dell Technologies recommends taking a snapshot of the original destination for backup purposes, before restoring replication.

1. On the menu bar, click **Protection** > **RCGs**.
2. Click the relevant replication consistency group checkbox, and click **More Actions** > **Restore**.
3. In the **Restore Replication RCG** dialog box, click **Apply**.

4. Verify that the operation has finished and was successful, and click **Dismiss**.

Test Failover

Test failover of the latest copy of snapshots of source and target systems before performing a failover.

Replication is still running and is in a healthy state.

1. On the menu bar, click **Protection** > **RCGs**.
2. Click the relevant RCG checkbox, and then click **More Actions** > **Test Failover**.
3. In the **RCG Test Failover** dialog box, click **Start Test Failover**.
4. In the **RCG Test Failover using target volumes** dialog box, click **Proceed**.
5. Verify that the operation has finished and was successful, and click **Dismiss**.

Stop a failover test

Stop a failover test that is currently in progress.

1. On the menu bar, click **Protection** > **RCGs**.
2. Click the relevant RCG checkbox, and click **More** > **Test Failover Stop**.
3. Click **Approve**.
4. Verify that the operation has finished and was successful, and click **Dismiss**.

Freeze an RCG

The freeze command stops the writing of data from the target journal to the target volume. This option is used while creating a snapshot or copy of the replicated volume.

1. On the menu bar, click **Protection** > **RCGs**.
2. Click the relevant RCG checkbox, and click **More Actions** > **Freeze Apply**.
3. Click **Freeze Apply**.
4. Verify that the operation has finished and was successful, and click **Dismiss**.

Unfreeze an RCG

The unfreeze apply command resumes data transfer from the target journal to the target volume.

1. On the menu bar, click **Protection** > **RCGs**.
2. Click the relevant RCG checkbox, and click **More Actions** > **Unfreeze Apply**.
3. Click **Unfreeze Apply**.
4. Verify that the operation has finished and was successful, and click **Dismiss**.

Activate an RCG

The activate command starts the initial copy process of replication for the replication consistency group (RCG), and activates replication of the RCG.

1. On the menu bar, click **Protection** > **RCGs**.
2. Click the relevant RCG checkbox, and click **More Actions** > **Activate**.
3. In the **Activate RCG** dialog box, click **Apply**.
4. Verify that the operation has finished and was successful, and click **Dismiss**.
5. Click **View Details**. On the **Topology** tab, check that the replication status is Active.

Click **View Details**. On the **Topology** tab, the replication status should be **Active**.

Terminate replication in an RCG

The terminate command stops replication in the replication consistency group (RCG). The configuration of the RCG is maintained.

1. On the menu bar, click **Protection** > **RCGs**.
2. Click the relevant RCG checkbox, and click **More Actions** > **Terminate**.
3. In the **Terminate RCG** dialog box, click **Apply**.
4. Verify that the operation has finished and was successful, and click **Dismiss**.
5. Click **View Details**. On the **Topology** tab, check that the replication status is **Inactive**.

Delete a replication consistency group

If you no longer require replication of the pairs in a replication consistency group, you can delete it.

1. On the menu bar, click **Protection** > **RCGs**.
2. Click the relevant replication consistency group checkbox, and click **More Actions** > **Delete**.
3. In the **Delete RCG** dialog box, verify that you have selected the desired RCG, and click **Delete**.
4. Verify that the operation has finished and was successful, and click **Dismiss**.

Pause creation of an initial copy

Pause replication of an initial copy from source to target.

1. On the menu bar, click **Protection** > **RCGs**.
2. Click the relevant RCG checkbox, and click **View Details**.
3. Click the **Volume Pairs** tab.
4. Select the volume pair for which you want to pause copying.
5. Click **Initial Copy** > **Pause Initial Copy**.
6. In the **Pause Initial Copy** dialog box, click **Pause Initial Copy**.
7. Verify that the operation has finished and was successful, and click **Dismiss**.

Resume creation of initial copy

Resume replication of an initial copy from source to target.

1. On the menu bar, click **Protection** > **RCGs**.
2. Click the relevant RCG checkbox, and click **View Details**.
3. Click the **Volume Pairs** tab.
4. Select the volume pair for which you want to resume copying.
5. Click **Initial Copy** > **Resume Initial Copy**.
6. In the **Resume Initial Copy** dialog box, click **Resume Initial Copy**.
7. Verify that the operation has finished and was successful, and click **Dismiss**.

Set copying priority

Set the priority order for copying volume pairs. Set the highest priority for pairs to be copied first, or set the lowest priority for pairs to be copied last.

1. On the menu bar, click **Protection** > **RCGs**.
2. Click the relevant RCG checkbox, and click **View Details**.
3. Click the **Volume Pairs** tab.
4. Select the relevant volume pair.
5. Click **Initial Copy** > **Set Priority**.
6. In the **Set Priority for Pair** dialog box, select **Default** or **High** and click **Save**.
7. Verify that the operation has finished and was successful, and click **Dismiss**.

Unpair from RCG

Remove a pair from a replication consistency group (RCG).

1. On the menu bar, click **Protection** > **RCGs**.
2. Click the relevant RCG checkbox, and click **View Details**.
3. Click the **Volume Pairs** tab.
4. Select the volume pair that you want to remove from the RCG.
5. Click **Unpair**.
6. In the **Remove Pair from RCG** dialog box, click **Remove Pair**.
7. Verify that the operation has finished and was successful, and click **Dismiss**.

15

Performing life cycle operations for a resource group (erasure coding)

This section provides tasks for deploying resource groups and building templates.

This capability is available on the PowerFlex appliance and PowerFlex rack offerings only.

Deploying a resource configuration in a resource group

The following section includes tasks for deploying and managing system resources in a resource group.

A resource group is a PowerFlex Manager object representing the infrastructure that is configured during a deployment. This object persists in PowerFlex Manager if the infrastructure is required. A resource group is viewed and interacted with to perform compliance, health, and life cycle management on the deployed infrastructure. PowerFlex Manager supports the deployment of two types of PowerFlex resource groups:

- Storage-only resource groups
- Compute-only resource groups

NOTE: Standard users see only the details of the resource groups that they created or for which they have permission.

The **Resource Groups** page displays the resource groups that are in the following states in both **Tile** and **List** view.

Table 38. Resource group states

State	Description
Healthy	The resource group is successfully deployed and is healthy.
Warning	One or more resources in the resource group requires corrective action.
Critical	The resource group is in a severely degraded or nonfunctional state and requires attention.
Pending	The deployment is scheduled for a later time or date.
In Progress	The resource group deployment is in progress, or has other actions currently in process, such as a node expansion or removal.
Cancelled	The resource group deployment has been stopped. You can update the resources or retry the deployment, if necessary.
Incomplete	The resource group is not fully functional because it has no volumes that are associated with it. Click **Add Resources** to add volumes.
Service Mode	The resource group is in service mode.
Lifecycle Mode	The resource group is in life cycle mode. Resource groups in life cycle mode are enabled with health and compliance monitoring, and non-disruptive upgrade features only.
Managed Mode	The resource group is in managed mode. Resource groups in managed mode are enabled with health and compliance monitoring, non-disruptive update, automated resource addition, and automated resource replacement features.

On the **Resource Groups** page, you can:

- Click **Deploy New Resource Group** to deploy a new resource group.

NOTE: Standard users are allowed to deploy only those resource groups they have created or for which they have permissions.

- Click **Add Existing Resource Group** to add an existing resource group.

To switch views, click the **Tile View** icon or **List View** icon .

To view the resource groups based on a particular resource group state, select an option from the **Filter By** drop-down list. Alternately, in the **Graphical** view, click the graphic in a particular state.

In the **Tile** view, each square tile represents a resource group and has the status of the resource group at the bottom of the graphic. The state icon on the graphic indicates the state of the resource group. The components in blue indicate the component types that are in the deployment. The components that are in gray indicate the component types that are not in the resource group.

In the **List** view, the following information displays:

- **Status**—Status of the resource group.
- **Name**—Name of the resource group.
- **Deployed By**—Name of the user who deployed the resource group.
- **Deployed On**—Date and time when the resource group is deployed.
- **Components**—Components used in the resource group.

Click the resource group in the **List** view or **Tile** view to view the following information about the resource group in the right pane:

- Resource group name and description to identify the resource group.
- Name of the user who deployed the resource group.
- Date and time when the resource group is deployed.
- Name of the reference template that is used in the resource group.
- Number of resources that are in the resource group for deployment, based on component type (cluster or node).

Click **View Details** to view more details about the resource group. You can also generate troubleshooting bundles from the resource group details page.

Click the resource group name in the **List** view to open the **Resource Group Details** page.

Click **Update Resources** to update the firmware of all nodes in the resource group that are not compliant.

Related information

Configuring block storage
Deploying and provisioning

Basic tasks

This section provides basic tasks for resource group management.

Deploying a resource group

Deployment is the automated process of selecting and configuring specific resource requirements that are outlined in a template using the integrated automation workflows that are provided with PowerFlex Manager. You cannot use a template that is in a draft state to deploy a resource group. Publish the template before using it to deploy a resource group.

1. On the menu bar, click one of the following:
   - **Lifecycle** > **Resource Groups**. Then, click **Deploy New Resource Group**.
   - **Lifecycle** > **Templates**. Then, click **Deploy**.

   The **Deploy Resource Group** wizard opens.
2. On the **Deploy Resource Group** page, perform the following steps:
   a. From the **Select Published Template** list, select the template to deploy a resource group.
   b. Enter the **Resource Group Name** (required) and **Resource Group Description** (optional) that identifies the resource group.
   c. To specify the version to use for compliance, select the version from the **Firmware and Software Compliance** list or choose **Use PowerFlex Manager appliance default catalog**.

You cannot select a minimal compliance version when you deploy a new resource group, since it only includes server firmware updates. The compliance version for a new resource group must include the full set of compliance update capabilities. PowerFlex Manager does not show any minimal compliance versions in the **Firmware and Software Compliance** list.

PowerFlex Manager checks the VMware vCenter version to determine if it matches the VMware ESXi version for the selected compliance version. If the ESXi version is greater than the vCenter version, PowerFlex Manager blocks the resource group deployment and displays an error. PowerFlex Manager instructs you to upgrade vCenter first, or use a different compliance version that is compatible with the installed vCenter version.

NOTE: Changing the firmware repository might update the firmware level on nodes for this resource group. The global default firmware repository maintains the firmware on the shared devices.

d. Indicate **Who should have access to the resource group deployed from this template** by selecting one of the following options:
   - To restrict access to super users and specified life cycle administrators, drive replacers, and storage administrators select **Only PowerFlex SuperUser and specfied additional users**.
     i. Click **Add User(s)** to add one or more LifecycleAdmin or DriveReplacer or StorageAdmin users to the list displayed.
     ii. Select which users will have access to this resource group.
     iii. To delete a user from the list, select the user and click **Remove User(s)**.
     iv. After adding the users, select or clear the checkbox next to the users to grant or block access.
   - To grant access to super users and all life cycle administrators, drive replacers, and storage administrators, select **PowerFlex SuperUser and all additional users**.

3. Click **Next**.
4. On the **Deployment Settings** page for **Device Group Spare Capacity** select **Custom** > **Nodes**.

   You must set the spare capacity limit of the device group in terms of nodes for this release. The default spare capacity is one node, and the maximum is two nodes.

   Utilizing spare devices is a feature that is recommended only for advanced configurations.

5. Click **Next**.
6. On the **Schedule Deployment** page, select one of the following options and click **Next**:
   - **Deploy Now**—Select this option to deploy the resource group immediately.
   - **Deploy Later**—Select this option and enter the date and time to deploy the resource group.
7. Review the **Summary** page.
   The **Summary** page gives you a preview of what the resource group will look like after the deployment.
8. Click **Finish** when you are ready to begin the deployment. If you want to edit the resource group, click **Back**.

   PowerFlex Manager displays a message indicating that a new job is being created for this resource group.

   PowerFlex Manager creates a configuration file that is called `/etc/sysctl.d/powerflex-so.conf` for a storage-only resource group. PowerFlex Manager maintains default values in a conf file for resource group deployment settings. If you have an old `sysctl.conf` file from a previous release, it overrides these settings. If you must customize these settings, you can do it in `sysctl.conf`.

Related information

Life cycle
Viewing resource group details
Adding components to a resource group
Build and publish a template
Component types
Removing a resource group

Adding an existing resource group

You can add an existing resource group to discover and import hardware resources that were not originally deployed with PowerFlex Manager.

Ensure that the following conditions are met before you add an existing resource group:

- The PowerFlex gateway, CloudLink Center, and hosts must be discovered in the resource list.
- You must have a resource group that includes the manager MDMs before importing other resource groups. The first resource group that you import must include the manager MDMs so that PowerFlex Manager has the manager MDM credentials.

1. On the menu bar, click **Lifecycle** > **Resource Groups** and then click **+ Add Existing Resource Group**.
2. On the **Add Existing Resource Group** page, enter a resource group name in the **Name** field.
3. Enter a description in the **Description** field.
4. Select the **Type** for the resource group.

   The choices are **Compute Only**, **Storage Only**, and **PowerFlex management cluster (KVM)**.

5. To specify the compliance version to use for compliance, select the version from the **Firmware and Software Compliance** list or select **Use PowerFlex Manager appliance default catalog**.

   You cannot specify a minimal compliance version when you add an existing resource group, since it only includes server firmware updates. The compliance version for an existing resource group must include the full set of compliance update capabilities. PowerFlex Manager does not show any minimal compliance versions in the **Firmware and Software Compliance** list.

   NOTE: Changing the compliance version might update the firmware level on nodes for this resource group. The global default firmware repository maintains firmware on shared devices.

6. Specify the resource group permissions under **Who should have access to the resource group deployed from this template?** by performing one of the following actions:
   - To restrict access to super users and specified life cycle administrators, drive replacers, and storage administrators select **Only PowerFlex SuperUser and specfied additional users**.
     a. Click **Add User(s)** to add one or more LifecycleAdmin or DriveReplacer or StorageAdmin users to the list displayed.
     b. Select which users will have access to this resource group.
     c. To delete a user from the list, select the user and click **Remove User(s)**.
     d. After adding the users, select or clear the checkbox next to the users to grant or block access.
   - To grant access to super users and all life cycle administrators, drive replacers, and storage administrators, select **PowerFlex SuperUser and all additional users**.
7. Click **Next**.
8. Choose one of the following network automation types:
   - **Full Network Automation**
   - **Partial Network Automation**

   When you choose **Partial Network Automation**, PowerFlex Manager skips the switch configuration step, which is performed for a resource group with **Full Network Automation**. Partial network automation allows you to work with unsupported switches. However, it also requires more manual configuration before a deployment can proceed successfully. If you choose to use partial network automation, you give up the error handling and network automation features that are available with a full network configuration that includes supported switches.

   In the **Number of Instances** box, provide the number of component instances that you want to include in the template.

9. On the **Cluster Information** page, enter a name for the cluster component in the **Component Name** field.
10. Select values for the cluster settings:

    For a compute-only resource group, select values for these cluster settings:

    a. **Target Virtual Machine Manager**—Select the vCenter name where the cluster is available.
    b. **Data Center Name**—Select the data center name where the cluster is available.

       NOTE: Ensure that the selected vCenter has unique names for clusters in case there are multiple clusters in the vCenter.

    c. **Cluster Name**—Select the name of the cluster that you want to discover.
    d. **OS Image**—Select the image or choose **Use Compliance File ESXi image** if you want to use the image provided with the target compliance version. PowerFlex Manager filters the operating system image choices to show only ESXi images for a compute-only resource group.

    For a storage-only resource group, select values for these cluster settings:

    a. **Target PowerFlex Gateway**—Select the gateway where the cluster is available.

b. **Protection Domain**—Select the name of the protection domain in PowerFlex.
c. **OS Image**—Select the image or choose **Use Compliance File Linux image**. If you want to use the image provided with the target compliance version. PowerFlex Manager filters the operating system image choices to show only Linux images for a storage-only resource group.

For the PowerFlex management cluster (KVM) resource group, select values for the following cluster setting:

a. **Target PowerFlex Gateway**—Select the gateway where the cluster is available.
b. **Protection Domain**—Select the name of the protection domain in PowerFlex.
c. **OS Image**—Select the image or choose **Use Compliance File Linux image** if you want to use the image provided with the target compliance version. PowerFlex Manager filters the operating system image choices to show only Linux images for the resource group.

11. Click **Next**.
12. On **OS Credentials** page, select the operating system credential that you want to use for each node and MVM.

    You can select one credential for all nodes or MVMs, or choose credentials for each item separately. You can create the operating system credentials on the **Credentials Management** page under **Settings**.

    PowerFlex Manager validates the credentials before it creates the resource group. This validation makes it possible for PowerFlex Manager to run a full inventory on all nodes and MVMs before creating the resource group.

    PowerFlex Manager runs the inventory on all nodes and MVMs for which the credentials are valid. The resource group uses any nodes and MVMs for which it has a successful inventory. For example, if you have four nodes, and one node has an invalid operating system password, PowerFlex Manager adds the three nodes for which the credentials are valid and ignores the one with an invalid password.

13. Click **Next**.

    The list of resources available in the cluster is displayed on the **Inventory Summary** page.
14. Review the inventory on the **Inventory Summary** page.

    The summary shows all nodes that are available. If a node is not available, it might be because this node does not match the **Type** you selected for the resource group (**Compute only**, **Storage only**).

    Depending on how the node is configured, the summary might show additional inventory information. For example, for a node that has NVDIMM compression, the summary shows additional information about the acceleration pool and compression settings.

    If the resources are discovered and in an available state, the **Available Inventory** displays the components as **Yes**. An unavailable PowerFlex gateway is shown as **No**.

    If the credentials are invalid for a node or if you have a network connectivity problem, PowerFlex Manager displays **No** in the **Available Inventory** column for the node, and displays an error message to notify you about the problem.

    PowerFlex Manager cannot update firmware and software versions for PowerFlex clusters that do not have available PowerFlex gateways. If expected PowerFlex gateways are not shown as available, you can discover the gateways and run the wizard again.

    NOTE: PowerFlex Manager retrieves the hostname value from iDRAC and not the operating system. If the hostname field is not updated in iDRAC, an incorrect value can be displayed in PowerFlex Manager. Certain operating systems require extra packages that are installed for iDRAC to update the correct hostname.
15. Click **Next**.
16. On the **Network Mapping** page, review the networks that are mapped to port groups and make any required edits.

    PowerFlex Manager attempts to select the correct network based on the VLAN ID, subnet, or IP ranges entered in PowerFlex Manager. If PowerFlex Manager finds only one network for a given network type, it selects the network automatically. If it finds more than one, you must select the network from the **Network** drop-down list. The operating system Installation network does not get a VLAN ID.

    NOTE: If the operating system install VLAN is not already configured in your environment, add it. This network is required to perform node expansions. This network is typically added during PowerFlex Manager configuration.

    If there are any port groups for which you do not want PowerFlex Manager to manage access, leave those port groups cleared. If no network is selected for a particular port group, PowerFlex Manager leaves it out of the deployment data and does not add it to the nodes.

17. To import many general-purpose VLANs from vCenter, perform these steps:
    a. Click **Import Networks** on the **Network Mapping** page.

PowerFlex Manager displays the **Import Networks** wizard. In the **Import Networks** wizard, PowerFlex Manager lists the port groups that are defined on the vCenter as **Available Networks**. You can see the port groups and the VLAN IDs.

b. Optionally, search for a VLAN name or VLAN ID.
PowerFlex Manager filters the list of available networks to include only those networks that match your search.

c. Click each network that you want to add under **Available Networks**. If you want to add all the available networks, click the checkbox to the left of the **Name** column.

d. Click the double arrow (>>) to move the networks you chose to **Selected Networks**.
PowerFlex Manager updates the **Selected Networks** to show the ones that you have chosen.

e. Click **Save**.

18. Click **Next**.

19. Review the **Summary** page and click **Finish** when you are ready to add the resource group.

PowerFlex Manager displays a message indicating that a new job is being created for this resource group.

The process of adding an existing resource group causes no disruption to the underlying hardware resources. It does not shut down any of the nodes or the vCenter.

For an existing resource group, the **Reference Template** field shows **Generated Existing Resource Group Template** on the **Resource Group Details** page. You can distinguish existing resource groups from new resource groups that were deployed with PowerFlex Manager.

When PowerFlex Manager must put a resource group in life cycle mode, the **Summary** page for the **Add Existing Resource Group** wizard displays a warning message indicating the reason.

In some situations, an imported configuration might not meet the minimal requirements for life cycle mode. In this case, PowerFlex Manager does not allow you to add the resource group.

When you add an existing resource group, PowerFlex Manager matches the hosts, vCenter, and other items it finds with discovered resources in the resource list. If you missed a component initially, you can change your resource inventory, and update the resource group to reflect these changes. Go back to the resources list, select the component, and mark it as **Managed** by selecting **Change resource state to Managed**. Then, perform an **Update Resource Group Details** operation on the resource group to pull in the missing component.

When you deploy an existing resource group, PowerFlex Manager reserves any IP addresses from vCenter or the PowerFlex gateway that it needs. If you later tear down the resource group, it releases those IP addresses so that they can be reused.

If you add an existing resource group that supports NSX-T, PowerFlex Manager displays a banner indicating that the resource group supports a limited set of actions. Most resource group actions are disabled for an NSX-T configuration, except the ability to update the firmware and software components, remove resources (or the resource group as a whole), and update resource group details.

PowerFlex Manager creates a configuration file that is called `/etc/sysctl.d/powerflex-so.conf` for a storage-only resource group. PowerFlex Manager maintains default values in a conf file for resource group deployment settings. If you have an old `sysctl.conf` file from a previous release, it overrides these settings. If you need to customize these settings, you can do it in `sysctl.conf`.

Related information

Getting started
Life cycle
Support for full and partial network automation
Defining a network
Getting started (mirroring)

Entering and exiting service mode

PowerFlex Manager enables you to put a node in service mode when you must perform maintenance operations on the node.

This capability is available on the PowerFlex appliance and PowerFlex rack offerings only.

PowerFlex Manager detects when a node is in instant maintenance mode. It automatically places the node in service mode and also ensures that the resource group itself goes into service mode.

If the node has a VMware NSX-T configuration, PowerFlex Manager does not enable you to enter service mode. PowerFlex Manager also does not enable you to enter service mode if the PowerFlex gateway used in the resource group is being updated on the **Resources** page.

1. On the menu bar, click **Lifecycle** > **Resource Groups**.

2. On the **Resource Groups** page, select a resource group and click **View Details** in the right pane.
3. Click **Enter Service Mode** under **More Actions**.
4. Select the node on the **Node Lists** page and click **Next**.
5. Click **Instant Maintenance Mode**.
6. Click **Finish**.

   PowerFlex Manager displays a yellow warning banner at the top of the Resource Groups page. The **Service Mode** icon displays for the **Deployment State** and **Overall Resource Group Health**, and for the **Resource Health** for the selected nodes.
7. When you are ready to leave service mode, click **More Actions** > **Exit Service Mode**.

Replacing a drive

You can replace a failed drive in a deployed node. PowerFlex Manager supports the replacement of SSD and NVMe drives for PowerFlex storage-only nodes.

Ensure that you have a replacement drive and can access the node.

This capability is available on the PowerFlex appliance and PowerFlex rack offerings only.

PowerFlex Manager supports drive replacement for:

- Nodes that have NVDIMM/SDPM enabled.
- Nodes that have NVMe drives for HBA SAS controllers
- **NOTE:** RAID controllers are not supported.
- CloudLink-enabled PowerFlex nodes with self-encrypting drives (SEDs) or software encryption.

If the node has an NSX-T configuration, PowerFlex Manager does not enable you to replace a drive.

1. On the menu bar, click **Lifecycle** > **Resource Groups**.
2. On the **Resource Groups** page, select a resource group and click **View Details** in the right pane.
3. Scroll down to the **Physical Nodes** section of the page.
4. Under **Physical Nodes**, click **Component Replacement** > **Drive Replacement**.
   PowerFlex Manager displays the **Node List** panel in the **Drive Replacement** wizard.
5. Click the node for which you want to perform a drive replacement.
6. Click **Next**.
   PowerFlex Manager displays the **Select Drive** panel. Any empty slots are shown in gray. The slots that have drives are shown in black.
7. Select the drive that you want to replace by clicking the drive slot in the hardware image, or select the drive from the table.

   Be sure to click the correct drive, because the drive replacement process is irreversible.

   The color for the selected drive changes to blue and the table below the hardware image is selected. The table shows details about the selected drive. To help you pick the correct drive, PowerFlex Manager provides the iDRAC name for the drive, the PowerFlex drive name, and the serial number.
8. Optionally, click **Launch iDRAC GUI** to see iDRAC details about the selected drive before proceeding.

   When you launch iDRAC, the iDRAC user interface opens in a different tab. Log in and go to the drive details.
9. Optionally, click **Blink LED** to cause the selected drive to blink.
   PowerFlex Manager turns on the blinking LED on the selected drive and keeps it blinking until you pull the disk out. A new disk is not in a blinking state. PowerFlex Manager is not able to determine whether the drive is blinking. If the drive is already blinking, you can click **Unblink LED** to turn off the blinking LED.
10. Click **Next** when you are ready to delete the drive.
    PowerFlex Manager displays the **Confirmation** page. PowerFlex Manager provides details about the drive and shows a hardware image again with the drive that is selected in blue.
11. Review the information carefully under **Confirm Drive Replacement** to be sure that you have selected the correct drive. If you are certain that you want to proceed, type **`REMOVE DRIVE`**.
12. Click **Confirm**.
    PowerFlex Manager puts the resource group into the **In Progress** state and updates the **Recent Activity** to show that a drive removal operation is started. After several minutes, if the node ejection is successful, PowerFlex Manager places the resource group and the node in service mode and deletes the drive from the storage pool. The resource group and the node remain in service mode until the drive is replaced in the node, and the drive replacement process has completed. During this

period, PowerFlex Manager does not enable you to enter service mode for the node, or perform another drive replacement within the same resource group.

PowerFlex Manager displays a banner at the top of the **Resource Group Details** page to notify you that a drive replacement is in process for the selected drive.

For NVMe disks, PowerFlex Manager initiates a job to instruct iDRAC to delete the NVMe disk safely from the operating system.

If the drive ejection process fails, PowerFlex Manager displays a banner at the top of the **Resource Group Details** page to notify you that the initial step to eject the drive failed. When the delete drive step fails in the drive replacement process, the banner displays a **Retry Drive Replacement** option to help you to restart the drive replacement process.

If there is a failure, the resource group and the node do not remain in service mode. Instead, they return to the previous state that the resource group was in before you attempted the drive replacement.

13. Pull the disk out of the slot, and insert the new disk.
14. On the disk replacement banner at the top of the **Resource Group Details** page, click **Component Replacement**.

    PowerFlex Manager opens the **Complete Drive Replacement** wizard and runs inventory on the node. It takes a couple of minutes to get the updated inventory for the node.

    After getting the updated inventory, PowerFlex Manager displays the **Found New Drive** page or the **Continue Without Drive** page. If PowerFlex Manager finds the new drive, it displays the drive that is selected in green on the **Found New Drive** page.

    If PowerFlex Manager does not find the new drive, it displays the **Continue Without Drive** page. If you are certain that you want to proceed without a new drive, type `CONTINUE WITHOUT DRIVE`. If you proceed, the node is taken out of service mode with fewer drives, which might impact the capacity and performance of the resource group. Click **Confirm** to finish the process without adding a drive.
15. Click **Complete** to finish the disk replacement process.
    PowerFlex Manager puts the resource group into the **In Progress** state and updates the **Recent Activity** to show that a complete drive replacement operation is started. Then, PowerFlex Manager replaces the drive in the resource group, and takes the resource group and the node out of service mode.

    When the drive addition is complete, PowerFlex Manager displays a banner at the top of the resource group to notify you that the drive replacement is complete. PowerFlex Manager enables the **Reconfigure MDM Roles** and **Drive Replacement** options again. The resource group returns to the previous state after the drive replacement is is successfully completed.

Reconfiguring MDM roles

PowerFlex Manager enables you to change the MDM role for a node in a PowerFlex cluster. For example, if you add a node to the cluster, you might want to switch the MDM role from an existing node to the new node.

NOTE: This is not applicable for PowerFlex in a PowerFlex management cluster (KVM), as there are only three nodes.

You can launch the wizard for reconfiguring MDM roles from the **Resource Groups** page or from the **Resources** page. The nodes that are listed and the operations available are the same regardless of where you launch the wizard.

1. To access the wizard from the **Resource Groups** page:
   a. On the menu bar, click **Lifecycle** > **Resource Groups**.
   b. Select a resource group that has the PowerFlex system for which you want to reconfigure MDM roles.
   c. In the right pane, click **View Details**.
      The **Resource Group Details** page is displayed.
   d. On the **Resource Group Details** page, click **Reconfigure MDM Roles** under **More Actions**.
      The **MDM Reconfiguration** page is displayed.
2. To access the wizard from the **Resources** page:
   a. On the menu bar, click **Lifecycle** > **Resources**.
   b. Select the PowerFlex system for which you want to reconfigure MDM roles.
   c. In the right pane, click **View Details**.
      The **Details** page is displayed for the selected PowerFlex gateway.
   d. On the **Details** page, click **Reconfigure MDM Roles**.
      The **MDM Reconfiguration** page is displayed.
3. Review the current MDM configuration for the cluster.
4. Find each MDM role that you want to reassign and choose the new hostname or IP address for the role in the **Select New Node for MDM Role** list.

You can reassign multiple roles at one time.

5. Click **Next**.
   The **Summary** page is displayed.
6. Type **CHANGE MDM ROLES** to confirm your changes.
7. Click **Finish**.

Related information

Viewing PowerFlex system details

Adding components to a resource group

You can add one or more node, volume, or network components to a new resource group that PowerFlex Manager deploys, or to an existing resource group not deployed by PowerFlex Manager.

You can add components to a resource group for which deployment was successful or to a failed resource group deployment. You do not need to go back to the original template from which the resource group was originally deployed. Standard users can add components only to a resource group for which they have permission.

1. On the menu bar, click **Lifecycle** > **Resource Groups**.
2. On the **Resource Groups** page, select a resource group and click **View Details** in the right pane.
3. On the **Resource Group Details** page, in the right pane, click one of the following options from the **Add Resources** menu:
   - **Add Nodes** —Add one or more nodes to the resource group
   - **Add Volumes**—Add one or more volumes to the resource group
   - **Add Network**—Add one or more networks to the resource group

   If a node has an NSX-T configuration, PowerFlex Manager removes the **Add Resources** button under **Resource Actions**.

   If the PowerFlex gateway used in the resource group is being updated on the **Resources** page, PowerFlex Manager also removes the **Add Resources** button.

   If the PowerFlex gateway used in the resource group is being updated on the **Resources** page, PowerFlex Manager does not allow you to add a node.

Related information

Component types
Deploying a resource group

Adding nodes to a resource group

You can add a node to a resource group. To add a node, you must define the PowerFlex, operating system, and hardware settings.

1. On the **Resource Groups** page, select the checkbox beside the resource group that want to add a node to and click **View Details**.
2. On the **Resource Groups** page, click **Add Resources** and choose **Add Nodes**.
   The **Duplicate Node** wizard opens.
3. From the **Duplicate Node** page, complete the following steps:
   a. From **Resource to Duplicate**, select a node.
      Select a node that is of the same type as the other nodes within the resource group.
   b. From **Number of Instances**, enter the number of node instances that you want to add to the resource group.
   c. Click **Next**.

   The **Additional Settings** window opens.
4. Under **PowerFlex Settings**, enter the **Protection Domain Name**, **Device Group Name**, **Device Group Spare Capacity**, and **Custom Spare Node Count**.

   You must set the spare capacity limit of the device group in terms of nodes for this release. The default spare capacity is one node, and the maximum is two nodes.

   Utilizing spare devices is a feature that is recommended only for advanced configurations.

5. Under **OS Settings**, set the **Host Name Selection** to **Auto-Generate**, **Specify at Deployment Time**, or **Reverse DNS Lookup**.
   - If you choose **Specify at Deployment**, provide a name for the host in the **Host Name** field.
   - If you choose **Auto-Generate**, specify a template for the name in the **Host Name Template** field.

   For an existing resource group that PowerFlex Manager did not deploy, the **Host Name Selection** option is automatically set to **Specify at Deployment Time** and you must type the hostname.

   In the **IP Source** box, provide an IP address. For an existing resource group that PowerFlex Manager did not deploy, the default choice is **User Entered IP** and the IP settings for each network default to **Manual Entry**. However, you can change the setting to **PowerFlex Manager Selected IP**.
   - If the **Multi-Network Selection** option is selected and the IP source is **PowerFlex Manager Selected IP**, the IPs are assigned sequentially from all the networks of that type.
   - If an IP cannot be assigned from any of the multiple networks of the selected type, an error is displayed.
   - If the IP source is **User Entered IP**, a generic name is displayed for the **IP Source** setting. For example, **PowerFlex Management IP Source**.
   - If multi networking is not enabled and the IP source is **User Entered IP**, the name of the network is displayed as on the **Network Settings** page. If the selected IP Source setting is of **Manual Entry** type, the IP address you entered is validated against all the networks of the selected type. An error is displayed when the IP is not validated in one of the networks.

   Under **Hardware Settings**, the **Target Boot Device** option is automatically set to **Use Local Flash Storage for Dell PowerFlex** for an existing compute-only resource group that PowerFlex Manager did not deploy.
6. Under **Hardware Settings**, in the **Node Source** box, select **Node Pool** or **Manual Entry**.

   For an existing resource group that PowerFlex Manager did not deploy, the node source defaults to **Manual Entry**, but you can change it to **Node Pool**.

   In the **Node Pool** box, select the node pool. Alternatively, if you chose **Manual Entry**, select the specific node in the **Choose Node** box.

   You can view all user-defined node pools and the global pool. Standard users can see only the pools for which they have permission.

   For an existing resource group that PowerFlex Manager did not deploy, the **Node Pool** defaults to **Global**. This is valid for compute-only and storage-only node deployments.
7. Click **Next**.
8. Click **Save**.
9. Review the **Summary** page and click **Finish**.

Related information

Add a storage node

Adding volumes to a resource group

You can create volumes for a storage-only resource group. You can add existing volumes to a compute-only resource group.

PowerFlex Manager allows you to create up to 50 new volumes at a time for a storage-only resource group.

Initially, there are no volumes for a new resource group. The resource group is in an incomplete state until you add volumes. Add at least one volume to a resource group to get it out of the incomplete state.

A compute-only resource group consumes volumes that the storage-only resource group provides.

After you create the volumes in a storage-only resource group, they are added to PowerFlex and are mapped to the storage node . When you add the volumes to a compute-only resource group, PowerFlex maps the volumes and creates the datastores.

PowerFlex Manager allows you to enable compression when adding volumes to a storage-only resource group.

1. On the **Resource Groups** page, select checkbox beside the resource group that you want to add a volume to and click **View Details**.
   The **Add Volume** wizard opens.
2. Ensure the **Create New Volumes** button is selected and click **Next**.
3. Click **+ Add New Volume** to add a volume section. PowerFlex Manager adds a volume section to the **Add Volume** wizard.
4. From the **Volume Name** dropdown, choose how you want the volume name to be created:
   - To enter a specific name for a volume.

- To automatically create multiple volumes that share a common naming pattern, select **Auto Generate Name**.

5. If you choose to create a volume with a specific name, complete the following steps:
   a. From the **Volume Name** field, select **Create New Volume** from the dropdown.
   b. To name the volume, enter a name in the **New Volume Name** field.
   c. To choose the storage pool where the volume will reside, select the location from the **Storage Pool** dropdown,
   d. To specify the size of the volume, enter the size in GB in the **Volume Size (GB)** field. The minimum size is 8 GB and the value that you specify must be divisible by eight.
   e. Click **Next**.
   f. Review the summary and click **Finish**.
6. If you choose to automatically create multiple volumes that share a common naming pattern, complete the following steps:
   a. From the **Volume Name** field, select **Auto Generate Name**.

      NOTE: The template used to generate volume names must contain a variable that will produce a unique volume name. The variables that are allowed are: *${num}* , *${num_5d}*(an auto generated unique number).

   b. In the **Volume Name Template** field, enter a template name that includes a variable which enables PowerFlex Manager to produce a unique volume name.
   c. In the **How Many Volumes** field, select the number of volumes you want to create.

      The number must be less than 50.
   d. To choose the storage pool where the volume will reside, select the location from the **Storage Pool** dropdown.
   e. To specify the size of the volume, enter the size in GB in the **Volume Size (GB)** field. The minimum size is 8 GB and the value that you specify must be divisible by eight.
   f. Click **Next**.
   g. Review the volume details on the summary page and click **Finish**.

The resource group moves to the **In Progress** state and the new volume icons appear on the **Resource Group Details** page. You may see multiple volume components while the `add` operation is still in progress. One the operation is complete, you see one volume component with the count updated.

After the deployment completes successfully, you can click **View Volumes** in the **Storage** list on the **Resource Group Details** page to search for volumes that are part of the resource group.

The new volumes are displayed under the storage pool.

For a storage-only resource group, the volumes are created and mapped to a storage node

For a compute-only resource group, the volumes are mapped to SDCs. In the vSphere client, you can see the volumes in the storage section and also see the hosts that are mapped to the volumes, once the mappings are in place.

Related information

Resize a volume
Viewing and selecting volumes

Adding networks to a resource group

If you have PowerFlex compute-only nodes, you can update the workload network by adding a general purpose LAN to a resource group. Alternatively, you can add a static route to allow nodes to communicate across different networks.

1. On the **Resource Group Details** page, click **Add Resources** and choose **Add Network**.
   The **Add Network** window is displayed. All used resources and networks are displayed under **Resource Name** and **Networks**.
2. Click **Add Additional Network** to add an additional network:
   a. From the **Available Networks** list, select the network, and click **Add**.
   The selected network is displayed under **Network Name**. You can define a new network by clicking **Define a New Network**.
   b. Select **Port Group** from the **Select Port Group** list.
   c. Click **Save**.

      NOTE: You cannot remove an added network using PowerFlex Manager.

3. Click **Add Additional Static Route** to add an additional static route:
   a. Click **Add New Static Route**.

b. Select a **Source Network**.

   The source network must be a PowerFlex data network.

c. Select a **Destination Network**.

   The destination network must be a PowerFlex data network.

d. Type the IP address for the **Gateway**.
e. Click **Save**.

Resize a volume

After adding volumes to a resource group, you can resize the volumes.

For a storage-only resource group, you can increase the volume size. For a VMware ESXi compute-only resource group, you can increase the size of the datastore that is associated with the volume.

If you resize a volume in a storage-only resource group, you must update the datastore size in the corresponding VMware ESXi compute-only resource group. The datastore size cannot exceed the size of the volume.

1. On the **Resource Groups** page, click the volume component and choose **Volume Actions** > **Resize**.
2. Choose the volume that you want to resize:
   a. Click **Select Volume**.
   b. Enter a volume or datastore name search string in the **Search Text** box.
   c. Optionally, apply additional search criteria by specifying values for the **Size**, **Type**, **Compression**, and **Storage** filters.
   d. Click **Search**.
   PowerFlex Manager updates the results to show only those volumes that satisfy the search criteria. If the search returns more than 50 volumes, you must refine the search criteria to return only 50 volumes.
   e. Select the row for the volume you want to resize.
   f. Click **Apply**.
3. Update the sizing information:

   If you are resizing a volume for a storage-only resource group, enter a value in the **New Volume Size (GB)** field. Specify a value that is greater than the current volume size. Values must be in multiples of eight, or an error occurs.

   If you are resizing a volume for a compute-only resource group, review the **Volume Size (GB)** field to see if the volume size is greater than **Current Datastore Size (GB)**. If it is, PowerFlex Manager expands the datastore size.

4. Click **Save**.

Related information

Adding volumes to a resource group

Removing resources from a resource group

You can remove resources that were previously added to a resource group.

If you are removing a node that is fulfilling a metadata manager (MDM) role, verify that you will have at least two MDMs and one tiebreaker in the cluster after the removal of the node. When you remove a node, PowerFlex Manager puts a standby node in place of the one that you removed. For a storage-only resource group, the platform minimum is four nodes in the cluster. This enables you to scale down to three nodes for a component replacement.

If the node has an NSX-T configuration, you can remove the node information that PowerFlex Manager displays for the resource group, but not delete the node from the resource group entirely. If the PowerFlex gateway used in the resource group is being updated on the **Resources** page, PowerFlex Manager does not enable you to remove resources from the resource group.

1. On the menu bar, click **Lifecycle** > **Resource Groups**.
2. Select a resource group from which you want to delete resources.
3. In the right pane, click **View Details**.
4. On the **Resource Group Details** page, in the right pane, under **More Actions**, click **Remove Resource**.
5. In the **Remove Resource** dialog box, select the node that you want to remove and click **Next**.
6. Select the **Resource removal type**:

- **Delete Resource** makes configuration changes to the nodes, switch ports, virtual machine managers, and PowerFlex to unconfigure those components, and returns the components to the available inventory.
- **Remove Resource** removes deployment information, but does not make any configuration changes to the nodes, switch ports, virtual machine managers, and PowerFlex, and returns the components to the available inventory.

7. If you choose **Remove Resource**, perform the following steps:
   a. To keep the node in the inventory, select **Leave resource in PowerFlex Manager inventory and set state to** and select the state:
      - **Managed**
      - **Unmanaged**
      - **Reserved**
   b. To remove the node, select **Remove resource from the PowerFlex Manager inventory**.
   c. Click **Remove**.
8. If you choose **Delete Resource**, perform the following steps:
   a. If you are certain that you want to proceed, type **`DELETE RESOURCE`**.
   b. Click **Delete**.

When you delete a node from a resource group, PowerFlex Manager performs the following steps to clean up the underlying components associated with the node:

1. Reconfigures the MDM cluster to ensure that the cluster is in good working order. This step is necessary only if the node is part of an MDM cluster.

   If necessary, PowerFlex Manager reconfigures the MDM role for each node to be a manager or tiebreaker when you perform a scale-down operation. PowerFlex Manager rebalances the roles as needed for a new resource group or for an existing resource group. PowerFlex Manager applies the required role changes for nodes that are within the same resource group or in a different resource group.

2. If the node is on the list of storage nodes, it removes the node from the list of storage nodes within PowerFlex. The node might only be removed from the storage node list if the spare capacity is sufficient, and the environment provides enough space.
3. Removes the operating system on the node.
4. Removes the node from the SDC list within PowerFlex.

If you remove a node that is CloudLink enabled from a storage-only resource group, PowerFlex Manager also performs these cleanup tasks:

- Unencrypts the drives on the server.
- Removes the machine from the machine group.
- Removes IP addresses from approved networks.

PowerFlex Manager does not remove the machine groups or keystores. The machine groups and keystores can be reused for subsequent deployments.

After a node is deleted from a resource group, the **Resource Group Details** page shows the cluster without the removed node, and updates the MDM roles to reflect the new configuration.

Adding NVMe over TCP storage access

PowerFlex Manager enables you to add NVMe over TCP storage access to nodes in a resource group. When you add storage access, it installs the SDT (storage data target) package on all nodes in the resource group.

NVMe storage access is only supported for a resource group that is on PowerFlex 4.x. In addition, the option to add NVMe storage access is only available if the resource group was deployed with the **SDC Only** option selected for **Client Storage Access** in the template.

1. On the menu bar, click **Lifecycle** > **Resource Groups**.
2. Select the resource group.
3. On the **Resource Group Details** page, under **More Actions**, click **Add NVMe Storage Access**.
4. Review the list of nodes.
5. Type **`ADD SDT TO NODES`** to confirm your changes.
6. Click **Confirm**.

Additional resource group management tasks

This section provides additional tasks for resource group management.

Redeploying a resource group

You can redeploy a resource group that failed to deploy because of errors. Standard users can redeploy only a failed resource group that they have deployed.

1. On the menu bar, click **Lifecycle** > **Resource Groups**.
2. On the **Resource Groups** page, select a resource group in an error state and click **View Details** in the right pane.
3. On the **Resource Group Details** page, in the right pane, under **More Actions**, click **Retry**.
4. Click **Yes** when a confirmation message appears.

   If necessary, PowerFlex Manager reconfigures the MDM role for each node to be a manager or tiebreaker when you perform a retry operation. PowerFlex Manager rebalances the roles as needed for a new resource group or for an existing resource group. PowerFlex Manager applies the required role changes for nodes that are within the same resource group or in a different resource group.

Edit resource group information

You can edit information about a resource group such as the resource group name, description, and who has access to the resource group information.

1. On the menu bar, click **Lifecycle** > **Resource Groups**.
2. On the **Resource Groups** page, click the resource group.
3. On the **Resource Group Details** page, in the right pane click **Modify**.
4. In the **Modify Resource Group Information** dialog box, modify the **Resource Group Name** and **Resource Group Description** that identifies the resource group.
   - To update the firmware and software running on the nodes when you deploy a resource group that uses this template, select **Firmware and Software Compliance**.

     If you select a minimal compliance version for a resource group, PowerFlex Manager puts the resource group in life cycle mode and restricts the actions that can be performed. In life cycle mode, the resource group supports monitoring, service mode, and compliance upgrade operations only. All other resource group operations are blocked.

     NOTE: Changing the firmware repository might update the firmware level on nodes for this resource group. The global default firmware repository maintains firmware on shared devices.

5. Set the **OS Image** for the resource group.

   You can optionally select a custom operating system image that will be used for future add node operations on the resource group. The custom image must be added to the **OS Image Repositories** tab on the **Settings** > **Compliance and OS Repositories** page.

   PowerFlex Manager filters the list of images available for selection to include only those images that are suitable for the current resource group.

6. Specify the permissions for this resource group under **Who should have access to the resource group deployed from this template?** by performing one of the following actions:
   - To restrict access to super users and specified life cycle administrators, drive replacers, and storage administrators select **Only PowerFlex SuperUser and specfied additional users**.
     a. Click **Add User(s)** to add one or more LifecycleAdmin or DriveReplacer or StorageAdmin users to the list displayed.
     b. Select which users will have access to this resource group.
     c. To delete a user from the list, select the user and click **Remove User(s)**.
     d. After adding the users, select or clear the checkbox next to the users to grant or block access.
   - To grant access to super users and all life cycle administrators, drive replacers, and storage administrators, select **PowerFlex SuperUser and all additional users**.
7. Click **Save**.

Viewing resource group details

You can view the state of a resource group at the component level. You can view the component topology and connections in a selected resource group template. You can view component logs, display the port view, place a component in service mode, and replace a disk.

If the resource group state is incomplete, a banner at the top of the page indicates that you must add volumes to make it fully functional. Sometimes, you may only need to run **Update Resource Group Details** to ensure that the resource group is complete.

To view the details of a resource group component, scroll down on the **Resource Group Details** page. The following information is displayed based on the resource types in the resource group:

Table 39. Resource group details

Resource type	Description
Clusters	View the following information about the clusters in the VMware vCenter environment: • **Health** • **Data Center Name** • **Cluster Name** • **Asset/Service Tag** • **Management IP**
PowerFlex Gateways	View the following information about the gateways that are part of the resource group: • **System Name** • **Primary MDM IP** • **Management IP** • **Virtual IP** • **Total Device Groups** Provides the number of device groups in the resource group. A device group consists of devices that come from four or more nodes that are contained in a single protection domain. The nodes that belong to a device group have the same drive type or media type. The capacity of a device group is derived from the capacity of the storage devices that it contains. Storage pools are allocated through the device group and consume the capacity of the device group. The device group details include the following information for each device group: ○ **Name** ○ **Device Group Type** ○ **Size** ○ **Total Storage Pools** The storage pool details include the name of each storage pool, size, device type, and RAID set. • **Protection Domain** A protection domain is a logical entity that consists of and manages a group of storage nodes each of which contains devices. A storage node contains a PowerFlex data server (PDS) and a Device Gateway Target (DGWT). These services enable the storage and processing of data to be separated. Data is logically isolated into specific datasets and performance capabilities within a specified protection domain. The effect of any specific device failure is limited.
CloudLink	View the following information about the CloudLink Centers participating in the resource group: • **Health** **NOTE:** An expected unknown health status is displayed for CloudLink because PowerFlex Manager does not have access to CloudLink health monitoring information. Since PowerFlex Manager is not able to determine the health status

Table 39. Resource group details (continued)

Resource type	Description
	of CloudLink, it is not updated. Log in to CloudLink Center to view detailed health status information. • **Hostname** • **Management IP** • **Machine Group** • **Keystore** When a resource group is deployed, if CloudLink has clustered CloudLink Centers, PowerFlex Manager shows all CloudLink Centers on the **Resource Group Details** page. PowerFlex Manager ensures that the resource group deployment succeeds as long as at least one CloudLink Center in the cluster is working.
Storage	View details about the storage volumes added for the resource group. Click **View Volumes** to search for volumes and see the following information about the volumes: • **Name** • **Size** • **Type** • **Compression** • **Storage Pool** • **Datastore**
Physical Nodes	View the following information about the nodes that are part of the resource group: • **Health** • **Asset/Service Tag** • **iDRAC IP/Hostname** • **PowerFlex Mode** The mode for each node is one of the following: ○ **Storage Only** includes only the storage node component. ○ **Compute Only** includes only the SDC component. • **MVM Hostname** • **Associated IPs** • **MDM Role** The MDM role is the metadata manager role. The MDM role applies only to those nodes that are part of a PowerFlex cluster. The MDM role is one of the following: ○ **Primary**: The MDM in the cluster that controls the and SDCs. The primary MDM contains and updates the MDM repository, the database that stores the storage node configuration, and how data is distributed between the . This repository is constantly replicated to the secondary MDMs, so they can take over with no delay. Every PowerFlex cluster has one primary MDM. ○ **Secondary**: An MDM in the cluster that is ready to take over the primary MDM role if necessary. ○ **Tie Breaker**: An MDM whose sole role is to help determine which MDM is the primary. ○ **Standby MDM**: A standby MDM can be called on to assume the position of a manager MDM when it is promoted to be a cluster member. ○ **Standby Tie Breaker**: A standby node that is prepared to take over as a tiebreaker.

View all settings

View All Settings allows you to view the settings configured on the resources in a resource group for deployment.

When you click **View All Settings**, the **Resource Group Deployment Settings** page shows each top-level component on a separate tab. Click a tab for a component to see the details for the component. By displaying the details on separate tabs,

PowerFlex Manager greatly improves the performance when you are working with a resource group that includes a large number of nodes.

For a PowerFlex resource group, the **VMware Cluster** section of the **Settings** page shows the **PowerFlex Settings** section, and the **Target PowerFlex Gateway** should specify the gateway that is included with the resource group.

For a PowerFlex resource group, the **Use Local Storage For Dell PowerFlex** property is set to **true** within the **OS Settings** section for any node in an existing PowerFlex resource group deployment, and for the nodes in a new PowerFlex resource group deployment.

Generate troubleshooting bundle

Generate Troubleshooting Bundle allows you to generate a compressed file of logs to use for troubleshooting. The following logs are included:

- ASM deployer
- iDRAC life cycle
- Dell PowerSwitch switch
- Cisco Nexus switch
- VMware ESXi
- CloudLink Center
- Standard output logs from all pods
- Kubernetes logs about pods, services, deployments, secrets, drivers, and volumes

View compliance report

View Compliance Report allows you to view the software or firmware compliance report.

Add resources

Add Resources allows you to add nodes, volumes, and networks to a resource group.

More actions

Under **More Actions**, you can perform the following tasks:

Table 40. More actions

Click...	To...
Update Resource Group Details	Update resource group definition.
Enter Service Mode	Put nodes in service mode so that maintenance operations can be performed.
Exit Service Mode	Take nodes out-of-service mode.
Reconfigure MDM Roles	Change the MDM role for a node in a PowerFlex cluster. For example, if you add a node to the cluster, you might want to switch the MDM role from an existing node to the new node.
Remove Resource Group	Remove a resource group that is no longer required. PowerFlex Manager supports two types of removal: • Delete the entire resource group, which deletes the deployment information and makes any required configuration changes for components that are associated with the resource group. This includes making destructive changes such as unconfiguring the nodes in the resource group and powering them down. • Delete the deployment information for the resource group without making any configuration changes to the deployed components.
Remove Resource	Removes resources from a resource group.

Table 40. More actions (continued)

Click...	To...
Retry	Redeploy a failed or canceled resource group. This button is only visible when a resource group is in a critical state. If you see a resource group fail for any reason, try clicking **Retry** to correct the problem. For example, a server deployment might fail with a `ChangeBootOrderByInstanceID failed` message. You can correct this failure by retrying the resource group.
Cancel	Cancel the resource group.

Resource actions

Under **Resource Actions**, you can perform the following tasks:

Table 41. Resource actions

Click...	To...
Add Resources	Select the type of the resources that you want to add to the resource group. If you must input data for the template that is used to create the running resource group, click **Confirm Resource Group Settings**. In the **Update Resource Group Component** window, enter values for all displayed fields. Click **Save**.
Remove Resource	Remove resources from a resource group.

Resource group information

In the **Resource Group Information** box on the right, you can view the following information:

- **Overall Resource Group Health**—Displays the health of the resource group. The following information determines the overall resource group health:
  - **Resource Health** —Displays the health monitoring of resources.
  - **Compliance**—Displays if the resources are firmware or software-compliant. This option is applicable only if you have enabled firmware or software update on the resource group.
  - **Deployment State**—Indicates if the deployment of the resource group completed successfully.
- **Deployed By**—Displays the name of the user who deployed the resource group.
- **Deployed On**—Displays the date and time when the resource group is deployed.
- **Reference Template**—Displays the name of the reference template that is used in the resource group.

  NOTE: For existing resource groups, the name displays as **User Generated Template** and not a template name from the inventory.
- **User Permissions**—Displays one of the following:
  - **Enabled**—Indicates that the permission is granted for one or more standard users to deploy this resource group.
  - **Disabled**—Indicates that the permission is not granted for the standard users to deploy this resource group.

Recent activity

Recent Activity displays component deployment status and information about the current deployed resource group.

You can also click the **Port View** tab to display port view details.

If you want to see the current firmware or software repository that is in use, look at **Target Version**. To change the compliance version, click **Change Target**.

If you select a minimal compliance version for a resource group, PowerFlex Manager puts the resource group in life cycle mode and restricts the actions that can be performed. In life cycle mode, the resource group supports monitoring, service mode, and compliance upgrade operations only. All other resource group operations are blocked.

Related information

Deploying a resource group

Viewing all settings

The **View All Settings** option lets you display the settings that are used to configure the resources in the deployment of a resource group.

When you click **View All Settings**, the **Resource Group Deployment Settings** page shows each top-level component on a separate tab. Click on a tab for a component to see the details for the component. By displaying the details on separate tabs, PowerFlex Manager greatly improves the performance when you are working with a resource group that includes a large number of nodes.

You can see all settings that are defined for the clusters and nodes, as well as storage. To search for volumes within the resource group, you can click **View Volumes** under **Storage**.

During volume operations, the **Resource Group Deployment Settings** page displays additional information about the volume operations that are in progress. If you create a new volume or add an existing volume, or resize a volume, the page provides details that can be helpful for troubleshooting. For example, if you specified an invalid name template string variable, the **Resource Group Deployment Settings** page indicates that this is the cause of a volume not being added.

Viewing and selecting volumes

Use this procedure to search for volumes in a resource group, and optionally select volumes for an add or resize operation.

You can use the volume search dialog to search for volumes using various types of search criteria. You can search by volume name or datastore name, as well as by volume type, compression setting, and storage pool name.

You can launch the volume search dialog from various places with the PowerFlex Manager user interface. You can launch the dialog from the following locations:

- **Storage** section of the **Resource Group Details** page
- **View All Settings** page
- **Add Volumes** wizard
- **Resize Volume** wizard

When you perform a search, PowerFlex Manager updates the results to show only those volumes that satisfy the search criteria. If the results would include more than 50 volumes, you must refine the search criteria to return only 50 volumes.

After performing a search in the **Add Volumes** wizard, you can select one or more existing volumes to add to a resource group. In the **Resize Volume** wizard, you can select the volume that you want to resize.

Identify the naming patterns that are used for the volumes and datastores within the resource group, so you can easily enter a search pattern.

1. To access the **Volumes** dialog from the **Storage** section of the **Resource Group Details** page:
   a. On the menu bar, click **Lifecycle** > **Resource Groups**.
   b. Select a resource group.
   c. On the **Resource Group Details** page, click **View Volumes** in the Storage section.
      You can also click **View Volumes** in the Storage section of the **View All Settings** page, which is also available from the **Resource Group Details** page.
      The **Volumes** dialog is displayed.
2. To access the **Volumes** dialog from the **Add Volumes** wizard:
   a. On the menu bar, click **Lifecycle** > **Resource Groups**.
   b. Select a resource group.
   c. On the **Resource Groups** page, click the **Add Resources** button and choose **Add Volumes**.
   d. When PowerFlex Manager displays the **Add Volume** wizard, click **Add Existing volumes**.
   e. Click **Select Volumes**.
      The **Select Existing Volumes** dialog is displayed.
3. To access the **Volumes** dialog from the **Resize Volume** wizard:
   a. On the menu bar, click **Lifecycle** > **Resource Groups**.
   b. Select a resource group.
   c. On the **Resource Groups** page, click the volume component and choose **Volume Actions** > **Resize**.

d. Click **Select Volume**.
   The **Select Volume** dialog is displayed.

4. Provide the search criteria needed to filter the results to show 50 volumes or fewer:
   a. Enter a volume or datastore name search string in the **Search Text** box.
   b. Optionally, select a **Size** range.
   c. Optionally, **Thick** or **Thin** for the **Type**.
   d. Optionally, select **Enabled** or **Disabled** for the **Compression** setting.
   e. Optionally, select a specific **Storage Pool**.
   f. Click **Search**.
      PowerFlex Manager updates the results to show only those volumes that satisfy the search criteria. If the search returns more than 50 volumes, you must refine the search criteria to return only 50 volumes.
   g. If you are viewing volumes, click **Close** when you are done looking at the search results.
5. If you are adding existing volumes, you can select one or more volumes to add:
   a. Select each volume you want to add.
   b. Click **>>** to add the volumes.

      To remove a selected volume, click the checkbox for the volume to remove and click **<<**.

   c. When you are ready to add the selected volumes, click **Add**.
6. If you are selecting a volume for a resize operation, you can select the volume that you want to resize and click **Apply**.

Related information

Adding volumes to a resource group
Adding volumes to a resource group (mirroring)

Updating the details for a resource group

When new resources are added or removed outside of PowerFlex Manager, you can update the resource group details to pull in those changes. If nodes are removed externally, updating the resource group details does not pull in the changes.

You can update the details for a new resource group that was deployed with PowerFlex Manager, or for an existing resource group that includes resources that were not originally deployed with PowerFlex Manager.

If the PowerFlex gateway used in the resource group is currently being updated on the **Resources** page, PowerFlex Manager does not enable you to update the resource group details.

Whenever you update the resource group details, PowerFlex Manager pulls in a representation of the components from the VMware vCenter cluster, PowerFlex gateway, or CloudLink Center that was specified for the resource group. These components might include new nodes, volumes, SVMs, or management virtual machines. When a new component is added to the definition of a resource group, PowerFlex Manager does not modify the component.

If the CloudLink Center for the resource group shuts down, PowerFlex Manager loses communication with the CloudLink Center. If the CloudLink Center is part of a cluster, PowerFlex Manager moves to another CloudLink Center when you update the resource group details.

Before you update the details for a resource group, ensure that you run inventory for VMware vCenter and the PowerFlex gateway.

1. On the menu bar, click **Lifecycle** > **Resource Groups**.
2. Select the resource group.
3. On the **Resource Group Details** page, under **More Actions**, click **Update Resource Group Details**.
4. Review the **OS Credentials** page, and click **Next**.

   PowerFlex Manager shows all nodes, SVMs, and management virtual machines, along with their credentials. This enables you to update the username and password if it has changed for any of these items.

5. Review the **Inventory Summary** page, and click **Next**.

   PowerFlex Manager does a rediscovery of the resource group, and tracks whether items are new to the resource group.

   In the **Inventory Update** column, PowerFlex Manager shows which items are added, removed, or kept as is within the resource group:

   - **Adding** indicates that an item was added externally and will now be added to the resource group.
   - **Removing** indicates that an item was removed externally and will now be removed from the resource group.

- **No Change** indicates that an item was not modified externally and will be kept in the resource group.

6. Review the **Summary** page, and click **Finish**.
   PowerFlex Manager checks the configuration and inventory of selected resources to ensure that everything is configured properly. The resource group is moved to the **In Progress** state, and any new component types are displayed on the **Resource Group Details** page. **Recent Activity** shows that a new deployment started.

Confirming resource group settings

If a PowerFlex Manager upgrade added new required fields to components within the template from which a resource group was deployed, the **Confirm Resource Group Settings** button is displayed on the **Resource Groups** page. Although confirming resource group settings is not mandatory, certain resource group or resource functions are not available until this step is complete.

Click **Confirm Resource Group Settings** to launch the **Upgrade Resource Group Components** window. Fields in this window vary depending on which components contain newly required settings. Complete all the displayed fields, and click **Save**.

Removing a resource group

You can remove a resource group that is no longer required. PowerFlex Manager supports two types of removal:

- Delete the entire resource group, which deletes the deployment information and also makes any required configuration changes for components that are associated with the resource group.
- Remove the deployment information for the resource group without making any configuration changes to the deployed components.

If the node has an NSX-T configuration, you can remove the deployment information for a resource group, but not delete the resource group entirely. PowerFlex Manager also does not allow you to delete a resource group if the PowerFlex gateway used in the resource group is being updated on the **Resources** page.

Standard users can delete only the resource groups that they have deployed.

To remove a resource group, perform the following steps:

1. On the menu bar, click **Lifecycle** > **Resource Groups**.
2. Select the resource group.
3. On the **Resource Group Details** page, under **More Actions**, click **Remove Resource Group**.
4. In the **Remove Resource Group** dialog box, select the **Resource group removal type**:
   - **Delete Resource Group** makes configuration changes to the nodes, switch ports, virtual machine managers, and PowerFlex to unconfigure those components. Also, it returns the components to the available inventory.
   - **Remove Resource Group** removes deployment information, but does not make any configuration changes to the nodes, switch ports, virtual machine managers, and PowerFlex. Also, it returns the components to the available inventory.
5. If you choose **Remove Resource Group**, perform the following steps:
   a. To keep the nodes in the inventory, select **Leave nodes in PowerFlex Manager inventory and set state to** and select the state:
      - **Managed**
      - **Unmanaged**
      - **Reserved**
   b. To remove the nodes, select **Remove nodes from the PowerFlex Manager inventory**.
   c. Click **Remove**.
6. If you choose **Delete Resource Group**, perform the following steps:
   a. Select **Delete Clusters(s) and Remove from vCenter** to delete and remove the clusters from VMware vCenter.
   b. Select **Remove Protection Domain and Storage Pools from PowerFlex** to remove the protection domain and storage pools that are created during the resource group deployment.

      If you select this option, you must select the target PowerFlex gateway. The PowerFlex gateway is not removed. PowerFlex Manager removes only the protection domain and storage pools that are part of the resource group. If multiple resource groups are sharing a protection domain, you might not want to delete the protection domain.

      For a compression enabled resource group, PowerFlex Manager deletes the acceleration pool and the DAX devices when you delete the resource group.

c. Select **Delete Machine Group and remove from CloudLink Center** to clean up the related components in CloudLink Center.

   CloudLink Center cleanup includes the deletion of the machine group, keystore, and approved network that is related to the resource group being deleted. These components are removed from CloudLink Center only if all the machines that are associated to the machine group are deleted first. If all the machines that are related to this machine group are not removed, the cleanup does not succeed because there are machines that are associated with the machine group.

d. If you are certain that you want to proceed, type **`DELETE RESOURCE GROUP`**.
e. Click **Delete**.

Related information

Deploying a resource group
Deploying a resource group (mirroring)

Canceling a resource group

Canceling a resource group stops a scheduled/pending, or in-process resource group deployment. After canceling a resource group, you can update the content and retry the deployment later.

1. On the menu bar, click **Lifecycle** > **Resource Groups**.
2. Select a resource group that is in **In Progress** or **Pending state**.
3. On the **Resource Group Details** page, click **Cancel** under **More Actions**.
4. On the **Cancel Deployment** page, click **Cancel Deployment**.

   You cannot cancel existing resource groups or resource groups that are used for managing the firmware.

   When you cancel a deployment, a banner displays at the top of the screen that displays a warning.

Preparing a template for a resource group deployment

This section includes tasks for preparing a template that will be used to deploy a set of infrastructure resources in a resource group.

A template is an object in PowerFlex Manager that represents resource types and their required configuration and topology. A template specifies requirements for the deployment of a set of infrastructure resources through PowerFlex Manager's automation workflows.

On the **Templates** page, you can access the default sample templates or create templates that meet your specific requirements. For most environments, you can clone one of the sample templates that are provided with PowerFlex Manager and edit as needed. Choose the sample template that is most appropriate for your environment.

PowerFlex Manager allows you to create templates that support storage-only and compute-only resource group deployments.

After you create a template, PowerFlex Manager saves the template in a draft state. You must publish the template before deploying it.

Life cycle administrators can view and use only those templates for which they have been granted permission by a super user.

Related information

Deploying and provisioning

Basic tasks

This section provides basic tasks for template management.

Clone a template

The **Clone** feature allows you to copy an existing template into a new template. A cloned template contains the components that existed in the original template. You can edit it to add additional components or modify the cloned components.

For most environments, you can clone one of the sample templates that are provided with PowerFlex Manager and edit as needed. Choose the sample template that is most appropriate for your environment. If you already created a template using the **Add a template** option, skip this task.

To clone an existing template:

1. On the menu bar, click **Templates**.
2. Click **Sample Template** > **Select** and click a template from the list.

   You can also click **Clone** on the **My Templates** page if you want to clone one of your own templates, rather than one of the sample templates.
3. In the **Clone Template** dialog box, enter a template name in the **Template Name** box.
4. Select a template category from the **Template Category** list. To create a template category, select **Create New Category**.
5. In the **Template Description** box, enter a description for the template.
6. To specify the version to use for compliance, select the version from the **Firmware and Software Compliance** list or choose **Use PowerFlex Manager appliance default catalog**.
7. Indicate **Who should have access to the resource group deployed from this template** by selecting one of the following options:
   - To restrict access to super users and specified life cycle administrators, drive replacers, and storage administrators select **Only PowerFlex SuperUser and specfied additional users**.
     a. Click **Add User(s)** to add one or more LifecycleAdmin or DriveReplacer or StorageAdmin users to the list displayed.
     b. Select which users will have access to this resource group.
     c. To delete a user from the list, select the user and click **Remove User(s)**.
     d. After adding the users, select or clear the checkbox next to the users to grant or block access.
   - To grant access to super users and all life cycle administrators, drive replacers, and storage administrators, select **PowerFlex SuperUser and all additional users**.
8. Click **Next**.
9. On the **Additional Settings** page, provide new values for the **Network Settings**, **PowerFlex Gateway Settings**, and **Node Pool Settings**.
10. Click **Finish**.
11. To modify the cluster components: Node (software only) and PowerFlex cluster, see *Edit a Template* for more information.

Related information

Configuring block storage
Edit a template

Sample templates

This topic lists the PowerFlex compute-only node and PowerFlex storage-only node sample templates that are provided with PowerFlex Manager.

To search for a sample template, click the filter icon and search for the template in the **Filter Templates** field.

PowerFlex compute-only node sample templates require that PowerFlex storage-only nodes are deployed first.

Table 42. Sample templates

Template name	Description	Minimum number of nodes required	Minimum number of data networks required
Compute ESXi 100Gb - 2 Data - LACP	An ESXi PowerFlex compute-only node 100 GB networking deployment with full network automation.	3	2

Table 42. Sample templates (continued)

Template name	Description	Minimum number of nodes required	Minimum number of data networks required
Compute ESXi 25Gb - 2 Data - LACP	A standard ESXi PowerFlex compute-only node deployment with full network automation.	3	2
Compute ESXi 25Gb - 2 Data - LACP - Partial Network Automation	A standard ESXi PowerFlex compute-only node deployment with partial network automation.	3	2
Compute ESXi 25Gb - 2 Data - Trunked	An ESXi PowerFlex compute-only node deployment with full network automation. ● Management and customer networks on two ports in mode1 bond (interface1 port1 and interface2 port2) ● Two networks (data 1 and data 2) as SDC-storage node traffic on the other two ports ● 25 GB NIC	3	2
Compute Linux 100Gb - 4 Data - Trunked	A Linux PowerFlex compute-only node deployment with full network automation. ● Management and customer networks on two ports in mode1 bond (interface1 port1 and interface2 port1) ● Two networks (data 2 and data 3) on interface1 port2 ● Two networks (data 1 and data 4) on interface2 port2 ● 100 GB NIC	3	4
Compute Linux 25Gb - 2 Data - LACP - Partial Network Automation	A Linux PowerFlex compute-only node deployment with partial network automation.	3	2
Compute-Linux-25Gb-2 Data-Trunked	A Linux PowerFlex compute-only node deployment with full network automation. ● Management and customer networks on two ports in mode1 bond (interface1 port1 and interface2 port2) ● Two networks (data 1 and data 2) as SDC-storage node traffic on the other two ports ● 25 GB NIC	3	2
Compute-Linux-25Gb-4 Data-Trunked	A Linux PowerFlex compute-only node deployment with full network automation. ● Management and customer networks on two ports in mode1 bond (interface1 port1 and interface2 port1) ● Two networks (data 2 and data 3) on interface1 port2 ● Two networks (data 1 and data 4) on interface2 port2 ● 25 GB NIC	3	4
Storage 100 Gb - 2 Data - LACP	A Linux PowerFlex storage-only node deployment using erasure coding with full network automation.	5	2
Storage 100 Gb - 2 Host Data - Trunked	A Linux PowerFlex storage-only node deployment using erasure coding with full network automation. ● Management network in mode1 bond on two ports ● Two networks (data 1 and data 2) as SDC-storage node traffic	5	2

Table 42. Sample templates (continued)

Template name	Description	Minimum number of nodes required	Minimum number of data networks required
	• Two networks (data 3 and data 4) as storage node-storage node traffic • 100 GB NIC • interface1 port1: management, data1 SDC-storage node traffic • interface1 port2: data 3 storage node-storage node traffic • interface2 port1: management, data 2 SDC-storage node traffic • interface2 port2: data 4 storage node-storage node traffic		
Storage 100 Gb - 4 Data - Trunked	A Linux PowerFlex storage-only node deployment using erasure coding with full network automation. • Management network in mode1 bond on the four ports • Four data networks, each on a different port • 100 GB NIC • interface1 port1: management, data 1 • interface1 port2: management, data 2 • interface2 port1: management, data 3 • interface2 port2: management, data 4	5	4
Storage 25 Gb - 2 Data - LACP - Compression Enabled with persistent memory	A Linux PowerFlex storage-only node deployment using erasure coding with full network automation.	5	2
Storage 25 Gb - 2 Data - LACP - Compression Enabled	A Linux PowerFlex storage-only node deployment using erasure coding with full network automation. Requires nodes with NVDIMM.	5	2
Storage 25 Gb - 2 Data - LACP - Encryption Enabled	A Linux PowerFlex storage-only node deployment using erasure coding with full network automation.	5	2
Storage 25 Gb - 2 Data - LACP - NVMe/TCP Enabled	A Linux PowerFlex storage-only node deployment using erasure coding with full network automation.	5	2
Storage 25 Gb - 2 Data - LACP - Partial Network Automation	A Linux PowerFlex storage-only node deployment using erasure coding with partial network automation.	5	2
Storage 25 Gb - 2 Host Data - Trunked	A Linux PowerFlex storage-only node deployment using erasure coding with full network automation. • Management network in mode1 bond on two ports • Two networks (data 1 and data 2) as SDC-storage node traffic • Two networks (data 3 and data 4) as storage node-storage node traffic • 25 GB NIC • interface1 port1: management, data 1 SDC-storage node traffic • interface1 port2: data 3 storage node-storage node traffic • interface2 port1: management, data 2 SDC-storage node traffic	5	2

Table 42. Sample templates (continued)

Template name	Description	Minimum number of nodes required	Minimum number of data networks required
	• interface2 port2: data 4 storage node-storage node traffic		
Storage 25 Gb - 4 Data - Trunked	A Linux PowerFlex storage-only node deployment using erasure coding with full network automation. • Management network in mode1 bond on the four ports • Four data networks, each not in a bond • 25 GB NIC • interface1 port1: management, data 1 • interface1 port2: management, data 2 • interface2 port1: management, data 3 • interface2 port2: management, data 4 This template was called **Storage-NoBonding** in previous releases.	5	4
Management - CloudLink Center	A two-node CloudLink Center cluster deployment, for use in templates with encryption.	2(VMs)	N/A

Add a template

You can create a template, clone the components of an existing template into a new template, or import a pre-existing template.

For most environments, you can clone one of the sample templates that are provided with PowerFlex Manager and edit as needed. Choose the sample template that is most appropriate for your environment.

1. On the menu bar, click **Templates**.
2. On the **Templates** page, click **Create**.
3. In the **Create** dialog box, select one of the following options:
   - **Clone an existing PowerFlex Manager template**
   - **Upload External Template**
   - **Create a new template**

   If you select **Clone an existing PowerFlex Manager template**, select the **Category** and the **Template to be Cloned**. The components of the selected template are in the new template.
4. Enter a **Template Name**.
5. From the **Template Category** list, select a template category. To create a category, select **Create New Category** from the list.
6. Enter a **Template Description** (optional).
7. To specify the version to use for compliance, select the version from the **Firmware and Software Compliance** list or choose **Use PowerFlex Manager appliance default catalog**.
8. Specify the resource group permissions for this template under **Who should have access to the resource group deployed from this template?** by performing one of the following actions:
   - To restrict access to super users and specified life cycle administrators, drive replacers, and storage administrators select **Only PowerFlex SuperUser and specfied additional users**.
     a. Click **Add User(s)** to add one or more LifecycleAdmin or DriveReplacer or StorageAdmin users to the list displayed.
     b. Select which users will have access to this resource group.
     c. To delete a user from the list, select the user and click **Remove User(s)**.
     d. After adding the users, select or clear the checkbox next to the users to grant or block access.
   - To grant access to super users and all life cycle administrators, drive replacers, and storage administrators, select **PowerFlex SuperUser and all additional users**.
9. Click **Next**.
10. In **Additional Settings** provide the new values for the following settings:

- **Network**
- **PowerFlex Gateway**
- **Node Pool**

11. Click **Finish**.

Build and publish a template

After creating a template using the **Create** feature, you use the template builder page to build and publish the customized template. Publishing a template indicates that a template is ready for deployment.

1. Click **Modify Template**.
2. To add a component type to the template, click **Add Node**, **Add Cluster**, or **Add VM** at the top of the template builder.

   The corresponding **<component type> component** dialog box appears.
3. If you are adding a node, choose one of the following network automation types:
   - **Full Network Automation**
   - **Partial Network Automation**

   When you choose **Partial Network Automation**, PowerFlex Manager skips the switch configuration step, which is normally performed for a resource group with **Full Network Automation**. Partial network automation allows you to work with unsupported switches. However, it also requires more manual configuration before deployments can proceed successfully. If you choose to use partial network automation, you give up the error handling and network automation features that are available with a full network configuration that includes supported switches.

   In the **Number of Instances** box, provide the number of component instances that you want to include in the template.
4. If you are adding a cluster, in the **Select a Component** box, choose the **PowerFlex Cluster** type.
5. If you are adding a VM, in the **Select a Component** box, choose one of the following cluster types:
   - **CloudLink Center**
   - **PowerFlex Gateway**
6. Under **Related Components**, perform one of the following actions:
   - To associate the component with all existing components, click **Associate All**.
   - To associate the component with only selected components, click **Associate Selected** and then select the components to associate.

   Based on the component type, specific settings and properties appear automatically that are required and can be edited.
7. Click **Save** to add the component to the template builder.
8. Repeat steps 1 through 6 to add additional components.
9. After you finish adding components to your template, click **Publish Template**.

   A template must be published to be deployed. It remains in draft state until published.

   After publishing a template, you can use the template to deploy a resource group on the **Resource Groups** page.

Related information

Support for full and partial network automation
Deploying a resource group
Support for full and partial network automation (mirroring)
Deploying a resource group (mirroring)

Edit a template

You can edit an existing template to change its draft state to "published" for deployment or to modify its components and their properties.

1. On the menu bar, click **Templates**.
2. Open a template, and click **Modify Template**.
3. Make changes as required to the settings for components within the template.

   Based on the component type, required settings and properties are displayed automatically. You can edit these settings by performing these steps:

NOTE: If an RCM associated with the template is modified, a wrench icon labelled **Modified** is displayed. However, if the update file is moved or deleted, the wrench icon with the text, Needs Attention, is displayed.

a. To edit PowerFlex cluster settings, select the **PowerFlex Cluster** component and click **Modify**. Make the necessary changes and click **Save**.
b. To edit node settings, select the **Node** component and click **Modify**. Make the necessary changes, and click **Save**.

4. Optionally, click **Publish Template** to make the template ready for deployment.

Related information

Component types

(Optional) Edit template information

You can edit the information assocaited with a template.

1. On the menu bar, click **Templates**.
2. On the **Templates** page, click the template that you want to edit and click **Modify Template** in the right pane.
3. On the template builder page, in the right pane, click **Modify**.
4. In the **Modify Template Information** dialog box, enter a template name in the **Template Name** field.
5. Select a template category from the **Template Category** list. To create a template category, select **Create New Category**.
6. In the **Template Description** box, enter a description for the template.
7. To specify the version to use for compliance, select the version from the **Firmware and Software Compliance** list or choose **Use PowerFlex Manager appliance default catalog**.
8. Indicate **Who should have access to the resource group deployed from this template** by selecting one of the following options:
   - To restrict access to super users and specified life cycle administrators, drive replacers, and storage administrators select **Only PowerFlex SuperUser and specfied additional users**.
     a. Click **Add User(s)** to add one or more LifecycleAdmin or DriveReplacer or StorageAdmin users to the list displayed.
     b. Select which users will have access to this resource group.
     c. To delete a user from the list, select the user and click **Remove User(s)**.
     d. After adding the users, select or clear the checkbox next to the users to grant or block access.
   - To grant access to super users and all life cycle administrators, drive replacers, and storage administrators, select **PowerFlex SuperUser and all additional users**.
9. Click **Save**.

Deploying a resource group

Deployment is the automated process of selecting and configuring specific resource requirements that are outlined in a template using the integrated automation workflows that are provided with PowerFlex Manager. You cannot use a template that is in a draft state to deploy a resource group. Publish the template before using it to deploy a resource group.

1. On the menu bar, click one of the following:
   - **Lifecycle** > **Resource Groups**. Then, click **Deploy New Resource Group**.
   - **Lifecycle** > **Templates**. Then, click **Deploy**.

   The **Deploy Resource Group** wizard opens.
2. On the **Deploy Resource Group** page, perform the following steps:
   a. From the **Select Published Template** list, select the template to deploy a resource group.
   b. Enter the **Resource Group Name** (required) and **Resource Group Description** (optional) that identifies the resource group.
   c. To specify the version to use for compliance, select the version from the **Firmware and Software Compliance** list or choose **Use PowerFlex Manager appliance default catalog**.

      You cannot select a minimal compliance version when you deploy a new resource group, since it only includes server firmware updates. The compliance version for a new resource group must include the full set of compliance update capabilities. PowerFlex Manager does not show any minimal compliance versions in the **Firmware and Software Compliance** list.

PowerFlex Manager checks the VMware vCenter version to determine if it matches the VMware ESXi version for the selected compliance version. If the ESXi version is greater than the vCenter version, PowerFlex Manager blocks the resource group deployment and displays an error. PowerFlex Manager instructs you to upgrade vCenter first, or use a different compliance version that is compatible with the installed vCenter version.

NOTE: Changing the firmware repository might update the firmware level on nodes for this resource group. The global default firmware repository maintains the firmware on the shared devices.

d. Indicate **Who should have access to the resource group deployed from this template** by selecting one of the following options:
   - To restrict access to super users and specified life cycle administrators, drive replacers, and storage administrators select **Only PowerFlex SuperUser and specfied additional users**.
     i. Click **Add User(s)** to add one or more LifecycleAdmin or DriveReplacer or StorageAdmin users to the list displayed.
     ii. Select which users will have access to this resource group.
     iii. To delete a user from the list, select the user and click **Remove User(s)**.
     iv. After adding the users, select or clear the checkbox next to the users to grant or block access.
   - To grant access to super users and all life cycle administrators, drive replacers, and storage administrators, select **PowerFlex SuperUser and all additional users**.

3. Click **Next**.
4. On the **Deployment Settings** page for **Device Group Spare Capacity** select **Custom** > **Nodes**.

   You must set the spare capacity limit of the device group in terms of nodes for this release. The default spare capacity is one node, and the maximum is two nodes.

   Utilizing spare devices is a feature that is recommended only for advanced configurations.

5. Click **Next**.
6. On the **Schedule Deployment** page, select one of the following options and click **Next**:
   - **Deploy Now**—Select this option to deploy the resource group immediately.
   - **Deploy Later**—Select this option and enter the date and time to deploy the resource group.
7. Review the **Summary** page.
   The **Summary** page gives you a preview of what the resource group will look like after the deployment.
8. Click **Finish** when you are ready to begin the deployment. If you want to edit the resource group, click **Back**.

   PowerFlex Manager displays a message indicating that a new job is being created for this resource group.

   PowerFlex Manager creates a configuration file that is called `/etc/sysctl.d/powerflex-so.conf` for a storage-only resource group. PowerFlex Manager maintains default values in a conf file for resource group deployment settings. If you have an old `sysctl.conf` file from a previous release, it overrides these settings. If you must customize these settings, you can do it in `sysctl.conf`.

Related information

Life cycle
Viewing resource group details
Adding components to a resource group
Build and publish a template
Component types
Removing a resource group

Additional template management tasks

This section provides additional tasks for template management.

Add cluster component settings to a template

To deploy a PowerFlex compute-only node , you must add VMware and PowerFlex cluster components settings to a template. For a PowerFlex storage-only node deployment, you must add cluster settings for the PowerFlex cluster component only.

1. On the template builder page, click **Add Cluster**.

2. In the **Cluster Component** dialog box, from the **Select a Component** list, select **VMware cluster**, or **PowerFlex Cluster**.
3. Under **Related Components**, select the components that you want to map to the selected cluster instance.
4. Click **Continue**.
5. Under **Cluster Settings**, specify the settings that you want to configure on the cluster components and click **Add**.

Related information

Cluster component settings

Add a new interface

Use the **Add New Interface** feature to create a network interface to match to a network card on a node when deploying a template.

PowerFlex Manager supports a network layout for compute-only, and storage-only deployments. This network layout allows you to use trunk ports and port channels for data networks, instead of access ports. In this type of configuration, both data networks are on both NICs, teamed or bonded together in the operating system.

- For a compute-only deployment, the first port on both interfaces is for trunk traffic, whereas the second port is for data 1 and data 2.
- For a storage-only deployment, the first port is for trunk traffic and data 1, whereas the second port is for data 2. PowerFlex Manager still supports the legacy network layout, but the sample templates use the new configuration.

1. On the node component page, under **Network Settings**, click **Add New Interface**.
2. Enter the following information for the new interface:
   - **Port Layout**—Select the NIC type from the list.

     For a VMware ESXi deployment with an NVIDIA card, you must select the two port, 25-gigabit NIC type.
3. Enter the network VLANs for each port.
   a. Click **Choose Networks** for a port.
   b. To add one or more networks to the port, select **Add Networks to this Port**, then click the check box for each network you want to add from the **Available Networks** list. Alternatively, click the check box in the upper left corner next to the **Name** label to select all the available networks.

      If you want to filter the list by network type, select a **Network Type**, then enter a name or VLAN ID to search.

      Click **>>** to move the selected items to the **Selected Networks** list on the right.

   c. To mirror network settings from another port for which you have already chosen the network VLANs, select **Mirror this Port with Another Port**. Select the other interface and port from which you want to mirror this port.
   d. Click **Save**.
4. To view the list of nodes that match the network configuration parameters, click **Validate Settings**.

   The list of nodes is filtered according to the target boot device and NIC type settings specified.

   When you enable PowerFlex settings for the node, the **Validate Settings** page filters the list of nodes according to the supported storage types (NVMe, All flash, and HDD). Within the section for each storage type, the nodes are also sorted by health, with the healthy (green) nodes displayed first and the critical (red) nodes displayed last.

NOTE: If you select the same network on multiple interface ports or partitions, PowerFlex Manager creates a team or bond on systems with the VMware ESXi operating system. This configuration enables redundancy.

Add a new static route

A static route allows nodes to communicate across different networks. Use the add a new static route feature to create a static route in a template.

1. On the node component page, under **Network Settings**, click **Enabled** under **Static Routes**.
2. Click **Add New Static Route**.
3. Enter the following information for the static route:
   - **Source Network**—Select the PowerFlex data network that is the source.

If you add or remove a network for a port, the **Source Network** list still shows the old networks. In order to see the changes, you must save the node settings and edit the node again.

- **Destination Network**—Select the PowerFlex data network that is the destination for the static route.
- **Gateway**—Enter the IP address for the gateway.

View template details

View more details about the topology of the components in the template builder.

1. On the **Templates** page, select a template.
2. In the right pane, click **View Details**.
3. To view all the component settings, click **More Actions** > **View All Settings** in the right pane.

   The **Template Settings** page shows each top-level component on a separate tab. Click a tab for a component to see the details for the component.

   NOTE: If an RCM is mapped to the template and is modified, a wrench icon with the text, **Modified,** is displayed. However, if the update file is moved or deleted, the wrench icon with the text, **Needs Attention**, is displayed.

Related information

Component types

Export a template

The export process packages a selected template into a ZIP file, optionally securing it with encryption so it can be shared or stored safely. Once the template is exported, it becomes a portable file that can be imported into another system instance or reused wherever needed.

1. On the menu bar, click **Templates**.
2. Select the template that you want to export.
3. Click **Export**.
4. In the **Export Template to ZIP File** window, enter values as follows:
   a. Enter a name for the template file in the **File Name**.
   b. If you have set an encryption password, select **Use Encryption Password from Backup Setting** to use that password. To set an encryption password, clear this option.
   c. If you clear **Use Encryption Password from Backup Setting**, two additional fields display. Enter a new password in the **Set File Encryption Password** field and enter the password again to confirm it.
5. Click **Export** to download the file. Select a location to save the file and click **OK**.

Import a template

The import template feature allows you to import the components of an existing template and its component configurations into a template. For example, you can create a template that defines a specific cluster and node topology and import this template definition into another template.

After importing, you can modify the component properties of the imported components. Editing an imported template does not affect the original template.

As an alternative, you can also start from a new template and import an existing template as you create it. This is a better approach to use. To do this, you select **Create**, then choose **Clone an existing VxFlex Manager template** in the wizard.

1. On the menu bar, click **Templates**.
2. On the **Templates** page, select the template into you want to import an existing template and click **Edit** in the right pane.
3. On the **Template Builder** page, in the right pane, click **Import Template**.
4. In the **Import Template** dialog box, select a specific template from the **Select a template** list and click **Import**.

Upload an external template

Uploading an external template involves bringing a template file into the system, providing its key identifying details, and defining how it should be categorized and accessed.

After the template is imported, you can further refine it by configuring additional settings such as network, operating system, cluster, and node-pool parameters so it is fully aligned with your system requirements.

1. On the **Templates** page, click **Create**.
2. In the **Create** dialog box, select **Upload External Template**.
   The **Upload External Template** page appears.
3. Click **Browse** to select the exported file from your directory.
4. Select the **Use Encryption Password from Backup Settings** checkbox, if you have set the encryption password in **Backup and Restore**.

   Clear the **Use Encryption Password from Backup Settings** checkbox if you have not set the encryption password in **Backup and Restore**.

   If you clear the **Use Encryption Password from Backup Settings**, you must enter the encryption password in the **Encryption Password** field.
5. Enter the template name in the **Template Name** field.
6. Select a template category from the **Template Category** list.

   If you want to create a template category, select **Create New Category**.
7. In the **Template Description** box, enter a description for the template.
8. To specify the version to use for compliance, select the version from the **Firmware and Software Compliance** list or choose **Use PowerFlex Manager appliance default catalog**.

   NOTE: Changing the firmware repository might update the firmware level on nodes for this resource group. Firmware on shared devices is maintained by the global default firmware repository.
9. Indicate **Who should have access to the resource group deployed from this template** by selecting one of the following options:
   - To restrict access to super users and specified life cycle administrators, drive replacers, and storage administrators select **Only PowerFlex SuperUser and specfied additional users**.
     a. Click **Add User(s)** to add one or more LifecycleAdmin or DriveReplacer or StorageAdmin users to the list displayed.
     b. Select which users will have access to this resource group.
     c. To delete a user from the list, select the user and click **Remove User(s)**.
     d. After adding the users, select or clear the checkbox next to the users to grant or block access.
   - To grant access to super users and all life cycle administrators, drive replacers, and storage administrators, select **PowerFlex SuperUser and all additional users**.
10. Click **Upload and Continue**.
    The **Additional Settings** page appears.
11. On the **Additional Settings** page, provide new values for the **Network Settings**, **OS Settings**, **Cluster Settings**, **PowerFlex Gateway Settings**, and **Node Pool Settings**.
12. Click **Finish**.

Deploy a CloudLink Center

Use this task to deploy a CloudLink Center. Although PowerFlex Manager supports up to three instances of CloudLink Center, two are recommended.

This capability is available on the PowerFlex appliance and PowerFlex rack offerings only.

Ensure the following:

- Hypervisor management or PowerFlex management networks are added to PowerFlex Manager.
- A VMware vCenter with a valid data center, cluster, network, and datastore is discovered.

1. On the menu bar, click **Templates** > **Sample Templates**.
2. On the **Sample Templates** page, click **Management - CloudLink Center**.
3. Click **More Actions** > **Clone**.
4. In the **Clone Template** wizard, do the following:
   a. Enter a template name.

b. From the **Template Category** list, select a template category. To create a category, select **Create New Category** from the list.
c. Enter a **Template Description** (optional).
d. To specify the version to use for compliance, select the version from the **Firmware and Software Compliance** list or choose **Use PowerFlex Manager appliance default catalog**.
e. Specify the resource group permissions for this template under **Who should have access to the resource group deployed from this template?**:
   - To restrict access to super users and specified life cycle administrators, drive replacers, and storage administrators select **Only PowerFlex SuperUser and specfied additional users**.
     i. Click **Add User(s)** to add one or more LifecycleAdmin or DriveReplacer or StorageAdmin users to the list displayed.
     ii. Select which users will have access to this resource group.
     iii. To delete a user from the list, select the user and click **Remove User(s)**.
     iv. After adding the users, select or clear the checkbox next to the users to grant or block access.
   - To grant access to super users and all life cycle administrators, drive replacers, and storage administrators, select **PowerFlex SuperUser and all additional users**.
f. Click **Next**.

5. On the **Additional Settings** page, do the following:
   a. Under **Network Settings**, select **Hypervisor Network** (PowerFlex management network).
   b. Under **OS Settings**, select **CLC** credential or click **+** to create a credential with root or CloudLink user.
   c. Under **CloudLink Settings**, do the following:
      i. Select the secadmin credential from the list or click **+** to create a secadmin credential and do the following:
         i. Enter **Credential Name**.
         ii. Enter **User Name** as **`secadmin`**.
         iii. Leave the **Domain** empty.
         iv. Enter the password for **`secadmin`** in **Password** and **Confirm Password**.
         v. Select **V2** in **SNMP Type**.
         vi. Click **Save**.
      ii. Select a license file from the list based on the types of drives or click **+** to upload a license through the **Add Software License** page.

      NOTE: For SSD/NVMe drives, upload a capacity-based license. For SED drives, upload an SED-based license.

   d. Under **Cluster Settings**, select **Management vCenter**.
   e. Click **Finish**.
6. Select the **VMware Cluster** and click **Edit** > **Continue**.
   a. Under **Cluster Settings**, select **Datacenter Name**, and then select **Cluster Name** from the drop-down list.
   b. Under **vSphere Network Settings**, select the hypervisor management port group or PowerFlex management port group.
   c. Click **Save**.

   NOTE: To deploy a CloudLink Center from PowerFlex Manager, you need a Management vCenter, data center, and cluster along with DvSwitch port groups for PowerFlex management or hypervisor management.

7. Select the **VM** and click **Edit** > **Continue** (by default, the number of CloudLink instances is two and PowerFlex Manager supports up to three instances).
   a. Under **VM Settings** select the **Datastore** and **Network** from the drop-down list.
   b. Under **CloudLink Settings** select the following:
      i. For **Host Name Selection**, either select **Specify At Deployment Time** to manually enter at deployment time or **Auto Generate** to have PowerFlex Manager generate the name.
      ii. Enter the vault passwords.

      NOTE: Other details such as operating system credentials, NTP, and secadmin credentials are auto populated.

8. Under **Additional CloudLink Settings**, you can choose either or both of the following settings:
   - **Configure Syslog Forwarding**
     a. Select the checkbox to configure syslog forwarding.
     b. For **Syslog Facility**, select the **syslog** remote server from the list.

- **Configure Email Notifications**
  a. Select the checkbox to configure email alerts.
  b. Specify the IP address of the email server.
  c. Specify the port number for the email server. The default port is 25. Enter the port numbers in a comma-separated list, with values between 1-65535.
  d. Specify the email address for the sender.
  e. Optionally, specify the username and password.

9. Click **Save**.
10. Click **Publish Template** and click **Yes** to confirm.
11. In the **Deploy Resource Group** wizard, do the following:
    a. Select the published template from the drop-down list, and enter **Resource Group Name** and description.
    b. Select who should have the access to the resource group and click **Next**.
    c. Provide a **Hostname** and click **Next**.
    d. Select **Deploy Now** or **Schedule deployment** and click **Next**.
    e. Review the details in the **Summary** page and click **Finish**.

Delete a template

The **Delete Template** option allows you to delete a template from PowerFlex Manager.

1. On the menu bar, click **Templates**.
2. Select the template that you want to delete. Click **More Actions** > **Delete Template** in the right pane.
3. Click **Yes** to confirm the deletion.

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Performing life cycle operations for a resource group (mirroring)

The following section provides tasks for deploying resource groups and building templates.

This capability is available on the PowerFlex appliance and PowerFlex rack offerings only.

Deploying a resource configuration in a resource group (mirroring)

The following section includes tasks for deploying and managing system resources in a resource group.

A resource group is a PowerFlex Manager object representing the infrastructure that is configured during a deployment. This object persists in PowerFlex Manager if the infrastructure is required. A resource group is viewed and interacted with to perform compliance, health, and life cycle management on the deployed infrastructure. PowerFlex Manager supports the deployment of three types of PowerFlex resource groups:

- Hyperconverged resource groups
- Storage-only resource groups
- Compute-only resource groups

NOTE: Standard users see only the details of the resource groups that they created or for which they have permission.

The **Resource Groups** page displays the resource groups that are in the following states in both **Tile** and **List** view.

Table 43. Resource group states

State	Description
Healthy	The resource group is successfully deployed and is healthy.
Warning	One or more resources in the resource group requires corrective action.
Critical	The resource group is in a severely degraded or nonfunctional state and requires attention.
Pending	The deployment is scheduled for a later time or date.
In Progress	The resource group deployment is in progress, or has other actions currently in process, such as a node expansion or removal.
Cancelled	The resource group deployment has been stopped. You can update the resources or retry the deployment, if necessary.
Incomplete	The resource group is not fully functional because it has no volumes that are associated with it. Click **Add Resources** to add volumes.
Service Mode	The resource group is in service mode.
Lifecycle Mode	The resource group is in life cycle mode. Resource groups in life cycle mode are enabled with health and compliance monitoring, and non-disruptive upgrade features only.
Managed Mode	The resource group is in managed mode. Resource groups in managed mode are enabled with health and compliance monitoring, non-disruptive update, automated resource addition, and automated resource replacement features.

On the **Resource Groups** page, you can:

- Click **Deploy New Resource Group** to deploy a new resource group.

NOTE: Standard users are allowed to deploy only those resource groups they have created or for which they have permissions.

- Click **Add Existing Resource Group** to add an existing resource group.

To switch views, click the **Tile View** icon or **List View** icon .

To view the resource groups based on a particular resource group state, select an option from the **Filter By** drop-down list. Alternately, in the **Graphical** view, click the graphic in a particular state.

In the **Tile** view, each square tile represents a resource group and has the status of the resource group at the bottom of the graphic. The state icon on the graphic indicates the state of the resource group. The components in blue indicate the component types that are in the deployment. The components that are in gray indicate the component types that are not in the resource group.

In the **List** view, the following information displays:

- **Status**—Status of the resource group.
- **Name**—Name of the resource group.
- **Deployed By**—Name of the user who deployed the resource group.
- **Deployed On**—Date and time when the resource group is deployed.
- **Components**—Components used in the resource group.

Click the resource group in the **List** view or **Tile** view to view the following information about the resource group in the right pane:

- Resource group name and description to identify the resource group.
- Name of the user who deployed the resource group.
- Date and time when the resource group is deployed.
- Name of the reference template that is used in the resource group.
- Number of resources that are in the resource group for deployment, based on component type (cluster or node).

Click **View Details** to view more details about the resource group. You can also generate troubleshooting bundles from the resource group details page.

Click the resource group name in the **List** view to open the **Resource Group Details** page.

Click **Update Resources** to update the firmware of all nodes in the resource group that are not compliant.

Basic tasks

This section provides basic tasks for resource group management.

Deploying a resource group (mirroring)

Deployment is the automated process of selecting and configuring specific resource requirements that are outlined in a template using the integrated automation workflows provided with PowerFlex Manager. You cannot use a template that is in a draft state to deploy a resource group. Publish the template before using it to deploy a resource group.

1. On the menu bar, click one of the following:
   - **Lifecycle** > **Resource Groups** click **Deploy New Resource Group**.
   - **Lifecycle** > **Templates** click **Deploy**.

   The **Deploy Resource Group** wizard opens.
2. On the **Deploy Resource Group** page, perform the following steps:
   a. From the **Select Published Template** list, select the template to deploy a resource group.
   b. Enter the **Resource Group Name** (required) and **Resource Group Description** (optional) that identifies the resource group.
   c. To specify the version to use for compliance, select the version from the **Firmware and Software Compliance** list or choose **Use PowerFlex Manager appliance default catalog**.

      You cannot select a minimal compliance version when you deploy a new resource group, since it only includes server firmware updates. The compliance version for a new resource group must include the full set of compliance update capabilities. PowerFlex Manager does not show any minimal compliance versions in the **Firmware and Software Compliance** list.

      PowerFlex Manager checks the VMware vCenter version to determine if it matches the VMware ESXi version for the selected compliance version. If the VMware ESXi version is greater than the VMware vCenter version, PowerFlex

Manager blocks the resource group deployment and displays an error. PowerFlex Manager instructs you to upgrade VMware vCenter first, or use a different compliance version that is compatible with the installed VMware vCenter version.

NOTE: Changing the firmware repository might update the firmware level on nodes for this resource group. The global default firmware repository maintains the firmware on the shared devices.

d. Indicate **Who should have access to the resource group deployed from this template** by selecting one of the following options:
   - To restrict access to super users and specified life cycle administrators, drive replacers, and storage administrators select **Only PowerFlex SuperUser and specfied additional users**.
     i. Click **Add User(s)** to add one or more LifecycleAdmin or DriveReplacer or StorageAdmin users to the list displayed.
     ii. Select which users will have access to this resource group.
     iii. To delete a user from the list, select the user and click **Remove User(s)**.
     iv. After adding the users, select or clear the checkbox next to the users to grant or block access.
   - To grant access to super users and all life cycle administrators, drive replacers, and storage administrators, select **PowerFlex SuperUser and all additional users**.

3. Click **Next**.
4. On the screens that follow the **Deployment Settings** page, configure the settings, as needed for your deployment.
5. Click **Next**.
6. On the **Schedule Deployment** page, select one of the following options and click **Next**:
   - **Deploy Now**—Select this option to deploy the resource group immediately.
   - **Deploy Later**—Select this option and enter the date and time to deploy the resource group.
7. Review the **Summary** page.
   The **Summary** page gives you a preview of what the resource group will look like after the deployment.
8. Click **Finish** when you are ready to begin the deployment. If you want to edit the resource group, click **Back**.

   PowerFlex Manager displays a message indicating that a new job is being created for this resource group.

   PowerFlex Manager creates a configuration file called /etc/sysctl.d/powerflex-so.conf for a storage-only resource group, and a configuration file called /etc/sysctl.d/powerflex-svm.conf for a hyperconverged resource group. PowerFlex Manager maintains default values in a conf file for resource group deployment settings. If you have an old sysctl.conf file from a previous release, it will override these settings. If you must customize these settings, you can do it in sysctl.conf.

Related information

Life cycle (mirroring)
Viewing resource group details (mirroring)
Adding components to a resource group (mirroring)
Build and publish a template
Component types
Removing a resource group

Adding an existing resource group (mirroring)

You can add an existing resource group to discover and import hardware resources that were not originally deployed with PowerFlex Manager.

Before you add an existing resource group, ensure that the following conditions are met:

- The vCenter, PowerFlex gateway, CloudLink Center, and hosts are discovered in the resource list.
- You must have a resource group that includes the manager MDMs before importing other resource groups. The first resource group that you import must include the manager MDMs so that PowerFlex Manager has the manager MDM credentials.

1. On the menu bar, click **Lifecycle** > **Resource Groups** and then click **+ Add Existing Resource Group**.
2. On the **Add Existing Resource Group** page, enter a resource group name in the **Name** field.
3. Enter a description in the **Description** field.
4. Select the **Type** for the resource group.

   The choices are **Hyperconverged**, **Compute Only**, **Storage Only**, and **PowerFlex Management Cluster**.

For compute only types, only the VMware compute only resource groups can be added to an existing resource group. Linux compute only resources groups cannot be added to existing resource groups.

PowerFlex Manager checks to see whether there are any vCLS VMs on local storage. If it finds any, it puts the resource group in life cycle mode and lets you migrate these VMs to shared storage.

5. To specify the compliance version to use for compliance, select the version from the **Firmware and Software Compliance** list or select **Use PowerFlex Manager appliance default catalog**.

   You cannot specify a minimal compliance version when you add an existing resource group, since it only includes server firmware updates. The compliance version for an existing resource group must include the full set of compliance update capabilities. PowerFlex Manager does not show any minimal compliance versions in the **Firmware and Software Compliance** list.

   NOTE: Changing the compliance version might update the firmware level on nodes for this resource group. The global default firmware repository maintains firmware on shared devices.

6. Specify the resource group permissions under **Who should have access to the resource group deployed from this template?** by performing one of the following actions:
   - To restrict access to super users and specfied life cycle administrators, drive replacers, and storage administrators select **Only PowerFlex SuperUser and specfied additional users**.
     a. Click **Add User(s)** to add one or more LifecycleAdmin or DriveReplacer or StorageAdmin users to the list displayed.
     b. Select which users will have access to this resource group.
     c. To delete a user from the list, select the user and click **Remove User(s)**.
     d. After adding the users, select or clear the checkbox next to the users to grant or block access.
   - To grant access to super users and all life cycle administrators, drive replacers, and storage administrators, select **PowerFlex SuperUser and all additional users**.
7. Click **Next**.
8. Choose one of the following network automation types:
   - **Full Network Automation**
   - **Partial Network Automation**

   When you select **Partial Network Automation**, PowerFlex Manager skips the switch configuration step, which is performed for a resource group with **Full Network Automation**. Partial network automation allows you to work with unsupported switches. However, it also requires more manual configuration before a deployment can proceed successfully. If you choose to use partial network automation, you give up the error handling and network automation features that are available with a full network configuration that includes supported switches.

   In the **Number of Instances** box, provide the number of component instances that you want to include in the template.

9. On the **Cluster Information** page, enter a name for the cluster component in the **Component Name** field.
10. Select values for the cluster settings:

    For a hyperconverged or compute-only resource group, select values for these cluster settings:

    a. **Target Virtual Machine Manager**—Select the VMware vCenter name where the cluster is available.
    b. **Data Center Name**—Select the data center name where the cluster is available.

       NOTE: Ensure that the selected VMware vCenter has unique names for clusters in case there are multiple clusters in VMware vCenter.

    c. **Cluster Name**—Select the name of the cluster that you want to discover.
    d. **OS Image**—Select the image or choose **Use Compliance File ESXi image** if you want to use the image provided with the target compliance version. PowerFlex Manager filters the operating system image choices to show only VMware ESXi images for a hyperconverged or compute-only resource group.

    For a storage-only resource group, select values for these cluster settings:

    a. **Target PowerFlex Gateway**—Select the gateway where the cluster is available.
    b. **Target Protection Domain**—Select the name of the protection domain in PowerFlex.
    c. **OS Image**—Select the image or choose **Use Compliance File Linux image** if you want to use the image provided with the target compliance version. PowerFlex Manager filters the operating system image choices to show only Linux images for a storage-only resource group.

    For the PowerFlex management cluster (KVM) resource group, select values for these cluster settings:

    a. **Target PowerFlex Gateway**—Select the gateway where the cluster is available.

b. **Target Protection Domain**—Select the name of the protection domain in PowerFlex.
c. **OS Image**—Select the image or choose **Use Compliance File Linux image** if you want to use the image provided with the target compliance version. PowerFlex Manager filters the operating system image choices to show only Linux images for the resource group.

11. Click **Next**.
12. On **OS Credentials** page, select the operating system credential that you want to use for each node, storage VM (SVM), and MVM.

    You can select one credential for all nodes, SVMs or management virtual machines, or choose credentials for each item separately. You can create the operating system credentials on the **Credentials Management** page under **Settings**.

    PowerFlex Manager validates the credentials before it creates the resource group. This validation makes it possible for PowerFlex Manager to run a full inventory on all nodes, SVMs, and management virtual machines before creating the resource group.

    PowerFlex Manager runs the inventory on all nodes, SVMs, and management virtual machines for which the credentials are valid. The resource group uses any nodes, SVMs, and management virtual machines for which it has a successful inventory. For example, if you have four nodes, and one node has an invalid operating system password, PowerFlex Manager adds the three nodes for which the credentials are valid and ignores the one node with an invalid password.

13. Click **Next**.
    The list of resources available in the cluster is displayed on the **Inventory Summary** page.
14. Review the inventory on the **Inventory Summary** page.

    The summary shows all nodes that are available. If a node is not available, it might be because this node does not match the **Type** you selected for the resource group (**Hyperconverged**, **Compute only**, or **Storage only**).

    Depending on how the node is configured, the summary might show additional inventory information. For example, for a node that has NVDIMM compression, the summary shows additional information about the acceleration pool and compression settings.

    If the resources are discovered and in an available state, the **Available Inventory** displays the components as **Yes**. An unavailable PowerFlex gateway is shown as **No**.

    If the credentials are invalid for a node or SVM, or if you have a network connectivity problem, PowerFlex Manager displays **No** in the **Available Inventory** column for the node, and displays an error message to notify you about the problem.

    PowerFlex Manager cannot update firmware and software versions for PowerFlex clusters that do not have available PowerFlex gateways. If expected PowerFlex gateways are not shown as available, you can discover the gateways and run the wizard again.

    NOTE: PowerFlex Manager retrieves the hostname value from iDRAC and not the operating system. If the hostname field is not updated in iDRAC, an incorrect value can be displayed in PowerFlex Manager. Certain operating systems require extra packages that are installed for iDRAC to update the correct hostname.

15. Click **Next**.
16. On the **Network Mapping** page, review the networks that are mapped to port groups and make any required edits.

    PowerFlex Manager attempts to select the correct network based on the VLAN ID, subnet, or IP ranges entered in PowerFlex Manager. If PowerFlex Manager finds only one network for a given network type, it selects the network automatically. If it finds more than one, you must select the network from the **Network** drop-down list. The operating system Installation network does not get a VLAN ID.

    NOTE: If the operating system installation VLAN is not already configured in your environment, add it. This network is required to perform node expansions. This network is typically added during PowerFlex Manager configuration.

    If there are any port groups for which you do not want PowerFlex Manager to manage access, leave those port groups cleared. If no network is selected for a particular port group, PowerFlex Manager leaves it out of the deployment data and does not add it to the nodes.

17. To import many general-purpose VLANs from VMware vCenter, perform these steps:
    a. Click **Import Networks** on the **Network Mapping** page.
    PowerFlex Manager displays the **Import Networks** wizard. In the **Import Networks** wizard, PowerFlex Manager lists the port groups that are defined on VMware vCenter as **Available Networks**. You can see the port groups and the VLAN IDs.
    b. Optionally, search for a VLAN name or VLAN ID.
    PowerFlex Manager filters the list of available networks to include only those networks that match your search.

c. Click each network that you want to add under **Available Networks**. If you want to add all the available networks, click the checkbox to the left of the **Name** column.
d. Click the double arrow (>>) to move the networks you chose to **Selected Networks**.
PowerFlex Manager updates the **Selected Networks** to show the ones that you have chosen.
e. Click **Save**.

18. Click **Next**.
19. Review the **Summary** page and click **Finish** when you are ready to add the resource group.

PowerFlex Manager displays a message indicating that a new job is being created for this resource group.

The process of adding an existing resource group causes no disruption to the underlying hardware resources. It does not shut down any of the nodes or VMware vCenter.

For an existing resource group, the **Reference Template** field shows **Generated Existing Resource Group Template** on the **Resource Group Details** page. You can distinguish existing resource groups from new resource groups that were deployed with PowerFlex Manager.

When PowerFlex Manager must put a resource group in life cycle mode, the **Summary** page for the **Add Existing Resource Group** wizard displays a warning message indicating the reason.

In some situations, an imported configuration might not meet the minimal requirements for life cycle mode. In this case, PowerFlex Manager does not allow you to add the resource group.

Related information

Getting started (mirroring)
Life cycle (mirroring)
Support for full and partial network automation (mirroring)
Migrating vCLS VMs to shared storage

Updating a resource group with new firmware and software (mirroring)

You can update resource group components with new firmware and software versions.

If you are updating the CloudLink Agent, it must be the same version as the CloudLink Center.

CAUTION: Check the Alerts page before performing the upgrade. Look for major and critical alerts related to PowerFlex Block to be sure the MDM cluster is healthy before proceeding.

NOTE: Dell Technologies recommends using the PowerFlex System Configuration Reporter to validate that configurations are ready to upgrade. PowerFlex System Configuration Reporter is a standalone application designed to provide comprehensive configuration reporting, health assessment, configuration checks, readiness/upgrade assessments checks, and log collection for PowerFlex systems.

If the new firmware and software catalog causes the PowerFlex gateway or CloudLink Agent to become non-compliant, you must upgrade the PowerFlex gateway or CloudLink Center first from the **Resources** page with the target RCM or IC. PowerFlex Manager blocks some of the resource group actions in this situation. For example, PowerFlex Manager disables scale up or scale down, firmware upgrade, the ability to delete the resource group, network scale up, and node actions.

1. On the **Resource Groups** page, click the **Non-Compliant** link, or click **View Compliance Report**.
2. On the **Node Compliance Report** page, click the **Change** button.
3. On the **Change Compliance File** page, choose the **Preferred Compliance File**.
4. Review the **Compatibility** setting. If it shows **Recommended** or **Supported**, you can proceed with the change. If it shows **Not Allowed**, PowerFlex Manager blocks the change. Review the alert for details about recommended or supported paths.
5. Type **`CHANGE COMPLIANCE FILE`** in the confirmation box and click **Save**.
6. On the **Node Compliance Report** page, choose the components to be upgraded.
7. Click **Update Resources**.

Related information

Upgrading a PowerFlex gateway (mirroring)

Viewing a compliance report (mirroring)

PowerFlex Manager enables you to view a compliance report and perform firmware and software updates as needed. You can update software and firmware components for a new resource group that PowerFlex Manager deploys or an existing resource group that was not originally deployed by PowerFlex Manager.

If you want to update only the firmware on the nodes, select a minimal compliance version with firmware only as the target version for the resource group before you proceed.

The update process handles node, BIOS, firmware, VMware ESXi driver updates, and VMware ESXi upgrades automatically.

PowerFlex Manager does not upgrade vCenter itself. However, it checks the vCenter version to determine if it matches the VMware ESXi version. If the VMware ESXi version is greater than the vCenter version, PowerFlex Manager blocks the VMware ESXi host upgrade and displays an error.

1. On the menu bar, click **Lifecycle** > **Resource Groups**.
2. Select a resource group to view the compliance report.
3. In the right pane, click **View Details**.
   The **Resource Group Details** page is displayed.
4. On the **Resource Group Details** page, click **View Compliance Report**.
   The **Node Compliance Report** page is displayed.

   You can also view the compliance status on this page.

   NOTE: Dell Technologies recommends using the PowerFlex System Configuration Reporter (SCR) to validate that configurations are ready to upgrade. PowerFlex SCR is a standalone application that is designed to provide comprehensive configuration reporting, health assessment, configuration checks, readiness/upgrade assessments checks, and log collection for PowerFlex systems.

   When you view a compliance report, PowerFlex Manager checks to see whether there are any vCLS VMs on local storage. If it finds any, it displays a banner at the top of the report and allows you to migrate the VMs to shared storage.
5. Click **Change** to modify the target compliance version.
   a. Select the **Preferred Compliance File**.
   b. Review the **Compatibility** setting. If it shows **Recommended** or **Supported**, you can proceed with the change. If it shows **Not Allowed**, PowerFlex Manager blocks the change. Review the alert for details that are about recommended or supported paths.
   c. If the change is either recommended or supported, type **`CHANGE COMPLIANCE FILE`** if you are ready to proceed.
   d. Click **Save**.
6. Click **Firmware Components** to view the firmware components.
7. Click **Software Components** to view the software components.

   For compute-only and storage-only resource groups, PowerFlex Manager lists the software components that are available for update under the **Components** drop-down list. For hyperconverged resource groups, PowerFlex Manager displays the following subcategories of software components that are available for update:

   - **Node Software** lists the ESXi drivers that are installed if you choose to update.
   - **SVM Software** lists the embedded operating system packages that are installed if you choose to update.
   - **MVM Software** lists the embedded operating system packages that are installed if you choose to update.
8. Select one or more noncompliant nodes, and click **Update Resources**.

   The PowerFlex gateway must first be compliant before you update the PowerFlex components for a node. If a node is already compliant, it is disabled and cannot be selected.

   If you are planning to perform an upgrade on all the nodes in a PowerFlex rack or PowerFlex appliance, click the box in the upper left corner of the **Firmware Components** tab to select all the nodes in the resource group.
9. Specify when to apply the resource updates and click **Apply**:
   - To perform an update right away, choose **Allow PowerFlex Manager to perform firmware and software updates now**.

     For a storage-only or hyperconverged resource group, specify the type of update you want to perform by selecting one of the following options:

     - **Instant Maintenance Mode** enables you to perform updates quickly. PowerFlex Manager does not migrate the data.
     - **Protected Maintenance Mode** enables you to perform updates that require longer than 30 minutes in a safe and protected manner. When you use protected maintenance mode, PowerFlex makes a temporary copy of the data so

that the cluster is fully protected from data loss. The protected maintenance mode applies only to hyperconverged and storage-only resource groups.

For a non-disruptive update, PowerFlex Manager performs updates for PowerFlex nodes within a single resource group in a serialized workflow. The workflow updates one node at a time.

To perform a disruptive update on a compute-only resource group, specify the number of nodes to update in parallel by choosing a number for **Nodes to update in parallel**. By updating nodes in parallel, you can make the update process go faster than it would if the nodes were updated one at a time. If you want to update nodes in parallel, verify that your cluster has been set up in such a way that it can handle multiple resources going into maintenance mode simultaneously . In addition, if you have any VMs that would block auto-migration, you must power these down manually.

- To perform an update later, choose **Schedule firmware and software updates**.

  For a storage-only or hyperconverged resource group, specify the type of update you want to perform by selecting one of the following options:

  - **Instant Maintenance Mode** enables you to perform updates quickly. PowerFlex Manager does not migrate the data.
  - **Protected Maintenance Mode** enables you to perform updates that require longer than 30 minutes in a safe and protected manner. When you use protected maintenance mode, PowerFlex makes a temporary copy of the data so that the cluster is fully protected from data loss. The protected maintenance mode applies only to hyperconverged and storage-only resource groups.

  To perform a disruptive update on a compute-only resource group, specify the number of nodes to update in parallel by choosing a number for **Nodes to update in parallel**. By updating nodes in parallel, you can make the update process go much faster than it would if the nodes were updated one at a time. If you want to update nodes in parallel, verify that your cluster has been set up in such a way that it can handle multiple resources going into maintenance mode at the same time. In addition, if you have any VMs that would block auto-migration, you'll need to power these down manually.

- To perform a disruptive update right away for a full hyperconverged deployment, choose **Allow PowerFlex Manager to perform disruptive updates now**.

  PowerFlex Manager does not support disruptive updates for the PowerFlex management platform (PFMP).

  The full system upgrade process is faster. This process should take a matter of hours. However, the nodes and all of the data are unavailable while the upgrade is in process.

  If you are certain that you want to proceed, type **`REBOOT ALL NODES AT ONCE`**.

  For a full system upgrade, PowerFlex Manager ensures that the PowerFlex gateway is used only by this resource group, and checks to see whether any other VMs are running in the cluster. PowerFlex Manager stops the process if these conditions are not met.

10. If you encounter any errors while performing firmware or software updates, you can view the PowerFlex Manager logs for the resource group. On the **Resource Group Details** page, click **Generate Troubleshooting Bundle**.

    This action creates a compressed file that contains:

    - PowerFlex Manager application logs
    - Dell Technologies connectivity services logs
    - PowerFlex gateway logs
    - iDRAC life cycle logs
    - Dell PowerSwitch switch logs
    - Cisco Nexus switch logs
    - VMware ESXi logs
    - CloudLink Center logs

    The logs are for the current resource group only.

    Alternatively, you can access the logs from a VMware ESXi console, or by using SSH to log in to PowerFlex Manager.

Related information

Life cycle (mirroring)

Entering and exiting service mode (mirroring)

PowerFlex Manager enables you to put a node in service mode when you must perform maintenance operations on the node.

This capability is available on the PowerFlex appliance and PowerFlex rack offerings only.

When you put a node in service mode, you can specify whether you are performing short-term maintenance or long-term maintenance work. The option that you use for long-term maintenance depends on the PowerFlex version you are using.

PowerFlex Manager detects when a node is in VMware ESXi or PowerFlex maintenance mode. It automatically places the node in service mode and also ensures that the resource group itself goes into service mode.

If the node has a VMware NSX-T configuration, PowerFlex Manager does not enable you to enter service mode. If the PowerFlex gateway used in the resource group is being updated on the **Resources** page, PowerFlex Manager also does not enable you to enter service mode.

1. On the menu bar, click **Lifecycle** > **Resource Groups**.
2. On the **Resource Groups** page, select a resource group and click **View Details** in the right pane.
3. Click **Enter Service Mode** under **More Actions**.
4. Select one or more nodes on the **Node Lists** page and click **Next**.

   You can only place multiple nodes in service mode simultaneously if all the nodes are in the same fault set.

   CAUTION: Selecting a PowerFlex node that hosts a PowerFlex Manager VM and putting the PowerFlex node into service mode causes PowerFlex Manager to be temporarily unavailable.

5. Specify the type of maintenance that you want to perform by selecting one of the following options:
   - **Instant Maintenance Mode** enables you to perform short-term maintenance that lasts less than 30 minutes. PowerFlex Manager does not migrate the data.
   - **Protected Maintenance Mode** enables you to perform maintenance that requires longer than 30 minutes in a safe and protected manner. When you use protected maintenance mode, PowerFlex makes a temporary copy of the data so that the cluster is fully protected from data loss. Protected maintenance mode applies only to hyperconverged and storage-only resource groups.
6. Click **Finish**.

   PowerFlex Manager displays a yellow warning banner at the top of the Resource Groups page. The **Service Mode** icon displays for the **Deployment State** and **Overall Resource Group Health**, and for the **Resource Health** for the selected nodes.
7. When you are ready to leave service mode, click **More Actions** > **Exit Service Mode**.

Replacing a drive (mirroring)

You can replace a failed drive in a deployed node. PowerFlex Manager supports the replacement of SSD and NVMe drives for PowerFlex storage-only nodes and hyperconverged nodes.

Ensure that you have a replacement drive and can access the node.

This capability is available on the PowerFlex appliance and PowerFlex rack offerings only.

PowerFlex Manager supports drive replacement for:

- Nodes that have NVDIMM compression enabled.
- SSD for HBA330 controllers.

  NOTE: RAID controllers are not supported.

- CloudLink-enabled PowerFlex nodes with self-encrypting drives (SEDs) or software encryption.

If the node has an NSX-T or NSX-V configuration, PowerFlex Manager does not enable you to replace a drive.

1. On the menu bar, click **Lifecycle** > **Resource Groups**.
2. On the **Resource Groups** page, select a resource group and click **View Details** in the right pane.
3. Scroll down to the **Physical Nodes** section of the page.
4. Under **Physical Nodes**, click **Drive Replacement**.
   PowerFlex Manager displays the **Node List** panel in the **Drive Replacement** wizard.
5. Click the node for which you want to perform a drive replacement.

6. Click **Next**.
   PowerFlex Manager displays the **Select Drive** panel. Any empty slots are shown in gray. The slots that have drives are shown in black.
7. Select the drive that you want to replace by clicking the drive slot in the hardware image, or select the drive from the table.

   Be sure to click the correct drive, because the drive replacement process is irreversible.

   The color for the selected drive changes to blue and the table below the hardware image is selected. The table shows details about the selected drive. To help you pick the correct drive, PowerFlex Manager provides the iDRAC name for the drive, the PowerFlex drive name, and the serial number.
8. Optionally, click **Launch iDRAC GUI** to see iDRAC details about the selected drive before proceeding.

   When you launch iDRAC, the iDRAC user interface opens in a different tab. Log in and go to the drive details.
9. Optionally, click **Blink LED** to cause the selected drive to blink.
   PowerFlex Manager turns on the blinking LED on the selected drive and keeps it blinking until you pull the disk out. A new disk is not in a blinking state. PowerFlex Manager is not able to determine whether the drive is blinking. If the drive is already blinking, you can click **Unblink LED** to turn off the blinking LED.
10. Click **Next** when you are ready to delete the drive.
    PowerFlex Manager displays the **Confirmation** page. PowerFlex Manager provides details about the drive and shows a hardware image again with the drive that is selected in blue.
11. Review the information carefully under **Confirm Drive Replacement** to be sure that you have selected the correct drive. If you are certain that you want to proceed, type **`REMOVE DRIVE`**.
12. Click **Confirm**.
    PowerFlex Manager puts the resource group into the **In Progress** state and updates the **Recent Activity** to show that a drive removal operation is started. After several minutes, if the node ejection is successful, PowerFlex Manager places the resource group and the node in service mode and deletes the drive from the storage pool. The resource group and the node remain in service mode until the drive is replaced in the node, and the drive replacement process has completed. During this period, PowerFlex Manager does not enable you to enter service mode for the node, or perform another drive replacement within the same resource group.

    PowerFlex Manager displays a banner at the top of the **Resource Group Details** page to notify you that a drive replacement is in process for the selected drive.

    For NVMe disks, PowerFlex Manager initiates a job to instruct iDRAC to delete the NVMe disk safely from the operating system.

    If the drive ejection process fails, PowerFlex Manager displays a banner at the top of the **Resource Group Details** page to notify you that the initial step to eject the drive failed. When the delete drive step fails in the drive replacement process, the banner displays a **Retry Drive Replacement** option to help you to restart the drive replacement process.

    If there is a failure, the resource group and the node do not remain in service mode. Instead, they return to the previous state that the resource group was in before you attempted the drive replacement.
13. Pull the disk out of the slot, and insert the new disk.
14. On the disk replacement banner at the top of the **Resource Group Details** page, click **Actions**>**Complete Drive Replacement**.

    PowerFlex Manager opens the **Complete Drive Replacement** wizard and runs inventory on the node. It takes a couple of minutes to get the updated inventory for the node.

    After getting the updated inventory, PowerFlex Manager displays the **Found New Drive** page or the **Continue Without Drive** page. If PowerFlex Manager finds the new drive, it displays the drive that is selected in green on the **Found New Drive** page.

    If PowerFlex Manager does not find the new drive, it displays the **Continue Without Drive** page. If you are certain that you want to proceed without a new drive, type **`CONTINUE WITHOUT DRIVE`**. If you proceed, the node is taken out of service mode with fewer drives, which might impact the capacity and performance of the resource group. Click **Confirm** to finish the process without adding a drive.
15. Click **Complete** to finish the disk replacement process.
    PowerFlex Manager puts the resource group into the **In Progress** state and updates the **Recent Activity** to show that a complete drive replacement operation is started. Then, PowerFlex Manager replaces the drive in the resource group, and takes the resource group and the node out of service mode.

    When the drive addition is complete, PowerFlex Manager displays a banner at the top of the resource group to notify you that the drive replacement is complete. PowerFlex Manager enables the **Reconfigure MDM Roles** and **Drive Replacement** options again. The resource group returns to the previous state after the drive replacement is completed successfully.

Replacing NVDIMM node or battery (mirroring)

PowerFlex Manager supports replacement of faulty NVDIMM and NVDIMM batteries.

This capability is available on the PowerFlex appliance and PowerFlex rack offerings only.

1. On the menu bar, click **Lifecycle** > **Resource Groups**.
2. On the **Resource Groups** page, select a resource group and click **View Details** in the right pane.
3. Scroll down to the **Physical Nodes** section of the page.
4. Under **Physical Nodes**, click **NVDIMM Replacement**.
   PowerFlex Manager displays the **Node List** panel in the **NVDIMM Replacement** wizard.
5. Click **Next**.
   PowerFlex Manager displays the **Select Component** panel. All the available NVDIMMs are displayed under the **NVDIMM** header while the available NVDIMM batteries are displayed under **NVDIMM Battery**.
6. Select the NVDIMM or battery that you want to replace.
   A message is displayed prompting you to ensure that the node selection is correct as the NVDIMM replacement process is irreversible.
7. Enter **`REMOVE NVDIMM`** and click **Confirm** to proceed with the NVDIMM replacement. To replace an NVDIMM battery, enter **`REMOVE NVDIMM BATTERY`**.
   A message stating the removal or addition of the NVDIMM or battery, with the node and slot numbers, is displayed on the **Resource Groups** page. The status of the resource group and the individual node is "In Progress." The log details are also displayed in the **Recent Activity** section on the right side of the page.

   A job for the replacement of the battery is created. Click the jobs icon at the upper right hand-side of the menu bar to view the details of the job.

   After the NVDIMM or battery is replaced, the host and SVM are turned off. After the physical replacement of the NVDIMM and the battery, the status of the host and node on the **Resource Groups** page display service mode. Also, the option **Discover Replacement NVDIMM** is displayed on the page. When the NVDIMM is replaced physically, click **Discover Replacement NVDIMM**. The **Discover Replacement NVDIMM** turns on the node and does a system erase of the NVDIMMs. On completion of the discovery, the log displays the status as "Complete" and the option **Complete NVDIMM Replacement** is displayed.
8. Click **Complete** to complete the replacement process.
   After the NVDIMM is replaced, you can create virtual hardware for the NVDIMM device, remove the SDS from maintenance or service mode, and turn on the SVM.

   After the new NVDIMM node or battery is added, a message indicating the completion of the NVDIMM node or battery is displayed on the **Resource Groups** page. And, **Dismiss** option is displayed for dismissing the task.

Completing NVDIMM replacement

On the **Complete NVDIMM Replacement** page, you can view details of the node at the top. The details are—the compliance status, inventory update, name, and size of the NVDIMM node. When you add a node, the **Inventory Update** column displays "Adding" next to the node that is added. A firmware update is initiated only if the NVDIMM device is noncompliant.

If an NVDIMM is removed and not added back, the **Inventory Update** column displays **Missing** next to the missing node name.

You also see the list of NVDIMM batteries that are replaced at the bottom of the page. The battery details are—name of the battery, device ID, enabled state, health, and operational status of the battery.

Reconfiguring MDM roles (mirroring)

PowerFlex Manager enables you to change the MDM role for a node in a PowerFlex cluster. For example, if you add a node to the cluster, you might want to switch the MDM role from an existing node to the new node.

You can launch the wizard for reconfiguring MDM roles from the **Resource Groups** page or from the **Resources** page. The nodes that are listed and the operations available are the same regardless of where you launch the wizard.

Each fault set can have a maximum of one MDM role (management or tiebreaker). PowerFlex Manager blocks your role assignments if the reconfigured roles do not conform to PowerFlex best practices.

1. To access the wizard from the **Resource Groups** page:
   a. On the menu bar, click **Lifecycle** > **Resource Groups**.
   b. Select a resource group that has the PowerFlex gateway for which you want to reconfigure MDM roles.

c. In the right pane, click **View Details**.
The **Resource Group Details** page is displayed.
d. On the **Resource Group Details** page, click **Reconfigure MDM Roles** under **More Actions**.
The **MDM Reconfiguration** page is displayed.

2. To access the wizard from the **Resources** page:
a. On the menu bar, click **Resources**.
b. Select the PowerFlex gateway for which you want to reconfigure MDM roles.
c. In the right pane, click **View Details**.
The **Details** page is displayed for the selected PowerFlex gateway.
d. On the **Details** page, click **Reconfigure MDM Roles**.
The **MDM Reconfiguration** page is displayed.
3. Review the current MDM configuration for the cluster.
4. Find each MDM role that you want to reassign and choose the new Host Name or IP address for the role in the **Select New Node for MDM Role** list.
You can reassign multiple roles at one time.
5. Click **Next**.
The **Summary** page is displayed.
6. Type **CHANGE MDM ROLES** to confirm your changes.
7. Click **Finish**.

Related information

Viewing PowerFlex system details (mirroring)

Migrating vCLS VMs to shared storage

When you add an existing resource group, PowerFlex Manager checks to see whether there are any vSphere Cluster Services (vCLS) VMs on local storage. If it finds any, it puts the resource group in life cycle mode and lets you migrate the VMs to shared storage.

This capability is available on the PowerFlex appliance and PowerFlex rack offerings only.

After you add an existing resource group, PowerFlex Manager displays a warning banner to notify you that the resource group is in life cycle mode. You can then migrate the vCLS VMs to shared storage by launching the **Migrate vCLS VMs** wizard.

You can launch the wizard for migrating vCLS VMs to shared storage from the **More Actions** section of the **Resource Group Details** page. You can also launch the wizard from the warning banner at the top of the **Resource Group Details** page or from the warning banner at the top of the **Node Compliance Report**. The operations available are the same regardless of where you launch the wizard.

When you migrate vCLS VMs, PowerFlex Manager automatically creates two resource group volumes and maps these volumes to datastores. The resource group volume names follow the convention of `powerflex-resource group-vol-<NUMBER>`, where *<NUMBER>* is the next available digit that has not been taken. The datastore names follow the convention of `powerflex-<CLUSTER_NAME>-ds-<NUMBER>`, where *<CLUSTER_NAME>* is the name of the vCenter cluster and the *<NUMBER>* is the next available digit. Examples of generated resource group volumes are: powerflex-resource group-vol-1 and powerflex-resource group-vol-2. Examples of generated resource group datastores are: powerflex-Hyperconverged-DKIM-ds-1 and powerflex-Hyperconverged-DKIM-ds-2.

CAUTION: Do not rename the automatically generated resource group volumes and datastores. If for some reason, the names have been changed, PowerFlex Manager creates new resource group volumes and datastores, as needed, that use the correct naming conventions. Also, be sure not to migrate vCLS VMs manually to a nonheartbeat datastore. This action activates life cycle mode and requires a migration. In this case, volumes and datastores may be created with new names and digits (for example, powerflex-resource group-vol-3, powerflex-resource group-vol-4, and so forth).

1. To access the wizard from the **More Actions** section of the **Resource Group Details** page:
a. On the menu bar, click **Lifecycle** > **Resource Groups**.
b. Select a resource group that is in life cycle mode because it includes vCLS VMs that are on local storage.
c. In the right pane, click **View Details**.
The **Resource Group Details** page is displayed.
d. In the right pane, click **Migrate vCLS VMs** under **More Actions**.
The **Migrate vCLS VMs** wizard is displayed.

2. For a compute-only resource group, select a protection domain from the **Protection Domain** field.

   The **Protection Domain** field is not displayed for a hyperconverged resource group, since it is already defined by the resource group.

3. For a hyperconverged or compute-only resource group, select a storage pool in the **Storage Pool** dropdown.

   The list of storage pools available for selection is filtered to list the storage pools in the selected protection domain.

4. Review the **Destination Datastores**. The destination datastores are the two heartbeat datastores that PowerFlex Manager creates automatically when you migrate the vCLS VMs to shared storage.

   PowerFlex Manager also creates two resource group volumes and maps these volumes to the destination datastores.

   Only two datastores and resource group volumes are created. If you already have existing datastores, PowerFlex Manager adds the new datastores to the list. If you already have datastores with the same names in the same cluster, PowerFlex Manager does not create new ones, but simply uses the ones that exist already.

5. Type **`MIGRATE VCLS VIRTUAL MACHINES`** to confirm.
6. Click **Confirm**.

After the migration, PowerFlex Manager takes the resource group out of life cycle mode, and the full set of resource group actions are available.

Related information

Adding an existing resource group (mirroring)
Viewing PowerFlex system details (mirroring)

Adding components to a resource group (mirroring)

You can add one or more node, volume, or network components to a new resource group that PowerFlex Manager deploys, or to an existing resource group not deployed by PowerFlex Manager.

You can add components to a resource group for which deployment was successful or to a failed resource group deployment. You do not need to go back to the original template from which the resource group was originally deployed. Standard users can add components only to a resource group for which they have permission.

1. On the menu bar, click **Lifecycle** > **Resource Groups**.
2. On the **Resource Groups** page, select a resource group and click **View Details** in the right pane.
3. On the **Resource Group Details** page, in the right pane, click one of the following options from the **Add Resources** menu:
   - **Add Nodes**—Add one or more nodes to the resource group
   - **Add Volumes**—Add one or more volumes to the resource group
   - **Add Network**—Add one or more networks to the resource group
   - **Add PowerFlex**—Add a PowerFlex configuration to an integrated PowerFlex management (KVM) for PowerFlex appliance or PowerFlex management cluster (KVM) for PowerFlex rack or PowerFlex appliance.

   If a node has an NSX-T or NSX-V configuration, PowerFlex Manager removes the **Add Resources** button under **Resource Actions**.

   If the PowerFlex gateway used in the resource group is being updated on the **Resources** page, PowerFlex Manager also removes the **Add Resources** button.

   If the PowerFlex gateway used in the resource group is being updated on the **Resources** page, PowerFlex Manager does not allow you to add a node.

Related information

Component types
Deploying a resource group (mirroring)

Adding nodes to a resource group (Mirroring)

You can add a node to a resource group. To add a node, you must define the PowerFlex, operating system, and hardware settings.

1. On the **Resource Groups** page, click **Add Resources** and choose **Add Nodes**.
2. In the **Duplicate Node** wizard:
   a. From the **Resource to Duplicate** list, select a node.

      Select a node that is of the same type as the other nodes within the resource group.

   b. In the **Number of Instances** box, enter the number of node instances that you want to add to the resource group.

   c. Click **Next**.

   d. Under **PowerFlex Settings**, specify the **PowerFlex Storage Pool Spare Capacity** setting by choosing one of the following options:

      - **Recommended Spare Capacity <n>%** sets the spare capacity to 1 divided by the current number of Storage Data Servers (SDSs) in the protection domain, plus the number of nodes that you want to duplicate. For example, if you have three SDSs and you want to add one more node instance, the recommended spare capacity is set to 25 percent based on the formula 1/4.
      - **Current Spare Capacity <n>%** sets the spare capacity to 1 divided by the current number of SDSs in the protection domain. For example, if you have three SDSs in the protection domain, the current spare capacity is set to 34 percent, based on the formula 1/3, rounded up.

   e. Under **OS Settings**, set the **Host Name Selection** to **Auto-Generate**, **Specify at Deployment Time**, or **Reverse DNS Lookup**.

   f. If you choose **Specify at Deployment**, provide a name for the host in the **Host Name** field. If you choose **Auto-Generate**, specify a template for the name in the **Host Name Template** field.

      For an existing resource group that was not deployed by PowerFlex Manager, the **Host Name Selection** option is automatically set to **Specify at Deployment Time** and you must type the hostname.

   g. If you are adding a node to a hyperconverged resource group, specify the **Host Name Selection** under **SVM OS Settings** and provide details about the hostname, as you did for the **OS Settings**.

   h. In the **IP Source** box, provide an IP address. For an existing resource group that was not deployed with PowerFlex Manager, the default choice is **User Entered IP** and the IP settings for each network default to **Manual Entry**. However, you can change the setting to **PowerFlex Manager Selected IP**.

      If the **Multi-Network Selection** option is selected and the IP source is **PowerFlex Manager Selected IP**, the IPs are assigned sequentially from all the networks of that type. If an IP cannot be assigned from any of the multiple networks of the selected type, an error is displayed. If the IP source is **User Entered IP**, a generic name is displayed for the IP Source setting. For example: "PowerFlex Management IP Source".

      If multi networking is not enabled and the IP source is **User Entered IP**, the name of the network is displayed as on the **Network Settings** page. If the selected IP Source setting is of **Manual Entry** type, the IP you entered is validated against all the networks of the selected type. An error is displayed when the IP is not validated in one of the networks.

      Under **Hardware Settings**, the **Target Boot Device** option is automatically set to **Use Local Flash Storage for Dell PowerFlex** for an existing hyperconverged or compute-only resource group that was not deployed by PowerFlex Manager.

   i. Under **Hardware Settings**, in the **Node Source** box, select **Node Pool** or **Manual Entry**.

      For an existing resource group that was not deployed by PowerFlex Manager, the node source defaults to **Manual Entry**, but you can change it to **Node Pool**.

   j. In the **Node Pool** box, select the node pool. Alternatively, if you chose **Manual Entry**, select the specific node in the **Choose Node** box.

      You can view all user-defined node pools and the global pool. Standard users can see only the pools for which they have permission.

      For an existing resource group that was not deployed by PowerFlex Manager, the **Node Pool** defaults to **Global**.

   k. Under **PowerFlex Settings**, specify the **Fault Set** for a node:
      - Select **PowerFlex Manager Selected Fault Set** to have PowerFlex Manager choose the fault set name based on the template settings.
      - Select ***fault-set-name*** to choose one of the fault sets in the resource group.

You can add nodes within a fault set, but PowerFlex Manager does not allow you to add a new fault set within the same resource group. To add a new fault set, you must deploy a separate resource group with settings for the fault set you want to create.

l. Click **Next**.

m. Review the **Summary** page and click **Finish**.

If the node you are adding has a different type of disk than the base deployment, PowerFlex Manager displays a banner at the top of the **Summary** page to inform you of the different disk types. You can still complete the node expansion. However, your resource group may have suboptimal performance.

Based on the component type, the required settings and properties are displayed automatically and can be edited as permitted for a node expansion.

3. Click **Save**.

Adding volumes to a resource group (mirroring)

Use the following procedure to create new volumes or add existing volumes to a resource group.

You can create new volumes for a storage-only or hyperconverged resource group. Alternatively, you can add existing volumes to a compute-only resource group or hyperconverged resource group. PowerFlex Manager allows you to create up to 50 new volumes at a time for a storage-only or hyperconverged resource group.

A compute-only resource group consumes volumes that are provided by a storage-only or hyperconverged resource group. Initially, there are no volumes for a new resource group. The resource group is in an incomplete state until you add volumes. Add at least one volume to a resource group to get it out of the incomplete state. If the resource group is a VMware ESXi cluster that has VMware High Availability (HA) enabled, you need at least two volumes.

NOTE: For a hyperconverged resource group, PowerFlex Manager does not put the resource group in an incomplete state, because it automatically creates two resource group volumes that are mapped to two heartbeat datastores.

After you create the volumes in a storage-only resource group, they are added to PowerFlex, but not mapped. When you add the volumes to a compute-only resource group, PowerFlex maps the volumes and creates the datastores.

For a hyperconverged resource group, the added volumes are mapped to the datastore. PowerFlex Manager requires you to enter the datastore name for each new volume that must be added. This is because PowerFlex Manager also creates a VMware ESXi cluster and VMware vCenter datastore for a hyperconverged resource group.

PowerFlex Manager allows you to enable compression when adding volumes to a storage-only or hyperconverged resource group.

To add volumes to a resource group:

1. On the **Resource Groups** page, click **Add Resources** and choose **Add Volumes**.
2. To add existing volumes to a compute-only or hyperconverged resource group, select **Add Existing Volumes** and follow these steps:
   a. Click **Select Volumes** to choose the volumes you want to add.
   b. Enter a volume name search string in the **Search Text** box.
   c. Optionally, apply additional search criteria by specifying values for the **Size**, **Type**, **Compression**, and **Storage Pool** filters.
   d. Click **Search**.
   PowerFlex Manager updates the results to show only those volumes that satisfy the search criteria. If the search returns more than 50 volumes, refine the search criteria to return only 50 volumes.
   e. Click the checkbox for each volume you want to add.
   f. Click the **>>** button to select the volumes.

   To remove a selected volume, click the checkbox for the volume to remove and click the **<<** button.

   g. When you are ready to add the selected volumes, click **Add Volumes**.
   h. After you have selected the volumes that you want to add, the **Datastore Name Template** field generates a unique datastore name. Click **Next**.
   i. In the **Summary** screen, click **View Volumes** to review the volumes you added.

   Click **Close** to return to the **Summary** screen.
   j. Click **Finish**.

3. To create new volumes for a storage-only or hyperconverged resource group, select **Create New Volumes** and follow these steps:

   Create a single volume with a specified name or multiple volumes with a common naming platform:

   a. Click **+ Add Volume** to add a new volume section.

      PowerFlex Manager adds a new volume section to the **Add Volume** wizard.

   b. A new volume set opens, in the **Number of Volumes** select the number of volumes to create.
   c. For one volume, in the **Volume Name** field, enter the volume name.

      For multiple volumes, in the **Volume Name Template** field, define the template for volume names. Add the variable, `$(num)`, in any part of the volume name template which provides a unique volume name and flexibility in the naming format. For example, you can add `volume-$(num)` or `vol-$(num)-data`.
   d. In the **Storage Pool** drop-down, select the storage pool name and protection domain for the volume.
   e. In the **Volume Size (GB)** field, select the size in GB. The default size is 8 GB and the value that you specify must be divisible by eight.
   f. In the **Volume Type** field, select thick or thin.

      A thick volume provides a larger amount of storage in advance, whereas a thin volume provides on-demand storage and faster setup and startup times.

      If you enable compression on a hyperconverged or storage-only resource group with the granularity of the storage pool set to fine, the only option for **Volume Type** is thin. This is the case regardless of whether you deploy a compressed or non-compressed volume.

   g. On the **Summary** screen, review the volume details.
   h. Click **Finish**.
4. Optionally, click **Add Volume Set** again to add another volume section. Then, provide the required information for that section.
5. Click **Next** once you have included information about all the volumes you want to add.
6. On the **Summary** screen, review the volume details.
7. Click **Finish**.

   The resource group moves to the **In Progress** state and the new volume icons appear on the **Resource Group Details** page. You may see multiple volume components while the add operation is still in progress. One the operation is complete, you will see just one volume component with the count updated.

   After the deployment completes successfully, you can click **View Volumes** in the **Storage** list on the **Resource Group Details** page to search for volumes that are part of the resource group.

   The PowerFlex user interface shows the new volumes under the storage pool. For a storage-only resource group, the volumes are created, but not mapped. For a compute-only or hyperconverged resource group, the volumes are mapped to SDCs. In the vSphere client, you can see the volumes in the storage section and also see the hosts that are mapped to the volumes, once the mappings are in place.

Related information

Resize a volume (mirroring)
Viewing and selecting volumes

Adding networks to a resource group (Mirroring)

You can update the workload network by adding a network to a resource group. Alternatively, you can add a static route to allow nodes to communicate across different networks. The static route can also be used to support replication in storage-only and hyperconverged resource groups.

1. On the **Resource Group Details** page, click **Add Resources** and choose **Add Network**.
   The **Add Network** window is displayed. All used resources and networks are displayed under **Resource Name** and **Networks**.
2. Click **Add Additional Network** to add an additional network:
   a. From the **Available Networks** list, select the network, and click **Add**.
      The selected network is displayed under **Network Name**. You can define a new network by clicking **Define a New Network**.
   b. Select **Port Group** from the **Select Port Group** list.

c. Click **Save**.

NOTE: You cannot remove an added network using PowerFlex Manager.

3. Click **Add Additional Static Route** to add an additional static route:
   a. Click **Add New Static Route**.
   b. Select a **Source Network**.

      The source network must be a PowerFlex data network or a replication network.

   c. Select a **Destination Network**.

      The destination network must be a PowerFlex data network or a replication network.

   d. Type the IP address for the **Gateway**.
   e. Click **Save**.

Related information

Defining a network

Add PowerFlex settings to a PowerFlex management cluster (KVM)

You can add PowerFlex settings to a PowerFlex management cluster (KVM) for PowerFlex rack or PowerFlex appliance.

1. On the **Resource Groups** page, select the checkbox beside the resource group to add PowerFlex settings and click **View Details**.
2. On the **Resource Groups** page, click **Add Resources** and choose **Add PowerFlex**.
   The **Add PowerFlex** wizard opens. Options include auto generating or specifying protection domain and storage pool names.
3. From the **Add PowerFlex** page, complete the following steps for the PowerFlex management cluster (KVM):
   a. In **PowerFlex Management Cluster Settings** enter a name for the PowerFlex management cluster (KVM).
   b. In **PowerFlex Settings**
   c. In **PowerFlex Settings** for **Protection Domain Name**, the default option is to auto generate a protection domain name and use variables to produce a unique protection domain.
   d. For **Protection Domain Template** defines the template for unique protection domain names.

      The variable, `$(num)` auto generates a unique number `$(num)_2d`, or `$(num)_3d` auto generates a number that is two or three digits respectively. You can also use `$(esxiClusterName)` for the VMware ESXi cluster.
   e. For **Storage Pool Name** the default option, auto generates the storage pool name which uses the storage pool name template.
   f. For the **Number of Storage Pools**, you can select up to six storage pools per protection domain. Dell Technologies recommends that you select one storage pool.
   g. For **Storage Pool Template** the default option is to auto generate the storage pool names.

      The variable, `$(diskType)-$(num)`, auto generates the type of storage pool disk (SSD or HDD) and a unique number to generate a unique storage pool name.
   h. For **PowerFlex System Name** enter a system name.
   i. For **PowerFlex MDM Virtual IP Source** select either **PowerFlex Manager Selected IP**, where PowerFlex Manager automatically selects virtual IP addresses or **User Entered IP**, where you enter the virtual IP addresses for each data source. The virtual IP addresses are used for the MDM communication and are added to each MDM data 1 and data 2 network.
   j. Click **Next**.
4. Alternatively in **PowerFlex Settings** specify the protection domain name and storage pool name by completing the following steps:
   a. For **Protection Domain Name** select **Specify a new protection domain now**. Select an existing protection domain from your selected target PowerFlex gateway.
   b. For **Storage Pool Name** select a storage pool.
   c. For the **Number of Storage Pools**, you can select up to six storage pools per protection domain. Dell Technologies recommends that you select one storage pool.
   d. For a single storage pool, select **Create new storage pool**.
   e. Enter a **New Storage Pool name**.
   f. Select the **Storage Pool Disk Type**.

g. For **PowerFlex MDM Virtual IP Source** select either **PowerFlex Manager Selected IP**, where PowerFlex Manager automatically selects virtual IP addresses or **User Entered IP**, where you enter the virtual IP addresses for each data source. The virtual IP addresses are used for the MDM communication and are added to each MDM data 1 and data 2 network.
h. Click **Next**.

5. Click **Save**.
6. Review the **Summary** page and click **Finish**.

Related information

Next steps

Add PowerFlex settings to an integrated PowerFlex management (KVM) environment

You can add PowerFlex settings to an integrated PowerFlex management (KVM) environment for PowerFlex appliance.

1. On the **Resource Groups** page, select the checkbox beside the resource group to add PowerFlex settings and click **View Details**.
2. On the **Resource Groups** page, click **Add Resources** and choose **Add PowerFlex**.
   The **Add PowerFlex** wizard opens.
3. From the **Add PowerFlex** page, complete the following steps for an integrated PowerFlex management (KVM) deployment:
   a. In **PowerFlex Settings**, **Compression** is selected by default because it is required for PowerFlex node deployment. All integrated PowerFlex management (KVM) environments on PowerFlex appliance deployments require 64 GB or more of SDPM memory.
   b. In **PowerFlex Settings**, you can optionally enable encryption.
   c. In **PowerFlex Configuration** for **Protection Domain Name**, the default option is to auto generate a protection domain name and use variables to produce a unique protection domain.
   d. For **Protection Domain Template** defines the template for unique protection domain names.
      The variable, `$(num)` auto generates a unique number `$(num)_2d`, or `$(num)_3d` auto generates a number that is two or three digits respectively. You can also use `$(esxiClusterName)` for the VMware ESXi cluster.
   e. For **New Acceleration Pool** verify that **Create new acceleration pool** is selected.
   f. Enter a name for the **New Acceleration Pool**.
   g. For **Storage Pool Name** the default option auto generates the storage pool name which uses the storage pool name template.
   h. For the **Number of Storage Pools**, you can select up to six storage pools per protection domain. Dell Technologies recommends that you select one storage pool.
   i. For **Storage Pool Template** the default option is to auto generate the storage pool names.
      The variable, `$(diskType)-$(num)`, auto generates the type of storage pool disk (SSD or HDD) and a unique number to generate a unique storage pool name.
   j. For **PowerFlex System Name** enter a system name.
   k. For **PowerFlex MDM Virtual IP Source** select either **PowerFlex Manager Selected IP**, where PowerFlex Manager automatically selects virtual IP addresses or **User Entered IP**, where you enter the virtual IP addresses for each data source. The virtual IP addresses are used for the MDM communication and are added to each MDM data 1 and data 2 network.
   l. Click **Next**.
4. Alternatively in **PowerFlex Settings** specify the protection domain name and storage pool name by completing the following steps:
   a. For **Protection Domain Name** select **Specify a new protection domain now**. Select an existing protection domain from your selected target PowerFlex gateway.
   b. For **Storage Pool Name** select a storage pool.
   c. For the **Number of Storage Pools**, you can select up to six storage pools per protection domain. Dell Technologies recommends that you select one storage pool.
   d. For a single storage pool, select **Create new storage pool**.
   e. Enter a **New Storage Pool name**.
   f. Select the **Storage Pool Disk Type**.
   g. For **PowerFlex MDM Virtual IP Source** select either **PowerFlex Manager Selected IP**, where PowerFlex Manager automatically selects virtual IP addresses or **User Entered IP**, where you enter the virtual IP addresses for each data source. The virtual IP addresses are used for the MDM communication and are added to each MDM data 1 and data 2 network.

h. Click **Next**.

5. Click **Save**.
6. Review the **Summary** page and click **Finish**.

Related information

Next steps

Resize a volume (mirroring)

After adding volumes to a resource group, you can resize the volumes.

For a storage-only resource group, you can increase the volume size. For a VMware ESXi compute-only resource group, you can increase the size of the datastore that is associated with the volume. For a hyperconverged resource group, you can increase the size of both the volume and the datastore.

If you resize a volume in a storage-only resource group, you must update the datastore size in the corresponding VMware ESXi compute-only resource group. The datastore size cannot exceed the size of the volume.

1. On the **Resource Groups** page, click the volume component and choose **Volume Actions** > **Resize**.
2. Choose the volume that you want to resize:
   a. Click **Select Volume**.
   b. Enter a volume or datastore name search string in the **Search Text** box.
   c. Optionally, apply additional search criteria by specifying values for the **Size**, **Type**, **Compression**, and **Storage** filters.
   d. Click **Search**.
   PowerFlex Manager updates the results to show only those volumes that satisfy the search criteria. If the search returns more than 50 volumes, you must refine the search criteria to return only 50 volumes.
   e. Select the row for the volume you want to resize.
   f. Click **Apply**.
3. Update the sizing information:

   If you are resizing a volume for a hyperconverged resource group, perform these steps:

   a. In the **New Volume Size (GB)** field, specify a value that is greater than the current volume size.
   b. Optionally, select **Resize Datastore** to increase the size of the datastore.

   If you are resizing a volume for a storage-only resource group, enter a value in the **New Volume Size (GB)** field. Specify a value that is greater than the current volume size. Values must be in multiples of eight, or an error occurs.

   If you are resizing a volume for a compute-only resource group, review the **Volume Size (GB)** field to see if the volume size is greater than **Current Datastore Size (GB)**. If it is, PowerFlex Manager expands the datastore size.

4. Click **Save**.

Related information

Adding volumes to a resource group (mirroring)

Removing resources from a resource group (mirroring)

You can remove resources that were previously added to a resource group.

If you are removing a node that is fulfilling a metadata manager (MDM) role, verify that you will have at least two MDMs and one tiebreaker in the cluster after the removal of the node. When you remove a node, PowerFlex Manager puts a standby node in place of the one that you removed. For a storage-only resource group, the platform minimum is four nodes in the cluster. This enables you to scale down to three nodes for a component replacement.

If the node has an NSX-T configuration, you can remove the node information that PowerFlex Manager displays for the resource group, but not delete the node from the resource group entirely. If the PowerFlex gateway used in the resource group is being updated on the **Resources** page, PowerFlex Manager does not enable you to remove resources from the resource group.

1. On the menu bar, click **Lifecycle** > **Resource Groups**.
2. Select a resource group from which you want to delete resources.
3. In the right pane, click **View Details**.

4. On the **Resource Group Details** page, in the right pane, under **More Actions**, click **Remove Resource**.
5. In the **Remove Resource** dialog box, select the node that you want to remove and click **Next**.
6. Select the **Resource removal type**:
   - **Delete Resource** makes configuration changes to the nodes, switch ports, virtual machine managers, and PowerFlex to unconfigure those components, and returns the components to the available inventory.
   - **Remove Resource** removes deployment information, but does not make any configuration changes to the nodes, switch ports, virtual machine managers, and PowerFlex, and returns the components to the available inventory.
7. If you choose **Remove Resource**, perform the following steps:
   a. To keep the node in the inventory, select **Leave resource in PowerFlex Manager inventory and set state to** and select the state:
      - **Managed**
      - **Unmanaged**
      - **Reserved**
   b. To remove the node, select **Remove resource from the PowerFlex Manager inventory**.
   c. Click **Remove**.
8. If you choose **Delete Resource**, perform the following steps:
   a. If you are certain that you want to proceed, type **`DELETE RESOURCE`**.
   b. Click **Delete**.

When you delete a node from a resource group, PowerFlex Manager performs the following steps to clean up the underlying components associated with the node:

1. Reconfigures the MDM cluster to ensure that the cluster is in good working order. This step is necessary only if the node is part of an MDM cluster.

   If necessary, PowerFlex Manager reconfigures the MDM role for each node to be a manager or tiebreaker when you perform a scale down operation. PowerFlex Manager rebalances the roles as needed for a new resource group or for an existing resource group. PowerFlex Manager applies the required role changes for nodes that are within the same resource group or in a different resource group.

2. If the node is on the list of SDSs, it removes the node from the list of SDSs within PowerFlex, The node might only be removed from the SDS list if the spare capacity is sufficient, and the environment provides enough space.
3. Deletes the SVM for the node.
4. Deletes the virtual networking configuration .
5. Removes the host from VMware vCenter (hyperconverged resource groups only) .
6. Removes the operating system on the node.
7. Removes the node from the SDC list within PowerFlex .

When deleting a node for a hyperconverged resource group that has compression, PowerFlex Manager removes the DAX from the acceleration pool.

If you remove a node that is CloudLink enabled from a hyperconverged or storage-only resource group, PowerFlex Manager also performs these cleanup tasks:

- Unencrypts the drives on the server.
- Removes the machine from the machine group.
- Removes IP addresses from approved networks.

PowerFlex Manager does not remove the machine groups or keystores. The machine groups and keystores can be reused for subsequent deployments.

After a node is deleted from a resource group, the **Resource Group Details** page shows the cluster without the removed node, and updates the MDM roles to reflect the new configuration.

Adding NVMe over TCP storage access

PowerFlex Manager enables you to add NVMe over TCP storage access to nodes in a resource group. When you add storage access, it installs the SDT (storage data target) package on all nodes in the resource group.

NVMe storage access is only supported for a resource group that is on PowerFlex 4.x. In addition, the option to add NVMe storage access is only available if the resource group was deployed with the **SDC Only** option selected for **Client Storage Access** in the template.

1. On the menu bar, click **Lifecycle** > **Resource Groups**.

2. Select the resource group.
3. On the **Resource Group Details** page, under **More Actions**, click **Add NVMe Storage Access**.
4. Review the list of nodes.
5. Type **ADD SDT TO NODES** to confirm your changes.
6. Click **Confirm**.

Additional resource group management tasks

This section provides additional tasks for resource group management.

Redeploying a resource group

You can redeploy a resource group that failed to deploy because of errors. Standard users can redeploy only a failed resource group that they have deployed.

1. On the menu bar, click **Lifecycle** > **Resource Groups**.
2. On the **Resource Groups** page, select a resource group in an error state and click **View Details** in the right pane.
3. On the **Resource Group Details** page, in the right pane, under **More Actions**, click **Retry**.
4. Click **Yes** when a confirmation message appears.

   If necessary, PowerFlex Manager reconfigures the MDM role for each node to be a manager or tiebreaker when you perform a retry operation. PowerFlex Manager rebalances the roles as needed for a new resource group or for an existing resource group. PowerFlex Manager applies the required role changes for nodes that are within the same resource group or in a different resource group.

Edit resource group information (mirroring)

You can edit information about a resource group such as the resource group name, description, and who has access to the resource group information.

1. On the menu bar, click **Lifecycle** > **Resource Groups**.
2. On the **Resource Groups** page, click the resource group.
3. On the **Resource Group Details** page, in the right pane click **Modify**.
4. In the **Modify Resource Group Information** dialog box, modify the **Resource Group Name** and **Resource Group Description** that identifies the resource group.
   - To update the firmware and software running on the nodes when you deploy a resource group that uses this template, select **Firmware and Software Compliance**.

     If you select a minimal compliance version for a resource group, PowerFlex Manager puts the resource group in life cycle mode and restricts the actions that can be performed. In life cycle mode, the resource group supports monitoring, service mode, and compliance upgrade operations only. All other resource group operations are blocked.

     NOTE: Changing the firmware repository might update the firmware level on nodes for this resource group. The global default firmware repository maintains firmware on shared devices.

     CAUTION: Selecting a PowerFlex node that hosts a PowerFlex Manager VM and upgrading the PowerFlex node causes PowerFlex Manager to be temporarily unavailable.
5. Set the **OS Image** for the resource group.

   You can optionally select a custom operating system image that will be used for future add node operations on the resource group. The custom image must be added to the **OS Image Repositories** tab on the **Settings** > **Compliance and OS Repositories** page.

   PowerFlex Manager filters the list of images available for selection to include only those images that are suitable for the current resource group. For example, if you are editing a hyperconverged resource group, PowerFlex Manager only allows you to choose VMware ESXi images.
6. Specify the permissions for this resource group under **Who should have access to the resource group deployed from this template?** by performing one of the following actions:

- To restrict access to super users and specified life cycle administrators, drive replacers, and storage administrators select **Only PowerFlex SuperUser and specfied additional users**.
  a. Click **Add User(s)** to add one or more LifecycleAdmin or DriveReplacer or StorageAdmin users to the list displayed.
  b. Select which users will have access to this resource group.
  c. To delete a user from the list, select the user and click **Remove User(s)**.
  d. After adding the users, select or clear the checkbox next to the users to grant or block access.
- To grant access to super users and all life cycle administrators, drive replacers, and storage administrators, select **PowerFlex SuperUser and all additional users**.

7. Click **Save**.

Viewing resource group details (mirroring)

You can view the state of a resource group at the component level. You can view the component topology and connections in a selected resource group template. You can view component logs, display the port view, place a component in service mode, and replace a disk.

If the resource group state is incomplete, a banner at the top of the page indicates that you must add volumes to make it fully functional. Sometimes, you may only need to run **Update Resource Group Details** to ensure that the resource group is complete.

To view the details of a resource group component, scroll down on the **Resource Group Details** page. The following information is displayed based on the resource types in the resource group:

Table 44. Resource group details

Resource type	Description
Clusters	View the following information about the clusters in the VMware vCenter environment: • **Health** • **Data Center Name** • **Cluster Name** • **Asset/Service Tag** • **Management IP**
PowerFlex Management Clusters	View the following information about the PowerFlex management cluster (KVM) environments in the resource group: • **Deployment State** • **PowerFlex Management Cluster Name** • **Management IP**
PowerFlex Gateways	View the following information about the gateways that are part of the resource group: • **Health** • **System Name** • **Primary MDM IP** • **Management IP** • **Virtual IP** • **Total Storage Pools** Provides the number of storage pools in the resource group. If the resource group has storage pools, PowerFlex Manager also lists the granularity setting and acceleration pool for each storage pool. In PowerFlex, the storage pools enable the generation of different storage tiers. A storage pool is a physical storage device in a protection domain. Each storage device belongs to one (and only one) storage pool. The resource group deployment results in the creation of a separate storage pool for each type of disk (SSD/NVMe or hard drive) found in the nodes. The deployment process adds the disks from the nodes to the appropriate storage pools based on the expected types for each pool. • **Protection Domain**

Table 44. Resource group details (continued)

Resource type	Description
	In PowerFlex, a protection domain is a logical entity that contains a group of Storage Data Servers (SDSs) that provide backup to each other. Each SDS belongs to one (and only one) protection domain. Each protection domain is a unique set of SDSs. It may also contain SDTs and SDRs.
PFMP	View the following information about the PowerFlex management platform components in the resource group: • **Health** • **System Name** • **Management IP** PowerFlex Manager also lists all the associated management and data networks, ingress IP address, and all the IP addresses in the IP address pool.
CloudLink	View the following information about the CloudLink Centers participating in the resource group: • **Health** **NOTE:** An expected unknown health status is displayed for CloudLink because PowerFlex Manager does not have access to CloudLink health monitoring information. Since PowerFlex Manager is not able to determine the health status of CloudLink, it is not updated. Log in to CloudLink Center to view detailed health status information. • **Hostname** • **Management IP** • **Machine Group** • **Keystore** When a resource group is deployed, if CloudLink has clustered CloudLink Centers, PowerFlex Manager shows all CloudLink Centers on the **Resource Group Details** page. PowerFlex Manager ensures that the resource group deployment succeeds as long as at least one CloudLink Center in the cluster is working.
Storage	View details about the storage volumes added for the resource group. Click **View Volumes** to search for volumes and see the following information about the volumes: • **Name** • **Size** • **Type** • **Compression** • **Storage Pool** • **Datastore**
Physical Nodes	View the following information about the nodes that are part of the resource group: • **Health** • **Asset/Service Tag** • **iDRAC IP/Hostname** • **PowerFlex Mode** The mode for each node is one of the following: ○ **Hyperconverged** includes both SDS and SDC components. ○ **Storage Only** includes only the SDS component. ○ **Compute Only** includes only the SDC component. • **MVM Hostname** For a hyperconverged resource group, PowerFlex Manager adds IP addresses for the MVM to the node IP list. PowerFlex Manager appends `(MVM)` to all network names

Table 44. Resource group details (continued)

Resource type	Description
	for the MVM. If there are no MVM hostnames, the MVM Hostname column is not shown. • **Associated IPs** • **MDM Role** The MDM role is the metadata manager role. The MDM role applies only to those nodes that are part of a PowerFlex cluster. The MDM role is one of the following: ○ **Primary**: The MDM in the cluster that controls the SDSs and SDCs. The primary MDM contains and updates the MDM repository, the database that stores the SDS configuration, and how data is distributed between the SDSs. This repository is constantly replicated to the secondary MDMs, so they can take over with no delay. Every PowerFlex cluster has one primary MDM. ○ **Secondary**: An MDM in the cluster that is ready to take over the primary MDM role if necessary. ○ **Tie Breaker**: An MDM whose sole role is to help determine which MDM is the primary. ○ **Standby MDM**: A standby MDM can be called on to assume the position of a manager MDM when it is promoted to be a cluster member. ○ **Standby Tie Breaker**: A standby node that is prepared to take over as a tiebreaker. • **Fault Set**: A logical group of SDSs within a protection domain that defines by the way it is grouped where the copies of data exist. If there is no fault set, this column is not shown.

View all settings

View All Settings allows you to view the settings configured on the resources in a resource group for deployment.

When you click **View All Settings**, the **Resource Group Deployment Settings** page shows each top-level component on a separate tab. Click a tab for a component to see the details for the component. By displaying the details on separate tabs, PowerFlex Manager greatly improves the performance when you are working with a resource group that includes a large number of nodes.

For a PowerFlex resource group, the **VMware Cluster** section of the **Settings** page shows the **PowerFlex Settings** section, and the **Target PowerFlex Gateway** should specify the gateway that is included with the resource group.

For a PowerFlex resource group, the **Use Local Storage For Dell PowerFlex** property is set to **true** within the **OS Settings** section for any node in an existing PowerFlex resource group deployment, and for the nodes in a new PowerFlex resource group deployment.

Generate troubleshooting bundle

Generate Troubleshooting Bundle allows you to generate a compressed file of logs to use for troubleshooting. The following logs are included:

- ASM deployer
- iDRAC life cycle
- Dell PowerSwitch switch
- Cisco Nexus switch
- VMware ESXi
- CloudLink Center
- Standard output logs from all pods
- Kubernetes logs about pods, services, deployments, secrets, drivers, and volumes

View compliance report

View Compliance Report allows you to view the software or firmware compliance report.

Add resources

Add Resources allows you to add nodes, volumes, and networks to a resource group.

More actions

Under **More Actions**, you can perform the following tasks:

Table 45. More actions

Click...	To...
Update Resource Group Details	Update resource group definition.
Enter Service Mode	Put nodes in service mode so that maintenance operations can be performed.
Exit Service Mode	Take nodes out-of-service mode.
Reconfigure MDM Roles	Change the MDM role for a node in a PowerFlex cluster. For example, if you add a node to the cluster, you might want to switch the MDM role from an existing node to the new node.
Remove Resource Group	Remove a resource group that is no longer required. PowerFlex Manager supports two types of removal: • Delete the entire resource group, which deletes the deployment information and makes any required configuration changes for components that are associated with the resource group. This includes making destructive changes such as unconfiguring the nodes in the resource group and powering them down. • Delete the deployment information for the resource group without making any configuration changes to the deployed components.
Remove Resource	Removes resources from a resource group.
Retry	Redeploy a failed or canceled resource group. This button is only visible when a resource group is in a critical state. If you see a resource group fail for any reason, try clicking **Retry** to correct the problem. For example, a server deployment might fail with a `ChangeBootOrderByInstanceID failed` message. You can correct this failure by retrying the resource group.
Cancel	Cancel the resource group.

Resource actions

Under **Resource Actions**, you can perform the following tasks:

Table 46. Resource actions

Click...	To...
Add Resources	Select the type of the resources that you want to add to the resource group. If you must input data for the template that is used to create the running resource group, click **Confirm Resource Group Settings**. In the **Update Resource Group Component** window, enter values for all displayed fields. Click **Save**.
Remove Resource	Remove resources from a resource group.

Resource group information

In the **Resource Group Information** box on the right, you can view the following information:

- **Overall Resource Group Health**

—Displays the health of the resource group. The following information determines the overall resource group health:

  - **Resource Health** —Displays the health monitoring of resources.
  - **Compliance**—Displays if the resources are firmware or software-compliant. This option is applicable only if you have enabled firmware or software update on the resource group.
  - **Deployment State**—Indicates if the deployment of the resource group completed successfully.
- **Deployed By**—Displays the name of the user who deployed the resource group.
- **Deployed On**—Displays the date and time when the resource group is deployed.
- **Reference Template**—Displays the name of the reference template that is used in the resource group.

  NOTE: For existing resource groups, the name displays as **User Generated Template** and not a template name from the inventory.
- **User Permissions**—Displays one of the following:
  - **Enabled**—Indicates that the permission is granted for one or more standard users to deploy this resource group.
  - **Disabled**—Indicates that the permission is not granted for the standard users to deploy this resource group.

Recent activity

Recent Activity displays component deployment status and information about the current deployed resource group.

You can also click the **Port View** tab to display port view details.

If you want to see the current firmware or software repository that is in use, look at **Target Version**. To change the compliance version, click **Change Target**.

If you select a minimal compliance version for a resource group, PowerFlex Manager puts the resource group in life cycle mode and restricts the actions that can be performed. In life cycle mode, the resource group supports monitoring, service mode, and compliance upgrade operations only. All other resource group operations are blocked.

Related information

Deploying a resource group (mirroring)
Upgrade the firmware components
Upgrade the software components

Viewing all settings (mirroring)

The **View All Settings** option lets you display the settings that are used to configure the resources in the deployment of a resource group.

When you click **View All Settings**, the **Resource Group Deployment Settings** page shows each top-level component on a separate tab. Click a tab for a component to see the details for the component. By displaying the details on separate tabs, PowerFlex Manager greatly improves the performance when you are working with a resource group that includes a large number of nodes.

Review all settings that are defined for the clusters, nodes, and storage. To search for settings within the research group perform the following tasks:

Table 47. View settings actions

Settings	Click….
Volumes	**View Volumes** under **Storage**
Node information	The node name under **Physical Nodes**
Platform resources	**Platform** under **PFMP**
PowerFlex Management Cluster information	The cluster name under **PowerFlex Management Clusters**

During volume operations, the **Resource Group Deployment Settings** page displays additional information about the volume operations that are in progress. If you create a new volume or add an existing volume, or resize a volume, the page provides details that can be helpful for troubleshooting. For example, if you specified an invalid name template string variable, the **Resource Group Deployment Settings** page indicates that this is the cause of a volume not being added.

Viewing and selecting volumes

Use this procedure to search for volumes in a resource group, and optionally select volumes for an add or resize operation.

You can use the volume search dialog to search for volumes using various types of search criteria. You can search by volume name or datastore name, as well as by volume type, compression setting, and storage pool name.

You can launch the volume search dialog from various places with the PowerFlex Manager user interface. You can launch the dialog from the following locations:

- **Storage** section of the **Resource Group Details** page
- **View All Settings** page
- **Add Volumes** wizard
- **Resize Volume** wizard

When you perform a search, PowerFlex Manager updates the results to show only those volumes that satisfy the search criteria. If the results would include more than 50 volumes, you must refine the search criteria to return only 50 volumes.

After performing a search in the **Add Volumes** wizard, you can select one or more existing volumes to add to a resource group. In the **Resize Volume** wizard, you can select the volume that you want to resize.

Identify the naming patterns that are used for the volumes and datastores within the resource group, so you can easily enter a search pattern.

1. To access the **Volumes** dialog from the **Storage** section of the **Resource Group Details** page:
   a. On the menu bar, click **Lifecycle** > **Resource Groups**.
   b. Select a resource group.
   c. On the **Resource Group Details** page, click **View Volumes** in the Storage section.

      You can also click **View Volumes** in the Storage section of the **View All Settings** page, which is also available from the **Resource Group Details** page.

      The **Volumes** dialog is displayed.
2. To access the **Volumes** dialog from the **Add Volumes** wizard:
   a. On the menu bar, click **Lifecycle** > **Resource Groups**.
   b. Select a resource group.
   c. On the **Resource Groups** page, click the **Add Resources** button and choose **Add Volumes**.
   d. When PowerFlex Manager displays the **Add Volume** wizard, click **Add Existing volumes**.
   e. Click **Select Volumes**.
      The **Select Existing Volumes** dialog is displayed.
3. To access the **Volumes** dialog from the **Resize Volume** wizard:
   a. On the menu bar, click **Lifecycle** > **Resource Groups**.
   b. Select a resource group.
   c. On the **Resource Groups** page, click the volume component and choose **Volume Actions** > **Resize**.
   d. Click **Select Volume**.
      The **Select Volume** dialog is displayed.
4. Provide the search criteria needed to filter the results to show 50 volumes or fewer:
   a. Enter a volume or datastore name search string in the **Search Text** box.
   b. Optionally, select a **Size** range.
   c. Optionally, **Thick** or **Thin** for the **Type**.
   d. Optionally, select **Enabled** or **Disabled** for the **Compression** setting.
   e. Optionally, select a specific **Storage Pool**.
   f. Click **Search**.
      PowerFlex Manager updates the results to show only those volumes that satisfy the search criteria. If the search returns more than 50 volumes, you must refine the search criteria to return only 50 volumes.
   g. If you are viewing volumes, click **Close** when you are done looking at the search results.
5. If you are adding existing volumes, you can select one or more volumes to add:
   a. Select each volume you want to add.
   b. Click **>>** to add the volumes.

      To remove a selected volume, click the checkbox for the volume to remove and click **<<**.

   c. When you are ready to add the selected volumes, click **Add**.
6. If you are selecting a volume for a resize operation, you can select the volume that you want to resize and click **Apply**.

Related information

Adding volumes to a resource group
Adding volumes to a resource group (mirroring)

Updating the details for a resource group

When new resources are added or removed outside of PowerFlex Manager, you can update the resource group details to pull in those changes. If nodes are removed externally, updating the resource group details does not pull in the changes.

You can update the details for a new resource group that was deployed with PowerFlex Manager, or for an existing resource group that includes resources that were not originally deployed with PowerFlex Manager.

If the PowerFlex gateway used in the resource group is currently being updated on the **Resources** page, PowerFlex Manager does not enable you to update the resource group details.

Whenever you update the resource group details, PowerFlex Manager pulls in a representation of the components from the VMware vCenter cluster, PowerFlex gateway, or CloudLink Center that was specified for the resource group. These components might include new nodes, volumes, SVMs, or management virtual machines. When a new component is added to the definition of a resource group, PowerFlex Manager does not modify the component.

If the CloudLink Center for the resource group shuts down, PowerFlex Manager loses communication with the CloudLink Center. If the CloudLink Center is part of a cluster, PowerFlex Manager moves to another CloudLink Center when you update the resource group details.

Before you update the details for a resource group, ensure that you run inventory for VMware vCenter and the PowerFlex gateway.

1. On the menu bar, click **Lifecycle** > **Resource Groups**.
2. Select the resource group.
3. On the **Resource Group Details** page, under **More Actions**, click **Update Resource Group Details**.
4. Review the **OS Credentials** page, and click **Next**.

   PowerFlex Manager shows all nodes, SVMs, and management virtual machines, along with their credentials. This enables you to update the username and password if it has changed for any of these items.

5. Review the **Inventory Summary** page, and click **Next**.

   PowerFlex Manager does a rediscovery of the resource group, and tracks whether items are new to the resource group.

   In the **Inventory Update** column, PowerFlex Manager shows which items are added, removed, or kept as is within the resource group:

   - **Adding** indicates that an item was added externally and will now be added to the resource group.
   - **Removing** indicates that an item was removed externally and will now be removed from the resource group.
   - **No Change** indicates that an item was not modified externally and will be kept in the resource group.

6. Review the **Summary** page, and click **Finish**.
   PowerFlex Manager checks the configuration and inventory of selected resources to ensure that everything is configured properly. The resource group is moved to the **In Progress** state, and any new component types are displayed on the **Resource Group Details** page. **Recent Activity** shows that a new deployment started.

Confirming resource group settings

If a PowerFlex Manager upgrade added new required fields to components within the template from which a resource group was deployed, the **Confirm Resource Group Settings** button is displayed on the **Resource Groups** page. Although confirming resource group settings is not mandatory, certain resource group or resource functions are not available until this step is complete.

Click **Confirm Resource Group Settings** to launch the **Upgrade Resource Group Components** window. Fields in this window vary depending on which components contain newly required settings. Complete all the displayed fields, and click **Save**.

Removing a resource group

You can remove a resource group that is no longer required. PowerFlex Manager supports two types of removal:

- Delete the entire resource group, which deletes the deployment information and also makes any required configuration changes for components that are associated with the resource group.
- Remove the deployment information for the resource group without making any configuration changes to the deployed components.

If the node has an NSX-T configuration, you can remove the deployment information for a resource group, but not delete the resource group entirely. PowerFlex Manager also does not allow you to delete a resource group if the PowerFlex gateway used in the resource group is being updated on the **Resources** page.

Standard users can delete only the resource groups that they have deployed.

To remove a resource group, perform the following steps:

1. On the menu bar, click **Lifecycle** > **Resource Groups**.
2. Select the resource group.
3. On the **Resource Group Details** page, under **More Actions**, click **Remove Resource Group**.
4. In the **Remove Resource Group** dialog box, select the **Resource group removal type**:
   - **Delete Resource Group** makes configuration changes to the nodes, switch ports, virtual machine managers, and PowerFlex to unconfigure those components. Also, it returns the components to the available inventory.
   - **Remove Resource Group** removes deployment information, but does not make any configuration changes to the nodes, switch ports, virtual machine managers, and PowerFlex. Also, it returns the components to the available inventory.
5. If you choose **Remove Resource Group**, perform the following steps:
   a. To keep the nodes in the inventory, select **Leave nodes in PowerFlex Manager inventory and set state to** and select the state:
      - **Managed**
      - **Unmanaged**
      - **Reserved**
   b. To remove the nodes, select **Remove nodes from the PowerFlex Manager inventory**.
   c. Click **Remove**.
6. If you choose **Delete Resource Group**, perform the following steps:
   a. Select **Delete Clusters(s) and Remove from vCenter** to delete and remove the clusters from VMware vCenter.
   b. Select **Remove Protection Domain and Storage Pools from PowerFlex** to remove the protection domain and storage pools that are created during the resource group deployment.

      If you select this option, you must select the target PowerFlex gateway. The PowerFlex gateway is not removed. PowerFlex Manager removes only the protection domain and storage pools that are part of the resource group. If multiple resource groups are sharing a protection domain, you might not want to delete the protection domain.

      For a compression enabled resource group, PowerFlex Manager deletes the acceleration pool and the DAX devices when you delete the resource group.

   c. Select **Delete Machine Group and remove from CloudLink Center** to clean up the related components in CloudLink Center.

      CloudLink Center cleanup includes the deletion of the machine group, keystore, and approved network that is related to the resource group being deleted. These components are removed from CloudLink Center only if all the machines that are associated to the machine group are deleted first. If all the machines that are related to this machine group are not removed, the cleanup does not succeed because there are machines that are associated with the machine group.

   d. If you are certain that you want to proceed, type **`DELETE RESOURCE GROUP`**.
   e. Click **Delete**.

Related information

Deploying a resource group
Deploying a resource group (mirroring)

Canceling a resource group

Canceling a resource group stops a scheduled/pending, or in-process resource group deployment. After canceling a resource group, you can update the content and retry the deployment later.

1. On the menu bar, click **Lifecycle** > **Resource Groups**.
2. Select a resource group that is in **In Progress** or **Pending state**.

3. On the **Resource Group Details** page, click **Cancel** under **More Actions**.
4. On the **Cancel Deployment** page, click **Cancel Deployment**.

   You cannot cancel existing resource groups or resource groups that are used for managing the firmware.

   When you cancel a deployment, a banner displays at the top of the screen that displays a warning.

Preparing a template for a resource group deployment (mirroring)

This section includes tasks for preparing a template that will be used to deploy a set of infrastructure resources in a resource group.

A template is an object in PowerFlex Manager that represents resource types and their required configuration and topology. A template specifies requirements for the deployment of a set of infrastructure resources through PowerFlex Manager's automation workflows.

On the **Templates** page, you can access the default sample templates or create templates that meet your specific requirements. For most environments, you can clone one of the sample templates that are provided with PowerFlex Manager and edit as needed. Choose the sample template that is most appropriate for your environment.

PowerFlex Manager allows you to create templates that support hyperconverged, storage-only, and compute-only resource group deployments.

After you create a template, PowerFlex Manager saves the template in a draft state. You must publish the template before deploying it.

Life cycle administrators can view and use only those templates for which they have been granted permission by a super user.

Basic tasks

This section provides basic tasks for template management.

Clone a template

The **Clone** feature allows you to copy an existing template into a new template. A cloned template contains the components that existed in the original template. You can edit it to add additional components or modify the cloned components.

For most environments, you can clone one of the sample templates that are provided with PowerFlex Manager and edit as needed. Choose the sample template that is most appropriate for your environment. If you already created a template using the **Add a template** option, skip this task.

To clone an existing template:

1. On the menu bar, click **Templates**.
2. Click **Sample Template** > **Select** and click a template from the list.

   You can also click **Clone** on the **My Templates** page if you want to clone one of your own templates, rather than one of the sample templates.

3. In the **Clone Template** dialog box, enter a template name in the **Template Name** box.
4. Select a template category from the **Template Category** list. To create a template category, select **Create New Category**.
5. In the **Template Description** box, enter a description for the template.
6. To specify the version to use for compliance, select the version from the **Firmware and Software Compliance** list or choose **Use PowerFlex Manager appliance default catalog**.
7. Indicate **Who should have access to the resource group deployed from this template** by selecting one of the following options:
   - To restrict access to super users and specified life cycle administrators, drive replacers, and storage administrators select **Only PowerFlex SuperUser and specfied additional users**.
     a. Click **Add User(s)** to add one or more LifecycleAdmin or DriveReplacer or StorageAdmin users to the list displayed.
     b. Select which users will have access to this resource group.
     c. To delete a user from the list, select the user and click **Remove User(s)**.

d. After adding the users, select or clear the checkbox next to the users to grant or block access.

- To grant access to super users and all life cycle administrators, drive replacers, and storage administrators, select **PowerFlex SuperUser and all additional users**.

8. Click **Next**.
9. On the **Additional Settings** page, provide new values for the **Network Settings**, **PowerFlex Gateway Settings**, and **Node Pool Settings**.
10. Click **Finish**.
11. To modify the cluster components: Node (software only) and PowerFlex cluster, see *Edit a Template* for more information.

Related information

Configuring block storage
Edit a template

Sample templates (mirroring)

This topic lists the sample templates that are provided with PowerFlex Manager.

Table 48. Sample templates

Template name	Description	Minimum number of nodes required	Minimum number of VMs required	Minimum number of data networks required
Compute - ESXi	A standard VMware ESXi compute-only node deployment. This template requires that hyperconverged or storage-only nodes be deployed first.	2	0	2
Compute - ESXi 100 Gb	A VMware ESXi compute-only node deployment. 100 GB networking deployment. This template requires that hyperconverged or storage-only nodes be deployed first.	2	0	2
Compute - ESXi - Partial Network	A standard VMware ESXi compute-only node deployment with partial network automation. The partial network feature allows you to work with unsupported switches. If you choose to use the partial network feature, you give up the error handling and network automation features that are available with a full network configuration that includes supported switches. This template requires that hyperconverged or storage-only nodes be deployed first.	2	0	2
Compute-Linux-25gb-2Data-Trunked	A Linux compute-only node deployment with these characteristics: • Management and customer networks on two ports in mode1 bond (interface1 port1 and interface2 port2) • Two networks (data 1 and data 2) as SDC-SDS traffic on the other 2 ports • 25 gig NIC This template requires that hyperconverged or storage-only nodes be deployed first.	2	0	2

Table 48. Sample templates (continued)

Template name	Description	Minimum number of nodes required	Minimum number of VMs required	Minimum number of data networks required
	This template was called **Compute - Linux** in previous releases.			
Compute-Linux-25gb-4Data-Trunked	A Linux compute-only node deployment with these characteristics: • Management and customer networks on two ports in mode1 bond (interface1 port1 and interface2 port1) • Two networks (data 2 and data 3) on interface1 port2 • Two networks (data 1 and data 4) on interface2 port2 • 25 gig NIC This template requires that hyperconverged or storage-only nodes be deployed first.	2	0	4
Compute-Linux-100gb-4Data-Trunked	A Linux compute-only node deployment with these characteristics: • Management and customer networks on two ports in mode1 bond (interface1 port1 and interface2 port1) • Two networks (data 2 and data 3) on interface1 port2 • Two networks (data 1 and data 4) on interface2 port2 • 100 gig NIC This template requires that hyperconverged or storage-only nodes be deployed first.	2	0	4
Compute-ESXi-25gb-2Data-Trunked	A Linux compute-only node deployment with these characteristics: • Management and customer networks on two ports set to "Route based on physical NIC load" (interface1 port1 and interface2 port1) • Two networks (interface1 port2: data2 and interface2 port2: data1) as SDC-SDS traffic on the other 2 ports • 25 gig NIC This template requires that hyperconverged or storage-only nodes be deployed first.	2	0	2
Compute - Linux - Partial Network	A standard Linux compute-only node deployment with partial network automation. The partial network feature allows you to work with unsupported switches. If you choose to use the partial network feature, you give up the error handling and network automation features that are available with a full network configuration that includes supported switches.	2	0	2

Table 48. Sample templates (continued)

Template name	Description	Minimum number of nodes required	Minimum number of VMs required	Minimum number of data networks required
	This template requires that hyperconverged or storage-only nodes be deployed first.			
Hyperconverged	A VMware ESXi standard hyperconverged node deployment.	3	0	2
Hyperconverged - 100 Gb	A VMware ESXi hyperconverged node deployment with 100 GB networking. Nodes require Mellanox CX5 or CX6 interface cards.	3	0	2
Hyperconverged - Compression	A standard VMware ESXi hyperconverged node deployment with compression-enabled. Requires NVDIMM in nodes.	3	0	2
Hyperconverged - Encryption	A standard VMware ESXi hyperconverged node deployment with CloudLink encryption. Requires CloudLink Center in environment.	3	0	2
Hyperconverged - Leaf-Spine	A VMware ESXi hyperconverged node deployment with networking without a vPC. This is the most common leaf-spine configuration.	3	0	2
Hyperconverged - Partial Network	A standard VMware ESXi hyperconverged node deployment with partial network automation. The partial network feature allows you to work with unsupported switches. If you choose to use the partial network feature, you give up the error handling and network automation features that are available with a full network configuration that includes supported switches.	3	0	2
Hyperconverged - Replication	A standard VMware ESXi hyperconverged node deployment with replication enabled.	3	0	3 Two networks for data. One network for replication.
Management - CloudLink Center	A two node CloudLink Center cluster deployment, for use in encryption templates.	2	2	N/a
Storage	A standard storage-only node deployment.	3	0	2
Storage-25gb-4Data-Trunked	A storage-only node deployment with these characteristics: • Management network in mode1 bond, on all 4 ports • Four Data networks, each not in a bond • 25 gig NIC • interface1 port1: management, data1	3	0	4

Table 48. Sample templates (continued)

Template name	Description	Minimum number of nodes required	Minimum number of VMs required	Minimum number of data networks required
	• interface1 port2: management, data2 • interface2 port1: management, data3 • interface2 port2: management, data4 This template was called **Storage-NoBonding** in previous releases.			
Storage-100gb-4Data-Trunked	A storage-only node deployment with these characteristics: • Management network in mode1 bond, on all 4 ports • Four Data networks, each on different port • 100 gig NIC • interface1 port1: management, data1 • interface1 port2: management, data2 • interface2 port1: management, data3 • interface2 port2: management, data4	3	0	4
Storage-25gb-2Host-Data-Trunked	A storage-only node deployment with these characteristics: • Management network in mode1 bond on 2 ports • Two networks (Data 1 and Data 2) as SDC-SDS traffic • Two networks (Data 3 and Data 4) as SDS-SDS traffic • 25 gig NIC • interface1 port1: management, data1 SDC-SDS traffic • interface1 port2: data3 SDS-SDS traffic • interface2 port1: management, data2 SDC-SDS traffic • interface2 port2: data4 SDS-SDS traffic	3	0	4
Storage-100gb-2Host-Data-Trunked	A storage-only node deployment with these characteristics: • Management network in mode1 bond on 2 ports • Two networks (Data 1 and Data 2) as SDC-SDS traffic • Two networks (Data 3 and Data 4) as SDS-SDS traffic • 100 gig NIC • interface1 port1: management, data1 SDC-SDS traffic • interface1 port2: data3 SDS-SDS traffic • interface2 port1: management, data2 SDC-SDS traffic • interface2 port2: data4 SDS-SDS traffic	3	0	4
Storage - Perf, No HA Management	A storage-only non-HA management node deployment.	3	0	2

Table 48. Sample templates (continued)

Template name	Description	Minimum number of nodes required	Minimum number of VMs required	Minimum number of data networks required
Storage with NVMe/TCP	A storage-only node deployment with NVMe/TCP.	3	0	2
Storage - 100 Gb	A storage-only node deployment with 100 GB networking. Nodes require Mellanox CX5 or CX6 interface cards.	3	0	2
Storage - Compression	A storage-only node deployment with compression enabled. Requires NVDIMM in nodes.	3	0	2
Storage - Encryption	A storage-only node deployment with encryption enabled.	3	0	2
Storage - Partial Network	A storage-only node deployment with partial network deployment. The partial network feature allows you to work with unsupported switches. If you choose to use the partial network feature, you give up the error handling and network automation features that are available with a full network configuration that includes supported switches.	3	0	2
Storage - Replication	A storage-only node deployment with replication enabled. Configures storage data replicator (SDR) and journal capacity.	3	0	2

Add a template

The **Create** feature allows you to create a template, clone the components of an existing template into a new template, or import a pre-existing template.

For most environments, you can clone one of the sample templates that are provided with PowerFlex Manager and edit as needed. Choose the sample template that is most appropriate for your environment.

1. On the menu bar, click **Templates**.
2. On the **Templates** page, click **Create**.
3. In the **Create** dialog box, select one of the following options:
   - **Clone an existing PowerFlex Manager template**
   - **Upload External Template**
   - **Create a new template**

   If you select **Clone an existing PowerFlex Manager template**, select the **Category** and the **Template to be Cloned**. The components of the selected template are in the new template.
4. Enter a **Template Name**.
5. From the **Template Category** list, select a template category. To create a category, select **Create New Category** from the list.
6. Enter a **Template Description** (optional).
7. To specify the version to use for compliance, select the version from the **Firmware and Software Compliance** list or choose **Use PowerFlex Manager appliance default catalog**.

8. Specify the resource group permissions for this template under **Who should have access to the resource group deployed from this template?** by performing one of the following actions:
   - To restrict access to super users and specified life cycle administrators, drive replacers, and storage administrators select **Only PowerFlex SuperUser and specfied additional users**.
     a. Click **Add User(s)** to add one or more LifecycleAdmin or DriveReplacer or StorageAdmin users to the list displayed.
     b. Select which users will have access to this resource group.
     c. To delete a user from the list, select the user and click **Remove User(s)**.
     d. After adding the users, select or clear the checkbox next to the users to grant or block access.
   - To grant access to super users and all life cycle administrators, drive replacers, and storage administrators, select **PowerFlex SuperUser and all additional users**.
9. Click **Next**.
10. In **Additional Settings** provide the new values for the following settings:
    - **Network**
    - **PowerFlex Gateway**
    - **Node Pool**
11. Click **Finish**.

Build and publish a template

After creating a template using the **Create** feature, you use the template builder page to build and publish the customized template. Publishing a template indicates that a template is ready for deployment.

1. Click **Modify Template**.
2. To add a component type to the template, click **Add Node**, **Add Cluster**, or **Add VM** at the top of the template builder.
   The corresponding **<component type> component** dialog box appears.
3. If you are adding a node, choose one of the following network automation types:
   - **Full Network Automation**
   - **Partial Network Automation**

   When you choose **Partial Network Automation**, PowerFlex Manager skips the switch configuration step, which is normally performed for a resource group with **Full Network Automation**. Partial network automation allows you to work with unsupported switches. However, it also requires more manual configuration before deployments can proceed successfully. If you choose to use partial network automation, you give up the error handling and network automation features that are available with a full network configuration that includes supported switches.

   In the **Number of Instances** box, provide the number of component instances that you want to include in the template.
4. If you are adding a cluster, in the **Select a Component** box, choose the **PowerFlex Cluster** type.
5. If you are adding a VM, in the **Select a Component** box, choose one of the following cluster types:
   - **CloudLink Center**
   - **PowerFlex Gateway**
6. Under **Related Components**, perform one of the following actions:
   - To associate the component with all existing components, click **Associate All**.
   - To associate the component with only selected components, click **Associate Selected** and then select the components to associate.

   Based on the component type, specific settings and properties appear automatically that are required and can be edited.
7. Click **Save** to add the component to the template builder.
8. Repeat steps 1 through 6 to add additional components.
9. After you finish adding components to your template, click **Publish Template**.
   A template must be published to be deployed. It remains in draft state until published.
   After publishing a template, you can use the template to deploy a resource group on the **Resource Groups** page.

Related information

Support for full and partial network automation
Deploying a resource group
Support for full and partial network automation (mirroring)
Deploying a resource group (mirroring)

Edit a template

You can edit an existing template to change its draft state to "published" for deployment or to modify its components and their properties.

1. On the menu bar, click **Templates**.
2. Open a template, and click **Modify Template**.
3. Make changes as required to the settings for components within the template.

   Based on the component type, required settings and properties are displayed automatically. You can edit these settings by performing these steps:

   NOTE: If an RCM associated with the template is modified, a wrench icon labelled **Modified** is displayed. However, if the update file is moved or deleted, the wrench icon with the text, Needs Attention, is displayed.

   a. To edit PowerFlex cluster settings, select the **PowerFlex Cluster** component and click **Modify**. Make the necessary changes and click **Save**.
   b. To edit node settings, select the **Node** component and click **Modify**. Make the necessary changes, and click **Save**.
4. Optionally, click **Publish Template** to make the template ready for deployment.

Related information

Component types

(Optional) Edit template information

You can edit the information assocaited with a template.

1. On the menu bar, click **Templates**.
2. On the **Templates** page, click the template that you want to edit and click **Modify Template** in the right pane.
3. On the template builder page, in the right pane, click **Modify**.
4. In the **Modify Template Information** dialog box, enter a template name in the **Template Name** field.
5. Select a template category from the **Template Category** list. To create a template category, select **Create New Category**.
6. In the **Template Description** box, enter a description for the template.
7. To specify the version to use for compliance, select the version from the **Firmware and Software Compliance** list or choose **Use PowerFlex Manager appliance default catalog**.
8. Indicate **Who should have access to the resource group deployed from this template** by selecting one of the following options:
   - To restrict access to super users and specified life cycle administrators, drive replacers, and storage administrators select **Only PowerFlex SuperUser and specfied additional users**.
     a. Click **Add User(s)** to add one or more LifecycleAdmin or DriveReplacer or StorageAdmin users to the list displayed.
     b. Select which users will have access to this resource group.
     c. To delete a user from the list, select the user and click **Remove User(s)**.
     d. After adding the users, select or clear the checkbox next to the users to grant or block access.
   - To grant access to super users and all life cycle administrators, drive replacers, and storage administrators, select **PowerFlex SuperUser and all additional users**.
9. Click **Save**.

Deploying a resource group (mirroring)

Deployment is the automated process of selecting and configuring specific resource requirements that are outlined in a template using the integrated automation workflows provided with PowerFlex Manager. You cannot use a template that is in a draft state to deploy a resource group. Publish the template before using it to deploy a resource group.

1. On the menu bar, click one of the following:
   - **Lifecycle** > **Resource Groups** click **Deploy New Resource Group**.
   - **Lifecycle** > **Templates** click **Deploy**.

   The **Deploy Resource Group** wizard opens.
2. On the **Deploy Resource Group** page, perform the following steps:

a. From the **Select Published Template** list, select the template to deploy a resource group.
b. Enter the **Resource Group Name** (required) and **Resource Group Description** (optional) that identifies the resource group.
c. To specify the version to use for compliance, select the version from the **Firmware and Software Compliance** list or choose **Use PowerFlex Manager appliance default catalog**.

   You cannot select a minimal compliance version when you deploy a new resource group, since it only includes server firmware updates. The compliance version for a new resource group must include the full set of compliance update capabilities. PowerFlex Manager does not show any minimal compliance versions in the **Firmware and Software Compliance** list.

   PowerFlex Manager checks the VMware vCenter version to determine if it matches the VMware ESXi version for the selected compliance version. If the VMware ESXi version is greater than the VMware vCenter version, PowerFlex Manager blocks the resource group deployment and displays an error. PowerFlex Manager instructs you to upgrade VMware vCenter first, or use a different compliance version that is compatible with the installed VMware vCenter version.

   NOTE: Changing the firmware repository might update the firmware level on nodes for this resource group. The global default firmware repository maintains the firmware on the shared devices.

d. Indicate **Who should have access to the resource group deployed from this template** by selecting one of the following options:
   - To restrict access to super users and specified life cycle administrators, drive replacers, and storage administrators select **Only PowerFlex SuperUser and specfied additional users**.
     i. Click **Add User(s)** to add one or more LifecycleAdmin or DriveReplacer or StorageAdmin users to the list displayed.
     ii. Select which users will have access to this resource group.
     iii. To delete a user from the list, select the user and click **Remove User(s)**.
     iv. After adding the users, select or clear the checkbox next to the users to grant or block access.
   - To grant access to super users and all life cycle administrators, drive replacers, and storage administrators, select **PowerFlex SuperUser and all additional users**.

3. Click **Next**.
4. On the screens that follow the **Deployment Settings** page, configure the settings, as needed for your deployment.
5. Click **Next**.
6. On the **Schedule Deployment** page, select one of the following options and click **Next**:
   - **Deploy Now**—Select this option to deploy the resource group immediately.
   - **Deploy Later**—Select this option and enter the date and time to deploy the resource group.
7. Review the **Summary** page.
   The **Summary** page gives you a preview of what the resource group will look like after the deployment.
8. Click **Finish** when you are ready to begin the deployment. If you want to edit the resource group, click **Back**.

   PowerFlex Manager displays a message indicating that a new job is being created for this resource group.

   PowerFlex Manager creates a configuration file called /etc/sysctl.d/powerflex-so.conf for a storage-only resource group, and a configuration file called /etc/sysctl.d/powerflex-svm.conf for a hyperconverged resource group. PowerFlex Manager maintains default values in a conf file for resource group deployment settings. If you have an old sysctl.conf file from a previous release, it will override these settings. If you must customize these settings, you can do it in sysctl.conf.

Related information

Life cycle (mirroring)
Viewing resource group details (mirroring)
Adding components to a resource group (mirroring)
Build and publish a template
Component types
Removing a resource group

Additional template management tasks

This section provides additional tasks for template management.

Add cluster component settings to a template (mirroring)

To deploy a PowerFlex compute-only node or a PowerFlex hyperconverged node, you must add VMware and PowerFlex cluster components. For a PowerFlex storage-only node deployment, you must add cluster settings for the PowerFlex cluster component only.

1. On the template builder page, click **Add Cluster**.
2. In the **Cluster Component** dialog box, from the **Select a Component** list, select **VMware cluster**, and **PowerFlex Cluster**.
3. Under **Related Components**, select the components that you want to map to the selected cluster instance.
4. Click **Save**.

Related information

Cluster component settings (mirroring)

Add a new interface (mirroring)

Use the **Add New Interface** feature to create a network interface to match to a network card on a node when deploying a template.

PowerFlex Manager supports a network layout for hyperconverged, compute-only, and storage-only deployments. This network layout allows you to use trunk ports and port channels for data networks, instead of access ports. In this type of configuration, both data networks are on both NICs, teamed or bonded together in the operating system. For a hyperconverged or compute-only deployment, the first port on both interfaces is for trunk traffic, whereas the second port is for data 1 and data 2. For a storage-only deployment, the first port is for trunk traffic and data 1, whereas the second port is for data 2. PowerFlex Manager still supports the legacy network layout, but the sample templates use the new configuration.

1. On the node component page, under **Network Settings**, click **Add New Interface**.
2. Enter the following information for the new interface:
   - **Port Layout**—Select the NIC type from the list.

     For a VMware ESXi deployment with an NVIDIA card, you must select the two port, 25-gigabit NIC type.

   The second interface ports 1 and 2 are automatically replicated to the first interface. This replication applies to sample templates as well. If you manually create a template from scratch and choose the networks for the interfaces, the second interface's port 1 and 2 are not automatically replicated to the first interface.
3. Enter the network VLANs for each port.
   a. Click **Choose Networks** for a port.
   b. To add one or more networks to the port, select **Add Networks to this Port**, then click the checkbox for each network you want to add from the **Available Networks** list. Alternatively, click the checkbox in the upper left corner next to the **Name** label to select all the available networks.

      If you want to filter the list by network type, select a **Network Type**, then enter a name or VLAN ID to search.

      Click **>>** to move the selected items to the **Selected Networks** list on the right.
   c. To mirror network settings from another port for which you have already chosen the network VLANs, select **Mirror this Port with Another Port**. Then, select the other interface and port from which you want to mirror this port.
   d. Click **Save**.
4. To view the list of nodes that match the network configuration parameters, click **Validate Settings**.

   The list of nodes is filtered according to the target boot device and NIC type settings specified.

   When you enable PowerFlex settings for the node, the **Validate Settings** page filters the list of nodes according to the supported storage types (NVMe, All flash, and HDD). Within the section for each storage type, the nodes are also sorted by health, with the healthy (green) nodes displayed first and the critical (red) nodes displayed last.

NOTE: If you select the same network on multiple interface ports or partitions, PowerFlex Manager creates a team or bond on systems with the VMware ESXi operating system. This configuration enables redundancy.

Add a new static route

A static route allows nodes to communicate across different networks. Use the add a new static route feature to create a static route in a template.

1. On the node component page, under **Network Settings**, click **Enabled** under **Static Routes**.
2. Click **Add New Static Route**.
3. Enter the following information for the static route:
   - **Source Network**—Select the PowerFlex data network that is the source.

     If you add or remove a network for a port, the **Source Network** list still shows the old networks. In order to see the changes, you must save the node settings and edit the node again.
   - **Destination Network**—Select the PowerFlex data network that is the destination for the static route.
   - **Gateway**—Enter the IP address for the gateway.

Configure VDS settings

Use the **Configure VDS Settings** feature to create the virtual distributed switch (VDS) settings for the VMware cluster component in a template.

You can specify the port group names or have PowerFlex Manager specify them for you.

1. On the VMware cluster component page, under **vSphere VDS Settings**, click **Configure VDS Settings**.
2. To specify the port group names, click **User Entered Port Groups** and click **Next**:
   a. Provide the name for each VDS under **VDS Naming**.
      For each VDS, click **Create VDS...** and type the VDS name.
   b. Click **Next**.
   c. On the **Port Group Select** page, for each VDS, click **Create Port Group...** and type the port group name. Initially, the port group name defaults to the name of the network, but you can type over the default to suit for your requirements.
      Alternatively, you can click **Select** and choose an existing port group.
   d. Click **Next**.
3. To have PowerFlex Manager assign the port group names for you, click **Auto Create All Port Groups** and click **Next**:

   PowerFlex Manager determines the VDS order based on the following criteria:
   a. PowerFlex Manager first considers the number of port groups on each VDS.
   b. PowerFlex Manager considers whether a management port group is present on a particular VDS.
   c. PowerFlex Manager considers the network type for port groups on a VDS.
   d. PowerFlex Manager considers the network name for port groups on a VDS.

   a. Provide the name for each VDS under **VDS Naming**.
      For each VDS, click **Create VDS...** and type the VDS name.
   b. Click **Next**.
   c. On the **Port Group Select** page, review the port group names automatically assigned for the networks.
   d. Click **Next**.
4. On the **Advanced Networking** page, optionally change the **MTU Size** for the Hypervisor Management and vMotion networks.

   PowerFlex Manager allows you to select 1500 or 9000 as the setting for either network. The default setting for the Hypervisor Management network is 1500. The default setting for the vMotion network is 9000.
5. Review the settings on the **Summary** page and click **Finish**.

View template details

View more details about the topology of the components in the template builder.

1. On the **Templates** page, select a template.
2. In the right pane, click **View Details**.
3. To view all the component settings, click **More Actions** > **View All Settings** in the right pane.

The **Template Settings** page shows each top-level component on a separate tab. Click a tab for a component to see the details for the component.

NOTE: If an RCM is mapped to the template and is modified, a wrench icon with the text, **Modified,** is displayed. However, if the update file is moved or deleted, the wrench icon with the text, **Needs Attention**, is displayed.

Related information

Component types

Export a template

The export process packages a selected template into a ZIP file, optionally securing it with encryption so it can be shared or stored safely. Once the template is exported, it becomes a portable file that can be imported into another system instance or reused wherever needed.

1. On the menu bar, click **Templates**.
2. Select the template that you want to export.
3. Click **Export**.
4. In the **Export Template to ZIP File** window, enter values as follows:
   a. Enter a name for the template file in the **File Name**.
   b. If you have set an encryption password, select **Use Encryption Password from Backup Setting** to use that password. To set an encryption password, clear this option.
   c. If you clear **Use Encryption Password from Backup Setting**, two additional fields display. Enter a new password in the **Set File Encryption Password** field and enter the password again to confirm it.
5. Click **Export** to download the file. Select a location to save the file and click **OK**.

Import a template

The import template feature allows you to import the components of an existing template and its component configurations into a template. For example, you can create a template that defines a specific cluster and node topology and import this template definition into another template.

After importing, you can modify the component properties of the imported components. Editing an imported template does not affect the original template.

As an alternative, you can also start from a new template and import an existing template as you create it. This is a better approach to use. To do this, you select **Create**, then choose **Clone an existing VxFlex Manager template** in the wizard.

1. On the menu bar, click **Templates**.
2. On the **Templates** page, select the template into you want to import an existing template and click **Edit** in the right pane.
3. On the **Template Builder** page, in the right pane, click **Import Template**.
4. In the **Import Template** dialog box, select a specific template from the **Select a template** list and click **Import**.

Upload an external template

Uploading an external template involves bringing a template file into the system, providing its key identifying details, and defining how it should be categorized and accessed.

After the template is imported, you can further refine it by configuring additional settings such as network, operating system, cluster, and node-pool parameters so it is fully aligned with your system requirements.

1. On the **Templates** page, click **Create**.
2. In the **Create** dialog box, select **Upload External Template**.
   The **Upload External Template** page appears.
3. Click **Browse** to select the exported file from your directory.
4. Select the **Use Encryption Password from Backup Settings** checkbox, if you have set the encryption password in **Backup and Restore**.

   Clear the **Use Encryption Password from Backup Settings** checkbox if you have not set the encryption password in **Backup and Restore**.

If you clear the **Use Encryption Password from Backup Settings**, you must enter the encryption password in the **Encryption Password** field.

5. Enter the template name in the **Template Name** field.
6. Select a template category from the **Template Category** list.

   If you want to create a template category, select **Create New Category**.
7. In the **Template Description** box, enter a description for the template.
8. To specify the version to use for compliance, select the version from the **Firmware and Software Compliance** list or choose **Use PowerFlex Manager appliance default catalog**.

   NOTE: Changing the firmware repository might update the firmware level on nodes for this resource group. Firmware on shared devices is maintained by the global default firmware repository.
9. Indicate **Who should have access to the resource group deployed from this template** by selecting one of the following options:
   - To restrict access to super users and specified life cycle administrators, drive replacers, and storage administrators select **Only PowerFlex SuperUser and specfied additional users**.
     a. Click **Add User(s)** to add one or more LifecycleAdmin or DriveReplacer or StorageAdmin users to the list displayed.
     b. Select which users will have access to this resource group.
     c. To delete a user from the list, select the user and click **Remove User(s)**.
     d. After adding the users, select or clear the checkbox next to the users to grant or block access.
   - To grant access to super users and all life cycle administrators, drive replacers, and storage administrators, select **PowerFlex SuperUser and all additional users**.
10. Click **Upload and Continue**.
    The **Additional Settings** page appears.
11. On the **Additional Settings** page, provide new values for the **Network Settings**, **OS Settings**, **Cluster Settings**, **PowerFlex Gateway Settings**, and **Node Pool Settings**.
12. Click **Finish**.

Deploy a CloudLink Center

Use this task to deploy a CloudLink Center. Although PowerFlex Manager supports up to three instances of CloudLink Center, two are recommended.

This capability is available on the PowerFlex appliance and PowerFlex rack offerings only.

Ensure the following:

- Hypervisor management or PowerFlex management networks are added to PowerFlex Manager.
- A VMware vCenter with a valid data center, cluster, network, and datastore is discovered.

1. On the menu bar, click **Templates** > **Sample Templates**.
2. On the **Sample Templates** page, click **Management - CloudLink Center**.
3. Click **More Actions** > **Clone**.
4. In the **Clone Template** wizard, do the following:
   a. Enter a template name.
   b. From the **Template Category** list, select a template category. To create a category, select **Create New Category** from the list.
   c. Enter a **Template Description** (optional).
   d. To specify the version to use for compliance, select the version from the **Firmware and Software Compliance** list or choose **Use PowerFlex Manager appliance default catalog**.
   e. Specify the resource group permissions for this template under **Who should have access to the resource group deployed from this template?**:
      - To restrict access to super users and specified life cycle administrators, drive replacers, and storage administrators select **Only PowerFlex SuperUser and specfied additional users**.
        i. Click **Add User(s)** to add one or more LifecycleAdmin or DriveReplacer or StorageAdmin users to the list displayed.
        ii. Select which users will have access to this resource group.
        iii. To delete a user from the list, select the user and click **Remove User(s)**.
        iv. After adding the users, select or clear the checkbox next to the users to grant or block access.

- To grant access to super users and all life cycle administrators, drive replacers, and storage administrators, select **PowerFlex SuperUser and all additional users**.

f. Click **Next**.

5. On the **Additional Settings** page, do the following:
   a. Under **Network Settings**, select **Hypervisor Network** (PowerFlex management network).
   b. Under **OS Settings**, select **CLC** credential or click **+** to create a credential with root or CloudLink user.
   c. Under **CloudLink Settings**, do the following:
      i. Select the secadmin credential from the list or click **+** to create a secadmin credential and do the following:
         i. Enter **Credential Name**.
         ii. Enter **User Name** as **`secadmin`**.
         iii. Leave the **Domain** empty.
         iv. Enter the password for **`secadmin`** in **Password** and **Confirm Password**.
         v. Select **V2** in **SNMP Type**.
         vi. Click **Save**.
      ii. Select a license file from the list based on the types of drives or click **+** to upload a license through the **Add Software License** page.

      NOTE: For SSD/NVMe drives, upload a capacity-based license. For SED drives, upload an SED-based license.

   d. Under **Cluster Settings**, select **Management vCenter**.
   e. Click **Finish**.
6. Select the **VMware Cluster** and click **Edit** > **Continue**.
   a. Under **Cluster Settings**, select **Datacenter Name**, and then select **Cluster Name** from the drop-down list.
   b. Under **vSphere Network Settings**, select the hypervisor management port group or PowerFlex management port group.
   c. Click **Save**.

   NOTE: To deploy a CloudLink Center from PowerFlex Manager, you need a Management vCenter, data center, and cluster along with DvSwitch port groups for PowerFlex management or hypervisor management.

7. Select the **VM** and click **Edit** > **Continue** (by default, the number of CloudLink instances is two and PowerFlex Manager supports up to three instances).
   a. Under **VM Settings** select the **Datastore** and **Network** from the drop-down list.
   b. Under **CloudLink Settings** select the following:
      i. For **Host Name Selection**, either select **Specify At Deployment Time** to manually enter at deployment time or **Auto Generate** to have PowerFlex Manager generate the name.
      ii. Enter the vault passwords.

      NOTE: Other details such as operating system credentials, NTP, and secadmin credentials are auto populated.

8. Under **Additional CloudLink Settings**, you can choose either or both of the following settings:
   - **Configure Syslog Forwarding**
     a. Select the checkbox to configure syslog forwarding.
     b. For **Syslog Facility**, select the **syslog** remote server from the list.
   - **Configure Email Notifications**
     a. Select the checkbox to configure email alerts.
     b. Specify the IP address of the email server.
     c. Specify the port number for the email server. The default port is 25. Enter the port numbers in a comma-separated list, with values between 1-65535.
     d. Specify the email address for the sender.
     e. Optionally, specify the username and password.
9. Click **Save**.
10. Click **Publish Template** and click **Yes** to confirm.
11. In the **Deploy Resource Group** wizard, do the following:
    a. Select the published template from the drop-down list, and enter **Resource Group Name** and description.
    b. Select who should have the access to the resource group and click **Next**.
    c. Provide a **Hostname** and click **Next**.
    d. Select **Deploy Now** or **Schedule deployment** and click **Next**.

e. Review the details in the **Summary** page and click **Finish**.

Delete a template

The **Delete Template** option allows you to delete a template from PowerFlex Manager.

1. On the menu bar, click **Templates**.
2. Select the template that you want to delete. Click **More Actions** > **Delete Template** in the right pane.
3. Click **Yes** to confirm the deletion.

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Managing resources (erasure coding)

This section includes tasks for discovering and managing resources in the PowerFlex Manager inventory.

A resource is a physical and virtual data center object that PowerFlex Manager interacts with, including but not limited to nodes, network switches, VM managers (for example, VMware vCenter), and element managers (for example, CloudLink Center).

The **Resources** page displays detailed information about all the resources and node pools that PowerFlex Manager has discovered and inventoried. You can perform various operations from the **All Resources** and **Node Pools** tabs.

NOTE: Depending upon the number of resources, it might take few minutes to display the discovered resources every time you run the inventory.

The **All Resources** tab displays the following information about the resources that are discovered and managed in PowerFlex Manager:

Table 49. All Resources tab

Field name	Description
Health	Indicates the resource health.
Compliance	Indicates if the resource firmware and software compliance state is **Compliant**, **Non-compliant**, **Update required**, or **Update failed**. Compliance is determined by the firmware/software version for the selected resource, based on the default compliance version. Click the compliance status to view the compliance report.
OS Hostname	Indicates the resource hostname. This capability is available on the PowerFlex appliance and PowerFlex rack offerings only. **NOTE:** PowerFlex Manager retrieves the hostname value from iDRAC and not the operating system. If the hostname field is not updated in iDRAC, an incorrect value might display in PowerFlex Manager. Certain operating systems require additional packages to be installed for iDRAC to update the correct hostname.
Resource Name	Indicates the name of the resource.
Management IP	Indicates the resource IP address. Click the IP address to open the **Element Manager**.
Asset/Service Tag	Indicates the resource service tag. This capability is available on the PowerFlex appliance and PowerFlex rack offerings only.
Deployment Status	Indicates if the resource deployment status is **In Use**, **Not in use**, **Available**, **Updating resource**, or **Pending Updates**. Click the deployment status to view resource group details.
Managed State	Indicates the resource state. If you did not upload a license file in the **Initial Configuration Wizard**, PowerFlex Manager is configured for monitoring and alerting only. All resources are restricted to the **Unmanaged** resource state, and you cannot change the state to **Managed** or **Reserved**.

Table 49. All Resources tab (continued)

Field name	Description
Model	Indicates the resource manufacturer name and model number. This capability is available on the PowerFlex appliance and PowerFlex rack offerings only.

To filter the resources that display, click the toggle filters icon on the **Resources** page.

Table 50. Resources page filters

To filter based on...	Select a value...
Resource Type	• **Element Manager** • **Node (Hardware/Software Management)** • **Switches** • **VM Manager** • **PowerFlex Gateway** • **Node (Software Management)**
Health	• **Healthy** • **Warning** • **Critical** • **Unknown** • **Service Mode**
Resource Groups	Allows you to choose one or more resource groups.
Resource State	**Managed**, **Unmanaged**, or **Reserved**
Node Pool	Lets you choose **Global** or one or more node pools.

On the **Resources** page, you can also perform the following tasks:

Table 51. Resources page tasks

Task	How to perform
Discover new resources	Click **Discover Resources**.
Manually run an inventory operation on a resource and update PowerFlex Manager with the latest resource details	Select a resource and click **Run Inventory**. After running the inventory, you can perform an **Update Resource Group Details** operation on any resource group that requires the updated resource data.
Remove a resource from PowerFlex Manager	Click **Remove**.
Update the firmware or software of a resource	Select one or more resources and click **Update Resources**.
Export a compliance details report for all resources to a CSV or PDF file	Click **Export Report** and select a report format.
Modify the current state of a resource	Select a resource and click **Change resource state to**. You can change the state to **Managed**, **Unmanaged** or **Reserved**.
View detailed information about a resource	Select the resource. In the right pane, click **View Details**.
View a firmware and software compliance report for a resource	Select the resource. In the right pane, click the link corresponding to **Compliance** field.
Update password for a resource	Select the resource and click **Update Password**.

Basic tasks

This section provides basic tasks for resource management.

Discover a resource

A resource must be discovered in PowerFlex Manager in order for PowerFlex Manager to manage it.

Before you start discovering a resource, complete the following:

- Gather the IP addresses and credentials that are associated with the resources.
- Ensure that both the resources and PowerFlex Manager are connected to the network.

A resource in PowerFlex Manager is categorized into one of the following groups: Element manager, node (hardware/software management), switch, VM manager, PowerFlex gateway, node (software management), or PowerFlex system.

1. Access the **Discovery Wizard** by performing either of the following actions:
   - On the **Getting Started** page, click **Discover Resources**.
   - On the menu bar, click **Resources**. On the **Resources** page, click **Discover Resources**.
2. On the **Welcome** page of the **Discovery Wizard**, read the instructions, and click **Next**.
3. On the **Identify Resources** page, click **Add Resource Type**, and perform the following steps:
   a. From the **Resource Type** list, select a resource that you want to discover.
      - **Element Manager**, for example, a CloudLink Center
      - **Node (Hardware / Software Management)**
      - **Switch**
      - **VM Manager**
      - **PowerFlex Gateway**
      - **Node (Software Management)**
      - **PowerFlex System**

      If you want to discover a PowerFlex software node or management VM (MVM) node with a credential that uses SSH key pairs, you must add the public keys to the nodes manually before performing the discovery. If you do not add the public keys first, the discovery fails.

      For a PowerFlex 5.x system, the PowerFlex instance that is used is a management cluster.

   b. Enter the management IP address (or hostname) of the resources that you want to discover in the **IP/Hostname Range** field.

      To discover one or more nodes by IP address, select **IP Address** and provide a starting and ending IP address.

      To discover one or more nodes by hostname, select **Hostname** and identify the nodes to discover in one of the following ways:

      - Enter the fully qualified domain name (FQDN) with a domain suffix.
      - Enter the FQDN without a domain suffix.
      - Enter a hostname search string that includes one of the following variables:

      Table 52. Hostname search strings

Variable	Description
${num}	Produces an automatically generated unique number.
${num_2d}	Produces an automatically generated unique number that has two digits.
${num_3d}	Produces an automatically generated unique number that has three digits.

      If you use a variable, you must provide a start number and end number for the hostname search.

   c. Select one of the following options from the **Resource State** list:

      Table 53. Resource state

Option	Description
Managed	Select this option to monitor the firmware version compliance, upgrade firmware, and deploy resource groups on the discovered resources. A managed state is the default option for the switch, vCenter, element manager, and PowerFlex gateway resource types.

Table 53. Resource state (continued)

Option	Description
	Resource state must be set to **Managed** for PowerFlex Manager to send alerts to Dell Technologies connectivity services.
Unmanaged	Select this option to monitor the health status of a device and the firmware version compliance only. The discovered resources are not available for a firmware upgrade or deploying resource groups by PowerFlex Manager. This option is the default for the node resource type. If you have yet not uploaded a license, PowerFlex Manager is configured for monitoring and alerting only. In this case, **Unmanaged** is the only option available.
Reserved	Select this option to monitor firmware version compliance and upgrade firmware. The discovered resources are not available for deploying resource groups by PowerFlex Manager.

d. To discover resources into a selected node pool instead of the global pool (default), select an existing or create a node pool from the **Discover into Node Pool** list. To create a node pool, click the **+** sign to the right of the **Discover into Node Pool** box.

e. Select an existing or create a credential from the **Credentials** list to discover resource types. To create a credential, click the **+** sign to the right of the **Credentials** box. PowerFlex Manager maps the credential type to the type of resource that you are discovering. The credential types are as follows:

- Element manager
- Node (hardware/software management)
- Switch
- VM manager
- PowerFlex gateway
- Node (software management)
- PowerFlex system

The default node (hardware/software management) credential type is **Dell PowerEdge iDRAC Default**.

f. To have PowerFlex Manager automatically configure iDRAC nodes to send alerts to PowerFlex Manager, select the **Auto configure nodes to send alerts to PowerFlex Manager** checkbox.

4. Click **Next**.

You might have to wait while PowerFlex Manager locates and displays all the resources that are connected to the managed networks.

To discover multiple resources with different IP address ranges, repeat steps 2 and 3.

5. On the **Discovered Resources** page, select the resources from which you want to collect inventory data and click **Finish**. The discovered resources are listed on the **Resources** page.

Related information

Getting started
Configuring block storage
Resource health status
Compliance status

Discover a PowerFlex system

Perform this task to discover a PowerFlex system.

This capability is available on the PowerFlex appliance and PowerFlex rack offerings only.

Before you start discovering a PowerFlex system, complete the following:

- Gather the IP addresses and credentials that are associated with the resource.
- Ensure that both the resource and PowerFlex Manager are connected to the network.

1. Access the **Discovery Wizard** by performing either of the following actions:
   - On the **Getting Started** page, click **Discover Resources**.
   - On the menu bar, click **Resources**. On the **Resources** page, click **Discover Resources** on the **All Resources** tab.
2. On the **Welcome** page of the **Discovery Wizard**, read the instructions, and click **Next**.
3. On the **Identify Resources** page, click **Add Resource Type**, and perform the following steps:
   a. From the **Resource Type** list, select **PowerFlex System**.
   b. Enter the management IP addresses of the LIA nodes in the **MDM Cluster IP Address** field. You must provide the IP addresses for all the nodes in a comma-separated list. The list should include a minimum of three nodes and a maximum of five nodes.

      If you forget to add a node, the node will not be reachable after discovery. To fix this problem, you can rerun the discovery later to provide the missing node. You can enter just the one missing node, or all the nodes again. If you enter IP addresses for any nodes that were previously discovered, these nodes are ignored on the second run.

   c. Select an existing credential or create a new one from the **Credentials** list. The credential must be a **PowerFlex Management System** credential. Be sure to provide the LIA password that was used for the original setup. The LIA password is required for the mTLS configuration.
4. Click **Next**.
5. On the **Summary** page, click **Finish**.
   After you complete the discovery, you should see a **PowerFlex System** resource on the **Resources** page.
6. SSH to the two SVMs associated with the **`powerflex-mds`** system. Run `scli --add_certificate --certificate_file /opt/emc/scaleio/mdm/cfg/mgmt_CA.pem`.

Viewing resource details

Standard users can view details only for resources that are part of node pools for which they have permissions.

1. On the menu bar, click **Resources**.
2. On the **All Resources** tab, click a resource for which you want to view details.
3. In the right pane, click **View Details**. The **Details** page displays detailed information about the resource and associated components.

   For the PowerFlex rack Intelligent Physical Infrastructure (IPI) cabinet, you can view information including system name and location, model number, IP address, cabinet temperature, and humidity details.

   For PowerEdge servers, you can also see performance details, including system usage, CPU usage, memory usage, and I/O usage. Performance usage values are updated every five minutes.

   For a PowerFlex gateway, you can view MDM cluster, storage, and node details.

   For supported switches, you can view packet count by port for all ports or a specific port. The packet count for all ports shows the average packet count for all the ports in that switch.

   For storage only, you can view details of the NIC interfaces, port connections, firmware/software revisions, CPUs, memory, and local storage.

Related information

Next steps

Viewing node details

From the node details page, you can open the remote console of the node's Integrated Dell Remote Access Controller (iDRAC). You can also view recent activities that are performed on the node.

This capability is available on the PowerFlex appliance and PowerFlex rack offerings only.

1. On the menu bar, click **Resources**.
2. On the **All Resources** tab, select a node from the resources list to view its details.
3. In the **Details** pane, click **View Details**.
   The **View Details** page displays the detailed information about the node on the following tabs:
   - **Network Interfaces**
   - **Port View**

- **Firmware/Software Revisions**
- **CPUs**
- **Memory**
- **Local Storage**

  NOTE: You can select one of the following options from the **Filter By** list.

  - **Logical Disks**
  - **Physical Disks**

    To see which disks are self-encrypting drives (SED), look at the **Security Status**. The value **Encryption Capable** indicates that the disk is an SED. The value **Not Capable** indicates that the disk is not an SED.

    On the **Resources** page for the node, filter by **Physical Disks** to view the details of the Vault Optimized Storage Solution (VOSS) under the PCIe Extender section.

Viewing VMware vCenter details

You can view details for VMware vCenter.

This capability is available on the PowerFlex appliance and PowerFlex rack offerings only.

1. On the menu bar, click **Resources**.
2. On the **All Resources** tab, click VMware vCenter from the resource list to view the details.
   In the right pane, you can see the basic information about the VMware vCenter, such as power state, management IP address, data centers, clusters, hosts, and virtual machines.
3. In the **Details** pane, under **vCenter Details**, expand **vCenter** > **Datacenter** > **Cluster** to view the lists of nodes and applications.

Opening the iDRAC remote console

To simplify routine node maintenance, you can open a remote console to the node's Integrated Dell Remote Access Controller (iDRAC) directly from PowerFlex Manager.

This capability is available on the PowerFlex appliance and PowerFlex rack offerings only.

1. On the menu bar, click **Resources**.
2. On the **All Resources** tab, click a node.
3. In the **Details** pane, click **View Details**.
4. Click **Launch iDRAC GUI**.

Viewing PowerFlex system details

You can view performance metrics and storage details for a PowerFlex system.

PowerFlex Manager stores up to 15 GB of PowerFlex gateway PowerFlex system metrics. Once this threshold is exceeded, PowerFlex Manager automatically purges the oldest data to free up space.

1. On the menu bar, click **Resources**.
2. On the **All Resources** tab, click a PowerFlex system resource from the list to view its details.
   The right pane displays basic information about the resource, such as the unused and spare space for PowerFlex. It also shows the number of protection domains, volumes, storage data clients (SDC), and storage nodes.
3. In the right pane, click **View Details**.
   The **Details** page displays detailed information about the PowerFlex System resource on the following tabs:
   - **MDM Cluster**: Displays details about the MDM roles within the cluster.
   - **Storage**: Displays the protection domain, device groups, storage pools, volumes, and raw capacity (TB).
   - **Nodes**: Displays the protection domain, type (storage node or SDC), connection, and the name, IP address for the storage node or SDC.

   The **Details** page also allows you to launch a wizard to see the status of a recent upgrade or reconfigure the MDM roles in the PowerFlex cluster.

4. Click the **MDM Cluster** tab to see information about the MDM roles within the cluster.
5. Click the **Storage** tab. For each volume within a storage pool, PowerFlex Manager shows the **Compression** and **Type** settings.

   For volumes, you may see a different number in the **Volumes** column on the **Resources** tab than you see for the total number of reported volumes on the **Volumes** page on the **Block** tab. The count of volumes that are listed on the PowerFlex system details view is the "vTree" count (volume + snapshots). The count of volumes that are listed on the **Block** tab is just the count of volumes.

   To search for volumes associated with the gateway:

   a. Enter a volume or datastore name search string in the **Search Text** box.
   b. Optionally, select a **Size** range.
   c. Optionally, select **Thick** for the **Type**.
   d. Optionally, select **Enabled** or **Disabled** for the **Compression** setting.
   e. Optionally, select a specific **Storage Pool**.
   f. Click **Search**.
   PowerFlex Manager updates the results to show only those volumes that satisfy the search criteria. If the search returns more than 50 volumes, you must refine the search criteria to return only 50 volumes.

Related information

Reconfiguring MDM roles

Viewing PowerFlex rack IPI cabinet details

The IPI cabinet contains an appliance that is configured to provide information about power, thermals, alerts, and all components in the cabinet's physical infrastructure. PowerFlex Manager uses SNMPv2 to poll the IPI appliance status and send results about temperature and humidity to be viewed in PowerFlex Manager.

This capability is available on the PowerFlex rack offering only.

- If your cluster is running on several nodes, you must add the IP address of each node to the SNMP network management system (NMS) section in the Panduit Appliance UI.
- Ensure that PowerFlex Manager discovers the IPI cabinet. You can verify by clicking the list of resources and checking that the IPI cabinet is present.

You can view additional cabinet details in the IPI appliance management application.

To view the details for the IPI cabinet, perform the following steps:

1. On the menu bar, click **Resources**.
2. On the **All Resources** tab, click the row containing the IPI cabinet.
3. In the right pane, click **View Details**. The **Details** page displays detailed information about the IPI cabinet, including system name and location, model number, and IP address. It also displays cabinet temperature and humidity details.
4. To view additional details, click **Launch IPI Cabinet Manager** to launch the IPI appliance management application.

Viewing CloudLink Center details

You can view details for CloudLink Center.

This capability is available on the PowerFlex appliance and PowerFlex rack offerings only.

To view the details for CloudLink Center, perform the following steps:

1. On the menu bar, click **Resources**.
2. On the **All Resources** tab, click the row containing the CloudLink Center.
3. In the right pane, click **View Details**.
   The **Details** page displays detailed information about the CloudLink Center resource on the following tabs:
   - **Summary**: Displays cluster details, system performance, alarms, pending machines, and security events. If CloudLink has clustered CloudLink Centers, PowerFlex Manager shows all CloudLink Centers.
   - **Machines**: Displays machine groups and machines.
   - **Approved Networks**: Displays a list of the names and IP addresses for all approved networks going in and out of the CloudLink Center.

The **Details** page displays information that is similar to information provided in the CloudLink application. All the data is based on the last inventory, as indicated by the **Last Inventory** timestamp shown on the **Details** page under **Resource Information**. Whenever the inventory is updated, either automatically or manually, any changes that are made in the CloudLink Center are reflected in the **Details** page.

4. To view additional details, click **Launch Console** to launch the CloudLink application.

Viewing NVDIMM details

To view NVDIMM details for a node, perform the following steps:

1. On the menu bar, click **Resources**.
2. On the **All Resources** tab, click the row containing a node that has NVDIMM compression.
3. In the right pane, click **View Details**.

   The **Details** page displays detailed information about the node.
4. Click the **Memory** tab.

   The **Memory** tab shows the NVDIMM modules and all DIMMs in the server. The DIMMs with rank 1 are the NVDIMM devices. The NVDIMMs shown in the top section of the **Memory** tab provide the name of each **NVDIMM Module**, as well as the **Battery Enabled State**, **Battery Health State**, **Battery Operational Status**, **Remaining Rated Read Write Endurance**, and **Size** for the module.

   All of the data is based on the last inventory.

Update resource inventory

You can run inventory to incorporate changes that are made to resource data outside of PowerFlex Manager. After running the inventory to incorporate these changes into the **Resources** page, you can update the details on any resource group that needs to include the new resource data.

Superusers can run the inventory on any resources. Life cycle administrators can run the inventory only on resources that are part of a node pool for which they have permission.

1. On the menu bar, click **Resources**.
2. On the **Resources** page, click the **All Resources** tab.
3. From the list of resources, click a resource, and in the **Details** pane, click **Run Inventory**.

   The resource state changes to **Pending**. When the inventory is complete, the resource state changes to **Available**. See PowerFlex Manager logs to view the start time and end time of the resource inventory operation.

Viewing a compliance report for a resource

A compliance report provides an overview of a resource's firmware and software status, highlighting whether its components align with defined baselines. It helps users quickly identify non-compliant areas and determine which updates are required to bring the resource back into compliance.

Perform the following steps to view a compliance report for a single resource:

1. On the menu bar, click **Resources**.
2. Select the resource for which you want to view the compliance report.
3. In the right pane, click the link corresponding to the **Compliance** option.
   The **Compliance Report** page displays.

   NOTE: If an RCM is mapped to the resource group and is modified, a wrench icon with the text, **Modified**, is displayed. However, if the update file is moved or deleted, the wrench icon with the text, **Needs Attention**, is displayed.
4. Select the **Firmware Components** option to view the details of the firmware available on the selected resource. For example, a switch.
5. Select the **Software Components** option to view the software components in the compliance report or to update the software components for a resource. If the resource is in the **In Use** state, it redirects you to the **Resource Groups** page.
6. To update the non-compliant resources, click **Update Resource**.

NOTE: If there is no compliance version that is attached to the resource, you cannot perform the firmware update or software update.

Related information

Exporting a compliance report for all resources

Removing resources

Only super users can remove resources from PowerFlex Manager.

To remove a resource from PowerFlex Manager, perform the following steps:

1. On the menu bar, click **Resources**.
2. On the **Resources** page, click the **All Resources** tab.
3. From the list of resources, select one or more resources, and click **Remove**.
4. Click **OK** when the confirmation message appears.

If you remove a node, the node state changes to **Pending** and it powers off.

NOTE: Before you add new nodes using an existing iDRAC IP address in inventory, ensure that you remove the old nodes from PowerFlex Manager before discovering the new nodes.

Related information

Resource health status
Compliance status

Additional resource management tasks

This section provides additional tasks for resource management.

Creating a node pool

You can create a node pool on the **Resources** page.

This capability is available on the PowerFlex appliance and PowerFlex rack offerings only.

1. On the menu bar, click **Resources**. Click the **Node Pools** tab.
2. On the **Node Pools** page, click **Create**.
   The **Create Node Pool** wizard starts.
3. On the **Welcome** page, read the instructions, and click **Next**.
4. On the **Node Pool Information** page, enter a name and description for the node pool. Click **Next**.
5. On the **Add Nodes** page, select the nodes that you want to add to the node pool. Click **Next**.
6. On the **Assign Users** page, select the users that you want to grant access rights to the node pool. Click **Next**.
7. On the **Summary** page, review the node pool configuration and click **Finish**.

Modifying a node pool

You can modify the details for a node pool on the **Resources** page.

This capability is available on the PowerFlex appliance and PowerFlex rack offerings only.

1. On the menu bar, click **Resources**. Click the **Node Pools** tab.
2. On the **Node Pools** page, click **Modify**.
   The **Modify Node Pool** wizard starts.
3. To change the name and description of the node pool, in the left pane, click **Node Pool Information**. Make the updates and click **Save**.

4. To add or remove nodes from the node pool, in the left pane, click **Add Nodes**. Make the updates and click **Save**.
5. To add or remove the access rights to the node pool, in the left pane, click **Assign Users**. Make the updates and click **Save**.

Removing a node pool

You can remove a node pool on the **Resources** page.

This capability is available on the PowerFlex appliance and PowerFlex rack offerings only.

1. On the menu bar, click **Resources**. Click the **Node Pools** tab.
2. On the **Node Pools** page, select one or more node pools to delete. Click **Remove**.
3. Click **OK** when the confirmation message appears.

Viewing the port view details

Port view is a PowerFlex Manager feature that is associated with a resource group deployment. It enables you to view the physical network configuration characteristics and topology from the network interface cards to upstream switches, highlighting such details as switch ports, VLANs, and network identifiers such as MAC addresses.

This capability is available on the PowerFlex appliance and PowerFlex rack offerings only.

You can access port view from both the **Resources** and **Resource Groups** pages. Port view provides connectivity information for both new and existing resource groups.

The **Port View** page displays the following information:

- Topology information for all networks and VLANs deployed in a resource group
- Network connections between the devices
- Health of the resources
- Connection details section with information about the network devices

Port View is only available for PowerFlex nodes and is not available for management VMs, such as CloudLink Center.

PowerFlex Manager does not show switch connectivity for deployments with partial network automation.

To view the port view details:

1. On the **Resource Groups** page, view the details for a resource group and click **Port View**. If you are accessing the port view from the **Resources** page, select a node that is in a **Deployed** state and click **View Details** on the right pane first.

   If you select a node that is not in a deployed state, only the interface card information is displayed.

2. To view device details such as hostname, model name, and management IP address, or information about associated devices, click the specific ports or devices.
3. To view information about intermediate devices in port view, ensure that the devices are discovered and available in the inventory. Sometimes, connectivity cannot be determined for an existing resource group because the switches have not yet been discovered. In this case, you see only the node in port view, but you do not see connectivity information. You can correct this by going back and discovering the switches, and updating the resource group again.

   PowerFlex Manager cannot discover interface cards that do not have integrated LLDP support (such as Intel X520).

4. To filter information based on the connectivity, select an option from the **Display Connections** list. **Show All Connections** is the default option.

Exporting a configuration report for all resources and resource groups

You can download a PDF report that lists configuration details for all resources and resource groups.

The configuration report shows the result of various configuration checks that PowerFlex Manager performs against the system. You can use the report to troubleshoot your resources and resource groups.

The report shows the following kinds of information:

Table 54. Configuration report

Column name	Description
Resource Name	Provides the name of the resource for which the checks have been run. This column corresponds to the **Resource Name** shown on the **Resources** page.
Asset/Service Tag	Provides the asset or service tag of the resource for which the checks have been run. This column corresponds to the **Asset/Service Tag** shown on the **Resources** page.
Management IP	Specifies the **Management IP** of the resource for which the check is run. For a PowerFlex cluster, the management IP address is the MDM cluster IP address. For a PowerFlex gateway, the management IP address is the PowerFlex gateway IP address.
Result	Shows the result of the check (`PASS` or `FAIL`).
Severity	Indicates the severity of the check. The severity is based on the result. The severity levels are INFO, WARNING, HIGH, or CRITICAL. If the result is PASS, the severity is INFO. If the result is FAIL, the severity depends on the type of check. PowerFlex Manager supports only CRITICAL checks.
Details	Provides a description of the check that was run.
Affected Resources	Gives a list of the IP addresses or unique identifiers of resources that are impacted by the check. The list of affected resources helps with troubleshooting.

1. On the menu bar, click **Resources**.
2. Click **Export Report**.
3. Select **Export Configuration PDF Report** from the drop-down list.

   The selected report downloads.

Related information

Configuration checks

Importing networks

You can import many general-purpose VLANs from VMware vCenter.

This capability is available on the PowerFlex appliance and PowerFlex rack offerings only.

After importing networks, you can add them to templates or resource groups. You can also import networks when you add an existing resource group.

1. On the menu bar, click **Resources**.
2. On the **All Resources** tab, click the VMware vCenter from which you want to import networks.

   NOTE: Adding a standard vSwitch network is not supported and leads to unsupported scenarios.

   For compute-only or hyperconverged add one of the supported networks:

   - Hypervisor management
   - PowerFlex management
   - PowerFlex data
   - vMotion
   - Out-of-band management

   Importing networks from VMware vCenter is not supported for PowerFlex software nodes.

3. In the right pane, click **Import Networks**.
   PowerFlex Manager displays the **Import Networks** wizard. In the **Import Networks** wizard, PowerFlex Manager lists the port groups that are defined on the VMware vCenter as **Available Networks**. You can see the port groups and the VLAN IDs.

4. Optionally, search for a VLAN name or VLAN ID.
   PowerFlex Manager filters the list of available networks to include only those networks that match your search.
5. Click each network that you want to add under **Available Networks**. If you want to add all the available networks, click the checkbox to the left of the **Name** column.
6. Click the double arrow (>>) to move the networks you chose to **Selected Networks**.
   PowerFlex Manager updates the **Selected Networks** to show the ones you have chosen.
7. Click **Save**.

PowerFlex Manager creates a job to add the networks. The job may finish quickly, so you may not see on the **Jobs** page.

Go to **Settings** > **Networks** to see the imported networks.

Updating passwords for system components

You can update the passwords for some system components from PowerFlex Manager.

1. On the menu bar, click **Resources**.
2. On the **All Resources** tab, select one or more resources of the same type for which you want to change passwords.
   For example, you could select one or more iDRAC nodes or you could select one or more PowerFlex gateway components.
3. Click **Update Password**.
   PowerFlex Manager displays the **Update Password** wizard.
4. On the **Select Components** page, select one or more components for which you want to update a password and click **Next**.
   The component choices vary depending on which resource type you initially selected on the **Resources** page.
5. On the **Select Credentials** page, create a credential or change to a different credential having the same username.
6. Click **Finish** and click **Yes** to confirm the changes.

Updating passwords for nodes

You can update the passwords for one or more nodes from PowerFlex Manager.

1. On the menu bar, click **Resources**.
2. On the **All Resources** tab, select one or more nodes for which you want to change the passwords.
3. Click **Update Password**.
   PowerFlex Manager displays the **Update Password** wizard.
4. On the **Select Components** page, specify which passwords you want to update for the selected nodes by clicking one or more of the following check boxes.
   - **iDRAC Password**
   - **Node Operating System Password**
   - **MVM Operating System Password**

   The option to update passwords for SVMs and MVMs will only appear if a node is in a resource group.
5. Click **Next**.
6. On the **Select Credentials** page, create a credential with a new password or change to a different credential.
   a. Open the **iDRAC**, **OS Password**, or **MVM Password** object under the **Type** column to see credential details for each node you selected on the **Resources** page.

      The **MVM Password** sections do not appear if there is nothing to show for SVMs or MVMs.

   b. To create a credential that has the new password, click the plus sign (**+**) under the **Credentials** column.

      Specify the **Credential Name** and the **User Name** for which you want to change the password. Enter the new password in the **Password** and **Confirm Password** fields.

   c. To modify the credential, click the pencil icon for the nodes under the **Credentials** column and select a different credential.
   d. Click **Save**.

   You must perform the same steps for the node operating system and SVM operating system password changes. For a node operating system credential, only the OS admin credential type is updated.

7. Click **Finish**.
8. Click **Yes** to confirm.

PowerFlex Manager starts a new job for the password update operation, and a separate job for the device inventory. The node operating system, SVM, and MVM operating components are updated only if PowerFlex Manager is managing a cluster with these components. If PowerFlex Manager is not managing a cluster with these components, these components are not displayed and their credentials are not updated. Credential updates for iDRAC are allowed for managed and reserved nodes only. Unmanaged nodes do not provide the option to update credentials.

Updating passwords for PowerFlex gateway components

You can update the passwords for one or more PowerFlex gateway components from PowerFlex Manager.

1. On the menu bar, click **Resources**.
2. On the **All Resources** tab, select one or more PowerFlex gateway components for which you want to change the passwords.
3. Click **Update Password**.

   PowerFlex Manager displays the **Update Password** wizard.
4. On the **Select Components** page, select **PowerFlex Password**.
5. Click **Next**.
6. On the **Select Credentials** page, create a credential with a new password or change to a different credential.
   a. Open the **PowerFlex** object under the **Type** column to see details about each gateway you selected on the **Resources** page.
   b. To create a credential that has the new password, click the plus sign (**+**) under the **Credentials** column.

      Specify the **Credential Name**, as well as the **Gateway Admin User Name** and **Gateway OS User Name** for which you want to change passwords. Enter the new passwords for both users and confirm these passwords.

   c. To modify the credential, click the pencil icon for one of the nodes under the **Credentials** column and select a different credential.
   d. Click **Save**.
7. Click **Finish**.
8. Click **Yes** to confirm.

PowerFlex Manager starts a new job for the password update operation, and a separate job for the device inventory. If PowerFlex Manager is managing a cluster for any of the selected PowerFlex gateway components, it updates the credentials for the **Gateway Admin User** and **Gateway OS User**, as well as any related credentials, such as the LIA and lockbox credentials. If PowerFlex Manager is not managing the cluster, it only updates the credentials for the **Gateway Admin User** and **Gateway OS User**.

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Managing resources (mirroring)

The following section includes tasks for discovering and managing resources in the PowerFlex Manager inventory.

A resource is a physical and virtual data center object that PowerFlex Manager interacts with, including but not limited to nodes, network switches, VM managers (for example, VMware vCenter), and element managers (for example, CloudLink Center).

The **Resources** page displays detailed information about all the resources and node pools that PowerFlex Manager has discovered and inventoried. You can perform various operations from the **All Resources** and **Node Pools** tabs.

The PowerFlex management platform object represents the platform resource used to operate and manage the PowerFlex management cluster (KVM).

NOTE: Depending upon the number of resources, it might take a few minutes to display the discovered resources every time you run the inventory.

The **All Resources** tab displays the following information about the resources that are discovered and managed in PowerFlex Manager:

Table 55. All Resources tab

Field name	Description
Health	Indicates the resource health.
Compliance	Indicates if the resource firmware and software compliance state are **Compliant**, **Non-compliant**, **Update required**, or **Update failed**. Compliance is determined by the firmware/software version for the selected resource, based on the default compliance version. Click the compliance status to view the compliance report.
OS Hostname	Indicates the resource hostname. This capability is available on the PowerFlex appliance and PowerFlex rack offerings only. **NOTE:** PowerFlex Manager retrieves the hostname value from iDRAC and not the operating system. If the hostname field is not updated in iDRAC, an incorrect value might display in PowerFlex Manager. Certain operating systems require additional packages to be installed for iDRAC to update the correct hostname.
Resource Name	Indicates the name of the resource. The resource name for the PowerFlex platform is PowerFlex Manager.
Management IP	Indicates the resource IP address. Click the IP address to open the **Element Manager**.
Asset/Service Tag	Indicates the resource service tag. This capability is available on the PowerFlex appliance and PowerFlex rack offerings only.
Deployment Status	Indicates if the resource deployment status is **In Use**, **Not in use**, **Available**, **Updating resource**, or **Pending Updates**. Click the deployment status to view resource group details.
Managed State	Indicates the resource state.

Table 55. All Resources tab (continued)

Field name	Description
	If you did not upload a license file in the **Initial Configuration Wizard**, PowerFlex Manager is configured for monitoring and alerting only. All resources are restricted to the **Unmanaged** resource state, and you cannot change the state to **Managed** or **Reserved**.
Model	Indicates the resource manufacturer name and model number. This capability is available on the PowerFlex appliance and PowerFlex rack offerings only.

To filter the resources that display, click the toggle filters icon on the **Resources** page.

Table 56. Resources page filters

To filter based on...	Select a value...
Resource Type	• **Element Manager** • **Node (Hardware/Software Management)** • **Switches** • **VM Manager** • **PowerFlex Gateway** • **Node (Software Management)**
Health	• **Healthy** • **Warning** • **Critical** • **Unknown** • **Service Mode**
Resource Groups	Allows you to choose one or more resource groups.
Resource State	**Managed**, **Unmanaged**, or **Reserved**
Node Pool	Lets you choose **Global** or one or more node pools.

On the **Resources** page, you can also perform the following tasks:

Table 57. Resources page tasks

Task	How to perform
Discover new resources	Click **Discover Resources**.
Manually run an inventory operation on a resource and update PowerFlex Manager with the latest resource details	Select a resource and click **Run Inventory**. After running the inventory, you can perform an **Update Resource Group Details** operation on any resource group that requires the updated resource data.
Remove a resource from PowerFlex Manager	Click **Remove**.
Update the firmware or software of a resource	Select one or more resources and click **Update Resources**.
Export a compliance details report for all resources to a CSV or PDF file	Click **Export Report** and select a report format.
Modify the current state of a resource	Select a resource and click **Change resource state to**. You can change the state to **Managed**, **Unmanaged** or **Reserved**.
View detailed information about a resource	Select the resource. In the right pane, click **View Details**.
View a firmware and software compliance report for a resource	Select the resource. In the right pane, click the link corresponding to **Compliance** field.
Update password for a resource	Select the resource and click **Update Password**.

Basic tasks

This section provides basic tasks for resource management.

Discover a resource

A resource must be discovered in PowerFlex Manager in order for PowerFlex Manager to manage it.

Before you start discovering a resource, complete the following:

- Gather the IP addresses and credentials that are associated with the resources.
- Ensure that both the resources and PowerFlex Manager are connected to the network.

A resource in PowerFlex Manager is categorized into one of the following groups: Element Manager, Node (Hardware/Software Management), Switch, VM manager, PowerFlex gateway, Node (Software Management), or PowerFlex System.

1. Access the **Discovery Wizard** by performing either of the following actions:
   - On the **Getting Started** page, click **Discover Resources**.
   - On the menu bar, click **Resources**. On the **Resources** page, click **Discover Resources** on the **All Resources** tab.
2. On the **Welcome** page of the **Discovery Wizard**, read the instructions, and click **Next**.
3. On the **Identify Resources** page, click **Add Resource Type**, and perform the following steps:
   a. From the **Resource Type** list, select a resource that you want to discover.
      - **Element Manager**, for example, a CloudLink Center or an Intelligent Platform Interface (IPI) cabinet.
      - **Node (Hardware / Software Management)**
      - **Switch**
      - **VM Manager**
      - **PowerFlex Gateway**
      - **Node (Software Management)**
      - **PowerFlex System**

      If you want to discover a PowerFlex software node or management VM (MVM) node with a credential that uses SSH key pairs, you must add the public keys to the nodes manually before performing the discovery. If you do not add the public keys first, the discovery fails.

      The **PowerFlex System** resource type is used to discover a management data store gateway.

      For a PowerFlex 4.x system, indicate whether the PowerFlex instance will be used for **Production Storage** or a **Management Cluster**.

   b. Enter the management IP address (or hostname) of the resources that you want to discover in the **IP/Hostname Range** field.

      To discover one or more nodes by IP address, select **IP Address** and provide a starting and ending IP address.

      To discover one or more nodes by hostname, select **Hostname** and identify the nodes to discover in one of the following ways:

      - Enter the fully qualified domain name (FQDN) with a domain suffix.
      - Enter the FQDN without a domain suffix.
      - Enter a hostname search string that includes one of the following variables:

      Table 58. Hostname search strings

Variable	Description
${num}	Produces an automatically generated unique number.
${num_2d}	Produces an automatically generated unique number that has two digits.
${num_3d}	Produces an automatically generated unique number that has three digits.

      If you use a variable, you must provide a start number and end number for the hostname search.

   c. Select one of the following options from the **Resource State** list:

Table 59. Resource state

Option	Description
Managed	Select this option to monitor the firmware version compliance, upgrade firmware, and deploy resource groups on the discovered resources. A managed state is the default option for the switch, VMware vCenter, element manager, and PowerFlex gateway resource types. Resource state must be set to **Managed** for PowerFlex Manager to send alerts to Dell Technologies connectivity services.
Unmanaged	Select this option to monitor the health status of a device and the firmware version compliance only. The discovered resources are not available for a firmware upgrade or deploying resource groups by PowerFlex Manager. This option is the default for the node resource type. If you have yet not uploaded a license, PowerFlex Manager is configured for monitoring and alerting only. In this case, **Unmanaged** is the only option available.
Reserved	Select this option to monitor firmware version compliance and upgrade firmware. The discovered resources are not available for deploying resource groups by PowerFlex Manager.

d. To discover resources into a selected node pool instead of the global pool (default), select an existing or create a node pool from the **Discover into Node Pool** list. To create a node pool, click the **+** sign to the right of the **Discover into Node Pool** box.

e. Select an existing or create a credential from the **Credentials** list to discover resource types. To create a credential, click the **+** sign to the right of the **Credentials** box. PowerFlex Manager maps the credential type to the type of resource that you are discovering. The credential types are as follows:

- Element Manager
- Node (Hardware/Software Management)
- Switch
- VM Manager
- PowerFlex Gateway
- Node (Software Management)
- PowerFlex System

The default node (Hardware/Software Management) credential type is **Dell PowerEdge iDRAC Default**.

f. To have PowerFlex Manager automatically configure iDRAC nodes to send alerts to PowerFlex Manager, select the **Auto configure nodes to send alerts to PowerFlex Manager** checkbox.

4. Click **Next**.

You might have to wait while PowerFlex Manager locates and displays all the resources that are connected to the managed networks.

To discover multiple resources with different IP address ranges, repeat steps 2 and 3.

5. On the **Discovered Resources** page, select the resources from which you want to collect inventory data and click **Finish**. The discovered resources are listed on the **Resources** page.

Related information

Getting started (mirroring)
Configuring block storage
Resource health status
Compliance status

Discover a PowerFlex system

Perform this task to discover a PowerFlex system.

This capability is available on the PowerFlex appliance and PowerFlex rack offerings only.

Before you start discovering a PowerFlex system, complete the following:

- Gather the IP addresses and credentials that are associated with the resource.
- Ensure that both the resource and PowerFlex Manager are connected to the network.

1. Access the **Discovery Wizard** by performing either of the following actions:
   - On the **Getting Started** page, click **Discover Resources**.
   - On the menu bar, click **Resources**. On the **Resources** page, click **Discover Resources** on the **All Resources** tab.
2. On the **Welcome** page of the **Discovery Wizard**, read the instructions, and click **Next**.
3. On the **Identify Resources** page, click **Add Resource Type**, and perform the following steps:
   a. From the **Resource Type** list, select **PowerFlex System**.
   b. Enter the management IP addresses of the LIA nodes in the **MDM Cluster IP Address** field. You must provide the IP addresses for all the nodes in a comma-separated list. The list should include a minimum of three nodes and a maximum of five nodes.

      If you forget to add a node, the node will not be reachable after discovery. To fix this problem, you can rerun the discovery later to provide the missing node. You can enter just the one missing node, or all the nodes again. If you enter IP addresses for any nodes that were previously discovered, these nodes are ignored on the second run.

   c. Select an existing credential or create a new one from the **Credentials** list. The credential must be a **PowerFlex Management System** credential. Be sure to provide the LIA password that was used for the original setup. The LIA password is required for the mTLS configuration.
4. Click **Next**.
5. On the **Summary** page, click **Finish**.
   After you complete the discovery, you should see a **PowerFlex System** resource on the **Resources** page.
6. SSH to the two SVMs associated with the **`powerflex-mds`** system. Run `scli --add_certificate --certificate_file /opt/emc/scaleio/mdm/cfg/mgmt_CA.pem`.

Viewing resource details

Standard users can view details only for resources that are part of node pools for which they have permissions.

1. On the menu bar, click **Resources**.
2. On the **All Resources** tab, click a resource for which you want to view details.
3. In the right pane, click **View Details**. The **Details** page displays detailed information about the resource and associated components.

   For the PowerFlex rack Intelligent Physical Infrastructure (IPI) cabinet, you can view information including system name and location, model number, IP address, cabinet temperature, and humidity details.

   For PowerEdge servers, you can also see performance details, including system usage, CPU usage, memory usage, and I/O usage. Performance usage values are updated every five minutes.

   For a PowerFlex gateway, you can view MDM cluster, storage, and node details.

   For supported switches, you can view packet count by port for all ports or a specific port. The packet count for all ports shows the average packet count for all the ports in that switch.

   For storage only, you can view details of the NIC interfaces, port connections, firmware/software revisions, CPUs, memory, and local storage.

Related information

Next steps

Viewing node details

From the node details page, you can open the remote console of the node's Integrated Dell Remote Access Controller (iDRAC). You can also view recent activities that are performed on the node.

This capability is available on the PowerFlex appliance and PowerFlex rack offerings only.

1. On the menu bar, click **Resources**.
2. On the **All Resources** tab, select a node from the resources list to view its details.
3. In the **Details** pane, click **View Details**.

The **View Details** page displays the detailed information about the node on the following tabs:

- **Network Interfaces**
- **Port View**
- **Firmware/Software Revisions**
- **CPUs**
- **Memory**
- **Local Storage**

  NOTE: You can select one of the following options from the **Filter By** list.

  - **Logical Disks**
  - **Physical Disks**

    To see which disks are self-encrypting drives (SED), look at the **Security Status**. The value **Encryption Capable** indicates that the disk is an SED. The value **Not Capable** indicates that the disk is not an SED.

    On the **Resources** page for the node, filter by **Physical Disks** to view the details of the Vault Optimized Storage Solution (VOSS) under the PCIe Extender section.

Viewing VMware vCenter details

You can view details for VMware vCenter.

This capability is available on the PowerFlex appliance and PowerFlex rack offerings only.

1. On the menu bar, click **Resources**.
2. On the **All Resources** tab, click VMware vCenter from the resource list to view the details.
   In the right pane, you can see the basic information about the VMware vCenter, such as power state, management IP address, data centers, clusters, hosts, and virtual machines.
3. In the **Details** pane, under **vCenter Details**, expand **vCenter** > **Datacenter** > **Cluster** to view the lists of nodes and applications.

Opening the iDRAC remote console

To simplify routine node maintenance, you can open a remote console to the node's Integrated Dell Remote Access Controller (iDRAC) directly from PowerFlex Manager.

This capability is available on the PowerFlex appliance and PowerFlex rack offerings only.

1. On the menu bar, click **Resources**.
2. On the **All Resources** tab, click a node.
3. In the **Details** pane, click **View Details**.
4. Click **Launch iDRAC GUI**.

Viewing platform details

You can view the PowerFlex management platform details.

This capability is available on the PowerFlex appliance and PowerFlex rack offerings only.

1. On the menu bar, click **Resources**.
2. On the **All Resources** tab, click **Platform** from the resource list to view the details.
   The right pane displays the PowerFlex management platform object and information for the platform resource. Basic information is displayed, such as the cluster health, node count, member nodes and their VLAN IP addresses, and data sources with their related network IP addresses.

Viewing PowerFlex system details (mirroring)

You can view performance metrics and storage details for a PowerFlex system.

PowerFlex Manager stores up to 15 GB of PowerFlex gateway PowerFlex system metrics. Once this threshold is exceeded, PowerFlex Manager automatically purges the oldest data to free up space.

1. On the menu bar, click **Resources**.
2. On the **All Resources** tab, click a PowerFlex system resource from the list to view its details.
   The right pane displays basic information about the resource, such as the unused and spare space for PowerFlex. It also shows the number of protection domains, volumes, storage data clients (SDC), and storage data servers (SDS).
3. In the right pane, click **View Details**.
   The **Details** page displays detailed information about the PowerFlex System resource on the following tabs:
   - **MDM Cluster**: Displays details about the MDM roles within the cluster.
   - **Storage**: Displays the protection domain, storage pools, volumes, and mapped storage data clients.
   - **Nodes**: Displays the protection domain, type (SDS or SDC), connection, name, IP address for the SDS or SDC, and fault set.

   The **Details** page also allows you to launch a wizard to see the status of a recent upgrade or reconfigure the MDM roles in the PowerFlex cluster.
4. Click the **MDM Cluster** tab to see information about the MDM roles within the cluster.
5. Click the **Storage** tab. For each storage pool within a protection domain, PowerFlex Manager shows the granularity setting and acceleration pool. For each volume within a storage pool, PowerFlex Manager shows the **Compression** and **Type** settings.

   For a storage pool that has compression disabled, the granularity is set to medium. For a storage pool that has compression enabled, the granularity is set to fine.

   For volumes, you may see a different number in the **Volumes** column on the **Resources** tab than you see for the total number of reported volumes on the **Volumes** page on the **Block** tab. The count of volumes that are listed on the PowerFlex system details view is the "vTree" count (volume + snapshots). The count of volumes that are listed on the **Block** tab is just the count of volumes.

   To search for volumes associated with the gateway:

   a. Enter a volume or datastore name search string in the **Search Text** box.
   b. Optionally, select a **Size** range.
   c. Optionally, select **Thick** or **Thin** for the **Type**.
   d. Optionally, select **Enabled** or **Disabled** for the **Compression** setting.
   e. Optionally, select a specific **Storage Pool**.
   f. Click **Search**.
      PowerFlex Manager updates the results to show only those volumes that satisfy the search criteria. If the search returns more than 50 volumes, you must refine the search criteria to return only 50 volumes.

Related information

Reconfiguring MDM roles (mirroring)
Migrating vCLS VMs to shared storage

Viewing PowerFlex rack IPI cabinet details

The IPI cabinet contains an appliance that is configured to provide information about power, thermals, alerts, and all components in the cabinet's physical infrastructure. PowerFlex Manager uses SNMPv2 to poll the IPI appliance status and send results about temperature and humidity to be viewed in PowerFlex Manager.

This capability is available on the PowerFlex rack offering only.

- If your cluster is running on several nodes, you must add the IP address of each node to the SNMP network management system (NMS) section in the Panduit Appliance UI.
- Ensure that PowerFlex Manager discovers the IPI cabinet. You can verify by clicking the list of resources and checking that the IPI cabinet is present.

You can view additional cabinet details in the IPI appliance management application.

To view the details for the IPI cabinet, perform the following steps:

1. On the menu bar, click **Resources**.
2. On the **All Resources** tab, click the row containing the IPI cabinet.
3. In the right pane, click **View Details**. The **Details** page displays detailed information about the IPI cabinet, including system name and location, model number, and IP address. It also displays cabinet temperature and humidity details.
4. To view additional details, click **Launch IPI Cabinet Manager** to launch the IPI appliance management application.

Viewing CloudLink Center details

You can view details for CloudLink Center.

This capability is available on the PowerFlex appliance and PowerFlex rack offerings only.

To view the details for CloudLink Center, perform the following steps:

1. On the menu bar, click **Resources**.
2. On the **All Resources** tab, click the row containing the CloudLink Center.
3. In the right pane, click **View Details**.
   The **Details** page displays detailed information about the CloudLink Center resource on the following tabs:
   - **Summary**: Displays cluster details, system performance, alarms, pending machines, and security events. If CloudLink has clustered CloudLink Centers, PowerFlex Manager shows all CloudLink Centers.
   - **Machines**: Displays machine groups and machines.
   - **Approved Networks**: Displays a list of the names and IP addresses for all approved networks going in and out of the CloudLink Center.

   The **Details** page displays information that is similar to information provided in the CloudLink application. All the data is based on the last inventory, as indicated by the **Last Inventory** timestamp shown on the **Details** page under **Resource Information**. Whenever the inventory is updated, either automatically or manually, any changes that are made in the CloudLink Center are reflected in the **Details** page.
4. To view additional details, click **Launch Console** to launch the CloudLink application.

Viewing NVDIMM details

To view NVDIMM details for a node, perform the following steps:

1. On the menu bar, click **Resources**.
2. On the **All Resources** tab, click the row containing a node that has NVDIMM compression.
3. In the right pane, click **View Details**.

   The **Details** page displays detailed information about the node.
4. Click the **Memory** tab.

   The **Memory** tab shows the NVDIMM modules and all DIMMs in the server. The DIMMs with rank 1 are the NVDIMM devices. The NVDIMMs shown in the top section of the **Memory** tab provide the name of each **NVDIMM Module**, as well as the **Battery Enabled State**, **Battery Health State**, **Battery Operational Status**, **Remaining Rated Read Write Endurance**, and **Size** for the module.

   All of the data is based on the last inventory.

Update resource inventory

You can run inventory to incorporate changes that are made to resource data outside of PowerFlex Manager. After running the inventory to incorporate these changes into the **Resources** page, you can update the details on any resource group that needs to include the new resource data.

Superusers can run the inventory on any resources. Life cycle administrators can run the inventory only on resources that are part of a node pool for which they have permission.

1. On the menu bar, click **Resources**.
2. On the **Resources** page, click the **All Resources** tab.
3. From the list of resources, click a resource, and in the **Details** pane, click **Run Inventory**.

The resource state changes to **Pending**. When the inventory is complete, the resource state changes to **Available**. See PowerFlex Manager logs to view the start time and end time of the resource inventory operation.

Viewing a compliance report for a resource

A compliance report provides an overview of a resource's firmware and software status, highlighting whether its components align with defined baselines. It helps users quickly identify non-compliant areas and determine which updates are required to bring the resource back into compliance.

Perform the following steps to view a compliance report for a single resource:

1. On the menu bar, click **Resources**.
2. Select the resource for which you want to view the compliance report.
3. In the right pane, click the link corresponding to the **Compliance** option.
   The **Compliance Report** page displays.
   NOTE: If an RCM is mapped to the resource group and is modified, a wrench icon with the text, **Modified**, is displayed. However, if the update file is moved or deleted, the wrench icon with the text, **Needs Attention**, is displayed.
4. Select the **Firmware Components** option to view the details of the firmware available on the selected resource. For example, a switch.
5. Select the **Software Components** option to view the software components in the compliance report or to update the software components for a resource. If the resource is in the **In Use** state, it redirects you to the **Resource Groups** page.
6. To update the non-compliant resources, click **Update Resource**.
   NOTE: If there is no compliance version that is attached to the resource, you cannot perform the firmware update or software update.

Related information

Exporting a compliance report for all resources

Upgrading a PowerFlex gateway (mirroring)

Perform the following procedure to upgrade the PowerFlex gateway.

You can update the firmware and software of a PowerFlex gateway from the **Resources** page.

Upgrading the PowerFlex gateway is a two-step process. When you upgrade the PowerFlex gateway for a deployed resource group from the **Resources** page, PowerFlex Manager upgrades the PowerFlex gateway version and automatically upgrades all the storage nodes for all the resource groups that are tied to the gateway.

Any nodes that require reconfiguration prior to an upgrade are shown in the **Needs Attention** section of the wizard. If you choose to reconfigure all the nodes, you can proceed with the upgrade process. If you select only a few of the nodes, PowerFlex Manager reconfigures these nodes, but does not proceed with the upgrade process until you have reconfigured the remaining nodes. You must reconfigure all the nodes before you can complete the upgrade process.

PowerFlex Manager does not allow you to downgrade the PowerFlex gateway.

NOTE: PowerFlex Manager requires the LIA password to be the same as the MDM cluster or PowerFlex gateway admin password. If it is different, during the upgrade of PowerFlex storage-only components using PowerFlex Manager, PowerFlex Manager will reset the LIA password to be the same as the MDM cluster or PowerFlex gateway admin password.

CAUTION: Check the Alerts page before performing the upgrade. Look for major and critical alerts related to PowerFlex Block and File to be sure the MDM cluster is healthy before proceeding.

1. Choose the PowerFlex gateway from the **Resources** page.
   You cannot upgrade more than one PowerFlex gateway at a time.
2. Click **Update Resources**.
3. On the **Update Details** page, check the **Needs Attention** section to see whether any of the nodes must be reconfigured before upgrade. Select any nodes that you want to reconfigure. To select all nodes, click the box to the left of **SDS Name**.
4. Click **Next**.

5. On the **Summary** page, choose **Allow PowerFlex Manager to perform non-disruptive updates now** or **Schedule non-disruptive updates to run later**.

   Specify the type of update you want to perform by selecting one of the following options:

   - **Instant Maintenance Mode** enables you to perform updates quickly. PowerFlex Manager does not migrate the data.
   - **Protected Maintenance Mode** enables you to perform updates that require longer than 30 minutes in a safe and protected manner.

6. If you only selected a subset of the nodes for reconfiguration, confirm the reconfiguration by typing **RECONFIGURE NODES**. Otherwise, confirm the update action by typing **UPDATE POWERFLEX**.

   If you reconfigured only a subset of the nodes, you must restart the wizard later to reconfigure the remaining nodes before you can complete the upgrade process.

7. Click **Finish**.
8. Go to the **Resource Groups** page to update any resource groups that are not in compliance with the new version of PowerFlex.

The PowerFlex gateway upgrade process performs some health prechecks to confirm that the resource group is healthy before the upgrade. If the resource group is not healthy, the PowerFlex gateway upgrade is not successful.

After a successful upgrade, the PowerFlex gateway should be in compliance with the new target version. However, the nodes in the resource group may require additional maintenance. In this case, you must update any resource groups that are noncompliant from the **Resource Groups** page.

When you initiate a PowerFlex gateway update, PowerFlex Manager upgrades both the Gateway RPM and the software components that are non-compliant.

Related information

Updating a resource group with new firmware and software (mirroring)

Removing resources

Only super users can remove resources from PowerFlex Manager.

To remove a resource from PowerFlex Manager, perform the following steps:

1. On the menu bar, click **Resources**.
2. On the **Resources** page, click the **All Resources** tab.
3. From the list of resources, select one or more resources, and click **Remove**.
4. Click **OK** when the confirmation message appears.

If you remove a node, the node state changes to **Pending** and it powers off.

NOTE: Before you add new nodes using an existing iDRAC IP address in inventory, ensure that you remove the old nodes from PowerFlex Manager before discovering the new nodes.

Related information

Resource health status

Compliance status

Additional resource management tasks

This section provides additional tasks for resource management.

Creating a node pool

You can create a node pool on the **Resources** page.

This capability is available on the PowerFlex appliance and PowerFlex rack offerings only.

1. On the menu bar, click **Resources**. Click the **Node Pools** tab.

2. On the **Node Pools** page, click **Create**.
   The **Create Node Pool** wizard starts.
3. On the **Welcome** page, read the instructions, and click **Next**.
4. On the **Node Pool Information** page, enter a name and description for the node pool. Click **Next**.
5. On the **Add Nodes** page, select the nodes that you want to add to the node pool. Click **Next**.
6. On the **Assign Users** page, select the users that you want to grant access rights to the node pool. Click **Next**.
7. On the **Summary** page, review the node pool configuration and click **Finish**.

Modifying a node pool

You can modify the details for a node pool on the **Resources** page.

This capability is available on the PowerFlex appliance and PowerFlex rack offerings only.

1. On the menu bar, click **Resources**. Click the **Node Pools** tab.
2. On the **Node Pools** page, click **Modify**.
   The **Modify Node Pool** wizard starts.
3. To change the name and description of the node pool, in the left pane, click **Node Pool Information**. Make the updates and click **Save**.
4. To add or remove nodes from the node pool, in the left pane, click **Add Nodes**. Make the updates and click **Save**.
5. To add or remove the access rights to the node pool, in the left pane, click **Assign Users**. Make the updates and click **Save**.

Removing a node pool

You can remove a node pool on the **Resources** page.

This capability is available on the PowerFlex appliance and PowerFlex rack offerings only.

1. On the menu bar, click **Resources**. Click the **Node Pools** tab.
2. On the **Node Pools** page, select one or more node pools to delete. Click **Remove**.
3. Click **OK** when the confirmation message appears.

Viewing the port view details

Port view is a PowerFlex Manager feature that is associated with a resource group deployment. It enables you to view the physical network configuration characteristics and topology from the network interface cards to upstream switches, highlighting such details as switch ports, VLANs, and network identifiers such as MAC addresses.

This capability is available on the PowerFlex appliance and PowerFlex rack offerings only.

You can access port view from both the **Resources** and **Resource Groups** pages. Port view provides connectivity information for both new and existing resource groups.

The **Port View** page displays the following information:

- Topology information for all networks and VLANs deployed in a resource group
- Network connections between the devices
- Health of the resources
- Connection details section with information about the network devices

Port View is only available for PowerFlex nodes and is not available for management VMs, such as CloudLink Center.

PowerFlex Manager does not show switch connectivity for deployments with partial network automation.

To view the port view details:

1. On the **Resource Groups** page, view the details for a resource group and click **Port View**. If you are accessing the port view from the **Resources** page, select a node that is in a **Deployed** state and click **View Details** on the right pane first.

   If you select a node that is not in a deployed state, only the interface card information is displayed.

2. To view device details such as hostname, model name, and management IP address, or information about associated devices, click the specific ports or devices.

3. To view information about intermediate devices in port view, ensure that the devices are discovered and available in the inventory. Sometimes, connectivity cannot be determined for an existing resource group because the switches have not yet been discovered. In this case, you see only the node in port view, but you do not see connectivity information. You can correct this by going back and discovering the switches, and updating the resource group again.

   PowerFlex Manager cannot discover interface cards that do not have integrated LLDP support (such as Intel X520).

4. To filter information based on the connectivity, select an option from the **Display Connections** list. **Show All Connections** is the default option.

Exporting a compliance report for all resources

You can download a CSV or PDF report that lists compliance details for all resources.

The CSV report includes all information and can be imported into a database for querying.

The PDF format contains a subset of information to make it easier to read. It measures each resource against a specified compliance file version. For each resource, it lists the components, the current and expected software or firmware version, and whether the component is compliant.

To download a compliance report for all resources:

1. On the menu bar, click **Resources**.
2. Click **Export Report**.
3. Select either **Export Compliance PDF Report** or **Export Compliance CSV Report** from the drop-down list.
   The selected report downloads.

Related information

Viewing a compliance report for a resource
Resource health status
Compliance status

Exporting a configuration report for all resources and resource groups

You can download a PDF report that lists configuration details for all resources and resource groups.

The configuration report shows the result of various configuration checks that PowerFlex Manager performs against the system. You can use the report to troubleshoot your resources and resource groups.

The report shows the following kinds of information:

Table 60. Configuration report

Column name	Description
Resource Name	Provides the name of the resource for which the checks have been run. This column corresponds to the **Resource Name** shown on the **Resources** page.
Asset/Service Tag	Provides the asset or service tag of the resource for which the checks have been run. This column corresponds to the **Asset/Service Tag** shown on the **Resources** page.
Management IP	Specifies the **Management IP** of the resource for which the check is run. For a PowerFlex cluster, the management IP address is the MDM cluster IP address. For a PowerFlex gateway, the management IP address is the PowerFlex gateway IP address.
Result	Shows the result of the check (`PASS` or `FAIL`).
Severity	Indicates the severity of the check. The severity is based on the result. The severity levels are INFO, WARNING, HIGH, or CRITICAL. If the result is PASS, the severity is INFO. If the result is FAIL, the severity depends on the type of check. PowerFlex Manager supports only CRITICAL checks.

Table 60. Configuration report (continued)

Column name	Description
Details	Provides a description of the check that was run.
Affected Resources	Gives a list of the IP addresses or unique identifiers of resources that are impacted by the check. The list of affected resources helps with troubleshooting.

1. On the menu bar, click **Resources**.
2. Click **Export Report**.
3. Select **Export Configuration PDF Report** from the drop-down list.

   The selected report downloads.

Related information

Configuration checks

Importing networks

You can import many general-purpose VLANs from VMware vCenter.

This capability is available on the PowerFlex appliance and PowerFlex rack offerings only.

After importing networks, you can add them to templates or resource groups. You can also import networks when you add an existing resource group.

1. On the menu bar, click **Resources**.
2. On the **All Resources** tab, click the VMware vCenter from which you want to import networks.

   NOTE: Adding a standard vSwitch network is not supported and leads to unsupported scenarios.

   For compute-only or hyperconverged add one of the supported networks:

   - Hypervisor management
   - PowerFlex management
   - PowerFlex data
   - vMotion
   - Out-of-band management

   Importing networks from VMware vCenter is not supported for PowerFlex software nodes.

3. In the right pane, click **Import Networks**.
   PowerFlex Manager displays the **Import Networks** wizard. In the **Import Networks** wizard, PowerFlex Manager lists the port groups that are defined on the VMware vCenter as **Available Networks**. You can see the port groups and the VLAN IDs.
4. Optionally, search for a VLAN name or VLAN ID.
   PowerFlex Manager filters the list of available networks to include only those networks that match your search.
5. Click each network that you want to add under **Available Networks**. If you want to add all the available networks, click the checkbox to the left of the **Name** column.
6. Click the double arrow (>>) to move the networks you chose to **Selected Networks**.
   PowerFlex Manager updates the **Selected Networks** to show the ones you have chosen.
7. Click **Save**.

PowerFlex Manager creates a job to add the networks. The job may finish quickly, so you may not see on the **Jobs** page.

Go to **Settings** > **Networks** to see the imported networks.

Updating passwords for system components

You can update the passwords for some system components from PowerFlex Manager.

1. On the menu bar, click **Resources**.

2. On the **All Resources** tab, select one or more resources of the same type for which you want to change passwords.

   For example, you could select one or more iDRAC nodes or you could select one or more PowerFlex gateway components.
3. Click **Update Password**.

   PowerFlex Manager displays the **Update Password** wizard.
4. On the **Select Components** page, select one or more components for which you want to update a password and click **Next**.

   The component choices vary depending on which resource type you initially selected on the **Resources** page.
5. On the **Select Credentials** page, create a credential or change to a different credential having the same username.
6. Click **Finish** and click **Yes** to confirm the changes.

Updating passwords for nodes

You can update the passwords for one or more nodes from PowerFlex Manager.

1. On the menu bar, click **Resources**.
2. On the **All Resources** tab, select one or more nodes for which you want to change the passwords.
3. Click **Update Password**.

   PowerFlex Manager displays the **Update Password** wizard.
4. On the **Select Components** page, specify which passwords you want to update for the selected nodes by clicking one or more of the following check boxes.
   - **iDRAC Password**
   - **Node Operating System Password**
   - **MVM Operating System Password**

   The option to update passwords for SVMs and MVMs will only appear if a node is in a resource group.
5. Click **Next**.
6. On the **Select Credentials** page, create a credential with a new password or change to a different credential.
   a. Open the **iDRAC**, **OS Password**, or **MVM Password** object under the **Type** column to see credential details for each node you selected on the **Resources** page.

      The **MVM Password** sections do not appear if there is nothing to show for SVMs or MVMs.

   b. To create a credential that has the new password, click the plus sign (**+**) under the **Credentials** column.

      Specify the **Credential Name** and the **User Name** for which you want to change the password. Enter the new password in the **Password** and **Confirm Password** fields.

   c. To modify the credential, click the pencil icon for the nodes under the **Credentials** column and select a different credential.
   d. Click **Save**.

   You must perform the same steps for the node operating system and SVM operating system password changes. For a node operating system credential, only the OS admin credential type is updated.
7. Click **Finish**.
8. Click **Yes** to confirm.

PowerFlex Manager starts a new job for the password update operation, and a separate job for the device inventory. The node operating system, SVM, and MVM operating components are updated only if PowerFlex Manager is managing a cluster with these components. If PowerFlex Manager is not managing a cluster with these components, these components are not displayed and their credentials are not updated. Credential updates for iDRAC are allowed for managed and reserved nodes only. Unmanaged nodes do not provide the option to update credentials.

Updating passwords for PowerFlex gateway components

You can update the passwords for one or more PowerFlex gateway components from PowerFlex Manager.

1. On the menu bar, click **Resources**.
2. On the **All Resources** tab, select one or more PowerFlex gateway components for which you want to change the passwords.
3. Click **Update Password**.

PowerFlex Manager displays the **Update Password** wizard.

4. On the **Select Components** page, select **PowerFlex Password**.
5. Click **Next**.
6. On the **Select Credentials** page, create a credential with a new password or change to a different credential.
   a. Open the **PowerFlex** object under the **Type** column to see details about each gateway you selected on the **Resources** page.
   b. To create a credential that has the new password, click the plus sign (**+**) under the **Credentials** column.

      Specify the **Credential Name**, as well as the **Gateway Admin User Name** and **Gateway OS User Name** for which you want to change passwords. Enter the new passwords for both users and confirm these passwords.

   c. To modify the credential, click the pencil icon for one of the nodes under the **Credentials** column and select a different credential.
   d. Click **Save**.
7. Click **Finish**.
8. Click **Yes** to confirm.

PowerFlex Manager starts a new job for the password update operation, and a separate job for the device inventory. If PowerFlex Manager is managing a cluster for any of the selected PowerFlex gateway components, it updates the credentials for the **Gateway Admin User** and **Gateway OS User**, as well as any related credentials, such as the LIA and lockbox credentials. If PowerFlex Manager is not managing the cluster, it only updates the credentials for the **Gateway Admin User** and **Gateway OS User**.

19

Monitoring events and alerts

This section includes tasks for monitoring events and alerts from resources that are installed or automatically discovered. This section also includes management diagnostics which provide detailed information about the health of PowerFlex management platform pods, nodes, and diagnostics.

Event and alert monitoring provides information regarding occurrences in your system.

An event is a notification that something happened in the system. Events monitor configuration changes, activities of the system, faults, and errors in the system and send notifications accordingly to inform the administrator.

An event that requires attention generates an alert as well. Other events can update or clear an alert when the system detects a change in the condition that needs attention.

An alert is a state in the system which is usually on or off. Alerts monitor serious events that require user attention or action.

Although the events may be interesting for troubleshooting purposes, it is not necessary to monitor events. Whereas an alert is a summation of one or more events that need or needed attention.

Only superusers and support users can view the management diagnostics page.

Only superusers and support users can troubleshoot, monitor, and review the health of PowerFlex management platform pods, nodes, and diagnostics.

Events and alerts that are generated by PowerFlex management platform pods, nodes, and diagnostics are accessible to all users.

The **Monitoring** menu displays the events, alerts, jobs, and management diagnostics on your system.

Related information

Enabling Dell Technologies connectivity services

Events

An event is a notification that something happened in the system. An event happens at a single point in time and has a single timestamp. An event may be unique or be part of a series.

Each event message is associated with a severity level. The severity indicates the risk (if any) to the system, in relation to the changes that generated the event message.

PowerFlex Manager stores up to 3 million events or events that are up to 13 months old. Once this threshold is exceeded, PowerFlex Manager automatically purges the events to free up space. The threshold is reviewed daily.

NOTE: An event is published each day that lists the events which have been removed that day.

Events severity level

The severity levels are as follows:

Table 61. Events severity level

Severity level	Explanation	Example
Information	Informs you of events that you should be aware of, but that do not put the system at risk (no urgency).	`The volume is created.`
Minor	Indicates a failure that may result from an acceptable condition (for example user error), but can also indicate a possible failure.	`The Device media type is mismatched.`

Table 61. Events severity level (continued)

Severity level	Explanation	Example
Major	An error alarm raised by the system. This error requires your attention. The system is stable but could be degraded.	`The device failed.`
Critical	A major error alarm raised by system. The system requires immediate attention. Some data may be unavailable.	`The storage pool is running out of capacity.`

Event default and advanced fields

The default and advanced fields are as follows:

Table 62. Event default and advanced fields

Field	Description	Example
Timestamp	The time the event occurred in the following format: `YYYY-MM-DD hh:mm:ss.sss`	`2022-04-03T13:49:18.709Z`
Severity	A predefined severity level that is associated with an event.	`INFORMATION`
Event ID	A unique identifier specific to the event instance that is always 16 digits.	`B46b66c8c79e4a2a`
Code	A unique event code that is always eight digits and identifies a specific event.	`80120001`
Event Name	An alphanumeric descriptive name for the event that is unique.	`MISCELLANEOUS`
Description	A free text description of the event.	`Successfully initialized the Discovery Manager`
Sub system	A resource domain that identifies the area that the alert relates to. The resource domains are: • Management • Block • File • Compute • Network • Security	`MANAGEMENT`
Resource Type	The type of resource that is involved in the event.	`basic-system-config`
Resource Name	The name of the resource that is involved in the event.	`asm manager`
Resource ID	The system specific ID of the resource involved in the event.	`asmcore`
Category	There are six categories of events that help you to filter: • Security • Audit • Software • Hardware • Maintenance	`STATE_CHANGED`

Table 62. Event default and advanced fields (continued)

Field	Description	Example
	• State changed	
Service Name	The name of the service that generated the event.	`mdm_gw`
Details	Extra details that are relevant to the operation, for example, user command URL or a list of command parameters.	`[cmd: login, user: admin]`
Originator Application Name	The name of the application that generated the event. This field can be null.	`REST`
Message	A message that describes the event.	`Successfully initialized the Discovery Manager`

Related information

Enabling Dell Technologies connectivity services

Viewing an event

You can view an event in PowerFlex Manager.

1. Go to **Monitoring** > **Events**.
   The **Events** page opens which displays a list of system events.
2. Click **Add filter** to control the list of events displayed.
3. Enter case-sensitive free text and add additional filters by selecting **Timestamp**, **Severity**, or **Sub system**.
4. Select the event that you want to view.

   NOTE: You can reset the default and advanced field columns by clicking the **Reset Columns** icon.

5. Click **Export All** to download and review all the events from the events log.

Related information

Enabling Dell Technologies connectivity services

Alerts

An alert is a state in the system which is usually on or off. Alerts monitor serious events that require user attention or action.

When an alert is no longer relevant or is resolved, the system automatically clears the alerts with no user intervention. This action ensures that cleared alerts are hidden from the default view so that only relevant issues are displayed to administrators. Cleared alerts can be optionally displayed through table filtering options. Alerts can also be acknowledged which removes the alert from default view. Acknowledging an alert does not indicate that the issue is resolved. You can view acknowledged alerts through the table-filtering options.

Alert severity level

The severity levels are as follows:

Table 63. Alert severity levels

Severity level	Explanation	Example
Information	An information level informs you of things that happen in the system that do not require action.	`NDU`

Table 63. Alert severity levels (continued)

Severity level	Explanation	Example
Minor	A minor level informs you that action is required but is not urgent.	`This Storage Pool contains devices that are not being accelerated.`
Major	A major level informs you that action is required soon.	`The MDM certificate is about to expire.`
Critical	A critical level informs you that the system requires immediate attention.	`The MDM certificate has expired.`

Alert default and advanced fields

The default and advanced fields are as follows:

Table 64. Alert default and advanced fields

Field	Description	Example
Description	A message that describes the alert.	`The MDM cluster is degraded, and data is not protected.`
System impact	A free text, description of the system impact of the alert.	`Risk of cluster unavailability`
Repair Flow	Free text, actions that you can take in order to repair the issue.	`Check that all MDM cluster nodes are functioning correctly, and fix and replace faulty nodes, if necessary, to return to full protection`
Alert Details	Any extra details that are relevant to the object and or incident involved.	`Percentage of SP capacity usage`
Associated Events	A list of events that modified the life cycle of the alert.	`[object Object]`
Resource ID	The ID of the resource that is associated with the alert.	`86fb0000000000`
Resource Name	The name of the resource that is associated with the alert. This is usually the resource that is involved with raising the event.	`storagenode212`
Resource Type	The type of resource that is associated with the alert. this is usually the resource that is involved with raising the event.	`storagenodes`
Last Updated	The UTC date and time that the alert was last updated.	`2022-04-28T07:58:46.16Z`
Timestamp	The UTC date and time that the alert was initially raised.	`2022-04-28T07:58:46.16Z`
Acknowledged status	The acknowledged status indicates whether the alert is acknowledged or unacknowledged.	`Y`

The total number of alerts that can be sent to secure remote services gateway from the PowerFlex Manager dispatcher service is restricted to 200 per day. The threshold for the number of alerts for a particular event is set to three per hour. After the first three alerts are sent, if fourth alert is generated for the same event, it is not sent to the secure remote services. However, if the alerts are sent from two different systems running iDRAC, the alerts are sent to the secure remote services gateway.

A new threshold alert is triggered from PowerFlex Manager when the threshold of 200 alerts per 24 hours is crossed per day and is automatically sent to secure remote services gateway. Similarly, a new alert is triggered from PowerFlex Manager and sent to secure remote services gateway when threshold of three alerts per hour for same alert type, symptom code, or resource is reached.

Related information

Enabling Dell Technologies connectivity services

Viewing and acknowledging an alert

You can view and acknowledge an alert in PowerFlex Manager. When you acknowledge an alert, PowerFlex Manager removes the alert from the default view. Acknowledging an alert only indicates that you have seen the alert. It does not indicate that the issue is resolved. You can view acknowledged alerts through the table filtering options.

1. Go to **Monitoring** > **Alerts**.
   The **Alerts** page opens and displays a list of system alerts. You can add one or more filters to control the list of alerts displayed.
2. Select the alert that you want to view or acknowledge.

   NOTE: You can reset the default and advanced field columns by clicking the **Reset Columns** icon.

3. You can:
   - Click **Acknowledge** to acknowledge an alert.
   - Click **Unacknowledge** to remove an alert acknowledgment.

Jobs

In PowerFlex Manager, you can view the details of discovery, firmware update, inventory, and resource group deployment jobs. Only a user with an administrator role can view jobs.

The **Jobs** page displays the following information about the jobs that are scheduled or running in PowerFlex Manager:

- **State** — Displays one of the following states that is based on the job status:
  - **Error** —Job has completed with errors (job is complete but failed on one or more resources).
  - **Scheduled**—Job is scheduled to run at a specific time. It can be scheduled to run at a single time or at several times as a recurring job.
  - **In progress**—Job is running.
- **Job Name**—Identifies the name of the job.
- **Started By**—Displays the name of the user who started the job.
- **Start Time**—Displays the date and time when the job is scheduled to run.
- **Time Elapsed**—Displays the time that has elapsed from the start time to the end time of a job instance.
- **Details**—Displays the detail of a job instance.

If a job scheduled is a one-time job, after execution, it is not listed in the **Jobs** page.

To cancel the job, select the job that you want to cancel, and click **Cancel**.

To filter for a job, click the filter icon.

Management diagnostics

Management diagnostics provide detailed information about the health of PowerFlex management platform pods, nodes, and diagnostics. Only superusers and support users have access to troubleshoot, monitor, and review management diagnostics.

The PowerFlex management platform **Diagnostics** page provides an overview of the latest scheduled run details and manually run diagnostic results.

The **Nodes** page displays the overall health of all the PowerFlex management platform nodes.

The **Pods** page displays the overall health of all the PowerFlex management platform pods. You can restart specified pods from the user interface, copy and download logs from the current pod life cycle or download historical logs from all pod life cycles.

Diagnostics

The **Diagnostics** page displays an overview of the latest run details and diagnostic results.

The following table lists the scheduled frequency of the diagnostics.

Table 65. Scheduled diagnostics

Diagnostic	Description	Scheduled frequency
Pod Status	Monitors the health of all the PowerFlex management platform pods. If there are three consecutive pod failures for a specific pod a failure event is raised in the events log, which usually occurs during upgrades.	1 minute
Node Status	Monitors the health of all the PowerFlex management platform nodes.	2 minutes
Database Health	Monitors the database instance and data replication.	5 minutes
etcd Performance	Evaluates the etcd cluster performance and health under various loads. It provides metrics such as average latency, throughput, and variance. The `etcdctl check perf` command provides a performance measurement and benchmarking tool for etcd clusters. This etcd performance might fail if the cluster components are busy or not fast enough.	N/A
Internal DNS Lookup	Monitors network connectivity through internal DNS queries.	2 minutes
Internal Connectivity	Monitors the internal connectivity between nodes.	5 minutes
Disk Space	Monitors the capacity of the management and operations data under `/opt/platform-provisioner`.	5 minutes
Message Bus Health	Monitors the number of messages in each stream. Warning threshold: 100,000 messages Error threshold: 1 million messages	2 minutes
etcd Certificate	Monitors when one of the nodes secret was created or changed after the node started. The warning threshold is 40 days before the secret was created or changed on one of the nodes. The error threshold is 10 days before the secret was created or changed on one of the nodes.	24 hours

The diagnostic cards display the following details:

- Diagnostic name
- Result icon
- Last run date and time
- Run state

If a diagnostic run receives an error a failure message is displayed.

View the run overview and result details:

- Overview of the run state, start time, and end time.
- Diagnostic results display the result of the latest run.
- If there are errors a failure reason displays.
- Recommendations to remediate failures.
- For more detailed diagnostic result information, click **View Raw Output**.

Optionally, you can edit diagnostic warning and error thresholds and scheduled frequency intervals in the configuration map with *kubectl* or *K9s* command-line tools. After editing the configuration map, restart the Diagnostic Engine pod and the PowerFlex Management Maintenance Manager pod. If the core messaging system is down, only restart the PowerFlex Management Maintenance Manager pod.

NOTE:

- A diagnostic failure event is generated in the events log after a diagnostic fails three consecutive times.
- If the core messaging system for PowerFlex is unavailable, then events and alerts are not available and scheduled diagnostics will not run. You can manually run all diagnostics, except for the etcd performance diagnostic. Historic data will not be available for any of the diagnostics.

Running diagnostics manually

Superusers and support users can manually run the PowerFlex management platform diagnostics to troubleshoot and review their overall status.

1. Go to **Monitoring** > **Management Diagnostics** > **Diagnostics**.
   The **Diagnostic** page opens which displays PowerFlex management platform diagnostic cards.
2. The result icon, last run date, run time, and run state are listed for each diagnostic card. Click **Run Now** to manually start a diagnostic.

   A notification message confirms that the specified diagnostic has started.

   After a diagnostic run begins, you can cancel it.

   If there are errors, a failure message displays.
3. The overview displays the run state, start time, and end time. The diagnostic result displays failure reasons, recommendations to remediate failures, and raw output only if errors occur. Click **View Raw Output** for more information.

PowerFlex management platform node health

The **Nodes** page displays details of the health of the PowerFlex management platform nodes.

The following table describes the PowerFlex management platform node health details.

Table 66. Node health

Field	Description
Name	Displays the name of the node.
Status	Displays an icon indicating one of the following statuses: • Ready: Traffic can flow through the node • Not ready: Traffic cannot flow through the node because it is failing or unavailable.
IP Address	Displays the IP address of the node.
Disk Pressure	Displays the disk space available for the root file system of the node and image file system. If the disk space is low, then it affects Kubernetes operations such as Kubernetes stops scheduling new pods to the node. The system runs the garbage collector which deletes images which are based on the last time they were used, starting with the oldest.

Table 66. Node health (continued)

Field	Description
	Warnings occur when the disk space for the node or image file systems is less than 10%. Displays an icon indicating that disk pressure is one of the following: ● Greater than 10% ● Less than 10% Critical: Node file system < 5% or Image file system < 5%
Memory Pressure	The node memory pressure uses the default configuration. The pods are evicted when less than 100 MiB are available. Displays an icon indicating the memory pressure is one of the following: ● Greater than 100 MiB is available ● Less than 100 MiB is available
Network Unavailability	Displays an icon indicating the node on the network is one of the following: ● Available ● Unavailable Container Network Interfaces (CNI) are used for cluster networking on the nodes.
PID Pressure	The process ID pressure for nodes measures the ability to start more containers. If the node reaches the specified PID limits, there might be workload failures when trying to obtain a new process ID. Warnings occur when there are less than 10% of available process IDs. Displays an icon indicating that the PID pressure is one of the following: ● Greater than 10% ● Less than 10% Critical available process IDs < 5%

Viewing nodes

Superusers and monitor users can view node details on the **Nodes** page in PowerFlex Manager.

1. Go to **Monitoring** > **Management Diagnostics** > **Nodes**.
   The **Nodes** page opens which displays a list of PowerFlex management platform nodes.
2. Click **Add filter** by entering case-sensitive free text to control the list of nodes displayed.
3. Select the node that you want to view.

NOTE: You can reset the default and advanced field columns by clicking the **Reset Columns** icon.

PowerFlex management platform pod health

The **Pods** page displays the health of the PowerFlex management platform pods.

The following table describes the health details of the PowerFlex management platform pods.

Table 67. Pod health

Name	Description
Pod Name	Displays the name of the pod.
Status	Displays the status of the pod and its containers. The status values are in the pod health status table.
Containers Ready	Displays the number of pod containers that are ready to accept traffic.
Namespace	Isolates groups within a single cluster. The primary namespace for pods is PowerFlex.
Node	Displays the name of the node.
Container Restarts	Displays the number of pod container restarts.
Age	Displays the age of the pod in days, hours, minutes, and seconds.

The following table describes the pod health status values that might be displayed.

Table 68. Pods health status

Icon	Name	Definition
	Running	All the pod containers were created and at least one pod container is running, starting, or restarting.
	Succeeded	All containers within the pod finished their tasks successfully.
	Completed	All pod containers completed their tasks.
	Container creating	The pod is scheduled on a node, and the kubelet on that node is creating the pod containers.
	Pending	The Kubernetes system accepted the pod, but one or more of its containers is pending.
	Terminating	The pod is terminating.
	OOMKilled	The Out-Of-Memory Killed terminated one or more of the pod containers.
	Error	One or more of the pod containers has an issue and failed to start or run correctly.
	Container cannot run	The container is unable to start or run properly.
	Image pull back off	Kubernetes is encountering problems because of incorrect credentials or incorrect container names.
	Create container error	Kubernetes is encountering problems creating containers.
	Create container config error	Kubernetes is encountering problems creating containers.
	Failed	A minimum of one pod container terminated with a non-zero exit code.

Table 68. Pods health status (continued)

Icon	Name	Definition
	Err image pull	An error happened when Kubernetes tried to pull the specified container image from the registry.
	Crash loop back off	One or more pod containers are repeatedly failing to start and are stuck in a restart loop.
	Evicted	One or more pods terminate because the node is running out of resources.
	Deadline exceeded	The pod operation took longer than the specified deadline and was terminated.
	Preempted	The pod was preempted to make room for other high-priority pods.
	Unknown	The state of the pod is unknown.

Restarting and collecting pod logs

Superusers and monitor users can restart pods, collect logs, and view pod details on the Pods page in PowerFlex Manager.

1. Go to **Monitoring** > **Management Diagnostics** > **Pods**.
   The **Pods** page opens which displays a list of PowerFlex management platform pods.
2. Click **Add filter** to control the list of pods displayed.
3. Enter case-sensitive free text and add additional filters by selecting **Status**, **Namespace**, or **Node Name**.

   NOTE: You can reset the default and advanced field columns by clicking the **Reset Columns** icon.

4. Select the pod.
5. Restart one of the following pods from the user interface:
   - alcm
   - alert-service
   - asm-manager
   - asmui
   - block-legacy-gateway
   - block-presentation-server
   - cloud-iq-collector
   - historical-metrics-calculator
   - mdm-events-extractor
   - metrics-gateway
   - notifier
   - snmp-listener
   - sso
   - thin-deployer

   Click **Restart**, then click **Approve** to confirm.

   If a pod fails to restart, review the pod status.

   The remaining pods can be restarted using the REST API.
6. Click **Log Collection**.
7. Select one of the following log collection types:
   - **Current pod lifecycle**

     Click **Run**, then select **Copy** or **Download** to review the logs. The default maximum log size is 10 MiB, which can be edited in kubectl.
   - **Full historical logs**

Click **Download** to review the ZIP file. The historical logs include all pod life cycles.

NOTE: After downloading the historical logs, there might be empty log directories, which usually occur after system upgrades or when pod logs are unavailable in the system.

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Configuring system settings (erasure coding)

This section provides information about configuring settings for PowerFlex Manager.

Only a user with the administrator role can configure certain settings.

Go to **Settings** to complete the following tasks:

- Manage local users, LDAP users, and directory services
- Upgrade a single firmware or software component
- Manage compliance versions, operating system images, and compliance management versions
- Define networks and specify the port numbers that PowerFlex Manager should use to verify IP addresses
- Configure settings for events and alerts, including notification policies, inputs, destinations, and alert severities
- Upload a license for PowerFlex, and other software licenses
- Configure security settings, including SSL trusted certificates, appliance SSL certificates, and resource credentials
- Perform serviceability operations, such as generating a trouble-shooting bundle and performing a backup
- Upgrade the management software

User management

The **User Management** page allows you to manage local users, LDAP users, and directory services.

Under **Settings** > **User Management**, you can find three pages:

- Local Users
- LDAP Users
- Directory Services

User roles (erasure coding)

User roles control the activities that different types of users can perform when using PowerFlex Manager.

Ensure that you configure the active directory before assigning roles. The roles that can be assigned to local users and LDAP users are identical. Each user can only be assigned one role. If an LDAP user is assigned directly to a user role and also to a group role, the LDAP user is provided with permissions of both roles.

The following table summarizes the activities that can be performed for each user role:

Table 69. User roles and activities

Role	Description	Activities
SuperUser	A SuperUser can perform all system operations.	• Manage storage resources • Manage life cycle operations, resource groups, templates, deployment, backend operations • Manage snapshots, snapshot policies • Manage users, certificates • Replace drives • Hardware operations • View storage configurations, resource details • View platform configuration, resource details • System monitoring (events, alerts) • Perform serviceability operations

Table 69. User roles and activities (continued)

Role	Description	Activities
		• Update system settings
SystemAdmin	A SystemAdmin can perform all operations, except for user management and security ones.	• Manage storage resources • Manage life cycle operations, resource groups, templates, deployment, backend operations • Manage snapshots, snapshot policies • Replace drives • Hardware operations • View storage configurations, resource details • View platform configuration, resource details • System monitoring (events, alerts) • Perform serviceability operations • Update system settings
StorageAdmin	A StorageAdmin can perform all storage-related front-end operations including block system operations, such as create a volume. **NOTE:** Operations such as create storage pool cannot be performed by the StorageAdmin role, but can be performed by the LifecycleAdmin role.	• Manage storage resources • Replace drives • Hardware operations • View storage configurations, resource details • View platform configuration, resource details • System monitoring (events, alerts)
LifecycleAdmin	A LifecycleAdmin can manage the life cycle of hardware and PowerFlex systems.	• Manage life cycle operations, resource groups, templates, deployment, backend operations • Replace drives • Hardware operations • View resource groups and templates • System monitoring (events, alerts)
SnapshotManager	SnapshotManager is a subset of StorageAdmin, working only on existing systems. This role includes all operations that are required to set up and manage snapshots.	• View storage configurations, resource details • System monitoring (events, alerts)
SecurityAdmin	The SecurityAdmin manages PowerFlex role-based access control (RBAC), and LDAP user federation. It includes all security aspects of the system.	• Manage users, certificates • System monitoring (events, alerts)
Technician	The Technician can perform all hardware FRU operations on the system, including entering a node into maintenance mode.	• Replace drives • Hardware operations • System monitoring (events, alerts) • Perform serviceability operations
DriveReplacer	This is a subset of the technician role. The DriveReplacer can perform only the operations that are required for a drive replacement.	• Replace drives • System monitoring (events, alerts)
Monitor	The monitor role has read-only access to the system, including topology, alerts, events, and metrics.	• View storage configurations, resource details • View platform configuration, resource details • System monitoring (events, alerts)
Support	The support role is a special kind of SystemAdmin (all activities except for user/ security management operations) to be used only by Dell Technologies support staff and developers. This user role has access to undocumented, special operations and options for common operations, required only for support purposes. **NOTE:** This special role should be used only by Dell Technologies support. It opens special,	• Manage storage resources • Manage life cycle operations, resource groups, templates, deployment, backend operations • Replace drives • Hardware operations • View storage configurations, resource details • View platform configuration, resource details • System monitoring (events, alerts)

Table 69. User roles and activities (continued)

Role	Description	Activities
	often dangerous, commands for advanced troubleshooting.	• Perform serviceability operations • Special Dell Technologies Support operations

LDAP users and security permissions (erasure coding)

LDAP security permissions are managed through group memberships, allowing permissions to be assigned to groups instead of individual users. This streamlines administration and improves security by ensuring only authorized users can access specific resources.

You cannot assign multiple roles to a group. All roles in a group have a proximate hierarchy.

Table 70. LDAP hierarchy

Ranking	PowerFlex group role without security officer permissions	PowerFlex group role with security officer permissions
1	N/A (local user)	N/A (local user)
2	Admin	SuperUser
3	Admin	SuperUser
4	Admin	SuperUser
5	StorageAdmin or LifecycleAdmin	SuperUser
6	Monitor	SecurityAdmin
Not ranked	SecurityAdmin	SecurityAdmin

Local users

You can create and manage local users within PowerFlex Manager.

Creating a user

Perform this task to create a local user and assign a role to that user.

1. On the menu bar, click **Settings** and click **User Management**.
2. Click **Local Users**.
3. On the **Local Users** page, click **Create**.
4. Enter a unique **User Name** to identify the user account.
5. Enter the **First Name** and **Last Name** of the user.
6. Enter the **Email** address.
7. Enter a **New Password** that a user enters to access PowerFlex Manager. Confirm the password in the **Verify Password** field.

   The password must be at least 8 characters long and contain one lowercase letter, one uppercase letter, one number, and one special character. Passwords cannot contain a username or email address.
8. In the **User Role** box, select a user role. Options include:
   - SuperUser
   - SystemAdmin
   - StorageAdmin
   - LifecycleAdmin
   - SnapshotManager
   - SecurityAdmin
   - DriveReplacer
   - Technician

- Monitor
- Support

9. Select **Enable User** to create the account with an `Enabled` status, or clear this option to create the account with a `Disabled` status.
10. Click **Submit** and click **Dismiss**.

PowerFlex Manager creates the new user with the specified password. The first time you login with the new user and password, PowerFlex Manager asks you to change the password.

Modifying a user

Perform this task to edit a PowerFlex Manager user profile.

1. On the menu bar, click **Settings** and click **User Management**.
2. Click **Local Users**.
3. On the **Local Users** page, select the user account that you want to edit.
4. Click **Modify**. For security purpose, confirm your password before editing the user.
5. You can modify the following user account information from this window:
   - **First Name**
   - **Last Name**
   - **Username**
   - **User Role**
   - **Email**
   - **Enable User**

   If you select the **Enable user** checkbox, the user can log in to PowerFlex Manager. If you disable the checkbox, the user cannot log in.
6. Click **Submit** and click **Dismiss**.

Deleting a user

Perform this procedure to remove an existing local user.

1. On the menu bar, click **Settings** and click **User Management**.
2. Click **Local Users**.
3. On the **Local Users** page, select one or more user accounts to delete.
4. Click **Delete**.

   Click **Apply** in the warning message to delete the user. Click **Dismiss**.

Resetting the password for a user

You can reset the password for a local user in PowerFlex Manager.

1. On the menu bar, click **Settings** and click **User Management**.
2. Click **Local Users**.
3. On the **Local Users** page, select one user account to reset the password.
4. Enter the **Current Password**.

   If you are changing a password for another user entering the current password is not required.
5. Click **Reset Password**.
6. Enter **New Password** and **Verify Password**.
7. If you want to set the password as a temporary one, check **Temporary Password**.
8. Click **Apply** and click **Dismiss**.

Remote users and groups

You can add and modify remote users and groups in PowerFlex Manager. Remote users are users tied to an LDAP directory service or to an SSO identity provider.

For LDAP configurations, the LDAP realm name must not exceed the 155 character limit defined by Microsoft.

Add remote users or groups

Add remote users or groups to PowerFlex, and assign roles to them. Remote users are users tied to an LDAP directory service or to an SSO identity provider.

When you add users and groups, you can also assign roles to the users and groups. The roles control access permissions for the corresponding LDAP user or group. The users and groups must also be configured in the LDAP directory service or SSO identity provider. On the PowerFlex side, these users and groups are mapped to PowerFlex roles.

1. On the menu bar, click **Settings**.
2. In the left pane, click **User Management**, then in the right pane, click **Remote Users/Groups**.
3. Click **Add**.
4. In the **Add Remote User/Group** dialog box, select a **Type** option.
   - `User`—a user definition will be configured for an individual user.
   - `Group`—a group definition will be configured for a specific group.
5. Select the LDAP directory service or SSO identity provider in the **Provider** drop-down.
6. In the **User Name** box, enter the user or group name.
7. In the **User Role** box, select the role to be assigned to the user or group. Options include:
   - SuperUser
   - SystemAdmin
   - StorageAdmin
   - LifecycleAdmin
   - SnapshotManager
   - SecurityAdmin
   - DriveReplacer
   - Technician
   - Monitor
   - Support
8. Click **Apply**.

Modify a remote user or group

You can modify the user role of a user or group tied to an LDAP directory service or an SSO identity provider. User roles control access permissions to the corresponding user or group.

1. On the menu bar, click **Settings** > **Remote Users**.
2. Click **Modify**.
3. In the **Modify Remote User/Group** dialog box, change the user role by selecting one of the **User Role** options:
   - SuperUser
   - SystemAdmin
   - StorageAdmin
   - LifecycleAdmin
   - SnapshotManager
   - SecurityAdmin
   - DriveReplacer
   - Technician
   - Monitor
   - Support
4. Click **Apply**.

Directory services

You can create a directory service that PowerFlex Manager can access to authenticate users.

An Active Directory or Open LDAP user is authenticated against the specific directory domain to which a user belongs.

The **Directory Services** page displays the following information about PowerFlex Manager active directories:

- LDAP configuration
- User search settings
- Group search settings

From this page, you can:

- Add a directory service (only available when no service is defined in the system)
- Modify a directory service
- Remove a directory service

Add a directory service

Add a directory service for user authentication.

The hostname character limitation is 15 characters.

1. On the menu bar, click **Settings**.
2. In the left pane, click **User Management**, then in the right pane, click **Directory services**.
3. Click **Add**.
4. For **LDAP Configuration**, configure the following:
   a. In the **Address** box, enter the address of the authentication server.
      The address must be specified in URL-like format:
      - Enter **`ldap://`*`HOSTNAME or IP ADDRESS`*** for a plaintext LDAP connection.
      - Enter **`ldaps://`*`HOSTNAME or IP ADDRESS`*** for a secure LDAP connection.

      For example: **`ldap://100.68.68.1`**
   b. In the **Bind DN** box, enter the bind distinguished name attributes.
      The Bind Distinguished Name (DN) uniquely identifies an entry and its position in the hierarchy of entries contained in a directory server.

      The user account entered here is the user that PowerFlex Manager uses to connect to your directory service.

      For example, for the administrator account in the **Users** folder of **`asm.delllabs.com`**, enter the following: **`CN=administrator,CN=Users,DC=asm,DC=delllabs,DC=net`**.
   c. In the **Bind DN Password** box, enter the Bind DN password.
      This is the password used to access the LDAP server.
   d. In the **Timeout** box, enter a value in milliseconds.
      For example: **`1000`**
5. For **User Search Settings**, configure the following:
   a. In the **Username LDAP Attribute** box, enter the name of an LDAP attribute that is mapped as the username. For many LDAP servers, it can be **`uid`**. For Active Directory, it can be **`sAMAccountName`** or **`cn`**. The attribute should be filled in for all LDAP users you want to import from LDAP to PowerFlex.

      NOTE: If you are using Microsoft Windows Active Directory (AD) as an LDAP server, the hostname cannot exceed 15 characters. A hostname that exceeds the 15-character limit may cause issues during deployment and integration.

      For example: **`sAMAccountName`**
   b. In the **ID Attribute** box, enter the ID attribute for users.
      For example: **`sAMAccountName`**
   c. In the **Object Class** box, enter an object class.
      For example: **`top,person,organizationalPerson,user`**
   d. In the **Search Path** box, enter the search path.
      The search path is used to identify and retrieve entries in the directory information tree that match a set of criteria.

      For example: **`CN=Users,DC=asm,DC=delllabs,DC=net`**
6. For **Group Search Settings**, configure the following:

a. In the **Group Member Attribute** box, enter a group member name.
   For example: **member**
b. In the **Group ID Attribute** box, enter the group ID.
   For example: **name**
c. In the **Group Object Class** box, enter the group object class.
   For example: **group**
d. In the **Group Search Path** box, enter the group search path.

   The search path is used to identify and retrieve entries in the directory information tree that match a set of criteria for groups.

   For example: **CN=Users,DC=asm,DC=delllabs,DC=net**

7. Click **Test Connection**.

   If the test is successful, the **Submit** button will become active. If the test fails, you will not be able to proceed until you fix the connectivity issue.

8. When you have finished making your changes, click **Submit**.

Modify a directory service

The **Modify** option allows you to edit the existing directory service settings.

1. On the menu bar, click **Settings**.
2. In the left pane, click **User Management**, then in the right pane, click **Directory services**.
3. Click **Modify**.
4. In the **LDAP Settings** dialog box, edit the desired fields.

   Note that the **Bind DN Password** must be reentered.

5. When you have finished making your changes, click **Test Connection**.

   If the test is successful, the **Submit** button becomes active. If the test fails, you will not be able to proceed until you fix the connectivity issue.

6. Click **Submit**.

Remove a directory service

The **Remove** option allows you to remove the directory service configuration from PowerFlex.

1. On the menu bar, click **Settings**.
2. In the left pane, click **User Management**, then in the right pane, click **Directory services**.
3. Click **Remove**.
4. In the warning dialog box, click **Submit**.

Single sign-on identity provider configuration

You can configure single sign-on (SSO) authentication through an identity provider (IdP) using the SAML 2.0 and OIDC protocols. This configuration makes it possible for users to take advantage of SSO capabilities.

You can add up to three IdPs for SSO authentication through PowerFlex Manager.

The supported IdPs and protocols are described in the following table:

Table 71. Supported IdPs and protocols

IdP	SAML	OIDC
Microsoft Active Directory Federation Services	Yes	No
Microsoft Azure	Yes	Yes
Keycloak	No	Yes
Okta	Yes	Yes
PingOne	Yes	No

OIDC instructions

This guide describes how to use the SAML 2.0 protocol. For instructions for using the OIDC protocol, see the Dell Knowledge Base article PowerFlex: Setting up PowerFlex single sign-on with Microsoft Azure using OpenID Connect (OIDC).

Add an identity provider

Add an SSO Identity Provider (IdP) for PowerFlex Manager to enable users to take advantage of single sign-on (SSO) capabilities through other applications.

1. On the menu bar, click **Settings**.
2. In the left pane, click **User Management**, then in the right pane, click **SSO Identity Provider (IdP) Configuration**.
3. Click **Add**.
   The **Add Identity Provider (IdP)** window opens.
4. Under **Name**, provide a name for the identity provider and click **Next**.
5. Under **Service Provider**, choose a method to download the service provider metadata for PowerFlex:
   - To download the metadata as a file, select **Download File** and click **Download SP Metadata (XML)**.
   - To copy the metadata, click **Manual Copy** and click **Copy** for each piece of metadata.
6. Configure PowerFlex as the service provider within the IdP.

   NOTE: To configure PowerFlex as the service provider within the IdP, you must log in to the portal of the IdP.

   To configure PowerFlex as the service provider within Active Directory Federation Services, see *Set up PowerFlex single sign-on with Microsoft Active Directory Federation Services* in the *PowerFlex and PowerFlex Manager Software Administration Guide*.

   To configure PowerFlex as the service provider within Microsoft Azure, see *Set up PowerFlex single sign-on with Microsoft Azure* in the *PowerFlex and PowerFlex Manager Software Administration Guide*.
7. Under **Service Provider**, select the **I have configured PowerFlex as a SP in my IdP using the metadata above** checkbox and click **Next**.
8. Under **IdP Setup**, upload the identity provider metadata so that PowerFlex can establish a connection to the identity provider.
   a. To upload the metadata as a file, select **Upload File** and specify the file location.
   b. To retrieve the metadata from a URL, select **URL** and specify the following URL: `https://hostname/FederationMetadata/2007-06/FederationMetadata.xml`

      The URL is always the same, except for the hostname, which you must specify for your environment.
   c. Click **Next**.
9. Under **Claims Mapping**, review the attribute mappings that are imported from the identity provider.
   a. Check the attribute mappings to ensure that they are correct.
   b. Click **Next**.
10. Under **Summary**, review the details and click **Finish**.

After you add an identity provider, PowerFlex Manager adds it to the list of identity providers on the **SSO Identity Provider (IdP) Configuration** page.

In addition, PowerFlex Manager updates the login page to show a login button with the new identity provider. You can see this button the next time you log in to PowerFlex Manager.

Set up PowerFlex single sign-on with Microsoft Active Directory Federation Services

To set up PowerFlex single sign-on with Microsoft Active Directory Federation Services (ADFS), you must first add ADFS as an IdP through PowerFlex Manager.

You must then download the service provider metadata for PowerFlex. After you complete this step, log in to the ADFS portal, configure PowerFlex as the service provider, and choose an access control policy. You must then return to PowerFlex Manager to complete the setup of single sign-on.

Add Active Directory Federation Services as an identity provider

Add an SSO Identity Provider (IdP) for PowerFlex Manager to enable users to take advantage of single sign-on (SSO) capabilities through other applications.

1. On the menu bar, click **Settings**.
2. In the left pane, click **User Management**, then in the right pane, click **SSO Identity Provider (IdP) Configuration**.
3. Click **Add**.
   The **Add Identity Provider (IdP)** window opens.
4. Under **Name**, provide a name for the identity provider and click **Next**.
5. Under **Service Provider**, choose a method to download the service provider metadata for PowerFlex:
   - To download the metadata as a file, select **Download File** and click **Download SP Metadata (XML)**.
   - To copy the metadata, click **Manual Copy** and click **Copy** for each piece of metadata.

Configure PowerFlex as a service provider for Active Directory Federation Services

To configure PowerFlex as a service provider for Active Directory Federation Services, upload the root CA certificate and log in with appropriate permissions. Then, establish trust by adding a relying party, configuring claim rules, importing metadata, and setting access control policies.

You must upload the root CA certificate for ADFS.

1. Log in to the Active Directory Federation Services portal with the security or administrator role.
2. Under **ADFS**, go to the **Relying Party Trusts** folder.
3. Under **Actions**, click **Add Relying Party Trust...**.
4. On the **Welcome** screen, select **Claims aware** and click **Start**.
5. On the **Select Data Source** screen, select **Import data about the relying party from a file**.
6. In the **Federation metadata file location** field, specify the location of the downloaded service provider metadata file and click **Next**.

   NOTE: If you copied the metadata instead of downloading the file, select **Enter data about the relying party manually** and input the copied metadata.

7. On the **Specify Display Name** screen, enter the name that you would like to use for the service provider and click **Next**.
8. On the **Choose Access Control Policy** screen, choose a policy and click **Next**.
9. On the **Ready to Add Trust** screen, click **Next**.
10. On the **Finish** screen, select **Configure claims issuance policy for this application** and click **Close**.
11. On the **Relying Party Trusts** screen, under **Actions** select the display name for the newly created service provider, and click **Edit Claim Issuance Policy...**.
12. Click **Add Rule...** to add the following LDAP attribute rule:
    a. For the **Claim rule template**, select **Send LDAP Attributes as Claims**.
    b. For the **Claim rule name**, enter `LDAP attributes`.
    c. For the **Attribute store**, select **Active Directory**.
    d. For the Mapping of **LDAP attributes to outgoing claim types**, select the following attributes:

Table 72. LDAP attributes and outgoing claim types

LDAP attribute	Outgoing claim type
Email-Addresses	Email Address
SAM-Account-Name	Name ID
Surname	Surname
Given-Name	Given Name

    e. Click **Finish**.
13. Click **Add Rule...** to add the following custom rule:
    a. For the **Claim rule template**, select **Send Claims Using a Custom Rule**.
    b. For the **Claim rule name**, enter `Get groups`.

c. For the **Custom rule**, paste in the following string:

```
c:[Type == "http://schemas.microsoft.com/ws/2008/06/identity/claims/
windowsaccountname", Issuer == "AD AUTHORITY"]  => add(store = "Active Directory",
types = ("http://schemas.xmlsoap.org/claims/Group"), query = ";tokenGroups;{0}",
param = c.Value);
```

d. Click **Finish**.

14. Click **Add Rule...** to add another custom rule:
    a. For the **Claim rule template**, select **Send Claims Using a Custom Rule**.
    b. For the **Claim rule name**, enter **`Claim of groups membership`**.
    c. For the **Custom rule**, paste in the following string:

```
c:[Type == "http://schemas.xmlsoap.org/claims/Group"]  => issue(Type = "Group",
Issuer = c.Issuer, OriginalIssuer = c.OriginalIssuer, Value = c.Value, ValueType =
c.ValueType);
```

    d. Click **Finish**.
15. Click **OK**.

Upload the PowerFlex metadata file

You must upload the metadata file through PowerFlex Manager to ensure that PowerFlex can establish a connection to the IdP.

1. In PowerFlex Manager, go back to the **Add Identity Provider (IdP)** window.
2. Under **Service Provider**, select the **I have configured PowerFlex as a SP in my IdP using the metadata above** checkbox and click **Next**.
3. Under **IdP Setup**, upload the identity provider metadata so that PowerFlex can establish a connection to the identity provider.
    a. To upload the metadata as a file, select **Upload File** and specify the file location.
    b. To retrieve the metadata from a URL, select **URL** and specify the following URL: `https://`*`hostname`*`/FederationMetadata/2007-06/FederationMetadata.xml`

    The URL is always the same, except for the hostname, which you must specify for your environment.
    c. Click **Next**.
4. On the **Settings** page, review the attribute mappings that are imported from the identity provider.
    a. Check the attribute mappings to be sure they are correct.
    b. Click **Next**.
    c. Review the **Summary** page and click **Finish**.

After you add ADFS as an IdP, it is added to the list of IdPs on the **SSO Identity Provider (IdP) Configuration** page in PowerFlex Manager.

In addition, PowerFlex Manager updates the login page to show a login button with ADFS. You can see this button the next time you log in to PowerFlex Manager.

Set up PowerFlex single sign-on with Microsoft Azure

To set up PowerFlex single sign-on with Microsoft Azure, you must first add Microsoft Azure as an IdP through PowerFlex Manager.

You must then download the service provider metadata for PowerFlex. After you complete this step, log in to the Microsoft Azure portal and configure PowerFlex as the service provider. You must then return to PowerFlex Manager to complete the setup of single sign-on.

Upload the PowerFlex metadata file

You must upload the metadata file through PowerFlex Manager to ensure that PowerFlex can establish a connection to the IdP.

1. In PowerFlex Manager, go back to the **Add Identity Provider (IdP)** window.
2. Under **Service Provider**, select the **I have configured PowerFlex as a SP in my IdP using the metadata above** checkbox and click **Next**.
3. Under **IdP Setup**, select **Upload File**.

4. Browse and select the `Federated Metadata XML` file and click **Next**.

You must now leave PowerFlex Manager. Go back to the Microsoft Azure portal to create a group claim.

Add Microsoft Azure as an identity provider

You must add Microsoft Azure as an IdP and download the service data for PowerFlex through PowerFlex Manager.

1. On the menu bar, click **Settings**.
2. In the left pane, click **User Management**, then in the right pane, click **SSO Identity Provider (IdP) Configuration**.
3. Click **Add**.
   The **Add Identity Provider (IdP)** window opens.
4. Under **Name**, provide a name for the identity provider and click **Next**.
5. Under **Service Provider**, choose a method to download the service provider metadata for PowerFlex:
   - To download the metadata as a file, select **Download File** and click **Download SP Metadata (XML)**.
   - To copy the metadata, click **Manual Copy** and click **Copy** for each piece of metadata.

Configure PowerFlex as a service provider for Microsoft Azure

You must export the PowerFlex metadata file and edit it to include additional attributes.

You have 15 minutes from when the SP Metadata (XML) file is downloaded from PowerFlex until the `Federation Metadata XML` file is uploaded to complete the procedure, or a timeout may occur. If there is a timeout, see *Troubleshooting a timeout error*.

1. Using an editor, open the `SP Metadata XML` file, replace any occurrence of the string `keycloak-http` with the PowerFlex Manager IP address, and save the file.
2. Log in to the Microsoft Azure Portal.
3. Go to **App registrations**.
4. Under **Owned applications**, select the Microsoft Azure application that PowerFlex uses.
5. Click the link beside **Managed Application in local directory**.
6. Go to **Manage** > **Single sign-on**.
7. Select **SAML**.
8. From the menu, click **Upload metadata file**.
9. To import the metadata file, browse and select the file, and then click **Add**.
10. Check that the URL contains the PowerFlex system IP and click **Save**.
11. Exit the import screen.
12. In the **SAML-based Sign-On screen** window, go to the **SAML Certificates** section, and click **Download.** to download the `Federation Metadata XML` file.
    The file name that is created contains the application name with an `.xml` extension. The file is downloaded to the `Downloads` directory. In this case, the `Federation Metadata XML` directory.
13. Go to the `Federation Metadata XML` file, open it in a text editor and edit it to include the additional attributes as follows:
    a. Before `</IDPSSODescriptor>`, add the following lines:

```
<Attribute xmlns="urn:oasis:names:tc:SAML:2.0:assertion"
Name="http://schemas.xmlsoap.org/ws/2005/05/identity/claims/emailaddress"
NameFormat="urn:oasis:names:tc:SAML:2.0:attrname-format:uri" FriendlyName="E-Mail
Address"/>
<Attribute xmlns="urn:oasis:names:tc:SAML:2.0:assertion"
Name="http://schemas.xmlsoap.org/ws/2005/05/identity/claims/givenname"
NameFormat="urn:oasis:names:tc:SAML:2.0:attrname-format:uri" FriendlyName="Given
Name"/>
<Attribute xmlns="urn:oasis:names:tc:SAML:2.0:assertion"
Name="http://schemas.xmlsoap.org/ws/2005/05/identity/claims/surname"
NameFormat="urn:oasis:names:tc:SAML:2.0:attrname-format:uri"
FriendlyName="Surname"/>
```

    b. Click **Save**.

You must now leave the Microsoft Azure portal. Go back to PowerFlex Manager and upload the `Federation Metadata XML` file to PowerFlex.

Create a group claim

You must create a group claim to manage user authorization to the application.

1. In the Microsoft Azure portal, go back to the **SAML-based Sign on** window.
2. From the left navigation, click **Single sign-on**.
   The **Set up Single Sign On with SAML** window opens.
3. From the **Attributes & Claims** window, click **Edit**.
4. Select **+ Add a group claim**.
5. From the **Group Claims** window, complete the following steps:
   a. For **Which groups associated with the user should be returned to the claim?**, select **Groups assigned to the application**.
   b. For **Source attribute**, select **Group ID**.
   c. Expand **Advanced options**.
   d. Select the **Customize the name of the group claim** checkbox.
   e. For **Name (required)**, enter `Group`.
   f. Click **Save**.

Assign a user or group to the application

The user or group you assign is given access to authenticate to PowerFlex through PowerFlex Manager.

1. In the Microsoft Azure portal, from the **SAML-based Sign on** window, go to the left side navigation, and click **Users and groups**.
2. In the **Add Assignment** window, click **Users and groups**.
3. Select the checkbox beside the users and groups that should be assigned.
4. Click **Assign**.

You must now leave the Microsoft Azure portal. Go back to PowerFlex Manager and provide permissions to the remote users and groups you created.

Provide permissions to remote users and groups

You can provide permissions to users and groups that are based on their role. The roles control access permissions for the corresponding group.

You must not connect to SSO from PowerFlex Manager before you assign permissions.

1. Log in to PowerFlex Manager using the local administrator rights.
2. Go to Settings > **User Management** > **Remote users/Groups**.
3. Click **Add**.
   a. Select the type.

      If you select the **Group** type for the **Group Name**, enter the group-id that is taken from Microsoft Azure (not the Group Name).

      If you select the **User** type for the **User Name**, enter the username that is defined in Microsoft Azure.
   b. Select the provider.
   c. Select a role.
   d. Click **Apply**.

After you add Microsoft Azure as an IdP, it is added to the list of IdPs on the **SSO Identity Provider (IdP) Configuration** page in PowerFlex Manager.

In addition, PowerFlex Manager updates the login page to show a login button with Microsoft Azure. You can see this button the next time you log in PowerFlex Manager.

Troubleshooting a timeout error

If a timeout error occurs, you can log in to PowerFlex Manager and create the IdP again.

1. Log in to PowerFlex Manager.
2. On the menu bar, click **Settings**.
3. In the left pane, click **User Management**, then in the right pane, click **SSO Identity Provider (IdP) Configuration**.
4. Select the IdP and click **Delete**.
5. Add the IdP.

6. Export the `Federation Metadata XML` file.
7. Import the `Federation Metadata XML` file that you downloaded from Microsoft Azure to PowerFlex.
8. Import the exported `Federation Metadata XML` file that you created in step 7 to Microsoft Azure.
   In the Microsoft Azure portal, update the entity-id by reimporting the PowerFlex metadata file through the application.

Modify an identity provider

The **Modify** option allows you to edit the existing identity provider settings.

1. On the menu bar, click **Settings**.
2. In the left pane, click **User Management**, then in the right pane, click **SSO Identity Provider (IdP) Configuration**.
3. Select an identity provider.
4. Optionally, change the name for the identity provider.
5. Optionally, enable or disable the identity provider.
6. Click **Apply**.

Enable or disable an identity provider

You can disable an identity provider after the initial configuration. If needed, you can also reenable the provider after it has been disabled.

Perform the following procedure to enable or disable an identity provider:

1. On the menu bar, click **Settings**.
2. In the left pane, click **User Management**, then in the right pane, click **SSO Identity Provider (IdP) Configuration**.
3. Click **Enable** or **Disable**.

Delete an identity provider

You can delete an identity provider if you no longer need it.

Perform the following procedure to delete an existing identity provider:

1. On the menu bar, click **Settings**.
2. In the left pane, click **User Management**, then in the right pane, click **SSO Identity Provider (IdP) Configuration**.
3. Click **Delete**.

Update the certificate for an identity provider

You can update the certificate for an identity provider if it is about to expire.

1. On the menu bar, click **Settings**.
2. In the left pane, click **User Management**, then in the right pane, click **SSO Identity Provider (IdP) Configuration**.
3. Check the **Status** column to see if any of the identity providers are about to expire.
4. If an identity provider is about to expire, select the identity provider and click **Update Certificate** in the details pane to the right of the list of providers.

Repositories

On the **Repositories** page, you can load compliance files, operating system images, and compatibility management files.

You can load compliance files and specify a default version for compliance checking on the **Compliance Versions** tab. You can load multiple compliance files into PowerFlex Manager with different operating system images included. The compliance file includes operating system images for VMware ESXi and PowerFlex. When you add a compliance file, these images are automatically added to the **OS Images** tab. If necessary, you can also create your own operating system image repositories. You can also load the compatibility management file on the **Compatibility Management** tab to allow PowerFlex Manager to determine which upgrade paths are valid and which are not.

On the **Compliance Versions** tab, click **Add** to load a compliance version. Click **Change** to change the default compliance version to a new file.

You cannot make a minimal compliance version the default version for compliance checking, since it includes only server firmware updates. The default version must include the full set of compliance update capabilities. PowerFlex Manager does not show any minimal compliance versions in the **Default Version** drop-down list.

The **Compliance Versions** tab displays the following information:

Table 73. Compliance version fields

Field	Description
State	Displays an icon indicating one of the following states: ● Pending—The compliance file download process is in progress. ● Downloading—The compliance file is being downloaded and PowerFlex Manager provides the percentage complete for the download operation. ● Unpacking—The compliance file is being unpacked and provides the percentage complete for the unpacking operation. ● Synchronizing—The compliance file is being synchronized with the management software after unpacking. ● Available—The compliance file is downloaded and copied successfully. ● Error—There is an issue downloading the compliance file. ● Needs Approval—The compliance file is unsigned and needs approval for use.
Version	Displays the compliance version.
Source	Displays the share path of the compliance version in a file share.
Signature	Displays whether the digital signature on the compliance file is signed or unsigned.
File Size	Displays the size of the compliance file in GB.
Type	Displays **Minimal** if the compliance file only contains firmware updates, or **Full** if it contains firmware and software updates.
View bundles	Displays details about any bundles added for the compliance version.
Available Actions	Select an option for a compliance file that is in an **Error** state: ● Delete the compliance version ● Resynchronize a compliance version
	Select an option for a compliance file that is in a **Needs Approval** state: ● Allow unsigned file ● Delete

Use the **OS Image Repositories** tab to create operating system image repositories and view the following information:

Table 74. Operating system image repositories

Field	Description
State	Displays the following states: ● Available—The operating system image repository is downloaded and copied successfully. ● Pending—The operating system image repository download process is in progress. ● Error—There is an issue downloading the operating system image repository.
Repository	Displays the repository name.
Image Type	Displays the operating system type.
Source Path	Displays the share path of the repository in a file share.
In Use	Displays **True** (the operating system image repository is in use) or **False** (the operating system image repository is not in use).
Available Actions	Select an option to delete or resynchronize.

You cannot perform any actions on repositories that are in use. However, you can delete repositories that are in an **Available** state, but not in use and not set as a default version.

All the options are available only for repositories in an **Error** state. The **Resynchronize** option appears only when you must perform a backup and restore of a previous image.

Use the **Compatibility Management** tab to load the compatibility management file. To facilitate upgrades, PowerFlex Manager uses information that is provided in the compatibility matrix file to determine which upgrade paths are valid and which are not. When you attempt an upgrade, PowerFlex Manager warns you if the current version of the software is incompatible with the target version, or if any of the RCM or IC versions that are currently loaded are incompatible with the target compliance versions.

Related information

Configuring block storage
Deploying and provisioning

Compliance versions

PowerFlex Manager displays the following information about the compliance versions:

Table 75. Compliance versions

Field	Description
State	Displays an icon indicating one of the following states: • Pending—The compliance file download process is in progress. • Downloading—The compliance file is being downloaded and PowerFlex Manager provides the percentage complete for the download operation. • Unpacking—The compliance file is being unpacked and provides the percentage complete for the unpacking operation. • Synchronizing—The compliance file is being synchronized with the management software after unpacking. • Available—The compliance file is downloaded and copied successfully. • Error—There is an issue downloading the compliance file. • Needs Approval—The compliance file is unsigned and needs approval for use.
Version	Displays the compliance version. If one or many components in the RCM are modified, a wrench icon and the text, **Modified** is displayed next to the name of the RCM. But, if the file is deleted from the file share after a component is updated, the wrench ion with the text, **Modified** or **Needs Attention**, is displayed.
Source	Displays the share path of the compliance version in a file share.
Signature	Displays whether the digital signature on the compliance file is signed or unsigned.
File Size	Displays the size of the compliance file in GB.
Type	Displays **Minimal** if the compliance file only contains firmware updates, or Full if it contains firmware and software updates.
View bundles	Displays details about any bundles added for the compliance version.
Available Actions	Select an option if the compliance file is in an **Error** or **Needs Approval** state: • Delete the compliance version • Resynchronize a compliance version • Allow unsigned file

Select the compliance version to view the following information about the firmware package:

- **Bundles**—Displays the number of bundles available in the firmware package.
- **Components**—Displays the number of software components available in the firmware package.
- **Created On**—Displays the date when the compliance version was created.
- **Last Updated**—Displays the date when the compliance version was last updated.
- **Resource Groups Affected**—Displays the resource groups in which the firmware is used.
- **Additional Components**:

- **VMware ISO**—Displays the date and time when the VMware ISO component was added.
- **Filename**—Displays the file name of the VMware ISO component added.
- **VMware Upgrade Depot**—Displays the date and time when the VMware patch depot file component was added.
- **Filename**—Displays the file name of the VMware patch depot file component added.

From this page, you can:

- Add a compliance file.

  A compliance file can be large and take some time to download and unpack. To help you monitor the progress of the downloading and unpacking operations, PowerFlex Manager provides percentage complete information for these operations in the **State** column on the **Compliance Versions** tab.

- Select **Default Version** and choose a version from the drop-down list to set it as the default for compliance.
- Select a compliance version from the list, and click the delete icon in the same row to remove the version.

  If you remove a compliance version, the version is deleted from the appliance not from the file share.

- Select a version from the list, and, in the right pane, click **View Bundles** to view the firmware and software bundles available in the compliance version.

Add a compliance file

A compliance file must be in an available state for PowerFlex Manager to add the file.

1. From the **Getting Started** page, click **Upload Compliance File**.

   Alternatively, click **Settings** > **Repositories**. On the **Repositories** page, select the **Compliance Versions** tab, and click **Add**.

2. In the **Add Compliance File** dialog, select one of the following options:
   - **Download from Dell Technologies connectivity services (Recommended)**—Select this option to import the compliance file that contains the firmware bundles you need.
   - **Download from local network path**—Select this option to download the compliance file from a CIFS file share. This option is intended for sites that do not have Internet accessibility to Dell Technologies connectivity services.
3. If you selected **Download from Dell Technologies connectivity services (Recommended)**, click the **Available Compliance Files** drop-down list, and select the file.

   Before downloading a compliance file from Dell Technologies connectivity services, you must configure Dell Technologies connectivity services. To configure Dell Technologies connectivity services , you must enable Dell Technologies connectivity services in the **Initial Configuration** wizard.

   If you are downloading a compliance file from Dell Technologies connectivity services, the file type is a ZIP.

   If you are downloading a compliance file from the Dell Technologies support site, the file will be called `PowerFlex _Software_5.1.x_xxx.zip`

   NOTE: The XML file type is not supported as a download option.

4. If you selected **Download from local network path**, perform the following:
   - In the **File Path** box, enter the location of the compliance file. Use the following format:
     - CIFS share for ZIP file: **`\\host\share\filename.zip`**
   - If you are using a CIFS share, enter the **User Name** and **Password**.
   - Select **Test Connection** to test the connection between the PowerFlex Manager user interface and the location of the CIFS file.
   - Mark the target upgrade catalog as the default version for compliance checking.
5. Click **Save**.

   PowerFlex Manager unpacks the compliance file as a repository that includes the firmware and software bundles that are required to perform any updates.

   If the compliance file contains an embedded ISO file for an operating system image (such as for a VMware ESXi or PowerFlex image), the download process unpacks the file and adds it to the operating system image repository.

   A compliance file can be large and take some time to download and unpack. To help you monitor the progress of the downloading and unpacking operations, PowerFlex Manager provides percentage complete information for these operations in the **State** column on the **Compliance Versions** tab.

Add VMware ESXi files to the compliance file

The following procedure is applicable only when using ESXi hosts. The vCenter Server Appliance (VCSA) can be hosted either in your environment or through the PowerFlex management cluster (KVM) compute-only extension (ESXi-based).

In this process, the ESXi ISO image and patch depot file that are compatible with the IC are uploaded. This step applies only to PowerFlex compute-only deployments.

NOTE: You must use the VMware standard ISO (not the Dell-customized ISO).

The VMware ESXi image and patch depot file are no longer available on the Dell support site. You must obtain these files from the Broadcom support portal and upload them to the compliance file before creating a template, performing a deployment, running a compliance check, or carrying out a non-disruptive upgrade with VMware ESXi. After downloading the appropriate image and patch depot file, you should add these components (ESXi ISO and patch depot file) to the compatible compliance file.

1. On the menu bar, click **Settings** > **Repositories** > **Compliance Versions**.
2. From the **Compliance Versions** tab, select the **IC** and click **Add Component**.
3. In the **Add Compliance Component** window, leave the default **VMware Components** setting for **Type of components being added**.
4. In the **Add VMware Installation ISO** section, enter the file path of the VMware installation ISO file, username, and password and click **Test Connection** to verify that the file path is valid and can be accessed.
5. In the **Add VMware Upgrade Depot Zip** section, enter the file path of the VMware patch depot file, username, and password and click **Test Connection** to verify that the file path is valid and can be accessed.
6. Click **Add**.

   The VMware ISO and patch depot files are tracked under the **Hardware** category on the **Monitoring** > **Events** page. The **Additional Components** in the **Compliance and OS Repositories** window displays the VMware ISO and patch depot file name, date, and time at which the VMware ISO and patch depot files are added.

Changing the compliance file

To perform upgrades against a different compliance version, you can change the compliance file.

1. To change the default compliance version, click **Change** on the **Compliance and OS Repositories** page.

   To change the target version for a resource group, click **Change Target** on the **Resource Group Details** page or click **Change** on the **Node Compliance Report**.
2. Choose the **Preferred Compliance File**.
3. Review the **Compatibility** setting. If it shows **Recommended** or **Supported**, you can proceed with the change. If it shows **Not Allowed**, PowerFlex Manager blocks the change. Review the alert for details about recommended or supported paths.
4. If the change is either recommended or supported, type **`CHANGE COMPLIANCE FILE`** if you are to proceed.
5. Click **Save**.

Firmware repository types

The **Resources** page displays devices that are validated against the default repository.

This capability is available on the PowerFlex appliance and PowerFlex rack offerings only.

There are two types of firmware repositories:

Table 76. Firmware repository types

Repository	Description
Default	The default firmware repository is applied to all devices that are not in a resource group. To set a default firmware repository, you must download a compliance file from either Dell Technologies connectivity services or an internal share through PowerFlex Manager. If you set a default compliance version, you can view compliance with this repository on the Resources page.

Table 76. Firmware repository types (continued)

Repository	Description
	Devices with firmware levels below the minimum firmware that is listed in the default compliance version are viewed as non-compliant (out of compliance).
Service level	This repository is applied only to nodes that are in service and assigned the service level firmware repository. Devices with firmware levels below the minimum firmware level that is listed in the service level repository are marked as non-compliant. When a service level firmware repository is assigned to a resource group, the firmware validation is checked only against the service level firmware repository and the default firmware repository checks are no longer applied to the devices associated with this resource group.

Viewing firmware and software bundles details

The details that are provided for bundles include name, version, and date, allowing for management and tracking of updates.

1. On the menu bar, click **Settings** and click **Compliance and OS Repositories**.
2. Click the **Compliance Versions** tab, select a repository, and then click **View Bundles**.
   A list of system and software bundles is displayed.
3. Click **Firmware Bundles** to view the firmware or system bundles.
4. Click **Software Bundles** to view the software bundles.

You can see the following information regarding bundles:

- Name—Name of the firmware or software update package.
- Version—Version of the firmware or software update package.
- Date and Time—Date and time when the firmware or software update package was downloaded.

Performing a single component upgrade

The single component upgrade allows for upgrading the software and firmware on individual components, such as applying security patches, without performing a full RCM/IC upgrade. It is recommended to use this option for component-specific upgrades, such as patch updates, and to ensure compliance with the standard RCM/IC version.

Single component upgrades are not engineering-validated upgrades. You should understand the implication and risks. Dell Technologies recommends aligning with RCM/IC as the official-validated catalog. If you consider deviating to align with the security or functionality requirements, consider testing the compatibility.

A single component upgrade involves the following steps:

1. Upgrading the firmware and software components.
2. Extracting the compliance report.
3. Modifying and publishing the existing template.

CAUTION: Since Dell Technologies only validates the released RCM/IC code catalogs, you must be aware of the implications and risks that are involved in a single component upgrade. If any issues are identified related to the RCM/IC deviation, you may be asked to modify the code or firmware.

If an upgrade file, for example iDRAC FW 7.xx.xx.xx, exceeds 200 MB, use the SMB or CIFS share option. Browser uploads are not supported for upgrade files larger than 200 MB.

The use of single component upgrade enables the PowerFlex platform to remain in compliance, using the deviated code/firmware, which means future upgrades will not be blocked due to non-compliance.

The single component upgrade option enables the PowerFlex platform to remain in compliance. The future upgrades are not blocked due to non-compliance.

The following limitations are associated with the single component upgrade:

- VMware ESXi version changes are restricted. Upgrading from VMware ESXi 8.x to 9.x or downgrading is not supported.
- Updates to embedded operating systems, such as SLES or CentOS are not supported.

Upgrade the firmware components

To upgrade firmware on a PowerFlex node, authenticate as an administrator, choose a bundle, modify the component, and apply the updates.

You must verify that you are using administrator credentials.

An upgrade of the following firmware components is supported:

- Drive firmware
- iDRAC Lifecycle Controller (LCC)
- BIOS firmware
- Network Interface Card (NIC) firmware
- BOSS card firmware
- Switch firmware

The upgrade of the PowerFlex core is not supported.

1. On the menu bar, click **Settings** > **Repositories** > **Compliance Versions**.
2. From the **Compliance Versions** page, select the current RCM/IC bundle and click **View Bundles**.
3. From the list of bundles in the **Firmware Bundles** tab, verify the bundles that you want to modify.
   To modify components on the PowerFlex R660 node, select **System Bundle (Windows x64) PER660 v850** or select the appropriate components such as switches, CloudLink, or operating system bundles.
4. Select the appropriate bundle to upgrade and click **View**.

   NOTE: Only one component can be modified at a time.

5. Select the specific component to upgrade and click **Modify**.
6. On the **Update Bundle Component** page, perform the following:
   a. Select either **Download from local network path** or **Upload from local**.
   b. Provide the **New Bundle Version** (if it is not picked automatically) to match the existing bundle version.
   c. In the **Type "MODIFY COMPLIANCE FILE" to confirm** field, enter `Modify Compliance File` and click **Proceed**.
7. Repeat the steps through 5 and 7 for the remaining components.
8. Click **Settings** > **Repositories** > **Compliance Versions** and verify that a wrench icon and `Modified` is appearing for the modified RCM/IC.
9. To update the resources, perform the following:
   a. On the menu bar, click **Resources** and select the PowerFlex nodes to update and click **Run Inventory**.
   b. Click **Lifecycle** > **Resource Groups**.
   c. Select the resource group to be modified and click **View Details**.
   d. Click **View Compliance Report**.
   e. Select the modified PowerFlex node or other components and click **Update Resources**.
   f. From the **Apply Resource Updates** page, select **Allow PowerFlex Manager to perform firmware and software updates now**.
   g. Select the appropriate maintenance mode and click **Apply**.
10. To monitor and confirm that the component is updated with the new version, perform the following:
    a. Click **Lifecycle** > **Resource Groups** to verify that the required component is upgraded.
    b. Select the resource group that you already modified and click **View Details**.
    c. On the **Resource Group Information** page, verify that the resource group compliance status is `Compliant`.

       If an upgrade failure occurs, contact Dell Technologies Support for assistance.

Related information

Viewing resource group details (mirroring)

Upgrade the software components

Use the following procedures to upgrade the software components using PowerFlex Manager. These procedures are applicable only when you consider deviating from the standard upgrade procedure.

NOTE: Ensure you upload identical versions of the VMware ESXi patch bundle and the ESXi ISO file to PowerFlex Manager.

A software component upgrade involves operating system updates such as the following:

- Applying ESXi patches
- Upgrading the CloudLink virtual machine

The following limitations are associated with the single component upgrade:

- ESXi version changes are restricted; upgrading from ESXi 8.x to 9.x or downgrading is not supported.
- Updates to embedded operating systems, such as SLES or CentOS are not supported.

Related information

Viewing resource group details (mirroring)

Upgrade VMware ESXi patch and ISO images

You can upgrade the VMware ESXi patch and ISO images using PowerFlex Manager.

1. On the menu bar, click **Settings** > **Repositories** > **Compliance Versions**.
2. Verify that the required RCM/IC is uploaded and is available.
3. Select the RCM/IC bundle to be modified and click **View Bundles**.
4. From the **Software Bundles** tab, select the **ESXI Software Bundle** checkbox, clear the checkbox for other software bundles, and click **View**.
5. Select **Upload Software Depot** and select either of the following options to upload the ESXi path depot file:
   - **Upload from local**: Select this option when the depot file is smaller than 500 MB.
   - **Download from local network path**: Select this option when the depot file is larger than 500 MB. Enter the CIFS file path and user credentials to access the file on the local network drive.
6. For the **Upload from local** option, complete the following:
   a. To provide a local file, click **Choose File**.
   b. To confirm that you want to modify the existing file, enter **`Modify Compliance File`** and click **Proceed**.
   c. Wait for the compliance file to synchronize and verify that the status of the compliance file is `Available`.
   d. Verify that a wrench icon and `Modified` displays for the modified RCM/IC.
7. For the **Download from local network path** option, complete the following:
   a. Enter the `CIFS` file or the `HTTPS` address to access the file.
   b. Enter the username and password.
   c. Select **Test Connection** to test the connection between the PowerFlex Manager user interface and the location of the CIFS file or the file that is at the HTTPS address.
   d. To confirm that you want to modify the existing file, enter **`Modify Compliance File`** and click **Proceed**.
   e. Wait for the compliance file to synchronize and verify that the status of the compliance file is `Available`.
   f. Verify that a wrench icon and `Modified` displays for the modified RCM/IC.
8. **NOTE:** You must complete steps 5 through 7 before you start step 8.

   To upload the `ESXI.depot.zip` and `ESXi.iso` files, complete the following:
   a. From **Repositories** > **Compliance Versions**, select the RCM/IC bundle, and click **Add**.
   The **Add Compliance File** dialog box displays.
   b. Enter the file path, username, and password for both the installation ISO and the depot ZIP file. Ensure the files match with the existing VMware ESXi patch and ISO files.
   c. Select **Test Connection** to test the connection between the PowerFlex Manager user interface and the location of the CIFS file or the file that is at the HTTPS address.
   d. Click **Save** and wait for the synchronization and download to complete.
   e. On the **Compliance Versions** page, verify that the files are added and the RCM/IC is in `Available` state.

Update resource inventory and resource groups

Once you upgrade the VMware ESXi patch and ISO images, use this procedure to update the resource inventory and resource groups.

1. On the menu bar, click **Lifecycle** > **Resources**.
2. Select the PowerFlex node to update, and click **Run Inventory**.
3. Click **Lifecycle** > **Resource Groups** and select the service to upgrade.

4. Click **Non-Compliant** in the **Resource Group Information** tab.
5. From the **Software Components** tab, select all the listed nodes, and click **Update Resources**.
6. Select the appropriate **Maintenance Mode** and click **Apply**.
7. To monitor the job and verify that the required component is upgraded, perform the following:
   a. Click **Lifecycle** > **Resource Groups**.
   b. Select the resource group that you modified and click **View Details**.
   c. On the **Resource Group Information** tab, verify that the resource group compliance status is `Compliant`.

   If the upgrade fails, contact Dell Technologies Support for assistance.

Upgrade CloudLink Center

Using the single component option, you can upgrade the CloudLink Center version.

CloudLink ISO is required to start the CloudLink version upgrade using the single component upgrade option.

You must verify that you are using the administrator credentials.

1. On the menu bar, click **Settings** > **Repositories** > **Compliance Version**.
2. Select the RCM/IC bundle, and click **View Bundles**.
3. From the **Firmware Bundles** tab, select the **System Bundle CloudLink** and click **View**.
4. Select the component **clc** that you want to upgrade and click **Modify**.
5. From the **Update Bundle Component** page, select **Download from local network path** option and upload the CloudLink ISO file.
6. Enter the **New Bundle Version** if it is not picked automatically, to match with the existing bundle version.
7. Type **`Modify Compliance File`** in the field and click **Proceed**.
8. Click **Repositories** > **Compliance Versions** and verify that the modified RCM/IC appears with a wrench icon and `Modified`.
9. To update the resources, perform the following:
   a. From **Lifecycle** > **Resources**, select both the CloudLink resources and click **Run Inventory**. Wait for two minutes to complete the inventory and refresh your browser.
   b. Select the CloudLink Center to upgrade and click **Non-Compliant** and click **Update Resource**.
   c. From the **Apply Resource Updates** page, select **Allow PowerFlex Manager to perform firmware and software updates now** and click **Apply Updates**. Wait till the deployment status becomes `Available`.
   d. On the **Resources** page, click **CloudLink Center** > **View Details** to verify the upgraded CloudLink Center version. Wait for three to four minutes. Both CloudLink Centers are updated with the new version.
10. To manually upgrade the SecureVM agent on all the PowerFlex nodes from CloudLink Center, perform the following:
    a. In your web browser, type the URL for CloudLink Center in the following format: **`https://clc_address`**
    b. Log in with the **`secadmin`** credentials.
    c. Click **Agents** > **Machines** and select the node.
    d. Click **Actions** > **Upgrade**. The upgrade option is disabled if an upgrade is not required for the SecureVM agent.
    e. Repeat step 10a through 10d for every PowerFlex node that requires a CloudLink agent upgrade.
    f. Click **Agents** > **Machines**, select the node, and verify the SecureVM agent version.
    g. After the agents are updated on the nodes, from CloudLink Center, click **Agents** > **Agent Download** to download the `64-bit SUSE RPM` package or `64-bit CentOS RPM` package depending on the operating system running on the nodes and modify the SecureVM agent component in the RCM/IC bundle.
11. To modify the SecureVM agent component in the RCM/IC bundle, perform the following:
    a. Verify that you are using the administrator credentials.
    b. On the menu bar, click **Settings** > **Repositories** > **Compliance Versions**, select the current RCM/IC bundle and click **View Bundles**.
    c. Click **Software Bundles** > **CloudLink SecureVM** > **View**. You will notice that there are two CloudLink SecureVM bundles with the same name. One bundle is for SUSE and the other bundle is for CentOS. Select the bundle and click **View** to differentiate the bundle. Select the bundle based on the operating system installed on the PowerFlex nodes.
    d. Click **Modify**.
    e. From the **Update Bundle Component** page, select the **Download from local network path** or **Upload from local** and select the downloaded agent.
    f. Enter the **New Bundle Version** to match the existing bundle version if it is not picked automatically.
    g. Type **`Modify Compliance File`** in the field and click **Proceed**.
12. To update the resource group details, perform the following:

a. On the menu bar, click **Lifecycle** > **Resource Groups**.
b. Select the resource group to modify and click **View Details**.
c. Click **More Actions** > **Update Resource Group Details**.
d. On the **Resource Group Information** page, verify that the resource group compliance status is `Compliant`.
   If the upgrade fails, contact Dell Technologies Support for assistance.

Exit the PowerFlex nodes from service mode.

Export the compliance report

You can export a compliance report using PowerFlex Manager. The compliance report identifies the list of modified components and their versions. It also compares component versions during upgrades from a modified RCM/IC to a standard RCM/IC.

Before exporting the compliance report, you must ensure that the default RCM/IC version is selected. To verify that the default version is selected, go to **Settings** > **Repositories** > **Compliance Versions**.

When transitioning from a modified RCM to a desired RCM, ensure the following:

- The component versions in the modified RCM are equal to or lower than the component versions in the desired RCM.
- If any component in the modified RCM has a higher version, use the **Extract Compliance** tool to identify the components.
- Use the single component upgrade to add those components with a higher version to the desired RCM to ensure version alignment and compatibility,

1. On the menu bar, click **Lifecycle** > **Resources** > **Export Report**.
2. Select **Export Compliance PDF Report** or **Export Compliance CSV Report**.
3. Identify the modified components and verify the number of components.

Modify and publish the template

After completing a single component upgrade, you can modify and publish the template. This process involves updating the existing template by incorporating a modified ESXi ISO image. The updated template is then published and used for future system expansions.

Ensure the modified RCM/IC is the same version as the **Target Version** under the **Existing Resource Group** page. While adding a resource group on an existing PowerFlex system, ensure that you select the latest uploaded operating system images.

1. On the menu bar, click **Templates** > **My Templates**.
2. Select the template that you want to modify and click **Modify**.
3. Select the node and click **Modify**.
   The node details display.
4. Click **Continue**.
   The **OS Settings** display.
5. Under **OS Settings** > **OS Image**, ensure that the newly uploaded image is selected from the dropdown.
6. Click **Save** > **Publish Template**.
   A confirmation message displays asking whether you are sure that you want to publish the template, click **Yes**.

Operating system images

An operating system image is a complete packaged copy of an operating system that is stored as a single file. After you add the operating system, you can manage it later using the modify or remove options.

1. On the **Repositories** page, click **OS Images**, and then click **Add**.
2. In the **Add OS Image Repository** dialog box, enter the following:
   a. In the **Repository Name** box, enter the name of the repository. The repository name must be unique.
   b. In the **Image Type** box, enter the image type.
   c. In the **Source Path and Filename** box, enter the path of the operating system image file name in a file share.
   d. To enter the CIFS share, see the format that is used in the following example: `\\host\lab\isos\filename.iso`
   e. If you are using the CIFS share, enter the username and password to access the share.
3. Click **Add**.

After adding an operating system image, you can modify it or remove it by selecting the image, then clicking **Modify** or **Remove**.

Resynchronizing an OS image repository

Use the resynchronize option to restore the OS image from the database after a backup and restore.

If an operating system image was uploaded as a part of an ISO file, you must resynchronize the OS image repository from the **OS Image Repositories** tab. However, if the operating system image was uploaded as a part of a compliance ZIP file, go to the **Compliance Versions** tab and resynchronize the ZIP file. The following procedure provides steps for performing the resynchronization from the **OS Image Repositories** tab.

To resynchronize the OS image repository for an operating system image that was uploaded as part of an ISO file:

1. On the **Repositories** page, click **OS Images**.
2. Click **Resynchronize** for a repository in an **Error** state.
   The **Resynchronize OS Repository** page is displayed.
3. Enter the user credentials and click **Test Connection** to test the network connection.

   NOTE: You cannot edit the **Source Path** and **Filename**.

4. Click **Resynchronize**.
   The repository state changes to `Copying state`.

Compatibility management

To facilitate upgrades, PowerFlex Manager uses information that is provided in the compatibility matrix file to determine which upgrade paths are valid and which are not.

When you attempt an upgrade, PowerFlex Manager warns you if the current version of the software is incompatible with the target version, or if any of the RCM or IC versions that are currently loaded are incompatible with the target compliance versions. PowerFlex Manager uses information that is provided in the compatibility matrix file to determine which paths are valid. The compatibility matrix file maps all the known valid and invalid paths for all previous releases of the software.

When you first install PowerFlex Manager, the software does not have the compatibility matrix file, so you must upload the file before performing any updates. You must upload the latest compatibility matrix file to ensure that PowerFlex Manager has access to the latest upgrade information.

You can download the file from Dell Technologies connectivity services or upload it from a local directory path. The file has a GPG extension and an associated compatibility version number.

If you are uploading from a local directory path, ensure that you have access to a valid compatibility matrix file that has the GPG extension. If you are using Dell Technologies connectivity services, ensure that Dell Technologies connectivity services is correctly configured.

1. On the menu bar, click **Settings** and then click **Compatibility Management**.
2. Click **Edit Settings**.
3. Click **Download from Dell Technologies connectivity services** if you are using Dell Technologies connectivity services.
4. Click **Upload from Local** to use a local file. Then, click **Choose File** to select the GPG file.

Getting started

If you want to return to the **Getting Started** page, you can launch it from the **Settings** page.

The **Getting Started** page guides you through the common configurations that are required to prepare a new PowerFlex Manager environment. A green check mark on a step indicates that you have completed the step. Only super users have access to the **Getting Started** page.

Networking

On the **Networking** page, you can define, edit, or delete a network. You can also verify which IP addresses are in use before deployment.

Related information

Getting started
Getting started (mirroring)
Configuring block storage
Deploying and provisioning

Networks

The **Networks** page displays information about networks that are defined in PowerFlex Manager, including:

- **Name**
- **Network Type**
- **VLAN ID**
- **IP Address Setting**
- **Starting IP Address**
- **Ending IP Address**
- **Role**
- **IP Address in Use**

On the **Networks** page, you can:

- Define or modify an existing network.
- Delete an existing network.
- Click **Export All** to export all the network details to a CSV file.
- Export network details for a specific network. To export the specific network details, select a network, and then click **Export Network Details**.
- Click a network to see the following details in the **Summary** tab:
  - Name of the user who created and modified the network.
  - Date and time that the network was created and last modified.
- Sort the column by network names, click the arrow next to the column header. You can also refresh the information about the page.

If you select a network from the **Networks** list, the network details are displayed.

For a static network, the following information is displayed:

- **Subnet Mask**
- **Gateway**
- **Primary DNS**
- **Secondary DNS**
- **DNS Suffix**
- **Last Updated By**
- **Date Last Updated**
- **Created By**
- **Date Created**
- **Static IP Details**

For a DHCP network, the following information is displayed:

- **Last Updated By**
- **Date Last Updated**
- **Created By**
- **Date Created**

You can filter the IPs by selecting any of the following options from the **View** drop-down list, under the **Static IP Address Details** section:

- **All IP Addresses**
- **IP Addresses in Use**
- **Available IP Addresses**

You can also select the links under the **IP Address in Use** column. The IPs are automatically filtered based on the **IP addresses In Use** criteria.

Defining a network

Adding the details of an existing network enables PowerFlex Manager to automatically configure nodes that are connected to the network.

Ensure that the following conditions are met before you define the network:

- PowerFlex Manager can communicate with the out-of-band management network.
- PowerFlex Manager can communicate with the operating system installation network in which the appliance is deployed.
- PowerFlex Manager can communicate with the hypervisor management network.
- The DHCP node is fully functional with appropriate PXE settings to PXE boot images from PowerFlex Manager in your deployment network.

1. On the menu bar, click **Settings** and then click **Networking**.
2. Click **Networks**.
   The **Networks** page opens.
3. Click **Define**. The **Define Network** page opens.
4. In the **Name** field, enter the name of the network. Optionally, in the **Description** field, enter a description for the network.
5. From the **Network Type** drop-down list, select one of the following network types:
   - **General Purpose LAN**
   - **Hypervisor Management**
   - **Hypervisor Migration**
   - **Hardware Management**
   - **PowerFlex Data**
   - **PowerFlex Data (Client Traffic Only)**
   - **PowerFlex Data (Server Traffic Only)**
   - **PowerFlex Management**

   For a PowerFlex configuration that uses a two-layer architecture with four dedicated data networks, you typically have two PowerFlex (client traffic only) VLANs and two PowerFlex data (server traffic only) VLANs. These network types are used with storage-only and compute-only resource groups.
6. In the **VLAN ID** field, enter a VLAN ID between **1** and **4094**.

   NOTE: PowerFlex Manager uses the VLAN ID to configure I/O modules that enable network traffic to flow from the node to configured networks during deployment.
7. Optionally, select the **Configure Static IP Address Ranges** check box, and then do the following:

   NOTE: The **Configure Static IP Address Ranges** check box is not available for all network types. For example, you cannot configure a static IP address range for the operating system installation network type. You cannot select or clear this check box to configure static IP address pools after a network is created.

   a. In the **Subnet Mask** box, enter the subnet mask.
   b. In the **Gateway** box, enter the default gateway IP address for routing network traffic.
   c. Optionally, in the **Primary DNS** and **Secondary DNS** fields, enter the IP addresses of primary DNS and secondary DNS.
   d. Optionally, in the **DNS Suffix** field, enter the DNS suffix to append for hostname resolution.
   e. To add an IP address range, click **Add IP Address Range**. In the row, indicate the role in PowerFlex nodes for the IP address range and then specify a starting and ending IP address for the range. For the **Role**, select either:
      - **Server or Client**: Range is assigned to the server and client roles.

      NOTE: IP address ranges cannot overlap. For example, you cannot create an IP address range of 10.10.10.1–10.10.10.100 and another range of 10.10.10.50–10.10.10.150.
8. Click **Save**.

Related information

Adding networks to a resource group (Mirroring)

Modifying a network

If a network is not associated with a template or resource group, you can edit the network name, the VLAN ID, or the IP address range.

1. On the menu bar, click **Settings** and then click **Networking**.
2. Click **Networks**.
   The **Networks** page opens.
3. Select the network that you want to edit and click **Modify**. The **Modify Network** page opens.
4. Edit the information in any of the following fields: **Name**, **VLAN ID**, **IP Address Range**.

   For a PowerFlex data network, you can specify a **Subnet** IP address for a static route configuration. The subnet is used to support static routes for data networks.

5. Click **Save**.

You can also view details for a network or export the network details by selecting the network, then clicking **View Details** or **Export Network Details**.

(Optional) Deleting a network

You cannot delete a network that is associated with a template or resource group.

1. On the menu bar, click **Settings** and then click **Networking**.
2. Click **Networks**.
3. In the **Networks** page, click the network that you want to delete, and click **Delete**.
4. Click **OK** when the confirmation message is displayed.

Adding an IP verification port number

You must verify what IP addresses are in use before they are deployed. PowerFlex Manager uses default ports 22, 80, and 135 for verification.

1. On the menu bar, click **Settings** and click **Networking**.
2. Under **IP Verification Port Number**, click **Edit**.
3. In the **Port Numbers Used for IP Verification** box, add or remove the port numbers used by PowerFlex Manager to verify the IP addresses.

   Port numbers must be entered in comma-separated list, and must be between 1 and 65535.
4. Click **Save**.

To remove a port number, click **X** to the right of the port number.

System data networks

The **System Data Networks** page displays information about system data networks for use with NVMe/TCP. This information includes:

- **Name**
- **Network**
- **Network Mask**

On the **System Data Networks** page, you can:

- Add a system data network.
- Rename a system data network.
- Remove a system data network.

Adding a system data network

Use this procedure to add a new data network for NVM/TCP host connectivity.

To add a system data network, complete the following steps:

1. On the menu bar, click **Settings** and then click **Networking**.
2. Click **System Data Networks**.
   The **System Data Networks** page opens.
3. Click **Add**. The **Create System Data Network** page opens.
4. In the **Name** field, enter the name of the network.
5. In the **Network** field, enter the IP address of the network.
6. In the **Network Mask** field, enter the network mask.
7. Click **Create**.

Renaming a system data network

After you have added a system data network, you can edit the network name.

1. On the menu bar, click **Settings** and then click **Networking**.
2. Click **System Data Networks**.
   The **System Data Networks** page opens.
3. Select the network that you want to edit and click **Rename**. The **Rename System Data Network** page opens.
4. Edit the information in the **System Data Network Name**.
5. Click **Apply**.

Removing a system data network

After you have added a system data network, you can delete it, if necessary.

1. On the menu bar, click **Settings** and then click **Networking**.
2. Click **System Data Networks**.
   The **System Data Networks** page opens.
3. Click the network that you want to delete, and then click **Remove**.
4. Click **OK** when the confirmation message is displayed.

Events and alerts

A source is used to configure the receiving of external events and syslog content.

A destination is used to configure the ability to send events and alerts information out. A destination is always external. Dell Technologies connectivity services, email, SNMP, and remote syslog are considered destinations.

Notification policies define what information is sent to each destination. Events and alerts exist irrespective of whether notification policies are created.

SNMP sources are not automatically discovered and must be configured to receive events about these sources. PowerFlex Manager is preconfigured and events, and alerts are automatically available. Resources in the PowerFlex rack, PowerFlex appliance, and Dell VxFlex Ready Node are automatically discovered. Any future resources, for example, switch replacements or additional nodes, are considered external and must be added manually as sources.

Related information

Enabling Dell Technologies connectivity services

Configuring an external source

You must define a source to enable PowerFlex Manager to receive an external event. You need not create a source to receive PowerFlex events.

External resources such as iDRAC, CloudLink, and Dell PowerSwitch or Cisco Nexus switches have to be discovered on the **Resources** page either in a managed or unmanaged state.

Alerts received from external resources like iDRAC, CloudLink, and Dell PowerSwitch or Cisco Nexus switches are seen only on **Events** page.

PowerFlex related alerts can be seen on both the **Alerts** and **Events** pages.

A syslog source can only go to a syslog destination and does not display in events. An SNMP source, either V2 or V3, displays in events even without a defined notification policy.

1. Go to **Settings** > **Events and Alerts** > **Notification Policies**.
2. From the **Sources** pane, click **Add**.
   The **Add Source** window opens.
3. Enter a source name and description.
4. Configure either SNMP, syslog, or audit log forwarding:
   - If you select **SNMPV2c**:
     a. Enter the community string by which the source forwards traps to destinations.
     b. Enter the same community string for the configured resource. During discovery, if you selected PowerFlex Manager to automatically configure iDRAC nodes to send alerts to PowerFlex Manager, enter the community string that is used in that credential here.
   - If you select **SNMP V3**:
     a. Enter the username, which identifies the ID where traps are forwarded on the network management system.

        NOTE: The username must be at most 16 characters.

     b. Select a security level from the following:

        Table 77. Security levels

Security Level	Details	Description	authPassword	privPassword
Minimal	noAuthNoPriv	No authentication and no encryption	Not required	Not required
Moderate	authNoPriv	Messages are authenticated but not encrypted (Password-based encryption with message digest (MD5) with at least eight characters)	Required	Not required
Maximum	authPriv	Messages are authenticated and encrypted (MD5 and Data Encryption Standard (DES) both with at least eight characters)	Required	Required

        PowerFlex supports SNMP v3 alerts only for iDRAC.

   - If you select **Syslog**, switch **Enable Syslog** right to on.
   - If you select **Audit Log**, click **Enable Audit Log Retention**. You can retain audit logs locally for 15 days.
5. Click **Submit**.

Related tasks

Modifying an external source
Configuring a destination
Modifying a destination
Add a notification policy
Modify a notification policy
Delete a notification policy
Configuring an audit log policy
Add a notification policy

Modifying an external source

You can edit the information about how PowerFlex Manager receives an event.

1. Go to **Settings** > **Events and Alerts** > **Notification Policies**.
2. From the **Sources** pane, click the source that you want to modify.
   The **Edit Source** window opens.
3. Edit the information and click **Submit**.

Related tasks

Configuring an external source
Configuring a destination
Modifying a destination
Add a notification policy
Modify a notification policy
Delete a notification policy
Configuring an audit log policy
Add a notification policy

Configuring a destination

Define a location where the event and alert data that PowerFlex Manager processes should be sent.

1. Go to **Settings** > **Events and Alerts** > **Notification Policies**.
2. From the **Destinations** pane, click **Add**.
   The **Create New Destination Protocol** window opens.
3. From the **Destination Properties** window:
   a. Enter the destination name and description.
   b. From the **Destination Type** menu, select either **SNMP V2c**, **SNMP V3**, **Syslog**, **Email (SMTP)**, **Webhook**, or **Audit Log**.
4. Click **Next**.
   The **Protocol Settings** window opens.
5. Depending on the destination type, enter the following information:

Destination type	Protocol settings
SNMP V2c	• Network name/IP address • Port • Community string
SNMP V3	• Network name/IP address • Port • Username • Security level: ○ Minimal - no more information required

Destination type	Protocol settings
	○ Moderate - MD5 authentication password required ○ Maximum - MD5 authentication and DES privacy passwords required
Syslog	● Network name/IP address ● Port ● Protocol ○ UDP ○ TCP ● Facility ○ All ○ Authentication ○ Security and authentication
Email (SMTP)	● Destination name ● Description ● Destination type ○ Server type: ■ SMTP ■ SMTP over SSL ■ SMTPS SMARTTLS ○ Server IP address or FQDN ○ Port ○ Sender address and up to five recipient addresses ○ If you choose to use credentials enter: ■ Username, password, sender address and up to five recipient addresses ● Send test email ● Test email server connection
Webhook	● Destination name ● Description ● Destination type ○ Select **Webhook**. ○ You can have up to three webhook destinations. ● Webhook destination API URL ○ For example, if you want to configure a webhook destination for BigPanda, you must specify the BigPanda endpoint here. ○ You can get this URL from the BigPanda site. ● Enable credentials ● Credentials ○ For example, for BigPanda, the credential you use must have an App Key and a Token string. You can generate these values on the BigPanda site and then copy them to the App Key and Token fields. ● Test webhook
Audit Log	● Destination name ● Description ● Destination type ○ Select **Audit Log**. ○ You can have up to four audit log destinations. ● Audit log network name or IP address ● Port ● Protocol ○ User Datagram Protocol (UDP) ○ Transmission Control Protocol (TCP)

6. Click **Save**.

Related tasks

Configuring an external source
Modifying an external source
Modifying a destination
Add a notification policy
Modify a notification policy
Delete a notification policy
Configuring an audit log policy
Add a notification policy

Related information

Migrating from key-based licensing to dynamic licensing (connected method)

Modifying a destination

You can edit the information about where event and alert data that is processed by PowerFlex Manager should be sent.

1. Go to **Settings** > **Events and Alerts** > **Notification Policies**.
2. From the **Destinations** pane, click the destination whose information you want to modify.
   The **Edit Source** window opens.
3. Edit the information and click **Submit**.

Related tasks

Configuring an external source
Modifying an external source
Configuring a destination
Add a notification policy
Modify a notification policy
Delete a notification policy
Configuring an audit log policy
Add a notification policy

Add a notification policy

When you add a notification policy, you define the rules for processing events or alerts from sources, and to which destinations that information should be sent.

1. Go to **Settings** > **Events and Alerts** > **Notification Policies**.
2. Click **Create New Policy**.
3. Enter a name and a description for the notification policy.
4. From the **Source Type** menu, select how you want events and alerts to be received. The source type options are:
   - **Snmpv2c**
   - **Snmpv3**
   - **Syslog**
   - **Powerflex_alerts**
   - **Powerflex_events**
   - **Auditlog**

   PowerFlex related alerts are generated and forwarded to a webhook destination when the notification policy is configured with the **Source Type** set to **Powerflex**.
5. From the **Resource Domain** menu, select the resource domain that you want to add a notification policy to. The resource domain options are:
   - **All**
   - **Management**
   - **Block (Storage)**

- **Compute (Servers, Operating Systems, virtualization)**
- **Network (Switches, connectivity etc.)**
- **Security (RBAC, certificates, CloudLink etc.)**

6. Select the checkbox beside the severity levels that you want to associate with this policy. The severity indicates the risk (if any) to the system, in relation to the changes that generated the event message.
7. Select the required destination. You can choose one or more destinations. For a webhook, you can have up to three destinations defined. For audit logs, you can configure up to four destinations.

   NOTE: The audit log policy is disabled by default. Click **Configure Now** to configure and activate the audit log policy.

8. Click **Submit**.

After adding a notification policy, you might need to perform additional configuration steps. For example, If you are setting up a notification policy for a webhook destination that uses BigPanda, you can optionally configure BigPanda to show the severity levels from PowerFlex Manager. To do this, you must configure the status mapping on the BigPanda site. Map the `major` and `minor` severity values from PowerFlex Manager to the `Warning` status for BigPanda.

Related tasks

Configuring an external source
Modifying an external source
Configuring a destination
Modifying a destination
Modify a notification policy
Delete a notification policy
Configuring an audit log policy

Related information

Enabling Dell Technologies connectivity services

Modify a notification policy

You can modify certain settings that are associated with a notification policy.

You cannot modify the source type or destination once it is assigned to a notification policy.

1. Go to **Settings** > **Events and Alerts** > **Notification Policies**.
2. Select the notification policy that you want to modify.
3. You can choose to modify the notification policy in the following ways:
   - To activate or deactivate the policy, click **Active**.
   - To modify the policy, click **Modify**. The **Edit Notification Policy** window opens.
4. Click **Submit**.

Related tasks

Configuring an external source
Modifying an external source
Configuring a destination
Modifying a destination
Add a notification policy
Delete a notification policy
Configuring an audit log policy
Add a notification policy

Delete a notification policy

Use this procedure to delete a notification policy.

NOTE: Once a notification policy is deleted, it cannot be recovered.

1. Go to **Settings** > **Events and Alerts** > **Notification Policies**.
2. Select the notification policy that you want to delete.
3. Click **Delete**.
   You receive a message to confirm if you want to delete the policy.
4. Click **Submit** and click **Dismiss**.

Related tasks

Configuring an external source
Modifying an external source
Configuring a destination
Modifying a destination
Add a notification policy
Modify a notification policy
Configuring an audit log policy
Add a notification policy

Configuring an audit log policy

Audit log is a collection of various events that occur in a system. Audit logs facilitate detection and response to cyber threats. By default, an audit log policy is disabled.

In PowerFlex Manager, the audit log contains details of the following events:

- User authentication and authorization
- System configuration and security policy changes
- Security events

Security Information and Event Manager (SIEM) manages log collection, log analysis, log forensics, real-time alerting, user activity monitoring and retention. About four SIEM can be sent over a secured connection.

Administrators can collect events and alerts from remote systems, also called sources and store them on a central server. The Audit Log Converter receives the events and alerts and forwards them in an encrypted form to the remote SIEM.

The Audit Logging Enabler is required to send logs in real-time on an automated basis. The audit logs are persistent for 10 to 90 days. Audit logs sent through SIEM have a retention period of 12 to 18 months.

1. Go to **Events and Alerts** > **Notification Policies** > **Policies** > **Audit Log Policy**.
2. Click **Configure Now**.
   The **Audit Log Policy** window opens.
3. Select and existing destination to forward the audit logs, from the **Destination** drop-down or click **Create Destination** to create a destination in the **Add Destination** window.

   You can create up to four audit log destinations.

   NOTE: You can skip this step if you do not want to forward the audit logs.
4. Click **Enable Audit Logs** to create a source of the type **Audit Log**.
5. Click **Save**.

Related tasks

Configuring an external source
Modifying an external source
Configuring a destination
Modifying a destination
Add a notification policy

Modify a notification policy
Delete a notification policy
Add a notification policy

License management

PowerFlex offers two approaches to license management—dynamic licensing and key-based licensing. Each approach governs how PowerFlex system functionality is activated and maintained. Whether you have key-based or dynamic licensing you have a free evaluation license for the first 90 days.

Dynamic licensing

The dynamic licensing approach provides an ability to manage license allocations more easily across product registrations, enabling a flexible and keyless experience. It automates and simplifies license management activities, such as registrations, capacity expansions, and renewals. The dynamic licensing approach is applicable for all systems with PowerFlex Manager 5.1.x and later.

The following methods are used to activate and use your products with dynamic licensing:

- Connected method
- Offline method

For more detailed information about dynamic licensing, see the *PowerFlex Family Licensing Guide*.

Key-based licensing

The key-based licensing approach is a traditional method of activating software using a unique product key. Each key is manually applied to a system, and once activated, the license remains tied to that installation. This approach works well for environments that prefer offline activation or need strict per-device licensing control. However, it requires manual management, offers limited automation, and does not adapt to changes in system configuration or usage.

For more detailed information about migrating from key-based to dynamic licensing, see the *PowerFlex Family Licensing Guide*.

Evaluation license

When you deploy PowerFlex, a free evaluation license is automatically provided for 90 days whether you are using dynamic licensing or key-based licensing. This evaluation license allows for full use of the storage cluster for 90 days. During this period, PowerFlex runs in trial mode.

To track when the license expires, go to **License Management** to view the start and end dates for a license.

After the 90-day evaluation period ends:

- Your storage remains intact.
- Configuration changes are disabled until a valid license is applied.

To continue using PowerFlex with full functionality, you must purchase a license that is based on the system capacity.

Related information

Enabling Dell Technologies connectivity services

Licensing PowerFlex for key-based licensing

For the first 90 days after you deploy PowerFlex, no license is required. During this period, you are running PowerFlex in trial mode, and all features are enabled. After the trial period expires, you must upload a key-based paid license through PowerFlex Manager.

Before you upload a key-based license, the MDM cluster must be deployed. If you experience an issue expanding you key-based licensing, contact Dell Technologies Support.

When you are running PowerFlex in trial mode, PowerFlex Manager displays an alert on the **Monitoring** > **Alerts** page.

1. On the menu bar, click **Settings** > **License Management**.
2. Click **PowerFlex License**.

   If you are not migrating to dynamic licensing now, ignore the blue banner that displays at the top of the screen. You can continue to use a key-based license. However, if you need to expand or your license expires, you cannot purchase additional licensing with a key-based license.
3. To upload a license for PowerFlex, click **Choose File** in the **Production License** section,
4. Select the license file, and click **Open**.
5. Click **Save**.

When you upload a license file, PowerFlex Manager checks the license file to ensure that it is valid.

After the upload is complete, PowerFlex Manager stores the license details and displays them on the **PowerFlex Manager License** page. You can see the installation ID, system name, SWID, and license registration method.

In addition, you can see the total licensed capacity and the license capacity left.

Licensing PowerFlex for dynamic licensing (connected method)

To use the connected method for dynamic licensing, first register your PowerFlex instance in the Dell Dynamic Licensing Portal. After registration, you will receive an access key and select a pin, which PowerFlex Manager uses to authenticate with Dell Technologies connectivity services.

Ensure that you registered your PowerFlex instance in the Dell Dynamic Licensing Portal. To register your PowerFlex instance, you must have the license authorization code (LAC) that was emailed to the primary email contact. Or alternatively you can look up the LAC on the Dell Dynamic Licensing Portal.

If you skipped the initial setup wizard and did not enable Dell Technologies connectivity services, go to **Settings** > **Events and Alerts** > **Notification Policies** to configure it.

1. On the menu bar, click **Settings** and then click **License Management** > **PowerFlex License**.
2. From the blue information banner, select **Migrate to Dynamic Licensing**.
   The **Migrate to Dynamic Licensing** dialog opens.
3. Enter the license authorization code (LAC).
4. Accept the license agreement and click **Migrate**.

The **PowerFlex License** page provides the following details that are associated with the license:

- **Researvation Pool Name**: The name of the reservation pool that links one or more installations, which are identified by their cSWID, to one or more entitlements.
- **cSWUID**: The customer software ID that is received as part of registering for dynamic licensing.
- **Installation ID**: The installation ID is a 16-character hexadecimal value that uniquely identifies a PowerFlex system.
- **Product Type**: The type of device and the deployment environment if the device type is software. For example, **PowerFlex software Amazon Web Services**.
- **Connection status**: The method that you used to register your license. For example, connected or offline.

Using the connected method, PowerFlex Manager sends a consumption report to the Dell Dynamic Licensing portal every four hours. If an error occurs or the compliance state changes, a report is sent every 15 minutes until the system is compliant.

Related information

Enabling Dell Technologies connectivity services
Migrating from key-based licensing to dynamic licensing (connected method)

Licensing PowerFlex for dynamic licensing (offline method)

You can manually license your PowerFlex instance for dynamic licensing when you are offline. When you license using the offline method, the system completes the process without connecting to the Dell licensing servers. You must download the license registration file from the Dell Dynamic Licensing portal.

You must manually register your PowerFlex instance in the Dell Dynamic Licensing Portal. Download the license registration file from the *Register Your Product* section, and save it locally. The license registration file is the dynamic license file that you will upload through PowerFlex Manager.

1. On the menu bar, click **Settings** and then click **License Management** > **PowerFlex License**.

2. From the blue information banner, select **Migrate to Dynamic Licensing**.
   The **Migrate to Dynamic Licensing** dialog opens.
3. To upload the license registration file, under **License Registration**, drag and drop the license registration file or browse to the location where the license registration file is stored locally.
4. Once the license registration file is uploaded, accept the license agreement and click **Migrate**.

Your existing license and entitlements are automatically migrated to the Dell Dynamic Licensing portal. No further action is required.

Migrating from key-based licensing to dynamic licensing (connected method)

You can migrate from key-based licensing to dynamic licensing through PowerFlex Manager when you are connected to Dell Technologies connectivity services. You must first deregister and then reregister with Dell Technologies connectivity services.

You must be registered with Dell Technologies connectivity services to migrate to dynamic licensing.

If you are already registered with Dell Technologies connectivity services, you must reregister with Dell Technologies connectivity services before you migrate.

Once you migrate to dynamic licensing, you cannot revert to key-based licensing.

1. On the menu bar , click **Settings** > **Events and Alerts** > **Notification Policies**.
2. Before deleting the existing policy and deregistering Dell Technologies connectivity services, go to **Policies**, select the policy, and click **Delete**.
3. Click **Submit** to confirm deletion of the policy.
4. Click **Dell Technologies connectivity services**.
5. From the **Destination Protocol Settings (Dell Technologies connectivity services)** window, click **Deregister Dell Technologies connectivity services**.
6. From the **Warning** window, click **Submit**.
7. Under **Notification Policies**, click **Dell Technologies connectivity services**.
8. Click **I accept this agreement** to accept the telemetry agreement and click **Next**.
9. From **Connect to Dell Technologies connectivity services**, click **Next**.
10. Under **Connection Type**, select either of the following connection type:
    - To connect directly to Dell Technologies, click **Connect Directly**. Click **Next**.
    - To connect to a Dell Technologies connectivity services gateway server by remote access, click **Connect via Gateway Server** and provide the primary gateway IP address or **Primary Gateway IP/Hostname** and **Gateway Port**.Click **Next**.
11. Under **Authentication Details**, enter the **Access Key** and **Pin**. You can obtain the access key and self-select a pin from Support for PowerFlex Software .
12. Select **I have a dynamic license authorization code (LAC)**.
13. Enter the **Support Contact** details and click **Finish**.
14. From **Settings**, click **License Management** and click **Migrate to Dynamic Licensing**.
15. Accept the license agreement and click **Migrate**.

Your existing license and entitlements are automatically migrated. No further action is required.

Related information

Licensing PowerFlex for dynamic licensing (connected method)
Configuring a destination
Enabling Dell Technologies connectivity services

Migrating from key-based licensing to dynamic licensing (offline method)

You can migrate from key-based licensing to dynamic licensing through PowerFlex Manager when you are offline.

You must manually register your PowerFlex instance. Log in to the Dell Dynamic Licensing Portal, download the license registration file from the **Register Your Product** section, and save it locally. The license registration file is the dynamic license file that you upload through PowerFlex Manager to migrate.

Once you migrate to dynamic licensing, you cannot revert to key-based licensing.

1. On the menu bar, click **Settings** and then click **License Management** > **PowerFlex License**.
2. From the blue information banner, select **Migrate to Dynamic Licensing**.
   The **Migrate to Dynamic Licensing** dialog opens.
3. To upload the license registration file, under **License Registration**, drag and drop the license registration file or browse to the location where the license registration file is stored locally. For example: **<dynamic_license_file>.dl**.
4. Once the license registration file is uploaded, accept the license agreement, and click **Migrate**.

Your existing license and entitlements are automatically migrated to the Dell Dynamic Licensing Portal. No further action is required.

Security

On the **Security** page, you can upload SSL trusted certificates to enable secure browser access to PowerFlex Manager and to establish secure LDAP connection to Active Directory. You can also define credentials for the resources that PowerFlex Manager accesses and manages.

Adding an appliance SSL certificate

To enable secure communication between external resources (such as a web browser) and PowerFlex Manager, you must replace the defualt appliance SSL certificate with a certificate that is signed by a trusted certificate authority (CA).

The process to add an appliance SSL certificate involves the following steps:

1. Generate a certificate signing request (CSR) in PowerFlex Manager. The generation of a CSR is supported through PowerFlex Manager only.
2. Download the CSR generated by PowerFlex Manager.
3. Submit the CSR to a trusted CA for signing and retrieve the signed certficate.
4. Upload the signed SSL certificate to PowerFlex Manager.

A root certificate is a self-signed certificate and serves as the trust anchor in a certificate chain. It is not signed by any other certificate authority (CA).

An intermediate certificate is signed by either a root CA or another intermediate CA, creating a chain of trust between the root CA and the end certificate.

If the appliance SSL certificate is signed by an intermediate CA, upload both the root and intermediate CA certificates. To ensure that the complete certificate chain can be validated, upload the intermediate certificates seperately through the **SSL Trusted Certificates** screen. Upload each intermediate certificate one at a time.

Once the CA-signed certificate(s) is uploaded, PowerFlex Manager allows external resources (such as a web browser) that trust the CA to securely verify the server's identity and establish an encrypted HTTPS connection without certificate warnings.

Generating a certificate signing request

A certificate signing request (CSR) includes information that the certificate authority (CA) requires to provide a valid SSL certificate. After generating the CSR, download the encrypted text, and then submit it to the CA. The CA validates the text, and once approved, issues a signed SSL certificate for you to upload.

You can select a **Standard** or a **Stringent** security level when you generate the certificate signing request.

- Use the **Standard** security level when the certificate's Subject Alternative Name (SAN) includes both IP addresses and FQDNs in the certificate.

- Use the **Stringent** security level if your signing authority does not allow IP addresses in the certificate. The FQDNs used in the certificate must resolve to DNS records.

  If you use the Stringent security level, you must enter an FQDN for each listed ingress IP addresses. If the data networks do not use FQDNs, you can reuse the management FQDN for the data networks. If the out-of-band (OOB) Ingress is physically isolated from management, you must include an FQDN specific to OOB and the FQDN must also resolve to DNS records.

1. On the menu bar, click **Settings** and then click **Security**.
2. Click **Appliance SSL Certificates**.
   The **Appliance SSL Certificates** page opens.
3. Click **Generate Certificate Signing Request**.
   a. Under **Security Level**, select either **Standard** or **Stringent**. The default is **standard**.
   b. In the **MGMT Distinguished Name (www.domain.com)** box, enter a distinguished name in the format `www.example.com`.
   c. In the **Business Name** box, enter a business name where the certificate is recorded.
   d. In the **Department Name** box, enter a department name of the organizational unit (for example, IT, HR, or Sales) for which the certificate is generated.
   e. In the **Locality (Town/City)** box, enter a locality name in which the organization is located.
   f. In the **State (Province/Region)** box, enter a state name in which the organization is located (do not abbreviate).
   g. From the **Country** list, select a country or region in which the organization is located.
   h. In the **Email** box, enter a valid email address.
   i. Click **Generate**.

Downloading the certificate signing request

After generating the certificate signing request (CSR) and the Certificate Authority (CA) approves the CSR, the CA approves the CSR, you can download the certifictae file.

1. On the menu bar, click **Settings** > **Security**.
2. Click **Appliance SSL Certificates**.
   The **Appliance SSL Certificates** page opens.
3. Click **Download Certificate Signing Request**.
4. To receive a valid appliance SSL certificate, copy the displayed text and then submit the text to a certificate authority.

CAUTION: Before uploading an appliance SSL certificate, verify that the certificate file meets the following requirements:

- **The file must contain an X.509 certificate in PEM format.**
- **The file must start with `---BEGIN CERTIFICATE---` and end with `---END CERTIFICATE---`.**
- **The file must have a `.pem` extension.**

Uploading an appliance SSL certificate

If the appliance SSL certificate is signed by an intermediate CA, upload both the root CA certificate and all the required intermediate CA certificates. To ensure that the complete certificate chain can be validated, upload the intermediate certificates separately through the **SSL Trusted Certificates** screen. Upload each intermediate certificate one at a time..

1. On the menu bar, click **Settings** > **Security**.
2. Click **Appliance SSL Certificates**.
   The **Appliance SSL Certificates** page opens.
3. Click **Choose File** under **SSL Certificate** and select the signed SSL certificate returned by your certificate authority (CA).
4. Click **Choose File** under **Trusted CA Certificate** and select the root CA certificate.
5. To upload the certificate, click **Save**.

Adding a trusted SSL certificate

You can add a trusted SSL certificate to validate external systems that PowerFlex Manager connects to, for example, Active Directory.

Before you upload a trusted SSL certificate, you must obtain the certificate file:

CAUTION: The certificate file must meet the following requirements:

- **The file must contain an X.509 certificate in PEM format.**
- **The file must start with `---BEGIN CERTIFICATE---` and end with `---END CERTIFICATE---`.**
- **The file must have a `.pem` extension.**

- Review and update your management node credentials. Go to **Settings** > **Security** > **Update Management Node Credentials**.

1. On the menu bar, click **Settings** > **Security**.
2. Click **SSL Trusted Certificates**.
   The **SSL Trusted Certificates** page opens.
   NOTE: To add a root certificate signed by a root authority, complete steps 3 through 6. To add an intermediate certificate signed by an intermediate authority, complete steps 3 through 6 to add the root certifcate and then repeat steps 3 through 6 to add the intermediate cerificate.
3. Click **Add**.
4. Click **Choose File** and select the SSL certificate.
5. Enter a **Name** for the certificate. The name must be a single word.
6. To upload the certificate, click **Save**.

To delete a trusted SSL certificate, select the certificate, and click **Delete**.

Adding resource credentials

PowerFlex Manager requires a root-level username and password to access and manage nodes, switches, VMware vCenter, element managers, PowerFlex gateway, and operating system resources.

The **Credentials Management** page displays the following information about the credentials:

- **Name**—A user-defined name that identifies the credentials.
- **Type**—A type of resource that uses the credential.
- **Resources**—The total number of resources to which the credential is assigned.

From the credential list, click a credential to view its details in the **Summary** tab:

- Name of the user who created and modified the credential.
- Date and time that the credential was created and last modified.

On the **Credentials Management** page, you can:

- Create credentials
- Edit existing credentials
- Delete existing credentials

Create credentials

Do the following to create credentials:

1. On the menu bar, click **Settings** and click **Security**.
2. Click **Resource Credentials**.
   The **Credentials Management** page opens.
3. Click **Create**.
4. In the **Create Credentials** dialog box, from the **Credential Type** drop-down list, select one of the following resource types for which you want to create the credentials:
   - **Node**
   - **Switch**
   - **vCenter**
   - **Element Manager**
   - **PowerFlex Gateway**
   - **OS Admin**
   - **OS User**
   - **PowerFlex Management System**

The **OS Admin** and **OS User** credential types apply to deployed items, not to PowerFlex Manager itself.

5. In the **Credential Name** field, enter the name to identify the credential.
6. Click **Enable Key Pairs** to enable login with SSH key pairs:

   To enable key pairs for the **Node** or **Switch** credential type:

   a. Import an existing key:
      i. Click **Import SSH Key Pair**.
      ii. Click **Choose File** and browse to the file that contains your public and private keys, and select the private key.
      iii. Type a name for the key pair.
      iv. Click **Import**.

   To enable key pairs for the **OS Admin** or **OS User** credential type:

   a. To create a key:
      i. Click **Create a new key**.
      ii. Click **Create & Download Key Pair**.
      iii. Type a name for the key pair.
      iv. Click **Create**.

      The private key file (id_rsa) is downloaded to the downloads folder. Click the **Download Public Key** button to download the public key file (id_rsa.pub).

   b. To import an existing key:
      i. Click **Import existing key**.
      ii. Click **Import SSH Key Pair**.
      iii. Click **Choose File** and browse to the file that contains your public and private keys.
      iv. Type a name for the **Key Pair Name field**.
      v. Click **Import**.

   If you enable SSH key pairs for a **Node** or **Switch** credential and use that credential for discovery, PowerFlex Manager uses public or private RSA key pairs to SSH into your node or switch securely, instead of using a username and password. If you enable SSH key pairs for an **OS user** or **OS Admin** credential and use that credential for a deployment, PowerFlex Manager uses RSA public or private key pairs for the deployment operations.

   NOTE: PowerFlex Manager does not consume SSH keys for all component types. For example, if you enable SSH key pairs for an admin credential, the SSH keys are not used for the deployment of a CloudLink Center VM. In this case, the username and password would be used instead for all communication.

7. In the **Domain** box, optionally specify an LDAP domain for the user.
8. In the **User Name** field, enter the username for the credential.

   root is the only valid username for root-level credentials on nodes (iDRAC). You can add iDRAC users with a username other than **root**.

   For the **OS User** credential type, you can enter a user other than root. For the embedded operating system, this user account must have SSH enabled and have sudo access. For VMware ESXi, the account must be configured with the administrator role on the local server permission setting, which should enable SSH and other tools like esxcli. You can add existing resource groups with a non-root user.

   The account on the SVM and/or storage-only nodes for the **OS User** credential type must have a `/home` directory and have the correct group permissions. For example, if the account were pfxm_admin, the home directory would be: `/home/pfxm_admin`

   Here is an example showing the requirements for a pxflex user in the pxflex-grp group on an SVM:

```
[root@svm-dkim-hc-node1 home]# ls -alhtr
        total 0
        drwxr-xr-x.  3 root   root        20 Feb 26 16:03 .
        drwxr-xr-x   3 pxflex pxflex-grp  22 Feb 26 16:10 pxflex
        dr-xr-xr-x. 17 root   root       224 Feb 26 16:37 ..
        [root@svm-dkim-hc-node1 home]# cd pxflex/
        [root@svm-dkim-hc-node1 pxflex]#
        [root@svm-dkim-hc-node1 pxflex]#
        [root@svm-dkim-hc-node1 pxflex]# pwd
        /home/pxflex
```

   For the **OS Admin** credential type, the **User Name** field is disabled because the user is assumed to be root. You must use the root user for new deployments.

Provide two usernames for the PowerFlex gateway credential type:

- **Gateway Admin User Name**
- **Gateway OS User Name**

The gateway admin user is the REST API administrator. The gateway operating system user is the SSH login user. The gateway admin user must be the **admin** user, and the gateway operating system user must be **root**.

9. In the **Password** and the **Confirm Password** boxes, enter the password for the credential.

   NOTE: When the SSH key pair feature is enabled, the switch credential does not require the **Password** option.

   For the PowerFlex gateway credential type, provide two passwords:

   - **Gateway Admin Password**
   - **Gateway OS Password**

   PowerFlex Manager allows you to supply different passwords for the gateway admin and gateway operating system users. The passwords are validated when you discover the PowerFlex gateway on the **Resources** page. The discovery fails if the passwords were not specified correctly on the **Credentials Management** page. PowerFlex Manager does not update the gateway admin and gateway operating system passwords on the node itself. However, if you change the password manually for either of these users, you can update the password on the **Credentials Management** page, and PowerFlex Manager ensures that subsequent operations on the resource group use the new password.

   Optionally, provide additional settings for VMware vCenter and element manager, and for switch credentials:

   - For VMware vCenter and element manager, in the **Domain** box, optionally enter the domain ID.
   - For switch credentials, under **Protocol**, optionally click one of the following connection protocols that are used to access the resource from remote:
     - **Telnet**
     - **SSH**

       NOTE: SSH is enabled on supported switches by default.

10. To configure trap receiving for SNMPv2:
    a. Under **SNMP Configuration**, select **V2** as the SNMP type.
    b. Click **+** beside the **SNMP v2 Community String** box.
    The **SNMP v2 Community String** page opens.
    c. Enter the community string by which PowerFlex Manager receives traps from devices and by which it forwards traps to destinations.
    d. Click **Save**.

    NOTE: You can add more than one community string. For example, add more than one if the community string by which PowerFlex Manager receives traps differs from the community string by which it forwards traps to a remote destination.

11. To configure trap receiving for SNMPv3:
    a. Under **SNMP Configuration**, select **V3** as the SNMP type.
    b. Click **+** beside the **SNMP V3 User** box.
    The **SNMP V3 User** page opens.
    c. Enter the **Username**, which identifies the ID where traps are forwarded on the network management system.

       NOTE: The username must be at least 16 characters.

    d. Select a **Security Level** from the following:

    - **Minimal** - No additional information is required.
    - **Moderate** - Provide an **MD5 Authentication Password** which is at least eight characters.
    - **Maximum** - Provide an **MD5 Authentication Password** and a **DES Privacy Password** which is at least eight characters.

    You can add more than one user.

    NOTE: You can only select SNMPv3 for nodes. Select the **Auto configure nodes to send alerts to PowerFlex Manager** checkbox to discover iDRACs.

12. Click **Save**.

Create credentials for root and non-root users

Use this procedure to create credentials for root and non-root users in PowerFlex Manager.

To import and create the SSH keys for a PowerFlex node, switch, **OS Admin**, **OS User**, ensure you generate SSH key pairs of RSA type without passphrase.

You can now use a non-root user instead of the root user for PowerFlex system administration functions. This enhances security by disabling the root user during node discovery, operating system installation, and non-disruptive updates. The default non-root user name is **`pflex`**.

NOTE: The credential type **OS Admin** is used for root users, and **OS User** is used for non-root user. **OS Admin** and **OS User** credential types apply for the deployment of resource groups.

For non-root user authentication, after the deployment of the resource group, the SSH access for the user root is disabled, but the password is still available for to use the console for troubleshooting.

PowerFlex Manager allows you to specify a non-root user when you configure a template for a compute-only, storage-only deployment.

1. On the menu bar, click **Settings** > **Security**.
2. Click **Resource Credentials**. The Credentials Management page opens.
3. Click **Create**.
4. In the **Create Credentials** dialog box, from the **Credential Type** drop-down list, select one of the following resource types for which you want to create non-root credentials:
   - **Node**
   - **Switch**
   - **OS Admin**
   - **OS User**

   The **OS Admin** and **OS User** credential types apply to deployed items, not to PowerFlex Manager. If you are creating an OS user credential set for the management virtual machines on a PowerFlex management controller resource group, select **OS User**.
5. In the **Credential Name** field, enter the name to identify the credential.

   NOTE: If you are creating an **OS User** credential set for the management virtual machines on a PowerFlex management controller resource group, do the following:

   a. Enter **`MVM delladmin`** to identify the credential.
   b. In the **User Name** field, enter **`delladmin`**.
   c. Enter the delladmin account password in the **Password** and **Confirm Password** fields.
6. Click **Enable Key Pairs** to enable log in with SSH key pairs and perform the following:

Table 78. Key pairs options

To...	Do this...
Enable key pairs for the **Node** or **Switch credential**:	a. Click **Import SSH Key Pair**. **NOTE:** Manually generate the SSH keys pairs. b. Click **Choose File** and browse to the file that contains the private key. c. Type a name for the key pair. d. Click **Import**.
Create keys using PowerFlex Manager for the **OS Admin** or **OS User** credential and enable key pairs:	a. Click **Create a new key**. b. Click **Create & Download Key Pair**. c. On **Key Pair Name**, type the name for key pair. d. Click **Create**. e. Click **Download Public Key**.
To manually generate and import an existing key pairs for the **OS Admin** or **OS User** credential	a. Click **Import SSH Key Pair**.

Table 78. Key pairs options (continued)

To...	Do this...
	NOTE: Manually generate the SSH keys pairs. b. Click **Choose File** and browse to the file that contains the public and private key. c. Type a name for the key pair. d. Click **Import**.

If you enable SSH key pairs for a **Node** or **Switch credential** and use that credential for discovery, PowerFlex Manager uses public or private RSA key pairs to SSH into your node or switch securely, instead of using a user name and password.

If you enable SSH key pairs for an **OS User** or **OS Admin** credential and use that credential for a deployment, PowerFlex Manager uses RSA public or private key pairs for the deployment operations.

NOTE: PowerFlex Manager does not consume SSH keys for all component types. For example, if you enable SSH key pairs for an admin credential, the SSH keys are not used for the deployment of a CloudLink Center VM. Instead, the user name and password are used instead for all communication.

7. In the **User Name** field, enter the username for the credential.

   For **Nodes** (iDRAC), `root` is the only valid username for root-level credentials. For a non-root user name, enter the default non-root user name.

   For the **OS Admin** credential type, the **User Name** field is disabled because the user is assumed to be root. You must use the root user for new deployments.

   For the **OS User** credential type, enter the default non-root user name.

   For the embedded operating system, this user account must have SSH enabled and have sudo access. For VMware ESXi, the account must be configured with the administrator role on the local server permission setting, which should enable SSH and other tools like esxcli. You can add existing resource groups with a non-root user. The account on the PowerFlex storage-only nodes for the **OS User** credential type must have a /home directory and have the correct group permissions.
8. In the **Password** and the **Confirm Password** boxes, enter the password for the credential.

Modify credentials

You can modify your credentials to update outdated information, improve security, and ensure that your system can correctly authenticate you for the resources you need.

1. On the menu bar, click **Settings** and click **Security**.
2. Click **Resource Credentials**.
   The **Credentials Management** page opens.
3. Select a credential that you want to edit, and click **Modify**.
4. Modify the credential information in the **Modify Credentials** dialog box.
5. Click **Save**.

Remove credentials

You can remove your credentials to eliminate outdated or unnecessary access and maintain stronger security.

1. On the menu bar, click **Settings** > **Security**.
2. Click **Resource Credentials**.
   The **Credentials Management** page opens.
3. On the **Credentials Management** page, select the credential that you want to delete, and click **Remove**.
4. Click **OK** when the confirmation message appears.

Converting a private key from PuTTY format to OpenSSH format

Perform this procedure to convert a private key from PuTTY to OpenSSH format.

1. Navigate to the folder where PuTTY was installed.

2. Double-click `puttygen.exe` to open the **PuTTY Key Generator**.
3. Click **File** > **Load private key** on the menu bar.
4. Select the private key that you want to convert to OpenSSH format and click **Open**.
5. Click **Conversions** > **Export open SSH key** on the menu bar.
6. Click **Save** to save the key.
   The key should now be in OpenSSH format.
7. Copy the public key content that is displayed on the **Key** tab in the **PuTTY Key Generator**.
   You need this content so you can copy it to the nodes.

Configuring SSH key pairs

Perform this procedure to configure SSH key pairs for nodes and switches.

1. To configure an SSH key pair for a software-only node, run these commands:

```
# below command will generate SSH Key pair in /tmp directory
ssh-keygen -m PEM -b 2048 -t rsa -f /tmp/sshkey-key1 -q -N ""

# To copy public key to software only node
ssh-copy-id -i /tmp/sshkey-key1.pub root@100.68.105.233

# Keys can be copied for both root and non-root users
```

2. To configure an SSH key pair for a Cisco Nexus switch, run these commands to add the public key:

```
username pfxmsvc password <password> role priv-15

username pfxmsvc sshkey ssh-rsa
AAAAB3NzaC1yc2EAAAADAQABAAABAQDWxvVQ9PgfFI3NMsg3TsR9G89ykZ3UEyciz47hcXH/
YXNakva5PV5l7ubHY4nUxKj3zpVHXg/Bq9zRK5lQFs4KGC5eaYB9nm2xEwhuhpvsCvROo8SzjjTNRcpok2EMD/
3VhuZAWdaXITSukq2D1Ek5W7qeE0LiLSC+2s0u6iPTU5KHs3mrELbRlxpVxnHQtrgUZnsLvkxwzuEnBKWNIEZ0
6HoDLP1vBCaYJslE5iDseUCZstWvFbvkaeE4FOrqohllAhVGVH19B6Q38ijVjLAwjXX03a6IQqeWj+JkpHpB61
vMsAqqje4pm9VFzAY2NAcvoBo7Oab+52qA1vJTCbG7
username pfxmsvc passphrase  lifetime 99999 warntime 14 gracetime 3
```

3. To configure an SSH key pair for an OS10 switch, run these commands to add the public key:

```
username pfxmsvc password <password> role sysadmin priv-lvl 15

username pfxmsvc sshkey "ssh-rsa
AAAAB3NzaC1yc2EAAAADAQABAAABAQDWxvVQ9PgfFI3NMsg3TsR9G89ykZ3UEyciz47hcXH/
YXNakva5PV5l7ubHY4nUxKj3zpVHXg/Bq9zRK5lQFs4KGC5eaYB9nm2xEwhuhpvsCvROo8SzjjTNRcpok2EMD/
3VhuZAWdaXITSukq2D1Ek5W7qeE0LiLSC+2s0u6iPTU5KHs3mrELbRlxpVxnHQtrgUZnsLvkxwzuEnBKWNIEZ0
6HoDLP1vBCaYJslE5iDseUCZstWvFbvkaeE4FOrqohllAhVGVH19B6Q38ijVjLAwjXX03a6IQqeWj+JkpHpB61
vMsAqqje4pm9VFzAY2NAcvoBo7Oab+52qA1vJTCbG7"
```

   Note that the quotes must be used as shown in the command line above.

4. To configure an SSH key pair for iDRAC, perform these steps:
   a. Connect to iDRAC.
   b. Choose **iDRAC Settings** > **Users** > **Edit/Add User** > **SSH Key Configuration** > **SSH Key (Edit)**.
   c. Add the public key.

Serviceability

On the **Serviceability** page, you can generate a troubleshooting bundle and also perform a backup of the appliance. To restore from a backup, you need to run a script outside of PowerFlex Manager. The user interface does not support the ability to restore from a backup.

Generating a troubleshooting bundle

You can generate a troubleshooting bundle which is a compressed file that contains logging information for PowerFlex Manager managed components.

If necessary, download the bundle and send it to Dell Technologies Support for issue debugging.

The troubleshooting bundle includes the following logs:

- ASM deployer
- iDRAC life cycle
- Dell PowerSwitch switch
- Cisco Nexus switch
- VMware ESXi
- CloudLink Center
- Standard output logs from all pods
- Kubernetes logs about pods, services, deployments, secrets, drivers, and volumes
- PowerFlex Block logs
- PowerFlex data server and device gateway target logs

NOTE: You can also generate troubleshooting bundles from the **Resource Groups** page.

1. On the menu bar, click **Settings** > **Serviceability**.
2. Click **Generate Troubleshooting Bundle**.
3. Select **Include PowerFlex Block Core Dump logs** to include core dump logs.
4. Select a **PowerFlex Log Level**:
   - Default Node Logs
   - Default Node Logs plus additional MDM information
   - Latest Logs only (Most recent copy of all logs)
5. Select a **Node Selection**:
   - Logs from all nodes
   - Select Specific Nodes

   If you select the **Select Specific Nodes** option and select the number of nodes for which you want to generate the log, the **View/Select Nodes** button is displayed. Click the button to view the list of nodes in the **Node Selection** window. Select the required nodes from the **Available Nodes** list and click **>>** to view them in the **Selected Nodes** list. Click **Save** to return to the **Generate Troubleshooting bundle** page.

   For the **Select Specific Nodes** option **Generate** is enabled only if a node is selected in the **Node Selection** window.
6. Choose a **Delivery Option**:

Option	Description
Send to Configured Dell Technologies connectivity services	If you are using Dell Technologies connectivity services, **Send to Configured Dell Technologies connectivity services** is selected by default. Leave this default setting. If Dell Technologies connectivity services is not configured, this option is disabled.
Download to Share	Provide a path using the following format with credentials: `CIFS:\\1.1.1.1\uploadDirectory`. Enter a username and password, and click **Test Connection** to verify the connection to the CIFS share before generating the bundle.
Download Locally	Download the troubleshooting bundle to a local file: `\\IP Address\Any folder`

7. Click **Generate**.

Sometimes, the troubleshooting bundle does not include log information for all the nodes. The log collection may appear to succeed, but the log for one or more of the nodes may be missing. You may see an error message in the scaleio.trace.log file that says `Could not run get_info script`. If you see this message, you must generate the troubleshooting bundle again to include information for all the logs.

Pausing an inventory job

An inventory job runs daily at midnight. When a PowerFlex management platform or a resource group/devices upgrade is in progress, both jobs compete for resources. The scheduled (nightly) inventory job is automatically disabled at the start of any upgrade and resumes immediately once the upgrade completes minimizing the load.

In addition to this automatic action on upgrade, you can manually pause scheduled inventory jobs for up to 24 hours using the PowerFlex Manager user interface. This manual block does not effect on-demand inventory actions—you can still run inventory for individual devices through **Lifecycle** > **Resources**.

The status of the inventory job displays on the upper right hand of the screen of the **Resources** page.

1. On the menu bar, click **Settings** > **Serviceability**.
2. Click **Inventory Job**.
3. Choose one of the following options:
   - If you want the inventory job to run at the next scheduled time, set the switch to **On**.
   - If you want to pause the inventory job, set the switch to **Off**.

   The name of the user who set the switch, the time it was set, and the time it expires is displayed under the switch.

Back up and restore

Creating a backup saves all user-created data to a remote share from which it can be restored. To restore from a backup, you need to run a script outside of PowerFlex Manager. The user interface does not support the ability to restore from a backup.

Perform frequent backups to guard against data loss and corruption. The best practice is to take a snapshot of PowerFlex Manager every time you perform a restore.

The **Backup** page displays information about the last backup operation that was performed on PowerFlex Manager. The information provided applies to both manual and automatically scheduled backups and includes the following:

- Last backup date
- Last backup status
- Back up directory path to a CIFS share
- Back up directory username

The **Backup** page also displays information about the status of automatically scheduled backups (enabled or disabled).

On this page, you can:

- Manually start an immediate backup
- Edit general backup settings
- Edit automatically scheduled backup settings

After performing a backup operation, you need to run a script outside of PowerFlex Manager to restore from the backup. The user interface does not support the ability to restore from a backup.

Editing backup settings and details

You can change the location where backup files are saved or the password that is required to access a backup file.

1. On the menu bar, click **Settings** and then click **Serviceability**.
2. Click **Backup**.
   The **Backup** page opens.
3. Click **Backup Settings**.
4. Indicate the network share location where the backup file is saved, enter a backup directory path in the **Backup Directory Path** box.

   Use the following format:

   - CIFS—`\\host\share`

5. If the username and password are required to access the network share, enter a username and password in the **Backup Directory User Name** and **Backup Directory Password** boxes.
6. Click **Test Connection** to confirm that the backup settings you provided are correct.
7. To open the backup file, enter a password in the **Encryption Password** box.
8. To verify the encryption password, enter the password in the **Confirm Encryption Password** box.
9. To schedule automatic backups, next to **Scheduled Backups**, select **Enabled**. To discontinue automatically scheduled backups, deselect **Enabled**.
10. To specify the days on which backup must occur, select the days under **Frequency**.
11. From the **Run Time** drop-down list, select the time.
12. Click **Save**.

Backing up

In addition to automatically scheduled backups, you can manually run an immediate backup.

1. On the menu bar, click **Settings** > **Serviceability**.
2. Click **Backup**.
   The **Backup** page opens.
3. Click **Backup Now**.
4. Select one of the following options:
   - To use the general settings that are applied to all backup files, select **Use Backup Directory Path and Encryption Password from Settings and Details**.
   - To use custom settings:
     a. In the **Backup Directory Path** box, enter a file path where the backup file is saved. Use this format:
        - CIFS—`\\host\share`

     b. Optionally, enter a username and password in the **Backup Directory User Name** and **Backup Directory Password** boxes, if they are required to access the location you provided.
     c. Click **Test Connection** to confirm that the backup settings you provided are correct.
     d. In the **Encryption Password** box, enter a password that is required to open the backup file, and verify the encryption password by entering the password in the **Confirm Encryption Password** box.
5. Click **Backup Now**.

Software upgrade

When a new version of the management software becomes available, you can upgrade to that version from the **Software Upgrade** page.

Upgrading PowerFlex Manager from a local repository path

You can upgrade the management software from a TGZ file on a local repository path.

Before you begin, ensure that you:

- Take a backup of PowerFlex Manager settings.
- Upload the latest compatibility matrix file to ensure that PowerFlex Manager has access to the latest upgrade information.

When you attempt an upgrade, PowerFlex Manager warns you if the current version of the software is incompatible with the target version, or if any of the RCM or IC versions that are currently loaded on the management software are incompatible with the target compliance versions. To determine which paths are valid and which are not, PowerFlex Manager uses information that is provided in the compatibility matrix file. The compatibility matrix file maps all the known valid and invalid paths for all previous releases of the software. Before proceeding with an upgrade, PowerFlex Manager notifies you if any of the following situations is detected:

- No upgrade is required. This situation occurs when the current and target versions are the same. You can still force an upgrade, if needed.
- The upgrade path that you are attempting to perform is a valid path. The source version of PowerFlex Manager is compatible with the target version and the source version of the RCM or IC is also compatible with the target version.

- The source version of PowerFlex Manager is compatible with the target version. However, the source version of the RCM or IC is not compatible with the target version. Your resource groups may go into life cycle mode until you perform an RCM or IC upgrade. PowerFlex Manager puts a resource group in life cycle mode if the resource group's RCM or IC is not within a year of the target PowerFlex Manager launch. You can mitigate a resource group that is in life cycle mode by upgrading to a compliance version that is within a year of the current PowerFlex Manager launch.
- The upgrade path that you are attempting to perform is not supported.

The compatibility matrix file is required for all upgrades. When you first install PowerFlex Manager, the software does not have the compatibility matrix file, so you must upload the file before performing any upgrades.

1. If you are using PowerFlex rack, log in to Dell Technologies Download Center. Then, go to the section for PowerFlex Manager.
2. If you are using PowerFlex appliance, log in to Dell Technologies Support. Then, enter your service tag to be taken to available downloads.
3. Download the PowerFlex management platform .TGZ file.
4. Copy the PowerFlex management platform .TGZ file to a share drive location.
5. On the menu bar, click **Settings**, and then click **Software Upgrade**.
6. Click **Management Software Upgrade**.

   NOTE: Dell Technologies recommends using the PowerFlex System Configuration Reporter (SCR) to validate that configurations are ready to upgrade. PowerFlex SCR is a standalone application that is designed to provide comprehensive configuration reporting, health assessment, configuration checks, readiness/upgrade assessments checks, and log collection for PowerFlex systems.

7. On the **Management Software Upgrade** page, you can see the **Current Management Software Version** and check the **Available Management Software Version** field to see if a newer version of PowerFlex Manager is available.
8. Click **Edit Settings**.
   a. To upgrade from a specific version on a local repository path, select **Update Software from local repository path**, and enter the path in the **Repository Path** box.
   b. Enter the username and password.
   c. Click **Save**.
9. Click **Upgrade Now**.
   After you click **Upgrade Now**, a dialog box displays with a warning message indicating whether the upgrade path is valid.
10. Under **Management Credentials**, optionally override the password for the nodes hosting PowerFlex Manager.

    The upgrade requires access to the nodes hosting PowerFlex Manager. Provide a single user that has superuser privileges for all nodes. Enter the password for each of the nodes.
11. Optionally, click **Enable Key Pairs** to enable login with SSH key pairs:

    To enable key pairs for the **Node** credential type:

    a. Import an existing key:
       i. Click **Import SSH Key Pair**.
       ii. Click **Choose File** and browse to the file that contains your public and private keys, and select the private key.
       iii. Type a name for the key pair.
       iv. Click **Import**.
12. Type **`UPDATE POWERFLEX MANAGER`** if you want to proceed with the update. If you want to perform an update even if none is required, type **`FORCE UPDATE`**.
13. Click **Upgrade** to upgrade PowerFlex Manager.
    The update process displays messages indicating the progress of the update. Depending on which services were upgraded, you may be required to log in once again, after the upgrade process is complete.

After upgrading to PowerFlex Manager, you may notice that resource groups that are tied to the PowerFlex gateway are no longer in compliance. To bring the resource groups into compliance, you may need to upgrade them. In this case, you will need to change to a later RCM or IC and upgrade the resource groups.

If you do not want to upgrade the resource groups, you do not need to perform the upgrade.

Editing the upgrade settings

You can edit the upgrade settings to control where the system retrieves software updates from and ensure that the upgrade process is secure.

1. On the menu bar, click **Settings** and then click **Software Upgrade**.
2. Click **Management Software Upgrade**.
3. Click **Edit Settings**.
   a. To update to the latest version using Dell Technologies connectivity services, select **Update Software from configured Dell Technologies connectivity services**.
   b. To upgrade from a specific version on a local repository path, select **Update Software from local repository path**, and enter the path in the **Repository Path** box.

      The connection to the path that you provided is not secure. You must choose one of the following options:

      - To allow the insecure connection, select the **Allow an insecure connection to the specified path** checkbox.
      - To make the connection to the specified path secure, go to **Security** > **SSL Trusted Certificates** and click **Add** to upload the SSL certificate. Then, retry the operation.

      You can upgrade from a TGZ file on a network share. This option provides a way to update the software in a dark site. Provide a path using one of the following formats:

      Table 79. Path for network shares

For:	Use this path:
CIFS	`\\IP Address\Any folder\file-name.tgz`
FTP	`ftp://IP Address/Any folder/file-name.tgz`
HTTP	`http://IP Address/Any folder/file-name.tgz`
HTTPS	`http://IP Address/Any folder/file-name.tgz`

   c. Optionally, provide a **User Name** and **Password**.

      The username and password are only used with CIFS. All other local repository paths must specify public share locations.

   d. Click **Save**.
4. On the **Management Software Upgrade** page, you can view the updated settings for **Current Management Software Version**, **Available Management Software Version**, and **Repository Path**.
5. To perform the upgrade, click **Upgrade Now**. A message displays confirming that you want to update the software.
6. Click **Upgrade Software**.
7. Click **Yes**.
8. Depending on which services were upgraded, you may be required to log in again once the upgrade process is complete.

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Configuring system settings (mirroring)

This section provides information about configuring settings for PowerFlex Manager.

Only a user with the administrator role can configure certain settings.

Go to **Settings** to complete the following tasks:

- Manage local users, LDAP users, and directory services
- Upgrade a single firmware or software component
- Manage compliance versions, operating system images, and compliance management versions
- Define networks and specify the port numbers that PowerFlex Manager should use to verify IP addresses
- Configure settings for events and alerts, including notification policies, inputs, destinations, and alert severities
- Upload a license for PowerFlex, and other software licenses
- Configure security settings, including SSL trusted certificates, appliance SSL certificates, and resource credentials
- Perform serviceability operations, such as generating a trouble-shooting bundle and performing a backup
- Upgrade the management software

User management

The **User Management** page allows you to manage local users, LDAP users, and directory services.

Under **Settings** > **User Management**, you can find three pages:

- Local Users
- LDAP Users
- Directory Services

User roles (mirroring)

User roles control the activities that different types of users can perform when using PowerFlex Manager.

Ensure that you configure the active directory before assigning roles. The roles that can be assigned to local users and LDAP users are identical. Each user can only be assigned one role. If an LDAP user is assigned directly to a user role and also to a group role, the LDAP user is provided with permissions of both roles.

NOTE: User definitions are not imported from earlier versions of PowerFlex and must be configured again.

The following table summarizes the activities that can be performed for each user role:

Table 80. User roles and activities

Role	Description	Activities
SuperUser	A SuperUser can perform all system operations.	• Manage storage resources • Manage life cycle operations, resource groups, templates, deployment, backend operations • Manage replication operations, peer systems, replication consistent groups (RCGs) • Manage snapshots, snapshot policies • Manage users, certificates • Replace drives • Hardware operations • View storage configurations, resource details • View platform configuration, resource details • System monitoring (events, alerts)

Table 80. User roles and activities (continued)

Role	Description	Activities
		• Perform serviceability operations • Update system settings
SystemAdmin	A SystemAdmin can perform all operations, except for user management and security ones.	• Manage storage resources • Manage life cycle operations, resource groups, templates, deployment, backend operations • Manage replication operations, peer systems, replication consistent groups (RCGs) • Manage snapshots, snapshot policies • Replace drives • Hardware operations • View storage configurations, resource details • View platform configuration, resource details • System monitoring (events, alerts) • Perform serviceability operations • Update system settings
StorageAdmin	A StorageAdmin can perform all storage-related front-end operations including block systems, such as create volume. **NOTE:** Operations such as create storage pool cannot be performed by StorageAdmin, but can be performed by the LifecycleAdmin role.	• Manage storage resources • Manage replication operations, peer systems, replication consistent groups (RCGs) • Manage snapshots, snapshot policies • Replace drives • Hardware operations • View storage configurations, resource details • View platform configuration, resource details • System monitoring (events, alerts)
LifecycleAdmin	A LifecycleAdmin can manage the life cycle of hardware and PowerFlex systems.	• Manage life cycle operations, resource groups, templates, deployment, backend operations • Replace drives • Hardware operations • View resource groups and templates • System monitoring (events, alerts)
ReplicationManager	The ReplicationManager is a subset of the StorageAdmin role, for work on existing systems for setup and management of replication and snapshots.	• Manage replication operations, peer systems, replication consistent groups (RCGs) • Manage snapshots, snapshot policies • View storage configurations, resource details (volume, snapshot, replication views) • System monitoring (events, alerts)
SnapshotManager	The SnapshotManager is a subset of StorageAdmin, working only on existing systems. This role includes all operations that are required to set up and manage snapshots.	• Manage snapshots, snapshot policies • View storage configurations, resource details • System monitoring (events, alerts)
SecurityAdmin	The SecurityAdmin manages PowerFlex role-based access control (RBAC), and LDAP user federation. It includes all security aspects of the system.	• Manage users, certificates • System monitoring (events, alerts)
DriveReplacer	This is a subset of the Technician role. The DriveReplacer can perform only the operations that are required for a drive replacement.	• Replace drives • System monitoring (events, alerts)
Technician	The Technician can perform all hardware FRU operations on the system, including entering a node into maintenance mode.	• Replace drives • Hardware operations • System monitoring (events, alerts) • Perform serviceability operations

Table 80. User roles and activities (continued)

Role	Description	Activities
Monitor	The Monitor role has read-only access to the system, including topology, alerts, events, and metrics.	• View storage configurations, resource details • View platform configuration, resource details • System monitoring (events, alerts)
Support	The Support role is a special type of SystemAdmin (all activities except for user/security management operations) to be used only by Dell Technologies support staff and developers. This user role has access to undocumented, special operations and options for common operations, required only for support purposes. **NOTE:** This special role should be used only by Dell Technologies support. It opens special, often dangerous, commands for advanced troubleshooting.	• Manage storage resources • Manage life cycle operations, resource groups, templates, deployment, backend operations • Manage replication operations, peer systems, replication consistent groups (RCGs) • Manage snapshots, snapshot policies • Replace drives • Hardware operations • View storage configurations, resource details • View platform configuration, resource details • System monitoring (events, alerts) • Perform serviceability operations • Special Dell Technologies Support operations

Mapping PowerFlex 4.0.x or later roles to legacy roles (mirroring)

For reference, the tables below map user and group roles from earlier releases to the roles used in the current release.

Role mapping per legacy user interface

Table 81. Role mapping

Legacy role before version PowerFlex 4.0	New role after PowerFlex 4.0.x and later
PowerFlex Monitor	Monitor
PowerFlex Back-end configurator	LifecycleAdmin
PowerFlex frontend configurator	StorageAdmin
PowerFlex Configurator	SystemAdmin
PowerFlex Security	SecurityAdmin
PowerFlex Administrator	StorageAdmin
PowerFlex local Super User	SuperUser
PowerFlex technician commands	Support
PowerFlex Manager Administrator	SuperUser
PowerFlex Manager read-only	Monitor
PowerFlex Manager Standard owner	LifecycleAdmin
PowerFlex Manager Standard member	LifecycleAdmin
PowerFlex Manager Operator	DriveReplacer

Legacy LDAP users and groups

LDAP user role permissions were modified in PowerFlex 4.0. For reference purposes, the following table shows the mapping of legacy users and groups to the roles now supported by PowerFlex.

Before PowerFlex 4.0, PowerFlex mapped LDAP users and groups to roles. This mapping allowed assignment of multiple roles to a single group. Assignment of multiple roles to a group is no longer supported.

All legacy roles have a proximate hierarchy. The group's single role is the highest ranking role.

Table 82. LDAP hierarchy

Ranking	Legacy PowerFlex LDAP role earlier than PowerFlex 4.0	PowerFlex 4.0 or later group role without security officer permissions	PowerFlex 4.0 or later group role with security officer permissions
1	SuperUser	N/A (local user)	N/A (local user)
2	Administrator	Admin	SuperUser
3	Configurator	Admin	SuperUser
4	Frontend configuration and backend configuration	Admin	SuperUser
5	Frontend configuration or backend configuration	StorageAdmin or LifecycleAdmin	SuperUser
6	Monitor	Monitor	SecurityAdmin
Not ranked	Security Officer (SE)	SecurityAdmin	SecurityAdmin

Local users

You can create and manage local users within PowerFlex Manager.

Creating a user

Perform this task to create a local user and assign a role to that user.

1. On the menu bar, click **Settings** and click **User Management**.
2. Click **Local Users**.
3. On the **Local Users** page, click **Create**.
4. Enter a unique **User Name** to identify the user account.
5. Enter the **First Name** and **Last Name** of the user.
6. Enter the **Email** address.
7. Enter a **New Password** that a user enters to access PowerFlex Manager. Confirm the password in the **Verify Password** field.

   The password must be at least 8 characters long and contain one lowercase letter, one uppercase letter, one number, and one special character. Passwords cannot contain a username or email address.
8. In the **User Role** box, select a user role. Options include:
   - SuperUser
   - SystemAdmin
   - StorageAdmin
   - LifecycleAdmin
   - SnapshotManager
   - SecurityAdmin
   - DriveReplacer
   - Technician
   - Monitor
   - Support
9. Select **Enable User** to create the account with an `Enabled` status, or clear this option to create the account with a `Disabled` status.
10. Click **Submit** and click **Dismiss**.

PowerFlex Manager creates the new user with the specified password. The first time you login with the new user and password, PowerFlex Manager asks you to change the password.

Modifying a user

Perform this task to edit a PowerFlex Manager user profile.

1. On the menu bar, click **Settings** and click **User Management**.
2. Click **Local Users**.
3. On the **Local Users** page, select the user account that you want to edit.
4. Click **Modify**. For security purpose, confirm your password before editing the user.
5. You can modify the following user account information from this window:
   - **First Name**
   - **Last Name**
   - **Username**
   - **User Role**
   - **Email**
   - **Enable User**

   If you select the **Enable user** checkbox, the user can log in to PowerFlex Manager. If you disable the checkbox, the user cannot log in.
6. Click **Submit** and click **Dismiss**.

Deleting a user

Perform this procedure to remove an existing local user.

1. On the menu bar, click **Settings** and click **User Management**.
2. Click **Local Users**.
3. On the **Local Users** page, select one or more user accounts to delete.
4. Click **Delete**.

   Click **Apply** in the warning message to delete the user. Click **Dismiss**.

Resetting the password for a user

You can reset the password for a local user in PowerFlex Manager.

1. On the menu bar, click **Settings** and click **User Management**.
2. Click **Local Users**.
3. On the **Local Users** page, select one user account to reset the password.
4. Enter the **Current Password**.

   If you are changing a password for another user entering the current password is not required.
5. Click **Reset Password**.
6. Enter **New Password** and **Verify Password**.
7. If you want to set the password as a temporary one, check **Temporary Password**.
8. Click **Apply** and click **Dismiss**.

Remote users and groups

You can add and modify remote users and groups in PowerFlex Manager. Remote users are users tied to an LDAP directory service or to an SSO identity provider.

For LDAP configurations, the LDAP realm name must not exceed the 155 character limit defined by Microsoft.

Add remote users or groups

Add remote users or groups to PowerFlex, and assign roles to them. Remote users are users tied to an LDAP directory service or to an SSO identity provider.

When you add users and groups, you can also assign roles to the users and groups. The roles control access permissions for the corresponding LDAP user or group. The users and groups must also be configured in the LDAP directory service or SSO identity provider. On the PowerFlex side, these users and groups are mapped to PowerFlex roles.

1. On the menu bar, click **Settings**.
2. In the left pane, click **User Management**, then in the right pane, click **Remote Users/Groups**.
3. Click **Add**.
4. In the **Add Remote User/Group** dialog box, select a **Type** option.
   - `User`—a user definition will be configured for an individual user.
   - `Group`—a group definition will be configured for a specific group.
5. Select the LDAP directory service or SSO identity provider in the **Provider** drop-down.
6. In the **User Name** box, enter the user or group name.
7. In the **User Role** box, select the role to be assigned to the user or group. Options include:
   - SuperUser
   - SystemAdmin
   - StorageAdmin
   - LifecycleAdmin
   - SnapshotManager
   - SecurityAdmin
   - DriveReplacer
   - Technician
   - Monitor
   - Support
8. Click **Apply**.

Modify a remote user or group

You can modify the user role of a user or group tied to an LDAP directory service or an SSO identity provider. User roles control access permissions to the corresponding user or group.

1. On the menu bar, click **Settings** > **Remote Users**.
2. Click **Modify**.
3. In the **Modify Remote User/Group** dialog box, change the user role by selecting one of the **User Role** options:
   - SuperUser
   - SystemAdmin
   - StorageAdmin
   - LifecycleAdmin
   - SnapshotManager
   - SecurityAdmin
   - DriveReplacer
   - Technician
   - Monitor
   - Support
4. Click **Apply**.

Directory services

You can create a directory service that PowerFlex Manager can access to authenticate users.

An Active Directory or Open LDAP user is authenticated against the specific directory domain to which a user belongs.

The **Directory Services** page displays the following information about PowerFlex Manager active directories:

- LDAP configuration

- User search settings
- Group search settings

From this page, you can:

- Add a directory service (only available when no service is defined in the system)
- Modify a directory service
- Remove a directory service

Add a directory service

Add a directory service for user authentication.

The hostname character limitation is 15 characters.

1. On the menu bar, click **Settings**.
2. In the left pane, click **User Management**, then in the right pane, click **Directory services**.
3. Click **Add**.
4. For **LDAP Configuration**, configure the following:
   a. In the **Address** box, enter the address of the authentication server.
      The address must be specified in URL-like format:
      - Enter **`ldap://HOSTNAME or IP ADDRESS`** for a plaintext LDAP connection.
      - Enter **`ldaps://HOSTNAME or IP ADDRESS`** for a secure LDAP connection.

      For example: **`ldap://100.68.68.1`**
   b. In the **Bind DN** box, enter the bind distinguished name attributes.
      The Bind Distinguished Name (DN) uniquely identifies an entry and its position in the hierarchy of entries contained in a directory server.
      The user account entered here is the user that PowerFlex Manager uses to connect to your directory service.
      For example, for the administrator account in the **Users** folder of **`asm.delllabs.com`**, enter the following: **`CN=administrator,CN=Users,DC=asm,DC=delllabs,DC=net`**.
   c. In the **Bind DN Password** box, enter the Bind DN password.
      This is the password used to access the LDAP server.
   d. In the **Timeout** box, enter a value in milliseconds.
      For example: **`1000`**
5. For **User Search Settings**, configure the following:
   a. In the **Username LDAP Attribute** box, enter the name of an LDAP attribute that is mapped as the username. For many LDAP servers, it can be **`uid`**. For Active Directory, it can be **`sAMAccountName`** or **`cn`**. The attribute should be filled in for all LDAP users you want to import from LDAP to PowerFlex.
      NOTE: If you are using Microsoft Windows Active Directory (AD) as an LDAP server, the hostname cannot exceed 15 characters. A hostname that exceeds the 15-character limit may cause issues during deployment and integration.
      For example: **`sAMAccountName`**
   b. In the **ID Attribute** box, enter the ID attribute for users.
      For example: **`sAMAccountName`**
   c. In the **Object Class** box, enter an object class.
      For example: **`top,person,organizationalPerson,user`**
   d. In the **Search Path** box, enter the search path.
      The search path is used to identify and retrieve entries in the directory information tree that match a set of criteria.
      For example: **`CN=Users,DC=asm,DC=delllabs,DC=net`**
6. For **Group Search Settings**, configure the following:
   a. In the **Group Member Attribute** box, enter a group member name.
      For example: **`member`**
   b. In the **Group ID Attribute** box, enter the group ID.
      For example: **`name`**
   c. In the **Group Object Class** box, enter the group object class.
      For example: **`group`**
   d. In the **Group Search Path** box, enter the group search path.

The search path is used to identify and retrieve entries in the directory information tree that match a set of criteria for groups.

For example: **CN=Users,DC=asm,DC=delllabs,DC=net**

7. Click **Test Connection**.

   If the test is successful, the **Submit** button will become active. If the test fails, you will not be able to proceed until you fix the connectivity issue.

8. When you have finished making your changes, click **Submit**.

Modify a directory service

The **Modify** option allows you to edit the existing directory service settings.

1. On the menu bar, click **Settings**.
2. In the left pane, click **User Management**, then in the right pane, click **Directory services**.
3. Click **Modify**.
4. In the **LDAP Settings** dialog box, edit the desired fields.

   Note that the **Bind DN Password** must be reentered.

5. When you have finished making your changes, click **Test Connection**.

   If the test is successful, the **Submit** button becomes active. If the test fails, you will not be able to proceed until you fix the connectivity issue.

6. Click **Submit**.

Remove a directory service

The **Remove** option allows you to remove the directory service configuration from PowerFlex.

1. On the menu bar, click **Settings**.
2. In the left pane, click **User Management**, then in the right pane, click **Directory services**.
3. Click **Remove**.
4. In the warning dialog box, click **Submit**.

Single sign-on identity provider configuration

You can configure single sign-on (SSO) authentication through an identity provider (IdP) using the SAML 2.0 and OIDC protocols. This configuration makes it possible for users to take advantage of SSO capabilities.

You can add up to three IdPs for SSO authentication through PowerFlex Manager.

The supported IdPs and protocols are described in the following table:

Table 83. Supported IdPs and protocols

IdP	SAML	OIDC
Microsoft Active Directory Federation Services	Yes	No
Microsoft Azure	Yes	Yes
Keycloak	No	Yes
Okta	Yes	Yes
PingOne	Yes	No

OIDC instructions

This guide describes how to use the SAML 2.0 protocol. For instructions for using the OIDC protocol, see the Dell Knowledge Base article PowerFlex: Setting up PowerFlex single sign-on with Microsoft Azure using OpenID Connect (OIDC).

Add an identity provider

Add an SSO Identity Provider (IdP) for PowerFlex Manager to enable users to take advantage of single sign-on (SSO) capabilities through other applications.

1. On the menu bar, click **Settings**.
2. In the left pane, click **User Management**, then in the right pane, click **SSO Identity Provider (IdP) Configuration**.
3. Click **Add**.
   The **Add Identity Provider (IdP)** window opens.
4. Under **Name**, provide a name for the identity provider and click **Next**.
5. Under **Service Provider**, choose a method to download the service provider metadata for PowerFlex:
   - To download the metadata as a file, select **Download File** and click **Download SP Metadata (XML)**.
   - To copy the metadata, click **Manual Copy** and click **Copy** for each piece of metadata.
6. Configure PowerFlex as the service provider within the IdP.

   NOTE: To configure PowerFlex as the service provider within the IdP, you must log in to the portal of the IdP.

   To configure PowerFlex as the service provider within Active Directory Federation Services, see *Set up PowerFlex single sign-on with Microsoft Active Directory Federation Services* in the *PowerFlex and PowerFlex Manager Software Administration Guide*.

   To configure PowerFlex as the service provider within Microsoft Azure, see *Set up PowerFlex single sign-on with Microsoft Azure* in the *PowerFlex and PowerFlex Manager Software Administration Guide*.
7. Under **Service Provider**, select the **I have configured PowerFlex as a SP in my IdP using the metadata above** checkbox and click **Next**.
8. Under **IdP Setup**, upload the identity provider metadata so that PowerFlex can establish a connection to the identity provider.
   a. To upload the metadata as a file, select **Upload File** and specify the file location.
   b. To retrieve the metadata from a URL, select **URL** and specify the following URL: `https://hostname/FederationMetadata/2007-06/FederationMetadata.xml`

      The URL is always the same, except for the hostname, which you must specify for your environment.
   c. Click **Next**.
9. Under **Claims Mapping**, review the attribute mappings that are imported from the identity provider.
   a. Check the attribute mappings to ensure that they are correct.
   b. Click **Next**.
10. Under **Summary**, review the details and click **Finish**.

After you add an identity provider, PowerFlex Manager adds it to the list of identity providers on the **SSO Identity Provider (IdP) Configuration** page.

In addition, PowerFlex Manager updates the login page to show a login button with the new identity provider. You can see this button the next time you log in to PowerFlex Manager.

Set up PowerFlex single sign-on with Microsoft Active Directory Federation Services

To set up PowerFlex single sign-on with Microsoft Active Directory Federation Services (ADFS), you must first add ADFS as an IdP through PowerFlex Manager.

You must then download the service provider metadata for PowerFlex. After you complete this step, log in to the ADFS portal, configure PowerFlex as the service provider, and choose an access control policy. You must then return to PowerFlex Manager to complete the setup of single sign-on.

Add Active Directory Federation Services as an identity provider

Add an SSO Identity Provider (IdP) for PowerFlex Manager to enable users to take advantage of single sign-on (SSO) capabilities through other applications.

1. On the menu bar, click **Settings**.
2. In the left pane, click **User Management**, then in the right pane, click **SSO Identity Provider (IdP) Configuration**.

3. Click **Add**.
   The **Add Identity Provider (IdP)** window opens.
4. Under **Name**, provide a name for the identity provider and click **Next**.
5. Under **Service Provider**, choose a method to download the service provider metadata for PowerFlex:
   - To download the metadata as a file, select **Download File** and click **Download SP Metadata (XML)**.
   - To copy the metadata, click **Manual Copy** and click **Copy** for each piece of metadata.

Configure PowerFlex as a service provider for Active Directory Federation Services

To configure PowerFlex as a service provider for Active Directory Federation Services, upload the root CA certificate and log in with appropriate permissions. Then, establish trust by adding a relying party, configuring claim rules, importing metadata, and setting access control policies.

You must upload the root CA certificate for ADFS.

1. Log in to the Active Directory Federation Services portal with the security or administrator role.
2. Under **ADFS**, go to the **Relying Party Trusts** folder.
3. Under **Actions**, click **Add Relying Party Trust...**.
4. On the **Welcome** screen, select **Claims aware** and click **Start**.
5. On the **Select Data Source** screen, select **Import data about the relying party from a file**.
6. In the **Federation metadata file location** field, specify the location of the downloaded service provider metadata file and click **Next**.

   NOTE: If you copied the metadata instead of downloading the file, select **Enter data about the relying party manually** and input the copied metadata.

7. On the **Specify Display Name** screen, enter the name that you would like to use for the service provider and click **Next**.
8. On the **Choose Access Control Policy** screen, choose a policy and click **Next**.
9. On the **Ready to Add Trust** screen, click **Next**.
10. On the **Finish** screen, select **Configure claims issuance policy for this application** and click **Close**.
11. On the **Relying Party Trusts** screen, under **Actions** select the display name for the newly created service provider, and click **Edit Claim Issuance Policy...**.
12. Click **Add Rule...** to add the following LDAP attribute rule:
    a. For the **Claim rule template**, select **Send LDAP Attributes as Claims**.
    b. For the **Claim rule name**, enter **`LDAP attributes`**.
    c. For the **Attribute store**, select **Active Directory**.
    d. For the Mapping of **LDAP attributes to outgoing claim types**, select the following attributes:

Table 84. LDAP attributes and outgoing claim types

LDAP attribute	Outgoing claim type
Email-Addresses	Email Address
SAM-Account-Name	Name ID
Surname	Surname
Given-Name	Given Name

    e. Click **Finish**.
13. Click **Add Rule...** to add the following custom rule:
    a. For the **Claim rule template**, select **Send Claims Using a Custom Rule**.
    b. For the **Claim rule name**, enter **`Get groups`**.
    c. For the **Custom rule**, paste in the following string:

```
c:[Type == "http://schemas.microsoft.com/ws/2008/06/identity/claims/
windowsaccountname", Issuer == "AD AUTHORITY"]  => add(store = "Active Directory",
types = ("http://schemas.xmlsoap.org/claims/Group"), query = ";tokenGroups;{0}",
param = c.Value);
```

d. Click **Finish**.

14. Click **Add Rule...** to add another custom rule:
    a. For the **Claim rule template**, select **Send Claims Using a Custom Rule**.
    b. For the **Claim rule name**, enter `Claim of groups membership`.
    c. For the **Custom rule**, paste in the following string:

```
c:[Type == "http://schemas.xmlsoap.org/claims/Group"]  => issue(Type = "Group",
Issuer = c.Issuer, OriginalIssuer = c.OriginalIssuer, Value = c.Value, ValueType =
c.ValueType);
```

    d. Click **Finish**.
15. Click **OK**.

Upload the PowerFlex metadata file

You must upload the metadata file through PowerFlex Manager to ensure that PowerFlex can establish a connection to the IdP.

1. In PowerFlex Manager, go back to the **Add Identity Provider (IdP)** window.
2. Under **Service Provider**, select the **I have configured PowerFlex as a SP in my IdP using the metadata above** checkbox and click **Next**.
3. Under **IdP Setup**, upload the identity provider metadata so that PowerFlex can establish a connection to the identity provider.
    a. To upload the metadata as a file, select **Upload File** and specify the file location.
    b. To retrieve the metadata from a URL, select **URL** and specify the following URL: `https://hostname/FederationMetadata/2007-06/FederationMetadata.xml`

       The URL is always the same, except for the hostname, which you must specify for your environment.
    c. Click **Next**.
4. On the **Settings** page, review the attribute mappings that are imported from the identity provider.
    a. Check the attribute mappings to be sure they are correct.
    b. Click **Next**.
    c. Review the **Summary** page and click **Finish**.

After you add ADFS as an IdP, it is added to the list of IdPs on the **SSO Identity Provider (IdP) Configuration** page in PowerFlex Manager.

In addition, PowerFlex Manager updates the login page to show a login button with ADFS. You can see this button the next time you log in to PowerFlex Manager.

Set up PowerFlex single sign-on with Microsoft Azure

To set up PowerFlex single sign-on with Microsoft Azure, you must first add Microsoft Azure as an IdP through PowerFlex Manager.

You must then download the service provider metadata for PowerFlex. After you complete this step, log in to the Microsoft Azure portal and configure PowerFlex as the service provider. You must then return to PowerFlex Manager to complete the setup of single sign-on.

Upload the PowerFlex metadata file

You must upload the metadata file through PowerFlex Manager to ensure that PowerFlex can establish a connection to the IdP.

1. In PowerFlex Manager, go back to the **Add Identity Provider (IdP)** window.
2. Under **Service Provider**, select the **I have configured PowerFlex as a SP in my IdP using the metadata above** checkbox and click **Next**.
3. Under **IdP Setup**, select **Upload File**.
4. Browse and select the `Federated Metadata XML` file and click **Next**.

You must now leave PowerFlex Manager. Go back to the Microsoft Azure portal to create a group claim.

Add Microsoft Azure as an identity provider

You must add Microsoft Azure as an IdP and download the service data for PowerFlex through PowerFlex Manager.

1. On the menu bar, click **Settings**.
2. In the left pane, click **User Management**, then in the right pane, click **SSO Identity Provider (IdP) Configuration**.
3. Click **Add**.
   The **Add Identity Provider (IdP)** window opens.
4. Under **Name**, provide a name for the identity provider and click **Next**.
5. Under **Service Provider**, choose a method to download the service provider metadata for PowerFlex:
   - To download the metadata as a file, select **Download File** and click **Download SP Metadata (XML)**.
   - To copy the metadata, click **Manual Copy** and click **Copy** for each piece of metadata.

Configure PowerFlex as a service provider for Microsoft Azure

You must export the PowerFlex metadata file and edit it to include additional attributes.

You have 15 minutes from when the SP Metadata (XML) file is downloaded from PowerFlex until the `Federation Metadata XML` file is uploaded to complete the procedure, or a timeout may occur. If there is a timeout, see *Troubleshooting a timeout error*.

1. Using an editor, open the `SP Metadata XML` file, replace any occurrence of the string `keycloak-http` with the PowerFlex Manager IP address, and save the file.
2. Log in to the Microsoft Azure Portal.
3. Go to **App registrations**.
4. Under **Owned applications**, select the Microsoft Azure application that PowerFlex uses.
5. Click the link beside **Managed Application in local directory**.
6. Go to **Manage** > **Single sign-on**.
7. Select **SAML**.
8. From the menu, click **Upload metadata file**.
9. To import the metadata file, browse and select the file, and then click **Add**.
10. Check that the URL contains the PowerFlex system IP and click **Save**.
11. Exit the import screen.
12. In the **SAML-based Sign-On screen** window, go to the **SAML Certificates** section, and click **Download.** to download the `Federation Metadata XML` file.
    The file name that is created contains the application name with an `.xml` extension. The file is downloaded to the `Downloads` directory. In this case, the `Federation Metadata XML` directory.
13. Go to the `Federation Metadata XML` file, open it in a text editor and edit it to include the additional attributes as follows:
    a. Before `</IDPSSODescriptor>`, add the following lines:

```
<Attribute xmlns="urn:oasis:names:tc:SAML:2.0:assertion"
Name="http://schemas.xmlsoap.org/ws/2005/05/identity/claims/emailaddress"
NameFormat="urn:oasis:names:tc:SAML:2.0:attrname-format:uri" FriendlyName="E-Mail
Address"/>
<Attribute xmlns="urn:oasis:names:tc:SAML:2.0:assertion"
Name="http://schemas.xmlsoap.org/ws/2005/05/identity/claims/givenname"
NameFormat="urn:oasis:names:tc:SAML:2.0:attrname-format:uri" FriendlyName="Given
Name"/>
<Attribute xmlns="urn:oasis:names:tc:SAML:2.0:assertion"
Name="http://schemas.xmlsoap.org/ws/2005/05/identity/claims/surname"
NameFormat="urn:oasis:names:tc:SAML:2.0:attrname-format:uri"
FriendlyName="Surname"/>
```

    b. Click **Save**.

You must now leave the Microsoft Azure portal. Go back to PowerFlex Manager and upload the `Federation Metadata XML` file to PowerFlex.

Create a group claim

You must create a group claim to manage user authorization to the application.

1. In the Microsoft Azure portal, go back to the **SAML-based Sign on** window.
2. From the left navigation, click **Single sign-on**.
   The **Set up Single Sign On with SAML** window opens.
3. From the **Attributes & Claims** window, click **Edit**.
4. Select **+ Add a group claim**.
5. From the **Group Claims** window, complete the following steps:
   a. For **Which groups associated with the user should be returned to the claim?**, select **Groups assigned to the application**.
   b. For **Source attribute**, select **Group ID**.
   c. Expand **Advanced options**.
   d. Select the **Customize the name of the group claim** checkbox.
   e. For **Name (required)**, enter `Group`.
   f. Click **Save**.

Assign a user or group to the application

The user or group you assign is given access to authenticate to PowerFlex through PowerFlex Manager.

1. In the Microsoft Azure portal, from the **SAML-based Sign on** window, go to the left side navigation, and click **Users and groups**.
2. In the **Add Assignment** window, click **Users and groups**.
3. Select the checkbox beside the users and groups that should be assigned.
4. Click **Assign**.

You must now leave the Microsoft Azure portal. Go back to PowerFlex Manager and provide permissions to the remote users and groups you created.

Provide permissions to remote users and groups

You can provide permissions to users and groups that are based on their role. The roles control access permissions for the corresponding group.

You must not connect to SSO from PowerFlex Manager before you assign permissions.

1. Log in to PowerFlex Manager using the local administrator rights.
2. Go to Settings > **User Management** > **Remote users/Groups**.
3. Click **Add**.
   a. Select the type.

      If you select the **Group** type for the **Group Name**, enter the group-id that is taken from Microsoft Azure (not the Group Name).

      If you select the **User** type for the **User Name**, enter the username that is defined in Microsoft Azure.
   b. Select the provider.
   c. Select a role.
   d. Click **Apply**.

After you add Microsoft Azure as an IdP, it is added to the list of IdPs on the **SSO Identity Provider (IdP) Configuration** page in PowerFlex Manager.

In addition, PowerFlex Manager updates the login page to show a login button with Microsoft Azure. You can see this button the next time you log in PowerFlex Manager.

Troubleshooting a timeout error

If a timeout error occurs, you can log in to PowerFlex Manager and create the IdP again.

1. Log in to PowerFlex Manager.
2. On the menu bar, click **Settings**.

3. In the left pane, click **User Management**, then in the right pane, click **SSO Identity Provider (IdP) Configuration**.
4. Select the IdP and click **Delete**.
5. Add the IdP.
6. Export the `Federation Metadata XML` file.
7. Import the `Federation Metadata XML` file that you downloaded from Microsoft Azure to PowerFlex.
8. Import the exported `Federation Metadata XML` file that you created in step 7 to Microsoft Azure.

   In the Microsoft Azure portal, update the entity-id by reimporting the PowerFlex metadata file through the application.

Modify an identity provider

The **Modify** option allows you to edit the existing identity provider settings.

1. On the menu bar, click **Settings**.
2. In the left pane, click **User Management**, then in the right pane, click **SSO Identity Provider (IdP) Configuration**.
3. Select an identity provider.
4. Optionally, change the name for the identity provider.
5. Optionally, enable or disable the identity provider.
6. Click **Apply**.

Enable or disable an identity provider

You can disable an identity provider after the initial configuration. If needed, you can also reenable the provider after it has been disabled.

Perform the following procedure to enable or disable an identity provider:

1. On the menu bar, click **Settings**.
2. In the left pane, click **User Management**, then in the right pane, click **SSO Identity Provider (IdP) Configuration**.
3. Click **Enable** or **Disable**.

Delete an identity provider

You can delete an identity provider if you no longer need it.

Perform the following procedure to delete an existing identity provider:

1. On the menu bar, click **Settings**.
2. In the left pane, click **User Management**, then in the right pane, click **SSO Identity Provider (IdP) Configuration**.
3. Click **Delete**.

Update the certificate for an identity provider

You can update the certificate for an identity provider if it is about to expire.

1. On the menu bar, click **Settings**.
2. In the left pane, click **User Management**, then in the right pane, click **SSO Identity Provider (IdP) Configuration**.
3. Check the **Status** column to see if any of the identity providers are about to expire.
4. If an identity provider is about to expire, select the identity provider and click **Update Certificate** in the details pane to the right of the list of providers.

Repositories

On the **Repositories** page, you can load compliance files, operating system images, and compatibility management files.

You can load compliance files and specify a default version for compliance checking on the **Compliance Versions** tab. You can load multiple compliance files into PowerFlex Manager with different operating system images included. The compliance file includes operating system images for VMware ESXi and PowerFlex. When you add a compliance file, these images are

automatically added to the **OS Images** tab. If necessary, you can also create your own operating system image repositories. You can also load the compatibility management file on the **Compatibility Management** tab to allow PowerFlex Manager to determine which upgrade paths are valid and which are not.

On the **Compliance Versions** tab, click **Add** to load a compliance version. Click **Change** to change the default compliance version to a new file.

You cannot make a minimal compliance version the default version for compliance checking, since it includes only server firmware updates. The default version must include the full set of compliance update capabilities. PowerFlex Manager does not show any minimal compliance versions in the **Default Version** drop-down list.

The **Compliance Versions** tab displays the following information:

Table 85. Compliance version fields

Field	Description
State	Displays an icon indicating one of the following states: ● Pending—The compliance file download process is in progress. ● Downloading—The compliance file is being downloaded and PowerFlex Manager provides the percentage complete for the download operation. ● Unpacking—The compliance file is being unpacked and provides the percentage complete for the unpacking operation. ● Synchronizing—The compliance file is being synchronized with the management software after unpacking. ● Available—The compliance file is downloaded and copied successfully. ● Error—There is an issue downloading the compliance file. ● Needs Approval—The compliance file is unsigned and needs approval for use.
Version	Displays the compliance version.
Source	Displays the share path of the compliance version in a file share.
Signature	Displays whether the digital signature on the compliance file is signed or unsigned.
File Size	Displays the size of the compliance file in GB.
Type	Displays **Minimal** if the compliance file only contains firmware updates, or **Full** if it contains firmware and software updates.
View bundles	Displays details about any bundles added for the compliance version.
Available Actions	Select an option for a compliance file that is in an **Error** state: ● Delete the compliance version ● Resynchronize a compliance version
	Select an option for a compliance file that is in a **Needs Approval** state: ● Allow unsigned file ● Delete

Use the **OS Image Repositories** tab to create operating system image repositories and view the following information:

Table 86. Operating system image repositories

Field	Description
State	Displays the following states: ● Available—The operating system image repository is downloaded and copied successfully. ● Pending—The operating system image repository download process is in progress. ● Error—There is an issue downloading the operating system image repository.
Repository	Displays the repository name.
Image Type	Displays the operating system type.
Source Path	Displays the share path of the repository in a file share.

Table 86. Operating system image repositories (continued)

Field	Description
In Use	Displays **True** (the operating system image repository is in use) or **False** (the operating system image repository is not in use).
Available Actions	Select an option to delete or resynchronize.

You cannot perform any actions on repositories that are in use. However, you can delete repositories that are in an **Available** state, but not in use and not set as a default version.

All the options are available only for repositories in an **Error** state. The **Resynchronize** option appears only when you must perform a backup and restore of a previous image.

Use the **Compatibility Management** tab to load the compatibility management file. To facilitate upgrades, PowerFlex Manager uses information that is provided in the compatibility matrix file to determine which upgrade paths are valid and which are not. When you attempt an upgrade, PowerFlex Manager warns you if the current version of the software is incompatible with the target version, or if any of the RCM or IC versions that are currently loaded are incompatible with the target compliance versions.

Related information

Configuring block storage
Deploying and provisioning

Compliance versions

PowerFlex Manager displays the following information about the compliance versions:

Table 87. Compliance versions

Field	Description
State	Displays an icon indicating one of the following states: • Pending—The compliance file download process is in progress. • Downloading—The compliance file is being downloaded and PowerFlex Manager provides the percentage complete for the download operation. • Unpacking—The compliance file is being unpacked and provides the percentage complete for the unpacking operation. • Synchronizing—The compliance file is being synchronized with the management software after unpacking. • Available—The compliance file is downloaded and copied successfully. • Error—There is an issue downloading the compliance file. • Needs Approval—The compliance file is unsigned and needs approval for use.
Version	Displays the compliance version. If one or many components in the RCM are modified, a wrench icon and the text, **Modified** is displayed next to the name of the RCM. But, if the file is deleted from the file share after a component is updated, the wrench ion with the text, **Modified** or **Needs Attention**, is displayed.
Source	Displays the share path of the compliance version in a file share.
Signature	Displays whether the digital signature on the compliance file is signed or unsigned.
File Size	Displays the size of the compliance file in GB.
Type	Displays **Minimal** if the compliance file only contains firmware updates, or Full if it contains firmware and software updates.
View bundles	Displays details about any bundles added for the compliance version.
Available Actions	Select an option if the compliance file is in an **Error** or **Needs Approval** state: • Delete the compliance version • Resynchronize a compliance version • Allow unsigned file

Select the compliance version to view the following information about the firmware package:

- **Bundles**—Displays the number of bundles available in the firmware package.
- **Components**—Displays the number of software components available in the firmware package.
- **Created On**—Displays the date when the compliance version was created.
- **Last Updated**—Displays the date when the compliance version was last updated.
- **Resource Groups Affected**—Displays the resource groups in which the firmware is used.
- **Additional Components**:
  - **VMware ISO**—Displays the date and time when the VMware ISO component was added.
  - **Filename**—Displays the file name of the VMware ISO component added.
  - **VMware Upgrade Depot**—Displays the date and time when the VMware patch depot file component was added.
  - **Filename**—Displays the file name of the VMware patch depot file component added.

From this page, you can:

- Add a compliance file.

  A compliance file can be large and take some time to download and unpack. To help you monitor the progress of the downloading and unpacking operations, PowerFlex Manager provides percentage complete information for these operations in the **State** column on the **Compliance Versions** tab.

- Select **Default Version** and choose a version from the drop-down list to set it as the default for compliance.
- Select a compliance version from the list, and click the delete icon in the same row to remove the version.

  If you remove a compliance version, the version is deleted from the appliance not from the file share.

- Select a version from the list, and, in the right pane, click **View Bundles** to view the firmware and software bundles available in the compliance version.

Add a compliance file

A compliance file must be in an available state for PowerFlex Manager to add the file.

1. From the **Getting Started** page, click **Upload Compliance File**.

   Alternatively, click **Settings** > **Repositories**. On the **Repositories** page, select the **Compliance Versions** tab, and click **Add**.

2. In the **Add Compliance File** dialog, select one of the following options:
   - **Download from Dell Technologies connectivity services (Recommended)**—Select this option to import the compliance file that contains the firmware bundles you need.
   - **Download from local network path**—Select this option to download the compliance file from a CIFS file share. This option is intended for sites that do not have Internet accessibility to Dell Technologies connectivity services.
3. If you selected **Download from Dell Technologies connectivity services (Recommended)**, click the **Available Compliance Files** drop-down list, and select the file.

   Before downloading a compliance file from Dell Technologies connectivity services, you must configure Dell Technologies connectivity services. To configure Dell Technologies connectivity services , you must enable Dell Technologies connectivity services in the **Initial Configuration** wizard.

   If you are downloading a compliance file from Dell Technologies connectivity services, the file type is a ZIP.

   If you are downloading a compliance file from the Dell Technologies support site, the file will be called `PowerFlex _Software_5.1.x_xxx.zip`

   NOTE: The XML file type is not supported as a download option.

4. If you selected **Download from local network path**, perform the following:
   - In the **File Path** box, enter the location of the compliance file. Use the following format:
     - CIFS share for ZIP file: **`\\host\share\filename.zip`**
   - If you are using a CIFS share, enter the **User Name** and **Password**.
   - Select **Test Connection** to test the connection between the PowerFlex Manager user interface and the location of the CIFS file.
   - Mark the target upgrade catalog as the default version for compliance checking.
5. Click **Save**.

PowerFlex Manager unpacks the compliance file as a repository that includes the firmware and software bundles that are required to perform any updates.

If the compliance file contains an embedded ISO file for an operating system image (such as for a VMware ESXi or PowerFlex image), the download process unpacks the file and adds it to the operating system image repository.

A compliance file can be large and take some time to download and unpack. To help you monitor the progress of the downloading and unpacking operations, PowerFlex Manager provides percentage complete information for these operations in the **State** column on the **Compliance Versions** tab.

Add VMware ESXi files to the compliance file

The following procedure is applicable only when using ESXi hosts. The vCenter Server Appliance (VCSA) can be hosted either in your environment or through the PowerFlex management cluster (KVM) compute-only extension (ESXi-based).

In this process, the ESXi ISO image and patch depot file that are compatible with the IC are uploaded. This step applies only to PowerFlex compute-only deployments.

NOTE: You must use the VMware standard ISO (not the Dell-customized ISO).

The VMware ESXi image and patch depot file are no longer available on the Dell support site. You must obtain these files from the Broadcom support portal and upload them to the compliance file before creating a template, performing a deployment, running a compliance check, or carrying out a non-disruptive upgrade with VMware ESXi. After downloading the appropriate image and patch depot file, you should add these components (ESXi ISO and patch depot file) to the compatible compliance file.

1. On the menu bar, click **Settings** > **Repositories** > **Compliance Versions**.
2. From the **Compliance Versions** tab, select the **IC** and click **Add Component**.
3. In the **Add Compliance Component** window, leave the default **VMware Components** setting for **Type of components being added**.
4. In the **Add VMware Installation ISO** section, enter the file path of the VMware installation ISO file, username, and password and click **Test Connection** to verify that the file path is valid and can be accessed.
5. In the **Add VMware Upgrade Depot Zip** section, enter the file path of the VMware patch depot file, username, and password and click **Test Connection** to verify that the file path is valid and can be accessed.
6. Click **Add**.

   The VMware ISO and patch depot files are tracked under the **Hardware** category on the **Monitoring** > **Events** page. The **Additional Components** in the **Compliance and OS Repositories** window displays the VMware ISO and patch depot file name, date, and time at which the VMware ISO and patch depot files are added.

Changing the compliance file

To perform upgrades against a different compliance version, you can change the compliance file.

1. To change the default compliance version, click **Change** on the **Compliance and OS Repositories** page.

   To change the target version for a resource group, click **Change Target** on the **Resource Group Details** page or click **Change** on the **Node Compliance Report**.
2. Choose the **Preferred Compliance File**.
3. Review the **Compatibility** setting. If it shows **Recommended** or **Supported**, you can proceed with the change. If it shows **Not Allowed**, PowerFlex Manager blocks the change. Review the alert for details about recommended or supported paths.
4. If the change is either recommended or supported, type **`CHANGE COMPLIANCE FILE`** if you are to proceed.
5. Click **Save**.

Firmware repository types

The **Resources** page displays devices that are validated against the default repository.

This capability is available on the PowerFlex appliance and PowerFlex rack offerings only.

There are two types of firmware repositories:

Table 88. Firmware repository types

Repository	Description
Default	The default firmware repository is applied to all devices that are not in a resource group. To set a default firmware repository, you must download a compliance file from either Dell Technologies connectivity services or an internal share through PowerFlex Manager. If you set a default compliance version, you can view compliance with this repository on the Resources page. Devices with firmware levels below the minimum firmware that is listed in the default compliance version are viewed as non-compliant (out of compliance).
Service level	This repository is applied only to nodes that are in service and assigned the service level firmware repository. Devices with firmware levels below the minimum firmware level that is listed in the service level repository are marked as non-compliant. When a service level firmware repository is assigned to a resource group, the firmware validation is checked only against the service level firmware repository and the default firmware repository checks are no longer applied to the devices associated with this resource group.

Viewing firmware and software bundles details

The details that are provided for bundles include name, version, and date, allowing for management and tracking of updates.

1. On the menu bar, click **Settings** and click **Compliance and OS Repositories**.
2. Click the **Compliance Versions** tab, select a repository, and then click **View Bundles**.
   A list of system and software bundles is displayed.
3. Click **Firmware Bundles** to view the firmware or system bundles.
4. Click **Software Bundles** to view the software bundles.

You can see the following information regarding bundles:

- Name—Name of the firmware or software update package.
- Version—Version of the firmware or software update package.
- Date and Time—Date and time when the firmware or software update package was downloaded.

Uploading VMware ESXi software bundle

You can upload the ESXi software bundle as a single .zip format file, from the RCM details page.

1. Click **Settings** > **Compliance Versions**.
2. On the **Compliance Versions** page, select the RCM version for which you want to view and upload the ESXi software bundle.
3. On the right side of the **Compliance Versions** page, click **View Details**.
   The RCM details page is displayed.
4. Click the **Software Bundles** tab.
5. Select the ESXi version for which you want to upload the software bundle and click **View**.
   The ESXi software bundle details page is displayed.
6. Click **Upload Software Depot**.
   The **Upload Software Depot** window is displayed.
7. Select the **Download from local network path** option as size of the files exceed 500 MB.
8. Enter the path and user credentials to access the file on the local network drive.
9. Enter `Modify Compliance File` to confirm the upload and click **Proceed**.
   After the software is upgraded, the modified icon is displayed next to the ESXi software version you selected for the upgrade. The upgraded version number and the modified icon are also displayed next to all the components that are upgraded in the ESXi bundle. On completion of the upgrade process, a "modified" folder is created in the shared network location. The folder contains the VIBs and zipped format of the software depot bundle.
10. Verify if all the upgraded files are present in the "modified" folder in the shared network location.

For RCM version 3.7.5, the supported VMware ESXi version is 7.Ou3 and for RCM version 3.8, the supported ESXi version is 8.0.2.

Performing a single component upgrade

The single component upgrade allows for upgrading the software and firmware on individual components, such as applying security patches, without performing a full RCM/IC upgrade. It is recommended to use this option for component-specific upgrades, such as patch updates, and to ensure compliance with the standard RCM/IC version.

Single component upgrades are not engineering-validated upgrades. You should understand the implication and risks. Dell Technologies recommends aligning with RCM/IC as the official-validated catalog. If you consider deviating to align with the security or functionality requirements, consider testing the compatibility.

A single component upgrade involves the following steps:

1. Upgrading the firmware and software components.
2. Extracting the compliance report.
3. Modifying and publishing the existing template.

CAUTION: Since Dell Technologies only validates the released RCM/IC code catalogs, you must be aware of the implications and risks that are involved in a single component upgrade. If any issues are identified related to the RCM/IC deviation, you may be asked to modify the code or firmware.

If an upgrade file, for example iDRAC FW 7.xx.xx.xx, exceeds 200 MB, use the SMB or CIFS share option. Browser uploads are not supported for upgrade files larger than 200 MB.

The use of single component upgrade enables the PowerFlex platform to remain in compliance, using the deviated code/ firmware, which means future upgrades will not be blocked due to non-compliance.

The single component upgrade option enables the PowerFlex platform to remain in compliance. The future upgrades are not blocked due to non-compliance.

The following limitations are associated with the single component upgrade:

- VMware ESXi version changes are restricted. Upgrading from VMware ESXi 8.x to 9.x or downgrading is not supported.
- Updates to embedded operating systems, such as SLES or CentOS are not supported.

Upgrade the firmware components

To upgrade firmware on a PowerFlex node, authenticate as an administrator, choose a bundle, modify the component, and apply the updates.

You must verify that you are using administrator credentials.

An upgrade of the following firmware components is supported:

- Drive firmware
- iDRAC Lifecycle Controller (LCC)
- BIOS firmware
- Network Interface Card (NIC) firmware
- BOSS card firmware
- Switch firmware

The upgrade of the PowerFlex core is not supported.

1. On the menu bar, click **Settings** > **Repositories** > **Compliance Versions**.
2. From the **Compliance Versions** page, select the current RCM/IC bundle and click **View Bundles**.
3. From the list of bundles in the **Firmware Bundles** tab, verify the bundles that you want to modify.
   To modify components on the PowerFlex R660 node, select **System Bundle (Windows x64) PER660 v850** or select the appropriate components such as switches, CloudLink, or operating system bundles.
4. Select the appropriate bundle to upgrade and click **View**.

   NOTE: Only one component can be modified at a time.

5. Select the specific component to upgrade and click **Modify**.
6. On the **Update Bundle Component** page, perform the following:
   a. Select either **Download from local network path** or **Upload from local**.

b. Provide the **New Bundle Version** (if it is not picked automatically) to match the existing bundle version.
c. In the **Type "MODIFY COMPLIANCE FILE" to confirm** field, enter **`Modify Compliance File`** and click **Proceed**.

7. Repeat the steps through 5 and 7 for the remaining components.
8. Click **Settings** > **Repositories** > **Compliance Versions** and verify that a wrench icon and `Modified` is appearing for the modified RCM/IC.
9. To update the resources, perform the following:
   a. On the menu bar, click **Resources** and select the PowerFlex nodes to update and click **Run Inventory**.
   b. Click **Lifecycle** > **Resource Groups**.
   c. Select the resource group to be modified and click **View Details**.
   d. Click **View Compliance Report**.
   e. Select the modified PowerFlex node or other components and click **Update Resources**.
   f. From the **Apply Resource Updates** page, select **Allow PowerFlex Manager to perform firmware and software updates now**.
   g. Select the appropriate maintenance mode and click **Apply**.
10. To monitor and confirm that the component is updated with the new version, perform the following:
    a. Click **Lifecycle** > **Resource Groups** to verify that the required component is upgraded.
    b. Select the resource group that you already modified and click **View Details**.
    c. On the **Resource Group Information** page, verify that the resource group compliance status is `Compliant`.

    If an upgrade failure occurs, contact Dell Technologies Support for assistance.

Related information

Viewing resource group details (mirroring)

Upgrade the software components

Use the following procedures to upgrade the software components using PowerFlex Manager. These procedures are applicable only when you consider deviating from the standard upgrade procedure.

NOTE: Ensure you upload identical versions of the VMware ESXi patch bundle and the ESXi ISO file to PowerFlex Manager.

A software component upgrade involves operating system updates such as the following:

- Applying ESXi patches
- Upgrading the CloudLink virtual machine

The following limitations are associated with the single component upgrade:

- ESXi version changes are restricted; upgrading from ESXi 8.x to 9.x or downgrading is not supported.
- Updates to embedded operating systems, such as SLES or CentOS are not supported.

Related information

Viewing resource group details (mirroring)

Upgrade VMware ESXi patch and ISO images

You can upgrade the VMware ESXi patch and ISO images using PowerFlex Manager.

1. On the menu bar, click **Settings** > **Repositories** > **Compliance Versions**.
2. Verify that the required RCM/IC is uploaded and is available.
3. Select the RCM/IC bundle to be modified and click **View Bundles**.
4. From the **Software Bundles** tab, select the **ESXI Software Bundle** checkbox, clear the checkbox for other software bundles, and click **View**.
5. Select **Upload Software Depot** and select either of the following options to upload the ESXi path depot file:
   - **Upload from local**: Select this option when the depot file is smaller than 500 MB.
   - **Download from local network path**: Select this option when the depot file is larger than 500 MB. Enter the CIFS file path and user credentials to access the file on the local network drive.
6. For the **Upload from local** option, complete the following:
   a. To provide a local file, click **Choose File**.
   b. To confirm that you want to modify the existing file, enter **`Modify Compliance File`** and click **Proceed**.

c. Wait for the compliance file to synchronize and verify that the status of the compliance file is `Available`.
d. Verify that a wrench icon and `Modified` displays for the modified RCM/IC.

7. For the **Download from local network path** option, complete the following:
   a. Enter the `CIFS` file or the `HTTPS` address to access the file.
   b. Enter the username and password.
   c. Select **Test Connection** to test the connection between the PowerFlex Manager user interface and the location of the CIFS file or the file that is at the HTTPS address.
   d. To confirm that you want to modify the existing file, enter **`Modify Compliance File`** and click **Proceed**.
   e. Wait for the compliance file to synchronize and verify that the status of the compliance file is `Available`.
   f. Verify that a wrench icon and `Modified` displays for the modified RCM/IC.
8. **NOTE:** You must complete steps 5 through 7 before you start step 8.

   To upload the `ESXI.depot.zip` and `ESXi.iso` files, complete the following:
   a. From **Repositories** > **Compliance Versions**, select the RCM/IC bundle, and click **Add**.
   The **Add Compliance File** dialog box displays.
   b. Enter the file path, username, and password for both the installation ISO and the depot ZIP file. Ensure the files match with the existing VMware ESXi patch and ISO files.
   c. Select **Test Connection** to test the connection between the PowerFlex Manager user interface and the location of the CIFS file or the file that is at the HTTPS address.
   d. Click **Save** and wait for the synchronization and download to complete.
   e. On the **Compliance Versions** page, verify that the files are added and the RCM/IC is in `Available` state.

Update resource inventory and resource groups

Once you upgrade the VMware ESXi patch and ISO images, use this procedure to update the resource inventory and resource groups.

1. On the menu bar, click **Lifecycle** > **Resources**.
2. Select the PowerFlex node to update, and click **Run Inventory**.
3. Click **Lifecycle** > **Resource Groups** and select the service to upgrade.
4. Click **Non-Compliant** in the **Resource Group Information** tab.
5. From the **Software Components** tab, select all the listed nodes, and click **Update Resources**.
6. Select the appropriate **Maintenance Mode** and click **Apply**.
7. To monitor the job and verify that the required component is upgraded, perform the following:
   a. Click **Lifecycle** > **Resource Groups**.
   b. Select the resource group that you modified and click **View Details**.
   c. On the **Resource Group Information** tab, verify that the resource group compliance status is `Compliant`.

   If the upgrade fails, contact Dell Technologies Support for assistance.

Upgrade CloudLink Center

Using the single component option, you can upgrade the CloudLink Center version.

CloudLink ISO is required to start the CloudLink version upgrade using the single component upgrade option.

You must verify that you are using the administrator credentials.

1. On the menu bar, click **Settings** > **Repositories** > **Compliance Version**.
2. Select the RCM/IC bundle, and click **View Bundles**.
3. From the **Firmware Bundles** tab, select the **System Bundle CloudLink** and click **View**.
4. Select the component **clc** that you want to upgrade and click **Modify**.
5. From the **Update Bundle Component** page, select **Download from local network path** option and upload the CloudLink ISO file.
6. Enter the **New Bundle Version** if it is not picked automatically, to match with the existing bundle version.
7. Type **`Modify Compliance File`** in the field and click **Proceed**.
8. Click **Repositories** > **Compliance Versions** and verify that the modified RCM/IC appears with a wrench icon and `Modified`.
9. To update the resources, perform the following:

a. From **Lifecycle** > **Resources**, select both the CloudLink resources and click **Run Inventory**. Wait for two minutes to complete the inventory and refresh your browser.
b. Select the CloudLink Center to upgrade and click **Non-Compliant** and click **Update Resource**.
c. From the **Apply Resource Updates** page, select **Allow PowerFlex Manager to perform firmware and software updates now** and click **Apply Updates**. Wait till the deployment status becomes `Available`.
d. On the **Resources** page, click **CloudLink Center** > **View Details** to verify the upgraded CloudLink Center version. Wait for three to four minutes. Both CloudLink Centers are updated with the new version.

10. To manually upgrade the SecureVM agent on all the PowerFlex nodes from CloudLink Center, perform the following:
    a. In your web browser, type the URL for CloudLink Center in the following format: **`https://clc_address`**
    b. Log in with the **`secadmin`** credentials.
    c. Click **Agents** > **Machines** and select the node.
    d. Click **Actions** > **Upgrade**. The upgrade option is disabled if an upgrade is not required for the SecureVM agent.
    e. Repeat step 10a through 10d for every PowerFlex node that requires a CloudLink agent upgrade.
    f. Click **Agents** > **Machines**, select the node, and verify the SecureVM agent version.
    g. After the agents are updated on the nodes, from CloudLink Center, click **Agents** > **Agent Download** to download the `64-bit SUSE RPM` package or `64-bit CentOS RPM` package depending on the operating system running on the nodes and modify the SecureVM agent component in the RCM/IC bundle.
11. To modify the SecureVM agent component in the RCM/IC bundle, perform the following:
    a. Verify that you are using the administrator credentials.
    b. On the menu bar, click **Settings** > **Repositories** > **Compliance Versions**, select the current RCM/IC bundle and click **View Bundles**.
    c. Click **Software Bundles** > **CloudLink SecureVM** > **View**. You will notice that there are two CloudLink SecureVM bundles with the same name. One bundle is for SUSE and the other bundle is for CentOS. Select the bundle and click **View** to differentiate the bundle. Select the bundle based on the operating system installed on the PowerFlex nodes.
    d. Click **Modify**.
    e. From the **Update Bundle Component** page, select the **Download from local network path** or **Upload from local** and select the downloaded agent.
    f. Enter the **New Bundle Version** to match the existing bundle version if it is not picked automatically.
    g. Type **`Modify Compliance File`** in the field and click **Proceed**.
12. To update the resource group details, perform the following:
    a. On the menu bar, click **Lifecycle** > **Resource Groups**.
    b. Select the resource group to modify and click **View Details**.
    c. Click **More Actions** > **Update Resource Group Details**.
    d. On the **Resource Group Information** page, verify that the resource group compliance status is `Compliant`.

    If the upgrade fails, contact Dell Technologies Support for assistance.

Exit the PowerFlex nodes from service mode.

Export the compliance report

You can export a compliance report using PowerFlex Manager. The compliance report identifies the list of modified components and their versions. It also compares component versions during upgrades from a modified RCM/IC to a standard RCM/IC.

Before exporting the compliance report, you must ensure that the default RCM/IC version is selected. To verify that the default version is selected, go to **Settings** > **Repositories** > **Compliance Versions**.

When transitioning from a modified RCM to a desired RCM, ensure the following:

- The component versions in the modified RCM are equal to or lower than the component versions in the desired RCM.
- If any component in the modified RCM has a higher version, use the **Extract Compliance** tool to identify the components.
- Use the single component upgrade to add those components with a higher version to the desired RCM to ensure version alignment and compatibility,

1. On the menu bar, click **Lifecycle** > **Resources** > **Export Report**.
2. Select **Export Compliance PDF Report** or **Export Compliance CSV Report**.
3. Identify the modified components and verify the number of components.

Modify and publish the template

After completing a single component upgrade, you can modify and publish the template. This process involves updating the existing template by incorporating a modified ESXi ISO image. The updated template is then published and used for future system expansions.

Ensure the modified RCM/IC is the same version as the **Target Version** under the **Existing Resource Group** page. While adding a resource group on an existing PowerFlex system, ensure that you select the latest uploaded operating system images.

1. On the menu bar, click **Templates** > **My Templates**.
2. Select the template that you want to modify and click **Modify**.
3. Select the node and click **Modify**.
   The node details display.
4. Click **Continue**.
   The **OS Settings** display.
5. Under **OS Settings** > **OS Image**, ensure that the newly uploaded image is selected from the dropdown.
6. Click **Save** > **Publish Template**.
   A confirmation message displays asking whether you are sure that you want to publish the template, click **Yes**.

Operating system images

An operating system image is a complete packaged copy of an operating system that is stored as a single file. After you add the operating system, you can manage it later using the modify or remove options.

1. On the **Repositories** page, click **OS Images**, and then click **Add**.
2. In the **Add OS Image Repository** dialog box, enter the following:
   a. In the **Repository Name** box, enter the name of the repository. The repository name must be unique.
   b. In the **Image Type** box, enter the image type.
   c. In the **Source Path and Filename** box, enter the path of the operating system image file name in a file share.
   d. To enter the CIFS share, see the format that is used in the following example: `\\host\lab\isos\filename.iso`
   e. If you are using the CIFS share, enter the username and password to access the share.
3. Click **Add**.

After adding an operating system image, you can modify it or remove it by selecting the image, then clicking **Modify** or **Remove**.

Resynchronizing an OS image repository

Use the resynchronize option to restore the OS image from the database after a backup and restore.

If an operating system image was uploaded as a part of an ISO file, you must resynchronize the OS image repository from the **OS Image Repositories** tab. However, if the operating system image was uploaded as a part of a compliance ZIP file, go to the **Compliance Versions** tab and resynchronize the ZIP file. The following procedure provides steps for performing the resynchronization from the **OS Image Repositories** tab.

To resynchronize the OS image repository for an operating system image that was uploaded as part of an ISO file:

1. On the **Repositories** page, click **OS Images**.
2. Click **Resynchronize** for a repository in an **Error** state.
   The **Resynchronize OS Repository** page is displayed.
3. Enter the user credentials and click **Test Connection** to test the network connection.

   NOTE: You cannot edit the **Source Path** and **Filename**.

4. Click **Resynchronize**.
   The repository state changes to `Copying state`.

Compatibility management

To facilitate upgrades, PowerFlex Manager uses information that is provided in the compatibility matrix file to determine which upgrade paths are valid and which are not.

When you attempt an upgrade, PowerFlex Manager warns you if the current version of the software is incompatible with the target version, or if any of the RCM or IC versions that are currently loaded are incompatible with the target compliance versions. PowerFlex Manager uses information that is provided in the compatibility matrix file to determine which paths are valid. The compatibility matrix file maps all the known valid and invalid paths for all previous releases of the software.

When you first install PowerFlex Manager, the software does not have the compatibility matrix file, so you must upload the file before performing any updates. You must upload the latest compatibility matrix file to ensure that PowerFlex Manager has access to the latest upgrade information.

You can download the file from Dell Technologies connectivity services or upload it from a local directory path. The file has a GPG extension and an associated compatibility version number.

If you are uploading from a local directory path, ensure that you have access to a valid compatibility matrix file that has the GPG extension. If you are using Dell Technologies connectivity services, ensure that Dell Technologies connectivity services is correctly configured.

1. On the menu bar, click **Settings** and then click **Compatibility Management**.
2. Click **Edit Settings**.
3. Click **Download from Dell Technologies connectivity services** if you are using Dell Technologies connectivity services.
4. Click **Upload from Local** to use a local file. Then, click **Choose File** to select the GPG file.

Getting started

If you want to return to the **Getting Started** page, you can launch it from the **Settings** page.

The **Getting Started** page guides you through the common configurations that are required to prepare a new PowerFlex Manager environment. A green check mark on a step indicates that you have completed the step. Only super users have access to the **Getting Started** page.

Networking

On the **Networking** page, you can define, edit, or delete a network. You can also verify which IP addresses are in use before deployment.

Related information

Getting started
Getting started (mirroring)
Configuring block storage
Deploying and provisioning

Networks

The **Networks** page displays information about networks that are defined in PowerFlex Manager, including:

- **Name**
- **Network Type**
- **VLAN ID**
- **IP Address Setting**
- **Starting IP Address**
- **Ending IP Address**
- **Role**
- **IP Address in Use**

On the **Networks** page, you can:

- Define or modify an existing network.
- Delete an existing network.
- Click **Export All** to export all the network details to a CSV file.
- Export network details for a specific network. To export the specific network details, select a network, and then click **Export Network Details**.
- Click a network to see the following details in the **Summary** tab:
  - Name of the user who created and modified the network.
  - Date and time that the network was created and last modified.
- Sort the column by network names, click the arrow next to the column header. You can also refresh the information about the page.

If you select a network from the **Networks** list, the network details are displayed.

For a static network, the following information is displayed:

- **Subnet Mask**
- **Gateway**
- **Primary DNS**
- **Secondary DNS**
- **DNS Suffix**
- **Last Updated By**
- **Date Last Updated**
- **Created By**
- **Date Created**
- **Static IP Details**

For a DHCP network, the following information is displayed:

- **Last Updated By**
- **Date Last Updated**
- **Created By**
- **Date Created**

You can filter the IPs by selecting any of the following options from the **View** drop-down list, under the **Static IP Address Details** section:

- **All IP Addresses**
- **IP Addresses in Use**
- **Available IP Addresses**

You can also select the links under the **IP Address in Use** column. The IPs are automatically filtered based on the **IP addresses In Use** criteria.

Defining a network

Adding the details of an existing network enables PowerFlex Manager to automatically configure nodes that are connected to the network.

Ensure that the following conditions are met before you define the network:

- PowerFlex Manager can communicate with the out-of-band management network.
- PowerFlex Manager can communicate with the operating system installation network in which the appliance is deployed.
- PowerFlex Manager can communicate with the hypervisor management network.
- The DHCP node is fully functional with appropriate PXE settings to PXE boot images from PowerFlex Manager in your deployment network.

1. On the menu bar, click **Settings** and then click **Networking**.
2. Click **Networks**.
   The **Networks** page opens.
3. Click **Define**. The **Define Network** page opens.
4. In the **Name** field, enter the name of the network. Optionally, in the **Description** field, enter a description for the network.
5. From the **Network Type** drop-down list, select one of the following network types:
   - **General Purpose LAN**
   - **Hypervisor Management**

- **Hypervisor Migration**
- **Hardware Management**
- **PowerFlex Data**
- **PowerFlex Data (Client Traffic Only)**
- **PowerFlex Data (Server Traffic Only)**
- **PowerFlex Management**

For a PowerFlex configuration that uses a two-layer architecture with four dedicated data networks, you typically have two PowerFlex (client traffic only) VLANs and two PowerFlex data (server traffic only) VLANs. These network types are used with storage-only and compute-only resource groups.

6. In the **VLAN ID** field, enter a VLAN ID between **1** and **4094**.

   NOTE: PowerFlex Manager uses the VLAN ID to configure I/O modules that enable network traffic to flow from the node to configured networks during deployment.

7. Optionally, select the **Configure Static IP Address Ranges** check box, and then do the following:

   NOTE: The **Configure Static IP Address Ranges** check box is not available for all network types. For example, you cannot configure a static IP address range for the operating system installation network type. You cannot select or clear this check box to configure static IP address pools after a network is created.

   a. In the **Subnet Mask** box, enter the subnet mask.
   b. In the **Gateway** box, enter the default gateway IP address for routing network traffic.
   c. Optionally, in the **Primary DNS** and **Secondary DNS** fields, enter the IP addresses of primary DNS and secondary DNS.
   d. Optionally, in the **DNS Suffix** field, enter the DNS suffix to append for hostname resolution.
   e. To add an IP address range, click **Add IP Address Range**. In the row, indicate the role in PowerFlex nodes for the IP address range and then specify a starting and ending IP address for the range. For the **Role**, select either:
      - **Server or Client**: Range is assigned to the server and client roles.

      NOTE: IP address ranges cannot overlap. For example, you cannot create an IP address range of 10.10.10.1–10.10.10.100 and another range of 10.10.10.50–10.10.10.150.

8. Click **Save**.

Related information

Adding networks to a resource group (Mirroring)

Modifying a network

If a network is not associated with a template or resource group, you can edit the network name, the VLAN ID, or the IP address range.

1. On the menu bar, click **Settings** and then click **Networking**.
2. Click **Networks**.
   The **Networks** page opens.
3. Select the network that you want to edit and click **Modify**. The **Modify Network** page opens.
4. Edit the information in any of the following fields: **Name**, **VLAN ID**, **IP Address Range**.

   For a PowerFlex data network, you can specify a **Subnet** IP address for a static route configuration. The subnet is used to support static routes for data networks.

5. Click **Save**.

You can also view details for a network or export the network details by selecting the network, then clicking **View Details** or **Export Network Details**.

Run inventory

You can run IP address verification on your entire list of networks. This verification process refreshes your IP address data manually since the scheduled scan occurs every four hours.

This IP verification process sends a ping request for the whole network.

1. On the menu bar, click **Settings** and then click **Networking**.
2. Click **Networks**.
   The **Networks** page opens.
3. Select the network that you want to edit and click **Run Inventory**. Only one network can be selected at a time.
   Click **Close** when the alert notification is displayed.

You can also export the network details by selecting the network, then clicking **Export Network Details**.

View details

You can view details, export network details, and run inventory on static IP addresses within your network.

1. On the menu bar, click **Settings** and then click **Networking**.
2. Click **Networks**.
   The **Networks** page opens.
3. Select the network that you want to edit and click **View Details**. Only one network can be selected at a time.
   The network details page opens.
4. Click **Export Network Details** to download a ZIP file of your network.
5. In **Static IP Address Details** select a static IP address and click **Run Inventory** to send a ping request. Only one static IP address can be selected at a time.
   After the static IP address receives a ping response, a notification displays that the IP address was found and verified in the **Resource Using IP Address** column.

   Click the icon next to the IP verification notification to view the port which received the response.

(Optional) Deleting a network

You cannot delete a network that is associated with a template or resource group.

1. On the menu bar, click **Settings** and then click **Networking**.
2. Click **Networks**.
3. In the **Networks** page, click the network that you want to delete, and click **Delete**.
4. Click **OK** when the confirmation message is displayed.

Adding an IP verification port number

You must verify what IP addresses are in use before they are deployed. PowerFlex Manager uses default ports 22, 80, and 135 for verification.

1. On the menu bar, click **Settings** and click **Networking**.
2. Under **IP Verification Port Number**, click **Edit**.
3. In the **Port Numbers Used for IP Verification** box, add or remove the port numbers used by PowerFlex Manager to verify the IP addresses.

   Port numbers must be entered in comma-separated list, and must be between 1 and 65535.
4. Click **Save**.

To remove a port number, click **X** to the right of the port number.

System data networks

The **System Data Networks** page displays information about system data networks for use with NVMe/TCP. This information includes:

- **Name**
- **Network**
- **Network Mask**

On the **System Data Networks** page, you can:

- Add a system data network.

- Rename a system data network.
- Remove a system data network.

Adding a system data network

Use this procedure to add a new data network for NVM/TCP host connectivity.

To add a system data network, complete the following steps:

1. On the menu bar, click **Settings** and then click **Networking**.
2. Click **System Data Networks**.
   The **System Data Networks** page opens.
3. Click **Add**. The **Create System Data Network** page opens.
4. In the **Name** field, enter the name of the network.
5. In the **Network** field, enter the IP address of the network.
6. In the **Network Mask** field, enter the network mask.
7. Click **Create**.

Renaming a system data network

After you have added a system data network, you can edit the network name.

1. On the menu bar, click **Settings** and then click **Networking**.
2. Click **System Data Networks**.
   The **System Data Networks** page opens.
3. Select the network that you want to edit and click **Rename**. The **Rename System Data Network** page opens.
4. Edit the information in the **System Data Network Name**.
5. Click **Apply**.

Removing a system data network

After you have added a system data network, you can delete it, if necessary.

1. On the menu bar, click **Settings** and then click **Networking**.
2. Click **System Data Networks**.
   The **System Data Networks** page opens.
3. Click the network that you want to delete, and then click **Remove**.
4. Click **OK** when the confirmation message is displayed.

Events and alerts

A source is used to configure the receiving of external events and syslog content.

A destination is used to configure the ability to send events and alerts information out. A destination is always external. Dell Technologies connectivity services, email, SNMP, and remote syslog are considered destinations.

Notification policies define what information is sent to each destination. Events and alerts exist irrespective of whether notification policies are created.

SNMP sources are not automatically discovered and must be configured to receive events about these sources. PowerFlex Manager is preconfigured and events, and alerts are automatically available. Resources in the PowerFlex rack, PowerFlex appliance, and Dell VxFlex Ready Node are automatically discovered. Any future resources, for example, switch replacements or additional nodes, are considered external and must be added manually as sources.

Related information

Enabling Dell Technologies connectivity services

Configuring an external source

You must define a source to enable PowerFlex Manager to receive an external event. You need not create a source to receive PowerFlex events.

External resources such as iDRAC, CloudLink, and Dell PowerSwitch or Cisco Nexus switches have to be discovered on the **Resources** page either in a managed or unmanaged state.

Alerts received from external resources like iDRAC, CloudLink, and Dell PowerSwitch or Cisco Nexus switches are seen only on **Events** page.

PowerFlex related alerts can be seen on both the **Alerts** and **Events** pages.

A syslog source can only go to a syslog destination and does not display in events. An SNMP source, either V2 or V3, displays in events even without a defined notification policy.

1. Go to **Settings** > **Events and Alerts** > **Notification Policies**.
2. From the **Sources** pane, click **Add**.
   The **Add Source** window opens.
3. Enter a source name and description.
4. Configure either SNMP, syslog, or audit log forwarding:
   - If you select **SNMPV2c**:
     a. Enter the community string by which the source forwards traps to destinations.
     b. Enter the same community string for the configured resource. During discovery, if you selected PowerFlex Manager to automatically configure iDRAC nodes to send alerts to PowerFlex Manager, enter the community string that is used in that credential here.
   - If you select **SNMP V3**:
     a. Enter the username, which identifies the ID where traps are forwarded on the network management system.

        NOTE: The username must be at most 16 characters.

     b. Select a security level from the following:

        Table 89. Security levels

Security Level	Details	Description	authPassword	privPassword
Minimal	noAuthNoPriv	No authentication and no encryption	Not required	Not required
Moderate	authNoPriv	Messages are authenticated but not encrypted (Password-based encryption with message digest (MD5) with at least eight characters)	Required	Not required
Maximum	authPriv	Messages are authenticated and encrypted (MD5 and Data Encryption Standard (DES) both with at least eight characters)	Required	Required

        PowerFlex supports SNMP v3 alerts only for iDRAC.

   - If you select **Syslog**, switch **Enable Syslog** right to on.
   - If you select **Audit Log**, click **Enable Audit Log Retention**. You can retain audit logs locally for 15 days.
5. Click **Submit**.

Related tasks

Modifying an external source
Configuring a destination
Modifying a destination
Add a notification policy
Modify a notification policy
Delete a notification policy
Configuring an audit log policy
Add a notification policy

Modifying an external source

You can edit the information about how PowerFlex Manager receives an event.

1. Go to **Settings** > **Events and Alerts** > **Notification Policies**.
2. From the **Sources** pane, click the source that you want to modify.
   The **Edit Source** window opens.
3. Edit the information and click **Submit**.

Related tasks

Configuring an external source
Configuring a destination
Modifying a destination
Add a notification policy
Modify a notification policy
Delete a notification policy
Configuring an audit log policy
Add a notification policy

Configuring a destination

Define a location where the event and alert data that PowerFlex Manager processes should be sent.

1. Go to **Settings** > **Events and Alerts** > **Notification Policies**.
2. From the **Destinations** pane, click **Add**.
   The **Create New Destination Protocol** window opens.
3. From the **Destination Properties** window:
   a. Enter the destination name and description.
   b. From the **Destination Type** menu, select either **SNMP V2c**, **SNMP V3**, **Syslog**, **Email (SMTP)**, **Webhook**, or **Audit Log**.
4. Click **Next**.
   The **Protocol Settings** window opens.
5. Depending on the destination type, enter the following information:

Destination type	Protocol settings
SNMP V2c	• Network name/IP address • Port • Community string
SNMP V3	• Network name/IP address • Port • Username • Security level: ○ Minimal - no more information required

Destination type	Protocol settings
	○ Moderate - MD5 authentication password required ○ Maximum - MD5 authentication and DES privacy passwords required
Syslog	● Network name/IP address ● Port ● Protocol ○ UDP ○ TCP ● Facility ○ All ○ Authentication ○ Security and authentication
Email (SMTP)	● Destination name ● Description ● Destination type ○ Server type: ■ SMTP ■ SMTP over SSL ■ SMTPS SMARTTLS ○ Server IP address or FQDN ○ Port ○ Sender address and up to five recipient addresses ○ If you choose to use credentials enter: ■ Username, password, sender address and up to five recipient addresses ● Send test email ● Test email server connection
Webhook	● Destination name ● Description ● Destination type ○ Select **Webhook**. ○ You can have up to three webhook destinations. ● Webhook destination API URL ○ For example, if you want to configure a webhook destination for BigPanda, you must specify the BigPanda endpoint here. ○ You can get this URL from the BigPanda site. ● Enable credentials ● Credentials ○ For example, for BigPanda, the credential you use must have an App Key and a Token string. You can generate these values on the BigPanda site and then copy them to the App Key and Token fields. ● Test webhook
Audit Log	● Destination name ● Description ● Destination type ○ Select **Audit Log**. ○ You can have up to four audit log destinations. ● Audit log network name or IP address ● Port ● Protocol ○ User Datagram Protocol (UDP) ○ Transmission Control Protocol (TCP)

6. Click **Save**.

Related tasks

Configuring an external source
Modifying an external source
Modifying a destination
Add a notification policy
Modify a notification policy
Delete a notification policy
Configuring an audit log policy
Add a notification policy

Related information

Migrating from key-based licensing to dynamic licensing (connected method)

Modifying a destination

You can edit the information about where event and alert data that is processed by PowerFlex Manager should be sent.

1. Go to **Settings** > **Events and Alerts** > **Notification Policies**.
2. From the **Destinations** pane, click the destination whose information you want to modify.
   The **Edit Source** window opens.
3. Edit the information and click **Submit**.

Related tasks

Configuring an external source
Modifying an external source
Configuring a destination
Add a notification policy
Modify a notification policy
Delete a notification policy
Configuring an audit log policy
Add a notification policy

Add a notification policy

When you add a notification policy, you define the rules for processing events or alerts from sources, and to which destinations that information should be sent.

1. Go to **Settings** > **Events and Alerts** > **Notification Policies**.
2. Click **Create New Policy**.
3. Enter a name and a description for the notification policy.
4. From the **Source Type** menu, select how you want events and alerts to be received. The source type options are:
   - **Snmpv2c**
   - **Snmpv3**
   - **Syslog**
   - **Powerflex**
   - **Auditlog**

   PowerFlex related alerts are generated and forwarded to a webhook destination when the notification policy is configured with the **Source Type** set to **Powerflex**.
5. From the **Resource Domain** menu, select the resource domain that you want to add a notification policy to. The resource domain options are:
   - **All**
   - **Management**
   - **Block (Storage)**
   - **Compute (Servers, Operating Systems, virtualization)**

- **Network (Switches, connectivity etc.)**
- **Security (RBAC, certificates, CloudLink etc.)**

6. Select the checkbox beside the severity levels that you want to associate with this policy. The severity indicates the risk (if any) to the system, in relation to the changes that generated the event message.
7. Select the required destination. You can choose one or more destinations. For a webhook, you can have up to three destinations defined. For audit logs, you can configure up to four destinations.

   NOTE: The audit log policy is disabled by default. Click **Configure Now** to configure and activate the audit log policy.

8. Click **Submit**.

You can send a test event to PowerFlex Manager for a specific or multiple resources by completing the following steps:

1. Select the policy for which you want to send an event.
2. Go to **More Actions** > **Configure to send events in PowerFlex Manager**.
3. Search by text or filter for the resource that you want to send a test event for. The types of filter options available are: **Element Manager**, **Node (Hardware/Software Management)**, and **Switches**.
4. Select one or more checkboxes beside one or more resources that you want to send a test event(s) for from the **Available Resources** table and move them to the **Selected Resources** table.
5. Click **Send Test Event**.

After adding a notification policy, you might need to perform additional configuration steps. For example, If you are setting up a notification policy for a webhook destination that uses BigPanda, you can optionally configure BigPanda to show the severity levels from PowerFlex Manager. To do this, you must configure the status mapping on the BigPanda site. Map the `major` and `minor` severity values from PowerFlex Manager to the `Warning` status for BigPanda.

Related tasks

Configuring an external source
Modifying an external source
Configuring a destination
Modifying a destination
Modify a notification policy
Delete a notification policy
Configuring an audit log policy

Related information

Enabling Dell Technologies connectivity services

Modify a notification policy

You can modify certain settings that are associated with a notification policy.

You cannot modify the source type or destination once it is assigned to a notification policy.

1. Go to **Settings** > **Events and Alerts** > **Notification Policies**.
2. Select the notification policy that you want to modify.
3. You can choose to modify the notification policy in the following ways:
   - To activate or deactivate the policy, click **Active**.
   - To modify the policy, click **Modify**. The **Edit Notification Policy** window opens.
4. Click **Submit**.

Related tasks

Configuring an external source
Modifying an external source
Configuring a destination
Modifying a destination
Add a notification policy
Delete a notification policy
Configuring an audit log policy

Add a notification policy

Delete a notification policy

Use this procedure to delete a notification policy.

NOTE: Once a notification policy is deleted, it cannot be recovered.

1. Go to **Settings** > **Events and Alerts** > **Notification Policies**.
2. Select the notification policy that you want to delete.
3. Click **Delete**.
   You receive a message to confirm if you want to delete the policy.
4. Click **Submit** and click **Dismiss**.

Related tasks

Configuring an external source
Modifying an external source
Configuring a destination
Modifying a destination
Add a notification policy
Modify a notification policy
Configuring an audit log policy
Add a notification policy

Configuring an audit log policy

Audit log is a collection of various events that occur in a system. Audit logs facilitate detection and response to cyber threats. By default, an audit log policy is disabled.

In PowerFlex Manager, the audit log contains details of the following events:

- User authentication and authorization
- System configuration and security policy changes
- Security events

Security Information and Event Manager (SIEM) manages log collection, log analysis, log forensics, real-time alerting, user activity monitoring and retention. About four SIEM can be sent over a secured connection.

Administrators can collect events and alerts from remote systems, also called sources and store them on a central server. The Audit Log Converter receives the events and alerts and forwards them in an encrypted form to the remote SIEM.

The Audit Logging Enabler is required to send logs in real-time on an automated basis. The audit logs are persistent for 10 to 90 days. Audit logs sent through SIEM have a retention period of 12 to 18 months.

1. Go to **Events and Alerts** > **Notification Policies** > **Policies** > **Audit Log Policy**.
2. Click **Configure Now**.
   The **Audit Log Policy** window opens.
3. Select and existing destination to forward the audit logs, from the **Destination** drop-down or click **Create Destination** to create a destination in the **Add Destination** window.

   You can create up to four audit log destinations.

   NOTE: You can skip this step if you do not want to forward the audit logs.
4. Click **Enable Audit Logs** to create a source of the type **Audit Log**.
5. Click **Save**.

Related tasks

Configuring an external source
Modifying an external source
Configuring a destination

Modifying a destination
Add a notification policy
Modify a notification policy
Delete a notification policy
Add a notification policy

License management

PowerFlex offers two approaches to license management—dynamic licensing and key-based licensing. Each approach governs how PowerFlex system functionality is activated and maintained. Whether you have key-based or dynamic licensing you have a free evaluation license for the first 90 days.

Dynamic licensing

The dynamic licensing approach provides an ability to manage license allocations more easily across product registrations, enabling a flexible and keyless experience. It automates and simplifies license management activities, such as registrations, capacity expansions, and renewals. The dynamic licensing approach is applicable for all systems with PowerFlex Manager 5.1.x and later.

The following methods are used to activate and use your products with dynamic licensing:

- Connected method
- Offline method

For more detailed information about dynamic licensing, see the *PowerFlex Family Licensing Guide*.

Key-based licensing

The key-based licensing approach is a traditional method of activating software using a unique product key. Each key is manually applied to a system, and once activated, the license remains tied to that installation. This approach works well for environments that prefer offline activation or need strict per-device licensing control. However, it requires manual management, offers limited automation, and does not adapt to changes in system configuration or usage.

For more detailed information about migrating from key-based to dynamic licensing, see the *PowerFlex Family Licensing Guide*.

Evaluation license

When you deploy PowerFlex, a free evaluation license is automatically provided for 90 days whether you are using dynamic licensing or key-based licensing. This evaluation license allows for full use of the storage cluster for 90 days. During this period, PowerFlex runs in trial mode.

To track when the license expires, go to **License Management** to view the start and end dates for a license.

After the 90-day evaluation period ends:

- Your storage remains intact.
- Configuration changes are disabled until a valid license is applied.

To continue using PowerFlex with full functionality, you must purchase a license that is based on the system capacity.

Related information

Enabling Dell Technologies connectivity services

Licensing PowerFlex for key-based licensing

For the first 90 days after you deploy PowerFlex, no license is required. During this period, you are running PowerFlex in trial mode, and all features are enabled. After the trial period expires, you must upload a key-based paid license through PowerFlex Manager.

Before you upload a key-based license, the MDM cluster must be deployed. If you experience an issue expanding you key-based licensing, contact Dell Technologies Support.

When you are running PowerFlex in trial mode, PowerFlex Manager displays an alert on the **Monitoring** > **Alerts** page.

1. On the menu bar, click **Settings** > **License Management**.
2. Click **PowerFlex License**.

   If you are not migrating to dynamic licensing now, ignore the blue banner that displays at the top of the screen. You can continue to use a key-based license. However, if you need to expand or your license expires, you cannot purchase additional licensing with a key-based license.
3. To upload a license for PowerFlex, click **Choose File** in the **Production License** section,
4. Select the license file, and click **Open**.
5. Click **Save**.

When you upload a license file, PowerFlex Manager checks the license file to ensure that it is valid.

After the upload is complete, PowerFlex Manager stores the license details and displays them on the **PowerFlex Manager License** page. You can see the installation ID, system name, SWID, and license registration method.

In addition, you can see the total licensed capacity and the license capacity left.

Licensing PowerFlex for dynamic licensing (connected method)

To use the connected method for dynamic licensing, first register your PowerFlex instance in the Dell Dynamic Licensing Portal. After registration, you will receive an access key and select a pin, which PowerFlex Manager uses to authenticate with Dell Technologies connectivity services.

Ensure that you registered your PowerFlex instance in the Dell Dynamic Licensing Portal. To register your PowerFlex instance, you must have the license authorization code (LAC) that was emailed to the primary email contact. Or alternatively you can look up the LAC on the Dell Dynamic Licensing Portal.

If you skipped the initial setup wizard and did not enable Dell Technologies connectivity services, go to **Settings** > **Events and Alerts** > **Notification Policies** to configure it.

1. On the menu bar, click **Settings** and then click **License Management** > **PowerFlex License**.
2. From the blue information banner, select **Migrate to Dynamic Licensing**.
   The **Migrate to Dynamic Licensing** dialog opens.
3. Enter the license authorization code (LAC).
4. Accept the license agreement and click **Migrate**.

The **PowerFlex License** page provides the following details that are associated with the license:

- **Researvation Pool Name**: The name of the reservation pool that links one or more installations, which are identified by their cSWID, to one or more entitlements.
- **cSWUID**: The customer software ID that is received as part of registering for dynamic licensing.
- **Installation ID**: The installation ID is a 16-character hexadecimal value that uniquely identifies a PowerFlex system.
- **Product Type**: The type of device and the deployment environment if the device type is software. For example, **PowerFlex software Amazon Web Services**.
- **Connection status**: The method that you used to register your license. For example, connected or offline.

Using the connected method, PowerFlex Manager sends a consumption report to the Dell Dynamic Licensing portal every four hours. If an error occurs or the compliance state changes, a report is sent every 15 minutes until the system is compliant.

Related information

Enabling Dell Technologies connectivity services

Migrating from key-based licensing to dynamic licensing (connected method)

Licensing PowerFlex for dynamic licensing (offline method)

You can manually license your PowerFlex instance for dynamic licensing when you are offline. When you license using the offline method, the system completes the process without connecting to the Dell licensing servers. You must download the license registration file from the Dell Dynamic Licensing portal.

You must manually register your PowerFlex instance in the Dell Dynamic Licensing Portal. Download the license registration file from the *Register Your Product* section, and save it locally. The license registration file is the dynamic license file that you will upload through PowerFlex Manager.

1. On the menu bar, click **Settings** and then click **License Management** > **PowerFlex License**.
2. From the blue information banner, select **Migrate to Dynamic Licensing**.
   The **Migrate to Dynamic Licensing** dialog opens.
3. To upload the license registration file, under **License Registration**, drag and drop the license registration file or browse to the location where the license registration file is stored locally.
4. Once the license registration file is uploaded, accept the license agreement and click **Migrate**.

Your existing license and entitlements are automatically migrated to the Dell Dynamic Licensing portal. No further action is required.

Migrating from key-based licensing to dynamic licensing (connected method)

You can migrate from key-based licensing to dynamic licensing through PowerFlex Manager when you are connected to Dell Technologies connectivity services. You must first deregister and then reregister with Dell Technologies connectivity services.

You must be registered with Dell Technologies connectivity services to migrate to dynamic licensing.

If you are already registered with Dell Technologies connectivity services, you must reregister with Dell Technologies connectivity services before you migrate.

Once you migrate to dynamic licensing, you cannot revert to key-based licensing.

1. On the menu bar , click **Settings** > **Events and Alerts** > **Notification Policies**.
2. Before deleting the existing policy and deregistering Dell Technologies connectivity services, go to **Policies**, select the policy, and click **Delete**.
3. Click **Submit** to confirm deletion of the policy.
4. Click **Dell Technologies connectivity services**.
5. From the **Destination Protocol Settings (Dell Technologies connectivity services)** window, click **Deregister Dell Technologies connectivity services**.
6. From the **Warning** window, click **Submit**.
7. Under **Notification Policies**, click **Dell Technologies connectivity services**.
8. Click **I accept this agreement** to accept the telemetry agreement and click **Next**.
9. From **Connect to Dell Technologies connectivity services**, click **Next**.
10. Under **Connection Type**, select either of the following connection type:
    - To connect directly to Dell Technologies, click **Connect Directly**. Click **Next**.
    - To connect to a Dell Technologies connectivity services gateway server by remote access, click **Connect via Gateway Server** and provide the primary gateway IP address or **Primary Gateway IP/Hostname** and **Gateway Port**.Click **Next**.
11. Under **Authentication Details**, enter the **Access Key** and **Pin**. You can obtain the access key and self-select a pin from Support for PowerFlex Software .
12. Select **I have a dynamic license authorization code (LAC)**.
13. Enter the **Support Contact** details and click **Finish**.
14. From **Settings**, click **License Management** and click **Migrate to Dynamic Licensing**.
15. Accept the license agreement and click **Migrate**.

Your existing license and entitlements are automatically migrated. No further action is required.

Related information

Licensing PowerFlex for dynamic licensing (connected method)

Configuring a destination
Enabling Dell Technologies connectivity services

Migrating from key-based licensing to dynamic licensing (offline method)

You can migrate from key-based licensing to dynamic licensing through PowerFlex Manager when you are offline.

You must manually register your PowerFlex instance. Log in to the Dell Dynamic Licensing Portal, download the license registration file from the **Register Your Product** section, and save it locally. The license registration file is the dynamic license file that you upload through PowerFlex Manager to migrate.

Once you migrate to dynamic licensing, you cannot revert to key-based licensing.

1. On the menu bar, click **Settings** and then click **License Management** > **PowerFlex License**.
2. From the blue information banner, select **Migrate to Dynamic Licensing**.
   The **Migrate to Dynamic Licensing** dialog opens.
3. To upload the license registration file, under **License Registration**, drag and drop the license registration file or browse to the location where the license registration file is stored locally. For example: **<dynamic_license_file>.dl**.
4. Once the license registration file is uploaded, accept the license agreement, and click **Migrate**.

Your existing license and entitlements are automatically migrated to the Dell Dynamic Licensing Portal. No further action is required.

Uploading other software licenses

You can also upload other software licenses, in addition to the licenses required for PowerFlex. For example, you might want to upload a license for CloudLink.

1. On the menu bar, click **Settings** and then click **License Management**.
2. Click **Other Software Licenses**.
3. Click **Add**.
4. Click **Choose File** and select the license file.
5. Choose the **Type** of license that you want to upload.
6. Click **Save**.

To remove a license, select it and click **Remove**.

Security

On the **Security** page, you can upload SSL trusted certificates to enable secure browser access to PowerFlex Manager and to establish secure LDAP connection to Active Directory. You can also define credentials for the resources that PowerFlex Manager accesses and manages.

Adding an appliance SSL certificate

To enable secure communication between external resources (such as a web browser) and PowerFlex Manager, you must replace the defualt appliance SSL certificate with a certificate that is signed by a trusted certificate authority (CA).

The process to add an appliance SSL certificate involves the following steps:

1. Generate a certificate signing request (CSR) in PowerFlex Manager. The generation of a CSR is supported through PowerFlex Manager only.
2. Download the CSR generated by PowerFlex Manager.
3. Submit the CSR to a trusted CA for signing and retrieve the signed certficate.
4. Upload the signed SSL certificate to PowerFlex Manager.

A root certificate is a self-signed certificate and serves as the trust anchor in a certificate chain. It is not signed by any other certificate authority (CA).

An intermediate certificate is signed by either a root CA or another intermediate CA, creating a chain of trust between the root CA and the end certificate.

If the appliance SSL certificate is signed by an intermediate CA, upload both the root and intermediate CA certificates. To ensure that the complete certificate chain can be validated, upload the intermediate certificates seperately through the **SSL Trusted Certificates** screen. Upload each intermediate certificate one at a time.

Once the CA-signed certificate(s) is uploaded, PowerFlex Manager allows external resources (such as a web browser) that trust the CA to securely verify the server's identity and establish an encrypted HTTPS connection without certificate warnings.

Generating a certificate signing request

A certificate signing request (CSR) includes information that the certificate authority (CA) requires to provide a valid SSL certificate. After generating the CSR, download the encrypted text, and then submit it to the CA. The CA validates the text, and once approved, issues a signed SSL certificate for you to upload.

You can select a **Standard** or a **Stringent** security level when you generate the certificate signing request.

- Use the **Standard** security level when the certificate's Subject Alternative Name (SAN) includes both IP addresses and FQDNs in the certificate.
- Use the **Stringent** security level if your signing authority does not allow IP addresses in the certificate. The FQDNs used in the certificate must resolve to DNS records.

  If you use the Stringent security level, you must enter an FQDN for each listed ingress IP addresses. If the data networks do not use FQDNs, you can reuse the management FQDN for the data networks. If the out-of-band (OOB) Ingress is physically isolated from management, you must include an FQDN specific to OOB and the FQDN must also resolve to DNS records.

1. On the menu bar, click **Settings** and then click **Security**.
2. Click **Appliance SSL Certificates**.
   The **Appliance SSL Certificates** page opens.
3. Click **Generate Certificate Signing Request**.
   a. Under **Security Level**, select either **Standard** or **Stringent**. The default is **standard**.
   b. In the **MGMT Distinguished Name (www.domain.com)** box, enter a distinguished name in the format `www.example.com`.
   c. In the **Business Name** box, enter a business name where the certificate is recorded.
   d. In the **Department Name** box, enter a department name of the organizational unit (for example, IT, HR, or Sales) for which the certificate is generated.
   e. In the **Locality (Town/City)** box, enter a locality name in which the organization is located.
   f. In the **State (Province/Region)** box, enter a state name in which the organization is located (do not abbreviate).
   g. From the **Country** list, select a country or region in which the organization is located.
   h. In the **Email** box, enter a valid email address.
   i. Click **Generate**.

Downloading the certificate signing request

After generating the certificate signing request (CSR) and the Certificate Authority (CA) approves the CSR, the CA approves the CSR, you can download the certifictae file.

1. On the menu bar, click **Settings** > **Security**.
2. Click **Appliance SSL Certificates**.
   The **Appliance SSL Certificates** page opens.
3. Click **Download Certificate Signing Request**.
4. To receive a valid appliance SSL certificate, copy the displayed text and then submit the text to a certificate authority.

CAUTION: Before uploading an appliance SSL certificate, verify that the certificate file meets the following requirements:

- **The file must contain an X.509 certificate in PEM format.**
- **The file must start with `---BEGIN CERTIFICATE---` and end with `---END CERTIFICATE---`.**
- **The file must have a `.pem` extension.**

Uploading an appliance SSL certificate

If the appliance SSL certificate is signed by an intermediate CA, upload both the root CA certificate and all the required intermediate CA certificates. To ensure that the complete certificate chain can be validated, upload the intermediate certificates separately through the **SSL Trusted Certificates** screen. Upload each intermediate certificate one at a time..

1. On the menu bar, click **Settings** > **Security**.
2. Click **Appliance SSL Certificates**.
   The **Appliance SSL Certificates** page opens.
3. Click **Choose File** under **SSL Certificate** and select the signed SSL certificate returned by your certificate authority (CA).
4. Click **Choose File** under **Trusted CA Certificate** and select the root CA certificate.
5. To upload the certificate, click **Save**.

Adding a trusted SSL certificate

You can add a trusted SSL certificate to validate external systems that PowerFlex Manager connects to, for example, Active Directory.

Before you upload a trusted SSL certificate, you must obtain the certificate file:

CAUTION: The certificate file must meet the following requirements:

- **The file must contain an X.509 certificate in PEM format.**
- **The file must start with `---BEGIN CERTIFICATE---` and end with `---END CERTIFICATE---`.**
- **The file must have a `.pem` extension.**

- Review and update your management node credentials. Go to **Settings** > **Security** > **Update Management Node Credentials**.

1. On the menu bar, click **Settings** > **Security**.
2. Click **SSL Trusted Certificates**.
   The **SSL Trusted Certificates** page opens.
   NOTE: To add a root certificate signed by a root authority, complete steps 3 through 6. To add an intermediate certificate signed by an intermediate authority, complete steps 3 through 6 to add the root certifcate and then repeat steps 3 through 6 to add the intermediate cerificate.
3. Click **Add**.
4. Click **Choose File** and select the SSL certificate.
5. Enter a **Name** for the certificate. The name must be a single word.
6. To upload the certificate, click **Save**.

To delete a trusted SSL certificate, select the certificate, and click **Delete**.

Adding resource credentials

PowerFlex Manager requires a root-level username and password to access and manage nodes, switches, VMware vCenter, element managers, PowerFlex gateway, and operating system resources.

The **Credentials Management** page displays the following information about the credentials:

- **Name**—A user-defined name that identifies the credentials.
- **Type**—A type of resource that uses the credential.
- **Resources**—The total number of resources to which the credential is assigned.

From the credential list, click a credential to view its details in the **Summary** tab:

- Name of the user who created and modified the credential.
- Date and time that the credential was created and last modified.

On the **Credentials Management** page, you can:

- Create credentials
- Edit existing credentials
- Delete existing credentials

Create credentials

Do the following to create credentials:

1. On the menu bar, click **Settings** and click **Security**.
2. Click **Resource Credentials**.
   The **Credentials Management** page opens.
3. Click **Create**.
4. In the **Create Credentials** dialog box, from the **Credential Type** drop-down list, select one of the following resource types for which you want to create the credentials:
   - **Node**
   - **Switch**
   - **vCenter**
   - **Element Manager**
   - **PowerFlex Gateway**
   - **OS Admin**
   - **OS User**
   - **PowerFlex Management System**

   The **OS Admin** and **OS User** credential types apply to deployed items, not to PowerFlex Manager itself.
5. In the **Credential Name** field, enter the name to identify the credential.
6. Click **Enable Key Pairs** to enable login with SSH key pairs:

   To enable key pairs for the **Node** or **Switch** credential type:

   a. Import an existing key:
      i. Click **Import SSH Key Pair**.
      ii. Click **Choose File** and browse to the file that contains your public and private keys, and select the private key.
      iii. Type a name for the key pair.
      iv. Click **Import**.

   To enable key pairs for the **OS Admin** or **OS User** credential type:

   a. To create a key:
      i. Click **Create a new key**.
      ii. Click **Create & Download Key Pair**.
      iii. Type a name for the key pair.
      iv. Click **Create**.

      The private key file (id_rsa) is downloaded to the downloads folder. Click the **Download Public Key** button to download the public key file (id_rsa.pub).

   b. To import an existing key:
      i. Click **Import existing key**.
      ii. Click **Import SSH Key Pair**.
      iii. Click **Choose File** and browse to the file that contains your public and private keys.
      iv. Type a name for the **Key Pair Name field**.
      v. Click **Import**.

   If you enable SSH key pairs for a **Node** or **Switch** credential and use that credential for discovery, PowerFlex Manager uses public or private RSA key pairs to SSH into your node or switch securely, instead of using a username and password. If you enable SSH key pairs for an **OS user** or **OS Admin** credential and use that credential for a deployment, PowerFlex Manager uses RSA public or private key pairs for the deployment operations.

   NOTE: PowerFlex Manager does not consume SSH keys for all component types. For example, if you enable SSH key pairs for an admin credential, the SSH keys are not used for the deployment of a CloudLink Center VM. In this case, the username and password would be used instead for all communication.
7. In the **Domain** box, optionally specify an LDAP domain for the user.
8. In the **User Name** field, enter the username for the credential.

   `root` is the only valid username for root-level credentials on nodes (iDRAC). You can add iDRAC users with a username other than **`root`**.

For the **OS User** credential type, you can enter a user other than root. For the embedded operating system, this user account must have SSH enabled and have sudo access. For VMware ESXi, the account must be configured with the administrator role on the local server permission setting, which should enable SSH and other tools like esxcli. You can add existing resource groups with a non-root user.

The account on the SVM and/or storage-only nodes for the **OS User** credential type must have a `/home` directory and have the correct group permissions. For example, if the account were pfxm_admin, the home directory would be: `/home/pfxm_admin`

Here is an example showing the requirements for a pxflex user in the pxflex-grp group on an SVM:

```
[root@svm-dkim-hc-node1 home]# ls -alhtr
        total 0
        drwxr-xr-x.  3 root   root        20 Feb 26 16:03 .
        drwxr-xr-x   3 pxflex pxflex-grp  22 Feb 26 16:10 pxflex
        dr-xr-xr-x. 17 root   root       224 Feb 26 16:37 ..
        [root@svm-dkim-hc-node1 home]# cd pxflex/
        [root@svm-dkim-hc-node1 pxflex]#
        [root@svm-dkim-hc-node1 pxflex]#
        [root@svm-dkim-hc-node1 pxflex]# pwd
        /home/pxflex
```

For the **OS Admin** credential type, the **User Name** field is disabled because the user is assumed to be root. You must use the root user for new deployments.

Provide two usernames for the PowerFlex gateway credential type:

- **Gateway Admin User Name**
- **Gateway OS User Name**

The gateway admin user is the REST API administrator. The gateway operating system user is the SSH login user. The gateway admin user must be the **`admin`** user, and the gateway operating system user must be **`root`**.

9. In the **Password** and the **Confirm Password** boxes, enter the password for the credential.

   NOTE: When the SSH key pair feature is enabled, the switch credential does not require the **Password** option.

   For the PowerFlex gateway credential type, provide two passwords:

   - **Gateway Admin Password**
   - **Gateway OS Password**

   PowerFlex Manager allows you to supply different passwords for the gateway admin and gateway operating system users. The passwords are validated when you discover the PowerFlex gateway on the **Resources** page. The discovery fails if the passwords were not specified correctly on the **Credentials Management** page. PowerFlex Manager does not update the gateway admin and gateway operating system passwords on the node itself. However, if you change the password manually for either of these users, you can update the password on the **Credentials Management** page, and PowerFlex Manager ensures that subsequent operations on the resource group use the new password.

   Optionally, provide additional settings for VMware vCenter and element manager, and for switch credentials:

   - For VMware vCenter and element manager, in the **Domain** box, optionally enter the domain ID.
   - For switch credentials, under **Protocol**, optionally click one of the following connection protocols that are used to access the resource from remote:
     - **Telnet**
     - **SSH**

       NOTE: SSH is enabled on supported switches by default.

10. To configure trap receiving for SNMPv2:
    a. Under **SNMP Configuration**, select **V2** as the SNMP type.
    b. Click **+** beside the **SNMP v2 Community String** box.
       The **SNMP v2 Community String** page opens.
    c. Enter the community string by which PowerFlex Manager receives traps from devices and by which it forwards traps to destinations.
    d. Click **Save**.

NOTE: You can add more than one community string. For example, add more than one if the community string by which PowerFlex Manager receives traps differs from the community string by which it forwards traps to a remote destination.

11. To configure trap receiving for SNMPv3:
    a. Under **SNMP Configuration**, select **V3** as the SNMP type.
    b. Click **+** beside the **SNMP V3 User** box.
    The **SNMP V3 User** page opens.
    c. Enter the **Username**, which identifies the ID where traps are forwarded on the network management system.

    NOTE: The username must be at least 16 characters.

    d. Select a **Security Level** from the following:

- **Minimal** - No additional information is required.
- **Moderate** - Provide an **MD5 Authentication Password** which is at least eight characters.
- **Maximum** - Provide an **MD5 Authentication Password** and a **DES Privacy Password** which is at least eight characters.

You can add more than one user.

NOTE: You can only select SNMPv3 for nodes. Select the **Auto configure nodes to send alerts to PowerFlex Manager** checkbox to discover iDRACs.

12. Click **Save**.

Create credentials for root and non-root users

Use this procedure to create credentials for root and non-root users in PowerFlex Manager.

To import and create the SSH keys for a PowerFlex node, switch, **OS Admin**, **OS User**, ensure you generate SSH key pairs of RSA type without passphrase.

You can now use a non-root user instead of the root user for PowerFlex system administration functions. This enhances security by disabling the root user during node discovery, operating system installation, and non-disruptive updates. The default non-root user name is **`pflex`**.

NOTE: The credential type **OS Admin** is used for root users, and **OS User** is used for non-root user. **OS Admin** and **OS User** credential types apply for the deployment of resource groups.

For non-root user authentication, after the deployment of the resource group, the SSH access for the user root is disabled, but the password is still available for to use the console for troubleshooting.

PowerFlex Manager allows you to specify a non-root user when you configure a template for a compute-only, storage-only deployment.

1. On the menu bar, click **Settings** > **Security**.
2. Click **Resource Credentials**. The Credentials Management page opens.
3. Click **Create**.
4. In the **Create Credentials** dialog box, from the **Credential Type** drop-down list, select one of the following resource types for which you want to create non-root credentials:
   - **Node**
   - **Switch**
   - **OS Admin**
   - **OS User**

   The **OS Admin** and **OS User** credential types apply to deployed items, not to PowerFlex Manager. If you are creating an OS user credential set for the management virtual machines on a PowerFlex management controller resource group, select **OS User**.
5. In the **Credential Name** field, enter the name to identify the credential.

   NOTE: If you are creating an **OS User** credential set for the management virtual machines on a PowerFlex management controller resource group, do the following:

   a. Enter **`MVM delladmin`** to identify the credential.
   b. In the **User Name** field, enter **`delladmin`**.

c. Enter the delladmin account password in the **Password** and **Confirm Password** fields.

6. Click **Enable Key Pairs** to enable log in with SSH key pairs and perform the following:

Table 90. Key pairs options

To...	Do this...
Enable key pairs for the **Node** or **Switch credential**:	a. Click **Import SSH Key Pair**. **NOTE:** Manually generate the SSH keys pairs. b. Click **Choose File** and browse to the file that contains the private key. c. Type a name for the key pair. d. Click **Import**.
Create keys using PowerFlex Manager for the **OS Admin** or **OS User** credential and enable key pairs:	a. Click **Create a new key**. b. Click **Create & Download Key Pair**. c. On **Key Pair Name**, type the name for key pair. d. Click **Create**. e. Click **Download Public Key**.
To manually generate and import an existing key pairs for the **OS Admin** or **OS User** credential	a. Click **Import SSH Key Pair**. **NOTE:** Manually generate the SSH keys pairs. b. Click **Choose File** and browse to the file that contains the public and private key. c. Type a name for the key pair. d. Click **Import**.

If you enable SSH key pairs for a **Node** or **Switch credential** and use that credential for discovery, PowerFlex Manager uses public or private RSA key pairs to SSH into your node or switch securely, instead of using a user name and password.

If you enable SSH key pairs for an **OS User** or **OS Admin** credential and use that credential for a deployment, PowerFlex Manager uses RSA public or private key pairs for the deployment operations.

NOTE: PowerFlex Manager does not consume SSH keys for all component types. For example, if you enable SSH key pairs for an admin credential, the SSH keys are not used for the deployment of a CloudLink Center VM. Instead, the user name and password are used instead for all communication.

7. In the **User Name** field, enter the username for the credential.

   For **Nodes** (iDRAC), `root` is the only valid username for root-level credentials. For a non-root user name, enter the default non-root user name.

   For the **OS Admin** credential type, the **User Name** field is disabled because the user is assumed to be root. You must use the root user for new deployments.

   For the **OS User** credential type, enter the default non-root user name.

   For the embedded operating system, this user account must have SSH enabled and have sudo access. For VMware ESXi, the account must be configured with the administrator role on the local server permission setting, which should enable SSH and other tools like esxcli. You can add existing resource groups with a non-root user. The account on the PowerFlex storage-only nodes for the **OS User** credential type must have a /home directory and have the correct group permissions.

8. In the **Password** and the **Confirm Password** boxes, enter the password for the credential.

Modify credentials

You can modify your credentials to update outdated information, improve security, and ensure that your system can correctly authenticate you for the resources you need.

1. On the menu bar, click **Settings** and click **Security**.
2. Click **Resource Credentials**.
   The **Credentials Management** page opens.

3. Select a credential that you want to edit, and click **Modify**.
4. Modify the credential information in the **Modify Credentials** dialog box.
5. Click **Save**.

Remove credentials

You can remove your credentials to eliminate outdated or unnecessary access and maintain stronger security.

1. On the menu bar, click **Settings** > **Security**.
2. Click **Resource Credentials**.
   The **Credentials Management** page opens.
3. On the **Credentials Management** page, select the credential that you want to delete, and click **Remove**.
4. Click **OK** when the confirmation message appears.

Converting a private key from PuTTY format to OpenSSH format

Perform this procedure to convert a private key from PuTTY to OpenSSH format.

1. Navigate to the folder where PuTTY was installed.
2. Double-click `puttygen.exe` to open the **PuTTY Key Generator**.
3. Click **File** > **Load private key** on the menu bar.
4. Select the private key that you want to convert to OpenSSH format and click **Open**.
5. Click **Conversions** > **Export open SSH key** on the menu bar.
6. Click **Save** to save the key.
   The key should now be in OpenSSH format.
7. Copy the public key content that is displayed on the **Key** tab in the **PuTTY Key Generator**.
   You need this content so you can copy it to the nodes.

Configuring SSH key pairs

Perform this procedure to configure SSH key pairs for nodes and switches.

1. To configure an SSH key pair for a software-only node, run these commands:

```
# below command will generate SSH Key pair in /tmp directory
ssh-keygen -m PEM -b 2048 -t rsa -f /tmp/sshkey-key1 -q -N ""

# To copy public key to software only node
ssh-copy-id -i /tmp/sshkey-key1.pub root@100.68.105.233

# Keys can be copied for both root and non-root users
```

2. To configure an SSH key pair for a Cisco Nexus switch, run these commands to add the public key:

```
username pfxmsvc password <password> role priv-15

username pfxmsvc sshkey ssh-rsa
AAAAB3NzaC1yc2EAAAADAQABAAABAQDWxvVQ9PgfFI3NMsg3TsR9G89ykZ3UEyciz47hcXH/
YXNakva5PV5l7ubHY4nUxKj3zpVHXg/Bq9zRK5lQFs4KGC5eaYB9nm2xEwhuhpvsCvROo8SzjjTNRcpok2EMD/
3VhuZAWdaXITSukq2D1Ek5W7qeE0LiLSC+2s0u6iPTU5KHs3mrELbRlxpVxnHQtrgUZnsLvkxwzuEnBKWNIEZ0
6HoDLP1vBCaYJslE5iDseUCZstWvFbvkaeE4FOrqohllAhVGVH19B6Q38ijVjLAwjXX03a6IQqeWj+JkpHpB61
vMsAqqje4pm9VFzAY2NAcvoBo7Oab+52qA1vJTCbG7

username pfxmsvc passphrase  lifetime 99999 warntime 14 gracetime 3
```

3. To configure an SSH key pair for an OS10 switch, run these commands to add the public key:

```
username pfxmsvc password <password> role sysadmin priv-lvl 15

username pfxmsvc sshkey "ssh-rsa
```

```
AAAAB3NzaC1yc2EAAAADAQABAAABAQDWxvVQ9PgfFI3NMsg3TsR9G89ykZ3UEyciz47hcXH/
YXNakva5PV5l7ubHY4nUxKj3zpVHXg/Bq9zRK5lQFs4KGC5eaYB9nm2xEwhuhpvsCvROo8SzjjTNRcpok2EMD/
3VhuZAWdaXITSukq2D1Ek5W7qeE0LiLSC+2s0u6iPTU5KHs3mrELbRlxpVxnHQtrgUZnsLvkxwzuEnBKWNIEZ0
6HoDLP1vBCaYJslE5iDseUCZstWvFbvkaeE4FOrqohllAhVGVH19B6Q38ijVjLAwjXX03a6IQqeWj+JkpHpB61
vMsAqqje4pm9VFzAY2NAcvoBo7Oab+52qA1vJTCbG7"
```

Note that the quotes must be used as shown in the command line above.

4. To configure an SSH key pair for iDRAC, perform these steps:
   a. Connect to iDRAC.
   b. Choose **iDRAC Settings** > **Users** > **Edit/Add User** > **SSH Key Configuration** > **SSH Key (Edit)**.
   c. Add the public key.

Serviceability

On the **Serviceability** page, you can generate a troubleshooting bundle and also perform a backup of the appliance. To restore from a backup, you need to run a script outside of PowerFlex Manager. The user interface does not support the ability to restore from a backup.

Generating a troubleshooting bundle

You can generate a troubleshooting bundle which is a compressed file that contains logging information for PowerFlex Manager managed components.

If necessary, download the bundle and send it to Dell Technologies Support for issue debugging.

The troubleshooting bundle includes the following logs:

- ASM deployer
- iDRAC life cycle
- Dell PowerSwitch switch
- Cisco Nexus switch
- VMware ESXi
- CloudLink Center
- Standard output logs from all pods
- Kubernetes logs about pods, services, deployments, secrets, drivers, and volumes
- PowerFlex Block logs
- PowerFlex data server and device gateway target logs

NOTE: You can also generate troubleshooting bundles from the **Resource Groups** page.

1. On the menu bar, click **Settings** > **Serviceability**.
2. Click **Generate Troubleshooting Bundle**.
3. Select **Include PowerFlex Block Core Dump logs** to include core dump logs.
4. Select a **PowerFlex Log Level**:
   - Default Node Logs
   - Default Node Logs plus additional MDM information
   - Latest Logs only (Most recent copy of all logs)
5. Select a **Node Selection**:
   - Logs from all nodes
   - Select Specific Nodes

   If you select the **Select Specific Nodes** option and select the number of nodes for which you want to generate the log, the **View/Select Nodes** button is displayed. Click the button to view the list of nodes in the **Node Selection** window. Select the required nodes from the **Available Nodes** list and click **>>** to view them in the **Selected Nodes** list. Click **Save** to return to the **Generate Troubleshooting bundle** page.

   For the **Select Specific Nodes** option **Generate** is enabled only if a node is selected in the **Node Selection** window.
6. Choose a **Delivery Option**:

Option	Description
Send to Configured Dell Technologies connectivity services	If you are using Dell Technologies connectivity services, **Send to Configured Dell Technologies connectivity services** is selected by default. Leave this default setting. If Dell Technologies connectivity services is not configured, this option is disabled.
Download to Share	Provide a path using the following format with credentials: `CIFS:\\1.1.1.1\uploadDirectory`. Enter a username and password, and click **Test Connection** to verify the connection to the CIFS share before generating the bundle.
Download Locally	Download the troubleshooting bundle to a local file: `\\IP Address\Any folder`

7. Click **Generate**.

Sometimes, the troubleshooting bundle does not include log information for all the nodes. The log collection may appear to succeed, but the log for one or more of the nodes may be missing. You may see an error message in the scaleio.trace.log file that says `Could not run get_info script`. If you see this message, you must generate the troubleshooting bundle again to include information for all the logs.

Pausing an inventory job

An inventory job runs daily at midnight. When a PowerFlex management platform or a resource group/devices upgrade is in progress, both jobs compete for resources. The scheduled (nightly) inventory job is automatically disabled at the start of any upgrade and resumes immediately once the upgrade completes minimizing the load.

In addition to this automatic action on upgrade, you can manually pause scheduled inventory jobs for up to 24 hours using the PowerFlex Manager user interface. This manual block does not effect on-demand inventory actions—you can still run inventory for individual devices through **Lifecycle** > **Resources**.

The status of the inventory job displays on the upper right hand of the screen of the **Resources** page.

1. On the menu bar, click **Settings** > **Serviceability**.
2. Click **Inventory Job**.
3. Choose one of the following options:
   - If you want the inventory job to run at the next scheduled time, set the switch to **On**.
   - If you want to pause the inventory job, set the switch to **Off**.

   The name of the user who set the switch, the time it was set, and the time it expires is displayed under the switch.

Back up and restore

Creating a backup saves all user-created data to a remote share from which it can be restored. To restore from a backup, you need to run a script outside of PowerFlex Manager. The user interface does not support the ability to restore from a backup.

Perform frequent backups to guard against data loss and corruption. The best practice is to take a snapshot of PowerFlex Manager every time you perform a restore.

The **Backup** page displays information about the last backup operation that was performed on PowerFlex Manager. The information provided applies to both manual and automatically scheduled backups and includes the following:

- Last backup date
- Last backup status
- Back up directory path to a CIFS share
- Back up directory username

The **Backup** page also displays information about the status of automatically scheduled backups (enabled or disabled).

On this page, you can:

- Manually start an immediate backup
- Edit general backup settings
- Edit automatically scheduled backup settings

After performing a backup operation, you need to run a script outside of PowerFlex Manager to restore from the backup. The user interface does not support the ability to restore from a backup.

Editing backup settings and details

You can change the location where backup files are saved or the password that is required to access a backup file.

1. On the menu bar, click **Settings** and then click **Serviceability**.
2. Click **Backup**.
   The **Backup** page opens.
3. Click **Backup Settings**.
4. Indicate the network share location where the backup file is saved, enter a backup directory path in the **Backup Directory Path** box.

   Use the following format:

   - CIFS—`\\host\share`
5. If the username and password are required to access the network share, enter a username and password in the **Backup Directory User Name** and **Backup Directory Password** boxes.
6. Click **Test Connection** to confirm that the backup settings you provided are correct.
7. To open the backup file, enter a password in the **Encryption Password** box.
8. To verify the encryption password, enter the password in the **Confirm Encryption Password** box.
9. To schedule automatic backups, next to **Scheduled Backups**, select **Enabled**. To discontinue automatically scheduled backups, deselect **Enabled**.
10. To specify the days on which backup must occur, select the days under **Frequency**.
11. From the **Run Time** drop-down list, select the time.
12. Click **Save**.

Backing up

In addition to automatically scheduled backups, you can manually run an immediate backup.

1. On the menu bar, click **Settings** > **Serviceability**.
2. Click **Backup**.
   The **Backup** page opens.
3. Click **Backup Now**.
4. Select one of the following options:
   - To use the general settings that are applied to all backup files, select **Use Backup Directory Path and Encryption Password from Settings and Details**.
   - To use custom settings:
     a. In the **Backup Directory Path** box, enter a file path where the backup file is saved. Use this format:
        - CIFS—`\\host\share`
     b. Optionally, enter a username and password in the **Backup Directory User Name** and **Backup Directory Password** boxes, if they are required to access the location you provided.
     c. Click **Test Connection** to confirm that the backup settings you provided are correct.
     d. In the **Encryption Password** box, enter a password that is required to open the backup file, and verify the encryption password by entering the password in the **Confirm Encryption Password** box.
5. Click **Backup Now**.

Restoring PowerFlex Manager

Restoring PowerFlex Manager returns user-created data to an earlier configuration that is saved in a backup file. To restore from a backup, you must run a script outside of PowerFlex Manager. The user interface does not support the ability to restore from a backup.

Before you begin the restore procedure, you must satisfy these prerequisites:

- The restore cluster must be exactly the same PowerFlex version and Kubernetes version.
- The restore cluster must have exactly the same IP addresses and configuration.

  The cluster configuration must be the same as the cluster configuration where the backup was taken.
  - All Kubernetes nodes must have the same IP addresses.

- All Kubernetes nodes must have the same names.
- All LoadBalancer IP addresses must be the same.

CAUTION: Restoring an earlier configuration restarts PowerFlex Manager and deletes the data created after the backup file to which you are restoring. Any running jobs could be terminated as well.

1. Log in to the node where the PowerFlex Manager platform installer was initially run.
2. Run the restore script that is included with the installer bundle:

```
./restore_backup.sh
```

3. Provide details as needed to complete the execution of the script.

   You will be prompted to provide the SSH username and password. In addition, you will be asked to specify whether the passwords are the same for all nodes, and also provide the location of the backup file and the encrypted password for the file. You may also be asked to provide the CIFS username and password. The CIFS credentials may not be required for a CIFS share that allows for anonymous access.

   NOTE: The passwords must be in base64 encoded format.

   To complete the execution of the restore script, you must specify whether the restore operation will be performed on an existing cluster or a new cluster.

   Here is a snippet that shows a sample run of the restore script:

```
[root@sheetal-installer scripts]# ./restore_backup.sh

/usr/local/lib/python3.8/site-packages/paramiko/transport.py:236:
CryptographyDeprecationWarning: Blowfish has been deprecated "class":
algorithms.Blowfish, Installation logs are available at <Bundle root>/PFMP_Installer/
logs/ directory. More detailed logs are available at <Bundle root>/atlantic/logs/
directory. PFMP Installer is about to reset a PFMP cluster based on the configuration
specified in the PFMP_Config.json.

Please enter the ssh username for the nodes specified in the
PFMP_Config.json[root]:root

Are passwords the same for all the cluster nodes[Y]?:Y

Please enter the ssh password for the nodes specified in the PFMP_Config.json.
Password:

Please enter backup file location - Example CIFS: \\1.2.3.4\Lab\backup\file.tgz

Please enter CIFS username. Press enter to skip if username is not
required:administrator

Please enter CIFS password(base64 encoded). Press enter to skip if username is not
required: UmFpZDR1cyE=

Please enter encryption password for backup file (base64 encoded): UmFpZDR1cyE=

Perform restore on existing cluster? Please enter yes/y or no/n :
```

The restore process prints status information until the restore is complete.

Related information

Retrieve the configuration file

Back up and restore using VM snapshots

If the PowerFlex management platform (PFMP) is running on VMs, you can back up and restore using the VMware VM snapshots once you shut down the database and VMs.

If the cluster is deployed on VMware VMs, you can take the snapshots from the VMware vSphere Client user interface, or you can use a CLI command on the VMware ESXi system that is hosting the VMs.

For VM snapshots, the database and VMs must be shut down first before a reliable snapshot can be taken.

CAUTION: When you shut down the database and VMs, you shut down the PFMP. The user interface is not accessible and any running jobs fail when the system comes back on. Only after the database is shut down, should the VMs be shut down or the snapshots taken.

WARNING: In a PowerFlex hyperconverged deployment, it is imperative that snapshots never be used on storage VMs (SVM). Using snapshots is not supported and could cause data unavailability or data loss.

Follow the steps below to shut down the database and VMs.

1. To set the k alias and default namespace, enter:

```
# Run this to make it easier to run the rest of the code and set the default
namespace.
alias k="kubectl -n $(kubectl get pods -A | grep -m 1 -E 'platform|pgo|helmrepo' |
cut -d' ' -f1)"
kubectl config set-context default --namespace=$(kubectl get pods -A | grep -m 1 -E
'platform|pgo|helmrepo|docker' | cut -d' ' -f1)
```

2. To verify that the Postgres Operator controller pod and the database pods are up and running with no errors in the logs, enter:

```
# Get the PostgreSQL operator pod and PostgreSQL cluster pods to verify.
echo $(kubectl get pods -l="postgres-operator.crunchydata.com/control-plane=pgo" --no-
headers -o name && kubectl get pods -l="postgres-operator.crunchydata.com/instance"
--no-headers -o name) | xargs kubectl get -o wide
```

3. To gracefully shut down the database cluster, enter:

```
# Trigger a shutdown
k patch $(k get postgrescluster -o name) --type merge --patch '{"spec":{"shutdown":
true}}'
```

4. To take the snapshots and adjust the resources, shut down the VMs.
5. Power the VMs back on.
6. To set the k alias and default namespace, enter:

```
# Run this to make it easier to run the rest of the code and set the default
namespace.
alias k="kubectl -n $(kubectl get pods -A | grep -m 1 -E 'platform|pgo|helmrepo' |
cut -d' ' -f1)"
kubectl config set-context default --namespace=$(kubectl get pods -A | grep -m 1 -E
'platform|pgo|helmrepo|docker' | cut -d' ' -f1)
```

7. To verify that the shutdown occurred, enter:

```
# Verify the shutdown
# Only the PostgreSQL operator pod "pgo" should remain when running this command.
echo $(kubectl get pods -l="postgres-operator.crunchydata.com/control-plane=pgo" --no-
headers -o name && kubectl get pods -l="postgres-operator.crunchydata.com/instance"
--no-headers -o name) | xargs kubectl get -o wide
```

8. To start the cluster back up, enter:

```
# Restart the database cluster
k patch $(k get postgrescluster -o name) --type merge --patch '{"spec":{"shutdown":
false}}'
```

9. To verify that the startup occurred, enter:

```
# Verify the database pods started and are in a running state
echo $(kubectl get pods -l="postgres-operator.crunchydata.com/control-plane=pgo" --no-
headers -o name && kubectl get pods -l="postgres-operator.crunchydata.com/instance"
--no-headers -o name) | xargs kubectl get -o wide
```

To restore to the previous point in time, you can revert the VMs from the snapshots in the VMware vSphere Client user interface.

Software upgrade

When a new version of the management software becomes available, you can upgrade to that version from the **Software Upgrade** page.

Upgrading PowerFlex Manager from a local repository path

You can upgrade the management software from a TGZ file on a local repository path.

Before you begin, ensure that you:

- Take a backup of PowerFlex Manager settings.
- Upload the latest compatibility matrix file to ensure that PowerFlex Manager has access to the latest upgrade information.

When you attempt an upgrade, PowerFlex Manager warns you if the current version of the software is incompatible with the target version, or if any of the RCM or IC versions that are currently loaded on the management software are incompatible with the target compliance versions. To determine which paths are valid and which are not, PowerFlex Manager uses information that is provided in the compatibility matrix file. The compatibility matrix file maps all the known valid and invalid paths for all previous releases of the software. Before proceeding with an upgrade, PowerFlex Manager notifies you if any of the following situations is detected:

- No upgrade is required. This situation occurs when the current and target versions are the same. You can still force an upgrade, if needed.
- The upgrade path that you are attempting to perform is a valid path. The source version of PowerFlex Manager is compatible with the target version and the source version of the RCM or IC is also compatible with the target version.
- The source version of PowerFlex Manager is compatible with the target version. However, the source version of the RCM or IC is not compatible with the target version. Your resource groups may go into life cycle mode until you perform an RCM or IC upgrade. PowerFlex Manager puts a resource group in life cycle mode if the resource group's RCM or IC is not within a year of the target PowerFlex Manager launch. You can mitigate a resource group that is in life cycle mode by upgrading to a compliance version that is within a year of the current PowerFlex Manager launch.
- The upgrade path that you are attempting to perform is not supported.

The compatibility matrix file is required for all upgrades. When you first install PowerFlex Manager, the software does not have the compatibility matrix file, so you must upload the file before performing any upgrades.

1. If you are using PowerFlex rack, log in to Dell Technologies Download Center. Then, go to the section for PowerFlex Manager.
2. If you are using PowerFlex appliance, log in to Dell Technologies Support. Then, enter your service tag to be taken to available downloads.
3. Download the PowerFlex management platform .TGZ file.
4. Copy the PowerFlex management platform .TGZ file to a share drive location.
5. On the menu bar, click **Settings**, and then click **Software Upgrade**.
6. Click **Management Software Upgrade**.

   NOTE: Dell Technologies recommends using the PowerFlex System Configuration Reporter (SCR) to validate that configurations are ready to upgrade. PowerFlex SCR is a standalone application that is designed to provide comprehensive configuration reporting, health assessment, configuration checks, readiness/upgrade assessments checks, and log collection for PowerFlex systems.

7. On the **Management Software Upgrade** page, you can see the **Current Management Software Version** and check the **Available Management Software Version** field to see if a newer version of PowerFlex Manager is available.
8. Click **Edit Settings**.
   a. To upgrade from a specific version on a local repository path, select **Update Software from local repository path**, and enter the path in the **Repository Path** box.
   b. Enter the username and password.
   c. Click **Save**.
9. Click **Upgrade Now**.
   After you click **Upgrade Now**, a dialog box displays with a warning message indicating whether the upgrade path is valid.
10. Under **Management Credentials**, optionally override the password for the nodes hosting PowerFlex Manager.
    The upgrade requires access to the nodes hosting PowerFlex Manager. Provide a single user that has superuser privileges for all nodes. Enter the password for each of the nodes.

11. Optionally, click **Enable Key Pairs** to enable login with SSH key pairs:

    To enable key pairs for the **Node** credential type:

    a. Import an existing key:
       i. Click **Import SSH Key Pair**.
       ii. Click **Choose File** and browse to the file that contains your public and private keys, and select the private key.
       iii. Type a name for the key pair.
       iv. Click **Import**.

12. Type **UPDATE POWERFLEX MANAGER** if you want to proceed with the update. If you want to perform an update even if none is required, type **FORCE UPDATE**.
13. Click **Upgrade** to upgrade PowerFlex Manager.
    The update process displays messages indicating the progress of the update. Depending on which services were upgraded, you may be required to log in once again, after the upgrade process is complete.

After upgrading to PowerFlex Manager, you may notice that resource groups that are tied to the PowerFlex gateway are no longer in compliance. To bring the resource groups into compliance, you may need to upgrade them. In this case, you will need to change to a later RCM or IC and upgrade the resource groups.

If you do not want to upgrade the resource groups, you do not need to perform the upgrade.

Editing the upgrade settings

You can edit the upgrade settings to control where the system retrieves software updates from and ensure that the upgrade process is secure.

1. On the menu bar, click **Settings** and then click **Software Upgrade**.
2. Click **Management Software Upgrade**.
3. Click **Edit Settings**.
   a. To update to the latest version using Dell Technologies connectivity services, select **Update Software from configured Dell Technologies connectivity services**.
   b. To upgrade from a specific version on a local repository path, select **Update Software from local repository path**, and enter the path in the **Repository Path** box.

      The connection to the path that you provided is not secure. You must choose one of the following options:

      - To allow the insecure connection, select the **Allow an insecure connection to the specified path** checkbox.
      - To make the connection to the specified path secure, go to **Security** > **SSL Trusted Certificates** and click **Add** to upload the SSL certificate. Then, retry the operation.

      You can upgrade from a TGZ file on a network share. This option provides a way to update the software in a dark site. Provide a path using one of the following formats:

      Table 91. Path for network shares

For:	Use this path:
CIFS	`\\IP Address\Any folder\file-name.tgz`
FTP	`ftp://IP Address/Any folder/file-name.tgz`
HTTP	`http://IP Address/Any folder/file-name.tgz`
HTTPS	`http://IP Address/Any folder/file-name.tgz`

   c. Optionally, provide a **User Name** and **Password**.

      The username and password are only used with CIFS. All other local repository paths must specify public share locations.

   d. Click **Save**.
4. On the **Management Software Upgrade** page, you can view the updated settings for **Current Management Software Version**, **Available Management Software Version**, and **Repository Path**.
5. To perform the upgrade, click **Upgrade Now**. A message displays confirming that you want to update the software.
6. Click **Upgrade Software**.
7. Click **Yes**.
8. Depending on which services were upgraded, you may be required to log in again once the upgrade process is complete.

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PowerFlex Manager user interface reference (erasure coding)

This section provides reference details for various parts of the user interface for PowerFlex Manager.

The topics in this section are organized according to the main tabs within the user interface.

Life cycle

This section provides reference information for the **Resource Groups**, Templates, and **Storage Nodes** pages.

Related information

Managing components
Monitoring system health
Managing external changes
Managing components
Monitoring system health
Deploying a resource group
Adding an existing resource group

Resource groups

This section provides reference information for the **Resource Groups** page.

Related information

Managing components
Managing external changes
Managing components

Resource group states

The resource group states indicate the current condition or health of all resources within a group, such as whether they are active, updating, or experiencing issues. They provide a quick, high-level view that helps you to understand the overall system status and decide when action or troubleshooting is needed.

Table 92. Resource group states

Icon	Resource group state	Description
	**Healthy**	The resource group was deployed successfully, and the resources are firmware compliant and healthy.
	**Warning**	One or more resources in the resource group require corrective action. This state does not affect the overall system health. For example, the firmware running on the resource is not at the required level or not compliant.
	**Critical**	Resource group deployment has failed due to some issues.

Table 92. Resource group states (continued)

Icon	Resource group state	Description
	**Pending**	Resource group deployment is scheduled for a later time or date.
	**In Progress**	Resource group deployment is in progress.
	**Canceled**	Resource group deployment was canceled.
	**Incomplete**	The resource group is not fully functional because it has no associated volumes. Click **Add Resources** to add volumes.
	**Service Mode**	The resource group has been placed in service mode because one of the nodes within the resource group was put into service mode. If a node is in instant maintenance mode, PowerFlex Manager detects this situation and automatically places the node in service mode and also ensures that the resource group itself goes into service mode. If a node is in instant maintenance mode for more than 30 minutes, PowerFlex Manager sends a critical alert to Dell Technologies connectivity services.
	**Lifecycle Mode**	The resource group supports health and compliance monitoring, service mode, and non-disruptive updates. All other resource group operations are blocked. Life cycle mode controls the operations that can be performed for configurations that have limited support.
	**Managed Mode**	The resource group supports health and compliance monitoring, non-disruptive updates, automated resource addition, and automated resource replacement features.

Existing resource groups

If you already have a cluster deployed that includes Dell-based hardware, you can discover and import these hardware resources as an existing resource group. Even though these resources were not originally deployed with PowerFlex Manager, you can take advantage of the health monitoring, compliance, and update features.

When you add an existing resource group for a PowerFlex cluster, PowerFlex Manager matches up the hosts, vCenter, and other items it finds with discovered resources in the resource list.

Only managed or reserved resources are included in the resource group. Any resource discovered as **Unmanaged** is missing from the resource group. If a component is missing, you can change your resource inventory, and update the resource group to reflect these changes. Go to the resources list, select the component, and mark it as **Managed** by selecting **Change resource state to Managed**. Then perform an **Update Resource Group Details** operation on the resource group to pull in the missing component.

Each node is identified as a PowerFlex host if the inventory for the node shows that it is present in a gateway, and it is properly configured as a storage data client (SDC) and storage node. A PowerFlex host has the **Use Node for Dell PowerFlex** check box set to **True** in the operating system settings for the node within the resource group deployment settings.

You can perform most of the actions on an existing resource group that are available within PowerFlex Manager for a new resource group. For some configurations, PowerFlex Manager displays a yellow banner for the resource group to indicate that some features are not available.

PowerFlex Manager enables you to add an existing resource group for an environment using VMware NSX-T or NSX-V. Most resource group actions are disabled for an NSX-T or NSX-V configuration, except the ability to update the firmware and software components, remove resources (or the resource group as a whole), and update resource group details.

When you add an existing resource group for a CloudLink configuration, you choose a single CloudLink Center as the target for the resource group. If the CloudLink Center for the resource group shuts down, PowerFlex Manager loses communication with the CloudLink Center. If the CloudLink Center is part of a cluster, PowerFlex Manager moves to another CloudLink Center when you update the resource group details.

Life cycle mode

If you add an existing resource group that includes an unsupported configuration, PowerFlex Manager might put the resource group in life cycle mode. This mode limits the actions that you can perform within the resource group.

Life cycle mode allows the resource group to perform only monitoring, service mode, and compliance upgrade operations. All other resource group operations are blocked when the resource group is in life cycle mode. Life cycle mode is used to control the operations that can be performed for configurations that have limited support.

When you add an existing resource group, PowerFlex Manager puts the resource group in life cycle mode if the configuration you want to import includes any of the following:

- Invalid server inventory
- Missing network settings

  The minimum required IP addresses are one data IP and one PowerFlex IP for a storage-only resource group,

- Unsupported NIC (a 1 GB QLogic NIC, for example)
- Unsupported server configurations (an unsupported SD Boot device, for example)
- Unsupported NIC teaming policies, an unsupported switch port configuration, or an unsupported access facing port configuration
- Network configuration without a PXE VLAN setting
- No switch configuration
- PowerFlex MDM cluster without virtual IPs
- vSphere Cluster Services (vCLS) VMs on local storage

When you add an existing resource group, PowerFlex Manager checks to see whether there are any vSphere Cluster Services (vCLS) VMs on local storage. If it finds any, it puts the resource group in life cycle mode and lets you migrate the VMs to shared storage.

PowerFlex Manager also puts a resource group in life cycle mode if you select a minimal compliance version that includes firmware only for the resource group.

When PowerFlex Manager must put a resource group in life cycle mode, the **Summary** page for the **Add Existing Resource Group** wizard displays a warning message indicating the reason.

In some situations, an imported configuration might not meet the minimal requirements for life cycle mode. In this case, PowerFlex Manager does not allow you to add the resource group.

For missing virtual IPs or no switch connectivity, you can correct the error and use **Update Resource Group Details** to take the resource group out of life cycle mode.

If you have manually added NSX to a cluster outside of PowerFlex Manager, click **Update Resource Group Details** on the resource group, so it will correctly reflect the environment with NSX, which will be life cycle mode.

Templates

This section provides reference information for the **Templates** page.

Related information

Getting started
Getting started (mirroring)
Managing components
Managing components

Managing templates

On the **Templates** page, you can view information about templates in a list or tile view. To switch between views, click the tile icon or list icon at the top of the **Templates** page. When in tile view, you can view the templates under a category by clicking the graphic that represents the category.

Users with standard permissions can view details of only those templates for which the administrator has granted permission.

Templates - My Templates List View

The **My Templates** page displays the details of the templates that you have created.

Table 93. My Templates page

Field	Description
+ Add a Template	Allows you to add a template. After you add a template, you can add node, cluster, and VM components on the template builder page.
Export All	Allows you to export all templates to a CSV file.
Filter By	Allows you to filter and view templates based on the template category.
State	Displays the state of the template - **Draft** or **Published**.
Category	Displays the template category.
Name	Displays the template name.
Last Deployed On	Displays the date when the template was last deployed.
Components	Displays the components in the template.

Templates - Sample Templates

The **Sample Templates** page displays the default templates that you can use in your environment.

Table 94. Templates - Sample Templates page

Field	Description
Name	Displays the template name.
Components	Displays the components in the template.

Templates

On the **Templates** page, the right pane displays the name of the template, icons of components in the template, and the following details for a selected template:

Table 95. Templates page

Field	Description
Edit	Click to edit the template.
Delete	Click to delete the template.
View Details	Click to view the details of the template, such as components in the template.
Clone	Click to clone the template.
Export Template	Click to export the template to a GPG file. You can use the exported template to duplicate the settings.
Created On	Displays the date and time when the template was created.
Created By	Displays the name of the user who created the template.
Updated On	Displays the date and time when the template was updated.
Updated By	Displays the name of the user who updated the template.

Building a template overview

The template builder allows you to build a customized template by configuring both physical and virtual components. On the template builder page, you can set the component properties. For example, you can create a template that provisions only physical nodes with operating systems on them.

NOTE: A newly created or a cloned template appears in a draft state on the **Template** page and remains in the same state until published.

You can configure node, cluster, and VM components in a template.

The template builder page displays a graphical representation of the topology that is created within a particular template. From this page, you can:

- Add node, cluster, and VM components to a template
- Build and publish a template
- Delete a template
- Import a template
- Deploy a resource group (this feature is available only on published templates)

Component types

Components (physical or virtual or applications) are the main building blocks of a template.

PowerFlex Manager has the following component types:

- Node (Software/hardware and software only)
- Cluster
- VM

Related information

Adding components to a resource group
Edit a template
Deploying a resource group
View template details
Adding components to a resource group (mirroring)
Deploying a resource group (mirroring)

Node settings

This table describes the following node settings: hardware, BIOS, operating system, and network.

Table 96. Node settings

Setting	Description
Full network automation	Allows you to perform deployments with full network automation. This feature allows you to work with supported switches, and requires less manual configuration. Full network automation also provides better error handling since PowerFlex Manager can communicate with the switches and identify any problems that may exist with the switch configurations.
Partial network automation	Allows you to perform switchless deployments with partial network automation. This feature allows you to work with unsupported switches, but requires more manual configuration before a deployment can proceed successfully. If you choose to use partial network automation, you give up the error handling and network automation features that are available with a full network configuration that includes supported switches. For a partial network deployment, the switches are not discovered, so PowerFlex Manager does not have access to switch configuration information. You must ensure that the switches are configured correctly, since PowerFlex Manager can not configure the switches for you. If your switch is not configured correctly, the deployment may fail and PowerFlex Manager is not able to provide information about why the deployment failed. For a partial network deployment, you must add all the interfaces and ports, as you would when deploying with full network automation. However, you do not need to add the operating system installation network, since PXE is

Table 96. Node settings (continued)

Setting	Description
	not required for partial network automation. PowerFlex Manager uses virtual media instead for deployments with partial network automation. The **Switch Port Configuration** must be set to **Port Channel (LACP enabled)**. In addition, the LACP fallback or LACP ungroup option must be configured on the port channels.
Component Name	Indicates the node component name
Drive Encryption Type	Specifies the type of encryption to use when encryption is enabled. The encryption options are: • Software encryption • Self-encrypting drive (SED)
Number of Instances	Enter the number of instances that you want to add. If you select more than one instance, a single component representing multiple instances of an identically configured component are created. Edit the component to add extra instances. If you require different configuration settings, you can create multiple components.
Related Components	Select **Associate All** or **Associate Selected** to associate all or specific components to the new component.
Import Configuration from Reference Node	Click this option to import an existing node configuration and use it for the node component settings. On the **Select Reference Node** page, select the node from which you want to import the settings and click **Select**.
OS Settings	
Host Name Selection	If you choose **Specify At Deployment Time**, you must type the name for the host at deployment time. The hostname must not exceed 15 characters. If you choose **Auto Generate**, PowerFlex Manager displays the **Host Name Template** field to enable you to specify a macro that includes variables that produce a unique hostname. For details on which variables are supported, see the context-sensitive help for the field. If you choose **Reverse DNS Lookup**, PowerFlex Manager assigns the hostname by performing a reverse DNS lookup of the host IP address at deployment time.
OS Image	Specifies the location of the operating system image installation files. You can use the image that is provided with the target compliance file, or specify your own location, if you created additional repositories. To deploy a compute-only or storage-only resource group with the Linux image that is provided with a compliance file, choose **Use Compliance File Linux image**. To deploy a storage-only resource group with Red Hat Enterprise Linux, you must create a repository on the **Settings** page and specify the path to the Red Hat Enterprise Linux image on a file share. Dell Technologies recommends that you use one of your own images that are published from the customer portal at Red Hat Enterprise Linux. For Linux, you may include one node within a resource group. For ESXi, you must include at least two nodes. **NOTE:** If you select an operating system from the **OS Image** drop-down menu, the field **NTP Server** displays. This field is optional, but it is highly recommended that you enter an NTP server IP to ensure proper time synchronization with your environment and PowerFlex Manager.

Table 96. Node settings (continued)

Setting	Description
	Sometimes when time is not properly synchronized, resource group deployment failure can occur.
OS Credential	Select an **OS Admin** or **OS User** credential that you created on the **Credentials Management** page. Alternatively, you can create a credential while you are editing a template. If you select a credential that was created on the **Credentials Management** page, you do not need to type the username and password, since they are part of the credential definition. PowerFlex Manager allows to specify a non-root user instead of the root user when you configure a template for a compute-only or storage-only deployment.
NTP Server	Specifies the IP address of the NTP server for time synchronization. If adding more than one NTP server in the operating system section of a node component, be sure to separate the IP addresses with commas.
Use Node For Dell PowerFlex	Indicates that this node component is used for a PowerFlex deployment. When this option is selected, the deployment installs the MDM, storage node, and SDC components, as required for a PowerFlex deployment in a VMware environment. The MDM and storage node components are installed on a dedicated PowerFlex VM (SVM), and the SDC is installed directly on the ESXi host. To deploy a PowerFlex cluster successfully, include at least three nodes in the template. PowerFlex Manager uses the following logic to determine the MDM roles for the nodes: 1. Checks the PowerFlex gateway inventory to see how many primary MDMs, secondary MDMs, and tiebreakers are present, and the total number of storage node components. 2. Adds the number of components being deployed to determine the overall PowerFlex cluster size. For example, if there are three storage node components in the PowerFlex gateway inventory, and you are deploying two more, you will have a five node cluster after the deployment. 3. Adds a single primary MDM and determines how many secondary MDMs and tiebreakers should be in the cluster by looking at the overall cluster size. The configuration varies depending on the size of the cluster: • A three-node cluster has one primary, one secondary, and one tiebreaker. • A five-node cluster has one primary, two secondaries, and two tiebreakers. 4. Determines the roles for each of the new components being added, based on the configuration that is outlined above, and the number of primary, secondary, and tiebreakers that are already in the PowerFlex cluster. For each storage node in the cluster, the deployment adds all the available disks from the nodes to the storage pools created. For each compute-only, the deployment installs the SDC VIB. When you select this option, the teaming and failover policy for the cluster are automatically set to **Route based on IP hash**. Also, the uplinks are configured as active and active, instead of active and standby. Teaming is configured for all port groups, except for the PowerFlex data 1 and PowerFlex data 2 port groups. If you select the option to **Use Node For Dell PowerFlex**, the **Local Flash Storage for Dell PowerFlex** option is automatically selected as the **Target Boot Device** under **Hardware Settings**.
PowerFlex Role	Specifies one of the following deployment types for PowerFlex:

Table 96. Node settings (continued)

Setting	Description
	• **Compute Only** indicates that the node is only used for compute resources. • **Storage Only** indicates that the node is only used for storage resources. If you select an ESXi image type in the **OS Image** field, the **PowerFlex Role** must be set to **Compute Only**. If you add a compute-only node, only the SDC is added. If you select a Red Hat Enterprise Linux image type in the **OS Image** field, the **PowerFlex Role** must be set to **Storage Only**. If you add a storage-only node, only the storage node is added. The only prerequisites for a storage-only node are that the iDRAC must have an IP address and a credential. PowerFlex Manager takes care of all other configuration steps that are required for the node. For each node, PowerFlex Manager configures the MDM roles as needed and configures the storage node RPMs. Once the cluster is set up, PowerFlex Manager adds every node as an storage node. Then, it adds all available disks for the storage node, adds a storage pool, and adds all disks to the storage pool. If you are creating a compute-only template, be sure to include both the **VMware Cluster** and **PowerFlex Cluster** components in the template builder. If you are creating a storage-only template, do not include a **VMware Cluster** component in the template builder. Only the **PowerFlex Cluster** component is required for a storage-only template.
Protection Domain Type	Erasure coding is the protection domain default type.
Persistent Memory	Specifies whether persistent memory is required.
Client Storage Access	Determines how clients access storage. For a storage-only role, select one of the following options: • Storage Data Client (SDC) Only • SDC and NVMe/TCP Initiator For a compute-only role the Client Storage Access control is not displayed, and the client access is set to SDC automatically.
Enable Compression	Enables compression on the protection domain. The default setting is for compression to be enabled. You can enable compression on a storage-only resource group. Compression is supported for new resource group deployments, and existing resource groups. You can also enable compression for storage-only nodes when performing a scale up of a resource group. **NOTE:** Irrespective of whether you have persistent memory,(SDPM or NVDIMM), you can enable compression.
Enable Encryption	Enables disk encryption on the node. This option allows you to take advantage of CloudLink encryption. You can enable CloudLink encryption on a storage-only resource group. Encryption is supported for new resource group deployments, and existing resource groups. You can also enable encryption for storage-only nodes when performing a scale up of a resource group. A resource group cannot mix encrypted and unencrypted nodes. Scale up of a storage-only resource group is supported, with the new nodes using the same encryption settings as the nodes already in the resource group. If you select this option, PowerFlex Manager looks for servers with 12 cores or more for CloudLink deployments. After encryption is enabled, the template displays a warning message indicating that the template is missing some data. The template adds the

Table 96. Node settings (continued)

Setting	Description
	Cloud Link Center Settings section to the PowerFlex cluster component to allow you to specify the required data. Some PowerFlex nodes might not be selected for deployment, depending on the encryption type that is selected. For example, if you choose **Software Encryption**, you cannot include a PowerFlex node with only SEDs. If you choose **Self Encrypting Drive**, you cannot include a PowerFlex node with only software encryption drives. PowerFlex Manager does not allow you to mix SEDs and software encryption drives in the same protection domain. Servers do not typically have this drive combination, but PowerFlex Manager verifies the drives and uses only servers of the specified type. **Validate Settings** detects PowerFlex nodes that do not match the specified **Drive Encryption Type**.
Drive Encryption Type	Specifies the type of encryption to use when encryption is enabled. The options are: • **Software Encryption** • **Self Encrypting Drive (SED)** Some nodes might not be selected for deployment, depending on the encryption type selected. For example, if you choose **Software Encryption**, you cannot include a node with only SEDs. Similarly, if you choose **Self Encrypting Drive**, you cannot include a node with only software encryption drives. PowerFlex Manager does not allow you to mix SEDs and software encryption drives in the same protection domain. Servers should not typically have this mix, but PowerFlex Manager checks for this and uses only servers of the type you specify. **Validate Settings** detects nodes that do not match the specified **Drive Encryption Type**.
Switch Port Configuration	Specifies whether Cisco virtual PortChannel (vPC) or Dell Virtual Link Trunking (VLT) is enabled or disabled for the switch port. For storage-only templates, the options are: • **Port Channel (LACP enabled)** turns on vPC or VLT with the link aggregation control protocol enabled. • **Trunk Port** enables a networking configuration that does not use vPC or VLT. For compute-only templates that use a Linux operating system image, the options are: • **Port Channel (LACP enabled)** turns on vPC or VLT with the link aggregation control protocol enabled. • **Trunk Port** enables a networking configuration that does not use vPC or VLT. For compute-only templates that uses an ESXi operating system image, the **Switch Port Configuration** setting includes these options: • **Port Channel** turns on vPC or VLT. • **Port Channel (LACP enabled)** turns on vPC or VLT with the link aggregation control protocol enabled. • **Trunk Port** enables a networking configuration that does not use vPC or VLT.
Teaming And Bonding Configuration	The teaming and bonding configuration options depend on the switch port configuration selected.

Table 96. Node settings (continued)

Setting	Description
	For compute-only templates, the following options are available: ● If you choose **Port Channel (LACP enabled)** or **Port Channel** as the switch port configuration, the only teaming and bonding option is **Route Based on IP hash**. ● If you choose **Trunk Port** as the switch configuration, there are three teaming and bonding options: ○ **Route Based on originating virtual port** ○ **Route Based on physical NIC load** ○ **Route Based on Source MAC hash** For storage-only templates, the following options are available: ● If you choose **Port Channel (LACP enabled)** as the switch port configuration, the only teaming and bonding option is **Mode 4 (IEEE 802.3ad policy)**. ● If you choose **Trunk Port** as the switch configuration, you have the following teaming and bonding options: ○ **Mode 1 (active-backup policy)** ○ **Mode 6 (ALB Adaptive policy)** For a non-bonded storage-only template, you need to select **Trunk Port** for **Switch Port Configuration** setting, not **Port Channel (LACP enabled)**. For the **Teaming and Bonding Configuration** setting, you typically select **Mode 1** for most deployments.
Transmit Hash Policy	The option is available for storage-only and compute-only Linux nodes only when the **Teaming And Bonding Configuration** is set to Mode 4 (EEE 802 3ed policy). Mode 4 is the bond with LACP and supports the following load balancing options: ● LACP in layer 2 & 3 ● LACP in layer 3 & 4 LACP in layer 3 & 4 is the default option
Hardware Settings	
Target Boot Device	Specifies the target boot device. ● **Local Flash Storage**: Installs the operating system to either the SATADOM or the BOSS flash storage device present in the node. With the **Local Flash Storage** option, only nodes with a BOSS storage controller and two attached hard drives or SATADOM are selected to be deployed as part of the resource group, depending on the Dell PowerEdge servers used. For PowerEdge servers that support BOSS, during deployment PowerFlex Manager creates RAID 1 with the two hard drives attached to the BOSS controller. ● **Local Flash Storage for Dell PowerFlex**: Installs the operating system to either the SATADOM or the BOSS flash storage device that is present in the node and configures the node to support PowerFlex. If you select the option to **Use Node for Dell PowerFlex** under **OS Settings**, the **Local Flash Storage for Dell PowerFlex** option is automatically selected as the target boot device.
Node Pool	Specifies the pool from which nodes are selected for the deployment.
BIOS Settings	
System Profile	Select the system power and performance profile for the node.

Table 96. Node settings (continued)

Setting	Description
User Accessible USB Ports	Enables or disables the user-accessible USB ports.
Number of Cores per Processor	Specifies the number of enabled cores per processor.
Virtualization Technology	Enables the additional hardware capabilities of virtualization technology.
Logical Processor	Each processor core supports up to two logical processors. If enabled, the BIOS reports all logical processors. If disabled, the BIOS reports only one logical processor per core.
Execute Disable	Enables or disables execute disable memory protection.
Node Interleaving	Enable or disable the interleaving of allocated memory across nodes. ● If enabled, only nodes that support interleaving and have the read/write attribute for node interleaving set to enabled are displayed. Node interleaving is automatically set to enabled when a resource group is deployed on a node. ● If disabled, any nodes that support interleaving are displayed. Node interleaving is automatically set to disabled when a resource group is deployed on a node. Node interleaving is also disabled for a resource group with NVDIMM compression. ● If not applicable is selected, all nodes are displayed irrespective of whether interleaving is enabled or disabled. This setting is the default.
Network Settings	
Multi-Network Selection	Select the check-box to include multiple management networks of the same type. If you select multiple networks of the same type without selecting the check box, an error is displayed when you publish the template. The multiple network selection is supported on the following networks: ● Hypervisor Management ● PowerFlex Management ● Hypervisor Migration
Add New Interface	Click **Add New Interface** to create a network interface in a template component. Under this interface, all network settings are specified for a node. This interface is used to find a compatible node in the inventory. For example, if you add **`Two Port, 100 gigabit`** to the template, when the template is deployed PowerFlex Manager matches a node with a two-port 100-gigabit network card as its first interface. To add one or more networks to the port, select **Add Networks to this Port**. Then, choose the networks to add, or mirror network settings defined on another port. To see network changes that are previously made to a template, you can click **View/Edit** under **Interfaces**. Or, you can click **View All Settings** on the template, and then click **View Networks**. To see network changes at resource group deployment time, click **View Networks** under **Interfaces**.
Add New Static Route	Click **Add New Static Route** to create a static route in a template. To add a static route, you must first select **Enabled** under **Static Routes**. A static route allows nodes to communicate across different networks. A static route requires a **Source Network** and a **Destination Network**, and a **Gateway**. The source and destination network must each be a PowerFlex data network that has the **Subnet** field defined. If you add or remove a network for one of the ports, the **Source Network** drop-down list does not get updated and still shows the old networks. In order to see the changes, save the node settings and edit the node again.

Table 96. Node settings (continued)

Setting	Description
Validate Settings	Click **Validate Settings** to determine what can be chosen for a deployment with this template component. The **Validate Settings** wizard displays a banner to when one or more resources in the template do not match the configuration settings that are specified in the template. The wizard displays the following tabs: • **Valid (*number*)** lists the resources that match the configuration settings. • **Invalid (*number*)** lists the resources that do not match the configuration settings. The reason for the mismatch is shown at the bottom of the wizard. For example, you might see `Network Configuration Mismatch` as the reason for the mismatch if you set the port layout to use a 100-Gb network architecture, but one of the nodes is using a 25 GB architecture. If you set the encryption method to use self-encrypting drives (SEDs), but the nodes do not have these drives, you might see `Self Encrypting Drives are required but not found on the node, or software encryption requested but only available drives are SED.`

After entering the information about operating system Installation (PXE) network in the respective field as described in the table above, PowerFlex Manager untags vLANs entered in the operating system installation network on the switch node facing port. For vMotion and hypervisor networks, PowerFlex Manager tags these networks on the switch node-facing ports for the entered information. For rack node, PowerFlex Manager configures the vLANs on node facing ports (untag PXE vLANs, and tag other vLANs).

If you select **Import Configuration from Reference Node**, PowerFlex Manager imports basic settings, BIOS settings, and advanced RAID configurations from the reference node and enables you to edit the configuration. Some BIOS settings might no longer apply once new BIOS settings are applied. PowerFlex Manager does not correct these setting dependencies. When setting advanced BIOS settings, use caution and verify that BIOS settings on the hardware are applicable when not choosing **Not Applicable** as an option. For example, when disabling SD card, the settings for internal SD card redundancy become not applicable.

You can edit any of the settings visible in the template, but keep in mind that many settings are hidden when using this option. For example, only ten out of many BIOS settings that you can see and edit when using a template are displayed. However, you can configure all BIOS settings. If you want to edit any of the settings that are not visible through the template feature, edit them before importing or uploading the file.

For storage only nodes, the memory required by the node for SDPM or NVDIMM is specified in the following table::

Table 97. SDPM or NVDIMM storage capacity

Node memory (in GB)	SDPM/NVDIMM capacity required (in GB)
512	128 or 64

Cluster component settings

This table describes the cluster component settings.

Table 98. Cluster component settings

Field name	Description
Select a Component	Select **VMware Cluster** or **PowerFlex Cluster**.
Component Name	Indicates the cluster component name.
Related Components	Select **Associate All** or **Associate Selected** to associate all or specific components to the new component.
Cluster settings (for the VMware cluster component)	

Table 98. Cluster component settings (continued)

Field name	Description
Target Virtual Machine Manager	Select virtual machine manager from the **Target Virtual Machine Manager** list.
Data Center Name	Select the data center name from **Data Center Name** list.
Cluster Name	Select a new cluster name from **Cluster Name** list.
New Cluster Name	Select new cluster name from **New Cluster Name** list.
Cluster HA Enabled	Enables or disables a highly available cluster. You can either select or clear (default) the check box.
Cluster DRS Enabled	Enables or disables distributed resource scheduler (DRS). You can either select or clear (default) the check box.
PowerFlex settings (for the PowerFlex cluster component)	
Target PowerFlex Gateway	Select a target PowerFlex gateway, which acts as an endpoint for PowerFlex API calls. During the provisioning process, PowerFlex Manager connects to the PowerFlex gateway and uses its APIs to configure all storage node and SDC parameters.
Protection Domain Name	Provide a name for the protection domain. You can automatically generate the name (recommended), or specify a new name explicitly. A protection domain is a logical entity that contains a group of storage nodes that provide backup for each other. Each storage node belongs to only one protection domain. Each protection domain is a unique set of storage nodes. A protection domain may also contain SDTs. You can automatically generate the protection domain name or specify a new name explicitly for a storage-only template. The PowerFlex cluster must have at least three nodes before you can publish the template and deploy it as a resource group. However, if you select an existing protection domain that is associated with another previously deployed storage-only resource group, and this protection domain has at least three nodes, PowerFlex Manager recognizes the new template as valid if the cluster has fewer than three nodes. You can publish the template successfully and deploy it as a resource group, since the protection domain it uses has enough nodes. **NOTE:** A compute-only resource group only requires a minimum of two nodes, since it does not have a protection domain. If you choose **Compute Only** as the **PowerFlex Role** for the node component, the **PowerFlex** settings do not include the **Protection Domain Name** field.
New Protection Domain Name	Specify a new name for the protection domain.
Protection Domain Name Template	If you choose to generate the protection domain name automatically, PowerFlex Manager fills this field with a default template that combines static text with variables for several pieces of the autogenerated name. If you modify the template, be sure to include the *${num}* variable to ensure that the name is unique. For details on the rules for defining a template, see the contextual help that appears when you hover over the field. If you choose **Compute Only** as the **PowerFlex Role** for the node component, the **PowerFlex** settings do not include the **Protection Domain Name Template** field.
Device Group Name	Provide a name for the device group. You can automatically generate the name (recommended), or specify a new name explicitly. A device group contains devices from four or more nodes in a single protection domain. The nodes that belong to a device group have the same drive type or media

Table 98. Cluster component settings (continued)

Field name	Description
	type. The total capacity of the device group is derived from the capacity of the storage devices that it contains. Storage pools are allocated through the device group and consume the capacity of the device group.
Device Group Name Template	Define a template for generating the device group name automatically. PowerFlex Manager fills this field with a default template that combines static text with variables for several pieces of the autogenerated name. If you modify the template, be sure to include the *${num}* variable to ensure that the name is unique. For details on the rules for defining a template, see the contextual help that appears when you hover over the field.
Persistent Memory Device Group Name	Provide a name for the persistent memory device group. You can automatically generate the name (recommended), or specify a new name explicitly.
Persistent Memory Device Group Template	Define a template for generating the persistent memory device group name automatically. PowerFlex Manager fills this field with a default template that combines static text with variables for several pieces of the autogenerated name. If you modify the template, be sure to include the *${num}* variable to ensure that the name is unique. For details on the rules for defining a template, see the contextual help that appears when you hover over the field.
Device Group Spare Capacity	The device group spare capacity is the additional storage reserved within the system to handle failures and ensure data is available, You must set the spare capacity limit of the device group in terms of nodes for this release. The default spare capacity is one node, and the maximum is two nodes.
Custom Spare Node Count	The spare capacity is set to the size of the largest device in the node multiplied by the number of devices in the node multiplied by the number of nodes specified. For example, if the largest device within a PowerFlex node is 900 GB and there are five devices in the PowerFlex node and two PowerFlex nodes in total, the custom spare node count is 9000 GB (900 GB x 5 devices per node x 2 nodes in total = 9000 GB).
Storage Pool Name	Provide a name for the storage pool. You can automatically generate the name (recommended), select from a list of existing storage pools for the selected protection domain, or specify a unique storage pool name. If you choose to automatically generate a name, you are prompted to define a storage pool name template. Storage pools allow the generation of different storage tiers in PowerFlex. A storage pool is a set of physical storage devices in a protection domain. Each storage device belongs to one (and only one) storage pool. The number of storage pools that are created at deployment time depends on the number of disks available.
CloudLink Center Settings	
Target CloudLink Center	Specifies the target CloudLink Center for an encrypted node. The **Cloud Link Center Settings** are only available when the **Enable Encryption** option is selected in the **OS Settings** for the node component. For any CloudLink Center that has encryption, you must either add an existing resource group to PowerFlex Manager, or use PowerFlex Manager to create the CloudLink encryption machine groups and keystores. PowerFlex Manager is only able to create the machine groups and keystores if the CloudLink Center exists in the management cluster. Ensure that CloudLink Center is available on the resource page.

Table 98. Cluster component settings (continued)

Field name	Description
	If you select an existing protection domain under the **PowerFlex Settings**, you must choose a **Target CloudLink Center**. The other CloudLink Center settings are not available, and PowerFlex Manager uses the machine group and keystore settings for the selected target. If you automatically generate the protection domain name or specify a new protection name explicitly, you can set the **Machine Group Name** and **Keystore Name** settings with the template. When you deploy a template with CloudLink settings, PowerFlex Manager creates the machine groups and keystores, and then assigns the machine groups to keystores and networks. Next, it adds the machines to the machine groups, and encrypts the drives. After deployment, you can see the machines that are added to machine groups by clicking **View Details** on the **Resources** page. When defining a template, you choose a single CloudLink Center as the target for the deployed resource group. If the CloudLink Center for the resource group shuts down, PowerFlex Manager loses communication with the CloudLink Center. If the CloudLink Center is part of a cluster, PowerFlex Manager moves to another CloudLink Center when you update the resource group details.
Machine Group Name	Allows you to provide a name for the machine group. The **Machine Group Name** is only available if you automatically generate the protection domain name or specify a new protection name explicitly. You can automatically generate the machine group name or specify a new machine group name explicitly.
New Machine Group Name	Specify a new name for the machine group.
Keystore Name	Allows you to provide a name for the keystore. The **Keystore Name** is only available if you automatically generate the protection domain name or specify a new protection name explicitly. You can automatically generate the keystore name or specify a new keystore name explicitly. Alternatively, you can choose an existing keystore from the list.
Keystore Name Template	Specify a new name for the keystore.

Related information

Add cluster component settings to a template

VM settings

The table describes VM settings for the CloudLink Center and the PowerFlex gateway.

Table 99. VM settings

Field name	Description
Select a Component	Select **CloudLink Center** or **PowerFlex Gateway**.
Number of Instances	Select the number of instances of the VM that you want to deploy.
Related Components	Select **Associate Selected** and select the **VMware Cluster** checkbox to associate the selected VMware cluster with the CloudLink Center.
VM Settings	
Specify Datastore	Specify the storage location.
Specify Network	Select the network type.
CloudLink Settings	
Host Name Selection	Defines how the hostname is selected at deployment time. There are two options:

Table 99. VM settings (continued)

Field name	Description
	• Auto Generate - Select to autogenerate the hostname at deployment time. • Specify at Deployment Time - Select to provide a unique hostname at deployment time.
OS Credential	Select the operating system credential used to set the username and password on the operating system that is installed.
NTP Server	Specify the IP address of the NTP server for time synchronization. If adding more than one NTP server, ensure that you separate the IP addresses with commas.
Secadmin Credential	Select the secadmin credential, an element manager credential, which is used to set the password for the secadmin user of the CloudLink Center. The password should be minimum of ten characters with at least one special character.
Vault Password	Specify up to three unique passwords for the CloudLink Vault. You must specify at least one. The password should be minimum of ten characters with at least one special character.
Confirm Vault Password	Retype the vault password.
License File	Upload a new or additional software license file. If additional, the resource is added to your current total resources.
Additional CloudLink Settings	
Configure Syslog Forwarding	Select the checkbox to send syslog messages from the CloudLink to a remote network management system.
Syslog Facility	Select the remote server where the syslog messages are forwarded to.
Configure Email Notifications	Select the checkbox to configure email alerts.
Server Address	Specify the IP address of the email server.
Port	Specify the port number for the email server. The default port is 25. The port numbers must be entered in comma-separated list and must be 1–65535.
Sender Address	Specify the email address for the sender.
User Name	Specify the required username.
Password	Specify the required password.
Confirm Password	Retype the password.
PowerFlex Gateway Settings	
Host Name Selection	Defines how the hostname is selected at deployment time. There are two options: • Auto Generate - Select to autogenerate the hostname at deployment time. • Specify at Deployment Time - Select to provide a unique hostname at deployment time.
PowerFlex Credential	Select the operating system credential used to set the username and password.
NTP Server	Specify the IP address of the NTP server for time synchronization. If adding more than one NTP server, ensure that you separate the IP addresses with commas.

Support for full and partial network automation

This topic lists the supported configurations for full and partial network automation.

Table 100. Supported configurations for full and partial network automation

Template components	Full network automation	Partial network automation (Requires manual switch configuration)
OS Images	ESXi, PowerFlex embedded operating system For more information, see PowerFlex Operating System OS Support Matrix	ESXi, PowerFlex embedded operating system For more information, see PowerFlex Operating System OS Support Matrix
PowerFlex roles	Compute-only, storage-only	Compute-only, storage-only (CentOS only)
Switch Port Configuration	• Port channel (LACP enabled) • Port channel • Trunk port	• Port channel (LACP enabled) • Trunk Port
Target Boot Device	• Local flash storage for Dell PowerFlex	Local flash storage for Dell PowerFlex
Network Settings	• Static bonding NIC port design or LACP bonding NIC port design • 10 GB, 25 GB, 100 GB • Required PXE network • Network Automation Type: Full	• LACP bonding NIC port design • 25 GB • No PXE network (using iDRAC virtual media) • Network Automation Type: Partial

Related information

Adding an existing resource group
Build and publish a template

Storage nodes

This topic provides reference information for the **Storage Nodes** page.

Add a storage node

You can add a storage node either through a resource group or the PowerFlex Installer.

1. On the menu bar, click **Lifecycle** > **Storage Nodes**.
2. Click **+ Add Storage Node**.
   An **Add Storage Node** dialog box opens.
3. If there are resource groups that are installed on your PowerFlex system, add a storage node through the resource groups.
   a. In the **Add Storage Node** window, go to the resource groups by clicking **Continue to Resource Groups**.
   You are redirected to the **Resource Groups** page.
   b. Select the resource group that you want to add a storage node and click **View Details** > **Add Resources** > **Add Nodes**.
4. If the nodes in your environment are not deployed through the resource groups, add a storage node through the getting started page.
   a. In the **Add Storage Node** dialog box, click **Continue to Getting Started**.
   The **PowerFlex Installer** wizard opens.
   b. Use the **PowerFlex Installer** wizard to add a storage node.

Related information

Getting started
Adding nodes to a resource group

View objects related to a storage node

You can view the protection domain, device groups, and devices associated with a specific storage node. You can view the overall health of the storage node and devices.

1. On the menu bar, click **Lifecycle** > **Storage Nodes**.
2. From the list of storage nodes, select the check box beside the storage node whose related objects you want to view.
3. In the right hand pane, click on any of the objects related to the storage node to view the object details in the storage system.

View component details of a storage node

You can view the details of the MDM, PowerFlex data server (PDS), device gateway target (DGWT), and devices associated with a storage node.

1. On the menu bar, click **Lifecycle** > **Storage Nodes**.
2. From the list of storage nodes, select the check box beside the storage node whose component details you want to view, and click **View Details**.
   A **Storage Nodes** page opens to display the details of the MDM, PowerFlex data server (PDS), device gateway target (DGWT), and devices.
3. To view the MDM, click **View Details**.
   You are redirected to **Resources**, where you can view the MDM cluster details, including the MDM role, name, management IP address, IP addresses, port, and status.
4. If you want to view details of other components, you must go back to **Lifecycle** > **Storage Nodes** and select the check box beside the storage node whose component details you want to view, and click **View Details**.

Add a device to a storage node

You can add a device to a storage node.

1. On the menu bar, click **Lifecycle** > **Storage Nodes**.
2. From the list of storage nodes, select the check box beside the storage node that you want to add a device to, and click **More Actions** > **Add Devices**.
   A **Add Device to Storage Node** dialog box opens.
3. From the **Add Device to Storage Node** dialog box, enter the following information:
   - **Path**: Enter a path for the device.
   - (Optional)**Name**: Enter a name for the device.
   - **Device Group**: From the drop-down, select the device group that the device will be part of.
   - **Media Type**: The media type is automatically populated based on the device group that you selected.
4. Click **Add devices**.
5. (Optional) You can add additional devices by clicking **Add row**.
6. (Optional) You can choose to set up device tests through **Advanced Options**. You can choose from the following settings:
   - **Test and activate device**: This operation tests whether the device is readable and writable. If it is readable and writable, the device is activated.
   - **Test only**: This operation tests whether the device is readable and or writable.
   - **Activate without test**: This operation activates the device without testing the devices.
   - **Timeout**: This setting specifies the time-out for each operation.
   - **Device Takeover**: This setting enables the MDM to use the device for the operations.
7. Verify that the operation is finished and is successful, and click **Dismiss**.

Remove a device from a storage node

You can remove a device from a storage node. However, removing a device reduces the capacity, and potentially enters the storage node into a state where the storage node does not have the same amount of capacity for failover and resiliency.

1. On the menu bar, click **Lifecycle** > **Storage Nodes**.
2. From the list of storage nodes, select the check box beside the storage node that you want to remove a device from and click **More Actions** > **Remove Device**.
3. In the **Remove Device** dialog box, verify you selected the wanted device and that the removal impacts the correct device group.
4. Type **REMOVE DEVICE** in the box and click **Remove**.
5. Verify that the operation is finished and is successful, and click **Dismiss**.

Remove a storage node

When you remove a storage node, the devices that are associated with the storage node are removed.

You cannot remove a storage node with an active MDM role.

1. On the menu bar, click **Lifecycle** > **Storage Nodes**.
2. From the list of storage nodes, select the check box beside the storage node that you want to remove , and click **More Actions** > **Remove Storage Node**.
3. In the **Remove Storage Node** dialog box, verify that you selected the wanted storage node.
4. Type **REMOVE STORAGE NODE** in the box and click **Remove**.
5. Verify that the operation is finished and is successful, and click **Dismiss**.

Related information

Enter a storage node into maintenance mode

Abort a storage node removal

You can stop the removal of a storage node during the remove operation.

You must start the remove storage node operation before you can stop the removal of the storage node.

1. On the menu bar, click **Lifecycle** > **Storage Nodes**.
2. From the list of storage nodes, select the check box beside the storage node that you want to stop the removal of, and click **More Actions** > **Abort Remove Storage Node**.
3. In the **Abort Removal for Storage Node** dialog box, verify that you want to stop the removal of the storage node and click **Abort**.
4. Verify that the operation is finished and is successful, and click **Dismiss**.

Enter a storage node into maintenance mode

When a storage node is put into maintenance both the PowerFlex data server and the device gateway target go into maintenance mode.

If the storage node that is entering into maintenance mode is serving as an NVMe host, you must go to NVMe hosts and put the node into maintenance mode before starting a reboot of this storage node.

1. On the menu bar, click **Lifecycle** > **Storage Nodes**.
2. From the list of storage nodes, select the check box beside the storage node that you want to enter into maintenance mode and click **More Actions** > **Enter Maintenance Mode**.
3. In the **Enter Storage Node into Maintenance Mode** dialog box, verify that you are selecting the wanted storage node, and click **Enter Maintenance Mode**.
4. Verify that the operation is finished and is successful, and click **Dismiss**.

Related information

Enter an NVMe target into maintenance mode

Exit a storage node from maintenance mode

If the devices associated with the storage node are error free, you can exit a storage node from maintenance mode.

1. On the menu bar, click **Lifecycle** > **Storage Nodes**.
2. From the list of storage nodes, select the checkbox beside the storage node that you want to exit from maintenance mode and click **More Actions** > **Exit Maintenance Mode**.
3. In the **Exit Maintenance Mode on Storage Node** dialog box, verify that you are selecting the wanted storage node, and click **Exit Maintenance Mode**.
4. Verify that the operation is finished and is successful, and click **Dismiss**.

Rename a storage node

You can change the name of a storage node.

1. On the menu bar, click **Lifecycle** > **Storage Nodes**.
2. From the list of storage nodes, select the checkbox beside the storage node that you want to rename and click **More Actions** > **Rename Storage Node**.
3. In the **Rename Storage Node** dialog box, verify that you are selecting the wanted storage node, and enter the new name. Click **Apply**.
4. Verify that the operation is finished and is successful.

Modify an IP address of a storage node

You can update the IP addresses and assign roles for a storage node. Any IP address changes must be consistently applied across all nodes in the same protection domain to maintain connectivity. You can modify up to a maximum of eight IP addresses.

Any update that is made changes the IP address configuration within the PowerFlex cluster. However, the IP address on the operating system of the storage node is not modified.

Modifying these settings can impact cluster operations. Depending on the changes, you must reconfigure the MDM cluster and perform additional follow-up actions.

1. On the menu bar, click **Lifecycle** > **Storage Nodes**.
2. From the list of storage nodes, select the checkbox beside the storage node whose IP address you want to modify and click **More Actions** > **Edit IP Storage Node**.
3. In the **Modify IP Addresses for Storage Node** dialog box,
   a. To delete an IP address, click **X** and then click **Apply**.
   b. To add an IP address, click **Add**.
   c. To assign a role to the network, select one of the following roles from the dropdown:
      - **All**
      - **Storage Only**
      - **Application Only**
4. Verify that the operation is finished and is successful.

Resources

This section provides reference information for the **Resources** page.

Related information

Deploying and provisioning
Managing components
Monitoring system health
Managing components
Monitoring system health

Resource health status

PowerFlex Manager assigns health status to the resources based on the conditions described in the following table.

Table 101. Resource health status

Icon	Health status	Description
	Healthy	Indicates that there is no issue with the resource and it is working as expected.
	Warning	Indicates that the resource is in a state that requires corrective action, but does not affect overall system health. For example, the firmware running on the resource is not at the required level or not compliant. For supported switches, a warning health status might be because the SNMP community string is invalid or not set correctly. In this case, PowerFlex Manager is unable to perform health monitoring for the switch. **NOTE:** If the resource group has VMware ESXi and is set to maintenance mode, a message is displayed in the **View Details** pane.
	Critical	Indicates that an issue requiring immediate attention exists in one of the following hardware or software components in the device: • Battery • CPU • Fans • Power supply • Storage devices • Licensing For supported switches, a critical health status might indicate that the power supply is not working properly or a CPU has overheated. **NOTE:** If the resource group is in a power off state, a message displays in the **View Details** pane.
	Unknown	Indicates that the resource status is unknown. For example, for a supported switch, an unknown status might indicate that the switch has been turned off or is still being discovered.
	Pending	Indicates that the resource status is pending.
	Service Mode	Indicates that the resource is in **Service Mode**. **NOTE:** If the resource fails to exit **Service Mode**, an informational message displays in the recent activity.

Related information

Discover a resource
Removing resources
Discover a resource
Exporting a compliance report for all resources

Resource operational state

After initiating resource discovery, PowerFlex Manager assigns one or more of the states to the resources. These operational states display on the **Resources** page, in the **Deployment Status** column of the **All Resources** tab.

Table 102. Deployment status

Deployment status	Description
Not In Use	Resource is available for deployment.
Available	Resource is available. Applies only to switches, virtual machine, or gateway.
Deploying	Resource is in the process of being deployed in a resource group.
In Use	Resource is deployed in a resource group.
Pending Updates	One or more of the following tasks are in progress: • Discovering resource • Determining resource details, including firmware version • Applying template to the resource • Updating firmware • Removing resource from PowerFlex Manager inventory
Deployment Failed	Resource group deployment failed.
Cancelled	Resource group deployment is cancelled.

PowerFlex Manager keeps track of which resources it is managing. These operational states display on the **Resources** page, in the **Managed State** column of the **All Resources** tab.

Table 103. Managed state

Managed state	Description
Managed	Indicates that PowerFlex Manager manages the firmware on that node, and the node can be used for deployments.
Unmanaged	Indicates that the resource is not managed by PowerFlex Manager. By default, the operational state for all discovered nodes is **Unmanaged**. If you want to perform firmware updates or deployments on a discovered node, you need to select the node and change the operational state to **Managed**. If you have not yet uploaded a license file, PowerFlex Manager is configured for monitoring and alerting only. All resources are restricted to the **Unmanaged** resource state, and you cannot change the state to **Managed** or **Reserved**.
Reserved	Indicates that PowerFlex Manager only manages the firmware on that particular node, but that node cannot be used for deployments. You can assign a host to the reserved state only if the host has been discovered, but is not part of a resource group.

Compliance status

PowerFlex Manager assigns one of the following firmware statuses to the resources:

Table 104. Compliance status

Firmware status	Description
Compliant	The firmware running on the resource is compliant with the firmware version that is specified in the default compliance version.
Non–Compliant	The firmware running on the resource is less than or greater than the firmware version specified in the default compliance version. Indicates that firmware update is required.
Update Required	The firmware running on the resource is less than the minimum firmware version recommended in the default compliance version. Indicates that firmware update is required.

Related information

Discover a resource
Removing resources
Discover a resource
Exporting a compliance report for all resources

Discovery overview

You can discover new resources or existing resources that are already configured within your environment. After discovery, you can deploy resource groups on these resources from a template. Only administrator-level users can discover resources.

By default, the operational state for all discovered nodes is **Unmanaged**. If you want to perform firmware updates or deployments on a discovered node, select the node and change the operational state to **Managed**.

If you have not yet uploaded a license, PowerFlex Manager is configured for monitoring and alerting only. In this case, all the resources are restricted to the **Unmanaged** resource state, and you cannot change the state to **Managed** or **Reserved**.

For some resources such as nodes, the default credentials are prepopulated in PowerFlex Manager. If the credentials are changed from the defaults, add the credential to PowerFlex Manager with the new login information.

Element Manager discovery

The IPI cabinet is discovered as an element manager. Once PowerFlex Manager discovers the IPI cabinet, you can view cabinet details or you can open the IPI appliance management application to view more details.

The CloudLink Center is discovered as an element manager. If you create your own username in CloudLink Center, you need the SecAdmin access role and the local user type for the new user to complete discovery. The client user type does not work for discovery in PowerFlex Manager. If one member of a cluster of CloudLink Center is shut down, the inventory still succeeds because PowerFlex Manager falls back to inventorying the other CloudLink Center members if the first one is down.

Node discovery

PowerFlex Manager supports PowerFlex node discovery and allows you to onboard nodes by configuring the initial management IP address and iDRAC credentials. To perform initial discovery and configuration, verify that the management IP address is set on the node and that PowerFlex Manager can access the IP address through the network. While configuring IP addresses on the node, verify that PowerFlex Manager can access any final IP address in a range used for hardware management, to complete discovery of these nodes.

PowerFlex Manager also allows you to use name-based searches to discover a range of nodes that were assigned IP addresses by DHCP to iDRAC. You can search for a range of DNS hostnames or a single hostname within the **Discovery Wizard**. After you perform a name-based discovery, PowerFlex Manager operations to the IDRAC continue to use name-based IP resolution, since DHCP may assign alternate addresses.

Switch discovery

If you attempt to discover a Cisco switch with terminal color configured, the discovery fails. To discover the switch successfully, disable the terminal color option by running `configure no terminal color persist`.

VM manager discovery

A VMware vCenter is discovered as a VM manager. PowerFlex Manager users with the administrator role can discover a vCenter in PowerFlex Manager. A vCenter read-only user can discover a vCenter in PowerFlex Manager only after the following requirements are met:

- The vCenter user who is specified in the vCenter credential is granted the `VirtualMachine.Provisioning.ReadCustSpecs` and `StorageProfie.View` privileges.
- The permission containing these privileges is granted to that user on the root vCenter object and the `Propogate to children` property is set to `True`.

PowerFlex Manager allows you to deploy new compute-only resource groups and add existing resource groups to a vCenter that has an Enhanced Linked Mode (ELM) configuration. ELM connects multiple vCenter servers, allowing you to search across servers and perform other vCenter management functions. ELM does not provide clustering or redundancy.

PowerFlex Manager allows you to deploy new compute-only resource groups and add existing resource groups to a vCenter that has an Enhanced Linked Mode (ELM) configuration. ELM connects multiple vCenter servers, allowing you to search across servers and perform other vCenter management functions. ELM does not provide clustering or redundancy.

If you are working with a vCenter that has an enhanced linked mode configuration, you must discover only the vCenter to which you want to deploy or add an existing resource group. You do not need to discover the other vCenters that are connected through the enhanced linked mode configuration.

Node pools

In PowerFlex Manager, a node pool is made up of nodes that are grouped for specific use cases such as business units or workload purposes. An administrator can specify which users can access these node pools.

Standard users can view details only for node pools for which they have permission.

From the **Node Pools** tab, you can view existing node pools.

Users with an administrator role can create, edit, or delete node pools.

Click a node pool from the list to view detailed information about the following tabs:

- **Nodes**: Displays the number of nodes that are associated with the node pool.
- **Users**: Displays the number of users with access rights to the node pool.

Configuration checks

When you initiate a resource group action, PowerFlex Manager performs critical configuration checks to ensure that the system is healthy before proceeding. A configuration check failure may result in the resource group action being blocked. You can export a PDF report at any time that shows detailed system health and configuration data. When you generate the report, PowerFlex Manager initiates the full set of configuration checks and reports the results.

Configuration checks for switches

Dell Networking OS9 switches are not covered by the configuration checks.

Table 105. Configuration checks

Configuration check	Resource group actions blocked if the check fails
PowerFlex gateway is powered off	None
PowerFlex cluster state is not normal	• Add node • Add volume

Table 105. Configuration checks (continued)

Configuration check	Resource group actions blocked if the check fails
	• Delete node • Delete resource group • Add resource group with an existing PowerFlex cluster • Compliance upgrade • Enter service mode • Drive replacement
Secondary MDMs have a status not equal to normal	• Add nodes • Add volumes • Delete node • Delete resource group • Add resource group with an existing PowerFlex cluster • Compliance upgrade • Enter service mode • Drive replacement
storage node registered data network IP address count does not equal 2 or 4	• Compliance upgrade • Enter service mode • Drive replacement • Reconfigure MDM roles
storage node has a state not equal to connected	• Add volume • Compliance upgrade • Reconfigure MDM roles
storage node interfaces have incorrect MTU sizes	• Add node • Add resource group with an existing PowerFlex cluster • Compliance upgrade • Reconfigure MDM roles
VMware ESXi vmks are not configured with an MTU size of 9000	• Add node • Add resource group with an existing PowerFlex cluster • Compliance upgrade
Interfaces are not configured with an MTU size of 9216, and QOS is enabled with MTU size	• Add node • Add resource group with an existing PowerFlex cluster • Compliance upgrade
storage node has a Performance Profile not equal to High Performance	None
SDC has a Performance Profile not equal to High Performance	None

Related information

Exporting a configuration report for all resources and resource groups

Settings

This section provides reference information for the **Settings** page.

Backup details

PowerFlex Manager backup files include the following information:

- Activity logs
- Credentials

- Deployments
- Resource inventory and status
- Events
- Initial setup
- IP addresses
- Jobs
- Licensing
- Networks
- Templates
- Users and roles
- Resource module configuration files
- Performance metrics

Network types

You can manage various network types in PowerFlex Manager.

- **General Purpose LAN**—Used to access network resources for basic networking activities.
- **Hypervisor Management**—Used to identify the management network for a hypervisor or operating system that is deployed on a node.
- **Hypervisor Migration**—Used to manage the network that you want to use for live migration. Live migration enables you to move running virtual machines from one node of the failover cluster to different node in the same cluster.
- **OS Installation**—Allows static or DHCP network for operating system imaging on nodes.
- **Hardware Management**—Used for out-of-band management of hardware infrastructure
- **PowerFlex Data**—Used for data traffic between storage node and storage data clients (SDC)
- **PowerFlex Data (Client Traffic Only)**—Used for storage data client traffic only
- **PowerFlex Data (Server Traffic Only)**—Used for storage node traffic only
- **PowerFlex Management**—Used for PowerFlex system management

23

PowerFlex Manager user interface reference (mirroring)

This section provides reference details for various parts of the user interface for PowerFlex Manager.

The topics in this section are organized according to the main tabs within the user interface.

Life cycle (mirroring)

This section provides reference information for the **Resource Groups** and **Templates** pages.

Related information

Deploying a resource group (mirroring)
Adding an existing resource group (mirroring)
Viewing a compliance report (mirroring)

Resource groups

This section provides reference information for the **Resource Groups** page.

Related information

Managing components
Managing external changes
Managing components

Resource group states (mirroring)

The resource group states indicate the current condition or health of all resources within a group, such as whether they are active, updating, or experiencing issues. They provide a quick, high-level view that helps you to understand the overall system status and decide when action or troubleshooting is needed.

Table 106. Resource group states

Icon	Resource group state	Description
	Healthy	The resource group was deployed successfully, and the resources are firmware compliant and healthy.
	Warning	One or more resources in the resource group require corrective action. This state does not affect the overall system health. For example, the firmware running on the resource is not at the required level or not compliant.
	Critical	Resource group deployment has failed due to some issues.
	Pending	Resource group deployment is scheduled for a later time or date.
	In Progress	Resource group deployment is in progress.

Table 106. Resource group states (continued)

Icon	Resource group state	Description
	Canceled	Resource group deployment was canceled.
	Incomplete	The resource group is not fully functional because it has no associated volumes. Click **Add Resources** to add volumes.
	Service Mode	The resource group has been placed in service mode because one of the nodes within the resource group was put into service mode. If a node is in VMware ESXi maintenance mode or PowerFlex maintenance mode, PowerFlex Manager detects this situation and automatically places the node in service mode and also ensures that the resource group itself goes into service mode. If a node is in PowerFlex maintenance mode for more than 30 minutes, PowerFlex Manager sends a critical alert to Dell Technologies connectivity services.
	Lifecycle Mode	The resource group supports health and compliance monitoring, service mode, and non-disruptive updates. All other resource group operations are blocked. Life cycle mode controls the operations that can be performed for configurations that have limited support.
	Managed Mode	The resource group supports health and compliance monitoring, non-disruptive updates, automated resource addition, and automated resource replacement features.

Existing resource groups (mirroring)

If you already have a cluster deployed that includes Dell-based hardware, you can discover and import these hardware resources as an existing resource group. Even though these resources were not originally deployed with PowerFlex Manager, you can take advantage of the health monitoring, compliance, and update features.

When you add an existing resource group for a PowerFlex cluster, PowerFlex Manager matches up the hosts, VMware vCenter, and other items it finds with discovered resources in the resource list.

Only managed or reserved resources are included in the resource group. Any resource discovered as **Unmanaged** is missing from the resource group. If a component is missing, you can change your resource inventory, and update the resource group to reflect these changes. Go to the resources list, select the component, and mark it as **Managed** by selecting **Change resource state to Managed**. Then perform an **Update Resource Group Details** operation on the resource group to pull in the missing component.

Each node is identified as a PowerFlex host if the inventory for the node shows that it is present in a gateway, and it is properly configured as a storage data client (SDC) and storage data server (SDS) node. A PowerFlex host has the **Use Node for Dell PowerFlex** checkbox set to **True** in the operating system settings for the node within the resource group deployment settings.

You can perform most of the actions on an existing resource group that are available within PowerFlex Manager for a new resource group. For some configurations, PowerFlex Manager displays a yellow banner for the resource group to indicate that some features are not available.

PowerFlex Manager enables you to add an existing resource group for an environment using VMware NSX-T or NSX-V. Most resource group actions are disabled for an NSX-T or NSX-V configuration, except the ability to update the firmware and software components, remove resources (or the resource group as a whole), and update resource group details.

When you add an existing resource group for a CloudLink configuration, you choose a single CloudLink Center as the target for the resource group. If the CloudLink Center for the resource group shuts down, PowerFlex Manager loses communication with the CloudLink Center. If the CloudLink Center is part of a cluster, PowerFlex Manager moves to another CloudLink Center when you update the resource group details.

Life cycle mode (mirroring)

If you add an existing resource group that includes an unsupported configuration, PowerFlex Manager might put the resource group in life cycle mode. This mode limits the actions that you can perform within the resource group.

Life cycle mode allows the resource group to perform only monitoring, service mode, and compliance upgrade operations. All other resource group operations are blocked when the resource group is in life cycle mode. Life cycle mode is used to control the operations that can be performed for configurations that have limited support.

When you add an existing resource group, PowerFlex Manager puts the resource group in life cycle mode if the configuration you want to import includes any of the following:

- Invalid server inventory
- Missing network settings

  The minimum required IP addresses are one data IP address and one PowerFlex IP address for a storage-only resource group, and one data IP address, one PowerFlex IP address, and one VMware ESXi IP address for a hyperconverged resource group.
- Unsupported NIC (a 1 GB QLogic NIC, for example)
- Unsupported server configurations (an unsupported SD Boot device, for example)
- Unsupported NIC teaming policies, an unsupported switch port configuration, or an unsupported access facing port configuration
- Network configuration without a PXE VLAN setting
- No switch configuration
- PowerFlex MDM cluster without virtual IP addresses
- vSphere Cluster Services (vCLS) VMs on local storage

When you add an existing resource group, PowerFlex Manager checks to see whether there are any vSphere Cluster Services (vCLS) VMs on local storage. If it finds any, it puts the resource group in life cycle mode and lets you migrate the VMs to shared storage.

PowerFlex Manager also puts a resource group in life cycle mode if you select a minimal compliance version that includes firmware only for the resource group.

When PowerFlex Manager must put a resource group in life cycle mode, the **Summary** page for the **Add Existing Resource Group** wizard displays a warning message indicating the reason.

In some situations, an imported configuration might not meet the minimal requirements for life cycle mode. In this case, PowerFlex Manager does not allow you to add the resource group.

For missing virtual IP addresses or no switch connectivity, you can correct the error and use **Update Resource Group Details** to take the resource group out of life cycle mode.

If you have manually added NSX to a cluster outside of PowerFlex Manager, click **Update Resource Group Details** on the resource group, so it will correctly reflect the environment with NSX, which will be life cycle mode.

Templates

This section provides reference information for the **Templates** page.

Related information

Getting started
Getting started (mirroring)
Managing components
Managing components

Managing templates

On the **Templates** page, you can view information about templates in a list or tile view. To switch between views, click the tile icon or list icon at the top of the **Templates** page. When in tile view, you can view the templates under a category by clicking the graphic that represents the category.

Users with standard permissions can view details of only those templates for which the administrator has granted permission.

Templates - My Templates List View

The **My Templates** page displays the details of the templates that you have created.

Table 107. My Templates page

Field	Description
+ Add a Template	Allows you to add a template. After you add a template, you can add node, cluster, and VM components on the template builder page.
Export All	Allows you to export all templates to a CSV file.
Filter By	Allows you to filter and view templates based on the template category.
State	Displays the state of the template - **Draft** or **Published**.
Category	Displays the template category.
Name	Displays the template name.
Last Deployed On	Displays the date when the template was last deployed.
Components	Displays the components in the template.

Templates - Sample Templates

The **Sample Templates** page displays the default templates that you can use in your environment.

Table 108. Templates - Sample Templates page

Field	Description
Name	Displays the template name.
Components	Displays the components in the template.

Templates

On the **Templates** page, the right pane displays the name of the template, icons of components in the template, and the following details for a selected template:

Table 109. Templates page

Field	Description
Edit	Click to edit the template.
Delete	Click to delete the template.
View Details	Click to view the details of the template, such as components in the template.
Clone	Click to clone the template.
Export Template	Click to export the template to a GPG file. You can use the exported template to duplicate the settings.
Created On	Displays the date and time when the template was created.
Created By	Displays the name of the user who created the template.
Updated On	Displays the date and time when the template was updated.
Updated By	Displays the name of the user who updated the template.

Building a template overview

The template builder allows you to build a customized template by configuring both physical and virtual components. On the template builder page, you can set the component properties. For example, you can create a template that provisions only physical nodes with operating systems on them.

NOTE: A newly created or a cloned template appears in a draft state on the **Template** page and remains in the same state until published.

You can configure node, cluster, and VM components in a template.

The template builder page displays a graphical representation of the topology that is created within a particular template. From this page, you can:

- Add node, cluster, and VM components to a template
- Build and publish a template
- Delete a template
- Import a template
- Deploy a resource group (this feature is available only on published templates)

Component types

Components (physical or virtual or applications) are the main building blocks of a template.

PowerFlex Manager has the following component types:

- Node (Software/hardware and software only)
- Cluster
- VM

Related information

Adding components to a resource group
Edit a template
Deploying a resource group
View template details
Adding components to a resource group (mirroring)
Deploying a resource group (mirroring)

Node settings

This table describes the following node settings: hardware, BIOS, operating system, and network.

Table 110. Node settings

Setting	Description
Full network automation	Enables you to perform deployments with full network automation. This feature allows you to work with supported switches, and requires less manual configuration. Full network automation also provides better error handling since PowerFlex Manager can communicate with the switches and identify any problems that may exist with the switch configurations.
Partial network automation	Enables you to perform switchless deployments with partial network automation. This feature allows you to work with unsupported switches, but requires more manual configuration before a deployment can proceed successfully. If you choose to use partial network automation, you give up the error handling and network automation features that are available with a full network configuration that includes supported switches. For a partial network deployment, the switches are not discovered, so PowerFlex Manager does not have access to switch configuration information. You must ensure that the switches are configured correctly,

Table 110. Node settings (continued)

Setting	Description
	since PowerFlex Manager does not configure the switches for you. If your switch is not configured correctly, the deployment may fail and PowerFlex Manager is not able to provide information about why the deployment failed. For a partial network deployment, you must add all the interfaces and ports, as you would when deploying with full network automation. However, you do not need to add the operating system installation network, since PXE is not required for partial network automation. PowerFlex Manager uses virtual media instead for deployments with partial network automation. The **Switch Port Configuration** can be set to **Port Channel (LACP enabled)** or **Trunk**. If **Port Channel (LACP enabled)** is used, the LACP fallback or LACP ungroup option must be configured on the port channels.
Component Name	Indicates the node component name
Drive Encryption Type	Specifies the type of encryption to use when encryption is enabled. The encryption options are: • Software encryption • Self-encrypting drive (SED)
Number of Instances	Enter the number of instances that you want to add. If you select more than one instance, a single component representing multiple instances of an identically configured component is created. Edit the component to add extra instances. If you require different configuration settings, you can create multiple components.
Related Components	Select **Associate All** or **Associate Selected** to associate all or specific components to the new component.
Import Configuration from Reference Node	Click this option to import an existing node configuration and use it for the node component settings. On the **Select Reference Node** page, select the node from which you want to import the settings and click **Select**.
OS Settings	
Host Name Selection	If you choose **Specify At Deployment Time**, you must type the name for the host at deployment time. If you choose **Auto Generate**, PowerFlex Manager displays the **Host Name Template** field to enable you to specify a macro that includes variables that produce a unique hostname. For details on which variables are supported, see the context-sensitive help for the field. If you choose **Reverse DNS Lookup**, PowerFlex Manager assigns the hostname by performing a reverse DNS lookup of the host IP address at deployment time.
OS Image	Specifies the location of the operating system image install files. You can use the image that is provided with the target compliance file, or specify your own location, if you created additional repositories. To deploy a compute-only or storage-only resource group with the Linux image that is provided with a compliance file, choose **Use Compliance File Linux image**. To deploy a storage-only resource group with Red Hat Enterprise Linux, you must create a repository on the **Settings** page and specify the path to the Red Hat Enterprise Linux image on a file share. Dell Technologies recommends that you use one of your own images that are published from the customer portal at Red Hat Enterprise Linux. For Linux, you may include one node within a resource group. For VMware ESXi, you must include at least two nodes.

Table 110. Node settings (continued)

Setting	Description
	NOTE: If you select an operating system from the **OS Image** drop-down menu, the field **NTP Server** displays. This field is optional, but it is highly recommended that you enter an NTP server IP to ensure proper time synchronization with your environment and PowerFlex Manager. Sometimes when time is not properly synchronized, resource group deployment failure can occur.
OS Credential	Select an **OS Admin** or **OS User** credential that you created on the **Credentials Management** page. Alternatively, you can create a credential while you are editing a template. If you select a credential that was created on the **Credentials Management** page, you do not need to type the username and password, since they are part of the credential definition. PowerFlex Manager allows to specify a non-root user instead of the root user when you configure a template for a compute-only, storage-only, or hyperconverged deployment.
NTP Server	Specifies the IP address of the NTP server for time synchronization. If adding more than one NTP server in the operating system section of a node component, be sure to separate the IP addresses with commas.
Use Node For Dell PowerFlex	Indicates that this node component is used for a PowerFlex deployment. When this option is selected, the deployment installs the MDM, SDS, and SDC components, as required for a PowerFlex deployment in a VMware environment. The MDM and SDS components are installed on a dedicated PowerFlex VM (SVM), and the SDC is installed directly on the VMware ESXi host. To deploy a PowerFlex cluster successfully, include at least three nodes in the template. The deployment process adds an SVM for each hyperconverged node. PowerFlex Manager uses the following logic to determine the MDM roles for the nodes: 1. Checks the PowerFlex gateway inventory to see how many primary MDMs, secondary MDMs, and tiebreakers are present, and the total number of SDS components. 2. Adds the number of components being deployed to determine the overall PowerFlex cluster size. For example, if there are three SDS components in the PowerFlex gateway inventory, and you are deploying two more, you will have a five node cluster after the deployment. 3. Adds a single primary MDM and determines how many secondary MDMs and tiebreakers should be in the cluster by looking at the overall cluster size. The configuration varies depending on the size of the cluster: • A three-node cluster has one primary, one secondary, and one tiebreaker. • A five-node cluster has one primary, two secondaries, and two tiebreakers. 4. Determines the roles for each of the new components being added, based on the configuration that is outlined above, and the number of primary, secondary, and tiebreakers that are already in the PowerFlex cluster. At deployment time, PowerFlex Manager automatically sets up the DirectPath I/O configuration on each hyperconverged node. This setup makes the devices available for direct access by the virtual machines on the host and also sets up the devices to run in PCI passthrough mode. For each SDS in the cluster, the deployment adds all the available disks from the nodes to the storage pools created. For each compute-only or hyperconverged node, the deployment installs the SDC VIB.

Table 110. Node settings (continued)

Setting	Description
	When you select this option, the teaming and failover policy for the cluster are automatically set to **Route based on IP hash**. Also, the uplinks are configured as active and active, instead of active and standby. Teaming is configured for all port groups, except for the PowerFlex data 1 and PowerFlex data 2 port groups. If you select the option to **Use Node For Dell PowerFlex**, the **Local Flash Storage for Dell PowerFlex** option is automatically selected as the **Target Boot Device** under **Hardware Settings**.
PowerFlex Role	Specifies one of the following deployment types for PowerFlex: ● **Compute Only** indicates that the node is only used for compute resources. ● **Storage Only** indicates that the node is only used for storage resources. ● **Hyperconverged** indicates that the node is used for both compute and storage resources. If you select a VMware ESXi image type in the **OS Image** field, the **PowerFlex Role** must be set to **Compute Only** or **Hyperconverged**. If you add a compute-only node, only the SDC is added. If you add a hyperconverged node, both the SDC and SDS are added. If you select a Red Hat Enterprise Linux image type in the **OS Image** field, the **PowerFlex Role** must be set to **Storage Only**. If you add a storage-only node, only the SDS is added. The only prerequisites for a storage-only node are that the iDRAC must have an IP address and a credential. PowerFlex Manager takes care of all other configuration steps that are required for the node. For each node, PowerFlex Manager configures the MDM roles as needed and configures the SDS RPMs. Once the cluster is set up, PowerFlex Manager adds every node as an SDS. Then, it adds all available disks for the SDS device, adds a storage pool, and adds all disks to the storage pool. If you are creating a compute-only or hyperconverged template, be sure to include both the **VMware Cluster** and **PowerFlex Cluster** components in the template builder. If you are creating a storage-only template, do not include a **VMware Cluster** component in the template builder. Only the **PowerFlex Cluster** component is required for a storage-only template.
Client Storage Access	Determines how clients access storage. For a storage-only role, select one of the following options: ● Storage Data Client (SDC) Only ● SDC and NVMe/TCP Initiator For a compute-only role the Client Storage Access control is not displayed, and the client access is set to SDC automatically. For a hyperconverged role, the Client Storage Access control is not displayed, and the client access is set to SDC/SDS automatically.
Enable Compression	Enables compression on the protection domain. This option allows you to take advantage of PowerFlex NVDIMM (non-volatile inline memory module) compression. You can enable compression on a storage-only or hyperconverged resource group. Compression is supported for new resource group deployments, and existing resource groups. You can also enable compression for storage-only and hyperconverged nodes when performing a scale up of a resource group. If you select this option, PowerFlex Manager looks for nodes that have at least two NVDIMMs installed, and SSD or NVMe, and have persistent memory. Fine granularity is not supported on HDDs.

Table 110. Node settings (continued)

Setting	Description
	If compression is enabled, the template adds fields to the **PowerFlex Cluster Settings** to allow you to specify an acceleration pool name and granularity setting for the storage pool. The storage pool must be set to fine granularity. The compression method at the storage pool level is overridden by the setting at the volume level. PowerFlex Manager creates the acceleration pool and sets the granularity according to **PowerFlex Cluster Settings** when you deploy the resource group.
Enable Encryption	Enables disk encryption on the node. This option allows you to take advantage of CloudLink encryption. You can enable CloudLink encryption on a storage-only or hyperconverged resource group. Encryption is supported for new resource group deployments, and existing resource groups. You can also enable encryption for storage-only and hyperconverged nodes when performing a scale up of a resource group. A resource group cannot mix encrypted and unencrypted nodes. Scale up of a storage-only or hyperconverged resource group is supported, with the new nodes using the same encryption settings as the nodes already in the resource group. If you select this option, PowerFlex Manager looks for servers with 12 cores or more for CloudLink deployments. After encryption is enabled, the template displays a warning message indicating that the template is missing some data. The template adds the **Cloud Link Center Settings** section to the PowerFlex cluster component to allow you to specify the required data. Some PowerFlex nodes might not be selected for deployment, depending on the encryption type that is selected. For example, if you choose **Software Encryption**, you cannot include a PowerFlex node with only SEDs. If you choose **Self Encrypting Drive**, you cannot include a PowerFlex node with only software encryption drives. PowerFlex Manager does not allow you to mix SEDs and software encryption drives in the same protection domain. Servers do not typically have this drive combination, but PowerFlex Manager verifies the drives and uses only servers of the specified type. **Validate Settings** detects PowerFlex nodes that do not match the specified **Drive Encryption Type**.
Enable Replication	Enables replication for a storage-only or hyperconverged resource group. Replication allows you to mirror the data across different geographical sites using native volume-level asynchronous replication. PowerFlex Manager deploys and configures the storage data replicator (SDR) on all SDS nodes. PowerFlex Manager configures the journal capacity before adding the SDR. If you enable replication for a template, you must have two different replication networks attached to the template before you can publish it. When replication is enabled, PowerFlex Manager lets you set the **Journal Capacity** at the time you deploy the resource group, or when you add a node to the resource group.
Drive Encryption Type	Specifies the type of encryption to use when encryption is enabled. The options are: • **Software Encryption** • **Self Encrypting Drive (SED)**

Table 110. Node settings (continued)

Setting	Description
	Some nodes might not be selected for deployment, depending on the encryption type selected. For example, if you choose **Software Encryption**, you cannot include a node with only SEDs. Similarly, if you choose **Self Encrypting Drive**, you cannot include a node with only software encryption drives. PowerFlex Manager does not allow you to mix SEDs and software encryption drives in the same protection domain. Servers should not typically have this mix, but PowerFlex Manager checks for this and uses only servers of the type you specify. **Validate Settings** detects nodes that do not match the specified **Drive Encryption Type**.
Switch Port Configuration	Specifies whether Cisco virtual PortChannel (vPC) or Dell Virtual Link Trunking (VLT) is enabled or disabled for the switch port. For hyperconverged templates, the options are: ● **Port Channel** turns on vPC or VLT. ● **Port Channel (LACP enabled)** turns on vPC or VLT with the link aggregation control protocol enabled. For storage-only templates, the options are: ● **Port Channel (LACP enabled)** turns on vPC or VLT with the link aggregation control protocol enabled. ● **Trunk Port** enables a networking configuration that does not use vPC or VLT. For compute-only templates that use a Linux operating system image, the options are: ● **Port Channel (LACP enabled)** turns on vPC or VLT with the link aggregation control protocol enabled. ● **Trunk Port** enables a networking configuration that does not use vPC or VLT. For compute-only templates that use a VMware ESXi operating system image, the **Switch Port Configuration** setting includes these options: ● **Port Channel** turns on vPC or VLT. ● **Port Channel (LACP enabled)** turns on vPC or VLT with the link aggregation control protocol enabled. ● **Trunk Port** enables a networking configuration that does not use vPC or VLT.
Teaming And Bonding Configuration	The teaming and bonding configuration options depend on the switch port configuration selected. For hyperconverged and compute-only templates, the following options are available: ● If you choose **Port Channel (LACP enabled)** or **Port Channel** as the switch port configuration, the only teaming and bonding option is **Route Based on IP hash**. ● If you choose **Trunk Port** as the switch configuration, there are three teaming and bonding options: ○ **Route Based on originating virtual port** ○ **Route Based on physical NIC load** ○ **Route Based on Source MAC hash** For storage-only templates, the following options are available:

Table 110. Node settings (continued)

Setting	Description
	• If you choose **Port Channel (LACP enabled)** as the switch port configuration, the only teaming and bonding option is **Mode 4 (IEEE 802.3ad policy)**. • If you choose **Trunk Port** as the switch configuration, you have the following teaming and bonding options: ○ **Mode 1 (active-backup policy)** ○ **Mode 6 (ALB Adaptive policy)** For a non-bonded storage-only template, you need to select **Trunk Port** for **Switch Port Configuration** setting, not **Port Channel (LACP enabled)**. For the **Teaming and Bonding Configuration** setting, you typically select **Mode 1** for most deployments.
Transmit Hash Policy	The option is available for storage-only and compute-only Linux nodes only when the **Teaming And Bonding Configuration** is set to Mode 4 (EEE 802 3ed policy). Mode 4 is the bond with LACP and supports the following load balancing options: • LACP in layer 2 & 3 • LACP in layer 3 & 4 LACP in layer 3 & 4 is the default option
Hardware Settings	
Target Boot Device	Specifies the target boot device. • **Local Flash Storage**: Installs the operating system to either the SATADOM or the BOSS flash storage device present in the node. With the **Local Flash Storage** option, only nodes with a BOSS storage controller and two attached hard drives or SATADOM are selected to be deployed as part of the resource group, depending on the PowerEdge servers used. For PowerEdge servers that support BOSS, during deployment PowerFlex Manager creates RAID 1 with the two hard drives attached to the BOSS controller. • **Local Flash Storage for Dell PowerFlex**: Installs the operating system to either the SATADOM or the BOSS flash storage device that is present in the node and configures the node to support PowerFlex. If you select the option to **Use Node for Dell PowerFlex** under **OS Settings**, the **Local Flash Storage for Dell PowerFlex** option is automatically selected as the target boot device.
Node Pool	Specifies the pool from which nodes are selected for the deployment.
BIOS Settings	
System Profile	Select the system power and performance profile for the node.
User Accessible USB Ports	Enables or disables the user-accessible USB ports.
Number of Cores per Processor	Specifies the number of enabled cores per processor.
Virtualization Technology	Enables the additional hardware capabilities of virtualization technology.
Logical Processor	Each processor core supports up to two logical processors. If enabled, the BIOS reports all logical processors. If disabled, the BIOS reports only one logical processor per core.
Execute Disable	Enables or disables running disable memory protection.
Node Interleaving	Enable or disable the interleaving of allocated memory across nodes.

Table 110. Node settings (continued)

Setting	Description
	• If enabled, only nodes that support interleaving and have the read/write attribute for node interleaving set to enabled are displayed. Node interleaving is automatically set to enabled when a resource group is deployed on a node. • If disabled, any nodes that support interleaving are displayed. Node interleaving is automatically set to disabled when a resource group is deployed on a node. Node interleaving is also disabled for a resource group with NVDIMM compression. • If not applicable is selected, all nodes are displayed irrespective of whether interleaving is enabled or disabled. This setting is the default.
Network Settings	
Multi-Network Selection	Select the checkbox to include multiple management networks of the same type. If you select multiple networks of the same type without selecting the checkbox, an error is displayed when you publish the template. The multiple network selection is supported on the following networks: • Hypervisor Management • PowerFlex Management • Hypervisor Migration • Replication Networks
Number of Replication Networks Per Node	This option is displayed only if the **Multi-Network Selection** and **Enable Replication** checkboxes are enabled. Select the number of networks you want to add to the port. For example, if the selected number is 2, you can assign one network each to two ports—Port 1 and Port 2. It is recommended that the number of selected networks is always even.
Add New Interface	Click **Add New Interface** to create a network interface in a template component. Under this interface, all network settings are specified for a node. This interface is used to find a compatible node in the inventory. For example, if you add **`Two Port, 100 gigabit`** to the template, when the template is deployed PowerFlex Manager matches a node with a two-port 100-gigabit network card as its first interface. To add one or more networks to the port, select **Add Networks to this Port**. Then, choose the networks to add, or mirror network settings defined on another port. To see network changes that are previously made to a template, you can click **View/Edit** under **Interfaces**. Or, you can click **View All Settings** on the template, and then click **View Networks**. To see network changes at resource group deployment time, click **View Networks** under **Interfaces**.
Add New Static Route	Click **Add New Static Route** to create a static route in a template. To add a static route, you must first select **Enabled** under **Static Routes**. A static route allows nodes to communicate across different networks. The static route can also be used to support replication in a storage-only or hyperconverged resource group. A static route requires a **Source Network** and a **Destination Network**, and a **Gateway**. The source and destination network must each be a PowerFlex data network or replication network that has the **Subnet** field defined. If you add or remove a network for one of the ports, the **Source Network** drop-down list does not get updated and still shows the old networks. In order to see the changes, save the node settings and edit the node again.
Validate Settings	Click **Validate Settings** to determine what can be chosen for a deployment with this template component.

Table 110. Node settings (continued)

Setting	Description
	The **Validate Settings** wizard displays a banner to when one or more resources in the template do not match the configuration settings that are specified in the template. The wizard displays the following tabs: • **Valid (*number*)** lists the resources that match the configuration settings. • **Invalid (*number*)** lists the resources that do not match the configuration settings. The reason for the mismatch is shown at the bottom of the wizard. For example, you might see `Network Configuration Mismatch` as the reason for the mismatch if you set the port layout to use a 100-Gb network architecture, but one of the nodes is using a 25 GB architecture. If you set the encryption method to use self-encrypting drives (SEDs), but the nodes do not have these drives, you might see `Self Encrypting Drives are required but not found on the node, or software encryption requested but only available drives are SED.`

After entering the information about operating system Installation (PXE) network in the respective field as described in the table above, PowerFlex Manager untags vLANs entered in the operating system installation network on the switch node facing port. For vMotion and hypervisor networks, PowerFlex Manager tags these networks on the switch node-facing ports for the entered information. For rack node, PowerFlex Manager configures the vLANs on node facing ports (untag PXE vLANs, and tag other vLANs).

If you select **Import Configuration from Reference Node**, PowerFlex Manager imports basic settings, BIOS settings, and advanced RAID configurations from the reference node and enables you to edit the configuration. Some BIOS settings might no longer apply once new BIOS settings are applied. PowerFlex Manager does not correct these setting dependencies. When setting advanced BIOS settings, use caution and verify that BIOS settings on the hardware are applicable when not choosing **Not Applicable** as an option. For example, when disabling an SD card, the settings for internal SD card redundancy become not applicable.

You can edit any of the settings visible in the template, but keep in mind that many settings are hidden when using this option. For example, only ten out of many BIOS settings that you can see and edit using template are displayed. However, you can configure all BIOS settings. If you want to edit any of the settings that are not visible through the template feature, edit them before importing or uploading the file.

For storage only and hyperconverged nodes, the required SDPM or NVDIMM storage capacity is specified in the following table.

Table 111. SDPM or NVDIMM storage capacity

Node storage capacity (in TB)	SDPM/NVDIMM capacity required (in GB)
Less than or equal to 46.08	32
Less than or equal to 92.16	64
Less than or equal to 122.88	96
Less than or equal to 145	128

Cluster component settings (mirroring)

This table describes the cluster component settings.

Table 112. Cluster component settings

Field name	Description
Select a Component	Select **VMware Cluster** or **PowerFlex Cluster** or **PowerFlex Management Platform**
Component Name	Indicates the cluster component name.
Related Components	Select **Associate All** or **Associate Selected** to associate all or specific components to the new component.
Cluster settings (for the VMware cluster component)	
Target Virtual Machine Manager	Select a virtual machine manager from the **Target Virtual Machine Manager** list.
Data Center Name	Select the data center name from **Data Center Name** list.
Cluster Name	Select a new cluster name from **Cluster Name** list.
New Cluster Name	Select a new cluster name from **New Cluster Name** list.
Cluster HA Enabled	Enables or disables a highly available cluster. You can either select or clear (default) the checkbox.
Cluster DRS Enabled	Enables or disables a distributed resource scheduler (DRS). You can either select or clear (default) the checkbox.
PowerFlex settings (for the PowerFlex cluster component)	
Target PowerFlex Gateway	Select a target PowerFlex gateway, which acts as an endpoint for PowerFlex API calls. During the provisioning process, PowerFlex Manager connects to the PowerFlex gateway and uses its APIs to configure all SDS and SDC parameters.
Protection Domain Name	Provide a name for the protection domain. You can automatically generate the name (recommended), or specify a new name explicitly. A protection domain is a logical entity that contains a group of SDSs that provide backup for each other. Each SDS belongs to only one protection domain. Each protection domain is a unique set of SDSs. A protection domain may also contain SDTs and SDRs. If you automatically generate the protection domain name or specify a new name explicitly for a hyperconverged or storage-only template, the PowerFlex cluster must have at least three nodes before you can publish the template and deploy it as a resource group. However, if you select an existing protection domain that is associated with another previously deployed hyperconverged or storage-only resource group, and this protection domain has at least three nodes, PowerFlex Manager recognizes the new template as valid if the cluster has fewer than three nodes. You can publish the template successfully and deploy it as a resource group, since the protection domain it uses has enough nodes. **NOTE:** A compute-only resource group only requires a minimum of two nodes, since it does not have a protection domain. If you choose **Compute Only** as the **PowerFlex Role** for the node component, the **PowerFlex** settings do not include the **Protection Domain Name** field.
New Protection Domain Name	Specify a new name for the protection domain.
Protection Domain Name Template	If you choose to generate the protection domain name automatically, PowerFlex Manager fills this field with a default template that combines static text with variables for several pieces of the autogenerated name. If you modify the template, be sure to include the *${num}* variable to ensure that the name is unique. For details on the rules for defining a template, see the contextual help that appears when you hover over the field. If you choose **Compute Only** as the **PowerFlex Role** for the node component, the **PowerFlex** settings do not include the **Protection Domain Name Template** field.

Table 112. Cluster component settings (continued)

Field name	Description
Acceleration Pool Name	Provide a name for the acceleration pool. You can automatically generate the name (recommended) or select from a list of existing acceleration pools. You can add acceleration pools to a protection domain to accelerate storage pool performance. An acceleration pool is a group of acceleration devices within a protection domain.
Acceleration Pool Name Template	Define a template for generating the acceleration pool name automatically. PowerFlex Manager fills this field with a default template that combines static text with variables for several pieces of the autogenerated name. If you modify the template, be sure to include the *$(num)* variable to ensure that the name is unique. For details on the rules for defining a template, see the contextual help that appears when you hover over the field.
Storage Pool Name	Provide a name for the storage pool. You can automatically generate the name (recommended), select from a list of existing storage pools for the selected protection domain, or specify a unique storage pool name. If you choose to automatically generate a name, you are prompted to define a storage pool name template. Storage pools allow the generation of different storage tiers in PowerFlex. A storage pool is a set of physical storage devices in a protection domain. Each storage device belongs to one (and only one) storage pool. The number of storage pools that are created at deployment time depends on the number of disks available.
Number Of Storage Pools	Select up to six storage pools per protection domain. The maximum number of nodes that can be added to a protection domain is 32. The maximum number of drives that can be added to a storage pool is 320.
Granularity	Set the granularity for compression by selecting **Fine** or **Medium**. The granularity setting applies to the storage pool.
Storage Pool Disk Type	Allows you to select the disk type - hard drive, SSD, or NVMe.
Storage Pool Name Template	Define a template for generating the storage pool name automatically. PowerFlex Manager fills this field with a default template that combines static text with variables for several pieces of the autogenerated name. If you modify the template, be sure to include the *${num}* variable to ensure that the name is unique. For details on the rules for defining a template, see the contextual help that appears when you hover over the field. If you choose **Compute Only** as the **PowerFlex Role** for the node component, the **PowerFlex** settings do not include the **Storage Pool Name Template** field.
Enable Fault Sets	Enables deployment with fault sets. This option allows you to deploy a fault set enabled resource group with a new protection domain or an existing protection domain.
Fault Set Name	Specify the fault set name. You can automatically generate the name (recommended), select from a list of existing fault sets for the selected protection domain, or specify a unique fault set name. If you choose to automatically generate a name, you are prompted to define a fault set name template.
Fault Set Name Template	If you choose to generate the fault set name automatically, PowerFlex Manager fills this field with a default template that combines static text with variables for several pieces of the autogenerated name. If you modify the template, be sure to include the *${num}* variable to ensure that the name is unique.

Table 112. Cluster component settings (continued)

Field name	Description
	For details on the rules for defining a template, see the contextual help that appears when you hover over the field.
Number of Fault Sets	Specifies the number of fault sets to create at deployment time. PowerFlex requires a minimum of three fault sets for a protection domain, with at least two nodes in each fault set. For a new protection domain name, you must specify a number 3–16. Each fault set acts as a single fault unit. If a fault set goes down, all nodes within the fault set go down as well. For an existing protection domain name, you must specify a number between 1 to 16 . This allows you to add more fault sets to an existing protection domain. If the selected protection domain already has three fault sets, for example, you can specify a number as low as 1, to include an additional fault set for this protection domain. PowerFlex Manager ensures that each new deployment has only one MDM role for each fault set. For example, if you deploy three fault sets, one has the primary MDM, another has the secondary MDM, and the third has the tiebreaker. You can use the **Reconfigure MDM Roles** wizard to change the MDM role assignments after deployment.
CloudLink Center Settings	
Target CloudLink Center	Specifies the target CloudLink Center for an encrypted node. The **Cloud Link Center Settings** are only available when the **Enable Encryption** option is selected in the **OS Settings** for the node component. For any CloudLink Center that has encryption, you must either add an existing resource group to PowerFlex Manager, or use PowerFlex Manager to create the CloudLink encryption machine groups and keystores. PowerFlex Manager is only able to create the machine groups and keystores if the CloudLink Center exists in the management cluster. Ensure that CloudLink Center is available on the resource page. If you select an existing protection domain under the **PowerFlex Settings**, you must choose a **Target CloudLink Center**. The other CloudLink Center settings are not available, and PowerFlex Manager uses the machine group and keystore settings for the selected target. If you automatically generate the protection domain name or specify a new protection name explicitly, you can set the **Machine Group Name** and **Keystore Name** settings with the template. When you deploy a template with CloudLink settings, PowerFlex Manager creates the machine groups and keystores, and then assigns the machine groups to keystores and networks. Next, it adds the machines to the machine groups, and encrypts the drives. After deployment, you can see the machines that are added to machine groups by clicking **View Details** on the **Resources** page. When defining a template, you choose a single CloudLink Center as the target for the deployed resource group. If the CloudLink Center for the resource group shuts down, PowerFlex Manager loses communication with the CloudLink Center. If the CloudLink Center is part of a cluster, PowerFlex Manager moves to another CloudLink Center when you update the resource group details.
Machine Group Name	Allows you to provide a name for the machine group. The **Machine Group Name** is only available if you automatically generate the protection domain name or specify a new protection name explicitly. You can automatically generate the machine group name or specify a new machine group name explicitly.
New Machine Group Name	Specify a new name for the machine group.

Table 112. Cluster component settings (continued)

Field name	Description
Keystore Name	Allows you to provide a name for the keystore. The **Keystore Name** is only available if you automatically generate the protection domain name or specify a new protection name explicitly. You can automatically generate the keystore name or specify a new keystore name explicitly. Alternatively, you can choose an existing keystore from the list.
Keystore Name Template	Specify a new name for the keystore.
vSphere VDS Settings	
Configure VDS Settings	Click **Configure VDS Settings** to create the virtual distributed switch (VDS) settings for the VMware cluster component in a template. PowerFlex Manager displays a wizard that allows you to provide a name for each VDS. The wizard also lets you specify how you want the port group names to be created. You can let PowerFlex Manager automatically create port group names based on the names of the networks, or you can enter the port group names yourself. To see VDS settings previously configured for a template, you can click **View VDS Settings** under **vSphere VDS Settings**. **NOTE:** When working with VMware vCenter in a multi data center configuration, VMware vSphere distributed switch names (vDS) must be unique. Work with the *Logical Configuration Survey* contact to plan vDSs based on the data center design.

Related information

Add cluster component settings to a template (mirroring)

VM settings

The table describes VM settings for the CloudLink Center and the PowerFlex gateway.

Table 113. VM settings

Field name	Description
Select a Component	Select **CloudLink Center** or **PowerFlex Gateway**.
Number of Instances	Select the number of instances of the VM that you want to deploy.
Related Components	Select **Associate Selected** and select the **VMware Cluster** checkbox to associate the selected VMware cluster with the CloudLink Center.
VM Settings	
Specify Datastore	Specify the storage location.
Specify Network	Select the network type.
CloudLink Settings	
Host Name Selection	Defines how the hostname is selected at deployment time. There are two options: • Auto Generate - Select to autogenerate the hostname at deployment time. • Specify at Deployment Time - Select to provide a unique hostname at deployment time.
OS Credential	Select the operating system credential used to set the username and password on the operating system that is installed.
NTP Server	Specify the IP address of the NTP server for time synchronization. If adding more than one NTP server, ensure that you separate the IP addresses with commas.

Table 113. VM settings (continued)

Field name	Description
Secadmin Credential	Select the secadmin credential, an element manager credential, which is used to set the password for the secadmin user of the CloudLink Center. The password should be minimum of ten characters with at least one special character.
Vault Password	Specify up to three unique passwords for the CloudLink Vault. You must specify at least one. The password should be minimum of ten characters with at least one special character.
Confirm Vault Password	Retype the vault password.
License File	Upload a new or additional software license file. If additional, the resource is added to your current total resources.
Additional CloudLink Settings	
Configure Syslog Forwarding	Select the checkbox to send syslog messages from the CloudLink to a remote network management system.
Syslog Facility	Select the remote server where the syslog messages are forwarded to.
Configure Email Notifications	Select the checkbox to configure email alerts.
Server Address	Specify the IP address of the email server.
Port	Specify the port number for the email server. The default port is 25. The port numbers must be entered in comma-separated list and must be 1–65535.
Sender Address	Specify the email address for the sender.
User Name	Specify the required username.
Password	Specify the required password.
Confirm Password	Retype the password.
PowerFlex Gateway Settings	
Host Name Selection	Defines how the hostname is selected at deployment time. There are two options: ● Auto Generate - Select to autogenerate the hostname at deployment time. ● Specify at Deployment Time - Select to provide a unique hostname at deployment time.
PowerFlex Credential	Select the operating system credential used to set the username and password.
NTP Server	Specify the IP address of the NTP server for time synchronization. If adding more than one NTP server, ensure that you separate the IP addresses with commas.

Support for full and partial network automation (mirroring)

This topic lists the supported configurations for full and partial network automation.

Table 114. Supported configurations for full and partial network automation

Template components	Full network automation	Partial network automation (Requires manual switch configuration)
OS Images	VMware ESXi, Red Hat Enterprise Linux	VMware ESXi, Red Hat Enterprise Linux
PowerFlex roles	Hyperconverged, compute-only, storage-only	Hyperconverged, compute-only
Switch Port Configuration	● Port channel (LACP enabled) ● Port channel ● Trunk port	● Port channel (LACP enabled) ● Trunk Port

Table 114. Supported configurations for full and partial network automation (continued)

Template components	Full network automation	Partial network automation (Requires manual switch configuration)
Target Boot Device	• Local flash storage for Dell PowerFlex	Local flash storage for Dell PowerFlex
Network Settings	• Static bonding NIC port design or LACP bonding NIC port design • 10 GB, 25 GB, 100 GB • Required PXE network • Network Automation Type: Full	• LACP bonding NIC port design • 25 GB • No PXE network (using iDRAC virtual media) • Network Automation Type: Partial

Related information

Adding an existing resource group (mirroring)
Build and publish a template

Resources

This section provides reference information for the **Resources** page.

Related information

Deploying and provisioning
Managing components
Monitoring system health
Managing components
Monitoring system health

Resource health status

PowerFlex Manager assigns health status to the resources based on the conditions described in the following table.

Table 115. Resource health status

Icon	Health status	Description
	Healthy	Indicates that there is no issue with the resource and it is working as expected.
	Warning	Indicates that the resource is in a state that requires corrective action, but does not affect overall system health. For example, the firmware running on the resource is not at the required level or not compliant. For supported switches, a warning health status might be because the SNMP community string is invalid or not set correctly. In this case, PowerFlex Manager is unable to perform health monitoring for the switch. **NOTE:** If the resource group has VMware ESXi and is set to maintenance mode, a message is displayed in the **View Details** pane.
	Critical	Indicates that an issue requiring immediate attention exists in one of the following hardware or software components in the device: • Battery • CPU • Fans

Table 115. Resource health status (continued)

Icon	Health status	Description
		• Power supply • Storage devices • Licensing For supported switches, a critical health status might indicate that the power supply is not working properly or a CPU has overheated. **NOTE:** If the resource group is in a power off state, a message displays in the **View Details** pane.
	Unknown	Indicates that the resource status is unknown. For example, for a supported switch, an unknown status might indicate that the switch has been turned off or is still being discovered.
	Pending	Indicates that the resource status is pending.
	Service Mode	Indicates that the resource is in **Service Mode**. **NOTE:** If the resource fails to exit **Service Mode**, an informational message displays in the recent activity.

Related information

Discover a resource
Removing resources
Discover a resource
Exporting a compliance report for all resources

Resource operational state

After initiating resource discovery, PowerFlex Manager assigns one or more of the states to the resources. These operational states display on the **Resources** page, in the **Deployment Status** column of the **All Resources** tab.

Table 116. Deployment status

Deployment status	Description
Not In Use	Resource is available for deployment.
Available	Resource is available. Applies only to switches, virtual machine, or gateway.
Deploying	Resource is in the process of being deployed in a resource group.
In Use	Resource is deployed in a resource group.
Pending Updates	One or more of the following tasks are in progress: • Discovering resource • Determining resource details, including firmware version • Applying template to the resource • Updating firmware • Removing resource from PowerFlex Manager inventory
Deployment Failed	Resource group deployment failed.

Table 116. Deployment status (continued)

Deployment status	Description
Cancelled	Resource group deployment is cancelled.

PowerFlex Manager keeps track of which resources it is managing. These operational states display on the **Resources** page, in the **Managed State** column of the **All Resources** tab.

Table 117. Managed state

Managed state	Description
Managed	Indicates that PowerFlex Manager manages the firmware on that node, and the node can be used for deployments.
Unmanaged	Indicates that the resource is not managed by PowerFlex Manager. By default, the operational state for all discovered nodes is **Unmanaged**. If you want to perform firmware updates or deployments on a discovered node, you need to select the node and change the operational state to **Managed**. If you have not yet uploaded a license file, PowerFlex Manager is configured for monitoring and alerting only. All resources are restricted to the **Unmanaged** resource state, and you cannot change the state to **Managed** or **Reserved**.
Reserved	Indicates that PowerFlex Manager only manages the firmware on that particular node, but that node cannot be used for deployments. You can assign a host to the reserved state only if the host has been discovered, but is not part of a resource group.

Compliance status

PowerFlex Manager assigns one of the following firmware statuses to the resources:

Table 118. Compliance status

Firmware status	Description
Compliant	The firmware running on the resource is compliant with the firmware version that is specified in the default compliance version.
Non–Compliant	The firmware running on the resource is less than or greater than the firmware version specified in the default compliance version. Indicates that firmware update is required.
Update Required	The firmware running on the resource is less than the minimum firmware version recommended in the default compliance version. Indicates that firmware update is required.

Related information

Discover a resource
Removing resources
Discover a resource
Exporting a compliance report for all resources

Discovery overview

You can discover new resources or existing resources that are already configured within your environment. After discovery, you can deploy resource groups on these resources from a template. Only administrator-level users can discover resources.

By default, the operational state for all discovered nodes is **Unmanaged**. If you want to perform firmware updates or deployments on a discovered node, select the node and change the operational state to **Managed**.

If you have not yet uploaded a license, PowerFlex Manager is configured for monitoring and alerting only. In this case, all the resources are restricted to the **Unmanaged** resource state, and you cannot change the state to **Managed** or **Reserved**.

For some resources such as nodes, the default credentials are prepopulated in PowerFlex Manager. If the credentials are changed from the defaults, add the credential to PowerFlex Manager with the new login information.

Element Manager discovery

The IPI cabinet is discovered as an element manager. Once PowerFlex Manager discovers the IPI cabinet, you can view cabinet details or you can open the IPI appliance management application to view more details.

The CloudLink Center is discovered as an element manager. If you create your own username in CloudLink Center, you need the SecAdmin access role and the local user type for the new user to complete discovery. The client user type does not work for discovery in PowerFlex Manager. If one member of a cluster of CloudLink Center is shut down, the inventory still succeeds because PowerFlex Manager falls back to inventorying the other CloudLink Center members if the first one is down.

Node discovery

PowerFlex Manager supports PowerFlex node discovery and allows you to onboard nodes by configuring the initial management IP address and iDRAC credentials. To perform initial discovery and configuration, verify that the management IP address is set on the node and that PowerFlex Manager can access the IP address through the network. While configuring IP addresses on the node, verify that PowerFlex Manager can access any final IP address in a range used for hardware management, to complete discovery of these nodes.

PowerFlex Manager also allows you to use name-based searches to discover a range of nodes that were assigned IP addresses by DHCP to iDRAC. You can search for a range of DNS hostnames or a single hostname within the **Discovery Wizard**. After you perform a name-based discovery, PowerFlex Manager operations to the IDRAC continue to use name-based IP resolution, since DHCP may assign alternate addresses.

Switch discovery

If you attempt to discover a Cisco switch with terminal color configured, the discovery fails. To discover the switch successfully, disable the terminal color option by running `configure no terminal color persist`.

VM manager discovery

A VMware vCenter is discovered as a VM manager. PowerFlex Manager users with the administrator role can discover VMware vCenter in PowerFlex Manager. A VMware vCenter read-only user can discover VMware vCenter in PowerFlex Manager only after the following requirements are met:

- The VMware vCenter user who is specified in the VMware vCenter credential is granted the `VirtualMachine.Provisioning.ReadCustSpecs` and `StorageProfie.View` privileges.
- The permission containing these privileges is granted to that user on the root VMware vCenter object and the `Propogate to children` property is set to `True`.

PowerFlex Manager allows you to deploy new hyperconverged and compute-only resource groups and add existing resource groups to VMware vCenter that has an Enhanced Linked Mode (ELM) configuration. ELM connects multiple VMware vCenter servers, allowing you to search across servers and perform other VMware vCenter management functions. ELM does not provide clustering or redundancy.

PowerFlex Manager allows you to deploy new compute-only resource groups and add existing resource groups to VMware vCenter that has an Enhanced Linked Mode (ELM) configuration. ELM connects multiple VMware vCenter servers, allowing you to search across servers and perform other VMware vCenter management functions. ELM does not provide clustering or redundancy.

If you are working with a VMware vCenter that has an enhanced linked mode configuration, you must discover only the VMware vCenter to which you want to deploy or add an existing resource group. You do not need to discover the other VMware vCenters that are connected through the enhanced linked mode configuration.

Node pools

In PowerFlex Manager, a node pool is made up of nodes that are grouped for specific use cases such as business units or workload purposes. An administrator can specify which users can access these node pools.

Standard users can view details only for node pools for which they have permission.

From the **Node Pools** tab, you can view existing node pools.

Users with an administrator role can create, edit, or delete node pools.

Click a node pool from the list to view detailed information about the following tabs:

- **Nodes**: Displays the number of nodes that are associated with the node pool.
- **Users**: Displays the number of users with access rights to the node pool.

Configuration checks (mirroring)

When you initiate a resource group action, PowerFlex Manager performs critical configuration checks to ensure that the system is healthy before proceeding. A configuration check failure may result in the resource group action being blocked. You can export a PDF report at any time that shows detailed system health and configuration data. When you generate the report, PowerFlex Manager initiates the full set of configuration checks and reports the results.

Configuration checks for switches

Dell Networking OS9 switches are not covered by the configuration checks.

Table 119. Configuration checks

Configuration check	Resource group actions blocked if the check fails
PowerFlex gateway is powered off	None
PowerFlex cluster state is not normal	• Add node • Add volume • Delete node • Delete resource group • Add resource group with an existing PowerFlex cluster • Compliance upgrade • Enter service mode • Drive replacement
Secondary MDMs have a status not equal to normal	• Add nodes • Add volumes • Delete node • Delete resource group • Add resource group with an existing PowerFlex cluster • Compliance upgrade • Enter service mode • Drive replacement
SDS registered data network IP address count does not equal 2 or 4	• Compliance upgrade • Enter service mode • Drive replacement • Reconfigure MDM roles
SDS has a state not equal to connected	• Add volume

Table 119. Configuration checks (continued)

Configuration check	Resource group actions blocked if the check fails
	• Compliance upgrade • Reconfigure MDM roles
SDS interfaces have incorrect MTU sizes	• Add node • Add resource group with an existing PowerFlex cluster • Compliance upgrade • Reconfigure MDM roles
VMware ESXi vmks are not configured with an MTU size of 9000	• Add node • Add resource group with an existing PowerFlex cluster • Compliance upgrade
Interfaces are not configured with an MTU size of 9216, and QOS is enabled with MTU size	• Add node • Add resource group with an existing PowerFlex cluster • Compliance upgrade
SDS has a Performance Profile not equal to High Performance	None
SDC has a Performance Profile not equal to High Performance	None

Settings

This section provides reference information for the **Settings** page.

Backup details

PowerFlex Manager backup files include the following information:

- Activity logs
- Credentials
- Deployments
- Resource inventory and status
- Events
- Initial setup
- IP addresses
- Jobs
- Licensing
- Networks
- Templates
- Users and roles
- Resource module configuration files
- Performance metrics

Network types (mirroring)

You can manage various network types in PowerFlex Manager.

- **General Purpose LAN**—Used to access network resources for basic networking activities.
- **Hypervisor Management**—Used to identify the management network for a hypervisor or operating system that is deployed on a node.
- **Hypervisor Migration**—Used to manage the network that you want to use for live migration. Live migration enables you to move running virtual machines from one node of the failover cluster to different node in the same cluster.
- **OS Installation**—Allows static or DHCP network for operating system imaging on nodes.
- **Hardware Management**—Used for out-of-band management of hardware infrastructure
- **PowerFlex Data**—Used for data traffic between storage data servers (SDS) and storage data clients (SDC)

- **PowerFlex Data (Client Traffic Only)**—Used for storage data client traffic only
- **PowerFlex Data (Server Traffic Only)**—Used for storage data server traffic only
- **PowerFlex Replication**—Used to support PowerFlex replication
- **PowerFlex Management**—Used for PowerFlex system management